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Montenegrosport - Podgorica

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## **EVROBAROMETAR – FIZIČKA AKTIVNOST STANOVNIŠTVA AP VOJVODINA**

### **UVOD**

Fizičko vežbanje dospelo je na listu aktuelnih problema savremene Evrope, ne samo kao potvrđeni metod poboljšanja zdravlja i fizičke kondicije u zemljama EU, nego i zbog sve veće uloge sporta u društvenom povezivanju i njegove edukativne vrednosti. Sagledavanju, merenju i postizanju poželjnog, normalnog nivoa fizičke aktivnosti stanovništva, posvećen je u EU napor širokog kruga istraživača kroz mnoge projekte. Jedan od takvih projekata je i Evrobarometar projekat, koji je zasnovao metodologiju statističkog praćenja obima vežbanja, motivacije i prepreka, uticaja demografskih faktora na sport i fizičke aktivnosti u zemljama Evropske unije<sup>1</sup>. Pomenuti projekat zaslužuje primenu u empirijskim istraživanjima fizičke aktivnosti i sporta u zemljama koje stupaju na put evropskih integracija, kakva je današnja Srbija, posebno njen region Vojvodina. Značaj navedenog istraživanja posebno se očituje u činjenici da stanovništvo savremenog sveta usled tehnološkog razvoja odlikuje manji stepen fizičke aktivnosti nego ikada pre – čak 2/3 populacije nedovoljno je fizički aktivno<sup>2</sup>. Takođe, normalni nivo fizičkog vežbanja, bez obzira na njegov organizacioni oblik (spontano vežbanje, sistematski stručno nadzirani trening, itd.), predstavlja jedan od najvažnijih uslova održivog zdravlja stanovništva u uslovima savremenog življenja.

### **PROBLEM I CILJ ISTRAŽIVANJA**

Problem istraživanja obuhvatio je uzrasne karakteristike fizičke aktivnosti stanovništva AP Vojvodine, sa posebnim osvrtnom na sledeća pitanja: koliko je fizički aktivno stanovništvo AP Vojvodine i koji stanovnici AP Vojvodine se najviše (i najmanje) bave fizičkim (sportskim) aktivnostima; kakav je kontekst u kojem ljudi vežbaju – da li se zvanično bave sportom ili su aktivni na neki drugi, manje formalan način; kakva je uporedna međugeneracijska „slika“ fizičkog vežbanja stanovništva regiona AP Vojvodina i evropskog okruženja (EU)?

Cilj ovog istraživanja je identifikovanje uzrasnih karakteristika fizičke aktivnosti stanovnika AP Vojvodina i njihovih uporednih vrednosti u odnosu stanovnike Evropske unije.

<sup>1</sup> Special Eurobarometer (2004): The Citizens of the European Union and Sport.

<sup>2</sup> Trost, G., Owen, N., Bauman, E., Sallis, F., & Brown W. (2002) Correlates of adults' participation in physical activity: review and update. *Medicine and Science in Sports and Exercise*, 34, 1996-2001

## METOD

Uzorakom ispitanika obuhvaćeno je ukupno 3132 ispitanika iz većih vojvođanskih mesta (Novi Sad, Subotica, Zrenjanin, Sombor, Kikinda, Sremska Mitrovica, Ruma, Šid, Vrbas, Indija, Kula, Apatin, Odžaci, Titel, Sečanj, Bačka Palanka, Beočin, Temerin, Bečej, Senta), podeljenih na pet subuzoraka: uzrast 15 – 24 godine, 25 – 39 godina, 40 – 54 godine, 55 – 69 godina i 70 i više godina. Prostor istraživanja ograničen je na četiri međusobno povezane tematske celine koje se odnose na: stav ispitanika o nivou i intenzitetu fizičke aktivnosti, stav ispitanika o mestu bavljenja fizičkom aktivnošću, stav ispitanika o motivima bavljenja fizičkom aktivnošću i stav ispitanika o barijerama bavljenja fizičkom aktivnošću. Instrument prikupljanja informacija za ovo istraživanje konstruisan je na osnovu upitnika projekta Eurobarometer (2010)<sup>3</sup>.

Prikupljenji podaci su najpre sređeni odvojeno za svaki od pet subuzoraka prema tematskim celinama, a potom obrađeni odgovarajućim matematičko – statističkim postupcima i to: multivarijantnom analizom varijanse (MANOVA) i diskriminativnom analizom, odnosno, univarijantnih postupaka kao što su Roy-ov test, Pirsonov koeficijent kontingencije ( $\chi$ ) i koeficijent multiple korelacije (R).

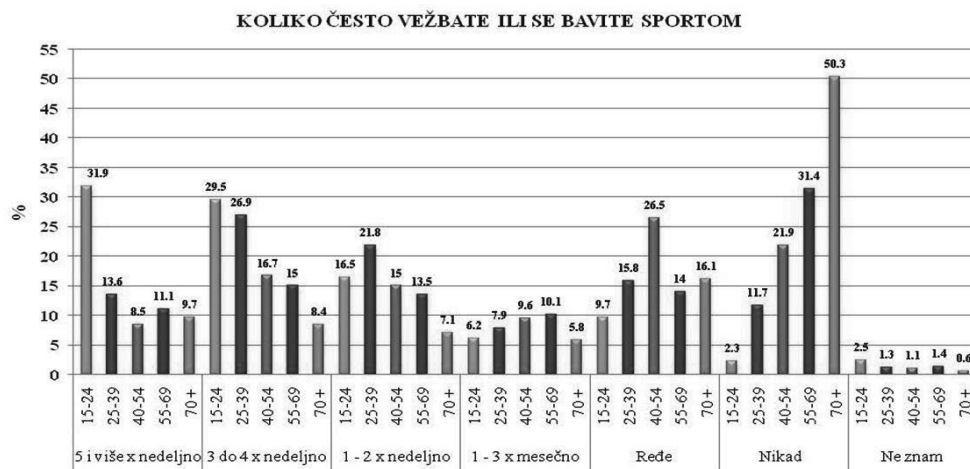
## REZULTATI

U ovom istraživanju analizirana je razlika između pet subuzoraka u okviru četiri tematske celine u odnosu na: stav ispitanika o nivou i intenzitetu fizičke aktivnosti (grafikon 1 i 2), stav ispitanika o mestu bavljenja fizičkom aktivnošću (grafikon 3), stav ispitanika o motivima bavljenja fizičkom aktivnošću (grafikon 4) i stav ispitanika o barijerama bavljenja fizičkom aktivnošću (grafikon 5).

Inspekcijom grafikona 1 moguće je zapaziti sledeće: Kod uzrasta 15-24 godine najviše je zastupljen stav **5 i > x nedeljno (31.9%)** što je značajno veće od učestalosti stava 3 do 4 x nedeljno (29.5%), stava 1 do 2 x nedeljno (16.5%), stava ređe (9.7%), stava 1 do 3 x mesečno (6.2%), stava ne znam (2.5%) i stava nikad (2.3%); Kod uzrasta 25-39 godina zastupljenost stava **3 do 4 x nedeljno (26.9%)**, je značajno veća od učestalosti stava ređe (15.8%), stava 5 i > x nedeljno (13.6%), stava nikad (11.7%), stava 1 do 3 x mesečno (7.9%) i stava ne znam (1.3%); Kod uzrasta 40-54 godine zastupljenost stava **ređe (26.5%)**, je značajno veća od učestalosti stava 3 do 4 x nedeljno (16.7%), stava 1 do 2 x nedeljno (15.0%), stava 1 do 3 x mesečno (9.6%), stava 5 i > x nedeljno (8.5%) i stava ne znam (1.1%); Kod uzrasta 55-69 godina zastupljenost stava **nikad (31.4%)**, je značajno veća od učestalosti stava 3 do 4 x nedeljno (15.0%), stava ređe (14.0%), stava 1 do 2 x nedeljno (13.5%), stava 5 i > x nedeljno (11.1%), stava 1 do 3 x mesečno (10.1%) i stava ne znam (1.4%); Kod uzrasta 70 i više godina zastupljenost stava **nikad (50.3%)**, je značajno veća od učestalosti stava ređe (16.1%), stava 5 i > x nedeljno (9.7%), stava 3 do 4 x nedeljno (18.4%), stava 1 do 2 x nedeljno (7.1%), stava 1 do 3 x mesečno (5.8%) i stava ne znam (.6%).

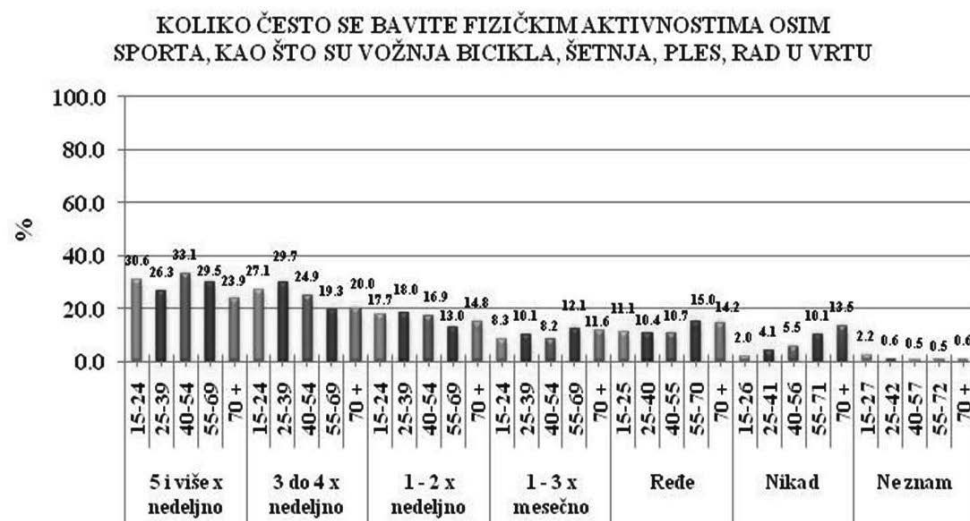
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<sup>3</sup> Special Eurobarometer, 334 (2010): Sport and Physical Activity, European Commission



Grafikon 1

Uvidom u fizičku aktivnost na nivou celog istraživnog uzorka zapaža se da redovno bavljenje sportom (pet puta nedeljno ili više) karakteriše 15% stanovništva u AP Vojvodina i da ona konstantno opada posle 24 godine života. Takođe se zapaža da se uzrast 15-24 godine najredovnije bavi sportom (32%), dok se kod uzrasta 70 i više godina čak polovina ispitanika nikada ne bavi sportom.



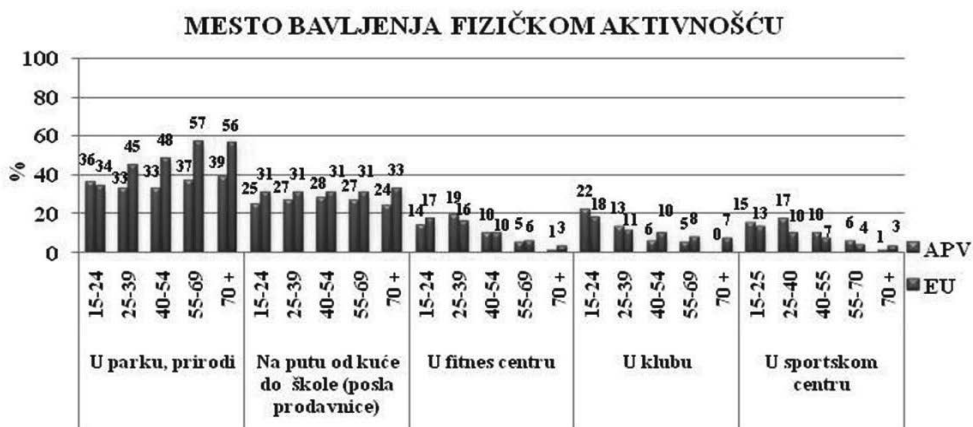
Grafikon 2

Inspekcijom grafikona 2 moguće je zapaziti sledeće: Kod uzrasta 15-24 godine najviše je zastupljen stav **5 i > x nedeljno (30.6%)**, što je značajno veće od učestalosti

stava 3 do 4 x nedeljno (27.1%), stava 1 do 2 x nedeljno (17.7%), stava ređe (11.1%), stava 1 do 3 x mesečno (8.3%), stava ne znam (2.2%) i stava nikad (2.0%); Kod uzrasta 25-39 godina zastupljenost stava **3 do 4 x nedeljno (29.7%)**, je značajno veća od učestalosti stava 5 i > x nedeljno (26.3%), stava 1 do 2 x nedeljno (18.0%), stava ređe (10.4%), stava 1 do 3 x mesečno (10.1%), stava nikad (4.1%) i stava ne znam (.6%); Kod uzrasta 40-54 godine zastupljenost stava **5 i > x nedeljno (33.1%)**, je značajno veća od učestalosti stava 3 do 4 x nedeljno (24.9%), stava 1 do 2 x nedeljno (16.9%), stava ređe (10.7%), stava 1 do 3 x mesečno (8.2%), stava nikad (5.5%) i stava ne znam (.5%); Kod uzrasta 55-69 godina zastupljenost stava **5 i > x nedeljno (29.5%)**, je značajno veća od učestalosti stava 3 do 4 x nedeljno (19.3%), stava ređe (15.0%), stava 1 do 2 x nedeljno (13.0%), stava 1 do 3 x mesečno (12.1%), stava nikad (10.1%), i stava ne znam (.5%); Kod uzrasta 70 i više godina zastupljenost **stava 5 i > x nedeljno (23.9%)**, je značajno veća od učestalosti stava 1 do 2 x nedeljno (14.8%), stava ređe (14.2%), stava nikad (13.5%), stava 1 do 3 x mesečno (11.6%) i stava ne znam (.6%).

Pregledom fizičke aktivnosti na nivou celog ispitivanog uzorka uočljivo je da se redovnim - „neformalnim“ fizičkim vežbanjem (više od 1-2 x nedeljno) bavi 29% stanovnika AP Vojvodine, dok se identičan procenat (29%) zapaža i kod onih koji se ili „nikada“ ne bave „neformalnom“ fizičkom aktivnošću, „ređe“ ili to čine vrlo retko (jedan do tri puta mesečno). Procenat ispitanika koji se redovno bave fizičkim aktivnostima, posebno u starijem uzrastu pokazuje da se stečene navike redovnog vežbanja u mladosti, zadržavaju tokom celog života, ali i da privremeni pad fizičke aktivnosti u uzrastu 25-39 godina verovatno potiče od preraspodele slobodnog u radno vreme, usled intenziviranja aktivnosti u zadovoljenju porodičnih (rađanje i podizanje dece) stambenih (stan, kuća) i socijalno-statusnih potreba (automobil).

Uvidom u grafikone 3, 4 i 5, pored pregleda mesta, motiva i barijera fizičke (ne)aktivnosti stanovništva AP Vojvodina, od posebnog značaja je i njihova uporedna analiza sa rezultatima navedenih obeležja u evropskom okruženju.



Grafikon 3

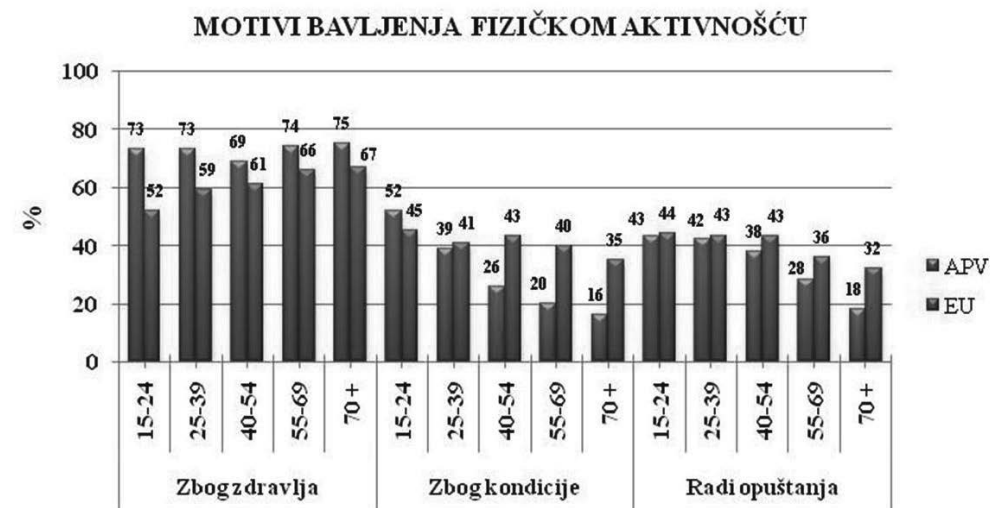
Inspekcijom grafikona 3 zapaža se da je na nivou celog uzorka ispitanika i uporednih pokazatelja stanovništva Evropske unije najviše zastupljeno **vežbanje u parku, odnosno prirodi** (APV 36%, EU 48%). Iza toga sledi vežbanje na putu od i do škole, posla ili prodavnice, (APV 26%, EU 31%). Manji deo stanovništva voli da vežba u zatvorenim sportskim objektima: u fitnes centru (APV 10%, EU 11%), zatim u klubu (APV 9%, EU 11%) i sportskom centru (APV 10%, EU 8%). U svim pomenutim sredinama, osim u sportskom centru, stanovništvo EU više procentualno vežba od stanovništva APV.

Gotovo da je identična situacija i kada su u pitanju pojedine uzrasne grupe stanovništva i vežbanje na otvorenom (u parku – prirodi i na putu od kuće do škole – posla, prodavnice), sa izuzetkom uzrasne grupe 15-24 godine, kada je u pitanju vežbanje u parku – prirodi, gde „prednost“ u odnosu na evropske vršnjake ima navedena uzrasna grupa stanovništva APV. Fitnes centar je privlačniji za tri uzrasne grupe stanovništva EU (15–24 godine, 55–69 godina, 70 i više godina). Za uzrast 25–39 godina fitnes centar je omiljeniji kod stanovništva APV, dok je kod uzrasne grupe 40-54 godine taj „odnos“ vežbanja u fitnes centru izjednačen. Vežbanje u klubu je više omiljeno kod mlađih uzrasnih grupa stanovništva APV (15 – 24 godine, 25 -39 godina), dok je obrnuto kod starijih uzrasnih grupa (40 – 54 godine, 55 – 69 godina i 70 i više godina), gde u klubu u proseku nešto više vežbaju stanovnici EU. U sportskom centru više vežbaju sve uzrasne grupe stanovništva APV, sa izuzetkom najstarije uzrasne grupe, gde određenu „prednost“ ima najstarija uzrasna grupa stanovništva EU.

Različite afinitete prema ambijentu u kome vežbaju imaju sve ispitivane (APV) i uporedne (EU) uzrasne grupe stanovništva, pa tako sa starenjem, uglavnom raste afinitet ka otvorenom prostoru za vežbanje (parkovi ili drugi otvoreni prostori) i obrnuto, sa starenjem konstantno slabi interes za vežbanje u fitnes centrima, klubovima i sportskim centrima, što se u značajnoj meri može pripisati „naturalizaciji“ potreba za kretanjem u kome dominiraju aerobne aktivnosti, kao sredstvo u realizaciji zdravstvenih, posebno „relaksacionih“ motiva.

Pregledom grafikona 4, u mnoštvu motiva koji se navode kao individualni razlozi za vežbanje, većina stanovništva, u oba uporedna uzorka, navodi **zdravlje** (APV 73%, EU 61%). Drugi razlog za vežbanje, ispitanici EU (41%) ističu „zbog kondicije“, a ispitanici APV (34%) navode „zbog opuštanja“. Na trećem mestu, kao najčešći razlog za fizičku aktivnost nalazi se „zbog opuštanja“ za ispitanike EU (40%), odnosno „zbog kondicije“ za ispitanike APV (31%). Pored pomenuta tri razloga zbog kojih ispitanici najčešće vežbaju, značajni su: „zbog zabave“, „zbog boljeg izgleda“, „zbog izdržljivosti“, „zbog regulisanja težine“, „zbog druženja“, „da bih usporio starenje“. Za zdravlje, kao najvažniji motivacioni faktor kod vežbanja, nešto više su zainteresovani stanovnici APV (73%) u odnosu na stanovnike EU (61%) i ta konstatacija važi i za svih pet uzrasnih grupa stanovništva, gde se zapaža procentualna razlika u „korist“ stanovništva APV. Kada je u pitanju „zdravlje“, u oba uporedna uzorka briga za zdravlje se uglavnom povećava sa starenjem. Za razliku od „zdravlja“, situacija je obrnuta kada su u pitanju „kondicija“ (APV 31%, EU 41%) i „opuštanje“ (APV 34%, EU 40%), gde su više zainteresovani stanovnici EU u odnosu na stanovnike APV. Interesovanje za

„kondiciju“ i „opuštanje“ više je izraženo kod mlađih uzrasnih kategorija stanovništva u oba posmatrana uzorka i opada sa starenjem.



**Grafikon 4**

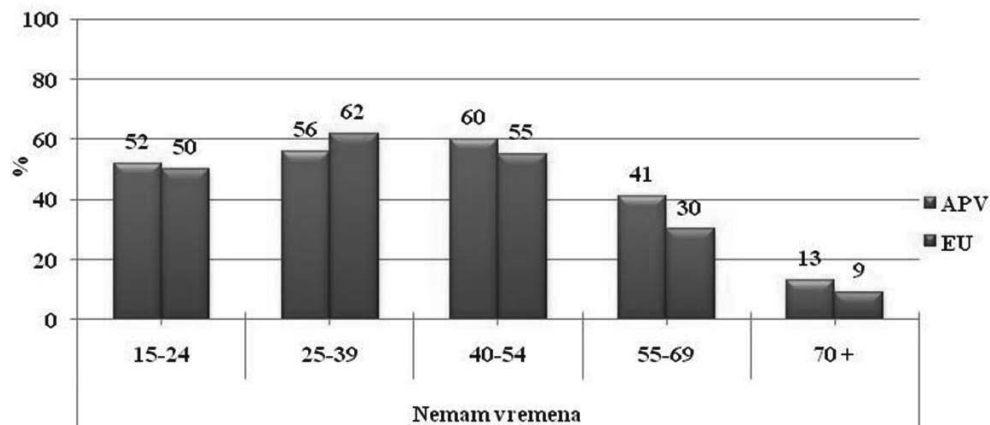
Dobijeni rezultati, nesumnjivo, ukazuju na postojanje visokog nivoa individualne i kolektivne svesti o fizičkom vežbanju kao nezaobilaznog faktora u očuvanju i unapređenju zdravlja i prevenciji većine bolesti savremenog doba (gojaznost, dijabetes, osteoporoza, kardiovaskularna i maligna oboljenja). Ne bez osnova je i „ekonomska strana“ bolesti, koja pojedinca, pored isključenja iz rada, lišava i drugih brojnih elemenata kvaliteta života.

Inspekcijom grafikona 5, među brojnim razlozima koji predstavljaju prepreku za redovnije bavljenje sportom ili ostalim neformalnim oblicima vežbanja, ispitanici oba uporedna uzorka najčešće navode **vreme**, odnosno njegov nedostatak, kao glavni razlog zašto se ne bave sportom. U AP Vojvodini 44%, a u Evropskoj uniji 41% stanovništva se neredovno bavi vežbanjem usled prezauzetosti, odnosno, malo vremena. Ostali faktori su relativno zanemarljivi na nivou oba uporedna uzorka, jer su u mnogo manjem procentu zastupljeni.

Posmatrano po uzrasnim grupama, kod oba uporedna uzorka, faktor vreme, odnosno njegov nedostatak, glavni je razlog koji sprečava uzrasne grupe 15-24 godine (52%), 25-39 godina (56%) i 40-54 godine (60%), da se redovnije bave sportom, čime je u okviru svakog navedenog uzrasta obuhvaćen natpolovičan deo stanovništva. Nedostatak vremena manje je izražen kod starijih uzrasta u oba uporedna uzorka.



### RAZLOG KOJI VAS TRENUTNO SPREČAVA ZA REDOVNIJE BAVLJENJE SPORTOM



Nemam vremena

Grafikon 5

### DISKUSIJA

Rezultatima istraživanja, kojim je obuhvaćeno pet uzrasnih grupa ispitanika, u okviru četiri tematske celine, analizirane su razlike između grupa u odnosu na: stav ispitanika o nivou i intenzitetu fizičke aktivnosti, stav ispitanika o mestu bavljenja fizičkom aktivnošću, stav ispitanika o motivima bavljenja fizičkom aktivnošću i stav ispitanika o barijerama bavljenja fizičkom aktivnošću. Generalno, kod sva četiri obeležja stava o fizičkoj aktivnosti utvrđena je razlika i jasno definisana granica između uzrasnih grupa, odnosno specifičnost fizičke (ne) aktivnosti regionalnog stanovništva (APV) u odnosu na stanovništvo evropskog okruženja (EU).

Najveći procenat aktivnog učešća u sportu karakterističan je za mlađe uzrasne kategorije stanovnika AP Vojvodina, dok je sa starenjem to interesovanje u osetnom opadanju. U odnosu na ispitivano obeležje, iako nadilaze evropski proseka bavljenja sportom, ispod je proseka nekih vodećih zemalja članica EU. Intenzitet prisustva sporta, posebno u uzrastu 15-24 godine, izraz je njegovog prirodnog "okupiranja" ličnosti u uzrastu koji pogoduje za realizaciju maksimalnih ostvarenja – rezultata. Međutim, pored navedenog, ne mogu se zanemariti i prenaplaćena očekivanja od sporta kao mehanizma socijalne pokretljivosti, koja su u odsustvu drugih "instrumenata" (obrazovanje, rad), komuniciranja među društvenim stratumima, posebno uočljiva u uslovima ekonomske krize na području AP Vojvodina i cele Srbije.

Procenat ispitanika AP Vojvodina, koji se redovno bave „neformalnim“ fizičkim aktivnostima, posebno u starijem uzrastu ukazuje da se stečene navike redovnog vežbanja u mladosti, zadržavaju tokom celog života, ali i da njihov pad u uzrastu 25-39 godina verovatno potiče od „gubitka“ slobodnog vremena na račun radnog. Tome posebno pogoduje intenzitet potrebe nastalih usled zadovoljenja porodičnih (rađanje i podizanje dece) stambenih (stan, kuća) i socijalno-statusnih potreba (automobil). Takođe, stano-

vnici AP Vojvodina, iako su iznad evropskog proseka bavljenja “neformalnim” fizičkim vežbanjem, ispod su proseka pojedinih razvijenih članica EU.

Različite afinitete prema ambijentu u kome vežbaju imaju sve ispitivane (APV) i uporedne (EU) uzrasne grupe stanovništva, pa tako sa starenjem, uglavnom raste afinitet ka otvorenom prostoru za vežbanje (parkovi ili drugi otvoreni prostori) i obrnuto, sa starenjem konstantno slabi interes za vežbanje u fitnes centrima, klubovima i sportskim centrima. Navedeni trendovi se u značajnoj meri mogu pripisati sve prisutnijoj kulturi „prirodnih“ potreba za kretanjem u kome dominiraju aerobne aktivnosti (hodanje, vožnja bicikla, plivanje), kao nezaobilaznih formi vežbanja u realizaciji zdravstvenih motiva – posebno onih za relaksacijom i opuštanjem.

U mnoštvu motiva koji se navode kao individualni razlozi za vežbanje, većina stanovništva, u oba uporedna uzorka, navodi zdravlje (AP Vojvodina 73%, EU 61%). Drugi razlog zbog kog se vežba, ispitanici EU (41%) ističu „zbog kondicije“, a ispitanici APV (34%) navode „zbog opuštanja“. Na trećem mestu, kao najčešći razlog za fizičku aktivnost nalazi se „zbog opuštanja“ za ispitanike EU (40%), odnosno „zbog kondicije“ za ispitanike APV (31%). Pored pomenuta tri razloga zbog kojih ispitanici najčešće vežbaju, značajni su: „zbog zabave“, „zbog boljeg izgleda“, „zbog izdržljivosti“, „zbog regulisanja težine“, „zbog druženja“, „da bih usporio starenje“. Za zdravlje, kao najvažniji motivacioni faktor kod vežbanja, nešto više su zainteresovani stanovnici APV (73%) u odnosu na stanovnike EU (61%) i ta konstatacija važi i za svih pet uzrasnih grupa stanovništva, gde se zapaža procentualna razlika u „korist“ stanovništva APV. Kada je u pitanju „zdravlje“, kod stanovništva AP Vojvodina briga za njegovo očuvanje i unapređenje kroz fizičko vežbanje uglavnom se povećava sa starenjem. Za razliku od „zdravlja“ situacija je obrnuta kada su u pitanju „kondicija“ (APV 31%, EU 41%) i „opuštanje“ (APV 34%, EU 40%), gde su više zainteresovani stanovnici EU u odnosu na stanovnike APV. Posmatrano po uzrastima, za „kondiciju“ i „opuštanje“ više brinu stanovnici EU u svim uzrastima, izuzev uzrasne grupe 15-24 godine kod „kondicije“, gde je odnos obrnut u korist stanovništva stanovništva APV. Interesovanje za „kondiciju“ i „opuštanje“ više je izraženo kod mlađih uzrasnih kategorija stanovništva u oba uporedna uzorka i opada sa starenjem.

Rezultati istraživanja uporednih uzoraka ukazuju na postojanje visokog nivoa individualne i kolektivne svesti o fizičkom vežbanju kao nezaobilaznog faktora zdravstvenog statusa i njegovoj ulozi u prevenciji većine bolesti savremenog doba (gojaznost, dijabetes, osteoporoza, kardiovaskularna i maligna oboljenja). Takođe, svest pojedinca sve više je pod uticajem i ekonomskih činilaca bolesti, koja ga, pored isključenja iz rada, lišava i drugih brojnih elemenata kvaliteta života – (ne)formalno obrazovanje, putovanje, zabava i dr.

Posmatrano po uzrasnim grupama, kod oba uporedna uzorka, faktor vreme, odnosno njegov nedostatak, glavni je razlog koji sprečava uzrasne grupe 15-24 godine, 25-39 godina i 40-54 godine da se redovnije bave sportom, čime je u okviru svakog uzrasta obuhvaćen natpolovičan deo stanovništva. Neredovno vežba zbog malo vremena: uzrast 15-24 godine APV (52%) i EU (50%), uzrast 25-39 godina APV (56%) i EU (62%), uzrast 40-54 godine APV (60%) i EU (55%). Nedostatak vremena manje je

izražen kod starijih uzrasta u oba uporedna uzorka. Nema sumnje da je vreme “novac”, kako u sferi njegovog pribavljanja – rada, tako i u “kulturi” trošenja. Njegov nedostatak za sportske aktivnosti i kulturu vežbanja najverovatnije je moguće potražiti u nezadovoljenim egzistencijalnim potrebama iz rezultata rada, ali i onim tzv. hedonističkim potrebama, izraslim na obrascima društvenih stereotipa - najčešće, kao materijalizovanih i uživalačkih (piće, jelo, moda i dr.) oblika ponašanja.

### ZAKLJUČAK

U ovom istraživanju izvršena je identifikacija karakteristika fizičke aktivnosti stanovnika AP Vojvodina u odnosu na njihovu generacijsku pripadnost i uporedne uzrasne pokazatelje u Evropskoj uniji. Evrobarometar – anketom je ispitan uzorak od 3.132 ispitanika iz većih vojvodanskih mesta, koji je podeljen na pet generacijskih subuzoraka (15 – 24 godine, 25 – 39 godina, 40 – 54 godine, 55 – 69 godina i 70 i više godina). Prostor istraživanja obuhvatio je četiri međusobno povezane tematske celine koje se odnose na stav ispitanika: 1) o nivou i intenzitetu fizičke aktivnosti, 2) o mestu bavljenja fizičkom aktivnošću, 3) o motivima bavljenja fizičkom aktivnošću, 4) o barijerama bavljenja fizičkom aktivnošću. Na osnovu dobijenih rezultata istraživanja utvrđeno je da u okviru sve četiri tematske celine postoji značajna razlika između pet generacijskih subuzoraka u odnosu na stav ispitanika prema ispitivanim područjima fizičke aktivnosti, posebno u odnosu na prevalenciju sportske aktivnosti u mlađim uzrasnim kategorijama. Relativno visok nivo i postojanost „neformalnih“ oblika fizičke aktivnosti i u starijem uzrastu, senzibilnost ka otvorenim prostorima za vežbanjem i zdravstvena motivisanost ispitanika AP Vojvodina, bitna su obeležja većine generacijskih subuzoraka ovog regiona Srbije. Takođe, ovim istraživanjem, pored uvida u trenutno stanje fizičke aktivnosti stanovništva AP Vojvodina, uporednom analizom sa stanovništvom zemalja EU, dobijen je uvid u osobenosti pojedinih uzrasnih kategorija oba entiteta u odnosu na ispitivana obeležja fizičke (ne)aktivnosti, posebno u odnosu na ispoljenost sportskih aktivnosti u uzrastu 15-24 godine stanovnika AP Vojvodina, odnosno, vežbanja u parku i prirodi stanovnika Evropske unije.

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#### EUROBAROMETER – PHYSICAL ACTIVITY OF INHABITANTS OF THE AP VOJVODINA

*In this paper we have identified the characteristics of physical activity of inhabitants of the AP Vojvodina compared to their generation group. We used the Eurobarometer survey to test the sample comprising of 3,132 participants from major places in Vojvodina, which was divided into five generation sub-samples (aged from 15 to 24, from 25 to 39, from 40 to 54, from 55 to 69, and older than 70). The area of this study comprises four themes that are mutually linked in a meaningful way and that refer to attitudes of participants: 1) about the level and intensity of physical activity, 2) about the place of dealing with physical activity, 3) about motives for getting involved in physical activity, 4) about barriers for getting involved in physical activity. A significant difference in the four given subject topics between the five AP Vojvodina population generation subsamples has been determined based on the research results that is, the specific physical (in)activity of the local population (the APV) compared to the population of the European surrounding (the EU).*

**Key words:** Physical activity, AP Vojvodina, Eurobarometer.



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## **ANALYSIS OF PERCEPTUAL PHENOMENON ON MOVEMENT OBSERVATION: INTERPRETATION OF MORPHOLOGICAL VIEWPOINT AS “DOPPELTER GESTALTKEIS”**

### **Introduction**

The purpose of this study is to clarify the Morphological Viewpoint (MOV) on Movement Observation as Perceptual Phenomenon (the relevance of consciousness and perception) from phenomenological standpoint. Furthermore, this study conceptualizes MOV's Perceptual Phenomenon as a hermeneutic figure (“Doppelter Gestaltkreis”) for enlarging the observation horizon of coaches and physical teachers.

Generally, Movement Observation is interpreted by two viewpoints. The first is the Morphological Viewpoint (MOV), which is based on phenomenological morphology. The MOV corresponds to the core ability of coaches and physical teachers (Meinel, 1981) because the MOV is focused on the generation of movement. The second is the mechanistic viewpoint, which is focused on partial elements and relevances. However some coaches cannot properly interpret the MOV (Meinel, 1981 ; Muraki, 1993). Because of this problem, this study focuses especially on MOV's Perceptual Phenomenon.

In some previous studies, the MOV has been interpreted from applying “Gestaltkreis” (Weizsäcker, 1940) which is a phenomenological hermeneutic figure for perceiving sound of “Melody of Movement” (Buytendijk, 1958). However, MOV's Perceptual Phenomenon has not been directly clarified as a hermeneutic figure.

Because of this, this study directly clarifies the “Doppelter Gestaltkreis” as the hermeneutic figure of MOV's Perceptual Phenomenon from the relevance of consciousness and Perception for enlarging the observation horizon of coaches and physical teachers.

### **Methods**

First, this study deductively analyzes MOV's Perceptual Phenomenon from the relevance of MOV's consciousness and perception. Second, this study conceptualizes the hermeneutic figure of MOV's Perceptual Phenomenon as the Doppelter Gestaltkreis from their relevance. Finally, this study discusses the utility of the Doppelter Gestaltkreis for enlarging the movement observation horizon of coaches and physical teachers.

### **Results**

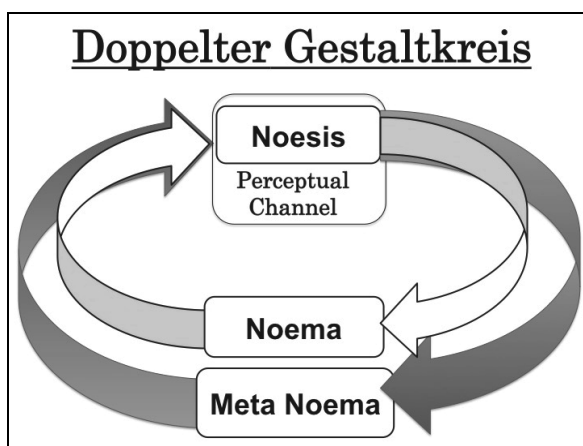
This study conceptualizes the Doppelter Gestaltkreis as the hermeneutic figure of MOV's Perceptual Phenomenon.

First, this study deductively analyzes MOV's Perceptual Phenomenon as the relevance of MOV's “Metacognition” (Fravell, 1976) and “Gestaltkreis” (Weizsäcker,

1940) which refers to the relevance of MOV's consciousness and perception in MOV's Perceptual Phenomenon. MOV's consciousness corresponds to MOV's Metacognition because subjective consciousness for perception deductively corresponds to Metacognition. In some previous studies, MOV's Perception, which perceives sound of "Melody of Movement" (Buytendijk, 1958), furthermore is interpreted from the hermeneutic figure of Gestaltkreis in phenomenological standpoint.

Moreover, Gestaltkreis is interpreted from the circulation of Noesis (the phenomenological concept of Noesis perceives a part of sound from movement) and Noema (the phenomenological concept of Noema perceives a real Melody). Accordingly, MOV's Metacognition deductively corresponds to "Meta Noema" (the concept of Meta Noema perceives an ideal Melody) in the circulation of Gestaltkreis because MOV's Metacognition corresponds to the consciousness for Perceiving Noema in Gestaltkreis. On the base of their relevance, this study adds the concept of Meta Noema (MOV's Metacognition) into the circulation of Noesis and Noema (Gestaltkreis) for the conceptualization of Doppelter Gestaltkreis.

As a result, this study conceptualizes the Doppelter Gestaltkreis (Figure 1) which includes the circulation of two circles. The first is the "Perception Circle" of Gestaltkreis which includes the circulation of Noesis and Noema. The second is the MOV's "Metacognition Circle" which includes the circulation of Noesis and Meta Noema. Therefore these two circles contain the same Noesis, which is interpreted as the phenomenological concept of the "Perceptual Channel" because Noesis comparatively perceives sound of the real and ideal Melody in the circulation (will be explain later).



*Figure 1. Doppelter Gestaltkreis*

## Discussion

### 1. The circulation of Doppelter Gestaltkreis

Doppelter Gestaltkreis has the utility that coaches and physical teachers can interpret MOV's Perceptual Phenomenon from phenomenological standpoint.

### 1) The Utility of Doppelter Gestaltkreis

MOV characteristics are to perceive sound of the real melody and furthermore comparatively perceive sound of real Melody from ideal Melody. In previous studies, perceiving sound of real Melody is interpreted from the hermeneutic figure of Gestaltkreis. However comparatively perceiving sound of real melody from ideal melody is indirectly interpreted from applying Gestaltkreis.

This study directly clarifies Doppelter Gestaltkreis as the hermeneutic figure of MOV's Perceptual Phenomenon. Therefore, this study discusses from this circulation and the concept of Perceptual Channel, that the Doppelter Gestaltkreis has the utility of directly interpreting MOV's characteristics in MOV's Perceptual Phenomenon.

### 2) How to interpret MOV's Perceptual Phenomenon from the circulation of Doppelter Gestaltkreis

This study summarizes how to interpret MOV's Perceptual Phenomenon in Figures 2 and 3.

#### First: Precondition (Figure 2)

- ❶ The coach must intend an ideal Melody for evaluation of real Melody.

#### Second: The Circulation of Doppelter Gestaltkreis (Figure 2)

- ❷ The coach perceives a part of the sound from the movement.
- ❸ The coach repeatedly refers sound to the real and ideal Melody.
- ❹ The coach again repeatedly refers the real and ideal Melody to the same sound in Perceptual Channel.
- ❺ The coach comparatively perceives sound of the real and ideal Melody in Perceptual Channel for the evaluation of sound of the real Melody.

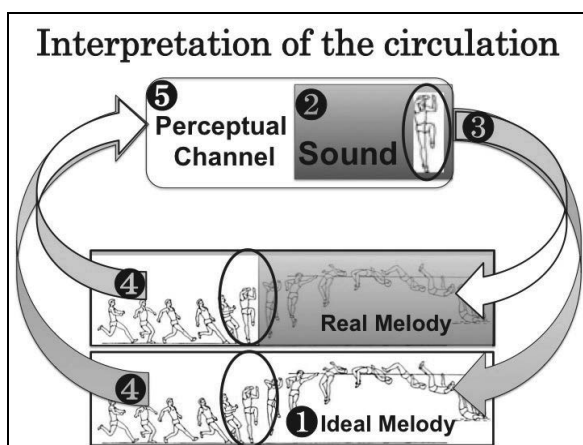
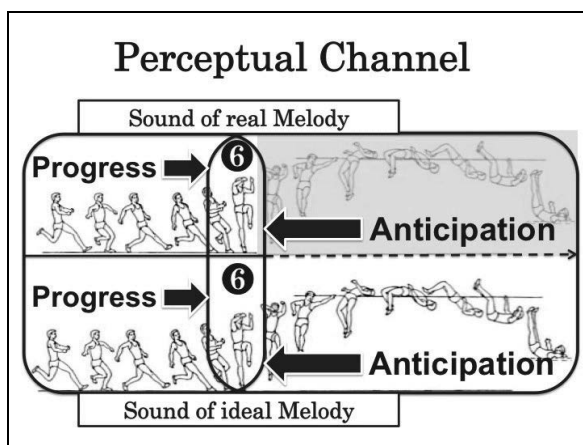


Figure 2. Interpretation of the circulation

**Third: Comparative Evaluation of Sound of the Real Melody (Figure 3)**

- ⑥ The coach comparatively evaluates and leads sound of the real Melody from sound of the ideal Melody in Perceptual Channel. In Perceptual Channel, the coach comparatively evaluates not only moment sound but also anticipation and past progress of the real and ideal Melody. On the base of the comparative perception, the ideal Melody furthermore leads the real Melody by comparing the real and ideal anticipation in sound.



*Figure 3. MOV's Perceptual Channel*

From the interpretation of the circulation and the concept of Perceptual Channel in Doppelter Gestaltkreis, this study clarifies the utility of Doppelter Gestaltkreis for enlarging the observation horizon of coaches and physical teachers.

## **2. About enlarging the observation horizon of coaches and physical teachers**

In order to apply the interpretation of the MOV to practical Movement Observation, coaches should perceive sound from anticipation and progress as intention of the purpose and cause of the sound. They furthermore should give motor learners some advice for the generation of a movement. Therefore coaches are required to be able to interpretatively perceive sound from "Knowledge and Value" and verbalize intention as advice. Because of this, in the future it needs to be clarified on how to train interpretative perception of the sound.



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### ANALYSIS OF PERCEPTUAL PHENOMENON ON MOVEMENT OBSERVATION: INTERPRETATION OF MORPHOLOGICAL VIEWPOINT AS “DOPPELTER GESTALTKREIS”

*The purpose of this study is to clarify the Morphological Viewpoint on Movement Observation from phenomenological standpoint.*

*Generally, Movement Observation is interpreted by two viewpoints. First is the Morphological Viewpoint (MOV) which is based on phenomenological morphology. Second is the Mechanistic Viewpoint, which is based on physics. The former corresponds to the core ability of coaches and physical teachers (Meinel, 1981) because the MOV is focused on the generation of movement. However some coaches and physical teachers cannot properly interpret the MOV (Meinel, 1981; Muraki, 1993).*

*In some previous studies, the MOV has been interpreted from applying Gestaltkreis (Weizsäcker, 1940) which is a phenomenological hermeneutic figure for perceiving sound of “Melody of Movement” (Buytendijk, 1958). However, MOV’s Perceptual Phenomenon has not been directly clarified as hermeneutic figure.*

*Because of this, this study deductively analyzes Perceptual Phenomenon of MOV from the relevance of consciousness and perception. On the base of their relevance, this study furthermore conceptualizes “Doppelter Gestaltkreis” which is a hermeneutic figure for directive interpretation of MOV’s Perceptual Phenomenon. This study further discusses the utility of Doppelter Gestaltkreis for enlarging the observation horizon of coaches and physical teachers.*

**Jana Nová***Masaryk University Brno, Faculty of Sport Studies (Brno, Czech Republic)*

## **INSTITUTIONALIZED SYSTEMATIC REVIEW AS A TOOL FOR KNOWLEDGE TRANSFER IN SPORT MANAGEMENT**

### **Introduction**

As in many scientific fields, the situation in sport management also takes note of the growth of scientific production. The abundance of knowledge raises interest in research synthesis as a complementary mode of research beside the primary research endeavour. However, so far systematic reviews have not been widely employed in the field of sport management (Weed, 2005). Currently, at national and international level no institutions exist to address the problem of systematic review in the field of sport management. Instead there are various organizations around the globe focusing on consultancy in sport management. The utility of research synthesis in sport management could develop a systematic and rigorous review of knowledge on a given topic and make sense of diverse research areas and ensure the full utilization of the existing research findings for the practice and further research development in sport. To achieve a situation similar to that in the health sector the institutionalization base is needed to enhance the conduction of systematic reviews in sport management. International importance and the necessity of the institutionalization of systematic review in sport management is linked to the fact that sport as a nationwide phenomenon has a significant social and economic dimension and these dimensions could achieve their full potential provided there is an appropriate organisation of sport in place (White Paper on Sport, 2007).

### **Methods**

The desk research and content analysis of various modes related to systematic reviews was conducted. A comparison of the models used for the transfer of knowledge from science to business practice, in particular in the health care sector, was realized. Based on this information the conceptual model for the systematic review and knowledge transfer in sport management which could take the institutional form was designed, describing the portfolio of services and expertise, development plan, the policy towards interaction and communication with the user community, and also the management strategy.

### **Results**

Currently the most developed approach in systematic review can be observed in the health sector. The proposal of an institutional basis for implementation of systematic reviews in sports management organizational structure of these institutions has been explored:

**Centre for Reviews and Dissemination (CRD)** is part of the National Institute for Health Research (NIHR) and is a department of the University of York. The CRD provides research-based information on the effects of health and social care interventions via databases and undertakes systematic reviews evaluating research evidence on health and public health questions of national and international importance. The findings of research outputs are widely disseminated and have impacted on health care policy and practice, both in the UK and internationally. The CRD databases are updated daily and provide decision-makers with access to quality assessed systematic reviews. Over 200 systematic reviews on DARE evaluate interventions related to sporting or physical activity.

**Cochrane Collaboration** is a registered charity in the United Kingdom. Cochrane works collaboratively with partners and funders to produce and disseminate authoritative, relevant, and reliable health evidence. Cochrane Reviews are systematic reviews of primary research in human health care and health policy, and are internationally recognized as the highest standard in evidence-based health care resources. They are published online in the Cochrane Library. Since February 2013, all new Cochrane Reviews have become free to access for all readers 12 months after publication. Cochrane systematic reviews are published in the Cochrane Database of Systematic Reviews (CDSR), a monthly publication within the Cochrane Library, which, in 2009, reached an impact factor of 5.65.

**The Campbell Collaboration** is a sibling organisation to Cochrane which prepares, maintains and promotes the accessibility of systematic reviews in areas such as education, criminal justice, social policy and social care. Campbell Systematic Reviews are a peer-reviewed online monograph series of systematic reviews prepared under the editorial control of the Campbell Collaboration. At Cochrane-Campbell meetings at the Rome Colloquium, support in principle was expressed for joint Cochrane-Campbell Methods Groups.

**The Joanna Briggs Institute (JBI)** is the international not-for-profit, research and development arm of the School of Translational Science based within the Faculty of Health Sciences at the University of Adelaide, South Australia. The Institute collaborates internationally with over 70 entities across the world. The Institute and its collaborating entities promote and support the synthesis, transfer and utilization of evidence through identifying feasible, appropriate, meaningful and effective healthcare practices to assist in the improvement of healthcare outcomes globally.

### **The conceptual framework of the institutionalized systematic review in sport management**

As mentioned earlier, currently, at national and international level no institutions exist to address the problem of knowledge transfer in the field of sport management. Instead there are various organizations around the globe focusing on consultancy in sport management. Based on the experience from the health sector it is suggested to create the non-profit, membership infrastructure “Centre for the Systematic Review and Knowledge Transfer in Sport Management” (hereinafter Centre).

**Mission of the Centre**

The Centre will be created as a network of collaborating entities around the world and will facilitate and strengthen international collaboration between systematic review groups, expert researchers, sport managers and other members of the network. Thus it will allow the development of the appropriate framework for analysing, producing and disseminating the best available research in sport management in order to provide and assure access of the information to various stakeholders in sport management. The information will be disseminated globally to inform higher education establishments, sport managers, sport organizations and a variety of stakeholders operating in the sport sector. The impact of the Centre's activities will be measured and assessed via changes in sport policy, sport practice and the advancement in sport management research.

**Users' structure** will have international scope and will include:

- Sport Management Researches;
- Higher Education Institutions;
- Sport Organizations (public, non-governmental, professional)
- Sports Associations
- Students

**Portfolio of services** provided by the Centre:

- methodological standards for research synthesis in sport management
- production of a high-quality, relevant and up-to-date systematic review and other synthesised research evidence in sport management
- development of the IT modes for effective storage and retrieval of the reviews by various users and stakeholders (Database of Systematic Reviews in Sport Management, Centre Library of Research Resources in Sport Management )
- open access to scholarly research via Internet for the Centre's members and the Centre's members of the network in the languages they speak and in an appropriate design
- methodological support for research in sport management
- educational programmes for researchers in sport management (research methodology in sport management, systematic review methodology, synthesis review methodologies)
- educational programmes for users, stakeholders and members of the network (utilization of the research reviews and Library sources and their practical utilization, case studies)
- measurement of the impact coming from the dissemination of knowledge via the Centre's services in sport management practice and research

The portfolio will evolve and it is envisaged that the Centre will develop into a multi-purpose platform and that the scope of the focus will widen to embrace all facets of sport, not only sport management. Collaborating entities will use the Centre's products on a regular basis and will interact with the Review groups via feedback. Regular conferences, workshops and educational programmes will be attended by

representatives from collaborating entities and members. The Centre's policy in terms of access will be to enable the approach and use of the systematic reviews to everybody, everywhere in the world via the Internet. The content of the Centre databases and Library will be accessible through licences or free access considering the users from different backgrounds. Different access policies would apply for the public, NGO and commercial institutions. Free access will be offered for collaborating institutions and members and also the funding institutions

To assure the feasibility and sustainability of the Centre, the following **development stages** are suggested:

**Stage 1**

Development of the Centre structures (Management – Steering committee. Establishment of the Systematic Review Groups, Methods Group, Editorial Board, and Development of the standards, IT tools and software for systematic reviews. Building the network)

**Stage 2**

Development of the systematic reviews in a selected range of sport management topics, building the Database of the Systematic Review and Library in Sport Management. Promotion of open access to the Centre's databases, promotion of effective communication with the people outside the Centre (stakeholder and public feedback). Further establishment of the Centre's structures including national centres and internal communication. Development of the business plan for the Centre's core activities.

**Stage 3**

Investigation and development of a responsive review programme. Developing marketing and communication strategies towards the various target and stakeholder groups. A formal membership establishment. Identification of the new functions and products of the Centre including educational and training programmes. Evaluation of the utilization of the Centre's products. Improvement of the functioning of the internal structures and activity planning.

**Stage 4**

Improvement in the quality, relevance and impact of the Centre's products. Expanding the training and capacity building programmes. Improving the technology and efficient editorial system. Extending the Centre's activities into various national centres. Developing further global participation and relevance of the Centre.

**Proposed management strategy and organizational structure**

Steering Group - shall determine the organizational structure of the Centre and decides on proposals for new Review Groups, and Collaborating Organizations; the scientific policies and strategic as well as operating procedures; sets guidelines and standards for systematic reviews;

Review Groups will be international in composition and shall primarily be responsible for the production, scientific merit and usefulness of systematic reviews;

Methods Group - shall hold responsibility for contributing to methodological issues, and generally shall support high quality of systematic reviews;

Collaborating Organizations - shall enter into formal agreements with the Centre to collaborate on an agreed-upon set of activities to further mutual interests;

Ad hoc working groups- shall be appointed on specific issues;

Administration of the Centre - shall be responsible for legal and fiscal matters related to the organization's status as a non-profit organization

### **Discussion**

The proposed institutionalized infrastructure of cooperating entities in sport management research and practice will address the problem of the research synthesis in sport management as described by Weed (2005). The suggested model thus represents the first attempt to conceptualize open access to international research results in sport management for sport practice in a manner which is unbiased, thanks to the used methodology (systematic review). The feedback from the various stakeholders from sports management practice could serve as a valuable tool for overcoming the gap between research and practice, a fact which has been heavily criticized recently. The research innovations which could be drawn from the proposed infrastructure are linked to the development of knowledge transfer science, implementation science and translational research in the context of sport management. Application of the principles of the translational research as described by Woolf (2009) would mean that two translational blocks, T1 and T2, will enable not only the transfer of the understanding of the new knowledge in sport management for the purpose of the further research but will also translate research results in sport management practice and everyday decision making. Thus the results of the proposed mode of the knowledge transfer can be used by a whole variety of institutions ranging from research institutions to the public, non – governmental but also commercial organisations operating in sport. The proposed conceptual model is also important for the further development of science and research in sport management. This is closely linked to the fact that, although the disciplinary aspects of the sport management have been discussed over the last 30 years in terms of the body of the knowledge, theory development and methodology appropriateness, there is still no clear explanation of sport management as a scientific discipline (Nová, 2014).

### **Practical implication of the paper**

The systematic review of the achievements of sport management research conducted at international level within the institutionalized framework of collaborating entities could create and enable easy access to the latest relevant information of the different facets of sport management for the various stakeholders in the sport sector at national and international level. Moreover research synthesis will provide a valuable source of information for the future direction in sport management research. So far in sport management there are various organizations around the globe focusing on paid consultancy in sport management. The proposed infrastructure should provide open access to international research results in sport management for sport practice in a manner which is unbiased thanks to the used methodology (systematic review). The feedback from the various stakeholders from sports practice could serve as a valuable tool

for overcoming the gap between research and practice, a fact which has been heavily criticized recently. Sport organizations are faced with the challenge of exploring for their own benefit, the vast body of past research that mostly remains ignored due to the lack of time of sport managers. The proposed institutionalization can overcome this shortage by providing open and comprehensive access to the relevant evidence-based information. Practitioners of sport management will be an integral part of the proposed institutionalized network and thereby will provide sport management research with valuable feedback related to the edifices of knowledge (so-called user involvement in the systematic review) which would help to direct future research in sport management. The uniqueness of the proposed institutionalized network is that it will operate in and for the sport sector. Usage of the evidence assessment for summarizing what is already known internationally about a particular topic in sport management and user involvement – all these features could be considered as the main competitive advantage of the proposed infrastructure. The knowledge base of the proposed infrastructure will serve also for educational purposes in so far as students will also have open access to the information resources. The student population will be considered as one of the network's stakeholder groups which will be allowed to raise its voice via the feedback framework.

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- <http://www.crd.york.ac.uk/crdweb/>
- <http://joannabriggs.org/>

### INSTITUTIONALIZED SYSTEMATIC REVIEW AS A TOOL FOR KNOWLEDGE TRANSFER IN SPORT MANAGEMENT

*Further development of research and knowledge transfer in sport management is limited by the lack of full utilization of the past research results. Considering the fact that sport is regarded as a crucial determinant of a region's social and economic development, the transfer of the latest knowledge from sport management research to sport in practice is of the utmost importance. The availability of research results in a*

usable form and the active interaction between academia and sport practitioners could help to a great extent enhance not only the knowledge base but also the performance of the sports organizations, thus contributing to their professionalization. Therefore the aim of the paper is to propose a conceptual model for the institutionalization of systematic knowledge exploration and consolidation in sports management so as to enable and simplify knowledge transfer to the practice of sports organizations. The design of a conceptual model is based on the comparison of the well-known models of institutionalized systematic review from health science.

„Vijesti“, 6. februar 2015.



„Najveći naučni skup iz oblasti sporta u regionu“: Duško Bjelica

Foto: L. ZEKOVIĆ

## SPORT I NAUKA: SKUP CSA

# Prezentacija radova i dvije radionice

**P**riredivački odbor Crnogorske sportske akademije (CSA) prihvatio je 110 radova za 12. Međunarodnu konferenciju CSA, koja će se od 2. do 5. aprila 2015. godine održati u Podgorici. Upravo sa konferencijom, održaću se i 11. kongres CSA sa oko 250 učesnika iz 28 zemalja.

- Ovo je najveći naučni skup iz oblasti sporta u regionu. I svi eminentni stručnjaci, koje smo najavili, potvrdili su učešće. Slike svih 110 radova biće objavljen na engleskom jeziku, a cjelokup-

no će biti publikovani u časopisu „Sport Mont“. Među plenarnim izlagачima biće **Henk Erik Maier** iz njemačkog Univerziteta u Munsteru, **Zdenko Reguli** i **Martin Zvonar** sa češkog Masarik univerziteta, **Višnja Đorđić** sa Univerziteta u Novom Sadu... - kazao je prof. dr **Duško Bjelica**, predsjednik priredivačkog odbora Kongresa.

Novina u 12. izdanju Međunarodne naučne konferencije iz oblasti sporta su dvije radionice.

- Savremeni pristup smanjenju jaza između is-

traživanja i potreba industrije, kroz praktične primjere prikazaće **Nejc Šarabon** sa Univerziteta Primorkoj u Sloveniji. Drugu prezentaciju o najnovijim dostignućima sistema za elektromiografiju uradiće predstavnici svjetski poznate kompanije „Desys“. Imaćemo i specijalnog gosta **Branislava Antal**, profesora sa Komenius univerziteta iz Slovačke, koji je predsjednik „FIEP Europe“, organizacije koja je osnovana 1923. godine u Briselu, sa ciljem da promovise razvoj širokog spektra djelatnosti u oblasti obrazovanja, sporta, fizičkog vaspitanja i školskog sporta, sporta za sve, fitnesa i zdravlja i rekreacije za ljude svih uzrasta i u svim zemljama svijeta. Ova federacija je, između ostalog, priznata i od strane Međunarodnog olimpijskog komiteta - zaključio je **Bjelica. S. Ž.**



**Sinku Kumar Singh***Swami Ramanand Teerth Marathwada University (Nanded, India)***CAUSES OF INJURIES AT THREE LEVELS IN COMPETITIVE FOOTBALL****Introduction**

Football is a most hazardous team sports and injury is a frequent event in football (Sinku 2012). Football requires a variety of physical attributes and specific playing skills, therefore participants need to train and prepare to meet at least a minimum set of physical, physiological and psychological requirements to cope with the demands of the game and to reduce the risk of injury. It is an enjoyable and social sport than can be played from childhood to old age, either at a recreational level or as a competitive sports. Football playing largely involves starting, running, slopping, twisting, jumping, kicking, and turning movements that place the players to greater risk of injury (Waston 1993). Football is a high risk sport dominated by overuse injuries while recovery time from injuries is relatively long, but only a few working days are lost by the players to return back to play, thus leading to abuse of the injured sites. In football only a few studies have been made in the literature regarding incidents of injury and pattern, possible risk factors and injury prevention (Winter Griffith, 1989; watan. 1993; Junge, 2004). In football overuse injuries are the most frequent occurrences of injury; and injuries are traditionally divided into contact and non contact mechanism in which case contact refers to players contact. Some of the forces involved in a non contact injury are transmitted from the playing surface to the injured body part.

**Methods**

The present study deals with comparison of causes of injuries among three groups of competitive footballers. The investigator has made an attempt to classify define of footballers based on the class of the games of the footballers. Accordingly three groups s of footballers were targeted. International, National and State groups footballers aged between 14 to 30 years. The data was collected with the help of questionnaires prepared by Cromwell, F.J. Walsh Gromley for Elite Gaelic footballers (2000) and it was modified by the investigator and utilized. In this study total 300 hundred players were selected; of out of 100 footballers of each groups. Total 300 questionnaires were administered out of 300 questionnaires 100 belongs to international, 100 belongs to national and 100 belongs to state groups footballers. This study involves a comparative survey of three groups of football players in a non-experimental, retrospective study design. One Way Analysis of Variance and scheffee post hoc test were used to assess overall differences of injuries among three groups.

## Results

The results and discussion have been presented in concise and comprehensive manner that is easy to comprehend. The results concerning this are presented in the form of tables. For the sake of convenience and methodical presentation of the results, following order has been adopted.

**Table 2.** Mean scores and Standard Deviations of occurrence of injuries with respect to cause among three groups of footballers

| Sr. No. | Causes            | Footballers   | Number | Mean Scores | Standard Deviations |
|---------|-------------------|---------------|--------|-------------|---------------------|
| 1       | Collision         | International | 10     | 1.4         | .46                 |
|         |                   | National      | 13     | 1.23        | .43                 |
|         |                   | State         | 12     | 1.66        | .59                 |
| 2       | Foul Play         | International | 21     | 1.42        | .53                 |
|         |                   | National      | 16     | 1.06        | .46                 |
|         |                   | State         | 22     | 1.04        | .43                 |
| 3       | Running           | International | 10     | 1.4         | .59                 |
|         |                   | National      | 20     | 1.4         | .62                 |
|         |                   | State         | 12     | 1.08        | .75                 |
| 4       | Contact with Ball | International | 08     | 1.37        | .76                 |
|         |                   | National      | 07     | 1.57        | .80                 |
|         |                   | State         | 04     | 1.25        | .47                 |
| 5       | Stumble           | International | 12     | 1.41        | .63                 |
|         |                   | National      | 05     | 1.2         | .34                 |
|         |                   | State         | 15     | 1.06        | .31                 |
| 6       | Tackle            | International | 17     | 1.41        | .43                 |
|         |                   | National      | 14     | 1.5         | .49                 |
|         |                   | State         | 05     | 1.2         | .34                 |
| 7       | Kicking the Ball  | International | 06     | 1.83        | .66                 |
|         |                   | National      | 07     | 1.28        | .52                 |
|         |                   | State         | 07     | .42         | .19                 |

As Table 1, indicates that the mean scores and standard deviations of occurrence of injuries with respect to causes among three groups of competitive footballers.

**Table 2.** Analysis of Variance of occurrence of injuries with respect to causes among three groups of footballers

| Sr. No. | Causes            | Source of Variance | SS | df.   | MSS | F-ratios           |
|---------|-------------------|--------------------|----|-------|-----|--------------------|
| 1.      | Collision         | Between groups     | 02 | 0.19  | .09 |                    |
|         |                   | Within groups      | 27 | 09.18 | .34 | .26 <sup>NS</sup>  |
| 2.      | Foul Play         | Between groups     | 02 | 01.90 | .95 |                    |
|         |                   | Within groups      | 56 | 15.05 | .26 | 3.65 *             |
| 3.      | Running           | Between groups     | 02 | 0.36  | .18 |                    |
|         |                   | Within groups      | 33 | 15.64 | .47 | .38 <sup>NS</sup>  |
| 4.      | Contact with Ball | Between groups     | 02 | 0.29  | .14 |                    |
|         |                   | Within groups      | 33 | 08.35 | .25 | .56 <sup>NS</sup>  |
| 5.      | Stumble           | Between groups     | 02 | 0.47  | .73 |                    |
|         |                   | Within groups      | 29 | 06.41 | .22 | 3.31 <sup>NS</sup> |
| 6.      | Tackle            | Between groups     | 02 | 0.30  | .15 |                    |
|         |                   | Within groups      | 24 | 18.22 | .75 | 1.20 <sup>NS</sup> |
| 7.      | Kicking the Ball  | Between groups     | 02 | 0.92  | .46 |                    |
|         |                   | Within groups      | 17 | 5.63  | .33 | 1.39 <sup>NS</sup> |

\* Significant at .05 level

NS = Not Significant

**Table 2** shows the, Analysis of Variance of occurrence of injuries with respect to causes among three groups of footballers. In order to find out the difference of occurrence of injuries with respect to causes among three groups of competitive footballers. F-ratio was computed for each cause separately. The data given in Table 2 shows that there was statistically significant difference of occurrence of injuries with respect to causes was found in Foul Play only(  $F=3.65, <.05$ ). However, there were no statistically significant difference of occurrence of injuries found in Collision ( $F=.26$ ), Running ( $F=.38$ ), Contact with Ball ( $F=.56$ ), Stumble ( $F=3.31$ ), Tackle ( $F=1.20$ ) and Kicking the Ball ( $F=1.39$ ). In order to locate the occurrence of injuries due to foul play among three groups of competitive footballers; Scheffe post hoc test was applied to comprise the occurrence of injuries; Table 3 shows the possible comparisons for three groups means.

**Table 3.** Scheffe post hoc Comparison for mean difference of occurrence of injuries due to Foul Play among three groups of competitive footballers

| International | Mean Scores |             | Mean difference | C.D.<br>at 5% level |
|---------------|-------------|-------------|-----------------|---------------------|
|               | National    | State       |                 |                     |
| 1.42          | 1.06        |             | .36             | .42                 |
| 1.42          |             | 1.04        | .38             | .36 *               |
|               | <b>1.06</b> | <b>1.04</b> | <b>.02</b>      | <b>.42</b>          |

\* Significant at .05 level.

**Table 3**, shows that the Scheffe post hoc statistically comparison for mean difference of occurrence of injuries due to Foul Play among three groups of competitive footballers.

### Discussion

The mean scores (S.Ds.) of injuries due to Collision of international group footballers was 1.4 (.46), national group footballers was 1.23 (.43) and State group footballers was 1.66 (.59). The mean scores (S.Ds.) of injuries due to Foul Play to international group footballers was 1.42 (.53), national group footballers was 1.06 (.46) and state group footballers was 1.04 (.43). The mean scores (S.Ds.) of injuries due to Running of international groups footballers was 1.49 (.59), national group footballers was 1.4 (.62) and state group footballers was 1.18 (.75). The mean scores (S.Ds.) of injuries due to Contact with Ball to international group footballers was 1.37 (.76), national group footballers was 1.57 (.80) and state group footballers was 1.25 (.47). The mean scores (S.Ds.) of injuries due to Stumble to international group footballers was 1.33 (.63), national group footballers was 1.2 (.34) and state group footballers was 1.06 (.31). The mean scores (S.Ds.) of injuries due to Tackle to international group footballers was 1.41 (.43), national group footballers was 1.5 (.49) and state group footballers was 1.2 (.34) and the mean scores (S.Ds.) of injuries due to Kicking the Ball to international group footballers was 1.33 (.66) , national group footballers was 1.28 (.52) and state group footballers was .85 (.19). Table 3, reveals that (i) No statistically significant difference of occurrence of injuries was found between international and national groups footballers due to foul play. (ii) Statistically significant difference of occurrence of injuries due to Foul Play was found between international and state groups footballers. International footballers got having more injuries due to Foul Play as compared to state groups footballers and (iii) No Statistically significant difference of occurrence of injuries was found between national and state groups footballers. This study sported the study of pagare, 2009 found that high level football players was found to have got more injuries due to foul play as compare than low performance football players.

Studies on foul play on injury are equivocal. In some of these studies foul play called by the referee are studied (Engstrom et al. 1990; Hawkins and Fuller 1996; Hawkins and Fuller 1999), while in other studies players reported whether it was foul or not Chomiak et al. 2000; . Junge et al. 2000a). Results have shown that foul play was the cause of 16-28% of all injuries (Nielsen and Yde 1989; Hawkins and Fuller 1999; Junge et al. 2000a), Other studies have found that 76-86% of the foul play injuries were caused by opponent and the rest by own foul (Ekstrand and Gillquist 1983b; Hawkins and Fuller 1999), and also that own foul play resulted in more serious injuries than opponent foul (Ekstrand and Gillquist 1983b). One study (Sinku et.al.2008) show that collision is the most usual mechanism of injuries. Studies reports that 58-67% of acute injuries in elite football occurred in non-contact situations, while 41-45% of acute injuries in players at lower or various level and 47-58% of acute injuries in youth players occurred in non-contact situation (Ostenberg and Roos 2000;

Heidt, Jr. et al. 2000; Chomiak et al. 2000; Hawkins et al. 2001) Running is the most usual non-contact causes of injury, accounting for 20-24% of acute injuries in elite players and 9-27% in youth players (Hawkins and Fuller 1999; Hawkins et al. 2001). Muscle strains occur most frequently during sprinting, especially hamstring strains (Nielsen and Yde 1989). Kicking the ball has found to be the causes of 9-10% of acute injuries in elite players and 8-13% in youth players (Nielsen 1990; Hawkins and Fuller 1999). The result of the research provides a useful insight in the injuries due to causes of football players and its ill effects of football performance.

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### CAUSES OF INJURIES AT THREE LEVELS IN COMPETITIVE FOOTBALL

The present study deals with comparison of causes of injuries among three groups of competitive footballers. Accordingly three groups of footballers were targeted. International, National and State groups footballers aged between 14 to 30 years. The data was collected with the help of questionnaires prepared by Cromwell, F.J. Walsh Gromley for Elite Gaelic footballers (2000) and it was modified by the investigator and utilized. In this study total 300 hundred players were targeted ; of which 100 footballers of each group. Total 318 injuries out of 300 hundred footballers were found out over the one year period; 125 injuries out of 84 footballers were found in international group footballers. 108 injuries out of 82 footballers were found in national group and 85 injuries out of 78 footballers were found in State group. The mean (SDs) age of International group to State group footballers were 21.25 (7.08), 23.33 (7.78) and 19.91 (6.29) in years respectively. Their weight were 58.35 (18.45), 58.23 (19.01), and 53.99 (17.33) kg. respectively, their height were 167.33 (55.33), 166.09 (55.10) and 164.87 (54.66) cm. respectively, their training were 4.61 (1.47), 4.31 (1.43), and 4.10 (1.33) days in a week respectively, their training durations were 2.74 (.58), 2.34 (.78), and 1.99 (.66) hours respectively, their warm-up were 28.53 (9.33), 36.05 (11.05) and 22.8 (7.8) minutes respectively, and competition was 8.67 (2.81), 8.68 (2.78), and 6.58 (2.11) in one year respectively. The results revealed that only significant difference of occurrence of injuries was found ( $F=3.65$ ,  $P < .05$ ) in foul play. International group footballers were found to have got more occurrences of injuries as compared to the state group footballers. The study suggest that their was no fair play at high level competition. However, no significant difference of occurrence of injuries were observed among three groups of competitive footballers with respect to causes like collision, running, contact with ball, stumble, tackle and kicking the ball.

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## **EVOLUCIJA REKLAMIRANJA SA POSEBNIM OSVRTOM NA REKLAMIRANJE U SPORTU**

### **UVOD**

Kako se reklamiranje u sportu nije razvijalo zasebno, već predstavlja sastavni dio opšteg reklamiranja koje ima mnogo dublje korjene u istoriji čovječanstva, u ovoj studiji je, prije svega obrađena evoluciju reklamiranja u cjelini a zatim je načinjen osvrt na elemente evolucije reklamiranja u sportu kao zasebne oblasti opšteg reklamiranja.

### **EVOLUCIJA REKLAMIRANJA**

Još u vrijeme prvobitnih ljudskih zajednica ljudi su međusobno komunicirali. Međutim, ta komunikacija se ogledala, isključivo u vidu verbalne komunikacije s obzirom na tadašnji način života i stepen civilizacijskog razvoja. Kako se ljudska zajednica dalje razvijala tako se, u jednom trenutku pojavila potreba da se određene poruke sačuvaju kako bi se kasnije mogle upotrijebiti i prenijeti u istom obliku na daljinu, većem broju zainteresovanih učesnika, ili da se identična poruka ponovi više puta u nepromijenjenom obliku. Da bi ostvarili ove želje i zadovoljili potrebe koje su se ispriječile pred njih, ljudi su, kao prvo izumjeli pismo, jedno od najznačajnijih otkrića u ljudskoj civilizaciji. Pojavom pisma ljudi su bili u stanju da, čak i prije pojave Gutenbergove štamparije, razmjenjuju poruke u istom obliku i na daljinu, da ih plasiraju većem broju zainteresovanih učesnika, te da identičnu poruku ponove više puta u nepromijenjenom obliku. Pojavom ovog novog komunikacijskog sredstva, sa sigurnošću se može tvrditi da se ovdje nalazi početak propagande, kao specifičnog oblika društvene komunikacije. Daljim razvojem komunikacijskih sredstava kao što su štampa, radio, televizija, zatim telegraf, telefon, film, kao i satelitski načini prenosa poruka, propaganda je dobila savremene oblike koje mi poznajemo danas. Propagandne aktivnosti su, u tom vremenu bile unaprijed osmišljene sa tačno određenim ciljem da vrše uticaj u skladu sa interesima pošiljaoca propagandne poruke. Međutim, tadašnja propaganda je, vrlo često bila u funkciji jedne vrste manipulacije.

Ukoliko bi se tražili, zaista najdublji korjeni savremenog reklamiranja, na žalost teško bi se došlo do tog podatka, budući da je propagandna djelatnost, kao jedan od oblika društvene komunikacije, stara koliko i sama ljudska komunikacija pa ne postoje pisani zapisi o prvim pojavama ovih aktivnosti. Ipak, činjenica je, da je reklamiranje bilo poznato i najstarijim narodima jer postoje mnogi predmeti dostupni savremenoj arheologiji koji svjedoče o njihovoj garderobi i ukrasima koji su se nalazili na njoj. Prema jednoj legendi (citirano u Popović, 2011) smatra se da je sve počelo kada je jedan nezadovoljni seljanin uvidio da su njegovi klipovi kukuruza manji u odnosu na kukuruz koji se nalazio na ostalim policama i da neće uspjeti da proda niti jedan klip uko-

liko bi čekao da potencijalni kupci samo dođu. Odlučio je reklamirati svoj proizvod tako što je počeo navoditi njegove kvalitete koji se nisu mogli provjeriti na licu mjesta. Ne sluteći, on je pokrenuo industriju koja, na početku XXI vijeka predstavlja jednu od najmoćnijih svjetskih industrija. Sa druge strane, prema mnogim izvorima koji su se bavili ovom problematikom, prva zabilježena pojava propagande (citirano u Popović, 2011), u značenju u kojem danas podrazumijevamo reklamiranje, veže se za 1622. godinu i papu Klementa VIII koji je uspostavio poseban organ katoličke crkve i nazvao ga „Sacra congregatio de propaganda fide“. Ovaj organ je bio sastavljen od kardinala i crkvenih velikodostojnika, a imao je za cilj da planski i efikasno širi katoličku vjeru (Slavujević, 1993). Shodno tome, pojam propaganda, prema Barašin (2007), se pojavio u vrijeme reformacije, i korišćen je, isključivo u vjerske, a kasnije se proširio i u političke, kulturne i druge svrhe. Propaganda je u to vrijeme bila glavni instrument kojim su se reformatorski pokreti služili kako bi širili svoju vjeru, a koliko je tadašnja propaganda bila dobro planirana svjedoče i propagandna sredstva u vidu simbola kao što je krst koji je označavao ideju žrtvovanja još od vremena Isusa Hrista. Od tadašnje propagande pa do specifičnog oblika komunikacije koji se danas prepoznaje kao reklamiranje nije prošlo mnogo vremena. Uplivom propagandnih aktivnosti u zakonske tokove, reklamiranje je dobilo i svoje zvanično ime i od tada se može govoriti o savremenom reklamiranju koje uz određene modifikacije i danas postoji. Na prvi pogled je izgledalo sputano strogim zakonskim aktima koji su obezbjedili zaštitu potencijalnih korisnika od manipulacije, ali je na taj način stvorena osnova za razvoj „reklamne industrije“, koja je u tijesnoj vezi sa ekonomijom, obećavala brz i uspješan razvoj. Rani oblici reklamiranja su, prema Džefkins-u (2003), bili znaci poput onih koji i danas postoje na krčmama, zatim stubovi sa crveno-belim prugama ispred berbernica, apotekarske posude ispunjene nekom obojenom tečnošću ili točak ispred kolarskih radionica.

Uparedo sa industrijalizacijom i razvojem masovne proizvodnje i pojavljivanjem viškova proizvoda koje je bilo veoma teško plasirati, stvorena je idealna prilika za razvoj najmlađe industrije u okviru ekonomije - „reklamne industrije“. Prvi narod koji je shvatio značaj reklamiranja su bili Francuzi kod kojih se ono razvijalo veoma brzo, dok su ga Njemci, koji su reklamiranje preuzeli od Francuza, razvijali postepeno i metodično (citirano u Popović, 2011). Dalji razvoj savremenih reklamnih aktivnosti u sferi ekonomije se veže za razvoj modernog industrijskog društva koje je bilo određeno ka proizvodnji i plasmanu proizvoda na tržište. Potreba za reklamiranjem se posebno pojačavala rastom populacije u gradovima, te pojavom sve većeg broja prodavnica i robnih kuća tj. pojavom konkurencije, kao i razvojem drumskog i železničkog saobraćaja potrebnog za prevoz proizvoda, i na kraju pojavom popularnih medija pogodnih za plasiranje reklamnih poruka u javnost. Ovaj način reklamiranja kojim su nepoznati potrošači na udaljenim lokacijama obavještavani o aktuelnoj proizvodnji velike količine proizvoda, pojavio se prvi put prije oko dvije stotine godina, a sa daljim razvojem medija, logično je da se razvijalo i reklamiranje.

Uparedo sa razvojem reklamiranja pojavile su se i prve reklamne agencije koje su bile, prema Džefkins-u (2003), isključivo zakupci reklamnog prostora koji je prodavan potencijalnim organizacijama. Međutim, vremenom su, uporedno sa razvojem



štampe reklamne agencije počele da se bore za posao nudeći organizacijama kreativnije usluge kao što su ilustrovane reklamne poruke, pa se smatra da su reklamne agencije od ovog momenta prešle put od zakupaca reklamnog prostora do mjesta gdje su postale uslužne agencije. Na konkretnom primjeru, istorijska je činjenica (citirano u Popović, 2011), da su kafanske novine iz XVII vijeka i reklamna agencija kao što je „White’s” iz 1800. godine bile zadužene za reklamiranje lutrije koju je organizovala britanska vlada. Zatim, isti autor navodi primjer da se na starim slikama Londona koje su oslikane krajem XIX vijeka, može primjetiti da se na konjskim zapregama nalaze reklamne poruke koje su promovisale neke proizvode koji još i danas postoje.

Kako se marketing razvijao u korak sa socijalnim, političkim i ekonomskim prilikama određenog perioda, najznačajniji razvoj je zabilježen u drugoj polovini XX vijeka kada se, nakon završetka Drugog svjetskog rata, kod ljudi počela javljati potreba za normalnim životom. Uporedo sa javljanjem ove potrebe, u prve dvije dekade nakon završetka rata, prepoznaje se period u kojem je stopa nataliteta bila u nevjerovatnom porastu (citirano u Popović, 2011), a novorođena populacija je polako postajala ključna ciljana grupa marketara u predstojećem periodu, s obzirom da dostupni statistički parametri svjedoče o neverovatno velikim zahtjevima ovih porodica za raznovrsnim proizvodima. Ubrzo, već u šestoj deceniji prošlog vijeka pojavljuju se savremeni imidž i savremena komunikacija. Međutim, prema Myerson-u, Vickers-u, i Bullmore-u (2004), kulturna komunikacija je u ovom periodu veoma spontana, a znakovi koje reklamne poruke sadrže sve više pokušavaju da postanu opšte prihvaćene predstave, ali ne obične predstave već njihovi ekstremni oblici. Ove promjene su, u šestoj deceniji prošlog vijeka snažno uzdrmale cijelu „reklamnu komunikaciju“, a vjerovalo se, prema Kodelupi-u (1995), da kolektivna predstava koja se pojavila može doprijeti do dna društvene zajednice i tako je navesti da aktivno učestvuje u društvenim procesima.

Reklamiranje, tek krajem šezdesetih godina prošlog vijeka postaje ozbiljnije i organizovanije, dok sedamdesetih godina doživljava svoje „zlatno doba“ (citirano u Popović, 2011). Glavne karakteristike reklamiranja u kasnim šezdesetim godinama prošlog vijeka, a koje se nastavljaju i u sedamdesetim godinama je jedinstvo snažne ideje i upečatljive slike. Međutim, sedamdesete godine je karakterisao i veliki optimizam, kada se činilo da je sve ostvarivo. U ovom periodu se sve više osnivaju nove reklamne agencije, istražuju se načini za formiranje što efikasnijih i zabavnijih novinskih reklama u kojima dominira isključivo pisani tekst. Javljaju se, čak i reklame protiv pušenja, ali i sve zanimljivije reklamne poruke koje promovišu prodaju cigareta. Reklamiranje je, tokom osamdesetih godina prošlog vijeka, poneseno mogućnostima kompjuterske grafike, bilo okruženo stalnim inovacijama i ambicijama na svim svojim poljima. Međutim, ovaj period su obilježile kontradiktornosti i složenost reklamnih poruka koje su bile oslikavane ekstremnom slavom na samom rubu dobrog ukusa tadašnje kulture. U tom periodu, takođe veliki broj organizacija obrađuje društvene probleme kao što su kampanje za slijepe i one sa oštećenim sluhom, zatim kampanje za prevenciju zloupotrebe djece, a sa ciljem skretanja pažnje javnosti.

Već početkom devedesetih godina prošlog vijeka, činjenica je da postoji više od 300 hiljada novina i časopisa (citirano u Popović, 2011), zatim više od 30 hiljada radio

stanica i da se, već tada koristi više od 6 hiljada baza podataka. Međutim, i pored navedenih podataka, devedesete godine su period u kome reklamne poruke postaju mračnije, realnije i snažnije predstavljaju stvarnost. Takođe se pojavljuje i snažnija konkurencija, pa se sve više sreću takmičenja u kreativnosti između proizvođača koja ih ohrabruju da smišljaju razne inovacije kako bi ostvarili primat na tržištu. Ovu dekadu je, takođe obilježila i pojava interneta, kao i prvi pokušaji elektronske trgovine te su srušene zabrane kada su u pitanju seksualni motivi, pa je vrijedno pomenuti da su promocione kampanje ženskog donjeg veša postigle nevjerovatne uspjehe. Novinske reklame, koje više nisu bile u pisanoj formi već su sve više ličile na postere, su postale veoma upečatljive a čitaoci nisu mogli da se ne zaustave na stranicama na kojima su se one nalazile. Prema tome, evidentna promjena koja se odigrala u ovom periodu se ogleda u činjenici da su se reklamne poruke sve više zasnivale na slikama a sve manje na pisanom tekstu.

Ulaskom u XXI vijek reklamne poruke su se razvijale u korak sa razvojem digitalne tehnologije, a reklamne agencije su tražile nove i neuobičajene načine da svojim nastupom obuhvate što veći broj ciljanih potrošača bez povećanja troškova za televizijsko reklamiranje. Marketari, sa ciljem da pronađu alternativu televizijskom reklamiranju, sve češće svoju pažnju posvećuju načinu na koji bi proizvodi trebalo da budu zapakovani i nose se idejom da sam proizvod svojim oblikom i dizajnom može efikasno prenijeti reklamnu poruku. Oni se, nakon što su se upoznali sa svim načinima putem kojih je moguće uvjeriti potrošače da kupe određeni proizvod, u ovom periodu odlučuju da promjene pristup potrošačima pa se više nije postavljalo pitanje da li su potrošačima potrebne npr. patike već se osmišljavala poruka koja bi postavljala pitanje da li potrošači koriste upravo njihov proizvod. Na konkretnom primjeru, vrlo je lako sjetiti se patika koje je promovisao košarkaš Majkl Džordan i robne marke kao što je „Nike“, a za koje se odomao naziv „Džordanke“.

Evolutivni proces razvoja reklamiranja je i danas prisutan, tako posmatrajući savremeni svijet i promjene koje se odvijaju uporedo sa procesom globalizacije, jasno naglašavaju ekspanziju „reklamne industrije“ sa kojom se susrećemo svakodnevno i na svakom koraku. Borba za potrošače, borba za tržište, za plasman raznih proizvoda nikada nije bila snažnija nego što je danas, ali isto tako ni reklamne aktivnosti nikada nisu bile duhovitije i sa originalnijim idejama nego što su danas. Kao što je već pomenuto, reklamiranje se, iako je marketinška tehnika, u sve većoj mjeri razvija u multidisciplinarnu i multimedijalnu disciplinu koja predstavlja kombinaciju mnogobrojnih nauka, kao što su komunikacija, sociologija i psihologija, a sa ciljem da se ostvari što bolji uticaj u komunikacijskom odnosu između organizacija, koji predstavljaju pošiljaoce reklamnih poruka, i potrošača, koji predstavljaju primaocce reklamnih poruka.

### **EVOLUCIJA REKLAMIRANJA U SPORTU**

„Reklamna industrija“ je zastupljena u svim segmentima društva, kako u sferama političkog života, posebno u političkim kampanjama u kojima neretko preuzima i lidersku poziciju i bez koje se današnje kampanje ne mogu ni zamisliti, tako i u ekonomskim, kulturnim i javnim dešavanjima, kao i socijalnim, psihološkim i drugih sistemima. Ona je, već toliko usavršena da je teško navesti sve njene oblike i sve oblasti u

kojima se primjenjuje. Danas, ona nije samo sredstvo poslovanja privatnih organizacija, nego i javnih ustanova, opština, gradova i zemalja. „Reklamna industrija“ se, kao dio svakodnevnog jezika i komunikacije među ljudima, neprekidno obogaćuje novim elementima i dimenzijama a, samim tim, ona prefinjeno sjedinjava u sebi različite jezičke i metajezičke kodove. Sumirajući saznanja iz različitih domena ljudskog djelovanja a u skladu sa naučnim i tehnološkim pronalascima, nije teško zaključiti da je reklamiranje postalo jedan od snažnijih komunikacijskih procesa koji se razvija u stalnom međusobnom odnosu između pojedinaca i društvene sredine.

Od svojih začetaka pa sve do danas reklamiranje se razvijalo, kako je već navedeno, ne samo po zakonima ekonomskog tržišta, što mu je bila primarna svrha, već se pretvorilo u jedan od najraširenijih jezika naše civilizacije prešavši put od verbalne, preko slikovne, pa sve do multimedijalne poruke koja danas predstavlja pravi praznik za oči potencijalnih primalaca reklamnih poruka. Sve više se u strukturi reklamnog jezika javlja značajan komunikacijski napredak između pošiljaoca i primaoca reklamnih poruka, a ove dvije grupe aktera u komunikacijskom lancu sve više grade odnos u kojem uspostavljaju međusobno povjerenje i na taj način stvaraju sisteme značenja, ugleda i identiteta putem kojih se poistovećuju proizvodi sa određenim načinom života, vrijednostima i zadovoljstvima.

U današnjem svijetu u kome sredstva javnog informisanja zauzimaju veliki dio informativnih tokova, reklama omogućava poruci da stigne direktno od pošiljaoca do primaoca, čak, u nekim slučajevima i bez posredovanja sredstava javnog informisanja. Međutim, ona ne predstavlja isključivo reklamiranje već ona sugerise na važnost načina života koji se vodi, a posjedovanje određenog predmeta podrazumeva dokaz takvog načina življenja. Reklamne poruke ne preporučuju direktno proizvod već, isključivo uspostavljaju vezu između određenog proizvoda i jednog načina života i društvenog položaja koji se nastoji zadobiti. Primjer sporta je najbolji dokaz za ovu tvrdnju a iz tog razloga oblast reklamiranja u sportu sve više postaje interesantan savremenim naučnicima.

Tokom poslednjih nekoliko decenija uporedo sa razvojem „industrije sporta“ razvija se i reklamiranje u sportu, posredstvom koga mnogobrojne poruke uspevaju da dođu do krajnjih potrošača a razne organizacije dolaze do mogućnosti da kroz reklamiranje u sportu ostvare čitav niz svojih ciljeva (Lyberger & McCarthy, 2001). Razlozi izučavanja reklamiranja u sportu su višestruki. Najvažniji je taj što reklamiranje u sportu ima dva pravca djelovanja. Prvi pravac djelovanja reklamiranja u sportu predstavlja kretanje ka sportu, kao specijalnom i posebnom medijskom prostranstvu, skupa povezanom snažnim odnosima i sa sredstvima javnog informisanja, dok drugi pravac djelovanja reklamiranja u sportu predstavlja kretanje od sporta, kao novog proizvodnog subjekta, prema drugim medijima, odnosno prema sredstvima javnog informisanja.

Reklamiranje u sportu se veoma brzo razvija širom svijeta, dominantan primjer ovog razvitka je najuočljiviji u „reklamnoj industriji“ u Sjedinjenim državama. Prema izveštaju Broughton-a i saradnika (1999), brojke pokazuju da su organizacije potrošile oko 45 milijardi dolara u toku 1999. godine na ime promocije u sportu u Sjedinjenim državama. Ova brojka čini samo 21% od 213 milijardi dolara koje su potrošene za

godinu dana u „industriji sporta“. Međutim, ono što je interesantno za ovu studiju je činjenica da većina promociionog ulaganja odlazi na reklamiranje, od 45 milijardi dolara potrošenih za promociju u sportu, približno 29,5 milijardi dolara je utrošeno na reklamiranje (Irwin i sar., 2002). Reklamiranje u sportu je postalo veoma važan medij za mnoge organizacije zbog svoje fleksibilnosti i širokih granica koje sportske platforme mogu ponuditi (Kropp, Lavack, Holden, & Dalakas, 1999). Potencijalni potrošači, kada prisustvuju sportskom događaju, amaterskom ili profesionalnom, ili ga gledaju posredstvom određenih sredstava javnog informisanja, izloženi su raznim vrstama reklamiranja. Za razliku od drugih vrsta reklama, potencijalni potrošači nehotice prihvataju reklamne poruke putem reklamiranja u sportu jer oni gledajući sportski događaj neprimjetno prihvataju reklamnu poruku, čak se u većini slučajeva i poistovećuju sa njom (Harshaw, & Turner, 1999).

### ZAKLJUČAK

Generalno, svjedoci smo, kada se govori o reklamiranju u sportu u Crnoj Gori, s vremena na vrijeme se mogu uočiti reklamne poruke u štampanim i elektronskim medijima. Međutim, iako su pozitivni pomaci prisutni, ipak se mora konstatovati da se ove aktivnosti još uvek odvijaju stihijski jer se ne može govoriti o planiranim aktivnostima kao što je to slučaj u Sjedinjenim državama, prije svega, ali i drugim razvijenim zemljama. Većina sponzorskih ugovora i plaćenih reklamnih poruka nemaju istinski komercijalni karakter, već su ugovorene sa željom da se pomogne određenoj sportskoj organizaciji te ne predstavljaju aktivnost koja bi bila dugoročnog karaktera i na obustrano zadovoljstvo svih učesnika u navedenom procesu. Stoga, krajnje je vrijeme da se određeni kadrovi u Crnoj Gori obrazuju u oblasti sportskog menadžmenta i zauzmu adekvatne pozicije u industriji sporta, sa ciljem da sa kolegama iz komercijalnih subjekata sačine strategiju koja bi reklamnu industriju u Crnoj Gori oživjela i da izvela je na pravi put kao novorođeno dijete. Stoga, ova studija može koristiti kao kamen temeljac daljim naporima, kako u teorijskom, tako i u praktičnom smisli, najviše iz razloga što je veoma bitno upoznati se sa evolucijom pitanja kojima želimo da se bavimo.

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#### *EVOLUTION OF ADVERTISING WITH A SPECIFIC RETROSPECTION AT SPORT ADVERTISING*

*The objective of this study represent the evolution of advertising, while the main goal will be directional to evolution of general advertising with a specific retrospection at sport advertising. The main tasks of this study are, the first explaining the interesting evolution of the advertising in general as well as drawing attention to the specific nature of the sport advertising as a separate field of advertising in general. Method: During the making of this study, the author used descriptive method with consulting of competent literature. The previous author' experience in this field was also so useful. Moreover, the author used the analytic method and parallel method that is the most productive if you make some inferences about some appearance. Results: Sport advertising did not develop separately, but it represents an integral part of the overall advertising that has much deeper roots in the history of mankind. This study made general retrospection of common characteristics and differences that might be relevant for many researchers in this area. Discussion: Sport advertising is developing rapidly around the world, the dominant example of this development is most evident in the "advertising industry" in the United States (Popović, 2011). However, the sport advertising is not at the expectable level in Montenegro, mostly due to the reason the "advertising industry" in the country is not developed yet. Hence, some further knowledge collected in this study might improve the contemporary situation and help the current and potential advertisers to invest in sport industry.*

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## **COMPETITIVE STATE ANXIETY: IMPACT OF POSITIVE SELF TALK TRAINING ON JUNIOR LEVEL FOOTBALL PLAYERS**

### **INTRODUCTION**

During the last few decades, coaches and athletes from a wide variety of sports have begun to realize the importance of the mental side of athletic performance. Sport specialists agree that athletic performance is influenced not only by physical skills but also by psychological ones. In order to achieve peak performance athletes need a “total package” including physical skills, psychological skills, fitness and injury prevention (Theodorakis et al 2012). An inherent aspect of competitive athletics is the need for players to meet the demands of competition and to perform well under pressure (Craft, L. L 2003). “The perception of a substantial imbalance between environmental demand and response capabilities under conditions which a failure to meet demands is perceived as having important consequences will respond to increase levels of cognitive and somatic state anxiety” (Martens, R., Vealey, R. S., & Burton, D. 1990). Competition can cause athletes to react both physically (somatic) and mentally (cognitive) in a manner which can negatively affect their performance abilities. Stress, arousal and anxiety are terms used to describe this condition.

The anxiety and performance relationship has been studied extensively in the past many years. Previous researches revealed that somatic and cognitive anxiety have different relationships with athletic performance. Athletic performance has been proven to be facilitated by moderate levels of arousal while high and low levels of arousal are associated with lower levels of performance (Martens & Landers, 1970; Sonstroem & Bemardo, 1982, Wrisberg 2010). Concerning the relationship between cognitive anxiety and performance, research has found a negative relationship involving these variables. Higher levels of cognitive anxiety are associated with lower levels of performance (Burton, 1988, Motowildo, Packard, & Manning, 1986). Klein & Taylor (2012) by using the technique of meta-analysis concluded that the negative relationship of anxiety and performance is: a) stronger in female athletes than in male athletes, b) stronger in young athletes than in older athletes, c) stronger in low-level athletes and d) stronger in team than in individual sports. On the other hand self-confidence shows a positive linear relationship, while the relationship between somatic anxiety and performance has the shape of an inverted U.

Theodorakis, Zourbanos (2013), Bandura A. (1977) stipulates that self-confidence is directly related to athletic performance. Increases in self-confidence are mirrored by improvements in performance. Prior experiences affect efficacy expectations; the probability performing to a high standard is much greater if you believe in your

abilities, therefore confidence has tremendous explanatory power when comparing fluctuations in performance ( Zeng 2008 ; Silva & Stevens, 2002).

Many different psychological skills training programs have been developed to teach athletes skills and techniques such as anxiety management, imagery, goal setting, concentration, self-talk, thought stopping and muscle relaxation technique (Weinberg & Williams, 1998). The implementation of a psychological program in the athlete's daily routine may result in the successful handling of pressure and anxiety which in turn enhance athletic performance. By establishing a psychological skills training (PST) program early, it may be possible for athletes to reach their potential more quickly by learning how to perform consistently through increased behavioral control (Balague, 2000). Psychological skills training programs have been shown to be effective for improving elite athletes' performances in golf putting (Cohen, Tenenbaum, & English, 2006; Thomas & Fogarty, 1997), tennis (Mamassis & Doganis, 2004) and football (Holm, Beckwith, Ehde, & Tinius, 1996).

Collectively, these findings suggest athletes may be utilizing more psychological skills in order to enhance self-confidence and protect against the potential debilitating effects of stress.

### **Positive self-talk**

Self-talk is said to be, "dialogue in which the individual interprets feelings and perceptions, regulates and changes evaluations and convictions, and gives him/herself instructions and reinforcement" (Hardy et al.2001). Basically, self-talk allows for an individual to take the perspective of another in their own mind and converse with themselves. When experiencing somatic and specific cognitive anxiety related symptoms, including doubts about performance and physical shaking, participants found that self-talk actually helped to control the anxiety responses. Self-talk helps to increase concentration on the task at hand. Increased levels of effort and motivation may be found by using constructive and adaptive statements regarding personal accomplishments and positive verbalizations about the training leading up to the competition (Hatzigeorgiadis, 2008).

Based on their findings, many researchers have thoroughly supported that self-talk can be an effective cognitive strategy for skill acquisition and performance enhancement," (Goltsois, Hatzigeorgiadis, Theodorakis & Zourbanos, 2008. Therefore, the purpose of the present study was to examine the effects of Positive Self-talk training program on the Competitive Anxiety and Self-Confidence of Junior Level Football Players.

## **METHODS**

### **Participants**

All the participants in this investigation were junior level football players who have participated in inter school level competitions. The total number of subjects in the experimental group who underwent the Psychological Intervention Program (PIP) was

18, aged between 15 and 18 years (mean age = 16.4, SD = 1.8). Their competition experience ranged from 1 to 3 years. The control group was comprised of 18 players aged between 15 and 19 years (mean age = 17.1, SD = 1.40) with competition experience ranged from 1 to 4 years.

### **Instruments**

Competitive State Anxiety Inventory-2 (Martens, Vealey, & Burton, 1990) was administered for the assessment of competitive anxiety of the players. They were advised to complete the CSAI-2 questionnaire twice, before and after Positive Self-Talk training program. The CSAI-2 is a 27-item self-reports having three subscales designed to measure cognitive and somatic anxiety, as well as self-confidence. Examples of somatic anxiety items include “I feel jittery” and “My heart is racing”; Cognitive items include “I am concerned about losing” and “I am concerned about performing poorly”; Self-confidence items include “I feel self-confident” and “I am confident I can meet the challenge”. Subjects responded on a 4-point Likert scale where categories vary from not at all (1), somewhat (2), moderately so (3) to very much so (4). The use of recall was used to ensure participants were cognisant of the thoughts and feelings they experience when excited or anxious, thereby facilitating accurate completion of the CSAI-2 in the appropriate scenario. This was the preferred methodology given the practical difficulties associated with the assessment of emotions at the time they actually occur (Ntoumanis & Biddle, 2000).

### **Procedure and intervention program**

Considering the fact that any psychological or mental training needs extensive practice before it starts blossoming (Theodorakis, I., Goudas, M., & Papaioannou, A. 2002), the Positive Self-talk training was administered for a period of eight weeks. Moreover, the training was carried out just before the competition period especially before a major tournament when they started the completion training to facilitate adequate time for the subjects to practice self-talk training program and to utilize it well before the competition. The daily mental practice lasted from 10 to 15 minutes before, during the training and also after practice, 3 to 5 days a week. The Pre-test data collection was done exactly before eight weeks of the actual competition and the post test data was collected just before 4 hours of the actual competition. Before the actual commencement of the intervention program participants in the experimental group were oriented about the use of Positive self-talk training and were informed that they were going to use this strategy for their training. The instructor explained and showed them how to use the technique during the training.

### **Results**

To test the effects of Positive Self-Talk Training on competitive anxiety, the initial and final test mean scores were tested treatment wise by applying the paired sample t-test using SPSS statistical package. Descriptive statistical values of the cognitive and somatic anxiety and self-confidence of the subjects in the experimental



group in table 2 and control group in table 3. The graphical representation of the mean values of cognitive anxiety, somatic anxiety and self-confidence were shown in diagrams 1, diagram 2 and diagram 3 respectively,

**Table 2.** Mean, standard deviation and t- ratio of the three subscales and total score of the Competitive State Anxiety Inventory of experimental group

|           | Cognitive anxiety |       |         | Somatic anxiety |       |         | Self-confidence |       |         |
|-----------|-------------------|-------|---------|-----------------|-------|---------|-----------------|-------|---------|
|           | Mean              | (±SD) | t-ratio | Mean            | (±SD) | t-ratio | Mean            | (±SD) | t-ratio |
| Pre test  | 22.6              | 2.4   | 6.79    | 23.8            | 2.8   | 9.51    | 20.4            | 3.1   | 11.2    |
| Post test | 18.1              | 2.1   |         | 17.6            | 1.9   |         | 24.8            | 2.3   |         |

The pre-test and post-test scores of the players in the experimental group shown in the Table – 1 indicates that the obtained ‘t’ ratios were: 6.79 for cognitive anxiety, 9.51 for somatic anxiety and 11.2 for self-confidence. While comparing these values with the critical value of 2.201 for degrees of freedom of 17 at 0.05 level of significance, it was found that the mean gains and mean losses are statistically significant. This indicates that eight week practice of positive self-talk training produced a significant improvement; that is decrease in cognitive anxiety and somatic anxiety and increase in self-confidence of the junior level football players in the experimental group under this study.

**Table 3.** Mean, standard deviation and t- ratio of the three subscales and total score of the Competitive State Anxiety Inventory of control group

|           | Cognitive anxiety |       |         | Somatic anxiety |       |         | Self confidence |       |        |
|-----------|-------------------|-------|---------|-----------------|-------|---------|-----------------|-------|--------|
|           | Mean              | (±SD) | t-ratio | Mean            | (±SD) | t-ratio | Mean            | (±SD) | t-atio |
| Pre test  | 22.3              | 2.1   | 7.63    | 23.4            | 2.9   | 8.62    | 20.5            | 1.9   | 10.9   |
| Post test | 21.8              | 1.9   |         | 22.6            | 2.1   |         | 21.2            | 1.9   |        |

Table 3 represents the pre-test and post- test mean values and t-ratios for the participants in the control group. It can be observed that the obtained t- ratios were 1.33 for cognitive anxiety, 1.62 for somatic anxiety and 1.90 for self-confidence. Since these obtained t-values were lower than the critical value of 2.201 for degrees of freedom of 17 at 0.05 level of significance, it was concluded that no significant difference existed between pre-test and post-test scores of cognitive and somatic anxiety and self-confidence of the players in the control group.

## DISCUSSION

The aim of the present investigation was to examine the effect of Positive Self-talk training on the competitive anxiety in junior level football players. The results revealed that the cognitive and somatic anxiety of the players in the experimental group

decreased significantly while self-confidence increased as an effect of a 8 week PST training programme.

Previous research has generally supported the beneficial effects of relaxation training on competitive anxiety and sports performance in various events. Fletcher and Hanton (2001) examined the intensity and direction of competitive state anxiety in swimmers who differed in their use of psychological skills. Findings showed that performers who reported a greater usage of relaxation strategies experienced lower levels of anxiety and interpreted symptoms as more beneficial to performance than their comparison groups. This is conversant with the findings of the present study. Maynard and colleagues found similar results when they employed an intervention approach with elite soccer players (Maynard et al., 1995a; 1995b). There are other studies revealing positive results of psychological training in non-sportspersons as well.

Regarding psychological skill usage, other studies predicted that elite athletes would use greater amounts of psychological skills, including goal setting, imagery, and self-talk while non-elite performers would report greater relaxation skill usage (Hanton and Jones, 1999; Richard Neil, Stephen D. Mellalieu and Sheldon Hanton, 2006). Anshel Porter (1996) and Bethany and Forrest (1998) demonstrated further collective evidence that application of psychological skill training programs can decrease competitive a-state anxiety as well as improve athletic performance.

## CONCLUSIONS

The present study provides preliminary evidence that Positive self-talk training for a period of eight weeks can be used as an effective tool to reduce cognitive and somatic anxiety and enhance self-confidence in junior level football players. It may be concluded that the psychological preparation of junior level players must be taken into serious consideration during the coaching procedure. The findings of the present study, hopefully, could help soccer coaches and sport psychologists to design more effective training plans, incorporating psychological skills. Professional help and programming of the psychological preparation of the athletes and observation of their emotional condition before and during a game is necessary to reduce competitive anxiety and contribute to better performance in volleyball players.

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#### *COMPETITIVE STATE ANXIETY: IMPCT OF POSITIVE SELF TALK TRAINING ON JUNIOR LEVEL FOOTBALL PLAYERS*

*During the last few decades, coaches and athletes from a wide variety of sports have begun to realize the importance of the mental side of athletic performance. Sport specialists agree that athletic performance is influenced not only by physical skills but also by psychological ones. In order to achieve peak performance athletes need a "total package" including physical skills, psychological skills, fitness and injury prevention (Singh 2011). Study was aimed to examine the effect of Positive Self-Talk training on the Competition anxiety and self-confidence of junior level football players. 36 junior level football players, aged  $18.7 \pm 2.8$  years, were randomly assigned into experimental group and control group. Competitive State Anxiety Inventory-2 (CSAI-2) was used to assess cognitive and somatic anxiety and self-confidence. Positive Self-Talk training was given to the experimental group for 8 weeks. Results of ANOVA revealed significant difference between the pre- test and post test scores of cognitive and somatic anxiety and self- confidence in the experimental group while no significant difference in the control group.*

**Key words:** Competitive anxiety, self-confidence, Positive Self-Talk

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## STAVOVI SRBIJANSKIH POTROŠAČA PREMA REKLAMIRANJU U SPORTU U ODNOSU NA UČESTALOST BAVLJENJA SPORTSKIM AKTIVNOSTIMA

### UVOD

Zaključci pionirskih studija iz navedene oblasti koji su objavljeni od strane Bauer-a i Greiser-a (1968) i Lutz-a (1985) pokrenuli su ideje koje se odnose na činjenicu da stavovi potrošača prema reklamiranju predstavljaju jedan od značajnih faktora koji mogu ojačati efikasnost reklamnih kampanja, uglavnom zbog razloga da kognitivna sposobnost potrošača prema reklamiranju jeste sadržana u njihovim osjećanjima i mislima (citirano u Muratović, Bjelica i Popović, 2014). Ako se nastavi sa istraživanjem ovog pitanja i ode se mnogo dublje, zanimljivo je da se primjećuje postojanje brojnih studija koje su istraživali stavove potrošača prema reklamiranju, ali većina njih smatra stavove potrošača prema reklamiranju samo uopšteno (Anderson, Engledov i Becker, 1978; Andrev, 1989; Bauer i Greiser, 1968; Lutz, 1985; Mittal, 1994; Muehling, 1987; Pollai i Mittal, 1993; Shavitt, Lovrei i Haefner, 1998; Zanut, 1984). U svakom slučaju, neki od njih su zaključili da su stavovi prema reklamiranju pokazali negativan trend tokom 1960-ih i 1970-ih (Andrevs, 1989; Muehling, 1987; Zanut, 1984), dok su drugi pronašli povoljnije stavove ispitanika u istom periodu (citirano u Shavitt i sar., 1998). Ove nedosljednosti su navele na sumnju da reklamne kampanje koje koriste elemente iz specifičnih medijuma utiču na stavove potrošača prema reklamiranju u skladu sa tim (Popović, 2011). Neki autori su vjerovali da je jedan od najrelevantnijih medija, upravo sport, odnosno sportisti i njihovi timovi, sportski događaji i sportski objekti (Bjelica i Popović, 2011; Pyun, 2006), uglavnom iz razloga što ovi elementi mogu da utiču na stavove potrošača i naprave značajne oscilacije kod stavova u poređenju sa ostalim elementima (Popović, Molnar i Radovanović, 2011; Pyun, Kvon, Chon i Wolf, 2012). Iz tog razloga, mnogi istraživači priznaju sport kao nezavisnu reklamnu platformu koju poslovni subjekti mogu koristiti kako bi prevazišli negativne stavove potrošača prema reklamiranju. Samim tim se postavilo i pitanje, kako bavljenje sportskim aktivnostima utiče na stavove potrošača prema reklamiranju u sportu kao jedan od niza problema sa kojim se nije susrećalo mnogo autora do sada (Bjelica i Popović, 2011; Popović, 2011; Popović, Molnar i Radovanović, 2011), a to upravo i jeste cilj ove studije.

## METOD

Populaciju u ovoj studiji su činili studenti Fakulteta za sport i fizičko vaspitanje na Univerzitetu u Novom Sadu koji su, u vrijeme anketiranja, imali boravište na teritoriji Srbije, dok je uzorak ispitanika organizovan putem kombinovanja ili raslojavanja, tako da su obrađena različita svojstva navedene populacije i različiti prostori na kojima je ona egzistirala.

Anketa je sprovedena tokom ljetnjeg semestra 2012/2013 školske godine a upitnici su distribuirani studentima osnovnih studija u štampanom i elektronskom obliku. Ukupno je prikupljeno 150 upitnika, ali je 29 upitnika bilo isključeno iz analize, budući da nisu bili adekvatno popunjeni, tako da je u istraživanju učestvovalo, ukupno 121 ispitanika. Instrument istraživanja je predstavljao standardizovani upitnik koji je bio prilagođen prema Pyun-u (2006) i sastojao se iz dva dijela, generalnih stavova prema reklamiranju u sportu i socio-demografskih karakteristika ispitanika kada je učestalost bavljenja sportskim aktivnostima tokom mjeseca u pitanju. Sistem promjenljivih u ovom upitniku je sadržao tri tvrdnje koje su ispitanici trebali da ocijene prema sedmo-stepenoj Likertovoj skali vrijednosti, kao i šest socio-demografskih karakteristika ispitanika (uopšte se ne bave, manje od jednom mjesečno, 1–4 puta mjesečno, 5–10 puta mjesečno, 11–20 puta mjesečno, i više od 20 puta mjesečno). Popunjavanje upitnika nije trajalo, u prosjeku više od 10 minuta a ispitanici su u anketi učestvovali na dobrovoljnoj osnovi. Važno je naglasiti da je anketa bila anonimna i da su svi odgovori bili strogo poverljivi. Vrijedno je napomenuti i da su ispitanici, i pored svega navedenog imali mogućnost da u svakom momentu opozovu svoje učešće u anketi, ali se niko od njih nije na tako nešto odlučio.

Empirijski podaci su analizirani putem statističkog paketa za društvene nauke (SPSS 20.0), a kao prvi korak, bila je primijenjena deskriptivna statistika kojom su izračunati, prije svega frekvencija, zatim aritmetička sredina, standardna devijacija, kao i mjere asimetrije (Skewness) i spljoštenosti (Kurtosis) za svaku od tvrdnji. Budući da su se promjenljive u ovoj studiji nalazile na naparametrijskim skalama, radi detaljnijih analiza koje su slijedile, bilo je neophodno da se one, primjenom Blomovog postupka, transformišu u skale višeg reda. Zatim su primjenom multivarijatne analize varijanse (MANOVA), univarijatne analize varijanse (ANOVA) i LSD Post Hoc testa utvrđene razlike u generalnim stavovima ispitanika prema reklamiranju u sportu u odnosu na učestalost bavljenja sportskim aktivnostima tokom mjeseca.

## REZULTATI

U prvoj tabeli su prikazani deskriptivni statistici za sve tri tvrdnje koje su se odnosile na generalne stavove ispitanika prema reklamiranju u sportu. Prije svega, prikazana je aritmetička sredina koja oslikava pozitivne vrijednosti stavova kada su sve tri tvrdnje u pitanju, dok vrijednosti standardne devijacije pokazuju da elementi skupa, u prosjeku ne odstupaju značajno od aritmetičke sredine. Kada se govori o mjerama asimetrije (Skewness) i spljoštenosti (Kurtosis), negativne vrijednosti asimetrije kod svih promjenljivih pokazuju da je većina rezultata desno od srednje vrijednosti, među većim vrijednostima, dok negativne vrijednosti spljoštenosti kod jedne promjenljive

(GSS1) pokazuju da je raspodjela pljosnatija od normalne tj. da ima više slučajeva na repovima a pozitivna vrijednost kod jedne promjenljive (GSS2 i GSS3) da je raspodjela šiljatija od normalne tj. da ima više rezultata nagomilanih oko centra raspodjele.

**Tabela 1. Generalni stavovi prema reklamiranju u sportu**

|      | Mean | S.D.  | Skewness  |      | Kurtosis  |      |
|------|------|-------|-----------|------|-----------|------|
|      |      |       | Statistic | S.E. | Statistic | S.E. |
| GSS1 | 5,09 | 1,845 | -,834     | ,215 | -,380     | ,427 |
| GSS2 | 5,39 | 1,840 | -1,203    | ,217 | ,387      | ,430 |
| GSS3 | 4,93 | 1,430 | -,693     | ,219 | ,259      | ,435 |

Legenda:

Mean – Aritmetička sredina

S.D. – Standardna devijacija

Skewness – Mjera asimetrije

Kurtosis – Mjera spljoštenosti

Statistic – Statistička vrijednost

S.E. – Standardna greška

GSS1 – Moje opšte mišljenje je naklonjeno reklamiranju u sportu

GSS2 – Sveobuhvatno, smatram da je reklamiranje u sportu dobra stvar

GSS3 – Sveobuhvatno, da li ne volite ili volite reklamiranje u sportu

U nastavku ove studije su prikazani komparativni statistici generalnih stavova prema reklamiranju u sportu koji su dobijeni primjenom multivarijatne analize varijanse (MANOVA), univarijatne analize varijanse (ANOVA) i LSD Post Hoc testa, a u cilju utvrđivanja razlike u generalnim stavovima ispitanika prema reklamiranju u sportu u odnosu na učestalost bavljenja sportskim aktivnostima.

Inspekcijom druge tabele u kojoj su prikazani rezultati multivarijatne analize, jasno se uočava da u cijelom sistemu upoređivanih parametara ne postoji statistički značajna razlika u generalnim stavovima prema reklamiranju u sportu u odnosu na učestalost bavljenja sportskim aktivnostima ( $p=,68$ ).

**Tabela 2. Multivarijatna značajnost razlika u sistemu generalnih stavova prema reklamiranju u sportu kod ispitanika sa različitim navikama kada je bavljenje sportskim aktivnostima u pitanju**

|      |            | N   | Mean | S.D.  |
|------|------------|-----|------|-------|
| GSS1 | Ne bave se | 2   | 4,50 | 2,121 |
|      | > 1        | 3   | 4,33 | 2,887 |
|      | 1 – 4      | 11  | 4,82 | 1,779 |
|      | 5 – 10     | 26  | 5,15 | 1,736 |
|      | 11 – 20    | 37  | 5,16 | 1,642 |
|      | < 20       | 42  | 5,29 | 1,929 |
|      | Ukupno     | 121 | 5,14 | 1,786 |
| GSS2 | Ne bave se | 2   | 3,50 | 2,121 |
|      | > 1        | 3   | 4,33 | 3,055 |
|      | 1 – 4      | 11  | 5,27 | 1,348 |



|      |            |     |      |       |
|------|------------|-----|------|-------|
|      | 5 – 10     | 26  | 5,62 | 1,745 |
|      | 11 – 20    | 37  | 5,59 | 1,771 |
|      | < 20       | 42  | 5,40 | 1,862 |
|      | Ukupno     | 121 | 5,44 | 1,793 |
| GSS3 | Ne bave se | 2   | 4,50 | ,707  |
|      | > 1        | 3   | 3,33 | 2,517 |
|      | 1 – 4      | 11  | 4,36 | 1,120 |
|      | 5 – 10     | 26  | 4,88 | 1,143 |
|      | 11 – 20    | 37  | 5,03 | 1,443 |
|      | < 20       | 42  | 5,12 | 1,533 |
|      | Ukupno     | 121 | 4,92 | 1,424 |

F=,798; p=,68

Inspekcijom treće tabele u kojoj su prikazani rezultati univarijatne analize, jasno se uočava da, takođe nije došlo do statistički značajnih razlika u generalnih stavovima prema reklamiranju u sportu u odnosu na učestalost bavljenja sportskim aktivnostima kod sve tri promjenljive.

**Tabela 3.** Univarijatna značajnost razlika u sistemu generalnih stavova prema reklamiranju u sportu kod ispitanika sa različitim navikama kada je bavljenje sportskim aktivnostima u pitanju

|      | F     | p    |
|------|-------|------|
| GSS1 | ,294  | ,916 |
| GSS2 | ,818  | ,539 |
| GSS3 | 1,344 | ,251 |

Inspekcijom naredne tri tabele gdje su prikazani rezultati Post Hoc testa koji ukazuju na ne postojanje značajnih razlika između parova pojedinačnih entiteta sa različitim navikama kada je bavljenje sportskim aktivnostima u pitanju za svaku promjenljivu. Iako je, sudeći po rezultatima koji su se pojavili na univarijatnom nivou, očekivano da ne postoje statistički značajne razlike pojedinih parametara, ipak je utvrđeno da ne razlike ne postoje niti u jednom pojedinačnom odnosu svih analiziranih parametara.

**Tabela 4.** Utvrđivanje značajnih razlika u sistemu generalnih stavova prema reklamiranju u sportu primjenom Post Hoc testa između pojedinačnih entiteta sa različitim navikama kada je bavljenje sportskim aktivnostima u pitanju kod tvrdnje „moje opšte mišljenje je naklonjeno reklamiranju u sportu“

| vs      | Ne bave se | > 1  | 1 – 4 | 5 – 10 | 11 – 20 |
|---------|------------|------|-------|--------|---------|
| > 1     | ,920       |      |       |        |         |
| 1 – 4   | ,820       | ,682 |       |        |         |
| 5 – 10  | ,624       | ,459 | ,608  |        |         |
| 11 – 20 | ,616       | ,448 | ,582  | ,986   |         |
| < 20    | ,550       | ,381 | ,448  | ,771   | ,763    |

**Tabela 5.** Utvrđivanje značajnih razlika u sistemu generalnih stavova prema reklamiranju u sportu primjenom Post Hoc testa između pojedinačnih entiteta sa različitim navikama kada je bavljenje sportskim aktivnostima u pitanju kod tvrdnje „sveobuhvatno, smatram da je reklamiranje u sportu dobra stvar“

| vs      | Ne bave se | > 1  | 1 – 4 | 5 – 10 | 11 – 20 |
|---------|------------|------|-------|--------|---------|
| > 1     | ,613       |      |       |        |         |
| 1 – 4   | ,203       | ,425 |       |        |         |
| 5 – 10  | ,112       | ,245 | ,598  |        |         |
| 11 – 20 | ,112       | ,245 | ,604  | ,964   |         |
| < 20    | ,146       | ,321 | ,829  | ,640   | ,641    |

**Tabela 6.** Utvrđivanje značajnih razlika u sistemu generalnih stavova prema reklamiranju u sportu primjenom Post Hoc testa između pojedinačnih entiteta sa različitim navikama kada je bavljenje sportskim aktivnostima u pitanju kod tvrdnje „sveobuhvatno, da li ne volite ili volite reklamiranje u sportu“

| vs      | Ne bave se | > 1  | 1 – 4 | 5 – 10 | 11 – 20 |
|---------|------------|------|-------|--------|---------|
| > 1     | ,368       |      |       |        |         |
| 1 – 4   | ,900       | ,265 |       |        |         |
| 5 – 10  | ,711       | ,074 | ,308  |        |         |
| 11 – 20 | ,609       | ,048 | ,174  | ,695   |         |
| < 20    | ,546       | ,037 | ,117  | ,508   | ,773    |

## DISKUSIJA

S obzirom da su rezultati pokazali da ispitanici imaju veoma pozitivan stav prema reklamiranju u sportu što potvrđuje visoka vrijednost aritmetičke sredine za sve tri promjenljive, kao i da skoro dvije trećine ispitanika imaju pozitivan stav prema reklamiranju u sportu što se ogleda u izrazito negativnim vrijednostima mjera asimetrije, trebalo bi naglasiti da su ovi rezultate usaglašeni sa rezultatima prethodnih istraživanja (Bjelica i Popović, 2011; Muratović, Bjelicu i Popović, 2014; Popović, 2011; Popović, Molnar i Radovanović, 2011; Popović, Jakšić, Matić, Bjelica i Maksimović, 2014; Pyun, 2006), i da ne postoje značajne razlike koje bi trebalo pomenuti. Dobijeni rezultati, takođe jasno ukazuju da ispitanici koji žive na različitim lokacijama, kao što su Sjedinjene države, Turska, Crna Gora i Srbiji, imaju pozitivne stavove prema reklamiranju u sportu, dok je, ipak, poređenja radi, vrijedno navesti da, prema Mittal-u (1994), različite studije ukazuju na negativne stavove kada je reklamiranje proizvoda u tradicionalnim industrijama u pitanju. Prema tome, više je nego evidentno da je primjena sporta u savremenoj poslovnoj komunikaciji uticala da se generalni stavovi potrošača značajno promijene kada je reklamiranje u pitanju, a prepoznavanje privlačnosti sporta je omogućilo poslovnim organizacijama da se približe sportskim potrošačima i na mnogo bezbolniji način utiču na njihovo ponašanje.

Na osnovu statističkih analiza, utvrđeno je da se značajne razlike ne javljaju na multivarijatom nivou, kao ni između sve tri varijable na univerzijatom nivou značajnosti od  $p=.05$ . Takođe, interesantno je istaći da nisu pronađene značajne razlike između potrošača koji učestvuju u raznim sportskim aktivnostima, što nije bio slučaj u

prethodnim studijama (Popović i sar., 2011). Ovi rezultati su veoma važni za poslovne subjekte u Srbiji, uglavnom iz razloga što mogu spojiti sve potencijalne potrošače koji učestvuju u raznim sportskim aktivnostima u jednu homogenu grupu, ali i za naučnu javnost, najviše iz razloga što do sada to nije bio slučaj u prethodnim istraživanjima (Popović i sar., 2011).

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*ATTITUDES OF SERBIAN CONSUMERS TOWARD ADVERTISING THROUGH  
SPORT WITH REGARD TO THE FREQUENCY OF THEIR PARTICIPATION IN  
SPORTS ACTIVITIES*

*Advertising is an attractive promotional tool for marketers who can use it to strengthen communication with consumers and persuade them to purchasing certain product or service (Bjelica et al., 2014; Muratović et al., 2014). Hence, there is necessity to analyse their general attitudes among various questions, while this research was aimed at gaining relevant knowledge about the attitudes of Serbian consumers toward advertising through sport among the question how often they participate in sports activities. Methods: The sample included 127 respondents, divided into six subsample groups: consumers who do not participate in sport activities at all, then consumers who participate in sport activities less than ones a month, next 1–4 a month, 5–10 a month, 11–20 a month, as well as consumers participate in sport activities more than 20 times a months. The sample of variables contained the system of three general attitudes which were modeled by seven-point Likert scale. The results of the measuring were analyzed by multivariate analysis (MANOVA) and univariate analysis (ANOVA) and Post Hoc test. Results: Based on the statistical analyses it was found that significant differences didn't occur at multivariate level, as well as between all three variables at a significance level of  $p=.05$ . Hence, it is interesting to highlight that it was found there are no significant differences showed up between the consumers who participate in various sports activities. Discussion: These results are so important for the marketers, mostly due to the reason they can merge all the potential consumers who participate in various sports activities into one homogenous group. This wasn't the case in previous investigations (Popović et al., 2011) and this observation presents relevant information.*

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## THE KINEMATIC ANALYSIS OF THE GRAB, REAR TRACK AND FRONT TRACK START IN SWIMMING

### Introduction

In competitive swimming, the fundamental goal is to cover a set distance in the least amount of time. When evaluating a swimmer's performance, several measures such as final time, strategy, and technical components are considered. The latter, important parameters are speed, stroke mechanics, starting, turning, and finishing (Galbraith *et al.*, 2008). The start in swimming, together with the other elements of the race, is very important for achieving better results in competitions. Faster racing start performances can provide a swimmer with a significant advantage over competitors with slower starting performances (Juergens, 1994). Starts account for approximately 10% of total time in swimming events of 50m and for approximately 5% in those of 100m (Blanksby *et al.*, 2002; Maglischo, 2003). Hay (1986) estimated that the start accounted for 11% of the total race time for the 50m freestyle (Thanopoulos *et al.*, 2012). The three main starting techniques currently used by elite swimmers include the track start (both rear and front weighted) and the grab start (Galbraith *et al.*, 2008).

The grab start was introduced and presented by Eric Hanauer in 1960 (Maglischo, 2003) and it has become popular among swimmers. The track start appeared later, and it was presented by Fitzgerald in 1973. The main difference between these two start techniques is the position of the feet on the starting block. In the grab start, both feet are positioned parallel on the front part of the starting block, with the toes curled over the front edge of the starting block. In the track start, the foot of the starting leg is placed forward, on the front part of the starting block, while other foot is behind it (Maglischo, 2003). The main objective of swim-start research has been to identify the most effective start technique in terms of performance (Vantorre *et al.*, 2014). In the literature, the following kinematic parameters have been measured: flight time, block time, start time, start reaction, start time that involves flight time and block time, flight length, angle of take-off, angle of entry, takeoff velocity, the centre of mass velocity (Welcher *et al.*, 2008; Takeda and Nomura, 2006). Thanopoulos *et al.* 2012 determined differences between these two starts technique in terms of gender and Blanksby *et al.*, 2002 have also examined the different start technique. Breed and Young (2003), investigated the effectiveness of a resistance training programme, designed to improve vertical jumping ability, on the grab, swing and rear-weighted track starts in swimming. Vantorre *et al.*, (2014) were analysed comparison of grab start between elite and

trained swimmers. The aim of this study was to examine the differences in the some kinematic parameters between the grab and track starts rear-weighted track and front-weighted track.

## Methodology

### Participants

Ten elite male swimmers (mean age  $\pm$  SD 11.3 $\pm$ 2.3 years, mean height 152.3 $\pm$ 5.6 cm, mean weight 48.2 $\pm$ 3.9 kg), from Yildizlar Swimming Club, Kocaeli, Turkey were participated to this study. The swimming performances of the subjects were measured in the indoor pool, at Kocaeli University. All subjects had no significant musculo-skeletal injury according to their recent medical history.

### Procedures

The current study was conducted consistently to comply with the recommendations of the declaration of Helsinki. Before participating to the study, the subjects were informed of the potential risks and benefits of the study. Swimmers performed three trials with three different starts technique (Figure 1-3) in indoor pool after 16.00 pm.



**Figure 1. Grab Start      Figure 2. Rear-Track Start      Figure 3. Front-Track Start**

Subjects had to refrain from using alcohol, caffeine and ergogenic aids the day before the test. All data were recorded directly to computer by using two Basler A602f cameras at 100 hz. and they were analyzed after the test using Simi Motion 8.5.7 program. For field calibration, Direct Linear Transformation technique that was developed by Abdel-Aziz and Karara (1971) and Shapiro (1978) was used. Four calibration points were calculated by using 0.3m x 0.3m x 0.5m calibration cube. Four kinematic parameters; Flight length (FL) - the distance from the swimming pool wall under the starting block to the first contact of the swimmer's fingers with the water (in meters); Flight time (FT) - the time between the last contact of the feet with the starting block and the first hand finger contact with water (in seconds); Entry time (ET)- the time between the start signal and first hand finger contact with water (in seconds); Block time (BT)- the time between the start signal and moment when leaving the foot of the start block were calculated.

### Statistical Analysis

SPSS version 11.5 for windows (SPSS, Chicago, IL) was used to analyse the data. The statistical analysis was performed using the Friedman test followed by Wilcoxon test to determine significant differences of kinematic and performance variables. The statistical significance level was set at 0.05.

### Results

According to the results, statistically significant difference in the parameter FL ( $p=0.024$ ) was found. The difference between other parameters (FT, BT, ET) is not statistically significant ( $p>0.05$ ). Longer flight length in grab start is obtained rather than front track starts (Table.1), ( $p=0.007$ ).

**Table 1.** Results of kinematic parameters of three starting positions and statistical differences between each other

| Parameters           | Grab start<br>Mean±St.D | Front-Track start<br>Mean±St.D | Rear-Track start<br>Mean ± St.D | p     |
|----------------------|-------------------------|--------------------------------|---------------------------------|-------|
| BT (Block time) sec  | 0.78±0.29               | 0.76±0.21                      | 0.72±0.35                       | 0.98  |
| FT (Flight time) sec | 0.26±0.09               | 0.20±0.08                      | 0.19±0.09                       | 0.07  |
| ET (Entry time) sec  | 1.05±0.36               | 0.95±0.31                      | 0.9 ±0.43                       | 0.48  |
| FL (Flight length) m | 2.83±0.25*              | 2.56±0.23*                     | 2.73±0.31                       | 0.02* |

\* $p<0.05$

### Discussion

The aim of this study was to determine the differences in the kinematic parameters between the grab, front-track and rear-track starts. The start in swimming is very important for achieving better performance. It takes up an increasing proportion of the total duration of the competition, especially if the distance is shorter (Nikodelis and Kollias, 2003). There are a few studies about different starting techniques in the literature and some of these were investigated similar parameters with the current study (Thanopoulos *et al.*, 2012; Kruger *et al.*, 2003; Takeda and Nomura, 2006). Flight distance is a good indicator in swimming start performance and with better flight distance; swimmers can get some choice of strategy to achieve a ‘good start’ at the beginning of the competition. Some studies investigated this parameter. Jorgiæ *et al.*, 2010 were found the flight length (FL) of the grab start is approximately 0, 23m bigger than in the track start in their study, but it had not statistically significant. Thanopoulos *et al.*, (2012) examined the differences in the kinematic parameters between the grab and track starts and the differences between these two starts in terms of gender and they indicated that the males had a statistically longer flight distance (FD) for both start techniques compared with the females ( $p = 0.00$ ), the FD was greater by 0.39 m for the grab start and 0.41 m for the track start. Blanksby *et al.*, (2002) studied the grab, track and handle swimming racing starts by elite level swimmers. Reaction, movement, block and flight times, flight distance and the centre of mass at the set position were measured. No significant differences between dive groups in time to 10 m were

revealed pre or post training. Kruger *et al.*, (2003) investigated the differences in the two starts on a sample of women and concluded that flight time did not differ between the two starts. Takeda and Nomura (2006) determined that there was no statistically significant difference in the flight distance between the grab and track starts, with the flight distance being greater for the grab start than the track start (3.25m vs. 3.15m). Yu Lee *et al.*, (2012) were found no significant differences between the two groups for flight distance. Another research was investigated that grab start has further FD (0,09 m.  $p=0,03$ ), in comparison with track start as statistically significant (Baykal, 2013). Miller *et al.*, (2003) determined a greater ( $p<0.001$ ) flight distance for the grab start compared with the track start, a difference that was statistically significant and measured 0.14m.

In the study of Jorgić *et al.*, (2010) flight time (FT) of the grab start were found approximately 0.10sec bigger than in the track start, but it had not statistically significant. Galbraith *et al.*, (2008) during the 10m swim trial, significant differences were found in time to 10m and total time. Nicholas and Watkins (2006) stated that the flight time recorded for the females in their sample was significantly shorter than the time recorded for the males. In addition to the FD and FT, the males also had a greater compared with the females, but this difference was not statistically significant. Yu Lee *et al.*, (2012) found no significant differences between the two groups for flight time, besides there was found the track start was faster than the grab start in block time. Blanksby *et al.*, (2002) determined that there was no difference in the flight time (FT) between the grab and track starts. Thanopoulos *et al.*, (2012) stated that the males also had a significantly longer flight time for the grab start compared with the females ( $p=0.04$ ). Baykal (2013) was found no significant differences between the two groups for flight time, block time and entry time. Welcher *et al.*, (2008) found significantly sooner using the front-weighted track start (0.80s) than the other two starts (both 0.87s;  $p<0.001$ ) for block time. Juergens (1994), statistical analysis also determined significant difference between the grab start and track start techniques for block time.

In conclusion, it was obtain longer flight length in grab start rather than front track starts. It is recommended that athletes experiment with different start types to find their preferred technique and coaches should consider individual differences to decided which of the track and grab technique will be better for the competition.

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## THE KINEMATIC ANALYSIS OF THE GRAB, REAR TRACK AND FRONT TRACK START IN SWIMMING

The aim of this study was to determine the differences in the kinematic parameters between the grab and track starts (rear-weighted track and front-weighted track). Ten elite male swimmers (mean age  $\pm$  sd 11.3  $\pm$  2.3 years, mean height 152.3  $\pm$  5.6 cm, mean weight 48.2  $\pm$  3.9 kg) participated to this study. Swimmers performed three trials with three different starts technique in indoor pool. Data were recorded directly to the computer by using two basler a602f cameras at 100 hz. and records were analyzed after the test using simi motion 8.5.7 program. The study examined the four kinematic parameters: flight length (FL), flight time (FT) block time (BT), entry time (ET). The kinematic data were compared with friedman test using spss 11.5 program. Based on the comparsion results, it was found statistically significant difference in the parameter FL ( $p < 0.05$ ). Also, the difference between other parameters (FT, ET, BT) is not statistically significant. The flight length during grab start is longer than during front track starts.

**Key words:** swimming, grab-track starts, kinematic analysis.

„Dan“, 7. februar 2015.

НАЈВЕЋИ НАУЧНИ СКУП ИЗ СПОРТСКИХ НАУКА У РЕГИОНУ  
ОДРЖАЋЕ СЕ ОД 2. ДО 5. АПРИЛА У ПОДГОРИЦИ

## Рекордан број учесника

За 12. међународну конференцију и 11. Конгрес Црногорске спортске академије, који ће од 2. до 5. априла у Подгорици окупити око 250 учесника из 28 земаља, рецензенти су одабрали 110 научних радова.

– Сви прихваћени радови биће објављени у зборнику сажетак на енглеском језику, а послије Конгреса биће штампани у специјализованом часопису Спортмонт. Што се тиче пленарних излагања учешће су потврдили Хенк Ерик Мајер, Зденко Ретули, Вишња Ђорђић и Мартин Звонар, рекао је председник приређивачког одбора Конгреса, проф. др Душко Бјелица. – Специјални гост биће Бранислав Антала, редовни професор



на Коменус Универзитету из Словачке. Он је и председник FIEP Europe, организације чији је циљ да промовише развој широ-

ког спектра дјелатности у области образовања, спорта и физичког васпитања и школског спорта, спорта за све, фитнеса и здравља и рекреације за људе свих узраста и у свим земљама свијета. Планиране су и двије радионице у оквиру Конгреса ЦСА. Нејц Шарабон са Универзитета у Приморској у Словенији приказаће савремени приступ смањењу јаза између истраживања и потреба индустрије, кроз практичне примјере, док ће представници свјетски познате компаније Десис за производњу система за електромиографију представити најновија достигнућа у тој области, рекао је Бјелица.

Т.Б.

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## TESTS TO DETERMINATE THE FITNESS LEVEL IN RHYTHMIC GYMNASTICS

### Introduction

Rhythmic gymnastics is not one of the most explored topic by the scientists compared to sports such as football or cyclic sports which are much more analysed and covered by science. What is neglected by coaches and specialists is the physiological side of that sport. The load in rhythmic gymnastics is considered wrongly to be not so demanding but the results show different prospective. The heart rate at the end of the routines goes up to 200 b/min and blood lactate concentration up 12-13 mmol/l (Andonov & Gateva 2014; Gateva, 2008). With the changes in rhythmic gymnastics for the last 2 decades, provoking higher intensity during the routines (Gancheva 2009), we have to follow the principles of the training load even more. Not many authors (Baldari & Guidetti 2001; Douda, 2008; Guidetti, 2000) were found in the literature working on the effect from the training load and changes on the rhythmic gymnast's body. That's why the aim of this study is to implement tests to determine the fitness level of the gymnasts.

### Methods

Twelve high level rhythmic gymnasts (some in the Bulgarian national team) aged 15,7 ( $\pm 2,1$ ) took part in this investigation during the preparation period. Every gymnast had her own training regime. A written consent was obtained from all individuals researched.

Two laboratory and three field tests were applied to determine the level of fitness – cardiovascular and specific endurance in rhythmic gymnastics.

1. **VO<sub>2</sub> max**– Increase of the incline of the treadmill with 0,6° (starting from 0°) on each 30 sec. The speed during the test was constant – 8,4 km/h.
2. **2 min Submaximal treadmill test (95% of the maximal test)(authors' test)** –The test requires a 2 min run on a treadmill with a constant speed of 8,4 km/h and individually set up incline of 95% of the maximum for each athlete. The maximal incline is based upon:
  - theoretical maximal pulse,
  - reached HR and incline during the maximal test,
  - extrapolation of the incline as 100% from the theoretical maximal pulse and calculation of the 95% of the incline for each gymnast.
3. **2 min Shuttle test** – set distance of 10 m with two lines. Each time the subject is asked to touch outside the lines. The covered distance in meters (with accuracy of up to 0,5 m) is measured for 2 min. Athletes start in pairs.
4. **2 min Specific Modified gymnastics routine (authors' tests)** – A 2 min routine executed with the typical basic exercises for rhythmic gymnastics.

All of the specific exercise groups are displayed in this routine with low level of difficulty and structure close to the competitive.

5. **Competitive routine** – each gymnasts performed her own routine (average duration 1,28).

Procedure and the protocol of the tests

The individuals undertook a maximal gradual exercise test on a treadmill enabling the assessment of their  $\text{VO}_2$  max. The 2 min submaximal laboratory test was performed 24 hours after the maximal one took place. The start of each of the laboratory tests was preceded by a three minute light warm up. After completing each test the subjects were asked to have a passive recovery period for 10 min.

All 3 field tests were done in one day on the same week of the laboratory testing. At the beginning of the field testing session, a basic warm up was performed, followed by a 10 min specific warm up prior to each of the three field tests. After the completion of each test the gymnasts were asked to have a passive recovery period for 10 min. The next field test was done after 20 min. The testing order was:

1. 2 min shuttle test
2. 2 min specific modified gymnastics routine
3. Competitive routine.

The methods used for each of the 5 tests are heart rate monitoring and concentration of the blood lactate. Based on those two indicators it is possible to compare the load during all the tests.

For laboratory testing gas exchange, blood pressure and other associated metabolic and cardiovascular functions indicators were used. Gas exchange and heart rate were monitored during and 10 min after the exercise (load). Breath by breath gas exchange was measured continuously using OxiconPro (Yeger, Germany). Heart rate was recorded by POLAR RCX3. The recording started at rest and continued until 10 min post exercise (recovery period). A drop of blood sample was taken from the fingertip to measure blood lactate using an Accutrend Plus Roche. Lactate was taken before and on the 3<sup>rd</sup>, 5<sup>th</sup> and the 9<sup>th</sup> min during the recovery period.

Descriptive statistics was used to characterize the physiological reaction during and after applying the tests and routines in rhythmic gymnastics. ANOVA was used to identify any differences between the tests and Student's paired t-test was applied to compare the results between the tests. The level of significance of  $p \leq 0.05$  was adopted in all cases.

## Results

The HR values at the end of the load of each test are summarised in table 1. The lowest values are achieved on the submaximal laboratory test – 186,2 b/min and the highest after rhythmic gymnastics routine – 194,1 ( $\pm 4,2$ ) b/min (excluding the maximal test). The other two tests submaximal and shuttle run test reach midway HR. When the statistical difference in ANOVA was found we applied paired t-test between the tests for greater accuracy (see table 2). We can see that all the tests have significant difference with the maximal test which is normal and we won't comment this as the reason to apply  $\text{VO}_2\text{max}$  was to identify the maximum for each gymnast. Submaximal laborato-

ry test with the lowest values also gives differences with the rest of the tests which means that we couldn't meet the expected load as in the gymnastics routine. Basically without difference – similar heart values were reached only in the shuttle run test and the modified gymnastics routine. Competitive routine is still not reached by any other test and seems to be the most demanding amongst the tests.

**Table 1.** Descriptive statistics of the HR and concentration of the blood lactate after the completed tests

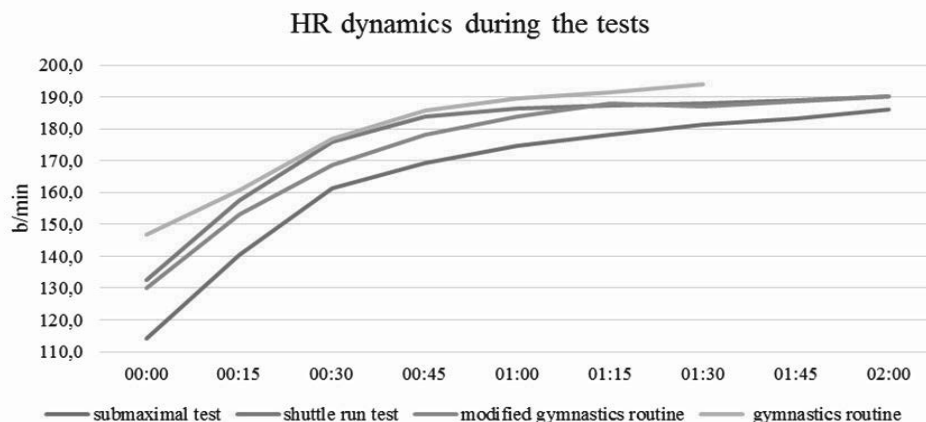
| test                                                       | HR mean (SD) (b/min)       | HR min. (b/min) | HR max. (b/min)           | La mean (SD) (mmol/L) | La min. (mmol/L) | La max. (mmol/L) |
|------------------------------------------------------------|----------------------------|-----------------|---------------------------|-----------------------|------------------|------------------|
| <b>VO2 max</b>                                             | 201,7<br>(±6,7)            | 191             | 211                       | 11,3 (±1,5)           | 9,4              | 13,6             |
| <b>Submaximal test</b>                                     | 186,2<br>(±5,8)            | 178             | 198                       | 10,0 (±1,6)           | 7,2              | 12,3             |
| <b>Shuttle run test</b>                                    | 190,1<br>(±6,6)            | 181             | 199                       | 10,9 (±1,8)           | 8,3              | 13,4             |
| <b>Modified routine</b>                                    | 190,1<br>(±5,4)            | 182             | 200                       | 7,2 (±2,5)            | 4,6              | 11,5             |
| <b>Gymnastics routine</b>                                  | 194,1<br>(±4,2)            | 187             | 200                       | 8,7 (±2,3)            | 5,7              | 12,8             |
| p < 0,05 significant difference between the groups (ANOVA) | F – 13,298;<br>Sig – 0,000 |                 | F – 8,317;<br>Sig – 0,000 |                       |                  |                  |

Concentration of the blood lactate varies between 7,2 to 11,3 mmol/l. The tests including running provoke higher values compared to gymnastics routines. And this is proven by no significant difference between the maximal, submaximal and shuttle run tests. The two gymnastics routines didn't show any difference between each other and this could explain the similarity and the structure of the tests.

**Table 2.** Student's paired t-test between tests – HR

|                                                   | VO2 max | Submaximal test | Shuttle run test | Modified routine | Gymnastics routine |
|---------------------------------------------------|---------|-----------------|------------------|------------------|--------------------|
| <b>VO2 max</b>                                    |         | 10,476**        | 5,694**          | 6,218**          | 3,939**            |
| <b>Submaximal test</b>                            | 2,105   |                 | -2,435*          | -2,326*          | -6,814**           |
| <b>Shuttle run test</b>                           | 0,868   | -1,994          |                  | -0,153           | -2,686*            |
| <b>Modified routine</b>                           | 4,655** | 5,223**         | 5,235**          |                  | -3,411*            |
| <b>Gymnastics routine</b>                         | 3,543** | 1,942           | 2,984*           | -1,937           |                    |
| <b>Student's paired t-test between tests – La</b> |         |                 |                  |                  |                    |

\* p ≤ 0.05/ \*\* p ≤ 0.01



Graph 1.

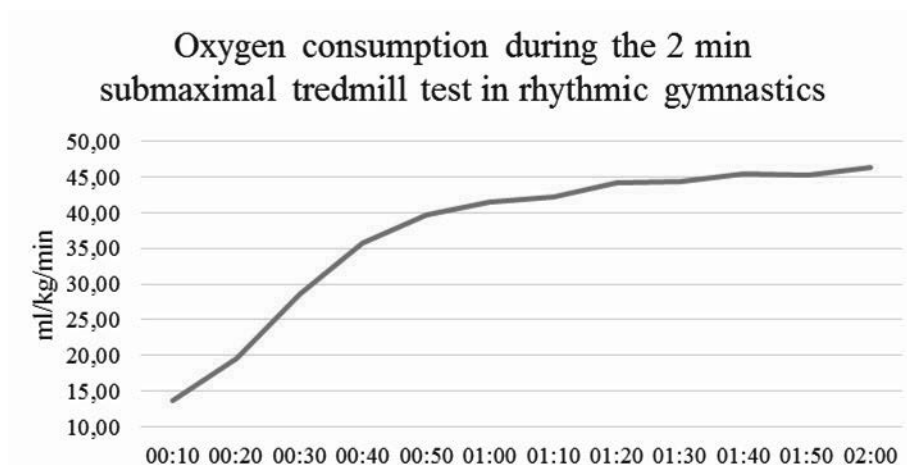
In the graphic (grap. 1) one can clearly see the cardio-dynamic and fast phases of the heart rate in the begginig of the tests. After the first 30 sec. of the load values continue to rise slowly just with 20 b/min. up to the end of the tests. At that point it is interesting to follow if there any difference. Submaximal test stayed with the lower HR all the way of the load with significant difference. Shuttle run test doesn't show differences with the gymnastics routine up to the last 30 sec of the routine. At the same point gymnastics routine clearly had highest heart rate and modified and suttle run test provoked similar response of the cardio-vascular system but lower than the gymnastics routine.

Table 3. Heart rate dynamics during the tests (means  $\pm$  SD)

| t (min) | Test 2<br>Submaximal test            | Test 3<br>Shuttle run test         | Test 4<br>Modified routine         | Test 5<br>Gymnastics routine          |
|---------|--------------------------------------|------------------------------------|------------------------------------|---------------------------------------|
| 00:00   | 114,2 ( $\pm$ 16,0)*                 | 132,5 ( $\pm$ 17,5) **             | 129,8 ( $\pm$ 17,1)                | 146,7 ( $\pm$ 10,9)**                 |
| 00:15   | 140,4 ( $\pm$ 13,3)*                 | 157,5 ( $\pm$ 13,4) **             | 153,1 ( $\pm$ 15,6)                | 160,8 ( $\pm$ 11,6) **                |
| 00:30   | 161,2 ( $\pm$ 11,9)*                 | 176,1 ( $\pm$ 7,1) **              | 168,8 ( $\pm$ 9,1)                 | 176,9( $\pm$ 7,5) **                  |
| 00:45   | 169,4 ( $\pm$ 9,9) *                 | 183,9 ( $\pm$ 6,1) **              | 178,3 ( $\pm$ 8,4) **              | 185,8 ( $\pm$ 5,3) **                 |
| 01:00   | 174,8 ( $\pm$ 8,7) *                 | 186,4 ( $\pm$ 6,1) **              | 184,0 ( $\pm$ 6,9) **              | 189,5 ( $\pm$ 4,9) **                 |
| 01:15   | 178,1 ( $\pm$ 7,6) *                 | 187,3 ( $\pm$ 6,2) **              | 187,9 ( $\pm$ 6,0) **              | 191,5 ( $\pm$ 4,2) **                 |
| 01:30   | <b>181,2 (<math>\pm</math>6,6) *</b> | <b>188,0 (<math>\pm</math>6,5)</b> | <b>187,2 (<math>\pm</math>5,5)</b> | <b>194,1 (<math>\pm</math>4,2) **</b> |
| 01:45   | 183,4 (6,4) *                        | 189,0 ( $\pm$ 6,5) **              | 188,6 ( $\pm$ 5,7)                 |                                       |
| 02:00   | 186,2 ( $\pm$ 5,8) *                 | 190,1 ( $\pm$ 6,6) **              | 190,1 ( $\pm$ 5,4) **              |                                       |

p < 0,05 significant difference between the groups (ANOVA); \* lowest of the groups/\*\* highest of the groups

The only test with oxygen consumption registry is the 2 min submaximal laboratory test. The highest consumption achieved at the end of the load is 46,4 ml/kg/min.

**Graph 2.**

### Discussion

VO<sub>2</sub>max was complete to calculate the individual 90-95% for the 2 min submaximal test. All the other tests were set to response to the same load as the competitive program in rhythmic gymnastics – a routine. It has been proven that in gymnastics the load during competitive routine is submaximal to maximal – in the anaerobic regime of work (Gateva, 2008; Guidetti, 2000; Jemni et al., 2011). That's why some adequate fitness tests responding to the specific load of the sport had to be created. Some other sports also suffer lack of assessment, and an identical study in climbing was found in the literature (Michailov et al., 2014).

The 2 min submaximal test and shuttle run test were likely to provoke absolutely the same load but the values showed other prospective. The curve of the dynamics of the HR appears to be absolutely identical but the field test had higher demands towards gymnasts probably because gymnasts started with maximal or close to the maximal speed, where in the laboratory test the load is limited. The modified routine also had lower HR values up to the first minute of the load compared to the gymnastics routine which is informative to make some changes in order to meet the same load like the competitive routine.

We can assume that the oxygen consumption during the competitive routine will be no far than the oxygen consumption during submaximal laboratory test. This means that, if we increase with 1-2% the values achieved on submaximal, the predicted oxygen consumption at the end of the gymnastics routine will be 47,4 ml/kg/min.

The differences between the tests based on the HR and blood lactate concentration gives us the information that more precise percentage of the maximal should be applied with the tests. The routine still showed higher demands on the gymnasts' body systems' compared to the other three applied tests. The shuttle run tests gave the clo-

sest values of HR to the routine load. Submaximal laboratory test have to be increased 1-2% in order to cause the same reaction to the gymnasts' body as a routine.

In the near future a more detailed comparison between the tests with Bland-Altman plot is planned to be made.

### Acknowledgement

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### TESTS TO DETERMINATE THE FITNESS LEVEL IN RHYTHMIC GYMNASTICS

*The need of implementing new tests for determining the fitness level of rhythmic gymnasts is vast. 12 high level rhythmic gymnasts were tested in preparatory period with 5 tests – two laboratory tests on a treadmill (VO<sub>2</sub>max and 2 min submaximal test) and three field tests (2 min shuttle run test, modified gymnastics routine and rhythmic gymnastics routine) to determine the fitness level of the gymnasts. VO<sub>2</sub>max was*



completed in order to calculate the individual 90-95% for the 2 min submaximal test. All the other tests were set to respond to the same load as the competitive program in rhythmic gymnastics – the routine.

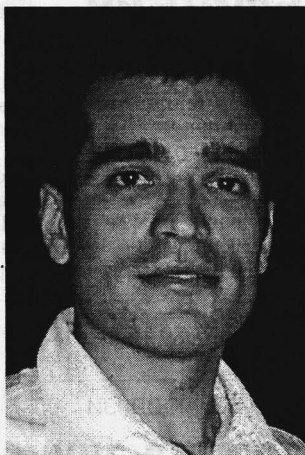
The differences between the tests, based on the HR and blood lactate give us reason to believe that the test loads should be selected more precisely. The routine still displayed higher demands upon the gymnasts' body systems compared to the other three applied tests. The shuttle run tests gave the closest values of HR compared to the routine load. Submaximal laboratory test have to be increased by 1-2% in order to cause the same reaction to the gymnasts' body as the routine.

„Dan“, 23. septembar 2014.

ПУБЛИКОВАН НОВИ БРОЈ СТРУЧНОГ ЧАСОПИСА MONTENEGRIN  
JOURNAL OF SPORTS SCIENCE AND MEDICINE

## У 25 међународних индексних база

Из штампе је изашао најновији број међународног часописа *Montenegrin journal of sports science and medicine*, који издаје Црногорска спортска академија из Подгорице. Уредништво је преузело иницијативу у смјеру индексирања часописа и његове што боље видљивости у широким научним круговима. Из тог разлога је формиран тим који је имао задатак да препознаје предности и слабости часописа те их представи уредништву, које је, одмах потом, доносило препоруке које су условиле да се повећају шансе за успешно индексирање. Тренутно, *Montenegrin journal of sports science and medicine* је препознат у 25 међународних индексних база и као такав представља реалног кандидата за добијање импакт фактора, који представља мјерило вриједности, у „Thomson Reyters“-овој бази у којој се још увијек процјењује. Као такав је лидер у Црној Гори, а главни уредник је проф. др Душко Бјелица, који има зајмјенике из пет држава.



Доц. др Стево Поповић

- Група полетних професора са Факултета за спорт и физичко васпитање на Универзитету Црне Горе и чланова Црногорске спортске академије из Подгорице запленили су путем откривања нових научних законитости у области спорте наука и спортске медицине и основали нови часопис чији је први број изашао у септембру 2012. године, при-

чећа се доц. др Стево Поповић одговорни уредник часописа. - Током протекле три године, часопис је доживио велике промјене. Дакле, у складу са својим континуираним растом, прије свега је осигурана уредна и правовремена публикација најновијих научних радова. У складу са тим, дошло је до значајног раста престижних чланака за објављивање који указују на све веће занимање научника из свих дјелова свијета за објављивање својих научних рада у нашем часопису. Часопис је у протеклом периоду објавио, просјечно по пет чланака, у сваком од својих бројева који су излазили шестомјесечно, док су досадашњи аутори чији су радови прошли ригорозну евалуацију били из Хрватске, САД-а, Шведске, Португала, Шпаније, Аустрије, Швајцарске, Француске, Чешке, Турске, Индије, Србије, Јапана, Бразила, као и из Црне Горе, рекао је доц. др Стево Поповић.

Т.Б.

UDC 796.015.53

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## **THE EFFECTS OF SINGLE LEG HOP PROGRESSION AND DOUBLE LEGS HOP PROGRESSION EXERCISE TO INCREASE SPEED AND EXPLOSIVE POWER OF LEG MUSCLE**

### **Introduction**

There are many training methods that can develop physical fitness of the athletes. One of them is plyometric training method. Plyometric is a training method that are used to increase strength and explosive power. Rimmer and Sleivert (2000) reported that plyometric is a type of training that can develop the ability of muscles to increase strength in high speed (power). They also found that plyometric can increase 40m sprint. In addition, Singh (2011) stated that plyometric has contribution to increase of jumping, speed, and strength. Faigenbaum (2007) reported that playometric can increase the ability of jumping and speed of the junior basketball. Furthermore, Shalla-by (2010) found that plyometric can improve physical fitness and skill of basketball players.

Radcliffe and Farentinos (1999) suggested that plyometric training should be started by using two legs and one leg to improve strength and power. McCurdy, et al (2005) reported that exercise using one leg and two legs were effective to improve strength and power for men and women. Kariyama, et al (2011) stated that one leg movement will give power output and hip relatively high and ankle relatively low, while two legs movement increased power output at joint ankle. Vaczi (2013) reported that one leg exercise in short training program will have advantages to get strength quickly. Base on those literature reviews, therefore the main purpose of this study was to determine the effect of single leg hop progression and double legs hop progression exercise to increase speed and explosive power of leg muscles.

### **Methods**

This research was conducted using quasi experiment with matching only design. The population of this study were 39 male students at state senior high school 1 Gerung West Lombok Indonesia who join basketball club in this school. There were three groups in this study: experiment group 1 (13 students), experiment group 2 (13 students), and control group (13 students). Each group was given different treatment during 8 weeks with 3 times training a week. Group 1 was trained single leg hop progression, group 2 double leg hop progression, and group 3 conventional exercise. Instruments of this research were sprint 30m (speed) and vertical jump (explosive power) that were tested on each student during pre test and post test. Data was analyzed using t test – paired sample test with  $\alpha=0,05$  and Analysis of Varians (Anova).

## Results

The result of this study showed that there were significant increased of speed and explosive power at group 1 and group 2 ( $\text{sig } 0,000 < \alpha = 0,05$ ). There were significant different between group 1, group 2 and group 3 ( $\text{sig } 0,000 < \alpha = 0,05$ ). The calculation of Post Hoc Test was presented in Table 1 and Table 2.

**Table 1. LSD of Speed**

| (I) groups                 | (J) groups                 | Mean             |            |      | 95% Confidence Interval |             |
|----------------------------|----------------------------|------------------|------------|------|-------------------------|-------------|
|                            |                            | Difference (I-J) | Std. Error | Sig. | Lower Bound             | Upper Bound |
| single leg hop progression | double leg hop progression | .05846*          | .02459     | .023 | .0086                   | .1083       |
|                            | control                    | .14077*          | .02459     | .000 | .0909                   | .1906       |
| double leg hop progression | single leg hop progression | -.05846*         | .02459     | .023 | -.1083                  | -.0086      |
|                            | control                    | .08231*          | .02459     | .002 | .0324                   | .1322       |
| Control                    | single leg hop progression | -.14077*         | .02459     | .000 | -.1906                  | -.0909      |
|                            | double leg hop progression | -.08231*         | .02459     | .002 | -.1322                  | -.0324      |

**Table 2. LSD of Explosive Power**

| (I) groups                 | (J) groups                 | Mean             |            |      | 95% Confidence Interval |             |
|----------------------------|----------------------------|------------------|------------|------|-------------------------|-------------|
|                            |                            | Difference (I-J) | Std. Error | Sig. | Lower Bound             | Upper Bound |
| single leg hop progression | double leg hop progression | 1.26846*         | .32296     | .000 | .6135                   | 1.9235      |
|                            | control                    | 2.52385*         | .32296     | .000 | 1.8689                  | 3.1788      |
| double leg hop progression | single leg hop progression | -1.26846*        | .32296     | .000 | -1.9235                 | -.6135      |
|                            | control                    | 1.25538*         | .32296     | .000 | .6004                   | 1.9104      |
| Control                    | single leg hop progression | -2.52385*        | .32296     | .000 | -3.1788                 | -1.8689     |
|                            | double leg hop progression | -1.25538*        | .32296     | .000 | -1.9104                 | -.6004      |

There were significant different between group 1 and group 2 on speed ( $\text{sig. } 0,023 < \alpha 0,05$ ) and explosive power ( $\text{sig. } 0,000 < \alpha 0,05$ ). There were significant different between group 1 and control group on speed ( $\text{sig. } 0,000 < \alpha 0,05$ ) and power ( $\text{sig. } 0,000 < \alpha 0,05$ ). There were significant different between group 2 and control group on speed ( $\text{sig. } 0,002 < \alpha 0,05$ ) and explosive power ( $\text{sig. } 0,000 < \alpha 0,05$ ). Base

on LSD above, it shown that there were significant different between single leg hop progression, double leg hop progression and conventional exercise. Single leg hop progression had better effect on speed and explosive power than double leg hop progression and conventional exercise.

### **Discussion**

Physical fitness is one of the important component for success to support the players during the competition. Speed and explosive power are needed by basketball player especially for lay up, jump shoot, rebound, and dribble. There are many kinds of exercise that can improve speed and explosive power, including single leg hop progression and double leg hop progression. Single leg hop progression has better result in improving speed and explosive power than double leg hop progression. It because of single leg hop progression can cause a heavier load compare to double leg hop progression. Although those exercise use the same their own body, however, single leg hop progression is more difficult. This finding is supported by some previous studies that reported one leg exercise in short training program will have advantages to get strength quickly; and one leg movement will give power output and hip relatively high and ankle relatively low, while two legs movement increased power output at joint ankle (Vaczi, 2013; Kariyama, et al, 2011). Single leg hop progression and double leg hop progression exercise have effect in increasing speed and explosive power of the players in this study. This is also supported by previous study that exercise using one leg and two legs were effective to improve strength and power for men and women (McCurdy et al, 2005).

Single leg hop progression is a better exercise to improve speed and explosive power. This finding is consistent with the result of relevant research which found that women who give one leg jumping exercise can improve jumping height quickly compare to two legs; plyometric training with one leg is better than two legs to improve explosive power women volleyball, therefore it can be used as reference for improving speed and explosive power (Makaruk, 2011; Dalacore et al, in McCurdy, 2005).

It can be concluded that single leg hop progression and double leg hop progression exercise can improve speed and explosive power, however, single leg hop progression is better than double leg hop progression in increasing speed and explosive power.

### **Acknowledgement**

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### THE EFFECTS OF SINGLE LEG HOP PROGRESSION AND DOUBLE LEGS HOP PROGRESSION EXERCISE TO INCREASE SPEED AND EXPLOSIVE POWER OF LEG MUSCLE

*The main purpose of this study was to determine the effect of single leg hop progression and double legs hop progression exercise to increase speed and explosive power of leg muscles. Plyometric is one of the training methods that can increase explosive power. There are many models of plyometric training including single leg hop progression and double leg hop progression. This research was experimental using match subject design techniques. The subjects of this study were 39 students who joined basketball school club. There were 3 groups in this study: Group 1 were 13 students who given single leg hop progression exercise, Group 2 were 13 students who given double legs hop progression exercise, Group 3 were 13 students who given conventional exercise. The*

data was collected during pre test and post test by testing 30m speed running and vertical jump. The data was analyzed using Analysis of Varians (Anova). It was found that there were significantly increased on speed and explosive power of leg muscles of Group 1 and Group 2. It can be stated that single leg hop progression exercise was more effective than double leg hop progression exercise. The recent findings supported the hypothesis that single leg hop progression and double legs hop progression exercise can increase speed and explosive power of leg muscles. These finding were supported by some previous studies (Singh, et al, 2011; Shallaby, H.K., 2010). The single leg hop progression is more effective than double legs hop progression. This finding was consistent with some previous evidences (McCurdy, et al, 2005; Makaruk et al, 2011).

**Key words:** single leg hop progression, double leg hop progression, speed, explosive power, and leg muscles.

„Dan“, 27. mart 2015.

ОДРЖАНА КОНСТИТУТИВНА СЈЕДНИЦА ПОЧАСНОГ ОДБОРА КОНФЕРЕНЦИЈЕ „СПОРТСКА ДОСТИГНУЋА“



## Развијати и научну димензију у области спорта

Већа међународна видљивост и оснаживање препознатљивог квалитета црногорског спорта заједнички је задатак академске, друштвене и спортске заједнице, а удруженим и посвећеним радом ти циљеви се могу и остварити. То је порука коју је на првој, конститутивној сједници упутио Почасни одбор међународне научне конференције „Спортска достигнућа“, којим предједава председник Владе Црне Горе Мило Ђукановић.

– Спорт представља најпрестижнију дјелатност у нашој држави, којом се Црна Гора данас најубједљивије представља на међународном тржишту. Зато морамо обезбиједити шире разумијевање и подршку укупне друштвене заједнице људима који су великим прегнућем допринијели тим резултатима. Важно је развијати и

научну димензију у области спорта, за шта су заслужне институције попут Факултета за спорт и физичко васпитање, рекао је Мило Ђукановић, поред којег су чланови Почасног одбора и: министарка науке Сања Влаховић, министар просвјете и председник РСЦГ Предраг Бошковић, градоначелник Подгорице Славољуб Стијеповић, председник ЦОК-а мр Душан Симоновић, директор Управе за младе и спорт Игор Вушуровић, председник АСЦГ Милорад Вулегић, председница Скијашког савеза Весна Меденица, председник ВИСЦГ Срђан Ковачевић, рукометни тренер Драган Ашић, ватерполо тренер Петар Поробић, фудбалски тренер Миодраг Божовић, рукометашица Катарина Булатовић, ватерполиста Никола Јановић, судиста Срђан Мрва-

љевић, спортски директор ЖРК Будућност Бојана Поповић, професор новосадског Факултета спорта и физичког васпитања Зоран Милошевић и професор Факултета спорта и тјелесног одгоја из Сарајева и потпредседник Олимпијског комитета БиХ Изет Рађо.

Међународна научна конференција о трансформационим процесима у спорту „Спортска достигнућа“, 12. по реду одржава се од 2. до 5. априла у Подгорици, и окупиће преко 160 водећих свјетских аутора и истраживача у области спортских, друштвених и хуманистичких наука и медицине из 22 земље. Конференцију организују Црногорска спортска академија и Факултет за спорт и физичко васпитање, под покровитељством Универзитета Црне Горе.

Т.Б.

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## **THE RELATIONSHIP AMONG COMMUNITY SPIRIT, COMMUNITY IDENTIFICATION, AND COMMUNITY LOYALTY OF ONLINE SPORTS COMMUNITY USER**

### **Introduction**

Online community is the group that develops social relations and connections by communicating consistently and repeatedly through computer among people who have similar interest and purpose on specific place and boundary of computer communication network (Kang, 2000). Online community is growing explosively to share similar interests, shopping, and information, which is supported by development of communications technology. These social and cultural phenomenon also appears on online sport community especially for among people who prefer sport by sharing information and by participating on sports. Accordingly, this study is to examine how community spirit of online sports community user affects community loyalty through community identification.

The importance of community spirit is led by Sarason (1977) and Corrigan (1996) who insist that it is most essential task to develop human community to recover into healthy society. With this point of view, developing sport community plays important role and recently various community spirit is built through online sport category. Generally community spirit in online sport emerges group phenomenon acting by organizing community voluntarily among people who have same interests (Park & Kang, 2007). In the community, people who prefer sports exchange information each other, enjoy leisure activities, and promote friendship by participating in sports.

In addition, community identification that is probable with community spirit is a medium connecting to community activities through process of having and accepting information by community members. Moreover, community loyalty that has relation with community spirit and community identification is the attitude and behavior of specific information and service user, which can be conceptualized as revisit attitude and revisit behavior of online and internet (Sohn, 2002). However, most of the studies implemented are focused on economic functions such as advertisement, customer satisfaction, and shopping mall and study related to sport participation is deficient. Accordingly the purpose of study is to provide a fundamental guideline through the process for which online sports become the sports socialization for generating academic advancement and establishing healthy online sport culture. To accomplish the purpose of the study, three subjects of the study are generated as follows.

First, study on the relationship between community spirit of online sports community user and community identification.

Second, study on the relationship between community spirit of online sports community user and community loyalty.

Third, study on the relationship between community identification of online sports community user and community loyalty.

## Methods

### 1. Research Subject

To accomplish the purpose of the study, adult male and female users of "N" research engine with category of sports and leisure are subjected to questionnaire based on convenience sampling. Total of 500 questionnaires are distributed and 426 collected after participants fill out the questionnaires by e-mail. Among the collected data, total of 383 are used for final data analysis excluding 43 questionnaires by using screen test and individual characteristics are shown on Table1.

**Table 1. Individual Characteristics**

| Division |           | Frequency | Ratio(%) |
|----------|-----------|-----------|----------|
| Sex      | Male      | 288       | 75.2     |
|          | Female    | 95        | 24.8     |
| Age      | 20's      | 135       | 35.2     |
|          | 30's      | 154       | 40.2     |
|          | Over 40's | 94        | 24.5     |
| Total    |           | 383       | 100.0    |

### 2. Research Method

The questionnaire is designed based on the theories and precedent studies of various useful disciplines and individual characteristics consist of question of sex and age. To examine community spirit, questionnaires from (Hong, 2011) is used after revising to fit for the study. Questionnaires from (Park & Lee, 2012) for community identification and questionnaires from (Kim, 2010) used to examine community loyalty are used after revising to fit for the study. The questionnaire is designed with 5 point Likert scale from 1 point of strongly disagree to 5 point of strongly agree.

### 3. Validity & Reliability

To verify validity and reliability of questionnaire, exploratory factor analysis is implemented. Eigenvalues that decides number of factor is based on over 1.0 and factor loading value set up over 0.4. For the better factor pattern matrix, varimax rotation is implemented.

As a result of the exploratory factor analysis of community spirit, 4 sub-factors, emotional relationship of over .741, need-satisfaction of over .698, membership of over .701, and influence of over .618 are extracted as shown on Table 2. Additionally, overall explanatory power shows 63.820% with reliability of .648 ~ .817.

For the community identification variables which are unidimensional consisting of five questions, exploratory factor analysis is not implemented.



**Table 2.** Result of Exploratory Factor Analysis for Community Spirit

| Question                 | Community Spirit |             |             |             |
|--------------------------|------------------|-------------|-------------|-------------|
|                          | Factor 1         | Factor 2    | Factor 3    | Factor 4    |
| Emotional relationship 3 | <b>.848</b>      | .080        | .021        | .038        |
| Emotional relationship 4 | <b>.776</b>      | .151        | .154        | .094        |
| Emotional relationship 1 | <b>.775</b>      | .151        | .130        | .107        |
| Emotional relationship 2 | <b>.741</b>      | .037        | .009        | .141        |
| Need-satisfaction 3      | .035             | <b>.883</b> | .006        | .054        |
| Need-satisfaction2       | .137             | <b>.801</b> | .126        | .176        |
| Need-satisfaction1       | .052             | <b>.744</b> | .079        | .091        |
| Need-satisfaction4       | .188             | <b>.698</b> | .074        | .052        |
| Membership 3             | .031             | .042        | <b>.811</b> | .062        |
| Membership 2             | .105             | .114        | <b>.811</b> | .023        |
| Membership 1             | .102             | .159        | <b>.701</b> | .124        |
| Influence 2              | .068             | .053        | .048        | <b>.866</b> |
| Influence 3              | .083             | .069        | .068        | <b>.746</b> |
| Influence 1              | .213             | .104        | .277        | <b>.618</b> |
| Eigen value              | 2.609            | 2.574       | 1.958       | 1.794       |
| Of variance(%)           | 18.633           | 18.383      | 13.988      | 12.816      |
| Cumulative(%)            | 18.633           | 37.016      | 51.005      | 63.820      |
| Reliability              | .817             | .806        | .701        | .648        |

**Table 3.** Result of Exploratory Factor Analysis for Community Loyalty

| Question             | Community Loyalty |             |
|----------------------|-------------------|-------------|
|                      | Factor 1          | Factor 2    |
| Attitudinal loyalty1 | <b>.852</b>       | .211        |
| Attitudinal loyalty4 | <b>.830</b>       | .086        |
| Attitudinal loyalty2 | <b>.825</b>       | .232        |
| Attitudinal loyalty3 | <b>.769</b>       | .223        |
| Behavioral loyalty 1 | .182              | <b>.807</b> |
| Behavioral loyalty 2 | .119              | <b>.774</b> |
| Behavioral loyalty 3 | .211              | <b>.722</b> |
| Behavioral loyalty 4 | .166              | <b>.711</b> |
| Eigen value          | 2.805             | 2.432       |
| Of variance(%)       | 25.065            | 30.405      |
| Cumulative(%)        | 35.065            | 65.469      |
| Reliability          | .861              | .776        |

For the exploratory factor analysis of community loyalty, 2 sub-factors, attitudinal loyalty of over .769 and behavioral loyalty of over .711 are extracted as specified in Table 3. In addition, overall explanatory power shows 65.496% with reliability of .776 ~ .861.

## Results

### 1. Relationship between community spirit and community identification

Table 4 shows the result of multiple regression analysis to examine the influence of community spirit on community identification. As a result, it shows that community spirit influences on community identification. In other words, it is identified that emotional relationship, need-satisfaction, membership, and influence factors influence on community identification. The explanatory power of community identification toward community spirit variables is 41.6%.

**Table 4.** *Influence of Community Spirit on Community Identification*

| Variables              | b    | SEB  | $\beta$ | t          | p    |
|------------------------|------|------|---------|------------|------|
| Emotional relationship | .290 | .037 | .330    | 7.732      | .001 |
| Need- satisfaction     | .136 | .037 | .153    | 3.673      | .001 |
| Membership             | .193 | .039 | .203    | 4.915      | .001 |
| Influence              | .267 | .041 | .275    | 6.535      | .001 |
| $F=67.222(p<.001)$     |      |      |         | $R^2=.416$ |      |

### 2. Relationship between community spirit and community loyalty

According to Table 5, community spirit influence on attitudinal loyalty. That is, it is identified that factors of emotional relationship, need-satisfaction, and membership influence on attitudinal loyalty and the explanatory power of attitudinal loyalty toward community spirit variables is 16.1%.

**Table 5.** *Influence of Community Spirit on Attitudinal Loyalty*

| Variables              | b    | SEB  | $\beta$ | t          | p    |
|------------------------|------|------|---------|------------|------|
| Emotional relationship | .126 | .054 | .118    | 2.318      | .021 |
| Need- satisfaction     | .181 | .054 | .168    | 3.374      | .001 |
| Membership             | .258 | .057 | .223    | 4.515      | .001 |
| Influence              | .109 | .059 | .093    | 1.838      | .067 |
| $F=18.169(p<.001)$     |      |      |         | $R^2=.161$ |      |

According to Table 6, community spirit influence on behavioral loyalty. That is, it is identified that emotional relationship, need-satisfaction, membership and influence factors influence on behavioral loyalty and the explanatory power of behavioral loyalty toward community spirit variables is 14.2%.

**Table 6.** *Influence of Community Spirit on Behavioral Loyalty*

| Variables              | b    | SEB  | $\beta$ | t          | p    |
|------------------------|------|------|---------|------------|------|
| Emotional relationship | .150 | .048 | .161    | 3.138      | .002 |
| Need- satisfaction     | .140 | .047 | .149    | 2.967      | .003 |
| Membership             | .157 | .050 | .156    | 3.125      | .002 |
| Influence              | .129 | .052 | .125    | 2.470      | .014 |
| $F=16.753(p<.001)$     |      |      |         | $R^2=.142$ |      |

### 3. Relationship between community identification and community loyalty

Table 7 shows the result of multiple regression analysis to examine the influence of community identification on community loyalty. As a result, it shows that community identification influences on attitudinal loyalty. The explanatory power of attitudinal loyalty toward community identification variables is 8.8%.

**Table 7.** *Influence of Community Identification on Attitudinal Loyalty*

| Variables                | b    | SEB  | $\beta$ | t          | p    |
|--------------------------|------|------|---------|------------|------|
| Community identification | .359 | .059 | .296    | 6.053      | .001 |
| $F=36.644(p<.001)$       |      |      |         | $R^2=.088$ |      |

As Table 8 shows, it is identified that community identification influences on behavioral loyalty and the explanatory power of behavioral loyalty toward community identification variables is 8.7%.

**Table 8.** *Influence of Community Identification on Behavioral Loyalty*

| Variables                | b    | SEB  | $\beta$ | t          | p    |
|--------------------------|------|------|---------|------------|------|
| Community identification | .287 | .052 | .272    | 5.507      | .001 |
| $F=38.956(p<.001)$       |      |      |         | $R^2=.8.7$ |      |

### Discussion

The purpose of the study is to examine the relationship among community spirit, community identification, and community loyalty of online sports community user. To accomplish the purpose of the study, analyses for the results are implemented based on the subjects of the study. Discussion is also implemented to identify the relationship among variables based on the facts verified through the analyses. First, according to the result of the study, community spirit of online sports community user influences on community identification. This result corresponds with the results of the study by Shu & Kim (2003) that insists social cohesion and social relation, characteristics of community spirit, influence on quality of relationships and identification of users. Also, the study by Coates(2004) that empathizes the importance of relation-oriented communication and identification among users for the success of online community partially supports the result of the study. Accordingly, community spirit of online sports community user has positive relationship with community identification. That is it is identified that subjective characteristics of online commitment and group consciousness perceived by community users are causing variables positively influencing on community identification.

Second, as a result of the study, it is identified that community spirit of online sports community user influences on community loyalty. This result corresponds with the results of the study by Park & Yu(2003) that insists there is relationship between community spirit and website loyalty of online community. This result also supports the result of the study by Mun & Choe(2003) that insists common interests, role, and dependence consciousness directly influence on revisit attitude and revisit behavior to

the community. Overall, it is identified that common interests and role of online sports community user directly influences on revisit attitude and revisit behavior.

Third, it is identified that community identification of online sports community user influences on community loyalty. This result supports the results of the study by Yu(2007) that insists if psychological commitment of group is higher, the rate of visit, participation, and contribution is greater. This result also supports the result of the study of loyalty model for sports' spectators by Trail, Anderson, & Fink(2005) that insists the if the level of identification of sports consumers is higher, the more positive influence is on active loyalty.

### **Conclusion and Suggestion**

The purpose of the study is to examine the relationship among community spirit, community identification, and community loyalty of online sports community user. To accomplish the purpose of the study, adult users of "N" research engine with category of sports and leisure are subjected to questionnaire. Based on convenience sampling, questionnaires for 383 users are used for final data analysis. By utilizing SPSS 21.0, frequency analysis, exploratory factor analysis, reliability analysis, simple regression analysis and multiple regression analysis are implemented and the results are as follows.

First, community spirit of online sports community user influences on community identification. Emotional relationship, need-satisfaction, membership, and influence factors influence on community identification.

Second, community spirit of online sports community user influences on community loyalty. Factors of emotional relationship, need-satisfaction, and membership influence on attitudinal loyalty while emotional relationship, need-satisfaction, membership and influence factors influence on behavioral loyalty.

Third, community identification of online sports community user influences on community loyalty. Community identification influences on attitudinal loyalty and behavioral loyalty.

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#### THE RELATIONSHIP AMONG COMMUNITY SPIRIT, COMMUNITY IDENTIFICATION, AND COMMUNITY LOYALTY OF ONLINE SPORTS COMMUNITY USER

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„Dan“, 4. april 2015.

МЕЂУНАРОДНА НАУЧНА КОНФЕРЕНЦИЈА ЦРНОГОРСКЕ СПОРТСКЕ  
АКАДЕМИЈЕ СВЕЧАНО ОТВОРЕНА ЈУЧЕ НА УНИВЕРЗИТЕТУ ЦРНЕ ГОРЕ

## Са четири континента 166 учесника



Са отварања Конференције

Међународна научна конференција о трансформационим процесима у спорту „Спортска достигнућа“, 12. по реду, као и 11. Конгрес Црногорске спортске академије свечано су отворени јуче у згради Ректората Универзитета Црне Горе. Присутне је на почетку свечаности отварања поздравио предједавајући Конференције проф. др Душко Бјелица, који је истакао да „заслуге за успјешну организацију и видљиве резултате адекватне и примјенљиве у нашој струци не припадају само нама већ уваженим и еминентним учесницима из 22 државе са четири континента, који ће у току тродневног рада на више нивоа у разним областима и са много аспеката дати своја виђења на тему трансформационих процеса у спорту“.

– На укупно 94 прихваћена и пажљиво одабрана рада потписује се 166 аутора и ко-

аутора. Изворност тематског оквира наше Конференције, који садржајношћу захвата ширину свих потребних спортских постулата указује да наша жеља да досагемо до тога шта су то данас спортска достигнућа у многим спортским категоријама и да им се са научним и списатељским сензибилитетом да оовременски значај, рекао је, између осталог проф. др Душко Бјелица.

На отварању је говорио и министар просвете Предраг Бошковић, као и потпредседник ФИЕП проф. др Георги Балинт, који је у име предсједника ФИЕП Европа проф. др Мирослава Анталe примио специјалну плакету Црногорске спортске академије. Након јучерашњег рада у пријеподневним сатима у двије сесије и једном радионицом и поподневних пленарних излагања, Конференција ће наставити са радом данас у 9 часова.

Т.Б.

UDC 796.42:338.48(497.16)

**Miomir Maros,**

**Zlatan Mujak**

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## **SUCCESS OF OUR ATHLETES AS A WAY OF PROMOTING MONTENEGRO**

### **Introduction**

Winning the independence of Montenegro by independence referendum in year of 2006 was a breaking point in many areas of not only political, but also social acting.

Besides engaging in all relevant international sports organizations, successes of our sportsman begin to appear – water polo players first of all, basket ball players, hand ball players, as well as football representation and individual sportsman. For the first time it was seen that the true feeling of national acceptance and national identity during sports competition and victories, begin to replace hereditary stereotypes that bonded people of Montenegro with glorious history, epic experience of honor and belonging to the governing political parties. But, following years came with economy crises that influenced the world. It is evident that we don't have enough capacity, infrastructure, experts, or financial resources, to fight for a better positioning of our sport relative to other areas in Montenegro and relative to other competitive states, on a higher level. We have young, healthy and talented sportsman, we have a sports public and states that supports them, but all of that is not enough for them to shine as they were shining at the beginning of country independence, because it is necessary to empower the infrastructure, create a strategy and better financing of competitive sports.

### **THEORY OF SPORT PROMOTION**

In the book "The image of Montenegro" dr Dervis Selhanovic gives a significant place to sport and he states: "Sport development and forming of good athletes represent the part of social, cultural and political marketing of the country. Top athletes (besides cultural ambassadors and popular artists) are the best ambassadors of every country and of Montenegro as well. That is exactly the trend and imperative of so called sport diplomacy." (Selhanovic, 2010:157). Selhanovic further states that the development of sport must include many subjects, in order to reach the top results, but also states that besides taking care of by the state, it is necessary for private sector to be engaged, for funding, thus building their own the image in the country and the world.

Author of the book "The image of Montenegro" further states that the best promoters of Montenegro has been footballers Dejan Savicevic and Predrag Mijatović (which are known by many in the world, but those many don't know about Montenegro, prim.aut.), and boxers Ivan Strugar and Goran Radonjic who were on the front pages of sports attachments Washington Post and the annual WAKO. He states further – the smaller the country is the more important sports achievements are.

How significant is potential of promoting successful athletes is demonstrated by the fact that they increasingly become "icons" and the youth is looking up to them, but they are also good commercial bait, and successful companies hire them for advertising. David Beckham is the best example.

"However, it is undeniable that these phenomena reveal the value of an epoch, circumstances and conditions not only of individual lives, but also of the nation. Johan Hujzinga impression that modern man is trying to save his personality by playing sports wasn't accidental. Sport is not just pure skill and strength development, but it is a medium of formation of style and youthful feelings of strength and life. "(Kokovic, 2000: 84).

In the process of good promotion media are irreplaceable, because they inform, animate and thus recreate the public. By displaying sports programs, with national or international content, they achieve record ratings and profit immediately or later.

"Although television network transmission Olympics often suffer financial losses, and they offer all Dalh large sums of money for future broadcast rights. Behind it is an attitude that conveys the Olympics attract a large number of viewers where they can advertise their entertainment shows. Therefore, if these ads can provide a larger audience shows will be disclosed after the Olympics, TV networks will make more than enough money to cover losses from the transfer of the Olympics and on an annual basis will achieve higher overall profit. "(Potter, 2011: 493).

"Although television networks that transmit Olympics often suffer financial losses, they still offer increasingly large sums of money for future broadcast rights. Behind it is an attitude that transmitting the Olympics attracts a large number of viewers for whom they can advertise their entertainment shows. Therefore, if these ads can provide a larger audience for the shows that will be aired after the Olympics, TV networks will make more than enough money to cover losses from transmitting the Olympics and on an annual basis they will achieve higher overall profit. "(Potter, 2011: 493).

So, the media use sport for self-promotion, and it is used the same way by a large advertisers. All this confirms the great potential of the sport, which is a good base for the promotion of others, through which sport is being simultaneously promoted.

### **Results and Discussion**

For this work we interviewed five renowned experts in the field of sports, media and management. The study involved the President of the Montenegrin Olympic Committee, Dusan Simonovic, an adviser in the Directorate of Sports of the Government of Montenegro Marko Begovic, president of the Athletic Federation of Montenegro Milorad Vuletic, a sports journalist Maksim Živaljević and Professor of Management Dr. Marija Jankovic.

### *SWOT ANALYSIS*

Method that is commonly used in strategic management is the SWOT analysis. Strengths and weaknesses are internal positive and negative elements that can be



affected, while opportunities and threats are external positive and negative factors that can't be affected but they can be predicted and possible they can be adapted. By analyzing conducted interviews and research intervention, we'll make a SWOT table, which will be commented.

| STRENGTHS           | WEAKNESSES                            | OPPORTUNITIES                                          | THREATS                               |
|---------------------|---------------------------------------|--------------------------------------------------------|---------------------------------------|
| Human resources     | Inadequate sport education            | Sport education of the youth                           | Depopulation                          |
| Interest in sport   | Low finances                          | Industrial investment in sport                         | Change of the youth's are of interest |
| Successes so far    | Inadequate legislation                | Huge stars as promoters                                | Competitive environment               |
| Existing capacities | Poor infrastructure                   | New trends in tourism                                  | Transfers of the top athletes         |
| Media support       | Weak management – disconnected system | Interested top players for national team of Montenegro | Lack of sponsors and disconnection    |

### STRENGTHS

Almost all interviewed people highlighted human resources as the most important factor affecting the success of Montenegrin sport.

"The genetic code is a code by which we can explain our sports successes. I think that the traditional Montenegrin magnification can't be match with the assertion that a similar comparative example of a smaller geographical area with a higher concentration of sports aces of world fame can't be found." (Dusan Simonovic).

"In Montenegro, we have a healthy population, healthy youth population, and we are predisposed for better results provided that we are better organized. If we can succeed in that effort, we have shown that we can." (Milorad Vuletic).

In regard to this, there is a large interest of young people in sports.

"The advantage of sport in Montenegro is a great interest for children to be athletes, as well as the physical qualities for many of the sports field."

Past successes of our athletes are certainly the best motive for the country, businessmen, sports workers and future athletes to commit to this field. Also existing infrastructure is in fact very important.

"Montenegro has good infrastructure when it comes to playing sports indoors, because almost every city has a hall or a space for handball, basketball, football, volleyball and many other sports. We also have good grounds for football, especially in the capital city. Although the best of our results are in water polo, we do not have enough indoor pools, which would enable young people qualitatively engagement in sport. What is most striking today, and it might not be sufficiently recognized by Montenegrin public, is that the capital of Montenegro has no athletic stadium. We are on the way to built it (for a maximum of two years), and we expect support from everyone, especially from the media, to provide it for our citizens." (Milorad Vuletic).

Media support is certainly important, not only for attracting sports in Montenegro, but also for the world promotion through satellite channels and internet portals.

“...all Montenegrin media gave an outstanding contribution for the development of Montenegrin sport and such success would not exist without such media support. The role of journalism and journalists is huge, the right way to promote both good and bad things in order to correct the mistakes, and I hope it will continue to be like that, so the Montenegrin sport can rise to the wanted and expected position.” (Milorad Vuletic).

### *WEAKNESSES*

The lack of adequate sports education from an early age is certainly a problem for the development of the sport.

"So, if we take the athletic is a basic sport, first of all, emphasis should be given in particular to the development of Montenegrin sport on a massive scale. This can be done by encouraging the development of school sport, from 5th grade to the end of secondary education. This is a trend that is increasingly popular in the world and what the IAAF is the most persistent about." (Milorad Vuletic).

"Currently hundreds of "schools" of various sports are organized in Montenegro, which in the junior categories take on the role of sports clubs. These "schools" are characterized by a large number of members, and however, there is no room for improvement after 11-12 years. Following is the transition to the clubs which are less and less by number, and which, at least most of them, do not have the money to allow proper development of gifted children. Montenegrin sport is more and more amateurish." (Maksim Živaljević).

Lack of finance is very large, perhaps the biggest problem.

"First of all, the financial crisis is something that is necessary reflected on sport and its functioning." (Dusan Simonovic).

"We must encourage the development of sports in Montenegro on a qualitatively different basis, where there is a talented base for the development of one type of sport, to provide financial structure." (Milorad Vuletic).

Related to this is inadequate normative provision and institutional linkages in Montenegro. Although the sport in our country is defined as a constitutional category and activity of public interest, followed by adopted the National Program for development of sport, still, better legal regulation and application of standards, would contribute to adequate financing of sport, which should be "mixed model" - public private.

"In addition, the system of institutional support is often inert, based on the legislation that is insufficiently adapted to the peculiar nature of Montenegrin sport." (Dusan Simonovic)

Poor infrastructure is a major problem for achieving major sporting success.

"The disadvantages are: lack of qualification structure of coaches, especially in football; insufficient financial investment in the training of coaches and athletes, insufficient investment in technical capacity, especially courts." (Marija Jankovic)

"This is supplemented by the fact that the sports infrastructure is quite outdated and inadequate for accrued social needs." (Dusan Simonovic)

Lack of good management in sport and poor linkages not only among sport ones, but also among other factors in the country that would promote sport, it is certainly a big problem.

"It is necessary to create an organizational scheme, setting out clear responsibilities for stakeholders in the structure of the sport (administration, local government, COK, ...) but also to show the level of centralization, ie., the level of intervention by public authorities in the system of sport in order to closer defining the terms for all participants in the system of sport "(Marko Begovic)

### *OPPORTUNITIES*

Better politic of youth's education would certainly encourage their greater interest in particular sports. Construction of the Athletics Stadium in Podgorica and other facilities, as well as looking up to the world in terms of development of the area, will be useful for education.

"We are talented and healthy nation, we have a lot of quality youth who want to engage in sports, but one of the biggest priorities is a relatively cheap athletic stadium in the capital of Montenegro. We're on our way to do it (for a maximum of two years). "(Milorad Vuletic)

Industrial investment in sport will certainly be an opportunity for encouraging the sports development.

"Firstly, I believe (and not just me) that the precondition for the successful functioning of any sport collectives as well as a successful company, is quality of the management. For us in the Olympic movement, although we believe that they are very much needed to each other, sport with its specific role as a social, educational and cultural activity should be in front and sponsorship should provide quality of the sports budget and to further maximize profit to the partner company. "(Dusan Simonovic)

Our big stars as promoters of sports are great potential that we have to use in Montenegro, and in the world.

"Also, giving a chance to our sports aces, which are recognized around the world, using their knowledge, experience and connections, their inclusion in sports forums, giving voting rights in decision-making is one of the modalities of success." (Dusan Simonovic)

Montenegro as an ideal destination, with beautiful nature and pleasant climate, so it would be a good bait for top athletes, but also for those who come for active recreation.

"The potential of the healthy climate, the meet of Mediterranean and harsh climate, gives the possibility of organizing mass camps for summer and winter preparations. In front of Montenegrin sport is a bright future, provided that we make a qualitative strategy for the development of Montenegrin sport. "(Milorad Vuletic).

"I see the activities of the Montenegrin Olympic Committee moving in that direction for the project of building two mini - Olympic center, one at the north and the

other at the south of the country. They would primarily improve the conditions for the preparation of our athletes, then, what is not less important, they would enriched our tourist facilities "(Dusan Simonovic)

These opportunities would add interest of famous athletes in the world to be part of our national team, helping spread the word about Montenegrin sport throughout the world. The fact that an Argentine footballer of Montenegrin origin Ariel Esteban Saveljic and Serbian football player Adam Marusic recently became part of the Montenegrin national team gave us this idea.

### **THREATS**

We need to add an imminent threat to SWOT analysis in order to complete the assessment of strengths and weaknesses. Depopulation and low population growth can certainly diminish the advantages as well as the change of interest of young people, if they are not systematically supported in their interest for the sport. Strong competition in the environment, can certainly affect professional and recreational athletes to decide for other destinations rather than Montenegro, if we don't improve the infrastructure and don't lower our prices. Transfers of athletes are inevitable, which certainly affects the outflow of the best, so we should try to keep them. But when they leave, they are the best advertisement for Montenegro and part of our national teams. Football players and basketball players confirm this fact.

Further systemic lack of connection, and therefore insufficient animation of businessmen for sponsorship, can affect the Montenegrin sport to stay where it is or to regress. We hope that this will not occur.

### **Conclusion**

The motive for writing this paper was to encourage the positive promotion of Montenegro, because we believe that this is insufficiently done so far and that numerous factors which are responsible for the promotion or that could help are poor connected, and therefore the result are very weak. However, by the actualization of this story, we must come to the problems plaguing the Montenegrin sport. We have detected advantages and disadvantages. We are aware that we have a healthy nation that is talented and interested in sport, as well as in tourist destination with an ideal climate for the training of athletes. We also have achievements to be proud of. Insufficient financial support should be strengthened by precise standardization, and by animating successful businessmen to invest in sport and thus promote themselves better. We also need sports capacities, and we should include the private sector into the construction of infrastructure, which would in fact find of interest in that. Adoption and implementation of the overall strategy, which would link all the available capacity, would certainly contribute to strategic performance in further sporting endeavors and successes, within their promotion. The attitude of the authors of this work is that the potential of our famous athletes is under-utilized for the purpose of promoting the sport and tourism of Montenegro. We have players in the world's most prestigious football clubs and in the strongest basketball league, and they could use their positions for

giving publicity to our country in the world, for which they should be marketing specifically advised. Why not say that, during a visit to Montenegro, their symbolic appearance at gatherings of young people would be a motive enough for them to do their best in sports efforts and results. Maybe we can't be like countries that are bigger than us, and whose players are the best products of Montenegrin descent. But it is our obligation to do as much as we can, for at least we owe that to ourselves and Montenegro. We can do more than what we are doing now!

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### SUCCESS OF OUR ATHLETES AS A WAY OF PROMOTING MONTENEGRO

*The purpose of this scientific work is to determine how the successes of our athletes affect the international media and tourism promotion of our country, since Montenegro independence 2006. The aim of this work is to consider theoretical assumptions and experts' opinions, in order to give recommendations for utilization of sport success into better promotion of Montenegro all over the world in the future.*

*The views of the interviewed experts confirm that sport is more than just a game and competition, that is part of the national identity, and as such, a window to the world of our country. The success of our athletes, especially the handball and water polo teams, confirm the thesis that this branch is one of the best ways for positive quotations of a small country and for being mentioned in international diplomacy, tourism and media circles. Our footballers before Savicevic and Mijatovic and now Vucinic, Savic, Jovetic and Kascelan, are often more famous than our touristic locations. The views of recognized experts suggest that this segment of promotion of Montenegro should be watched closely, and that it would be useful for marketing campaigns to include our famous athletes, so the Montenegro can be recognizable in the world, and thus become a destination that will attract more tourists.*

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## **RAZLIKE IZMEĐU MOTORIČKIH I SPECIFIČNO-MOTORIČKIH SPOSOBNOSTI CRNOGORSKIH RUKOMETAŠA KONTINENTALNE REGIJE UZRASTA 14 I 15 GODINA**

### **Uvod**

Savremeni pristup bazičnim i specifičnim antropološkim obilježjima rukometaša, te specifičnim zahtjevima i karakteristikama rukometne igre, zahtijeva naučno utemeljene analize i spoznaje do kojih se može doći jedino na temelju naučno istraživačkih projekata. Da bi igrači bili sposobni ostvariti zahtjeve igre modernog rukometa, od njih se očekuju apsolutno najveći dometi u tehničko-taktičkoj i fizičkoj pripremljenosti.

Osnovni cilj ovog istraživanja bio je da se utvrde razlike u opštim motoričkim i situaciono-motoričkim sposobnostima mladih rukometaša. Rukomet je jedan od najpopularnijih sportova u Crnoj Gori. S obzirom da se Crna Gora dijeli na kontinentalnu i mediteransku regiju, u ovom radu je realizovano istraživanje koje se odnosi na mlade rukometaše iz kontinentalnog područja. Savremeni rukomet je igra izuzetno brzih eksplozivnih i višestruko složenih pokreta razvoja akcija, situacionih rješavanja problema, a koje sve zajedno obilježava nivo intenziteta aktivnosti, što zahtijeva od igrača visok nivo sposobnosti, kako bazičnih tako i specifičnih (Czerwinski, 1995).

### **Metode**

Uzorak ispitanika ukupno čini 50 rukometaša podijeljenih u dvije grupe. Prvu grupu čini 25 rukometaša RK „Sutjeska” iz Nikšića, uzrasta 14 godina  $\pm$  6 mjeseci, dok drugu grupu čini 25 rukometaša RK „Berane” iz Berana, uzrasta 15 godina  $\pm$  6 mjeseci. Igrači se takmiče u crnogorskoj kadetskoj ligi.

Motorički prostor procjenjen je sa 10 testova: taping rukom (TAPRUK), taping nogom o zid (TAPNOZ), pretklon raskoračno (PRERAS), duboki pretklon na klupi (DUBPRKL), skok udalj s mjesta (SKUDMJ), skok uvis s mjesta (SKUVMJ), bacanje rukometne lopte iz sjedla raznožnog sa tla (BARUKL), bacanje košarkaške lopte s grudi iz sjeda na stolici (BAKOŠL), okretnost u zraku (OKRZRA) i koraci u stranu (KORSTR). Specifično-motorički prostor procjenjen je sa 3 testa: sposobnost u bacanju i hvatanju lopti odbijenih od zid (SPBHLOZ), brzina vođenja lopte u kvadratu (BVLUKV) i kretanje u trouglu osnovnim odbrambenim stavom (KRUTROS).

Podaci dobijeni testiranjem obrađeni su elementarnim statističkim postupcima. U manifestnom prostoru primijenjena je komparativna statistika: t-test aritmetičkih sredina između malih nezavisnih uzoraka.

### Rezultati

Za utvrđivanje značajnosti razlika između aritmetičkih sredina rukometaša uzrasta 14 i 15 godina, primijenjen je t-test za male nezavisne uzorke, dok je statistički značajna razlika tretirana na nivou 0,05 (5%) (tabela 1).

**Tabela 1.** T-test između aritmetičkih sredina 14 i 15-godišnjih rukometaša kontinentalne regije (grupe 1 i 2) – u opštim i specifično-biomotoričkim sposobnostima

| Variable | Mean<br>G_1 - 14 | Mean<br>G_2 - 15 | t-value | df | p    |
|----------|------------------|------------------|---------|----|------|
| TAPRUK   | 36,88            | 32,20            | 4,22    | 48 | 0,00 |
| TAPNOZ   | 22,84            | 21,68            | 1,67    | 48 | 0,10 |
| PRERAS   | 55,92            | 55,20            | 0,25    | 48 | 0,80 |
| DUBPRKL  | 44,24            | 46,20            | -0,92   | 48 | 0,36 |
| SKUDMJ   | 2,13             | 2,02             | 2,05    | 48 | 0,05 |
| SKUVMJ   | 44,00            | 38,68            | 3,04    | 48 | 0,00 |
| BARUKL   | 17,82            | 18,60            | -0,69   | 48 | 0,50 |
| BAKOŠL   | 9,19             | 9,25             | -0,09   | 48 | 0,93 |
| OKRZRA   | 4,51             | 4,93             | -1,75   | 48 | 0,09 |
| KORSTR   | 9,88             | 10,47            | -2,84   | 48 | 0,01 |
| SPBHLOZ  | 20,56            | 20,48            | 0,07    | 48 | 0,94 |
| BVLUKV   | 6,37             | 6,80             | -2,71   | 48 | 0,01 |
| KRUTROS  | 6,99             | 8,19             | -5,94   | 48 | 0,00 |

Legenda: Mean – aritmetička sredina, t-value- vrijednost, df –stepeni slobode, p -nivo značajnosti

Rezultati dobijeni sprovedenim analizama ukazuju na postojanje statističkih značajnih razlika između četrnaestogodišnjih i petnaestogodišnjih rukometaša kontinentalne regije.

Ispitanici iz grupe rukometaša uzrasta 14 godina (G\_1) postigli su statistički značajno bolje rezultate od grupe rukometaša uzrasta 15 godina (G\_2) u ukupno 6 varijabli: tapping rukom (TAPRUK), skok udalj s mjesta (SKUDMJ), skok uvis s mjesta (SKUVMJ), koraci u stranu (KORSTR), brzina vođenja lopte u kvadratu (BVLUKV) i kretanje u trouglu osnovnim odbrambenim stavom (KRUTROS). Utvrđene su i određene razlike u ostalih sedam varijabli između ove dvije grupe koje nijesu statistički značajne.

Ispitanici iz grupe rukometaša uzrasta 15 godina (G\_2) nijesu postigli statistički značajno bolje rezultate od grupe rukometaša uzrasta 14 godina (G\_1) u bilo (nijednoj) kojoj varijabli.

U ostalim varijablama u kojima nije utvrđena statistički značajna razlika, rukometaši uzrasta 14 godina (G\_1) imali su bolje rezultate od grupe rukometaša uzrasta 15 godina (G\_2) u 4 varijable: tapping nogom o zid (TAPNOZ), pretklon raskoračno (PRERAS), okretnost u zraku (OKRZRA) i sposobnost u bacanju i hvatanju lopti odbijenih od zid (SPBHLOZ).

Ispitanici iz grupe rukometaša uzrasta 15 godina (G\_2) imali su bolje rezultate od grupe rukometaša uzrasta 14 godina (G\_1) u 3 varijable: duboki pretklon na klupi

(DUBPRKL), bacanje rukometne lopte iz sijeda raznožnog sa tla (BARUKL) i bacanje košarkaške lopte s grudi iz sjeda na stolici (BAKOŠL).

### **Zaključak i diskusija**

Dosadašnja istraživanja nijesu pronađena u prostoru rukometnog sporta kada je u pitanju problematika utvrđivanja razlika prema regijama u kojoj mladi sportisti odrastaju, a isto tako mali je broj radova koji analiziraju razlike rukometaša različitog uzrasta ili pak različitog ranga takmičenja.

Na osnovu dobijenih rezultata primjenom t-testa između rukometaša četrnaestogodišnjaka i petnaestogodišnjaka iz kontinentalne regije, dolazimo do zaključka da su rukometaši četrnaestogodišnjaci postigli statistički značajno bolje rezultate u šest varijabli od ukupno trinaest primijenjenih. Dobijene rezultate možemo opravdati činjenicom da se djeca navedenog uzrasta razvijaju različitom dinamikom, tj. brzina rasta njihovog košanog sistema, mišića, organskih sistema različita je od stepena do stepena i taj razvoj uveliko diktira njihove fiziološke i motoričke sposobnosti. Upravo je i to razlog zašto program treninga mora uzeti u obzir pojedinačne razlike i potencijale treniranja (Bompa, 2005).

U prethodnim istraživanjama analizirane su razlike između rukometaša različitog uzrasta (Vuleta, Milanović & Nikolić, 2012), kao i razlike između različitog ranga takmičenja (Vuleta, Milanović & Jukić, 1999), gdje dobijeni rezultati pokazuju na postojanje statistički značajnih razlika u prostoru motoričkih i specifično-motoričkih sposobnosti.

U ovom istraživanju dobijeni rezultati ukazuju na to da kod mlađih rukometaša (14 godina  $\pm$  6mjeseci) mogu preovladavati i drugi faktori kao što je motivacija koja predstavlja veoma bitan domen psihološke pripreme sportista, a sve u cilju dobijanja na minutaži i izgaranju za odgovarajuću poziciju u svom timu.

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*DIFFERENCES BETWEEN THE MOTOR AND SPECIFIC MOTOR SKILLS OF  
MONTENEGRIN HANDBALL PLAYERS CONTINENTAL REGIONS AGED 14-15  
YEARS*

*The main objective of this study was to determine the differences in the general motor and specific-motor skills of young handball players. Handball is one of the most popular sports in Montenegro. Given that Montenegro is divided into the continental and mediterranean region in this paper was carried out research relating to young handball players continental areas.*

*Modern handball is exceptionally quick and explosive and is made up of varying physically intense movements which require high level motor skills, both basic and specific (Czerwinski, 1995).*

*The sample makes a total of 50 handball players divided into two groups. The first group consists of 25 handball HC "Sutjeska" from Niksic, aged  $14 \pm 6$  months, while the second group consists of 25 handball HC "Berane" Berane age  $15 \pm 6$  months. Players compete in the Montenegrin cadet league. Motoric space is treated with 10 variables that hypothetically cover the space of: segmented speed, flexibility, explosive leg power, explosive power of arm and shoulder and coordination. The space of specific motoric abilities is tested with 3 variables that hypothetically cover the space of: the ability of manipulating the ball, speed in ball control in movement and speed of movement without ball.*

*Data obtained from the tests is calculated on basic statistic approaches. Comparative statistics is used in the manifest space: t-test arithmetic between small independent samples.*

*Results obtained by the conducted analyzes indicate the existence of statistically significant difference between the fourteen year old and fifteen-year-handball players continental regions. A statistically significant difference was found in 6 variables (hand tapping, long jump, high jump with seats, side steps, ability in throwing and catching balls rejected from the wall and move the triangle basic defensive attitude) out of 13 applied. In previous studies the differences were analyzed between the handball players of different ages (Vuleta, Milanovic & Nikolic, 2012), as well as the differences between the different ranks of competition (Vuleta, Milanovic & Jukic, 1999), where the results show that there are statistically significant differences in the area of motor and specific -motoric abilities. In this paper, statistically significant difference was found in six tests in favor of the fourteen year old handball, what can we explain that in the training process may prevail and other factors such as motivation, which is a very important domain of psychological preparation of athletes.*

**Key words:** *handball, continental regions, motor skills, specific-motor skills, t-test*

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## **ZNANJE SPORTISTA O REDUKOVANOJ SPORTSKOJ ISHRANI**

### **Uvod**

Decenije istraživanja podržavaju teoriju da od kada postoje sportska takmičenja prisutno je pitanje šta jesti i piti da bi se poboljšao sportski rezultat. Optimalna ishrana može da smanji zamor muskulature i omogućava sportisti koji je duže pod trenažnim opterećenjem i takmiči se, da se brže oporavlja (Lin i Lee, 2005). Sportska ishrana ima direktan uticaj na trenutno fizičko stanje organizma, odnosno fizička priprema sportista zavisi i od nutritivnog statusa angažovanog sportiste (Beals i Manore, 1998). Prema većini istraživanja sprovedenih u svijetu sportisti najviše informacija dobijaju od svojih trenera kada je u pitanju sportska ishrana, a pogotovo od trenera koji se bave kondicionim treningom (Burns, Schiller, Merrick & Wolf, 2004). Ne postoji samo jedan plan ishrane koji će doprinijeti poboljšanju takmičarske sposobnosti. U planiranju modela ishrane treba uzeti u obzir energetske potrebe, makronutritivni sastav namirnica, unos mikronutrijenata i balans tečnosti sportiste. Cilj ovog istraživanja je utvrđivanje znanja aktivnih sportista o redukovanoj sportskoj ishrani.

### **Metod**

Uzorak ispitanika bio je sastavljen od 60 profesionalnih sportista iz Crne Gore (fudbal, košarka i džudo). Znanje o redukovanoj sportskoj ishrani utvrđeno je putem odgovarajućeg standardizovanog anketnog upitnika koji je preuzet i modifikovan (Matković, Knjaz, Cigrovski 2006). Upitnik je bio koncipiran tako da se utvrdi znanje sportista o redukovanoj sportskoj ishrani, o sastojcima koji su neophodni da bi se obezbijedila dovoljna količina energije za trening i takmičenja, o dodacima ishrani, obrocima prije takmičenja kao i dehidraciji i rehidraciji za vrijeme treninga i takmičenja. Anketa je bila anonimna. Dobijeni podaci su obrađeni statističkim postupcima, pomoću statističkog paketa STATISTICA kojim je dobijen uvid u kvantitativne i kvalitativne vrijednosti istraživanja.

### **Rezultati**

U tabeli 1. prikazan je anketni upitnik sa pitanjima na koja su ispitanici dali svoje odgovore kao i odnos tačnih i netačnih odgovora. Kada se sagledaju rezultati u cjelini, može se zaključiti je da je znanje sportista o sportskoj ishrani na zadovoljavajućem nivou. Od ukupno 1200 odgovora, ostvareno je 787 tačnih odgovora, odnosno 65,5%. Međutim kada se pogledaju pojedinačni odgovori onda se zadovoljstvo zbog relativnog

visokog procenta gubi jer se uočavaju veliki propusti o vrlo bitnim pitanjima vezanim za sportsku ishranu.

**Tabela 1.** Anketni upitnik sa tačnim odgovorima i odnosom tačnih i netačnih odgovora

| Pitanja koja su bila ponuđena ispitanicima u anketnom upitniku                                                               | Tačni odgovori | Odnos tačnih i netačnih odgovora |
|------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------|
| 1. Proteini su glavni izvor energije?                                                                                        | N              | 23 - 37                          |
| 2. Prekomjieran unos proteina hranom opterećuje rad bubrega i jetre?                                                         | T              | 39 - 21                          |
| 3. Sportistima je potrebno 3 puta više proteina od netreniranih osoba?                                                       | N              | 16 - 44                          |
| 4. Sportistima je potrebno više ugljenih hidrata od netreniranih osoba?                                                      | T              | 48- 12                           |
| 5. Žeđ nije adekvatan pokazatelj potrebe za vodom tokom vježbanja?                                                           | T              | 26 - 34                          |
| 6. Tečnost se mora unositi prije, za vrijeme i poslije takmičenja?                                                           | T              | 47 - 13                          |
| 7. Preskakanje obroka je opravdano kada se želi postići brzi gubitak težine?                                                 | N              | 43 - 17                          |
| 8. Brzi gubitak tjelesne težine s jako restriktivnom dijetom može negativno uticati na sportsko izvođenje?                   | T              | 48 - 12                          |
| 9. Gubitak težine restriktivnim kratkotrajnim dijetama najvećim je dijelom posljedica gubitka tečnosti?                      | T              | 42 - 18                          |
| 10. Nedostatak kalcijuma može uzrokovati prelom kostiju (fraktura) i osteoporozu?                                            | T              | 48 - 12                          |
| 11. Unos mješavine različitih aminokisjelina može izazvati nutritivni disbalans – višak jedne aminokisjeline utiče na drugu? | T              | 30 - 30                          |
| 12. Citrusi (limun, narandža) su jedini izvor vitamina C u hrani?                                                            | N              | 41 - 19                          |
| 13. Obrok poslije takmičenja važan je samo ako je sportista gladan?                                                          | N              | 42 - 18                          |
| 14. Uravnotežena ishrana važna je samo prije takmičenja?                                                                     | N              | 49 - 11                          |
| 15. Posljednji čvrsti obrok treba pojesti 3-4 sata prije takmičenja?                                                         | T              | 43 - 17                          |
| 16. Nadoknadu tečnosti i ugljenih hidrata treba započeti neposredno poslije takmičenja?                                      | T              | 41 - 19                          |
| 17. Nemoguće je pobijediti bez suplemenata?                                                                                  | N              | 47 - 13                          |
| 18. Voće i povrće su dobar izvor vitamina i minerala?                                                                        | T              | 56 - 4                           |
| 19. Velike doze vitamina i minerala mogu biti opasne po zdravlje?                                                            | T              | 28 - 32                          |
| 20. Vitaminu i minerali su izvor energije?                                                                                   | N              | 16 - 44                          |

## Diskusija

Analizirajući i upoređujući rezultate istraživanja (Matković, Knjaz & Cigrovski, 2006) koji su na uzorku od 56 trenera košarke i skijanja, dobili 77,8% ispravnih odgovora i istraživanja (Vasiljević, Bojanić, Petković i Muratović, 2014 ) od 30 licenciranih trenera iz Crne Gore (fudbal, rukomet, košarku, odbojku, tenis i atletika ) na uzorku dobili 78,1% tačnih odgovora. Uvidom u rezultate našeg istraživanja, jasno je da su odgovori koji su dobijeni kada su u pitanju sportisti, sa manjim procentom tačnih odgovora u odnosu na sportske trenere. Uvidom u dobijene rezultate našeg istraživanja, više je nego jasno da sportisti pokazuju manji nivo znanja kada je u pitanju redukovana sportska ishrana, u odnosu na sportske trenere. Sagledajući pojedinačne odgovore, zabrinjavajuće djeluje podatak da jedna trećina anketiranih ima izuzetno nizak nivo znanja o proteinskoj ishrani i smatra da su proteini glavni izvor energije, a još više to što gotovo 75 % ispitanika ne poznaje metabolizam proteina. Ako se uzme u obzir činjenica da sportisti često koriste aminokiseline kao dodatak ishrani, bilo bi za očekivati da osobe koje savjetuju uzimanje takvih dodataka znaju i o problemima ili negativnim pojavama koje mogu ugroziti zdravlje sportista. Međutim ispitivani sportisti pokazali su što je i očekivano da znaju pravilno rasporediti obroke i tečnosti prije, u toku i poslije treninga i takmičenja, ali nažalost nešto više od polovine anketiranih sportista ne zna da žed nije adekvatan pokazatelj potrebe za vodom tokom vježbanja. Kad je riječ o vitaminima i mineralima koji se vrlo često koriste kao dodatak u ishrani sportista anketirani sportisti su pokazali dosta visok nivo informisanosti, što je bilo i očekivano. Ipak, zabrinjavajući podatak je da približno 75% anketiranih sportista misli da su vitamini i minerali izvor energije. Vitamini i minerali sami po sebi nijesu izvor energije i nemaju energetska vrijednost kako se ponekad pogrešno misli. Svake sekunde u tijelu se zbiva na hiljade hemijskih procesa koji nam omogućavaju da mislimo, osjećamo, vidimo, čujemo, pokrećemo se i drugo. Za sve te procese potrebni su nam vitamini i drugi makro i mikro sastojci (ugljeni hidrati, proteini i masti ). Bez dovoljnih količina tih sastojaka u tijelu nastaju brojni poremećaji koji mogu uzrokovati čak i teška oboljenja.

## Zaključak

Kada se sagledaju rezultati u cjelini, može se zaključiti da je znanje sportista o redukovanoj sportskoj ishrani na zadovoljavajućem nivou. Od ukupno 1200 odgovora, ostvareno je 787 tačnih odgovora, odnosno 65,5%, međutim kada se pogledaju pojedinačni odgovori onda se zadovoljstvo zbog relativno visokog procenta ispravnih odgovora donekle gubi jer se mogu jasno uočiti značajni propusti u sveukupnom znanju i to o vrlo bitnim pitanjima vezanim za sportsku ishranu. Na osnovu svega navedenog zaključak je da bi svaki sportski kolektiv trebao da ima sportskog nutricionistu kako bi se propusti u sportskoj ishrani sveli na minimum jer u planiranju ishrane treba uzeti u obzir energetske potrebe, makronutritivni sastav namirnica, unos mikronutrijenata i balans tečnosti sportiste. Očuvanje i unapređenje zdravstvenog statusa, poboljšanje kondicije, redukcija masnih naslaga i postizanje optimalne tjelesne konstitucije, pražnjenje negativne energije, lakše podnošenje stresnih situacija su osnovni benefiti koji se

dobijaju redovnim treningom i optimalnom ishranom. Poznato je da su redovni treninzi i optimalna ishrana dva neraskidiva faktora i samo njihovom kombinacijom mogu se postići optimalni rezultati.

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Vasiljevic, I., Bojanic, D. Petkovic, J., & Muratovic, A. (2014). Knowledge about sports nutrition coach. *Sport Mont*, 40,41,42 / XII, 126-131.

### ATHLETES' KNOWLEDGE OF REDUCED SPORTS NUTRITION

*Decades of research support the theory that when there are sports competitions there is also the question of what to eat and drink in order to enhance sports performance. Optimal diet can reduce fatigue, and allow athletes who train longer and compete to recovering faster (Lin and Lee, 2005). Nutritional status has a direct impact on the level of physical effect. In other words, the physical condition of preparation much depends on the nutritional status of persons engaged in sport (Beals and Manore, 1998). Methods: The sample was composed of 60 professional athletes from Montenegro (football, basketball and judo). Knowledge of reduced sports nutrition was tested by means of a standardized questionnaire. The questionnaire was designed to determine the knowledge of sports nutrition, the ingredients that are necessary in order to provide a sufficient amount of energy for training and competition, the dietary supplements, a meal prior to the competition as well as dehydration and rehydration during training and competition. Results: According to the results as a whole, it can be concluded that the professional athletes' knowledge of sports nutrition is at a satisfactory level. Out of 1200 responses 787 correct answers were achieved, or 65.5%. However, when looking at the individual responses then the satisfaction with the relative high percentage is not equal since we observed large gaps on very important issues related to sports nutrition. Discussion: By analyzing and comparing research results (Matkovic, Prince & Cigrovski, 2006), in a sample of 56 basketball and skiing coaches, 77.8% of correct answers were received. From a survey (Vasiljevic, Bojanic, Petkovic & Muratovic, 2014) of 30 licensed coaches from Montenegro (football, handball, basketball, volleyball, athletics and tennis) 78.1% of correct answers*

were received. By looking into the results of our study, it is clear that the results indicate that athletes provided a lower percentage of correct answers in relation to sports coaches, which was expected.. If we take into account the fact that athletes often use a dietary supplement exactly as recommended by coaches, it would be expected that the people who advise taking these supplements should know about any problems.

„Vijesti“, 9. mart 2015.

SPORT I NAUKA: ŠESTI ČASOPIS CSA

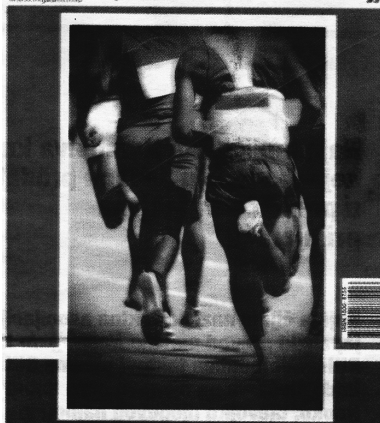
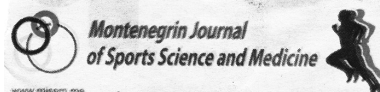
## Cenzuru prošao i rad naših naučnika

Crnogorska sportska akademija izdala je šesti broj međunarodnog stručnog časopisa na engleskom jeziku, „Montenegrin Journal of sports science and medicine“. To je prvi od dva broja časopisa koje će izdavač objaviti u ovoj godini.

Martovski broj „Montene-

urednik i direktor Crnogorske sportske akademije, prof. dr **Duško Bjelica**.

A da je prisustvo „Montenegrin Journal of sports science and medicine“ u najprestižnijim svjetskim naučnim bazama magnet za naučnike i sa najprestižnijih univerziteta, potvrđuje podatak da je prvi



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rad u martovskom časopisu objavio tim stručnjaka sa Harvarda, Državnog univerziteta u Bostonu i Zagrebačkog univerziteta.

Aktuelnost i atraktivnost časopisa obezbijedili smo međunarodnim uredničkim kolegijumom. U njemu su naučnici sa svih univerziteta u okruženju, ali i iz SAD-a, Australije, Njemačke, Španije, Italije, Novog Zelanda, Južne Afrike, Poljske Ujedinjenih Arapskih Emirata, Irana...- objasnio je Bjelica i dodao da je i rad njegovog stručnog tima prošao strogu cenzuru za objavu u ovom izdanju.

S. Z.

UDC 796.012.1-053.6

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## COMPARISON OF MORPHOLOGICAL CHARACTERISTICS AND SPECIFIC MOTOR OF JUNIOR SOCCER PLAYERS IN TWO PERIODS

### Introduction

Numerous researches (Verheijen, R 1997; Hoff, J 2005,) have shown that modern football is characterized by a dynamic higher than in the past decades as a result of the advancement of training processes has resulted in morphological changes and specific motor performances in favor of the current players from those decades earlier.

Gunnar, G. & Bizans, G. (2008) the importance of standardized testing technical skills and motor skills in different age players. Frank le Gall, F., Carling, C., Williams, M., & Reilly, T. (2010). These results Suggest That Anthropometric and Fitness Assessments of elite soccer players Youth Can play a part in determining the chances of proceeding to their Higher Achievement levels.

Molnar, Popović of Smajic (2009) compared the morphological characteristics in intervals of 10 years of the two generations in 1997 and 2007 soccer players of age 10 participating soccer school. Are certified morphological differences in favor of the new generation in 8 anthropometric variables.

The purpose of this paper is to compare the number of morphological characteristics and specific motor between players Junior (U-19) in two time periods in 2004 and 2014. The purpose of the paper was also to prove if we have positive developments in the researched areas in favor of Junior soccer players U19 2014 from the U19 junior soccer players a decade ago (2004).

### Methods

The research was conducted on a sample of 100 young soccer players (U-19 League first Kosovo) divided into two groups:

The first group comprised of 50 players U-19 to the five best clubs in 2003/04 season and the second group of 50 players U-19 also five best clubs 2013/14 season. In the two generations of players were conducted measurements of morphological characteristics and specific motor.

The sample contains 6 anthropometric variables, (AWEIGHT, AHEIGHT, ACHEC, APBELLY, APHIGH, ATHIGHBONE) eight specific motor variables; (M-JUMP, MR10m, MR30m, MR5x10m, MR300m, T-JUGGL, T-DRSLA, T-PASI).

To validate the differences between U-19 players two generations is used the method T-Test.

## Results

The results obtained have shown statistically significant differences in favor of players U19, 2013/14 generation to generation U-19, 2003/04 in two storage areas; anthropometric (in variables ACHEC-chest circumference, APBELLY-waist) and specific motor (in variables MR10m-“reaction speed”, M-R30m- “basic speed”, M-R300m “shuttle running Tempo- Test”, M-DRSLLA-slalom dribble the ball, M-PASI-precision technique of dynamic passing the ball).

**Table 1.** *T-test for each anthropometric variables*

|            | Group      | Group Statistics |           |                | Sig. (2-tailed) |
|------------|------------|------------------|-----------|----------------|-----------------|
|            |            | N                | Mean      | Std. Deviation |                 |
| AWEIGHT    | 2004 (U19) | 50               | 68.5600   | 4.65946        | .506            |
|            | 2014 (U19) | 50               | 67.8160   | 6.36465        |                 |
| AHEIGHT    | 2004 (U19) | 50               | 1778.7800 | 50.44912       | .505            |
|            | 2014 (U19) | 50               | 1771.3800 | 59.82624       |                 |
| ACHEC      | 2004 (U19) | 50               | 857.6600  | 30.99349       | .000            |
|            | 2014 (U19) | 50               | 898.1800  | 39.50339       |                 |
| APBELLY    | 2004 (U19) | 50               | 748.2400  | 30.77738       | .000            |
|            | 2014 (U19) | 50               | 782.0800  | 38.87596       |                 |
| APTHIGH    | 2004 (U19) | 50               | 533.4600  | 27.07451       | .638            |
|            | 2014 (U19) | 50               | 530.7800  | 29.69044       |                 |
| ATHIGHBONE | 2004 (U19) | 50               | 353.8800  | 19.84495       | .166            |
|            | 2014 (U19) | 50               | 359.4600  | 20.11417       |                 |

The results obtained have the shown very small changes in all anthropometric parameters between the two generations junior players, but changes statistically Significant Differences in favor of players U19, 2013/14 Generation to Generation U-19, 2003/04 in Areas Anthropometric (in ACHEC variable-Chest Circumference, APBELLY-waist), which gives to understand that in the last decade the program contents with players over the age of 16 are applied specific programs of force exercises that have resulted in the development of part of the body trunk.

**Table 2.** *T-test for each specific motor variables*

|         | Group | Group Statistics |          |                | Sig. (2-tailed) |
|---------|-------|------------------|----------|----------------|-----------------|
|         |       | N                | Mean     | Std. Deviation |                 |
| M-JUMP  | 2004  | 50               | 237.0800 | 12.57928       | .695            |
|         | 2014  | 50               | 236.1000 | 12.34744       |                 |
| MR10m   | 2004  | 50               | 1.9294   | .07369         | .007            |
|         | 2014  | 50               | 1.8800   | .10339         |                 |
| MR30m   | 2004  | 50               | 4.5536   | .17371         | .000            |
|         | 2014  | 50               | 4.3374   | .14416         |                 |
| MR5x10m | 2004  | 50               | 11.9322  | .46099         | .361            |
|         | 2014  | 50               | 12.0122  | .40955         |                 |
| MR300m  | 2004  | 50               | 61.1158  | 2.30623        | .000            |
|         | 2014  | 50               | 59.2792  | 1.96912        |                 |



|          |      |    |         |          |      |
|----------|------|----|---------|----------|------|
| T-JUGGL  | 2004 | 50 | 47.2000 | 31.36032 | .605 |
|          | 2014 | 50 | 49.9600 | 20.82675 |      |
| T-DRSLLA | 2004 | 50 | 28.5740 | 1.85752  | .000 |
|          | 2014 | 50 | 26.1040 | 1.66173  |      |
| T-PASI   | 2004 | 50 | 12.7212 | .88781   | .000 |
|          | 2014 | 50 | 11.7252 | .53837   |      |

The results obtained have shown statistically significant differences in favor of players U19, 2013/14 generation to generation U-19, 2003/04 in specific motor (in variables MR10m-“reaction speed”, M-R30m- “basic speed”, M-R300m “shuttle running Tempo- Test”, M-DRSLLA-salom dribble the ball, M-PASI-precision technique of dynamic passing the ball, very small differences, but not statistically significant, there have been other variables specific motor. The results show us that generation of soccer junior players (U19) 2014 are free and quick on the ball, compared with junior players (U19) 2004.

### Discussion

Research in both periods was made by the author of the paper (Sermahaj, S. 2004 and 2014) with the same testing methodology and sample selected in the same manner and the results obtained show antropometric changes in the upper body in favour of generation U19 junior players in 2014 to those in 2004 (in variable ACHEC-Chest Circumference, APBELLY-waist), which can assume that these changes have come about as a result of the application of the content specifically stated strength exercises that in the last decade we see an individual attendance increased fitness centers by soccer junior players (U19).

These changes are in favor of players U-19, 2014 generation to generation in 2004 hypothetically can justify as a result of vocational education framework coaches and completion of program content with additional exercises fitness to young soccer players. Development of upper body parts can assume that have had a positive reflection on improving motor skills namely speed with and without the ball (in variable MR10m- "reaction speed", M-R30m- "basic speed", M-R300m "shuttle running Tempo- Test ", M-DRSLLA-salom Dribble the ball, M-UPON-precision Technique of dynamic passing the ball), in favor of junior players U19-generation 2014 as compared to the generation U19 soccer players 2004.

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Sermahaj, S. & Telai, B. (2014) Influence of some anthropometric variables and the specific motoric on the success of the Football players of First Junior League of Kosovo, Research in Physical Education, Sport and Health, (111-115)

#### COMPARISON OF MORPHOLOGICAL CHARACTERISTICS AND SPECIFIC MOTOR OF JUNIOR SOCCER PLAYERS IN TWO PERIODS

*The purpose of this paper is to compare the number of of morphological characteristics and specific motor between players Junior (U-19) in two time periods in 2004 and 2014.*

*The research was conducted on a sample of 100 young soccer players (U-19 League first Kosovo) divided into two groups:*

*The first group comprised of 50 players U-19 to the five best clubs in 2003/04 season and the second group of 50 players U-19 also five best clubs 2013/14 season. In the two generations of players were conducted measurements of morphological characteristics and specific motor.*

*To validate the differences between U-19 players two generations is used the method T-Test. The results obtained have shown statistically significant differences in favor of players U19 2013/14 generation to generation U-19 2003/04 in two storage areas (anthropometric and specific motor) ..These changes are in favor of players U-19 2014 generation to generation in 2004 hypothetically can justify as a result of vocational education framework coaches and completion of program content with additional exercises fitness to young soccer players.*

**Key Words:** junior footballer, morphology, specific motor, comparing

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## **DIFFERENCES IN TECHNICAL MOVEMENT PRECISION WITH BALL TO NEW AGES SOCCER PLAYERS**

### **Introduction**

The primary goal of this research is to compare the accuracy of the collision of the ball to players of both age groups U-17 and U-19.

The second purpose is the evaluation of the accuracy of technical movements with the ball at young soccer players (U17 and U19) of Kosovo. Numerous researches have shown that the execution precise movements are characterized by high levels of special motor skills. Measuring Accuracy Among Young Soccer Players and have proved irrelevant differences between U17 and U19 players on tests of motor and precision does not depend much on the age of the players Petra et al. (2012). Bjelica, D. Georgiev, G. Popović S. (2011), Based on the results of their research it was affirmed that significant differences occur in the case of almost all the variables as it was expected and it was concluded that various intensities affect, while various states don't affect the accuracy of instep kicking by non-preferred leg at a significance level of  $p=.05$ .

### **Methods**

The research was conducted on a sample of 100 young soccer players from 10 clubs of Kosova that have become regular racing season 2013/14, divided into two groups: the first group comprised of 50 young players U17 of five teams (Kf. Ramiz Sadiku, Kf. Flamurtari, Kf. MK, FC Prishtina, ACD-IFA) and second group comprised of 50 young soccer players U-19 of five teams (Kf. Ramiz Sadiku, FC Prishtina, K.F. Trepça 89', Kf. Llapi, dhe Kf. Llamkos Kosova). Each club is represented with 10 players better. To assess the precision of the technical movement with the ball all the players they have subjected technical demonstration testing in four tests: T-JUGGL ball alternating juggling: food-head (technique – coordination under pressure time), T-PASI passing ball accuracy to two bounce walls (precision technique under time pressure), T-KROS- target kick as cross-over (precise technique cross the ball), T-SHOOT- goal shot after own pass ( timing and precise technique)..But the evaluation it's done based on a grading system which is used since 2002 in faculty of sports sciences, Uni – Leipzic.

## 1. T-JUGGL :alternating juggling:instep-head

(technique-coordination under pressure time), Instructions for realization:

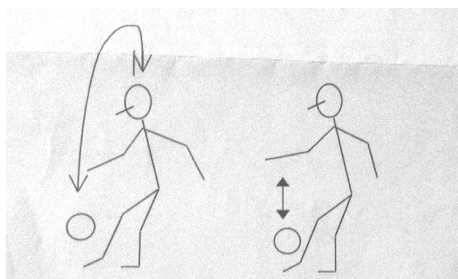
The player has to juggle the ball between instep and head for 30 seconds. It's allowed to touch the ball only once in each time.

Every ball contact performed correctly is counted (instep-head=1-2).

At the beginning, the ball is stationary on the ground; it is "taken up" with the foot and juggled.

When the instep touches the ball three times, this count as one contact; when the ball touches the head several times, it is counted as one contact, too.

Every player should endeavour to juggle the ball as often as possible within 30 seconds correctly (alternation: instep-head one time each). Whenever the ball drops to the ground or is touched with the hand, one ball contact is subtracted. When the ball is touched with the thigh, the test may continue. Yet, this contact doesn't count, and the ball must be played with the instep again before counting can be continued.



| assessment:<br>(number/ 30sec.) | Points |
|---------------------------------|--------|
| from 38                         | 5      |
| 35-37                           | 4      |
| 30-34                           | 3      |
| 25-29                           | 2      |
| 20-24                           | 1      |

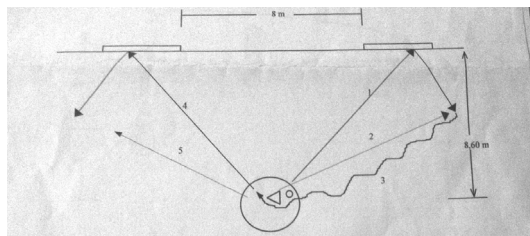
## 2. T-PASI, Shooting accuracy to two bounce walls

(precise technique under time pressure), Instructions for realization:

The stationary ball lies in a circle. Every player tries to shoot, as often as possible, to two bounce walls within 30 seconds (2 benches, each with a marked area of 1.20m).

He starts shooting the ball to the first wall, runs after the bounced ball, dribbles into the circle from where he tries to shoot with the other leg to the second target, to get the ball, to dribble back into the circle and to continue.

The ball must always be in motion when it is inside the circle. Shoot to the right target with the right leg and to the left target with the left leg. If the ball hits with its entire diameter the target, it counts as a valid hit. Also run after balls that hit the edge, but not the target. If the ball hits the edge so it bounces off very far or the ball is played behind the bounce wall, the attempt is repeated.



bounce wall = 1.20m

circle  $r = 1.80\text{m}$

|                 |          |
|-----------------|----------|
| Assessment:     | 5 points |
| 6 hits and more |          |
| 5               | 4 points |
| 4               | 3 points |
| 3               | 2 points |
| 2               | 1 points |

### 3. T-KROS: Target kick as cross-over

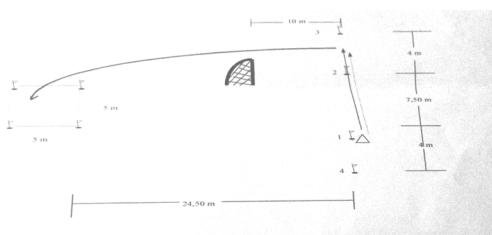
(precise technique in a combination of exercises), Instructions for realization:

The player plays the ball forward from flag 1, runs after it to cross it over between flag 2 and 3 over a handball goal (2x3m) into a target square. The player can touch the ball twice. If the ball is touched more than two times, the hit is not acknowledged.

The exercise must be performed in a way similar to a real match situation; the player must keep running, the ball mustn't rest, otherwise the hit doesn't count.

Left-leg players start at flag 2 and cross over between flag 1 and 4.

Every player has 5 attempts.



|             |          |
|-------------|----------|
| assessment: | 5 points |
| 5 hits      |          |
| 4 hits      | 4 points |
| 3 hits      | 3 points |
| 2 hits      | 2 points |
| 1 hit       | 1 point  |

### 4. T-SHOOT: Goal shot after own pass (timing and precise technique)

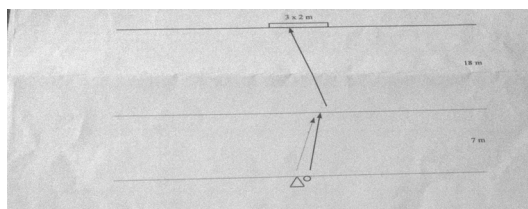
Instructions for realization:

The player stand 7m away from the shooting line at a position the suits him best. He passes the ball in direction of the goal so that, possibly near the shooting line, he shoot the ball directly into the goal. He has to keep moving. He's allowed to touch the ball a total of two times only (pass-goal shot); the ball has to be shot hard and with the instep, otherwise a possible hit is not acknowledged.

If the ball is shot from behind the shooting line, the hit cannot be accepted, either.

Every player has 5 attempts.

Cross-bar und goal-posts do not count as hits



|             |          |
|-------------|----------|
| assessment: | 5 points |
| 5 hits      | 5 points |
| 4 hits      | 4 points |
| 3 hits      | 3 points |
| 2 hits      | 2 points |
| 1 hit       | 1 point  |

The test exercises it's done in football field with artificial grass

Up to 5 points can be given for each of the four exercises.

The criteris are determined by the course of the exercises and emphasize following aspects (of different values):

-the ball sense rhythimization abilitytion a time unit, the accuracy of the course of the actions, the speed of an action, the exactness of targeting, timing.

Grading is done in a differentiated manner on the basis of the aspects mentioned above and uses the following grading system.

| Mark               | Points |
|--------------------|--------|
| Very good (1.0)    | 15     |
| (1.3)              | 14     |
| Good (1.7)         | 13     |
| (2.0)              | 12     |
| (2.3)              | 11     |
| satisfactory (2.7) | 10     |
| (3.0)              | 9      |
| (3.3)              | 8      |
| sufficient (3.7)   | 7      |
| (4.0)              | 6      |

## Results

To see the difference between the soccer players of age 17 and 19 it's used the T-test method. Through T-test method are proven differences in favor of players U-19 to U-17 in all tests of precision technical movements with the ball, but statistically significant differences are shown in the collision test at the gateway (T-SHOOT) and the test ball juggling (T-JUGGL). Results obtained through the scoring of each player in each variable demonstration showed that the overall assessment of the technical with the ball to U17 players at the "satisfactory" while at U19 level players "good".

**Table 2.** T-test for each precision with ball variables

|         |       | Group Statistics |         |                |                 |
|---------|-------|------------------|---------|----------------|-----------------|
|         |       | N                | Mean    | Std. Deviation | Sig. (2-tailed) |
| T-JUGGL | U- 17 | 50               | 27.0000 | 4.30472        | .003            |

|         |       |    |         |          |      |
|---------|-------|----|---------|----------|------|
|         | U-19  | 50 | 34.2000 | 15.90533 |      |
| T-PASI  | U- 17 | 50 | 4.5400  | 1.09190  | .593 |
|         | U-19  | 50 | 4.4200  | 1.14446  |      |
| T-KROS  | U- 17 | 50 | 2.4000  | 1.10657  | .100 |
|         | U-19  | 50 | 2.7600  | 1.06061  |      |
| T-SHOOT | U- 17 | 50 | 2.2600  | 1.08440  | .000 |
|         | U-19  | 50 | 3.0400  | 1.00934  |      |

U17, 10 poena =satisfactory,

U19, 13 poena= good

### Discussion

The accuracy is motoric skill on high coefficient of genetic. but in this research results have shown differences in the four variables with ball precision in favour of those U17 U19 players but statistically significant differences were observed only among variables (T-Juggle and T-SHOOT) with what could argue that the more experienced players (with more training sessions and more repetition) are more precise in the execution of technical movements with the ball.

The results obtained show us that more experienced players U-19 have a better precision in demonstrating the technical movements with the ball, compared to U-17 players. Assuming that the training process is very important to voice learning, mastering demonstration , acquisition, and precise technical movements with the ball, because the precision peak levels is very significant.

It may also conclude that the execution of precise movements with the ball is an ongoing process of young proceeding with excellence at later ages and practices have shown us that the most experienced players have higher precision compared to players with less experience.

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## DIFFERENCES IN TECHNICAL MOVEMENT PRECISION WITH BALL TO NEW AGES SOCCER PLAYERS

*The primary goal of this research is to compare the accuracy of the collision of the ball to players of both age groups U-17 and U-19.*

*The research was conducted on a sample of 100 young soccer players Kosovo divided into two groups: the first group comprised of 50 young players U17 and second group comprised of 50 young soccer players U-19. To assess the precision of the attack on the ball all the players they have subjected technical demonstration testing in four tests: (T-JUGGL, T-PASI, T-KROS, T-SHOOT).*

*Through T-test method are proven differences in favor of players U-19 to U-17 in all tests of precision technical movements with the ball, but statistically significant differences are shown in the collision test at the gateway (T-SHOOT) and the test ball juggling (T-JUGGL). The results obtained show us that more experienced players U-19 have a better precision in demonstrating the technical movements with the ball, compared to U-17 players. Assuming that the training process is very important to voice learning, mastering demonstration, acquisition, and precise technical movements with the ball, because the precision peak levels is very significant.*

**Key words:** young players, technical, demonstration, precision with ball

„Dan“, 12. april 2015.

НАУЧНИ ЧАСОПИС ЦРНОГОРСКЕ СПОРТСКЕ АКАДЕМИЈЕ У 48 ЦИТАТНИХ БАЗА

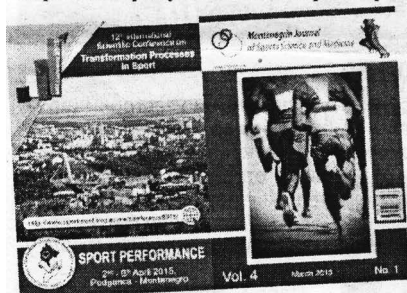
### Ускоро најпрестижнији у региону

Научни часопис Црногорске спортске академије „Montenegrin journal of sports science and medicine“ тренутно је најквалитетнија и најзначајнија публикација у Црној Гори када су у питању индексне међународне базе у којима се налази

– „Montenegrin journal of sports science and medicine“, чији је уредник проф. др Душко Бјелица, налази се у 48 индексних база и на посматрању је код Кобсона за улазак у Томсон Ројтерс базу. Часопис има међународни савјетодавни и уређивачки одбор који чине 48 међународних научника, три замјеника уредника, извршног и почасног уредника. Тренутно се преговара са 19 међународних факултета за спорт и физичко васпитање из

исто толико држава о потписивању споразума за објављивање научних радова у овом часопису који би на тај начин постао најпрестижнији у региону, рекао је извршни уредник доц. др Стево Поповић.

Т.Б.



Проф. др Душко Бјелица



UDC 796.012.47: 796.011.3

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## THE DIFFERENCE IN ANTHROPOMETRIC CHARACTERISTICS, MOTOR AND TECHNICAL SKILLS AMONG SUCCESSFUL SOCCER PLAYERS TO THOSE LESS SUCCESSFUL

### Introduction

Impact of specific motor and cognitive dimension to the success of the game of football youth aged 16-18 years has researched Petrić (1994) which among other things has proven that most influence variables in predicting criterion has been precision factor hitting the ball.

The primary goal of this research has been the assertion of differences in anthropometric parameters, motor and technical demonstration between players and successful and less successful ones.

Secondary purpose of this paper has been the assertion of anthropometric status, motor and technical skills of U-17 young players of Pristina which all the investigated spaces.

### Methods

The research was applied to a sample of 60 players U-17 (6 clubs Pristina region that have become regular racing season 2013/14) divided into two groups: The first group comprised 20 players U-17 both teams first (Kf. Ramiz Sadiku, Kf. Flamurtari) and second group 40 U-17 player of the four other teams (Kf. MK, FC Prishtina, ACD-IFA, KF. Beslidhja). Each club is represented with 10 players better. To all the soccer players were conducted anthropometric measurements, motor and technical demonstration.

The sample contains 12 anthropometric variables, (AWEIGHT-body weight, AHEIGHT-body height, AUAC- Upper arm circumference, ACHEC-chest circumference, APBELLY-waist, APHIGH-thigh circumference, ATHIGHBONE- calf circumference, AUASFT - upper arm skinfold thickness, ABSFT – back skinfold thickness, AASFT - abdominal skinfold thickness, ATHSFT-thigh skinfold thickness, ATHBSFT-mm thing bone skinfold thickness), four motor variables; (M-JUMP length from the place, MR10m-“sprint test” Start by 10m high, M-R30m- “sprint test” by senior fast start, M-R5x10m “shuttle running sprint Test”), and five technique variables (T-JUGGL ball juggling techniques, T-DRSLLA-slalom dribble the ball, T-PASI-

precision technique of dynamic passing the ball, T-KROS-precision technique to cross the ball, T-SHOOT-precision technique of ball busting at the gate).

Processing of results is refined SPSS statistical program with appropriate statistical methods: X-mid arithmetic, Ds - standard deviation, maximum score Max, Min-minimum score, und T-Test method.

## Results

*Table 1. T-test for each anthropometric variables*

|             |       | Group Statistics |           |                | Sig. (2-tailed) |
|-------------|-------|------------------|-----------|----------------|-----------------|
|             | group | N                | Mean      | Std. Deviation |                 |
| AWEIGHT     | 1     | 20               | 64.1400   | 5.45242        | .436            |
|             | 2     | 40               | 65.6175   | 7.47395        |                 |
| AHEIGHT     | 1     | 20               | 1748.4000 | 47.70016       | .250            |
|             | 2     | 40               | 1769.5250 | 73.78972       |                 |
| AUAC        | 1     | 20               | 260.5000  | 14.81642       | .895            |
|             | 2     | 40               | 259.7750  | 22.03434       |                 |
| ACHEC       | 1     | 20               | 883.8000  | 30.94749       | .842            |
|             | 2     | 40               | 881.7000  | 41.43528       |                 |
| APBELLY     | 1     | 20               | 769.2000  | 39.45497       | .908            |
|             | 2     | 40               | 767.9250  | 40.71627       |                 |
| APTHIGH     | 1     | 20               | 511.7500  | 18.93722       | .387            |
|             | 2     | 40               | 519.7250  | 38.50940       |                 |
| ATHIGH-BONE | 1     | 20               | 353.4000  | 19.20088       | .208            |
|             | 2     | 40               | 364.3500  | 35.89768       |                 |
| AUASFT      | 1     | 20               | 78.2000   | 22.34561       | .198            |
|             | 2     | 40               | 96.9000   | 61.95110       |                 |
| ABSFT       | 1     | 20               | 74.3000   | 13.31837       | .760            |
|             | 2     | 40               | 75.7000   | 18.01168       |                 |
| AASFT       | 1     | 20               | 83.8000   | 26.50839       | .906            |
|             | 2     | 40               | 82.9000   | 28.24417       |                 |
| ATHSFT      | 1     | 20               | 88.8000   | 36.90899       | .961            |
|             | 2     | 40               | 88.3750   | 29.14305       |                 |
| ATHBSFT     | 1     | 20               | 83.9000   | 21.44737       | .487            |
|             | 2     | 40               | 88.1250   | 22.37236       |                 |

In Table 1, through the T-test method are not proven statistically significant differences in any anthropometric variables between the first group (Successful players) to the second group (less Successful players). The results obtained show us that in the anthropometric space aren't differences among successful players from those less successful that mean that research is homogeneous samples selected.

**Table 2.** *T-test for each specific motor variables*

|          | Group | Group Statistics |          |                | Sig. (2-tailed) |
|----------|-------|------------------|----------|----------------|-----------------|
|          |       | N                | Mean     | Std. Deviation |                 |
| M-JUMP   | 1     | 20               | 228.8000 | 13.45793       | .116            |
|          | 2     | 40               | 222.2500 | 15.68561       |                 |
| M-R10m   | 1     | 20               | 1.8905   | .10262         | .093            |
|          | 2     | 40               | 1.9392   | .10511         |                 |
| M-R30m   | 1     | 20               | 4.4465   | .20090         | .818            |
|          | 2     | 40               | 4.4347   | .17736         |                 |
| M-R5x10m | 1     | 20               | 11.9565  | .41994         | <b>.000</b>     |
|          | 2     | 40               | 12.5492  | .60976         |                 |
| T-JUGGL  | 1     | 20               | 27.5000  | 3.98021        | .184            |
|          | 2     | 40               | 25.9250  | 4.41668        |                 |
| T-DRSLLA | 1     | 20               | 28.0060  | 3.00304        | .525            |
|          | 2     | 40               | 27.5702  | 2.19669        |                 |
| T-PASI   | 1     | 20               | 4.6000   | 1.23117        | .445            |
|          | 2     | 40               | 4.3750   | .97895         |                 |
| T-KROS   | 1     | 20               | 2.6500   | .87509         | .115            |
|          | 2     | 40               | 2.1750   | 1.17424        |                 |
| T-SHOOT  | 1     | 20               | 3.1000   | .71818         | <b>.000</b>     |
|          | 2     | 40               | 1.6500   | 1.05125        |                 |

To prove spaces investigates differences between players of both groups used the T-Test methods. Results obtained through the method of T-Test show us that statistically significant differences in favor of the first group (successful players) to the second group (less successful players) had agility (“shuttle running sprint Test” M-R5x10m) and precision shooting (T-SHOOT)., small difference but not statistically significant, are seen in all variables in favor of the players be successful against those less successful.

### Discussion

Other researchers have also put special emphasis on agility as an outspoken skills in the success of players (Figuerido und co author. 2009) and the precision of striking the ball with the foot as a coordinative skills very important in the game of football Finnoff et al. 2002). (Bjelica, D. 2008). With the method T-test of this research slightly higher (-2.92), it can be concluded that there exist a significant difference in favor of the variable between the results obtained by kicks performed by favored leg in the state of fatigue in relation to the kick performed by unfavored leg in the state of rest.

Verheijen, R (1997), prefers testing conditional performances (started speed 10m, inordinate 30m, agilitetin 5x10m) of players who have a special significance for programming and training processes are simultaneously a motivation tool for improving the performances conditional. In this research we have no differences between successful players to those less successful anthropometric parameter, which means that the research sample was selected and homogeneous, whereas statistically significant differences in

favor of the successful players are shown only in to two motor space variables (M-R5x10m agility, and precision shooting T-SHOOT)., which may conclude that agility and precision strike are relevant determinant in distinguishing high level players.

Therefore we can suggest that agility and precision ball kick taken a special place in the training process with young soccer players because they are a special significance in the difference of the level of successful players from those less successful.

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### *THE DIFFERENCE IN ANTHROPOMETRIC CHARACTERISTICS, MOTOR AND TECHNICAL SKILLS AMONG SUCCESSFUL SOCCER PLAYERS TO THOSE LESS SUCCESSFUL*

*The primary goal of this research has been the assertion of differences in anthropometric parameters, motor and technical demonstration between players and successful and less successful ones. The research was applied to a sample of 60 players U-17 (6 clubs Pristina region that have become regular racing season 2013/14) divided into two groups: The first group comprised 20 players U-17 both teams first and second group 40 U-17 player of the four other teams. To all the soccer players were conducted anthropometric measurements, motor and technical demonstration. To prove spaces investigates differences between players of both groups used the T-Test methods. Results obtained through the method of T-Test show us that statistically significant differences in favor of the first group (successful players) to the second group (less successful players) had agility ("shuttle running sprint Test" M-R5x10m) and precision shooting (T-SHOOT)., So we can suggest that agility and precision busy busting a special place in the training process with young soccer players since been an outspoken in distinguishing the level of successful players from those less successful.*

**Keywords:** young soccer players, agility, precision of shooting.

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## THE PROCESS OF CHANGE - PREDICTION OF SPORT ACHIEVEMENTS HISTORICAL TENDENCY

### 1. Introduction

Sports clearly are an important part of cultures and societies around the world. As we look around us, we see that the Olympic Games, Football's World Cup, the Tour de France, The Tennis Championships at Wimbledon, The NBA and rarely Paralympics, Special Olympics are now worldwide events capturing the interest of billions of people. These events are televised in over two hundred countries, including Kosovo. Children around the world grow up with vivid images of televised sports and sport figures, they play video games based on these sports and sport figures, and they are encouraged to participate in sports by parents, Physical Education teachers, and the elite athletes who often are presented as role models in their lives. Some people (sport researchers) are concerned more directly with learning about scientific sports prediction. Their involvement in the sport sciences focuses on understanding how sports organized and how changes in that organization might influence sports experiences for both athletes and coaches. The goal of these scholars is often to improve sport experiences and performance prediction for current participants and make sport participation more attractive and accessible for those who do not currently play sports (prospective athletes). They also may want to help athletes improve their performance, help coaches work effectively with athletes and win more games, help sport organizations grow and operate more efficiently and profitably, and improve sport achievement prediction.

Humans, who are rational beings, make and give meaning to social phenomena. Their thoughts and actions, as well as the individual character of experience and meaning, are neither predetermined nor predictable. Before one even can begin to answer the "how?" questions, there is a need to identify clearly all the variables, the agents the social factors and structures, that operated in the process of change - prediction of sport achievements. Questions important to this work include the following:

- What are the early stages of sports and athletics performance?
- Development of organized and competitive sport activities, ambiguity of thoughts in regard to sport achievements prediction, and which are environmental constraints on sport performance?

These questions have been raised and have been issues of interest to many researchers in the field of Kinesiology, Sports Psychology, Sociology of sport and Biomedical Sciences.

Whilst questions of the relative roles of genes and environment in the past may have been addressed from the standpoint of the academic interest, in recent times there have been increasing reports of medical interventions and attempts to use genetic knowledge for performance enhancement in sport.

The aim of this work is to summarize the different standpoints and different approaches in regard to the sport performance preparation and prediction. Put simply, the gene-environment interaction concerns the extent to which an individual is a product of her/ his genes or environment, generating questions over the role of genes and environmental influences and experience in a range of human behavioral contexts, including athletics and sport achievement.

## **2. THE EARLY STAGES OF SPORT AND ATHLETIC PERFORMANCE**

To understand sports as a social phenomenon in today's world, we should have a sense of what physical games, contests, and sport activities were like in past times? Therefore, this paper presents brief social overviews of sport activities in different cultural and historical settings. People in all cultures have engaged in physical activities and used human movement as a part of their ritual life.

### **Games in Ancient Greece and Roman contests and games (100 B.C. to A.D. 500)**

Competitors in these early games were from wealthy, respected Greek families. Events were based on the interests of young males. They consisted primarily of warrior sports, such as; wrestling and boxing, javelin and discus throwing, foot racing, archery, and long jumping. Violence and serious injuries were commonplace, in comparison with today's sport events (Elias, 1986; Kidd, 1984, 1996b). Greek contest and games were different from the organized competitive sports of today (Gutmann, 1978). First, they were grounded in religion, second, they lacked complex administrative structures; and, third, they did not involve measurements and record keeping from event to event as we do in 21<sup>st</sup> century. Roman leaders used physical contests and games to train soldiers and provide mass entertainment spectacles. As the power and influence of the Roman Empire grew, these spectacles, consisting of contests and games became increasingly important as diversions for the masses. Men and women were forced into the arena to engage in mortal combat with lions, tigers, and panthers. Gladiators armed with a variety of weapons, were pitted against one another in glory fights to the death. These spectacles achieved two purposes for the Romans: they entertained an idle populace and disposed of socially "undesirable" people, such as thieves, murderers, and Christians (Baker, 1988).

### **Sport activities during middle ages (500 to 1300)**

Sport activities during Medieval Europe consisted of folk games played by local peasants, tournaments staged for knights and nobles, archery contest, and activities in which animals were brutalized (Dunning, 1999). The contests and games of the medieval period, the tournaments and games of that time were not much like today's organized sports. They lacked specialization, they never involved the measurement and

recording of athletic achievements, and they were not based on a commitment to equal and open competition among athletes from diverse backgrounds.

### **Games and sport activities during the Renaissance and The Industrial Revolution (1300 to 1900)**

About the same time that the peasants were being subjected to increased controls in many locations, the “scholar –athlete” became the ideal among many of the aristocrats and the affluent. They saw the ‘Renaissance man’ as someone who was “socially adept, sensitive, to aesthetic values, skilled in weaponry, strong of body, and learned in letters” (Baker, 1988, p. 59). During the enlightenment period (1700 to 1800), many games and sports activities in parts of Europe and North America began to resemble sport forms that we are familiar with today (Guttmann, 1978).

Although opportunities for participation in organized, competitive sports between 1880 and 1920 were not equally distributed by social class, gender, skin color, ethnicity, age or ability, participation among most categories of people increased dramatically. In most Western cultures, the organizational attributes that we associate with today’s high-profile organized sports became clearly established during this time. The most heavily promoted sports were football, baseball, and basketball. Each was native to the US; each celebrated a form of masculinity, emphasizing aggression, domination, and emotional control; and national loyalties (Burstyn, 1999). Commercial interests had promoted an emphasis on competition, winning, and record setting. Basic organizational structures for professional sports had been established.

### **3. DEVELOPMENT OF ORGANIZED AND COMPETITIVE SPORTS**

The organized competitive sports so popular in many parts of the world today are very different from the folk games played before the industrial revolution. Allen Guttmann’s study of sport activities through history shows that dominant sports forms (DSF’s) today comprise seven interrelated characteristics, which have never before appeared together in past physical activities and games (Dunning and Sheard, 1979). Everything that can be reduced to a time, distance, or score is measured and recorded. Standards of achievements are discussed in measurable terms, and statistics are used as proof of achievements. Performances are compared from one event to another, and records are published for individuals, teams, leagues, events, communities, states, provinces, and continents. Most important, of course are world records. One or more of these traits have characterized physical games during previous historical periods, but not until the 19<sup>th</sup> century did all characteristics appear together in what might be called modern sports (Dunning, 1999).

#### **Modern sports in 21<sup>st</sup> century**

Modern sports are highly organized and competitive. Generally, they emphasize the following; the use of strength, speed, and power to push human limits and aggressively dominant opponents in the quest for victories and championships. The importance of the setting records, defining the body as a machine, and using the technology to control and monitor the body. Selection systems based on physical skills and competitive achievement.

#### 4. THE INTERACTION BETWEEN GENETICS/ENVIRONMENT ON SPORTS PERFORMANCE

How environmental and genetic constraints correlate or interact to shape performance variations in sport and exercise is a question of increasing interest in sport medicine, sports pedagogy, and sports psychology (Baker & Davis, 2006; Brutsaert & Parra, 2006; Pitsiladis, et al., 2006; Bjelica & Petković, 2011 ). In particular, there have been some efforts in molecular biology to identify single gene variants with the potential to profoundly impact on individual performance or the propensity to lead to a specific disease (e.g. a gene responsible for physical power, propensity to exercise). In sport, this ‘single gene as magic bullet’ philosophy, favored by some molecular geneticists, has led to claims that elite performers are born to succeed. However, the relative contribution of genes to sport performance is likely more varied than the 50% contribution indicated by Hopkins (2001).

This approach is exemplified by attempts to identify sprinters and endurance runners on the basis of differing alleles (i.e. forms) of a single gene known as  $\alpha$ -actinin-3 (Coghlar, 2003; Yang et al., 2003). There have been similar claims on the roles of different variants of the ACE gene in endurance and power events such as mountaineering and running (Coghlar, 1998).

There is growing awareness in science of ‘the complementary nature’, as Kelso and Engström (2006) so aptly entitled their critique of the historical tendency to attempt to explain phenomena on the basis of dualist positions and theoretical stances. Their thesis is that a systems biological approach can help categorize natural phenomena as exhibiting strong cooperative tendencies to interact as well as tendencies to function separately, the emergence of which can characterize the relationship of genes and environments. The ‘complementary nature’ of phenomena in the natural world suggests that it is highly important for geneticists to identify many single gene variants, although the role of these genes in regulating behaviour needs to be framed by their overarching tendencies to network and to cooperate or compete with environmental constraints.

##### **Environmental influences on sports performance**

While ‘heredity’ refers to the innate characteristics an individual contributes to their performance, ‘environment’ makes up those qualities that result from one’s experiences (Davis and Baker, 2007). In this paper we provide with the range of environmental factors that contribute to sport performance and more accurate sports achievement prediction.

***Quantity and Quality of Training.*** - Much of our understanding about environmental influences on performance comes from studies of sports expertise (Davids and Baker, 2007). In 1993, Ericsson et al. produced a seminal paper on the role of practice and expert development that shaped a great deal of the research that followed. Their position (based on the work of ex Ericsson et al., 1980; Simon and Chase, 1973 and others) was that, with proper attention to what they called ‘deliberate practice’ (i.e. relevant, effortful activities done with the specific goal of improving performance), healthy individuals could prevent performance improvements from tapering off. In



early development, activities that are inherently enjoyable and motivating may be necessary to provide an impetus to continue training during times when more diligent, effortful practice is required. Without this pleasurable involvement, athletes run the risk of dropping out of sport (Petlichkoff, 1993)

**Coaches.** - play a critical role in optimizing an athlete's training time, in some cases having complete control over the practice environment. Indeed, a key characteristic of expert coaches is meticulous planning (Voss, et al. 1983). Researchers have dissected practices in sports ranging from individual sports such as wrestling and figure skating, (Deakin & Cobley, 2003; Starkes, 2000) to team sports such as volleyball (Deakin & Cobley, 2003) and basketball (Bloom, et al., 1999; Tharp & Gallimore, 1976). What seems clear is that the structure and content of expert athletes' practice is superior to that of non-experts. For example, Cote et al.( 2003) suggested that, during early phases of development, coaches should emphasize opportunities to develop fundamental motor skills and general abilities that will form the basis for more advanced development in the future.

**Parents** -Researchers have highlighted the critical role that parents play in promoting the athletic development of their children. In an examination of exceptional individuals across a range of sports and other activities, (Bloom, 1985) reported that parental resources, such as support, were imperative in nurturing talent. During early sport development, parents typically provide the initial opportunity to get involved in sports. As the athlete matures, parental involvement decreases and the performer takes greater control of the decision-making process regarding their future career. Parents continue to provide support in a background role, as providers of financial support and more importantly, emotional support (Cote, 1999).

**Culture effect**-The importance that a nation or community places on a particular sport can have a dramatic influence on any success achieved. For example, in Canada, the sport of ice hockey has become a fundamental component of the national identity (Russell, 2000). Robinson ( 1998) suggest that Canada has three times more children playing ice hockey than Russia, Sweden, Finland, The Czech Republic and Slovakia combined. Given these factors, Canada's international success is not surprising. Other countries have similar sport-related associations including Austria with skiing and Kenya with distance running.

## 5. CONCLUSIONS AND RECOMMENDATIONS

When sport science scientists say that sports are contested activities, they mean that, through history, people have disagreed about what sports could and should be. Sports experiences vary with the conditions under which sports are organized and played. To assume that all sports are organized around the same goals and emphasize the same orientations and behaviors' is a mistake.

In all over the world, there are highly organized competitive sports, informal sports, recreational sports, extreme sports, alternative sports, cooperative sports, folk sports, contact sports, team sports, individual sports and so on, and there are various

combinations of these types. However at this time, it seems that the dominant sport form in many societies all over the world is organized power and performance sports.

Although power and performance sports have become standard for determining what sports should be in many countries, they have not been accepted by everyone. The alternative and oppositional sports forms are diverse, but many fit in so called pleasure and participation sports (recreational sport activities). Power and performance sports remain dominant today in the sense that they receive the most attention and support. Historically, sports domain offered the opportunity to participants involved in, athletes and their respective coaches for showing up the racing spirit explicitly, since the early stages of sport and athletic performance. The competition of involved athletes and coaches in games and sport performance activities – their respective team of experts from complementary sciences is ongoing process, athletes and experts are in forever relative racing of who is the best on sport competitions in the playing pitches (players and sport teams) and sport achievement prediction outside the playing pitches (coaches and sport experts).

Allied to this recommendation, the literature reviewed in this paper suggests that many clinicians and sports scientists may be expecting more from use and abuse of quantitative molecular studies than can be delivered. A requisite step forward would be to discard both the dualist approach to nature vs nurture and the ‘gene as magic bullet’ philosophy. It is becoming clear that sports performance is the result of interactions among a host of genes and environmental constraints.

Although challenging, an interactive approach for analyzing relative contributions of genes and environment in degenerate neurobiological systems may shed light on a number of significant issues in sport science and medicine related to the acquisition of expertise in sport.

Current data on genetic and environment constrains on motor skill acquisition are unclear due to a number of methodological weaknesses and conflicting findings, and there is a need for more work to identify genetic and environment mechanisms underlying performance variations, if we want to be more accurate sport achievement predictors.

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#### THE PROCESS OF CHANGE - PREDICTION OF SPORT ACHIEVEMENTS HISTORICAL TENDENCY

*The aim of this paper is to summarize the different standpoints and different approaches in regard to the sport performance preparation and achievement prediction. Sports researchers are concerned more directly with learning about scientific sports prediction. Their involvement in the sport sciences focuses on understanding how sports organized and how changes in that organization might influence sports experiences for both athletes and coaches. The goal of these scholars*

is often to improve sport experiences and performance prediction for current participants and make sport participation more attractive and accessible for those who do not currently play sports, prospective athletes. They also may want to help athletes improve their performance, help coaches work effectively with athletes and win more games. Sports researchers intention is also to assist and help sport organizations grow and operate more efficiently and profitably, and improve sport achievement prediction.

**Key words:** achievement, prediction, sport performance, athletes, coaches

„Dan“, 3. april 2015.

МЕЂУНАРОДНА НАУЧНА КОНФЕРЕНЦИЈА ЦРНОГОРСКЕ СПОРТСКЕ  
АКАДЕМИЈЕ ОКУПИЋЕ 166 АУТОРА ИЗ 22 ЗЕМЉЕ СА ТРИ КОНТИНЕНТА

## Свечано отварање данас у 16

Међународна научна конференција о трансформационим процесима у спорту, 12. по реду, као и 11. Конгрес Црногорске спортске академије биће свечано отворени данас у 16 часова у згради Ректората Универзитета Црне Горе. Учесници конференције почеће у 9 часова са радом у двије сесије и једном радионицом, у оквиру које ће проф. др **Нејц Шарабон** са Универзитета Приморска из Словеније приказати повезивање високошколског образовања са индустријом спорта. Након свечаног отварања планирана су четири пленарна излагања – о спорту и националном поносу у уједињеној Њемачкој говориће **Хенк Ерик Мајер** (Универзитет у Мунстеру, Њемачка), падове у спортским дисциплинама обрадиће **Зденко Регули** (Масарик Универзитет, Чешка), **Вишња Ђорђић** (Универзитет у Новом Саду) говориће о промоцији физичке активности а **Мартин Звонар** (Масарик Универзитет, Чешка) о промјени температуре стопала приликом коришћења специфичне обуће. Иначе, за овогодишњу конференцију прихваћена су 94 научна рада са 166 аутора и коаутора из 22 земље: Црне Горе, Турске, Србије, Чешке, Словеније, Албаније, Босне и Херцеговине, Њемачке, Гане, Индонезије, Јужне Кореје, Аустралије, Бугарске, Хрватске, Индије, Јапана, Македоније, Румуније, Саудијске Арабије, Тајвана и Велике Британије. Најбоља два рада биће објављена у часопису на енглеском језику „Montenegrin journal of sports science and medicine“, који од свог оснивања 2012. године излази два пута годишње и налази се у чак 46 индексних база. Остали радови ће бити објављени у часопису „Спорт Монт“, који је од 2003. године доживио своја 42 издања – троброја са преко 1.000 аутора.



Т.Б.

UDC 796.012.6:373.3

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## **MOTIVATIONAL LEVEL AND PARTICIPATION BARRIERS IN SCHOOL PHYSICAL EDUCATION AMONG ADOLESCENTS**

### **Introduction**

Physical inactivity is a major public health challenge in the today world of technology advancement. It is known that adolescence its the time when a person tends to act as an adult, they do see themselves as an adult and tend to act independently. Adolescence prepares you to become a young adut able to be independent and responsible for yourself. Research findings demonstrate that adolescence is an important period of time to investigate the variables that support physical activity (PA) participation. More specifically, there is evidence that childhood patterns of PA track into adulthood, meaning that active adolescents tend to be active adults (Telama et al., 2005). However, it has been well documented that engagement in PA decreases during adolescence (Currie et al., 2008; Dumith et al., 2011). During the adolescent developmental period, there are a multitude of psychological, physical and social factors which may influence the decrease in PA patterns (Sallis et al., 2000).

Time is the one resource that we all struggle to make the best of it at all time and schools are no exceptions for example Physical education right now we got students receiving at Kosovo schools two 45 minute period /a week. The reward is that children need a break in order to go back and concentrate and focus better on the task and in order to achieve better results in other school subjects. Anybody who has worked with children will know that if they have a chance to interact with each other, when let go some other energy, get their heart rate going and get themselves just activated and the brain is more alert and able to focus on the learning that than will happen.

So, PE classes it's an excellent way of connecting with each other and reducing anxiety and getting students ready for the work that they are supposed to do for the rest of the day.

How much activity is needed in a 24 h day?

The recommendations guide to for a healthy living children and youth need a minimum of 60 minute a day a physical activity, and that includes daily vigorous and moderate physical activity and also opportunities for flexibility, strength and endurance, and children just don't get enough of that in Kosovo schools unless is integrated in their lives in many ways so at school for sure at home as well, but school is an opportunity where can reach all children, so it's an ideal place to do it!, unfortunately we don't see it happening in our schools, due to lack of motivation and different types of barriers. Therefore, the purpose of this study is to examine two research questions:

- 1) What are student attitudes about their motivation regarding school based physical education (PE) and sport activities
- 2) What are student attitudes concerning the participation barriers into school based PE/Sport activities

### Methods

The participants of the study were ( $n = 305$ ) (155 males and 150 females) 10th , and 11th grade students from the two high schools in the northern Kosovo. The mean age of the participants was 15.57 ( $SD = 1.26$ ). All data were collected during the month of September and October 2014/'15 school year. Students carried out the motivational and participation self reported questionnaires to gather quantitative data regarding students' perceptions of motivation and participation in PE.

*The Physical Activity and Leisure Measurement Scale* (PALMS, Morris & Rogers, 2004) is a 40-item self-report measure of motives for participation in physical activity, answering the stem question "I undertake physical activity...". The scale consists of eight subscales: 1) mastery (e.g. "to get better in an activity"), 2) enjoyment (e.g. "because it's fun") , 3) affiliation (e.g. "to do something in common with friends"), 4) competition/ego (e.g. "to work harder than others"), 5) others' expectations (e.g. "because people tell me I need to"), 6) physical condition (e.g. "because it helps me maintain a healthy body"), 7) psychological condition (e.g. "because it acts as a stress release"), and 8) appearance (e.g. "to improve body shape"). Each item uses a 5-point Likert scale ranging from 1 = strongly disagree to 5 = strongly agree. Also, the participants were asked to state the number of days engaging in physical activity during a typical week and how important / useful / interesting exercise is for them, rated in a 5-point Likert scale ranging from 1 = nothing at all to 5 = very much.

#### Data collection

Questionnaire items were designed by the author to reflect the research questions. Items sought information on student perceptions about their motivation, participation barriers during their school based PE/Sport activities. Perceptions of time spent on a PE/ sport activities were discussed to stimulate student discussion about their activity likes or dislikes and their reasons. Additional items included information related to activities they wanted to learn in PE. Focus group interview questions were modeled after the questionnaire so that the same topics were probed and answered in greater detail. Students were informed of the purpose of the study and the confidential treatment of the data. The study was explained by the researcher at the beginning of each class and questionnaires were distributed to every student who participated in the study. In general, questionnaires were completed in about 10-15 minutes

#### Data analysis

Questionnaire data were analyzed using three computerized programs. Quantitative data was analyzed using SPSS 21, a predictive analytic applications software, to provide mean and standard deviations derived from the nine Likert-scaled questions. Excel was used to organize data from the open-ended questions.

## Results

Children and adolescents in the Republic of Kosovo schools, cannot become more physically active and fit if they don't have a wide range of accessible, safe and affordable opportunities to be active. Teens, therefore, will not increase their levels of PA unless they are sufficiently motivated to do so.

Secondary upper school students ranked from most ('10') to least influential ('1') the barriers to their participation in school based PE/ and sport activities (see table 1).

**Table 1.** Perceived barriers to student participation in PE and sport in Kosovo public upper secondary schools (N= 305, boys 155 and 150 girls)

| Barriers                                          | Number of students | M   | F   | M+F % | Barrier category |
|---------------------------------------------------|--------------------|-----|-----|-------|------------------|
| 1 Lack of facilities/ NO gym                      | 290                | 140 | 150 | 95    | I                |
| 2 Lack of sport equipment                         | 284                | 133 | 151 | 93    | I                |
| 3 Large class sizes                               | 276                | 129 | 147 | 90    | I                |
| 3 Outdated curriculum/Teacher centered            | 242                | 117 | 125 | 79    | T                |
| 4 PE/Sport not priorities in the school           | 238                | 135 | 103 | 78    | I                |
| 5 Focus on too many traditional sports/repetitive | 219                | 117 | 102 | 71.8  | T                |
| 7 Staff use outdated teaching methods             | 201                | 120 | 81  | 65.9  | I/T              |
| 8 Peer pressure                                   | 151                | 65  | 89  | 49,5  | S                |
| 9 Strict instructional strategies                 | 130                | 84  | 26  | 42.62 | T                |
| 10 Past negative experiences with PE              | 41                 | 26  | 17  | 13.44 | T                |
| Total                                             | 305                | 155 | 150 |       |                  |

Note: Of all the participants, 305 students made one or multiple comments, T=teacher related barriers, i-institutional related barriers, s=student related barriers

Students ranked from most ('10') to least influential ('1') the barriers to their participation in school based PE/and sports. Lack of facilities and lack of sport equipment (institutional related barriers)  $\bar{x}=9.10$  (2.3) were ranked the greatest barriers to their participation during PE classes, third most influential barrier perceived by students was large class size (40), another factor ranked higher on the list of barriers to participation was listed outdated PE curriculum, teacher centered teaching (79%) see table 1, One of the most frequently chosen barrier (ranked most within the top five) by respondents was PE/ Sport not priorities in the school. Previous research on children's and adolescents' self-reported barriers to participation in PE and physical activity has reported changing attitudes to activity and PE, adolescents' decision making favoring more sedentary activities, similar to our study findings, 71,8 percent (219), stated that their PE teachers, focus on too many traditional and team sports, like football,

basketball, and volleyball. Theoretically, the 21st century schools are supposed to be also the places where students are able to express their opinions and be part of creating a healthy safe supportive climate in their schools. PE classes are the perfect setting for implementation of the theory into practice, PE curriculum content it is not just physical activity, competitive sports, highly structured activities, lack of teacher support, it's kind of "puzzle" pieces. As PE teacher, it is our responsibility, co-workers, school principals, school board members, to join our efforts and enable adolescents and youth to become physically active and healthy, by making school attractive place and make them happy during the time they spent within the junior high school years. Given the difficulties associated with adolescence for both girls and boys, offering PE extra-curricular classes to less skilled and less athletic students may provide them with the opportunity to develop their abilities in a non-threatening environment in both environments indoor and outdoor. It is our duty to take into consideration students' needs and rights, adolescents' "voice" needs to be heard, they must be part of the solution and part of creating norm of acceptance amongst each other and support for each other rather than negative behavior, bullying, tobacco and alcohol use, and other socio-pathologic behavior.

### **Discussion -**

The idea for the current study stemmed from the curiosity and the recognition that the psychological side of the participation and level of motivation into PE classes is under researched, especially in our country.

Qualitative results of the study are in accordance with the research of Barker (2013), our results shows that females wanted to participate in Volleyball, Swimming and Dancing, and they also expressed that they do prefer a coed setting during PE classes due to the possibilities that they tend to broke the mentality of 'traditionally dominant gender' stereotype. On the other hand, males wanted to take part in more competitive physical sports including Football, Basketball and Karate. This supported the cultural assumption that males want a more competitive sport environment, while females want a cooperative environment (Couturier et al., 2005; Hill & Cleven, 2006). Hohepa et al. (2005) also reported similar results when it came to gender differences and safety. However, this study provides further insights, demonstrating that there is a long list of different barriers that teachers may encounter when providing PE in secondary schools. Participation barriers including those imposed by the students themselves; lack of affordable school facilities, sport gym, safety school yards, lack of sport equipment and requisites, large class sizes, traditional teaching methodologies (teacher centered teaching, PE/sports not priorities in the school (PE ranked on the bottom of list of priorities), focus on too many traditional sports, team sports, repetitive/ not attractive learning topics, peer pressure, strict PE teacher instructional strategies, past negative experience.

Fitness-based activities are essential to the purpose of physical education but we must abandon militaristic teaching strategies of the past and replace them with positive motivation and encouragement coupled with activities that match adolescents' need for



physical exertion and play. Ultimately, students will value physical activity that is meaningful to them as cited in the National Association for Sport and Physical Education Standard 6 (NASPE, 2004). It is essential that teachers be able to justify why physical education is in the school curriculum (beside the fact that it is ‘compulsory’) and be knowledgeable about the value of physical education to the school, to students, other colleagues, management and leadership teams and those who make public and school policy decisions (Le Masurier & Corbin, 2006).

Teachers must help battle sedentary lifestyle habits by instilling in students the value and desire to be active (Standard 3 and 6) (NASPE, 2004, p. 11). Increased levels of sedentary behaviors among adolescents indicate a critical need for PE programs to adopt curriculum that is inclusive of all students and that increases their interest and motivation levels for physical activity participation. The level of challenge or the task has to be adjusted by the PE teacher, depending on the skills of the student’s results of the evaluation done in the beginning of the academic year.

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#### MOTIVATIONAL LEVEL AND PARTICIPATION BARRIERS IN SCHOOL PHYSICAL EDUCATION AMONG ADOLESCENTS

*The purpose of the present study was to explore motivation in school physical education (PE), barriers and demotivation factors that influence student's participation in PE. The participants of the study were (n = 305) kosovan students (155 boys, 150 girls), 15.5 years at grade 10 and 11 of secondary high schools. Results of the study indicate that adolescents showed lack of enthusiasm for stereotypical activities and lack of motivation. Students were more likely to report two main barriers (a) **Institutional related barriers** to participation in school Physical education (PE) such as ;the lack of facilities and equipment (Gym and sport requisites), lower priority given to PE by schools, traditional PE curriculum not flexible enough to meet different student's needs and (b) **PE teacher related barriers**; low levels of confidence and competence in teaching PE, being unable to provide safely planned and structured lessons, lack of teacher support, PE uniforms, outdated curriculum focusing mostly on team sports rather than overall fitness by giving more importance to winning than to participation. Often or most of the times teachers use "direct instruction" to introduce a new movement or activity, which has a lack of student involvement, group work, opportunities to explore, etc*

**Key words:** motivation, participation, barriers, Physical Education (PE), Physical Activity (PA)

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## **IS THERE A PROMOTION OF HEALTHY LIFESTYLES PRACTICES IN HIGH SCHOOL PHYSICAL EDUCATION?**

### **Introduction**

In a healthy school, students learn-through lessons and through example-to value their own health and that of the environment. Achieving this vision will require: Providing safe and healthy places to learn and play. School should offer a qualitative PE classes with resources included, licensed and accredited PE teachers, reasonable class sizes, affordable playing fields, (indoor and outdoor), facilities and financial support. The inactive lifestyle of daily living is very unhealthful for population in general, but especially is dangerous for children and adolescents. The health habits the teenagers develop in adolescence time will have positive long-term effects on their health and wellness. School setting is very important source, when students learn basics of healthy lifestyle („Healthy behavior“). Practicing healthy behaviors by making a physical activity (PA) part of the daily routine, eating well-balanced meals and healthy snacks and getting plenty of sleep so they will have enough energy during the day are the key elements of choosing to live healthfully. Schools represent important institutions promoting PA as they reach all age cohorts of adolescents. It has been suggested that school PE, particularly, has been suggested to play an important role in motivating students towards a physically active lifestyle (McKenzie, 2007).

### **Methods**

(a) In-depth Interviews. Two types of semi-structured interviews were employed as the main mode of data collection: face-to-face interviews with four PE and sport teachers of public secondary schools in Kosovo.

(b) Document Analysis. This study used curriculum content, lesson plans, class time-tables, and other documents relevant to the research in order to gather more insight on the topic under investigation. This study have benefited immensely from direct observations of PE classes.

### **Sampling**

In this study; four public school PE teachers (Amy, Bob, John and Mike) were interviewed. Each of the teachers has taught for at least 5-42 years and in more than one school Patton (2002) confirms that qualitative studies normally use small samples and even single cases are accepted.

### **Analysis of data**

In current study was used Moustaka's modification of the Stevick-Collaizzi-Keen method as explained by Creswell (2007). The next step was to develop a list of significant statements from the interviews and other data. These are statements concerning how the informants are experiencing their teaching and addressing the challenges.

### **Overwhelmed challenges of The PE teachers in secondary schools (Kosovo case)**

While PE and sports are acclaimed as having immense benefits for young people as well as adults, it is ironic that PE continues to be a marginalized subject in school curricula throughout much of the world. In many schools in Kosovo, PE class time is used as a time to take a break from serious class work. Therefore, PE teachers feel they have to continually provide justification for the existence of their subject and to plead for actual control of the time they are allocated. In an international survey, which did not cover Kosovo but did investigate some African nations, Hardman (2008) discovered that the status of PE was low and the subject was in grave danger of being sidelined. The problems ranged from reduced curriculum time and a lack of adequately prepared teachers, to the poor state of facilities and a negative perception from teachers, students and parents. Although many school principals and teachers appear to understand the importance of PE, they are also aware of the immense pressure for students to perform well in high stakes examinations (Di Fiore, 2010). Often, PE programs in Kosovo schools are not high on a school's agenda because they are not considered to be very important academically. This survey indicated that while government had committed themselves through legislation to provide for PE, little had changed in many schools all over Kosovo. Unfortunately, school principals, school board members, and parents in many cases are very prejudiced against the subject and most PE periods are instead used for other activities. The current study analysis confirmed that children in schools in Kosovo were still being denied the opportunity to experience the benefits of PE. There are many factors-(emerging themes) from in depth -interviews done with four PE teachers, among others: (1) infrastructure constrains (gymnasium, facilities, equipments),(2) social norms-perceived status of PE and its Teachers, (3) financial constraints ( school budget allocated for PE) (4) time tabling (PE time allocations), and class sizes,(40-50 students per class) (5) Outdated curriculum (PE programs) and gender issues (equity challenges (cooperative vs. competitive activities and adapted physical activity APA (inclusive activities for special needs students) (6)School sport and extracurricular sport activities, (7) PE teachers networking (professional associations),

### **Barriers to the delivery of quality PE**

In most Kosovo schools, the status of PE is perceived to be lower than that of other school subjects. In the PE curriculum, of great consideration are the aims of PE, the range of activities offered in PE programmes and their quality and relevance. PE

resources include teachers, classes, playing fields, facilities and financial ability. Generally, this refers to all that is needed to ensure the provision of PE in schools. Additionally, equity issues deal with the inclusiveness of PE in the areas of gender, school sport and disabled students. To sum up, partnership pathways refer to the networks that teachers form within their schools and outside too. These may include links with other teachers in the same school and in other schools, links with sports clubs, sports organizations and other outside school community providers. These partnership pathways may, to a large extent, determine the success of PE programmes in school.

### **Social norms- Perceived status of PE and its teachers**

PE has been deemed to be a marginal subject in schools, with low status of subject and teachers. This inferior status is partly attributed to its practical nature in educational contexts that consider intellectual activity to be more superior (Macdonald & Brooker, 1997). Although PE is considered as being less important than the other subjects in Kosovo schools, it is still allocated the mandatory 80 minutes per week. Frequent cancellation of PE lessons is not reported because schools adhere to the government's mandate to ensure that all timetabled subjects are taught. Therefore, even though PE may have a low status, it is timetabled, taught and assessed in Kosovo schools. According to Hardman (2008), the actual legal and perceived status of PE and its teachers compared to other subjects and their teachers is a highly contentious issue. Even in countries where all subjects are weighted equally, somehow the status of PE and its teachers is placed at a level lower than in the traditionally academic subjects. Results of the current study indicated that this situation is especially worse in Kosovo education system where PE is ranked lower on the subjects list of priorities because school principals, parents and students concentrate more on the more important academically subjects.

### **Marginalization of PE in Kosovo Schools**

PE in Kosovo has a very low status and PE teachers are often regarded as being of a lower standard than other teachers. PE teachers in this study share this opinion about PE teacher status in Kosovo schools (Mike) PE teacher with 42 years of work experience stated

*“ PE teacher in Kosovo schools is considered as someone who is physically fit, but,... not necessary smart enough, because of that they are PE teachers, aren't they? strong and fit, that's all what they are supposed to be !... we do feel really very marginalized, because I must say that we do feel disadvantaged and underestimated teachers, especially we do feel this kind of discrimination from school principals too whenever I used to work I got more or less the same label, “..You are like a Tarzan...” , (my past experience from 4 different schools)... I felt really, on professionally basis for sure discriminated“, I can share with you my experience that PE teachers do not enjoy the same respect as teachers of compulsory academic subjects do, in conclusion the status of most PE teachers in all over Kosovo , particularly, in the secondary*

*schools with no gymnasium, leaves much to be desired* ” The frequency of cancellation of PE classes is very high in Kosovo and is done by academic subject teachers who use allocated PE time to supplement that of their subjects. During times of adverse conditions or when extra time is needed for academic or other school activities, PE periods are often the first to be re-allocated. Regarding timetabling, other subjects are positioned on the timetable with the highest priority, with PE often being placed on the timetable as the last lesson of the day, at a time when both students and teachers are tired and in many cases two (40 minutes) in row. In Kosovo PE is hardly taught and while in some cases it is taught, teachers of academic subjects especially at secondary high schools get angry with their students who attend sports activities (representing school sport teams), in regard to this issue (Bob), said *“..One of my best players of the school football team gets punishment with lower grade in Maths due to the fact that he was absent, due to the reason that he was part of football school team by representing school on tournament organized for country Independence day - Reason for punishment was; because they take away from real academic work and „waste their time playing sports they get punished, that’s our reality“!?* In Kosovo there is still much work to be done on developing teacher competencies, updating assessment methods and introducing new ways of organizing student centered teaching. Curriculum change is under way, and while good progress has been made since now, but this alone is not enough. Teachers and school principals are essential agents for curriculum implementation in the classroom. Their role represents a real challenge in the organization of learning where the capacities of teachers and school leaders are critical

### **Situation of PE in upper secondary public Schools-**

Schools were directed to schedule PE twice a week, separate from after class games and sports, PE lessons were largely ignored for a long time. In addition the syllabus did not clearly show which activities were to be taught and engaged in at each level. Greater emphasis was placed on team competitive sports like football, basketball, handball, and volleyball. Students in Kosovo public schools still receive only two (forty minutes) of PE instruction per week. PE in Kosovo faces many problems, chief of which is lack of affordable and safe environment indoors and outdoors and failure of new national curriculum policy implementation. Since academic subjects are seen as key to a bright future, PE is regarded as a non-productive educational activity, a view that is shared by some teachers too (Marshall & Hardman, 2000). In fact while PE is compulsory in the curriculum, there is a big disparity between policy and implementation because head teachers timetable it only to satisfy school inspectors (Chapell, 2001). There remains a vast difference between official policy and actual delivery of PE in practice. Whipp, Anderson, Yeo and Tan (2006) say that objective assessment is necessary if the goal of PE is to enhance a student’s wellbeing, motor skills and knowledge about (PA) However, this means that teachers must be willing to frequently check students’ learning and to modify instruction to satisfy the students’ needs (Huebner, 2009)

**PE timetabling (Time Allocation), Class sizes**

Most schools have PE slotted on the timetable but hardly implement it. Unfortunately, while there have been many education reforms since the end of the war (98'-99) these reforms have impacted negatively on PE because PE time allocation has continued to decrease in many schools in regard to inappropriate school infrastructure (gym, outdoor playing places). In Kosovo policy requires that secondary schools allocate two periods (40 minutes) a week to all pre university levels of education. Often these two periods 40 minutes are not utilized well because of the lack of school infrastructure, much is left to the teacher to decide regarding use of this time. Many PE teachers find it very challenging to teach meaningfully within such a limited time frame and class sizes of 40 to 50 students per class. PE teachers face the challenge of teaching these large classes with minimal facilities and equipment and other facilities to ensure that PE is taught adequately. To illustrate this situation (John) stated *“a class of 40-50 students would need a considerable investment in sport equipment actually my school principal said to me more than once,” ...Dear Mr. PE teacher I hope I've made myself clear in the beginning of the school year we got budget cut ,we have no more donations from any international organizations, please take care of these sport equipment what we actually have, I do believe that you understand that all over the world countries are facing financial crisis and we are affected too, we have no budget to support your request for some additional sport equipment and requisites that you asked for... we have to manage with actually school budget from the beginning of this school year until the next one, do your best and keep on eye on your students as they play, please ....?!).* However this depends on the financial resources at the disposal of the respective school and the willingness of the school principal- to spend money on such equipment.

**PE Curriculum in public secondary Schools**

Although there is an official school syllabus, it is often not adhered to and individual schools determine PE programmes with huge variations in content and practice depending on the resources at the disposal of each school. While this diversity may be regarded as a positive thing, lack of proper supervision makes it difficult to determine the relevance and effectiveness of such programmes. In the sporting arena schools in public sector can only provide the basic and less financially demanding sports like soccer, basketball, volleyball and athletics. Facilities and equipment are very necessary for PE to be taught properly because the level of provision of facilities can be detrimental to the quality of PE programmes. The facilities and equipment also need to be provided in the right quantity and quality, apart from being well maintained. However, all these factors depend on availability of adequate financial resources.

**Equity (Inclusion), Adapted Physical Education, School Sport issues**

Therefore, PE teachers in Kosovo acknowledge that girls enjoy PE just as boys do, but their ways of doing things are different and this needs to be valued and

respected within PE and sports. Current school playgrounds and classes in Kosovo do not provide teachers with opportunities to construct positive attitudes towards gender and members of the opposite sex. In addition PE teachers should balance between cooperative and competitive activities; provide environments that are safe; and make decisions about content in consultation with the students. Despite all International investments on teacher training Inclusive education these efforts, the biggest challenge that disabled students continue to face in Kosovo schools is inadequate provision of adapted facilities, equipment and trained personnel. School sport in Kosovo is elitist since only the best students get to play in a school team. This means that average students rarely get a chance to participate in inter-school competitions because winning is the ultimate goal, schools always strive to present their best teams for competitions. Kosovo's education system does not give students time to build on their sporting talents because as soon as competitions are over, the focus reverts to the strenuous academic curriculum. PE teacher (Amy) stated, *(sport in Kosovo has not proved that it can create employment and this forces most talented students to choose between sport and education... Therefore, most talented players cease to participate in sport once they leave school, especially if they are a female athlete, they must choose leave it or take it?)*. In Kosovo there is no any Sport body, professional organization, association governmental or nongovernmental one who coordinates talent identification and development programme which tests students and identifies which sport suits them and develop their talents.

### **Overcoming challenges faced in public secondary schools in Kosovo**

By listening to teachers and knowing their experiences, the study goal is to better understand the current position of PE in Kosovo secondary upper schools. Furthermore, if appropriate quantities of equipment are not available, School infrastructures has influence the way PE teachers, plan , design instruction and asses their students Well designed high quality PE curriculum require the following steps; proper equipment and facilities, reasonable class sizes, PE classes taught by qualified PE specialists according to international standards, adapting to the different types of barriers. Unfortunately, currently most in-service PE teachers in Kosovo do not have the competencies and knowledge in teaching health and wellbeing, it's hard to believe that PE can promote healthy, active lifestyles, provide children with a significant proportion of their recommended physical activity, increase physical fitness levels, and teach generalizable movement and behavioral skills. Schools have recently recognized as the most widely used and cost effective setting (especially in countries where compulsory education, primary and secondary level are cost free) for promoting of active healthy lifestyles among adolescents during PE. Knowledge and understanding of teachers' experiences may in turn help education officials, curriculum coordinators and school boards to know how best to support PE teachers. Further research for increasing teacher promotion of healthy lifestyles during PE is needed.



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**IS THERE A PROMOTION OF HEALTHY LIFESTYLES PRACTICES IN HIGH SCHOOL PHYSICAL EDUCATION?**

*The objective of this study is to better understand the current position of Physical education (PE) in Kosovo upper -secondary schools. The participants of this study were four PE teachers (males=3; females=1) from four secondary schools, with 5 to 42 years of PE teaching experience. This research used a qualitative approach to identify concerns, causes and possible teacher-proposed interventions to address the school –based PE issues. In-depth Interviews, document analysis and observations were the methods used to collect data. Results of the study indicates that the lack of facilities and sport equipment (Gym and sport requisites), lower priority given to PE by schools are main concerns of the PE teachers involved in Kosovo' Education system. Currently most in-service PE teachers do not have the competencies and knowledge in teaching health and wellbeing based on the new National Curriculum Framework and according to international standards. These findings of the study can assist all Kosovo schools in identifying the importance of PE in high schools, and students' needs in order to implement their healthy lifestyles.*

**Key words:** Healthy lifestyles, Physical education (PE), students, teachers

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## **MINERALNA GUSTINA KOSTIJU FIZIČKI AKTIVNIH ŽENA PROCENJENA ULTRAZVUČNOM DENZINTOMETRIJOM**

### **Uvod**

Mineralna gustina kostiju se smatra najvažnijim parametrom za procenu rizika osteoporoze i preloma. Osteoporoza je bolest koja se javlja sa godinama starosi, jer se mišićna masa, sila i snaga i koštana gustina smanjuju (Stolzenberg, 2013). Ovi efekti se posebno uočavaju na nosećim delovima tela. Osteoporoza se javlja kod osoba oba pola, a naročito su žene izložene opasnosti usled smanjene produkcije estrogena u periodu menopauze, tako da neke žene mogu izgubiti čak 15% ukupne mase kosti (Kohrt i sar, 2004). Identifikacija žena sa smanjenom koštanom gustinom je važna strategija u smanjenju učestalosti osteoporoznih preloma (Flöter i sar., 2011). Redovnom fizičkom aktivnošću se u premenopauzalnom periodu može povećati gustina koštane mase (Ahola i sar, 2009), a u postmenopauzalnom periodu se može sprečiti opadanje mineralne gustine kosti, te sprečiti pojavu osteoporoze (Borer, 2005).

Procena gustine gosti se može izvesti različitim dijagnostičkim metodama, najčešće je to dvostrukom ekspozicijom X zraka (DEXA). Kvantitativna ultrazvučna osteodenzitometrija predstavlja dijagnostički test kojim se određuje mineralna gustina na jednoj od perifernih kostiju, najčešće na petnoj kosti nedominantne noge (QUS BMD calcanei). Ovaj pregled je brz, jeftin, ne zahteva posebne uslove i obuku kadrova, a za razliku od DXA-e ne primenjuje jonizirajuće zračenje i aparat je prenosiv (Lazić i Spasojević, 2012, Flöter i sar., 2011). Savetodavni panel stručnjaka Svetske zdravstvene organizacije (SZO) je utvrdio kriterijumski prag za mere BMD za dijagnozu osteoporoze (WHO, 1994). BMD se izražava u gramima minerala po površini ( $\text{g}/\text{cm}^2$ ) ili po volumenu ( $\text{g}/\text{cm}^3$ ) i svaka osoba ima svoj maksimum koštane mase (Peak bone mass) i različit intenzitet koštanog gubitka. Rezultati merenja BMD mogu da se izraze kao odstupanje (broj standardnih devijacija) od srednje gustine kosti odraslih zdravih osoba i to se zove T-skor ili kao odstupanje od vrednosti koja odgovara istim godinama starosti zdravih osoba i zove se Z skor (Lazić i Spasojević, 2012).

Cilj ove studije je utvrditi mineralnu gustinu kosti kod žena koje se rekreativno bave vežbanjem bar šest meseci na Fakultetu sporta i fizičkog vaspitanja u Novom Sadu, u sklopu dugoročnog projekta pod nazivom „Uticaj fizičke aktivnosti na rizikofaktore radno aktivnog stanovništva“, koji sufinansira Pokrajinski sekretarijat za nauku i tehnološki razvoj.

### **Metod**

U cilju utvrđivanja mineralne gustine kostiju osoba koje se rekreativno bave vežbanjem, izmereno je 35 žena. Podeljene su u dve grupe, ispitanice koje nisu u perio-

du menopauze (N=20), prosečne starosti  $43.52 \pm 7.56$  god., i ispitanice koje su u manopauzi (N=15) prosečne starosti  $55.89 \pm 5.48$  godina. Ispitanice su vežbale Pilates na Fakultetu sporta i fizičkog vaspitanja u Novom Sadu. Za potrebe ispitivanja su izabrane žene koje su vežbale bar šest meseci, dva puta nedeljno. Mineralna gustina kostiju je merena ultrazvučnom denzitometrijom, odnosno metodom kvalitativnog ultrazvuka (qualitative ultrasound - QUS) uređajem „Sahara“ (Hologic, Inc., MA, USA). „Sahara“ je klinički sonometar koji koristi nejonizujući ultrazvuk za procenu gustine kosti. Rezultati dobijeni ovom metodom su u dovoljnoj korelaciji kako sa DEXA i drugim metodama (Yanagimoto i sar., 2000), te je pogodna za otkrivanje osoba sa visokim stepenom rizika za nastanak osteoporoze (Lazić i Spasojević, 2012). Ovaj uređaj meri širokopojasnu ultrazvučnu atenuaciju (broad-band ultrasound attenuation - BUA) i brzinu zvuka (speed of sound - SOS) petne kosti (calcaneusa), i na osnovu toga se mogu proceniti osteoporotične frakture. Kombinacijom izmerenih vrednosti SOS i BUA, koje se uvrštavaju u određenu formulu, dobija se „kvantitativni ultrazvučni indeks“ (QUS). Osim ovog indeksa, dobija se i procenjena BMD - izvedena vrednost iz SOS i BUA, i T-skor – gustina kosti izražena u vrednostima standardne devijacije ispod prosečne vrednosti odrasle osobe (pri maksimalnoj masi kosti), koji će biti opisani u radu, posebno za desnu i levu nogu. Referentna vrednost procenjene BMD iznosi 0,577.

Svetska zdravstvena organizacija (SZO) definiše osteoporozu kao T-score  $\leq -2,5$  za zdravu populaciju; smanjenu mineralnu gustinu kostiju (osteopeniju) kao  $-2,5 < \text{T-score} < -1$ ; a normalnu gustinu kostiju kao T-score  $\geq -1$  (WHO, 1994).

U cilju saznanja o mineralnoj gustoći kostiju žena koje se rekreativno bave vežbanjem, izračunati su deskriptivni statistici za pokazatelje procenjene mineralne gustine kostiju petne kosti, i T-scora posebno za desnu i levu nogu. Za obradu podataka korišćen je program SPSS 19.0.

## Rezultati

Na osnovu vrednosti centralnih i disperzionih statističkih parametara, kao i parametara normaliteta krive distribucije podataka, utvrđeno je da distribucije podataka svih primenjenih varijabli u ovom istraživanju ne odstupaju statistički značajno od normalne distribucije, na nivou značajnosti 0,01.

U tabeli 1 i tabeli 2 su prikazani deskriptivni parametri antropometrijskih varijabli telesne kompozicije: visine, težine i vrednosti BMI ispitanica. Prema vrednostima BMI ispitanice obe grupe pripadaju grupi normalno uhranjenih osoba.

**Tabela 1.** Centralni i disperzioni parametri antropometrijskih varijabli i varijabli telesne kompozicije žena koje su u premenopauzi

| N=20        | AM     | SD   | MIN    | MAX    | SKEW   | KURT   |
|-------------|--------|------|--------|--------|--------|--------|
| VISINA (cm) | 167.95 | 4.62 | 160.00 | 179.00 | 0.606  | 0.454  |
| TEŽINA (kg) | 64.00  | 6.29 | 58.00  | 75.00  | 0.633  | -1.123 |
| BMI         | 22.71  | 2.27 | 19.35  | 2.57   | -1.027 | 0.101  |

**Tabela 2.** Centralni i disperzioni parametri antropometrijskih varijabli i varijabli telesne kompozicije žena koje su u postmenopauzi

| N=15        | AM     | SD   | MIN    | MAX    | SKEW   | KURT   |
|-------------|--------|------|--------|--------|--------|--------|
| VISINA (cm) | 166.27 | 6.96 | 158.00 | 184.00 | 1.139  | 1.617  |
| TEŽINA (kg) | 66.1   | 8.25 | 46.00  | 83.00  | -0.436 | 2.234  |
| BMI         | 23.9   | 2.69 | 18.43  | 27.43  | -0.417 | -0.564 |

U tabeli 3 i 4 prikazani su sledeći deskriptivni parametri za grupe ispitanica u pokazateljima procenjene mineralne gustine kosti (Est. BMD) i T-scoru posebno za desnu i levu nogu: aritmetička sredina (AS), minimalna (MIN) i maksimalna (MAX) vrednost, te skewness i kurtosis. U grupi ispitanica koje nisu u menopauzi prosečna vrednost procenjene BMD iznosi za desno stopalo  $0.486 \pm 0.062 \text{ g/cm}^2$ , što u vrednostima T-scora iznosi  $-0.84 \pm 0.574$ . Vrednost BMD levog stopala je  $0.493 \pm 0.082 \text{ g/cm}^2$ , odnosno  $-0.781 \pm 0.761$  T-scora. Uzimajući u obzir vrednosti oblika distribucije vrednosti opisuju normalno distribuisanu krivu. T-scor desne i leve noge ukazuje na normalnu gustinu kostiju prema definiciji SZO.

**Tabela 3.** Deskriptivne statistike ispitanica koje su u premenopauzi

| VARIJABLA                     | AS     | SD    | MIN    | MAX   | SKEW  | KURT   |
|-------------------------------|--------|-------|--------|-------|-------|--------|
| Est.BMD_D ( $\text{g/cm}^2$ ) | 0.486  | 0.062 | 0.392  | 0.593 | 0.054 | -1.018 |
| Est.BMD_L ( $\text{g/cm}^2$ ) | 0.493  | 0.082 | 0.361  | 0.669 | 0.47  | -0.372 |
| T-score_D (st.dev)            | -0.84  | 0.574 | -1.714 | 0.146 | 0.054 | -1.018 |
| T-score_L (st.dev)            | -0.781 | 0.761 | -1.996 | 0.854 | 0.47  | -0.372 |

**Tabela 4.** Deskriptivne statistike ispitanica koje su u postmenopauzi

| VARIJABLA                     | AS     | SD    | MIN    | MAX   | SKEW  | KURT  |
|-------------------------------|--------|-------|--------|-------|-------|-------|
| Est.BMD_D ( $\text{g/cm}^2$ ) | 0.448  | 0.081 | 0.368  | 0.615 | 1.249 | 0.415 |
| Est.BMD_L ( $\text{g/cm}^2$ ) | 0.436  | 0.069 | 0.372  | 0.644 | 2.176 | 5.877 |
| T-score_D (st.dev)            | -1.195 | 0.753 | -1.931 | 0.348 | 1.249 | 0.415 |
| T-score_L (st.dev)            | -1.305 | 0.637 | -1.898 | 0.623 | 2.176 | 5.877 |

Kod ispitanica koje su u menopauzi (Tabela 4) procenjena gustina kosti desne noge iznosi  $0.448 \pm 0.081 \text{ g/cm}^2$ , odnosno  $-1.195 \pm 0.753$  SD u vrednostima T-scora. Rezultati BMD leve noge iznose  $0.436 \pm 0.069 \text{ g/cm}^2$ , što prema vrednostima T-scora iznosi  $-1.305 \pm 0.637$  SD. Prema podeli SZO, pokazalo se da je ispitanicama u ovoj grupi smanjena mineralna gustina kostiju, jer su vrednosti T-scora ispod -1. Na osnovu vrednosti oblika distribucije, rezultati ukazuju na pozitivan skewness, odnosno većina rezultata je u nižim vrednostima, što potvrđuje prethodni zaključak.

Najniža vrednost kod ispitanica u premenopauzi iznosi -1.996 SD za levo stopalo i -1.714 SD za desno stopalo. Kod osoba u postmenopauzi najniža vrednost je -1.931 SD za desno stopalo i -1.898 SD za levo stopalo. Prema ovim vrednostima ni jednoj ispitanici nije utvrđena vrednost koja ukazuje na osteoporozu merenih na petnoj kosti.

## Diskusija

Cilj ovog rada je prikazati gustinu kosti osoba koje se bave fizičkom aktivnošću, odnosno utvrditi njihovo stanje kosti, ultrazvučnom denzintometrijom. Kvantitativni ultrasonometri, kao Sahara, su razvijeni da precizno identifikuju osobe sa rizikom od preloma koji bi trebalo da se temeljnije ispituju korišćenjem DXA-e (Wüster i sar., 1998). U našem slučaju su izabrane žene koje vežbaju bar šest meseci u sklopu projekata „Uticaj fizičke aktivnosti na riziko faktore kod radno aktivnog stanovništva“. Prema rezultatima dobijenim metodom ultrazvučne denzintometrije, prema minimalnim vrednostima utvrđeno je da ni jedna učesnica nema vrednost T-scora ispod -2.5 SD, što predstavlja gornju granicu osteoporoze. Ipak, vrednosti su niže kod postmenopauzalnih ispitanica, s obzirom da nedostatak estrogena ima negativne efekte na skelet, posebno u prvih 5-10 godina posle menopauze, sa preferencijalnim gubitkom trabekularne u odnosu na kortikalnu kost (Shed i sar., 2007). Fizička aktivnost se odavno smatra kao deo vođenja zdravog načina života, bavljenje fizičkom aktivnošću u slobodno vreme je povezano sa visokim nivoom BMD kod premenopauzalnih i postmenopauzalnih žena (Wee i sar., 2013). Žene u postmenopauzi bi trebale upražnjavati umerenu fizičku aktivnost kao deo sekundarne prevencije od osteoporoze i preloma kosti (Wee i sar., 2013). Blanchet i sar (2003) su uočili da je nivo redovne fizičke aktivnosti u slobodno vreme žena u postmenopauzi povezano sa QUS parametrima, nezavisno od BMDa. Žene koje se u slobodno vreme redovno bave fizičkom aktivnošću se klasifikuju kao aktivne (tri i više puta nedeljno) i imaju povećane vrednosti BUA, SOS i SI i prilagođen BMD vrata butne kosti, dok umereni nivo predstavlja manje intenzivnu fizičku aktivnost (jednom do dva puta nedeljno) i povezano je samo sa SI i SOS. Dok su u istraživanju Kim i sar. (2012) žene u pre- i postmenopauzi koje su se bavile rekreativnom aktivnošću na umerenom ili visokom nivou imale bolje vrednosti BMD procenjene DXA-om, u odnosu na kontrolnu grupu koje slobodno vreme nisu koristile u te svrhe.

U našoj studiji ispitanice se bave umerenom fizičkom aktivnošću, vežbajući Pilates dva puta nedeljno. Prema Babayigit Irez (2009) Pilates ima pozitivan uticaj na BMD, s obzirom da u godinu dana praćenja nije primećeno smanjenje BMD-a. Pilates vežbanje uključuje vežbe snage i istezanja, a prema istraživanjima uticaj treninga snage na koštani status treba još ispitati, s obzirom da su tri od četiri studije koje pokazuju značajan efekat treninga snage na kosti randomizirane studije, u poređenju sa četiri od osam prethodnih studija koje su bile neutralne ili negativne (Singh i sar., 2009). U ovom slučaju ne možemo da uporedimo efekte Pilates programa na BMD, ali nam je bitno da smo ustanovili da nema ispitanica sa osteoporozom, da ispitanice u premenopauzi pripadaju grupi sa normalnim BMD prema srednjim vrednostima desnog i levog stopala. Ispitanice u postmenopauzi imaju smanjenu gustinu kostiju, što nije neuobičajeno s obzirom na promene koje donosi menopauza usled nedostatka estrogena. Ispitanice u postmenopauzi sa smanjenom gustinom kosti mogu povećati gustinu kosti u distalnom delu tibije, u periodu od devet meseci, vežbanjem dva puta nedeljno sa otporom (Stolzenberg i sar., 2013).

Kvalitativna ultrazvučna denzintometrija je dobar metod za procenu kvaliteta kosti i utvrđivanja stanja kosti osobe u svrhu prevencije od osteoporoze. Ispitanice vežbaju Pilates na umerenom nivou, koji uključuje vežbe snage i balansa, a nakon odre-

denog vremenskog perioda će biti utvrđivani efekti istih sa postojećim rezultatima gustine kosti i T-scora.

### Zahvale

Ova studija je sprovedena u sklopu projekta pod nazivom „Uticaj fizičke aktivnosti na riziko faktore kod radno aktivnog stanovništva“, br.pr. 114-451-2337/2011/04; 2011-2014), koji su sufinansirali Pokrajinski sekretarijat za nauku i tehnološki razvoj i Fakultet sporta i fizičkog vaspitanja Univerziteta u Novom Sadu.

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#### BONE MINERAL DENSITY IN PHYSICALLY ACTIVE WOMEN ASSESSED BY ULTRASOUND DENSITOMETRY

*Osteoporosis is a disease characterized by low bone mass and density. Physical activity has a positive effect on bone tissue, and it is recommended to prevent bone loss which comes with age. Methods: In purpose of determining bone mineral density in women who are physically active we examined 35 women divided into two groups – subjects who are premenopausal (n=20, 43.52 ± 7.56 years), and subjects who are postmenopausal (n=15, 55.89 ± 5.48). The subjects exercised Pilates method twice a week for one hour. Bone mineral density measurements were done by ultrasound densitometer „Sahara“ through the calcaneus. We get the data of the estimated bone density and T-score for right and left foot separate. Results: According to results premenopausal women have normal bone density, and postmenopausal values represent osteopenia, according to the WHO. There is no subjects who established osteoporosis. Discussion: Previous investigations have confirmed that physical activity is important for the preservation of the bone quality. Increasing steps, using simple everyday tasks, can prevent decrease in BMD in postmenopausal women (Muir et al., 2013, Ashe et al, 2008). There's difference in the density of the calcaneus between physically active and those who are not, measured by ultrasound densitometry (Vainionpää et al, 2005). We use quantitative ultrasound densitometry to describe BMI of women who exercise Pilates and support the finding that physical activity is one of the ways that may prevent the BMD loss. These research was done as a part of long-term project entitled „Impact of physical activity of the working population“ which is co-financed by Provincial Secretariat for Science and Technological Development.*

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## APPLICATION OF TAXONOMY ANALYSIS IN THE SELECTION OF PUPILS IN SPORT

### Introduction

Correct and timely selection of sport is important precondition for success of children. Correct selection of sport is important and difficult since there are no sure indicators based on which one can identify talent of a child for a specific sport. For correct selection of sport, testing the child by professionals is important, test in which child can demonstrate its motoric abilities important for a given sport. When selecting the sport, the motoric abilities are the most important. In some sports the results depend more on good technique and in others, depend more on functional endurance or anthropometric abilities. However, the age limit for commencing to practice a competitive sport moved its limits to an earlier age. Selection of a sport where children should have top results is for children a precondition for success because of early beginning of specialization in sport.

Correct selection of children fit for certain sport is the most demanding and responsible task for kinesiology professionals and sport trainers. Sport activities are, to a great extent, different in structure and content. Different sports are marked with authentic kinesiology structures and specific anthropologic symbols (Srhoj, Lj. i sur., 2006.).

The success of an individual in specific sport activity mostly depends on compatibility of their anthropological symbols i.e. anthropological model for the given sport (Viskić Štalec, 2006.). This is why it is important to select sportsmen and women whose anthropological symbols are to the biggest extent possible fitting specific kinesiology activities.

Starting from the importance of proper selection of children in the sport, the Kinesiology Faculty in Split, Croatia and in the Institute for Poly-technique of Natural and Mathematics Science within the scientific project “Discovering talents in sport” is developing an expert content in discovering talent in sport – TALENT (Rogulj, Papić 2006.). The knowledge base on the project is on normative orientation values of school kids in Croatia gained through a comprehensive research which was commissioned in 1992 by V. Findak, D. Metikoš and M. Mrakoviću within the scope of project dealing with methodology of monitoring and evaluation of body and health education in schools.

In selection of children for sports, very supportive are also specific quantitative methods that group entities based on anthropologic characteristics. One of quantitative methods that found its application in the selection of children for sports is cluster or taxonomy analysis.

Correct selection and orientation of children in sports represents a very important professional task of, not only the trainer but of a bigger team of professionals



that engage in developing sport in specific geographical area. In development of specific sports important place is given to as early as possible selection of youth for specific sport. Use of taxonomy analysis in selection of youth in specific sport based on physical development and motoric abilities is one of the goals of this article.

### **Methods**

Research is of transversal nature, meaning that it contains measurement of anthropometric indicators and motoric indicators in the sample of 26 pupils aged 15 years from primary and secondary lower level school from urban and a sample of 30 pupils from rural area. Measurements have been completed at the beginning of school year. Sample of variable are six anthropometric and four motoric variables have been applied (Kurelić et al., 1975).

#### **Anthropometric variables:**

- Body height (ATV),
- Foot length (ADS),
- Body mass (ATT),
- Volume of upper arm at relaxed position (AONL),
- Volume of upper leg (AONK),
- Volume of lower leg (AOPK).

#### **Motoric variables:**

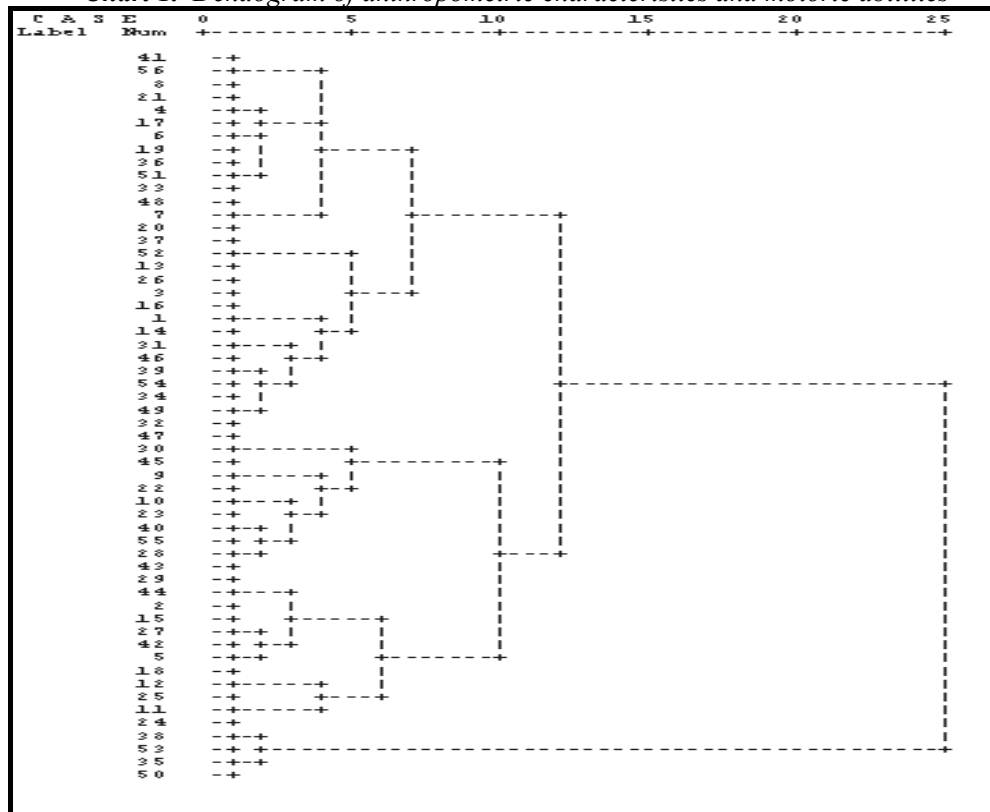
- Long jump from standing position (MESDM),
- 30 m distance running (MTR30V),
- Bench bending (MFLPRK), and
- Push-ups (MSKLEK).

For data analysis, the taxonomy analysis has been applied.

### **Results**

Taxonomy analysis was used for the purpose of forming sub samples with as much as homogenous characteristics and abilities as possible. A hierarchal method was used based on iterative process of connecting respondents into groups, thus in the given phase of research we connect respondents in previously formed groups, meaning that previously formed groups are only expended with new members according to their particularities and there is no possibility for respondents to move from one formed group to another.

Based on the analysis of dendogram (chart 1) true taxonomy analysis of anthropometric characteristics and motoric abilities we can notice visual configuration of collection of respondents into clusters according to similar structures of their characteristics and abilities. From total of ten applied variables, 6 anthropometric and 4 motoric abilities on the given sample, two taxonomy dimensions have been extracted and this is verification that we are dealing with relatively good sample based on anthropometric characteristics and motoric abilities.

**Chart 1.** Dendrogram of anthropometric characteristics and motoric abilities

Dendrogram using Average Linkage (Between Groups), scaled Distance Cluster Combine

**Table 1.** Taxonomy analysis of anthropometric characteristics and motoric abilities

| Final Cluster Centers |        |        |        |
|-----------------------|--------|--------|--------|
| Cluster               |        |        |        |
| 1                     |        | 2      |        |
| (N=22)                |        | (N=34) |        |
|                       | 167.20 |        | 172.24 |
|                       | 25.64  |        | 26.50  |
|                       | 64.15  |        | 66.82  |
|                       | 25.22  |        | 25.87  |
|                       | 54.77  |        | 54.85  |
|                       | 36.39  |        | 36.44  |
|                       | 157.64 |        | 210.47 |
|                       | 5.85   |        | 5.26   |
|                       | 28.09  |        | 42.71  |
|                       | 9.36   |        | 24.47  |

In the group of six anthropometric characteristics and four motoric abilities two taxonomy dimensions are extracted, where based on squared Euclidean distances and given arithmetic means (table 1) it can be seen that out of 56, 22 pupils are grouped in the first group while 34 pupils fit the second group.

Second taxonomy dimension (Table 1) in a satisfactory manner defines anthropometric variables: body height (ATV) and foot length (ADS), body mass (ATT), volume of upper arm in relaxed position (AONL), volume of upper leg (AONK), volume of lower leg (AOPK) and motoric variables: jump from standing position (MESDM), 30 meter distance running (MTR30V), bench bending (MFLPRK) and push-ups (MSKLEK).

**Table 2.** Taxonomy analysis of anthropometric characteristics and motoric abilities inside and outside isolated groups

|        | Cluster     |    | Error       |    | F       | Sig. |
|--------|-------------|----|-------------|----|---------|------|
|        | Mean Square | df | Mean Square | df |         |      |
| ATV    | 339.451     | 1  | 57.150      | 54 | 5.940   | .018 |
| ADS    | 9.963       | 1  | 1.687       | 54 | 5.906   | .018 |
| ATT    | 94.730      | 1  | 204.455     | 54 | .463    | .499 |
| AONL   | 5.685       | 1  | 9.563       | 54 | .595    | .444 |
| AONK   | 11.300      | 1  | 45.651      | 54 | .248    | .621 |
| AOPK   | .026        | 1  | 13.425      | 54 | .002    | .965 |
| MESDM  | 37285.867   | 1  | 349.659     | 54 | 106.635 | .000 |
| MTR30V | 4.677       | 1  | .181        | 54 | 25.910  | .000 |
| MFLPRK | 2853.052    | 1  | 198.979     | 54 | 14.338  | .000 |
| MSKLEK | 3048.367    | 1  | 72.177      | 54 | 42.235  | .000 |

With the insight of the size of calculated parameters of the descriptive statistics in the values of first and second cluster, it is obvious that pupils in the second cluster dominate with some anthropometric and all of the motoric abilities. In anthropometric variables pupils in the second cluster are dominant in longitudinal dimension: body height (ATV) and foot length (ADS). On statistically significant level variables selecting pupils for the second cluster (table 2) represent: variable defining explosive power of lower parts of body, distance jump from standing position (MESDM), defining speed and starting position 30 meter distance running from high start (MTR30V), followed by variable defining flexibility, bench bending (MFLPRK) and variable defining repetitive power, push-ups (MSKLEK).

### Discussion

Taxonomy analysis was used in order to group, namely classify sample of pupils based on anthropometric and motoric parameters. Goal of this analysis was to form sub samples in the given sample that are homogenous in as more parameters that are

common in terms of anthropometric dimension and motoric abilities. Results obtained indicated that we have gained two homogenous sub-groups where the second group of 34 pupils represents a group with best anthropometric and motoric parameters.

Based on taxonomy analysis of anthropometric characteristics we can conclude that classification of pupils occurred on two self-types out of which one is presented with longitudinal dimension while the second one is characterized with lower height and increased body mass. In general the second cluster in anthropometric sense has given numerous numbers of pupils that belong to ecto-mesomorph type. In the same cluster for basis motoric space, pupils that have motoric dimensions responsible for explosive and repetitive power as well as for flexibility stand out.

Taxonomy analysis as quantitative method contributes in proper selection and orientation of pupils for specific sports based on characteristics and anthropological abilities applied in this procedure.

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### APPLICATION OF TAXONOMY ANALYSIS IN THE SELECTION OF PUPILS IN SPORT

*Primary aim of physical and health education and of this education and upbringing activity, among other things, is to secure favorable conditions for physical education as modern civilization value and a need, in order to advance quality and effectiveness of life, and for advancement of health level especially in the adolescent phase of life, and to create opportunities for involvement of pupils in various sports. Methods: This research has the goal to evaluate importance of taxonomy analysis in*

*the selection of pupils in sports. This means involvement of a process of measurement of morphological and motoric indicators of 26 pupils involved in sport activities in the age group of 15 years in the elementary and lower level secondary school “Faik Konica” from Prishtina and 30 pupils of age group 15 years in the elementary and lower level school „Avdulla Tahiri“ from Malishevë. 6 anthropometric variables have been applied and 4 motoric variables (Kurelić et al., 1975). Anthropometric variables applied are: body height (ATV), length of foot (ADS), body mass (ATT), volume of upper arm in down position (AONL), volume of upper leg (AONK), volume of lower leg (AOPK). Motoric variables applied were: distance jump from standing position (MFESDM), running from 30 meter distance (MTR30V), bench bending (MFLPRK), and push-ups (MSKLEK). Results: Taxonomy analysis applied demonstrates that these groups of pupils has been divided in two groups, first group composed of 22 pupils and second group of 34 pupils. Results obtained in the table using Anova discriminative analysis have verified statistical significance of the difference between these groups. Discussion: Taxonomy analysis is used in order to group, namely classify sample of students on the basis of anthropometric and motoric parameters. Goal of this analysis was to create a homogenous sub sample on the basis of given sample, one that has as more as possible of a joint morphological and motoric parameters. Obtained results demonstrate that this research has resulted in two sub-groups of homogenous nature, the second group of 34 pupils' shows to be a group with better anthropometric and motoric parameters for selection for football. Similar results when it comes to selection have been obtained in other researches (Goletić, et al. 2012). The other group of 22 pupils has lower anthropometric and motoric parameters.*

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## DIFFERENCES BETWEEN PUPILS FROM URBAN AND RURAL AREAS IN MORPHOLOGICAL AND MOTORIC DEVELOPMENT

### Introduction

One of the most important preconditions for effective influence of physical exercises on pupils during ordinary sport education classes in the schools is increase of volume and quality of performance. In order to increase the volume and quality of learning and activities it is necessary to implement reforms in primary and secondary lower education system in Kosova toward advancement of staff and increase of number of teaching hours for sports and health education.

In different countries diverse programs and strategies are developed for organization of sport activities for youth. A hallmark example is England where from 2002 government has demonstrated clear commitment for developing partnership between body development education and sports and has committed investing 459 million pounds for physical education in schools and other developmental initiatives related to sport and thus dedicated 750 million pounds for development of supporting infrastructure. Reasons for lack of investments are diverse and they are founded on learning about the health condition of the nation, obesity, sitting life style and high rate of absence of sport activities. Government policy has made a strong echo in professional circles due to making strong links between body development and sports that is overcoming historical dualism between these two areas of human activity (Donovan et al. 2006).

Education system in the USA is to a large extent different from European system, and it is especially different from Kosova system. Differences are visible in all areas of school education and they are especially visible in upbringing and education of body and health culture in sports in the schools. In American state or private school one of the significant area is sport, which has significant place also in religious schools, for example in catholic secondary schools where religion is main subject. In this way the popularity of students among peers and professors is mostly determined based on achievements in sport. Also the reputation and popularity of schools grows with sport successes of its pupils. Good sport school draws attention of sponsors and this is increasing possibility for increased sports achievements (Đonlić, 2008).

Sport besides preserving health, creation of work and defense capabilities has also significant social role. This role is primarily directed toward upbringing and education values. Sport has significant role for formation of personality. Sport enables children to develop human qualities and transfers authentic human values (Đonlić, 2005).

In Kosova there is a big difference between urban and rural areas in connection with conditions for sport education and realization of curricula. Lack of infrastructure for sport, decreased hours for sport (only two per week), lack of physical activities outside of

school in rural areas is an important factor for physical and motoric development that presents disadvantageous environment compared to pupils in urban areas.

### Methods

This research is of transversal nature, meaning that there is measurement of anthropometric and motoric indicators in the sample of 26 students of 15 years age group from primary and lower level secondary school from urban center and the sample of 30 pupils from primary and lower level secondary school from rural area. Measurements were done at the beginning of school year. Six anthropometric and four motoric variables were applied (Kurelić et al., 1975).

#### Anthropometric variables:

- Body height (ATV),
- Foot length (ADS),
- Body mass (ATT),
- Volume of upper arm at relaxed position (AONL),
- Volume of upper leg (AONK).
- Volume of lower leg (AOPK).

#### Motoric variables:

- Standing long jump (MESDM),
- 30 meter distance running (MTR30V),
- Benchng (MFLPRK), i
- Push-ups (MSKLEK).

For data analysis T-test for dependent sample was applied.

### Results

Basic statistical parameters and parameters of distribution in the group of pupils from school in urban center that took part in this research are presented in table number 1. Starting from values of asymmetry (Skew) and coefficient of distribution curve (Kurt.) of applied variable, it was confirmed that there is no significant asymmetry. In majority of variables coefficient of asymmetry is positive (epikurtic) meaning that their arithmetic mean tends toward higher results and that majority of results of their variables tend toward lower values. Two anthropometric and one motoric variable are hipokurtic meaning that their mean arithmetic has lower values compared to majority that tend to higher values.

Through analysis of the basic descriptive statistical parameters and parameters of distribution in the group of pupils from school in rural environment that took part in this research as represented in table 2. It can be seen in the table that all applied motoric variables have normal distribution of results. Based on the values of asymmetry of skewness (Skew) and coefficient of distribution curve (Kurt.) of applied variable it was determined that there is no significant asymmetry. In majority of variables (7 variables) asymmetry coefficient indicates that their arithmetic mean tends toward higher results and majority of variable results tend toward lower values. In this table two anthropometric variables and one motoric are hipokurtic (negative) meaning that their arithmetic mean have lower values and majority of results of these variables tend toward higher values.

**Table 1.** Basic statistical parameters of motoric variables at the beginning for the group of pupils from school in urban area

|        | N  | Min    | Max    | Mean     | Std. Dev | Skew  | Kurt   |
|--------|----|--------|--------|----------|----------|-------|--------|
| ATV    | 26 | 157.00 | 184.00 | 171.6692 | 8.86064  | -.186 | -1.019 |
| ADS    | 26 | 24.00  | 30.00  | 26.7308  | 1.59519  | .535  | .002   |
| ATT    | 26 | 44.50  | 93.00  | 63.7308  | 15.54042 | .696  | -.656  |
| AONL   | 26 | 20.50  | 30.00  | 24.9231  | 3.15497  | -.017 | -1.326 |
| AONK   | 26 | 44.50  | 63.00  | 52.2692  | 5.88087  | .625  | -.468  |
| AOPK   | 26 | 30.00  | 42.00  | 35.3077  | 3.54140  | .474  | -.674  |
| MESDM  | 26 | 160.00 | 230.00 | 195.8462 | 22.08654 | -.024 | -.920  |
| MTR30V | 26 | 4.84   | 5.75   | 5.2562   | .26295   | .353  | -.376  |
| MFLPRK | 26 | 10.00  | 71.00  | 38.0769  | 18.92918 | .104  | -1.218 |
| MSKLEK | 26 | 9.00   | 40.00  | 21.9231  | 11.28866 | .682  | -1.222 |

**Table 2.** Basic statistical parameters of motoric variables at the beginning in the school group from school in rural area

|        | N  | Min    | Max    | Mean     | Std. Dev | Skew  | Kurt   |
|--------|----|--------|--------|----------|----------|-------|--------|
| ATV    | 30 | 156.00 | 183.00 | 169.0400 | 6.86448  | .162  | .027   |
| ADS    | 30 | 24.00  | 27.00  | 25.6667  | .86436   | -.127 | -.680  |
| ATT    | 30 | 51.00  | 96.00  | 67.5400  | 12.99283 | .785  | -.245  |
| AONL   | 30 | 21.00  | 31.00  | 26.2133  | 2.93630  | .014  | -.891  |
| AONK   | 30 | 46.50  | 67.50  | 55.9000  | 7.01771  | .134  | -1.269 |
| AOPK   | 30 | 32.00  | 42.80  | 37.3800  | 3.48221  | -.106 | -1.427 |
| MESDM  | 30 | 120.00 | 255.00 | 184.4000 | 38.12285 | .034  | -.655  |
| MTR30V | 30 | 4.83   | 6.78   | 5.6940   | .58802   | .608  | -.747  |
| MFLPRK | 30 | 10.00  | 53.00  | 36.0000  | 12.56706 | -.673 | -.137  |
| MSKLEK | 30 | 4.00   | 35.00  | 15.6000  | 10.50977 | .446  | -1.235 |

Third table contains all data related to T-test for independent sample. For each variable, primarily results with assumption of equality of group variance, and secondly inequality of variance.

In first two columns results of Leven's test of equality of variance its value (F) and significance (sig) are presented. In remaining columns of the table are results of testing equality of arithmetic mean of two groups: value of T-test (t), number of freedom degree (df), significance of two way testing of differences of arithmetic means (2-tailed), difference of arithmetic mean (mean difference), standard error difference (Std. Error difference), limits of upper and lower interval and differences of confidence from 95% (95% Confidence Interval Of the. of the difference –Lower, Uper).



**Table 3.** Test of arithmetic mean difference of independent sample

|        | F      | Sig. | T      | Df     | Sig.<br>(2-tailed) | Mean<br>Diff. | Std.<br>Err. Diff. | Lower     | Upper    |
|--------|--------|------|--------|--------|--------------------|---------------|--------------------|-----------|----------|
| ATV    | 2.586  | .114 | 1.250  | 54     | .217               | 2.62923       | 2.10390            | -1.58884  | 6.84730  |
|        |        |      | 1.227  | 46.846 | .226               | 2.62923       | 2.14251            | -1.68132  | 6.93978  |
| ADS    | 5.741  | .020 | 3.160  | 54     | .003               | 1.06410       | .33673             | .38900    | 1.73920  |
|        |        |      | 3.037  | 37.262 | .004               | 1.06410       | .35039             | .35431    | 1.77389  |
| ATT    | .620   | .434 | -.999  | 54     | .322               | -3.80923      | 3.81262            | -11.45308 | 3.83461  |
|        |        |      | -.986  | 48.972 | .329               | -3.80923      | 3.86209            | -11.57051 | 3.95205  |
| AONL   | .652   | .423 | -1.584 | 54     | .119               | -1.29026      | .81442             | -2.92307  | .34256   |
|        |        |      | -1.576 | 51.570 | .121               | -1.29026      | .81868             | -2.93338  | .35287   |
| AONK   | 1.773  | .189 | -2.080 | 54     | .042               | -3.63077      | 1.74596            | -7.13121  | -.13033  |
|        |        |      | -2.106 | 53.949 | .040               | -3.63077      | 1.72389            | -7.08703  | -.17451  |
| AOPK   | .307   | .582 | -2.204 | 54     | .032               | -2.07231      | .94042             | -3.95773  | -.18688  |
|        |        |      | -2.201 | 52.607 | .032               | -2.07231      | .94157             | -3.96119  | -.18342  |
| MESDM  | 8.245  | .006 | 1.347  | 54     | .184               | 11.44615      | 8.50002            | -5.59536  | 28.48767 |
|        |        |      | 1.396  | 47.541 | .169               | 11.44615      | 8.19800            | -5.04114  | 27.93345 |
| MTR30V | 17.413 | .000 | -3.502 | 54     | .001               | -.43785       | .12502             | -.68849   | -.18720  |
|        |        |      | -3.676 | 41.372 | .001               | -.43785       | .11910             | -.67831   | -.19738  |
| MFLPRK | 11.350 | .001 | .490   | 54     | .626               | 2.07692       | 4.24253            | -6.42883  | 10.58268 |
|        |        |      | .476   | 42.412 | .637               | 2.07692       | 4.36413            | -6.72772  | 10.88157 |
| MSKLEK | .202   | .655 | 2.170  | 54     | .034               | 6.32308       | 2.91453            | .47981    | 12.16635 |
|        |        |      | 2.158  | 51.577 | .036               | 6.32308       | 2.92970            | .44306    | 12.20310 |

Results obtained indicated that pupils from urban environment and pupils from rural environment have statistically significant differences in anthropometric variables such as the length of foot/sole (ADS) of  $p < 0.003$ , whereas pupils from urban area have better longitudinal parameters while the value of perimeter of upper leg (AONK) is  $p < 0.042$ , and perimeter of volume of lower leg (AOPK) is  $p < 0.003$  indicating that pupils from rural areas have higher body mass. Pupils from urban areas have better results in all motoric tests, especially the test of high start 30 meter running (MTR30V) with  $p < 0.001$  and push-ups (MSKLEK) with  $p < 0.036$ . Differences between children from urban and rural areas in the physical development as well as motoric, begs a question of factors in the environment that influence pupils from rural area to have lower results. Maybe it is not all up to conditions for implementation of physical education in the school. The role of parents in times of big social and economic changes is significant.

Involvement of children in sport activities today is dedicated more to parents than to children. While in the past children were dealing with body activities in their yard and open play grounds, social changes and fear by parents for safety of their kids have resulted in bigger involvement of children in organized indoor clubs and activities. In this

way from an informal game sport has gained a completely new character of activity supervised by parents. Besides their presence, parents get involved through their expectation and pressure on kids. However, if child is not successful in what he/she is doing it will need help and not expect too much just because they are exercising every day (Halpern, 2003). Experience of success and personal effectiveness are elements for developing an image of ourselves and prevention of undesired forms of behavior (Wigfield, 1992). Results indicate that promoting healthy life style among youth gains importance with the insight gained by Paavola and sur. (2004.) which emphasizes positive correlation between regular sport activities and correct nutrition habits.

### Discussion

In order to determine difference between pupils from urban and rural areas in anthropometric and motoric development, this research was implemented at the sample of 26 male pupils from urban area and 30 male pupils from rural area, all aged 15 years. Measurement of physical development has been carried out using 6 variables that cover longitudinal dimensions and mass dimension and body size, while motoric skills were valued with 4 motoric tests. After implementation of specified statistical procedures it was determined that in the group of urban and rural area there is a significant statistical difference in some anthropometric and motoric variables. Pupils from urban areas have higher longitudinal anthropometric parameters and lower body mass. When it comes to motoric skills pupils from urban areas have better results. Lack of infrastructure for sport, lack of physical activities outside of school and also maybe social issues are considerable factors for physical body development and motoric which for children in rural areas are not favorable compared to pupil from urban areas.

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## DIFFERENCES BETWEEN PUPILS FROM URBAN AND RURAL AREAS IN MORPHOLOGICAL AND MOTORIC DEVELOPMENT

*One of the most important preconditions for effective influence of physical body exercises among pupils during regular classes of physical education in schools is increased volume and quality of work and study. In order to increase volume and quality of work and study, it is necessary to carry out reform of education at elementary and lower level secondary education in Kosova in order to advance professional personnel and increase number of hours for sport and health education classes. Methods: This research is of a transversal nature, meaning that there has been a measurement of morphological and motoric indicators in the sample of 26 pupils of the age group of 15 years of the elementary and lower level school "Faik Konica" from Prishtina as an urbane center and sample of 30 pupils of elementary and lower level school „Avdulla Tahiri“ from Malisheva. 6 anthropometric and 4 motoric variables have been used (Kurelić et al., 1975). Anthropometric variables included: body height (ATV), length of foot (ADS), body mass (ATT), volume of upper arm in down position (AONL), volume of upper leg (AONK), volume of lower leg (AOPK). Motoric variables included: standing position distance jump (MFESDM), 30 meters distance running (MTR30V), bench bending (MFLPRK), and push-ups (MSKLEK). T-test analysis has been used for independent variables. Results: Obtained results from the statistical analysis demonstrate that anthropometric characteristics and motoric skills of two independent groups of pupils have normal distribution and no visible asymmetry and have tendency toward higher values (epikurtic). T-test analysis demonstrates that pupils from rural areas have lower muscular mass and lower motoric results. Discussion: Conditions for execution of physical education classes and lack of physical activities in the rural environment have strong influence on developments of morphological and motoric characteristics of pupils. Significant statistical differences obtained in these groups demonstrate that pupils in the elementary and lower level secondary education school „Avdulla Tahiri“ from Malisheva have lower muscular mass and motoric abilities at the level  $p < 0.05$ , due to economic and social conditions. Results indicate that promotion of healthy life style among adolescents has even greater importance due to insight gained by Paavola and sur. (2004.), that underlines positive correlation between regular sports and recreational activities and correct nutrition habits.*

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## INFLUENCE OF PHYSICAL BODY EXERCISES ON PUPILS

### Introduction

In Kosova education system, frequent changes that have occurred in past 14 years have given indication that the education system has strived to adjust to social, economic and cultural changes, etc. These changes, as they were not studied well, have affected and have continued to affect organization of the physical education in schools, its quality and quantity. These changes have not yielded necessary results and have often been regarded as experiments. Proposed reforms have been oriented in decrease of the number of study hours from two hours per week of physical activity and one hour of theory in health and wellbeing.

Commencing from social conditions of pupils in Kosova, primary schools and secondary are the only possibility for children to engage in physical activities. We need to understand that physical education in schools is based on learning and developing basic motoric skills that are fundamental for regular and harmonized development of children's bodies based on the natural shape of movements and game. In diverse activities knowing better its motoric skills, children develop and upgrade new motoric abilities and all of these happen during regular physical education. Therefore, enhancement of quality and quantity of physical activities in schools is very important obligation.

Selection of tools for development of physical activities of children during execution of programs of physical education in schools is mainly determined based on the research studies of various authors that have chosen to deal with issues of valuing physical activities in relation to psychosomatic status of children and adults. Specific value has research carried out by Klojcnik (1977) (as cited by Madić, 2000) which treated the issue of selection of exercises for children and adults. Some of the general conclusions of this research have to do with selection of tools within the program of physical education. Author has underlined the importance of the use of gymnastics exercises and athletics exercises in this very sensitive period of child's development, and in particular their positive influences on body position, development of anthropometric characteristics and motoric skills and also influence on cognitive skills and conative characteristics of children.

Starting from the fact that the growth of quality and quantity of physical activities influences strongly in a harmonized and normal body development and motoric development of children and youth is basic goals of this research. Goal of this research is to determine influence of one additional hour of physical education, in total three hours during a week for anthropometric and motoric development of pupils.

## Methods

In conditions of defining time, this study has longitudinal empirical character and is composed of two measures of morphological indicators and motoric of pupil of the primary and lower secondary school from urban locality. The study covered in total 26 pupils, aged 15 years. Measurements have been executed at the end of the school year.

We have applied 6 anthropometric variables and 4 motoric variables (Kurelić et al., 1975). **Anthropometric variables:**

- Body height (ATV),
- Foot length (ADS),
- Body mass (ATT),
- Size of upper arm in released position (AONL),
- Size of upper leg (AONK),
- Size of lower leg (AOPK) and

## Motoric variables:

- Long jump from standing position (MESDM),
- Running from 30 meters (MTR30V),
- Bench bending (MFLPRK), and
- Press-ups (MSKLEK).

In order to analyze the data we have used T-test analysis for dependent sample.

## Results

The first table below contains results of descriptive basic parameters and of distribution for analyzed variables in the sample. In the table, the columns present number of sample involved in study (N), minimal result (Min), maxi result (Skew) mal (Maks), arithmetic mean (Mean), standard deviation (Std. Dev), measures of asymmetric distribution (Skew), measures of length of distribution (Kurt) and variation coefficient (KV).

Results obtained indicate that all morphologic and motoric variables applied do not have highlighted asymmetry and have normal distribution of results. In a big number of variables, asymmetry coefficient indicated that results tend to lower values given that the arithmetic mean has higher value from medium and mod. In three variables asymmetry tests are hipokurtic (negative), meaning that results tend toward higher ones given that arithmetic mean has lower values.

Table 2 presents basic parameters of descriptive statistics of distribution in pupils after one year of treatment in the school with 3 hours of physical education, and all morphologic and motoric variables indicate a normal distribution of results. Based on the values of asymmetry (Skew) and coefficient of the distribution curve (Kurt.) in the applied variables confirm that there is no significant asymmetry. With majority of variables (7 variables) asymmetry of coefficient indicates that their arithmetic mean values tend toward higher results (epikurtic) and applied results tend toward lower

values. Variation coefficient indicates that pupils were at the beginning and at the end of the experiment a heterogeneous group in the execution of motoric test of flexibility.

**Table 1.** Basic statistical parameters of motoric variables at the beginning

|        | N  | Min    | Max    | Mean     | Std. Dev | Skew  | Kurt   | Kv     |
|--------|----|--------|--------|----------|----------|-------|--------|--------|
| ATV    | 26 | 157.00 | 184.00 | 171.6692 | 8.86064  | -.186 | -1.019 | 5.164  |
| ADS    | 26 | 24.00  | 30.00  | 26.7308  | 1.59519  | .535  | .002   | 5.967  |
| ATT    | 26 | 44.50  | 93.00  | 63.7308  | 15.54042 | .696  | -.656  | 24.384 |
| AONL   | 26 | 20.50  | 30.00  | 24.9231  | 3.15497  | -.017 | -1.326 | 12.658 |
| AONK   | 26 | 44.50  | 63.00  | 52.2692  | 5.88087  | .625  | -.468  | 11.251 |
| AOPK   | 26 | 30.00  | 42.00  | 35.3077  | 3.54140  | .474  | -.674  | 10.030 |
| MESDM  | 26 | 160.00 | 230.00 | 195.8462 | 22.08654 | -.024 | -.920  | 11.277 |
| MTR30V | 26 | 4.84   | 5.75   | 5.2562   | .26295   | .353  | -.376  | 5.002  |
| MFLPRK | 26 | 10.00  | 71.00  | 38.0769  | 18.92918 | .104  | -1.218 | 49.713 |
| MSKLEK | 26 | 9.00   | 40.00  | 21.9231  | 11.28866 | .682  | -1.222 | 51.492 |

**Table 2.** Basic statistical parameters of motoric variables at the end

|        | N  | Min    | Max    | Mean     | Std. Dev | Skew  | Kurt   | KV     |
|--------|----|--------|--------|----------|----------|-------|--------|--------|
| ATV    | 26 | 161.00 | 188.20 | 175.3231 | 8.93928  | -.200 | -1.194 | 5.098  |
| ADS    | 26 | 24.00  | 31.00  | 27.1923  | 1.87124  | .516  | -.106  | 6.881  |
| ATT    | 26 | 47.50  | 99.00  | 67.6923  | 15.03268 | .668  | -.250  | 22.207 |
| AONL   | 26 | 21.00  | 32.00  | 25.8846  | 3.01432  | .149  | -.314  | 11.645 |
| AONK   | 26 | 44.50  | 63.00  | 53.1538  | 5.34933  | .321  | -.009  | 10.060 |
| AOPK   | 26 | 30.00  | 43.00  | 36.7308  | 3.30221  | .070  | .157   | 8.990  |
| MESDM  | 26 | 171.00 | 232.00 | 203.4615 | 19.96643 | -.024 | -1.364 | 9.813  |
| MTR30V | 26 | 4.56   | 5.70   | 5.0992   | .30354   | .336  | -.115  | 5.952  |
| MFLPRK | 26 | 15.00  | 75.00  | 53.6154  | 20.03412 | -.770 | -.714  | 37.366 |
| MSKLEK | 26 | 10.00  | 49.00  | 28.3846  | 12.73445 | .236  | -1.147 | 44.863 |

Third table contains all data related to T-test procedure for dependent sample. For each variable results are indicated primarily based on assumption of the equality of group's variance and results, and second, the inequality of the variance is not presented due to the size of the table.

**Table 3.** Test of arithmetic mean differences with dependent sample

|          | Mean      | Std. Deviation | Std. Error Mean | Lower    | Upper   | T       | Df | Sig. (2-tailed) |
|----------|-----------|----------------|-----------------|----------|---------|---------|----|-----------------|
| ATVIF    | -3.65385  | 1.67791        | .32907          | -4.331   | -2.976  | -11.104 | 25 | .000            |
| ADSIF    | -.46154   | .50839         | .09970          | -.666    | -.256   | -4.629  | 25 | .000            |
| ATTIF    | -3.96154  | 3.10781        | .60949          | -5.2168  | -2.706  | -6.500  | 25 | .000            |
| AONLIF   | -.96154   | 1.00919        | .19792          | -1.36916 | -.55392 | -4.858  | 25 | .000            |
| AONKIF   | -.88462   | 2.07030        | .40602          | -1.72083 | -.04840 | -2.179  | 25 | .039            |
| AOPKIF   | -1.42308  | 1.32433        | .25972          | -1.95798 | -.88817 | -5.479  | 25 | .000            |
| MESDMIF  | -7.61538  | 8.78670        | 1.72321         | -11.164  | -4.066  | -4.419  | 25 | .000            |
| MTR30VIF | .15692    | .30409         | .05964          | .0340    | .279    | 2.631   | 25 | .014            |
| MFLPRKIF | -15.53846 | 10.94433       | 2.14636         | -19.958  | -11.117 | -7.239  | 25 | .000            |
| MSKLEKIF | -6.46154  | 9.01213        | 1.76742         | -10.101  | -2.821  | -3.656  | 25 | .001            |

Data analysis indicates that pupils tested at the end of 2013/2014 schools year aged 15 years have best results in all morphologic and motoric variables. Results gained indicate that physical activities and physical exercises applied three times per week influence strongly on development of morphologic and motoric parameters of pupils. Differences gained based on the analysis of T-test indicate significant statistical difference for all applied anthropometric and motoric variables for value of  $p < 0.05$ . The age of the sample represents a phase when pupils did not finalize their anthropometric and motoric aspects and this may be a reason for the gained difference of the applied variables. This is in compliance with the research of various authors where it is said that the time frame of child development is more intensive from the age of 13 to 16 (Medved et al., 1987.).

Compared to intensive physical development in this group age, physical activities should be increased in quality and quantity. It is well familiar that the physical activity has positive influence on the health condition of pupils and it represents important factor for health improvement. Therefore it is necessary to use diverse types of possibilities so that the bigger number of pupils can find their interest and voluntarily engage in some form of body activity (Rusch and Weineck 1998).

### Discussion

Results obtained based on the testing of differences of arithmetic means at the beginning and at the end of school year 2013/2014 in some anthropometric and motoric variables of pupils of 15 years of age, have indicated that the results at the end of school year have undergone positive changes and that these changes are manifested in a significant statistical difference in all variables applied, those anthropometric and motoric. However these changes can be attributed with a great certainty to additional hour of physical education having in mind the growth of pupils in this period of their life. Thus, it is certain that for obtaining a real picture of influence of additional

physical education hour on motoric and physical development in this group age, it is necessary to involve in the research control group compared to experimental group. Influence of physical activities on transformation and success at the end of the school year cannot be negated.

It was learned in this study that program applied and its content, selection and distribution of exercises for three times per week during school year result in a condition that was desired for anthropometric and motoric development of pupils in this age. In this way, with a certainty, we can conclude that increase of number of hours for physical education in schools of various levels in Kosova is necessary for proper and harmonized development of all anthropological aspect of youth.

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### INFLUENCE OF PHYSICAL BODY EXERCISES ON PUPILS

*In the education system of Kosova, frequent changes that have occurred in last 14 years, have demonstrated that the education system has tried to adjust to economic, social and cultural changes, etc. As these changes have not been studied well, they have influenced and are still influencing organization of physical education in schools as regards their quality and quantity. These changes have not provided desired results and are often regarded as experiments. Methods: When defining the time, this study has longitudinal empirical character and consists of two measures, the morphological and motoric indicators in the pupils of primary school and lower secondary school*



“Faik Konica”. This study involved 26 pupils in the age group of 15 years. 6 anthropometric and 4 motoric variables have been applied (Kurelić et al, 1975). Anthropometric variables included: body height (ATV), length of foot (ADS), body mass (ATT), volume of upper arm in down position (AONL), volume of thigh (AONK), volume of lower leg (AOPK). Motoric variables included: standing distance jump (MFESDM), 30 meters distance running (MTR30V), bench bending (MFLPRK) and push-ups (MSKLEK). For data analysis t-test for dependent sample has been used. Results: Basic statistical parameters of the obtained results before and after the application of additional class of physical education indicate that the results have normal distribution, have no visible asymmetry and have tendency toward higher values of distribution (epikurtic). T-test analysis for dependent variable demonstrates that the obtained differences within all variables in the groups are statistically significant. Discussion: Physical activities and body exercises during sport education classes of three times per week have visible influence in development of morphologic parameters and motoric skills of pupils. Differences obtained based on the t-test analysis demonstrate a strong statistical difference for all variables applied, anthropometric and motoric for the value of  $p < 0.05$ . It is well known that body activity has positive influence on health of pupils and represents significant factor for improvement of health. Therefore it is necessary to use different possibilities that sport offers so that the biggest number of pupils can find interest and motivation for voluntary involvement in some type of physical activity (Rusch and Weineck 1998).

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## ЦРНОГОРСКА СПОРТСКА АКАДЕМИЈА

### Признање Олимпијског покрета

Следећа, 12. Међународна научна конференција Црногорске спортске академије, која се традиционално одржава у првој недељи априла, већ је нашла место на званичној страници Олимпијског покрета. Тема конференције је „Трансформациони процеси у спорту“, а одржаће се под мотом „Спортска достигнућа“.

Тај догађај је сврстан међу 49 научних конференција у спорту, које је ова реномирана организација признала као најзначајније научне догађаје у предстојећих 12 месеци и објавила их на званичном сајту.

- Велико је признање за нашу научну конференцију. Препознат је квалитет рада и радова који се презентују, чак и ван европског континента. На последњој конференцији имали смо преко стотину стручњака и научника из европских земаља, али и ауторе радова из Азије - Јапана, Тајланда и Тајвана. Стандарди су поштрени, постављен је највиши научни праг за рецензију, па смо добили високи ранг квалитета - рекао је проф. др Душко Бјелица, председник научног одбора и оснивач Црногорске спортске академије.

**Б. Ђукановић**

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## **RELATION BETWEEN MOTORIC ABILITIES AND SPECIFIC MOTORIC ABILITIES WITH FOOTBALL PLAYERS FROM 14 TO 16 YEARS OLD**

### **Introduction**

Football is one of the most famous games that man has invented. It is a complex of collective organization and individual ingenuity. No doubt, top results in sport require a serious base of scientific researches and skills. Certainly, modern top results that have been achieved can be analyzed scientifically. It is exactly the scientific approach that is to establish the characteristics of basic importance to provide success in football. It is in favor of top football game to regularly conduct researches, which, as elements of special interest, would cover the basic motoric and specific motoric abilities with football players.

### **Methods**

The subject of the research is the motoric and specific motoric abilities with young football players aged from 14 to 16. The goal of the research is to establish how the explosive power influences the dribbling skills. The sample of respondents in the research consists of 39 school children at the age of 14 to 16, who play at FC "Gjilani" and have attended the club for at least two years. They underwent three tests for assessing the explosive power: 1. Standing long jump (SLJ); 2. 20-meter running (R20M) and 3. Sargent test (SARG). Along with it, two tests for assessing the specific motorics: 1. Dribbling 20 metres with start from the place (D20P) and 2. Dribbling 20 metres - fly start (D20F).

### **Results and Discussion**

The data obtained from the applied five tests is worked out with basic statistic parameters. The relation between motoric and specific motoric abilities is determined through linear regressive analyzes in manifest space.

According to the results presented in table 1, it is evident that the achieved results anticipated.

Normal distribution is recognized for all applied tests. It suggests a good selection of the tests. They fit properly with the tested football players.

**Table 1.** Basic descriptive parameters of the used tests

| Tests | Mean   | Std.Dev | Min    | Max    | Skew | Kurt | K-S  |
|-------|--------|---------|--------|--------|------|------|------|
| SLJ   | 181.94 | 18.82   | 153.00 | 224.00 | .40  | -.34 | .987 |
| R20M  | 4.14   | .32     | 3.45   | 4.90   | .37  | -.13 | .329 |
| SARG  | 15.29  | 9.31    | 3.00   | 40.00  | .93  | .08  | .063 |
| D20P  | 4.63   | .38     | 3.80   | 5.60   | .42  | .29  | .437 |
| D20F  | 3.89   | .40     | 3.10   | 5.00   | .29  | .28  | .243 |

Table 2 represents the results from the regressive analysis. We can conclude that the tests of explosive power, as a system of prediction, have a statistically significant effect on performing test 1 - Dribbling 20 metres with start from the place. They provide 66% of prediction.

**Table 2.** Regressive analysis of the test Dribbling 20 meters with start from the place

|                | <b>B</b>              | <b>Std. Error</b> | <b>Beta</b> | <b>t</b> | <b>Sig.</b> |
|----------------|-----------------------|-------------------|-------------|----------|-------------|
| SLJ            | .06                   | .04               | .43         | 1.48     | .15         |
| R20M           | -.04                  | .03               | -.46        | -1.59    | .12         |
| SARG           | -.14                  | .05               | -.75        | -2.48    | .02         |
| <b>R = .81</b> | <b>R square = .66</b> | <b>Sig = .00</b>  |             |          |             |

Table 3 suggests the conclusion that the tests have a significant influence on performing test 2 - Dribbling 20 metres - fly start as a prediction test.

The coefficient of multiple correlation is  $R=0.83$ , while the coefficient of determination is 0.69.

On the base of the obtained results, we can conclude that the selected tests of explosive power present a significant effect in successful achievement of specific motoric tests with football players.

**Table 3.** Regressive analysis of the test Dribbling 20 meters - fly start

|                | <b>B</b>              | <b>Std. Error</b> | <b>Beta</b> | <b>t</b> | <b>Sig.</b> |
|----------------|-----------------------|-------------------|-------------|----------|-------------|
| SLJ            | .02                   | .02               | .40         | 1.49     | .15         |
| R20M           | -.16                  | .05               | -.81        | -2.83    | .01         |
| SARG           | .06                   | .05               | .59         | 1.32     | .20         |
| <b>R = .83</b> | <b>R square = .69</b> | <b>Sig = .00</b>  |             |          |             |

The results suggest that there is a positive and significant relation between the applied motoric and specific motoric abilities. This confirms the close relation between the explosive power and specific motorics of football players expressed through dribbling.

Similar results are obtained in the researches of the following authors: Christou et al., 2006, Kraemer et al., 1998, and Young et al., 2001.

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### RELATION BETWEEN MOTORIC ABILITIES AND SPECIFIC MOTORIC ABILITIES WITH FOOTBALL PLAYERS FROM 14 TO 16 YEARS OLD

*Football is one of the most famous games that man has invented. It is a complex of collective organization and individual ingenuity. No doubt, top results in sport require a serious base of scientific researches and skills. Certainly, modern top results that have been achieved can be analyzed scientifically. It is exactly the scientific approach that is to establish the characteristics of basic importance to provide success in football. It is in favor of top football game to regularly conduct researches, which, as elements of special interest, would cover the basic motoric and specific motoric abilities with football players. The subject of the research is the motoric and specific motoric abilities with young football players aged from 14 to 16. The goal of the research is to establish how the explosive power influences the dribbling skills. The sample of respondents in the research consists of 39 school children at the age of 14 to 16, who play at FC "Gjilani" and have attended the club for at least two years. They underwent three tests for assessing the explosive power: 1. Standing long jump; 2. 20-meter running and 3. Sargent test. Along with it, two tests for assessing the specific motorics: 1. Dribbling 20 metres with start from the place and 2. Dribbling 20 metres - fly start. The data obtained from the applied five tests is worked out with basic statistic parameters. The relation between motoric and specific motoric abilities is determined through linear regressive analyzes in manifest space. The results suggest that there is a positive and significant relation between the applied motoric and specific motoric abilities. This confirms the close relation between the explosive power and specific motorics of football players expressed through dribbling. Similar results are obtained in the researches of the following authors: Christou et al., 2006, Kraemer et al., 1998, and Young et al., 2001.*

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## **RELATION BETWEEN LATENT SPECIFIC MOTOR ABILITIES AND SITUATION MOTOR SKILLS WITH VOLLEYBALL PLAYERS AGED FROM 16 TO 17**

### **Introduction**

The game of volleyball with its dynamic character is present in the world of the sport with permanent development and growing popularity and fans. Volleyball is part of a pollystructural complex sports activities. It is performed on a ground of a relatively small size (18 x 9 meters) and is a kind of game that requires of players a high level of advanced motoric abilities (speed, strength, endurance, a fast rate of visual reaction, explosivity), as well as specific motoric skills (precision etc.).

### **Methods**

Scientific conclusion as well as the growing number of conducted researches in the very game, have a real contribution to its modern development and level of popularity.

Situation-motoric skills make a significant dimension in the structure of volleyball game.

The subject of the research is specific-motoric abilities and situation-motoric skills of 52 volleyball players aged from 16 to 17.

The basic goal of the research is to establish the effect of specific-motoric abilities on situation-motoric skills of volleyball players in latent space. In order to assess the specific-motoric abilities 9 tests are used, and to assess the situation-motoric skills 3 precision tests are used.

The variables of specific motoric abilities are:

1. Hand tapping (HT);
2. Double hand tapping (DHT);
3. Leg tapping on the wall (LTW);
4. 20-meter sprint (S20M);
5. Japan – test (JT);
6. “T” – test (TT);
7. Sitting down on a ball (SDB);
8. 9-3-6-3-9 meter sprint (S936); and
9. 9-meter backward sprint (S9B)

The variables of situation-motoric skills are:

1. Passes the ball with fingers to zone 4 (PBZ4);
2. Passes the ball with fingers to zone 2 (PBZ2);
3. Service precision (SP).

### Results & Discussion

The results obtained from the 12 applied tests are worked out through the basic statistic parameters. Through component factor analysis 3 latent specific-motoric dimensions are isolated as well as one situation-motoric dimension.

By regressive analysis there is established a low but statistically significant relation between the criterion and predictor latent dimensions.

That confirms the dependence and relation between the specific-motoric abilities and situation-motoric skills.

Table 1 present the basic descriptive statistic parameters of variables of assessing specific-motoric abilities and situation-motoric skills. In addition, they show that the applied variables are normally distributed.

Observing the coefficient of variability (CV%), it is clear that there is different level of variation in achieved results

Aiming to determine the effect that latent specific-motoric factors have on the variables of assessing situation-motoric skills, first there were calculated factor scores of each respondent within the obtained latent factors, and later the linear regressive analysis was used, when latent specific-motoric factors were treated as predictors, and single variables of assessing situation-motoric skills were used as criteria.

*Table 1. Basic descriptive parameters*

| Variables | N  | X     | SD   | CV%   | Min   | Max   | K-S |
|-----------|----|-------|------|-------|-------|-------|-----|
| HT        | 52 | 34.94 | 3.65 | 10.45 | 25.00 | 43.00 | .27 |
| DHT       | 52 | 18.12 | 2.58 | 11.24 | 11.00 | 26.00 | .08 |
| LTW       | 52 | 23.01 | 2.28 | 9.91  | 14.00 | 29.00 | .00 |
| S20M      | 52 | 3.97  | .29  | 7.30  | 3.31  | 5.03  | .15 |
| JT        | 52 | 10.31 | .91  | 8.83  | 8.34  | 12.85 | .43 |
| TT        | 52 | 14.31 | 1.18 | 8.25  | 11.28 | 17.28 | .47 |
| SDB       | 52 | 23.90 | 3.60 | 15.06 | 9.00  | 33.00 | .10 |
| S936      | 52 | 20.13 | 6.93 | 34.43 | 4.00  | 40.00 | .69 |
| S9B       | 52 | 3.10  | .36  | 11.61 | 2.35  | 4.20  | .28 |
| PBZ4      | 52 | 38.84 | 8.10 | 20.85 | 13.00 | 50.00 | .01 |
| PBZ2      | 52 | 23.26 | 8.98 | 38.61 | 5.00  | 49.00 | .57 |
| SP        | 52 | 18.65 | 4.52 | 24.24 | 3.00  | 30.00 | .13 |

Analysing the results in table 2, which presents regressive analysis, it is obvious that between the predictor system and the criterion there is a statistically significant relation on level of  $Q(F)=0.00$  ( $R=0,356$ ). That suggests the specific-motoric latent factors can be explained 11% of variance of criterion ( $\Delta = 0,11$ ). The

unexplained 89% can be ascribed to other characteristics and skill of our respondents, which is not a case of interest of our study.

Following the partial regressive (Beta) coefficients of the predictor system, we can notice that only the first latent factor has a single statistically significant contribution in explaining criterion (DELTA=0,11) on level of Q (B)=0,00.

**Table 2.** Regression analysis

| Variables | r    | Part-R     | Beta  | t         | Q(B) |
|-----------|------|------------|-------|-----------|------|
| FACTOR_1  | .333 | .265       | .281  | 3.508     | .001 |
| FACTOR_2  | .211 | .141       | .137  | 1.824     | .070 |
| FACTOR_3  | .048 | -.009      | -.008 | -.109     | .913 |
| R=.365    |      | DELTA=.112 |       | Q(F)=.000 |      |

Researches in the field of similar questions have been conducted by the following authors: Jurko et al., 2013 and Nešić, et al., 2011.

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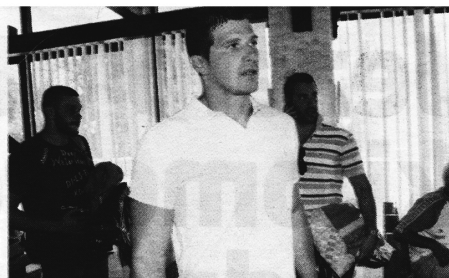
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### RELATION BETWEEN LATENT SPECIFIC MOTOR ABILITIES AND SITUATION MOTOR SKILLS WITH VOLLEYBALL PLAYERS AGED FROM 16 TO 17

*The game of volleyball with its dynamic character is present in the world of the sport with permanent development and growing popularity and fans. Volleyball is part of a pollystructural complex sports activities. It is performed on a ground of a relatively small size (18 x 9 meters) and is a kind of game that requires of players a high level of advanced motoric abilities (speed, strength, endurance, a fast rate of visual reaction, explosivity), as well as specific motoric skills (precision etc.). Scientific conclusion as well as the growing number of conducted researches in the very game, have a real contribution to its modern development and level of popularity. Situation-motoric skills make a significant dimension in the structure of volleyball game. The subject of the research is specific-motoric abilities and situation-motoric skills of 52 volleyball players aged from 16 to 17. The basic goal of the research is to establish the effect of specific-motoric abilities on situation-motoric skills of volleyball players in latent space. In order to assess the specific-motoric abilities 9 tests are used, and to assess the situation-motoric skills 3 precision tests are used. The results obtained from the 12 applied tests*

are worked out through the basic statistic parameters. Through component factor analysis 3 latent specific-motoric dimensions are isolated as well as one situation-motoric dimension. By regressive analysis there is established a low but statistically significant relation between the criterion and predictor latent dimensions. That confirms the dependence and relation between the specific-motoric abilities and situation-motoric skills. Researches in the field of similar questions have been conducted by the following authors: Jurko et al., 2013 and Nešić, et al., 2011.

„Vijesti“, 11. mart 2015.



Nikola Janović

Foto: B. PEJOVIĆ

NAUKA: U PODGORICI, 2 - 5. APRIL

## Radovi vodećih svjetskih autora

**P**odgorica će od 2. do 5. aprila biti domaćin 12. međunarodne naučne konferencije o transformacionim procesima u sportu „Sportska dostignuća“, a organizatori su Crnogorska sportska akademija i Fakultet za sport i fizičko vaspitanje, pod pokroviteljstvom Univerziteta Crne Gore.

Konferencija će okupiti preko 250 učesnika iz 27 zemalja, a svoje radove predstaviće vodeći svjetski autori i istraživači u oblasti sportskih, društvenih i humanističkih nauka i medicine.

U susret tom skupu, formiran je i Počasni odbor, kojim će predsjedavati premijer Crne Gore **Milo Đukanović**. Članovi su **Radmila Vojvodić** (rektorka UCG-a), ministarka nauke **Sanja**

**Vlahović**, gradonačelnik Podgorice **Slavoljub Stjepović**, **Dušan Simonović** (predsjednik COK-a), direktor Uprave za mlade i sport **Igor Vušurović**, predsjednik RSCG **Predrag Bošković**, predsjednik ASCG **Milorad Vuletić**, predsjednica SSCG **Vesna Medenica**, predsjednik VPSCG **Srdan Kovačević**...

Tu su i treneri **Dragan Adžić**, **Petar Porobić**, **Miodrag Božović**, rukometašica **Katarina Bulatović**, vaterpolista **Nikola Janović**,

džudista **Srdan Mrvaljević**, sportski direktor ŽRK Budućnost **Bojana Popović**, profesor Fakulteta sporta i fizičkog vaspitanja iz Novog Sada **Zoran Milošević** i profesor Fakulteta sporta i tjelesnog odgoja iz Sarajeva i potpredsjednik OKBiH **Izet Rado. D. M.**

**Konferencija  
će okupiti  
preko 250  
učesnika iz 27  
zemalja**



UDC 796-057.875

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The subject of physical education (kinesiology) at primary schools in Kosovo is present in school programs with three classes weekly.

In order to form a realistic idea of what students feel about their teacher in Physical Education as well as of the implementation of the educational program, we conducted a questionnaire survey.

**Methods**

The survey examines 210 respondents (118 school boys and 92 school girls) at the age of 14. It was conducted mainly to establish and compare the attitudes of boys and girls on the base of their answers to the following six questions:

1. Is the teacher in Physical Education always dressed in sports equipment in class?
2. Is the teacher always present and active in class performances?
3. Does the teacher form realistic marks to your performance?
4. Does the school possess the necessary facilities and equipment for the realization of the class program?
5. Do you regularly wear sports equipment in your Physical Education classes?
6. Are you happy with the way your Physical Education classes are conducted?

**Results & Discussion**

According to the answers to each question the groups of boys and girls are considered individually through methods of frequency and percentage of results. The differences between boys' and girls' attitudes are tested by non-parametric chi-square test, at the level of 0,05(5%).

On the first question - Is the teacher in Physical Education always dressed in sports equipment in class? (table 1) with "YES" answered 115 male students, and "NO" 3, while the female students with "YES" answered 90, and "NO" 2. The established difference in responses between male students and female students is not statistically significant different (chi-square=0,030 and p=0,862).

*Table 1. Chi-square test on the first question*

| Question 1 | YES | NO | Total | Chi-square | p     |
|------------|-----|----|-------|------------|-------|
| Male       | 115 | 3  | 118   | 0.030      | 0.862 |
| Female     | 90  | 2  | 92    |            |       |
| Total      | 205 | 5  | 210   |            |       |

On the second question - Is the teacher always present and active in class performances? (table 2) with "YES" answered 109 male students, and "NO" 9, while the female students with "YES" answered 80, and "NO" 12. The established difference in responses between male students and female students is not statistically significant different (chi-square=1,685 and  $p=0,194$ ).

**Table 2.** Chi-square test on the second question

| Question 2 | YES | NO | Total | Chi-square | p     |
|------------|-----|----|-------|------------|-------|
| Male       | 109 | 9  | 118   | 1.685      | 0.194 |
| Female     | 80  | 12 | 92    |            |       |
| Total      | 189 | 21 | 210   |            |       |

On the third question - Does the teacher form realistic marks to your performance? (table 3) with "YES" answered 100 male students, and "NO" 18, while the female students with "YES" answered 84, and "NO" 8. The established difference in responses between male students and female students is not statistically significant different (chi-square=2,050 and  $p=0,152$ ).

**Table 3.** Chi-square test on the third question

| Question 3 | YES | NO | Total | Chi-square | p     |
|------------|-----|----|-------|------------|-------|
| Male       | 100 | 18 | 118   | 2.050      | 0.152 |
| Female     | 84  | 8  | 92    |            |       |
| Total      | 184 | 26 | 210   |            |       |

On the fourth question - Does the school possess the necessary facilities and equipment for the realization of the class program? (table 4) with "YES" answered 113 male students, and "NO" 5, while the female students with "YES" answered 86, and "NO" 6. The established difference in responses between male students and female students is not statistically significant different (chi-square=0,544 and  $p=0,461$ ).

**Table 4.** Chi-square test on the fourth question

| Question 4 | YES | NO | Total | Chi-square | p     |
|------------|-----|----|-------|------------|-------|
| Male       | 113 | 5  | 118   | 0.544      | 0.461 |
| Female     | 86  | 6  | 92    |            |       |
| Total      | 199 | 11 | 210   |            |       |

On the fifth question - Do you regularly wear sports equipment in your Physical Education classes? (table 5) with "YES" answered 104 male students, and "NO" 14, while the female students with "YES" answered 56, and "NO" 36. The established difference in responses between male students and female students is statistically significant different (chi-square=21,186 and  $p=0,000$ ).

**Table 5.** Chi-square test on the fifth question

| Question 5 | YES | NO | Total | Chi-square | p     |
|------------|-----|----|-------|------------|-------|
| Male       | 104 | 14 | 118   | 21.186     | 0.000 |
| Female     | 56  | 36 | 92    |            |       |
| Total      | 160 | 50 | 210   |            |       |

On the sixth question - Are you happy with the way your Physical Education classes are conducted? (table 6) with "YES" answered 108 male students, and "NO" 10, while the female students with "YES" answered 64, and "NO" 28. The established difference in responses between male students and female students is statistically significant different (chi-square=16,821 and p=0,000).

**Table 6.** Chi-square test on the sixth question

| Question 6 | YES | NO | Total | Chi-square | p     |
|------------|-----|----|-------|------------|-------|
| Male       | 108 | 10 | 118   | 16.821     | 0.000 |
| Female     | 64  | 28 | 92    |            |       |
| Total      | 172 | 38 | 210   |            |       |

According to the final results, the conclusion is that school boys and girls have similar attitudes (do not differ) in their answers to questions 5 and 6.

To some extent, the results obtained from the survey show similarity with the research conducted by Memedi, Nika and Hajredini (2011).

### Conclusions

According to the realized survey that consisted of sixth questions using 210 students (118 male and 92 female), the following conclusions are possible:

1. In their opinions the male and female students do not differ based on the answers obtained on the first, second, third and fourth question;
2. A significant difference in terms of their opinions according to the gained answers is established on the fifth and sixth question.

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## SOME STUDENTS' ATTITUDES TOWARDS PHYSICAL EDUCATION

The subject of physical education (kinesiology) at primary schools in Kosovo is present in school programs with three classes weekly. In order to form a realistic idea of what students feel about their teacher in Physical Education as well as of the implementation of the educational program, we conducted a questionnaire survey. Methods: The survey examines 210 respondents (118 school boys and 92 school girls) at the age of 14. It was conducted mainly to establish and compare the attitudes of boys and girls on the base of their answers to the following six questions: 1. Is the teacher in Physical Education always dressed in sports equipment in class? 2. Is the teacher always present and active in class performances? 3. Does the teacher form realistic marks to your performance? 4. Does the school possess the necessary facilities and equipment for the realization of the class program? 5. Do you regularly wear sports equipment in your Physical Education classes? 6. Are you happy with the way your PhE classes are conducted? Results & Discussion: According to the answers to each question the groups of boys and girls are considered individually through methods of frequency and percentage of results. The differences between boys' and girls' attitudes are tested by non-parametric chi-square test, at the level of 0,05(5%). According to the final results, the conclusion is that school boys and girls have similar attitudes (do not differ) in their answers to questions 5 and 6. To some extent, the results obtained from the survey show similarity with the research conducted by Memedi, Nika and Hajredini (2011).

„Dan“, 9. mart 2015.

ПУБЛИКОВАН НОВИ БРОЈ СТРУЧНОГ ЧАСОПИСА  
„MONTENEGRIN JOURNAL OF SPORTS SCIENCE AND MEDICINE“

## У чак 30 индексних база

Ових дана из штампе је изашао најновији број међународног часописа „**Montenegrin Journal of Sports Science and Medicine**“ који издаје Црногорска спортска академија из Подгорице. Први број овог часописа, који излази два пута годишње и који се налази у чак 30 индексних међународних база објављен је у септембру 2012. године. У овом, шестом по реду, чији је тираж 1500 примјерака објављено је пет научних радова, чији су аутори из Сједињених Држава, Литваније, Хрватске, Турске, Португала, Енглеске, Грчке, Бугарске, Србије и Црне Горе. Часопис је објављен у штампаном и он-лајн издању. Т.Б.



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## UTICAJ NEKIH ANTROPOMETRIJSKIH KARAKTERISTIKA I MOTORIČKIH SPOSOBNOSTI NA AGILNOST MLADIH ODOJKAŠICA

### UVOD

Pošto u naukama, koje se bave ljudskim bićem, interdisciplinarni pristup izučavanja ličnosti predstavlja osnovnu metodološku orijentaciju, predmet nauke i u oblasti sporta je antropološki status sportista. Pod antropološkim statusom podrazumevaju se sledeće čovekove sposobnosti i karakteristike: morfološke karakteristike, funkcionalne sposobnosti, motoričke sposobnosti, specifično-motoričke karakteristike, kognitivne sposobnosti, konativne karakteristike, sociološke karakteristike (Dautbašić i Bradić, 2005).

Kako postoje određene specifičnosti pojedinih sportova i sportskih disciplina, koje se ogledaju u različitosti njihove takmičarske strukture, samim tim postoji i izrazita potreba za neprekidnim istraživanjem i u praksi proveravanjem specifičnosti pojedinih sportova, uključujući prvenstveno genetičku uslovljenost (ograničenost) pojedinih antropoloških sposobnosti i karakteristika, zatim njihovu hijerarhijsku vrednost po sportovima, kao i njihovu strukturu i razvoj pod uticajem određenih trenažnih sredstava, metoda i opterećenja (Malacko i Rado, 2004).

Uspeh u odbojci svakako zavisi i od morfoloških karakteristika formiranog odbojkaša/ice, od kojih su osnovne telesna visina i masa, a koje se valorizuju s obzirom na trenutni uzrast odbojkaša/ica (Marelić i saradnici, 2008). Kako u svakoj sportskoj aktivnosti, tako i u odbojci, nijedan tehnički element se ne može izvesti bez adekvatne motoričke sposobnosti i u punoj meri ispoljiti bez racionalne tehnike izvođenja kretanja.

Cilj ovog istraživanja je da se utvrdi da li postoji statistički značajna povezanost između određenih antropometrijskih karakteristika i motoričkih sposobnosti u odnosu na agilnost kao izolovanu motoričku sposobnost koja je predmet ovog istraživanja.

### METODE

Na uzorku od 16 selektovanih devojčica starosti od 14 do 16 godina (kadetski uzrast) učesnica kampa u Bečeju 2006. godine, izvršeno je merenje antropometrijskih mera za procenu morfoloških karakteristika (telesna masa i telesna visina), 3 testa za procenu motoričkih sposobnosti (taping rukom, pretklon na klupi i skok udalj iz mesta) i 2 testa za procenu agilnosti („Japan test“ i „Jelka test“). Antropometrijske mere i motorički testovi su korišćeni kao prediktor a testovi agilnosti kao kriterijum. Uticaj nekih antropometrijskih mera i motoričkih testova na agilnost mladih odbojkašica utvrđena je primenom regresione analize.

**REZULTATI**

Na osnovu prikazanih osnovnih deskriptivnih statistika za analizirane mere i testove (Tabela 1.), zapaža se nešto povećana i pozitivna asimetrija distribucije rezultata kod testova Pretklona na klupi, Japan test i Jelka test. Radi se o većem grupisanju rezultata u zoni nižih vrednosti, dok se kod testa Pretklon na klupi zapaža i povećana disperzija rezultata.

**Tabela 1.** Deskriptivni statistici antropometrijskih mera i motoričkih testova mladih odbojkašica

| Mere i testovi           | AS     | S     | Min    | Max    | Skew | Kurt  |
|--------------------------|--------|-------|--------|--------|------|-------|
| Telesna visina (cm)      | 176,31 | 4,85  | 165,50 | 184,50 | -,38 | ,34   |
| Telesna masa (kg)        | 62,56  | 7,99  | 50,00  | 79,00  | ,86  | ,39   |
| Taping rukom (frek)      | 43,94  | 3,99  | 38     | 51     | ,10  | -,83  |
| Pretklon na klupici (cm) | 53,34  | 6,59  | 44,0   | 72,0   | 1,28 | 3,60  |
| Skok udalj iz mesta (cm) | 210,75 | 15,87 | 189    | 240    | ,29  | -1,02 |
| Jelka test (0.01s)       | 26,34  | 1,40  | 24,66  | 30,04  | 1,14 | 2,16  |
| Japan test (0.01s)       | 14,96  | 1,141 | 13,67  | 17,49  | 1,21 | 1,00  |

AS – aritmetička sredina; S – standardna devijacija; Min. – minimalna vrednost; Max. – maksimalna vrednost; Skew. – skjunis, mera asimetrije distribucije; Kurt. – kurtosis, mera homogenosti distribucije

Na osnovu analize vrednosti koeficijenta multiple korelacije za test “Japan test”, kao i visine izračunate statističke značajnosti (Tabela 2.), može se zaključiti da sistem primenjenih prediktora nije pokazao statistički značajnu povezanost sa testom “Japan test”. Primenjeni sistem prediktora objašnjava 55% varijanse kriterija. Određen je i pojedinačni uticaj antropometrijskih mera i motoričkih testova na rezultate u testu “Japan test”. Statistički značajnu povezanost pokazali su testovi Skok udalj iz mesta i Taping rukom, koji pokazuju i dosta visoku parcijalnu povezanost sa kriterijskim testom “Japan test”.

**Tabela 2.** Uticaj prediktorskog sistema na varijablu Japan test

| Testovi i mere       | Beta  | t      | p    | R <sub>p</sub> |
|----------------------|-------|--------|------|----------------|
| Skok u dalj iz mesta | -,551 | -2,392 | ,038 | -,603          |
| Pretklon na klupici  | -,239 | -1,067 | ,311 | -,320          |
| Taping rukom         | -,627 | -2,623 | ,025 | -,638          |
| Telesna masa         | ,159  | ,503   | ,626 | ,157           |
| Telesna visina       | -,005 | -,015  | ,988 | -,005          |

R = 0,74    R<sup>2</sup> = 0,55    p = 0,11

Beta – standardizovani regresioni koeficijenti t – t test; p – statistička značajnost; R<sub>p</sub> – koeficijent parcijalne korelacije; R – koeficijent multiple korelacije; R<sup>2</sup> – koeficijent determinacije

Na osnovu vrednosti koeficijenta multiple korelacije za test “Jelka test” i koji je statistički značajan, može se zaključiti da sistem prediktora pokazuje statistički značajnu povezanost sa testom “Jelka test” uz 5% greške u zaključivanju (Tabela 3.). Prime-

njeni sistem prediktora objašnjava 64% varijanse kriterija. Regresionom analizom je utvrđen i pojedinačni uticaj antropometrijskih mera i motoričkih testova na ovaj test. Statistički značajnu povezanost pokazali su testovi Skok udalj iz mesta i Taping rukom, koji imaju i dosta visoke koeficijente parcijalne korelacije sa kriterijskim testom “Jelka test”.

**Tabela 3.** Uticaj prediktorskog sistema na varijablu Jelka test

| Testovi i mere       | Beta  | T      | p           | R <sub>p</sub> |
|----------------------|-------|--------|-------------|----------------|
| Skok u dalj iz mesta | -,679 | -3,273 | <b>,008</b> | -,719          |
| Pretklon na klupici  | -,051 | -,252  | ,806        | -,079          |
| Taping rukom         | -,535 | -2,486 | <b>,032</b> | -,618          |
| Telesna masa         | -,134 | -,473  | ,647        | -,148          |
| Telesna visina       | ,388  | 1,477  | ,170        | ,423           |

R = 0,80    R<sup>2</sup> = 0,64    p = 0,04

## DISKUSIJA

Prilikom definicije agilnosti kao motoričke sposobnosti potrebno je uvažiti da ona nastaje kao posledica sadejstva skupa motoričkih sposobnosti, koje se aktuelizuju u brzom kretanju sa promenom smera, u brzim promenama smera kretanja ruku i nogu i uz kognitivne procese i tehniku kretanja (Verstegen i Marcello, 2001). Za uspešnost kretanja na startu, značajne su karakteristike ispoljavanja jačine (maksimalna jačina i brzina ispoljavanja jačine), za ubrzanje i usporenje kretanja značajna je snaga, a za promenu smera, kao i za tehnikukretanja značajna je koordinacija.

Uspešnost u zadacima agilnosti zavisi od različite participacije jačine i snage, kao motoričkih sposobnosti u promenjenim uslovima kretanja, odnosno zavisi od koordinacije kao motoričkog svojstva. S druge strane definicije, u kojim je agilnost posmatrana kao kompleksna sposobnost, sugerišu da se ona sastoji od: a) perceptualnih faktora i faktora odluke – uočavanja, predviđanja i procene situacije i b) brzine promene smera kretanja – tehnika kretanja, brzine, karakteristike mišića nogu i antropometrijske karakteristike (Young i saradnici, 2002). U ovoj determinaciji može se zapaziti da su za uspešno ispoljavanje agilnosti, pored perceptivnih faktora, značajne i karakteristike mišića nogu, odnosno, značajne su i morfološke karakteristike. Objašnjenje kretanja bilo koje vrste (u odnosu na intenzitet, trajanje, složenost, kao i u odnosu na otvorenost i zatvorenost uslova kretanja), nije moguće bez uvažavanja ukupnih karakteristika čoveka (kognitivnih, konativnih, morfoloških i motoričkih), odnosno, bez uzimanja u obzir njihovih međusobnih interakcija. Isto tako, objašnjenje agilnosti, kao kompleksnog motoričkog svojstva, podrazumeva uvažavanje i morfoloških i psiholoških karakteristika i motoričkih sposobnosti i veštine.

S obzirom da je agilnost kompleksna motorička sposobnost i da se ispoljava u uslovima različite participacije motoričkih sposobnosti u različitim uslovima kretanja, ona je, na određeni način, pokazatelj individualnih motoričkih karakteristika sportista. Zbog toga sportisti, sa sličnim kvantitavnim pokazateljima motoričkih karakteristika (jačina, snaga, brzina i dr), imaju različitu uspešnost u uslovima ispoljavanja agilnosti,

odnosno, sportisti sa sličnom uspešnošću u ispoljavanju agilnosti, imaju različite kvantitativne pokazatelje pojedinih motoričkih sposobnosti. Sportisti u cikličnim aktivnostima (npr. trčanja u atletici), čije su motoričke sposobnosti slične motoričkim sposobnostima sportista u acikličnim aktivnostima (npr: sportske igre), imaju različitu uspešnost u zadacima tipa agilnosti – uspešniji su sportisti iz sportskih igara. Takođe, postoje razlike između sportista koji se bave istom sportskom igrom, odnosno, koji igraju u istoj ekipi, ali na različitim pozicijama – za sve njih je karakteristično da, u više ili manje složenim aktivnostima, uspešnost ostvaruju na osnovu različite participacije motoričkih sposobnosti (jačine, snage, brzine, gipkosti, okretnosti, izdržljivosti). Razlike između sportista u okviru iste sportske igre i sličnosti između sportista u različitim aktivnostima, idu u prilog osnovanosti da je njihovu uspešnost moguće objasniti kao posledicu uticaja određenog morfortipa, motoričkog tipa, sportskog tipa, pobedničkog tipa (model šampiona) i dr. U celini posmatrano, na osnovu rezultata istraživanja različitih autora, agilnost podrazumeva kompleksno ispoljavanje motoričkih sposobnosti od kojih zavisi brzo i efikasno povezivanje samo ubrzanja i usporenja, odnosno, promena smera i ponovnog ubrzanja i usporenja uz stalnu kontrolu kretanja u vertikalnom, odnosno u horizontalnom smeru (Drabik, 1996; Plisk, 2000; Verstegen i Marcello, 2001). Ubrzanje i usporenje kretanja, u horizontalnom i vertikalnom smeru, podrazumeva ispoljavanje snage kao motoričkog svojstva, pa uspešnost u kretanjima koja zahtevaju agilnost zavisi od karakteristika ispoljavanja jačine i snage – sportisti koji za kraće vreme ispolje silu, odnosno, koji ispolje veću snagu, imaju očekivane preduslove neophodne za ispoljavanje agilnosti. Sem dobrih karakteristika ispoljavanja jačine i snage, za ispoljavanje agilnosti neophodno je da sportisti imaju i dobre karakteristike funkcija organizma od kojih zavisi upravljanje kretanjem. Sportisti koji ispoljavaju dobru agilnost verovatno će imati dobre karakteristike u strukturi koordinacionih sposobnosti kao što su dinamička ravnoteža, snalaženje u prostoru, osećaj za ritam kao i vizuelnu kontrolu (Ellis i saradnici, 2000). Očigledno je da uspešnost sportista, u zadacima koja zahtevaju agilnost kao motoričko svojstvo, zavisi od mogućnosti ukupnog komplementarnog sadejstva motoričkih sposobnosti, u prvom redu jačine, snage i koordinacije. Kvalitativna povezanost jačine, snage, brzine, okretnosti i gipkosti, ispoljenih u kretanjima maksimalnim intenzitetom sa promenom pravca i smera, uslovljava i ispoljavanje agilnosti kao motoričkog svojstva. To znači da su aktivnosti koje zahtevaju agilnost povezane sa motoričkim sposobnostima (jačinom, snagom, brzinom, okretnošću i gipkošću), odnosno sa modalitetima njihovog ispoljavanja (brzinom i ispoljavanja jačine, brzinskom snagom, tačnošću - u odnosu na prostor, vreme, jačinu i dr).

Sem sa aspekta motoričkih sposobnosti, ispoljavanje agilnosti treba objasniti i sa aspekta uticaja morfoloških karakteristika tela (visina, masa tela, sastav tela i dr), kao i sa aspekta uticaja psiholoških karakteristika, kao što su perceptivni faktori i faktori odluke, koji utiču na kretanje i dr. Morfološke karakteristike obuhvataju aktivne komponente (mišiće) i inercione komponente (masa tela, visina tela, masno tkivo i dr).

U kretanjima maksimalno mogućim intenzitetom sa promenom pravca i smera morfološke karakteristike se ponašaju kao inercioni faktor, odnosno, one otežavaju promenu smera i pravca kretanja. Promene pravca i smera kretanja povezane su sa



odgovorajućim promenama brzine, a promene brzine kretanja (ubrzanje i usporenje) povezane su sa ispoljavanjem snage kao motoričkog svojstva. Na osnovu navedenih činjenica nije moguće objasniti uticaje morfoloških karakteristika na uspešnost u kretanjima sa promenom pravca i smera, bez povezivanja uticaja snage (eksplozivna snaga u ovom istraživanju), kao motoričkog svojstva, na brzinu kretanja.

Na osnovu celokupne analize dobijenih rezultata došlo se do saznanja da eksplozivna snaga (skok udalj iz mesta) i brzina pojedinačnih pokreta (taping rukom) imaju visok nivo korelacije sa agilnošću kod ispitivanog uzorka odbojkašica, što je na osnovu dosadašnjih istraživanja (Vuković, 1989) bilo i očekivano. Karakter povezanosti sugeriše da su bolji pokazatelji u testovima snage povezani sa kraćim vremenom u testovima za procenu agilnosti, to jest odgovaraju boljim rezultatima u testovima za procenu agilnosti. Do sličnih rezultata došli su i drugi istraživači (Webb i Lander, 1983; Young, i saradnici 1996; Negrete i Brophy, 2000; Djevalikian, 1993; Young i saradnici, 2002). Gipkost nema statistički značajnu povezanost sa agilnošću kod ispitivanog uzorka. Eksplozivna snage omogućava objašnjavanje uspešnosti u kretanjima sa brzom promenom stanja, odnosno, sa brzom promenom brzine, pravca i smera kretanja. Morfološke karakteristike imaju niske korelacije sa testovima agilnosti. Visina tela i masa, kao karakteristike morfološkog razvoja, nemaju izraženu povezanost sa rezultatima u testovima agilnosti. Iako je dobijen slab intenzitet veza, njihov pozitivan karakter sugeriše da povišena telesna masa otežava ispoljavanje agilnosti. Otežavanje brzine kretanja u testovima agilnosti, dakle u kretanjima sa promenom smera, može se povezati sa ispoljavanjem inercionih faktora analogno stepenu promena smera i brzine kretanja.

Motoričke sposobnosti imaju pozitivan uticaj na rezultate u testovima agilnosti koje neposredno, preko snage i brzine, utiču pozitivno na ispoljavanje agilnosti, a morfološke karakteristike, kao inercioni faktor, imaju negativan uticaj na rezultate u testovima agilnosti. Morfološke karakteristike neposredno, kao inercioni faktor, otežavaju ispoljavanje agilnosti. Otežavanje ispoljavanja agilnosti posledica je uloge mase tela u uslovima ubrzanja, usporenja, kao i u uslovima promene smera kretanja.

Ovo se donekle može objasniti specifičnošću ispitivanog uzorka, njihove uzrasne kategorije tj. poznavanjem zakonitosti razvoja motoričkih sposobnosti u odnosu na uzrasni period – heterohronost razvoja (Krsmanović i Berković, 1999). Starosna dob članica kadetske reprezentacije Vojvodine je u rasponu od dve godine (14-16 godina) što takođe može da bude jedan od razloga ovako dobijenih rezultata.

Cilj ovog istraživanja je bio da se utvrdi da li postoji statistički značajna povezanost između određenih antropometrijskih karakteristika i motoričkih sposobnosti u odnosu na agilnost kao izolovanu motoričku sposobnost.

Odbojka, kao i svi ostali sportovi, zahteva određeni nivo morfoloških i motoričkih karakteristika, radi što boljeg i uspešnijeg delovanja u situacionim uslovima i ostvarivanje rezultata. S obzirom na specifičnosti odbojke logičan je značajan uticaj specifičnih motoričkih sposobnosti. Najveći broj stručnjaka iz odbojke pod specifičnim motoričkim sposobnostima podrazumeva sledeće: eksplozivna snaga i agilnost, gipkost, koordinacija tela, brzina alternativnih pokreta, koordinacija ruku i nogu. S obzirom da je odbojka sport praćen skokovima i brzim promenama pravca kretanja,

eksplozivna snaga i agilnost su posebno dominantne sposobnosti u ovom sportu. Razvojem eksplozivne snage može se poboljšati celokupan rezultat kome se teži. Agilnost se ispoljava u fazi tranzicije, koja je za razliku od ostalih sportskih igara u odbojci vrlo kratka i profinjena, te je iz tog razloga veoma važna u igri.

Vrhunske ekipe karakteriše visoko efikasna i brza tranzicija u igri brzi prelaz iz početne u finalnu poziciju, kao i promene pravca kretanja naročito nakon odbijene lopte od bloka ili od saigrača. Na osnovu ovog istraživanja došlo se do saznanja da se radom na razvoju eksplozivne snage i brzine pojedinačnih pokreta, može posredno uticati i na poboljšanje agilnosti, motoričke sposobnosti koja je značajna za uspešno nadmetanje u odbojkaškoj igri. Pod eksplozivnom snagom se podrazumeva ispoljavanje maksimalne snage za što kraće vreme.

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#### THE INFLUENCE OF SOME ANTHROPOMETRIC CHARACTERISTICS AND MOTOR ABILITIES ON AGILITY IN YOUNG FEMALE VOLLEYBALL PLAYERS

*The success in volleyball certainly depends on the morphological characteristics of the formed volleyball players, of which the basic body height and weight, which can be valorized in view of the current age of volleyball players (Marelič et al, 2008). As in all sports activities, as well as volleyball, no technical element can not be performed without adequate motor abilities and fully manifested without rational techniques of performing motion. The aim of this research is to determine whether there is a statistically significant correlation between certain anthropometric characteristics and motor abilities in relation to the agility of an isolated motor abilities that are the subject of this research. Methods: In a sample of 16 selected girls aged 14 to 16 years (cadet age) participating in the camp Becej 2006, were measured two anthropometric measures for evaluation of morphological characteristics, three tests for evaluation of motor abilities (both as predictor) and two tests for assessing agility (as criterion). The influence of some anthropometric characteristics and motor abilities on agility in young female volleyball players lities was performed by regression analysis. Results: It can be concluded that the system applied predictor variables showed no statistically significant association with variable Japan test, while the variable Jelka test a statistically significant correlation. Discussion: Based on the overall analysis of the obtained results it was discovered that the explosive power (long jump from the place) and the speed of individual movements (hand tapping) have a high level of correlation with agility were detected in female volleyball players, which is based on previous research (Vukovic, 1989) was and expected. Similar results were also other researchers (Webb and Lander, 1983; Negrete and Brophy, 2000).*

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## EFEKTI PLIOMETRIJSKOG TRENINGA NA MOTORIČKE SPOSOBNOSTI TENISERA

### UVOD

Tenis je polistrukturalna aktivnost sa acikličnim tipom kretanja. Kretanja se sastoje od ponovljenih startova, ubrzanja i usporavanja, te brzih promena pravca kretanja od kojih su ishod udarca i taktička komponenta u direktnoj zavisnosti. Iz navedenog razloga najvažnije i najviše zastupljene motoričke sposobnosti su upravo startna brzina i ubrzanje, te brza promena pravca kretanja. Veliki broj kretnih struktura i situacija u teniskoj igri (tehničkih i taktičkih varijanti) ukazuje da je uspešnost tenisera određena nivoom i strukturom velikog broja motoričkih sposobnosti, znanja i osobina od kojih se neke mogu izmeriti i analizirati. Merenje tih sposobnosti i osobina omogućuje kvalitetnije planiranje, programiranje i kontrolu trenažnog procesa, te poboljšanje sportske forme (Baumgartner & Jakson, 1995; Groppe, Loehr, Melville & Quinn, 1989; Šentija, 1991). Pliometrija je trenažni metod u kome se podrazumeva da se pre kontrakcije aktuelni mišić stavlja u stanje povećane istegnutosti koja služi za razvoj snažnih sposobnosti, i to pre svega elastične i eksplozivne snage te ima značajnu ulogu u sportovima gde se zahteva visok nivo brzine i snage, kao na primer u sprintu, skokovima, udaranju lopte, bacanju lopte itd. Pliometrija je metoda treninga koja koristi preopterećenje (Chu, 1983), a osnovna svrha pliometrijskog treninga jeste razvoj što veće reaktivne sile (Allerheiligen & Rogers, 1995). Sportski fiziolozi se slažu da pliometrija obuhvata specifične vežbe koje izazivaju značajno istezanje mišića koji se nalazi pod ekscentričnom kontrakcijom i nakon koje sledi snažna koncentrična kontrakcija koja služi za razvoj snažnog pokreta u kratkom vremenskom periodu (LaChance, 1995). Pliometrijski trening ima veliki uticaj na kondicionu pripremljenost tenisera i to prvenstveno u segmentima promena pravca kretanja, startnih ubrzanja, te udaraca, a samim tim i na njihov uspeh u teniskoj igri. Kod pliometrijskog treninga moramo imati na umu princip usmerenosti što znači da se sportski trening, a u ovom slučaju pliometrijski trening treba temeljiti na specifičnim relacijama između ciljeva koji se žele postići i programa treninga koji treba osigurati postizanje željenih ciljeva. Zbog toga je bitno vrlo dobro poznavati mehanizme delovanja pliometrije i mesto koje zauzima u kondicionom treningu.

Cilj istraživanja je da se utvrde i analiziraju transformacioni efekti pliometrijskog programa vežbanja na manifestne dimenzije snage, agilnosti i brzine trčanja eksperimentalne grupe tenisera.

## METODE

Uzorak ispitanika činilo je 50 tenisera uzrasta 17.5 godina ( $\pm 6$  meseci), koji su bili podeljeni u dve grupe. Jednu grupu je činilo 25 tenisera TK „Gemaks“ iz Beograda koji su predstavljali eksperimentalnu grupu, dok je drugu grupu činilo 25 tenisera TK „AS“ iz Beograda koji su predstavljali kontrolnu grupu. Za procenu motoričkih sposobnosti korištena su 4 motorička testa koji su standardni testovi za procenu eksplozivne snage nogu (troskok iz mesta i skok udalj iz mesta), agilnosti („Lepeza“) i brzine trčanja (trčanje 20 m iz visokog starta). Radi utvrđivanja razlika između grupa ispitanika u inicijalnom i finalnom merenju bila je primenjena univarijatna analiza varijanse (ANOVA). Radi utvrđivanja efekta trenažnog programa između dva testiranja primenjena je univarijatna analiza kovarijanse (ANKOVA).

## REZULTATI

Rezultati deskriptivnih statistika, aritmetičke sredine i standardne devijacije ukazuju na homogenost ispitanika eksperimentalne i kontrolne grupe tenisera u sva četiri primenjena motorička testa na inicijalnom i finalnom merenju (Tabela 1.). Oni su bili na sličnom nivou razvoja eksplozivne snage nogu, agilnosti i brzine trčanja u inicijalnom merenju, kao i na kraju procesa njihovog praćenja nakon tri meseca (na finalnom merenju) posmatrajući ih u okviru svojih grupa.

**Tabela 1.** Osnovni deskriptivni statistici motoričkih testova inicijalnog i finalnog merenja

| Merenje    | Test                              | Grupa           | AS     | S     | MIN   | MAKS         |
|------------|-----------------------------------|-----------------|--------|-------|-------|--------------|
| Inicijalno | Troskok iz mesta (cm)             | Eksperimentalna | 495,68 | 64,91 | 410   | <b>690</b>   |
|            |                                   | Kontrolna       | 483,48 | 42,60 | 415   | <b>566</b>   |
|            | Skok udalj iz mesta (cm)          | Eksperimentalna | 201,96 | 13,22 | 185   | <b>234</b>   |
|            |                                   | Kontrolna       | 202,28 | 14,13 | 179   | <b>235</b>   |
|            | Trčanje 20m iz visokog starta (s) | Eksperimentalna | 3,60   | 0,26  | 3,20  | <b>3,99</b>  |
|            |                                   | Kontrolna       | 3,72   | 0,23  | 3,20  | <b>4,11</b>  |
|            | Lepeza (s)                        | Eksperimentalna | 19,50  | 1,84  | 16,59 | <b>22,83</b> |
|            |                                   | Kontrolna       | 19,08  | 0,70  | 17,65 | <b>20,35</b> |
| Finalno    | Troskok iz mesta (cm)             | Eksperimentalna | 509,32 | 61,07 | 423   | <b>700</b>   |
|            |                                   | Kontrolna       | 487,48 | 40,27 | 419   | <b>570</b>   |
|            | Skok udalj iz mesta (cm)          | Eksperimentalna | 209,48 | 15,34 | 190   | <b>239</b>   |
|            |                                   | Kontrolna       | 204,88 | 14,52 | 180   | <b>236</b>   |
|            | Trčanje 20m iz visokog starta (s) | Eksperimentalna | 3,57   | 0,22  | 3,20  | <b>3,98</b>  |
|            |                                   | Kontrolna       | 3,70   | 0,20  | 3,25  | <b>3,99</b>  |
|            | Lepeza (s)                        | Eksperimentalna | 19,24  | 1,78  | 16,24 | <b>22,58</b> |
|            |                                   | Kontrolna       | 19,03  | 0,72  | 17,42 | <b>20,15</b> |

Legenda: AS – aritmetička sredina; S – standardna devijacija; MIN – minimalne vrednosti rezultata merenja; MAKS – maksimalne vrednosti rezultata merenja

Rezultati univarijatne analize varijanse (Tabela 2.) ukazuju na izjednačenost ispitanika u inicijalnom merenju u sva četiri motorička testa jer nisu uočene statistički

značajne razlike između eksperimentalne i kontrolne grupe. Na finalnom merenju, statistički značajne razlike su uočene samo u testu Trčanje 20 m iz visokog starta u korist ispitanika eksperimentalne grupe tenisera. Teniseri iz eksperimentalne grupe su nakon tri meseca i upražnjavanja pliometrijskog treninga dva puta nedeljno, ostvarili statistički značajnije i bolje rezultate u odnosu na tenisere iz kontrolne grupe.

**Tabela 2.** Razlika između grupa ispitanika u inicijalnom i finalnom merenju

| Merenje           | Testovi                           | F    | p    |
|-------------------|-----------------------------------|------|------|
| <b>Inicijalno</b> | Troskok iz mesta (cm)             | 0,62 | 0,44 |
|                   | Skok udalj iz mesta (cm)          | 0,01 | 0,93 |
|                   | Trčanje 20m iz visokog starta (s) | 2,83 | 0,10 |
|                   | Lepeza (s)                        | 1,14 | 0,29 |
| <b>Finalno</b>    | Troskok iz mesta (cm)             | 2,23 | 0,14 |
|                   | Skok udalj iz mesta (cm)          | 1,19 | 0,28 |
|                   | Trčanje 20m iz visokog starta (s) | 4,58 | 0,04 |
|                   | Lepeza (s)                        | 0,30 | 0,59 |

Legenda: F-vrednost F-testa; p-nivo statističke značajnosti F-testa

Nakon neutralisanja efekata inicijalnog merenja (Tabela 3.) teniseri iz eksperimentalne grupe su postigli statistički značajnije i bolje rezultate u odnosu na tenisere iz kontrolne grupe u tri od četiri motorička testa: Troskok iz mesta, Skok udalj iz mesta i Lepeza, dok u testu Trčanje 20 m iz visokog starta, nisu uočene statistički značajne razlike.

Primena pliometrijskog treninga u trajanju od tri meseca i sprovođenjem ovakvog načina treninga dva puta nedeljno, kod eksperimentalne grupe tenisera dovela je do poboljšanja eksplozivne snage nogu i agilnosti. Upražnjavajući vežbe za razvoj eksplozivne snage nogu (razne skokove, razne vežbice koje su bile u planiranom programu rada i bile ponavljane u serijama), teniseri iz eksperimentalne grupe su poboljšali svoje motoričke sposobnosti (eksplozivnu snagu nogu i agilnost) u toku tromesečnog redovnog upražnjavanja pliometrijskog treninga uz redovne teniske treninge koji su se najviše bavili razvojem njihove tehnike teniske igre.

**Tabela 3.** Efekti trenažnog programa između dva testiranja

| Test                                     | Grupa           | AS*    | F     | p           |
|------------------------------------------|-----------------|--------|-------|-------------|
| <b>Troskok iz mesta (cm)</b>             | Eksperimentalna | 503,72 | 15,20 | <b>0,00</b> |
|                                          | Kontrolna       | 493,12 |       |             |
| <b>Skok udalj iz mesta (cm)</b>          | Eksperimentalna | 209,62 | 10,74 | <b>0,00</b> |
|                                          | Kontrolna       | 204,72 |       |             |
| <b>Trčanje 20m iz visokog starta (s)</b> | Eksperimentalna | 3,62   | 2,79  | 0,10        |
|                                          | Kontrolna       | 3,65   |       |             |
| <b>Lepeza (s)</b>                        | Eksperimentalna | 19,04  | 4,57  | <b>0,04</b> |
|                                          | Kontrolna       | 19,23  |       |             |

Legenda: AS\*-korigovana aritmetička sredina; F-vrednost odnosa statističke značajnosti razlika između grupa; p-nivo statističke značajnosti F-odnosa

## DISKUSIJA

Ovim istraživanjem je dokazano da je pliometrijski trening koji je bio primenjen na grupi tenisera iz Beograda u trajanju od tri meseca, juniorskog uzrasta doprineo poboljšanju njihovih motoričkih sposobnosti, pre svega eksplozivnoj snazi i agilnosti. Realizujući ovaj vid treninga dva puta nedeljno, kao dodatne treninge (teniseri su redovno trenirali kao i kontrolna grupa, razvijajući svoju tehniku tri puta nedeljno), teniseri iz eksperimentalne grupe su nakon tri meseca poboljšali svoju eksplozivnu snagu nogu i agilnost što se manifestovalo kroz rezultate u motoričkim testovima Troskok iz mesta, Skok udalj iz mesta i Lepeza na finalnom merenju u odnosu na tenisere iz kontrolne grupe.

Primena eksperimentalnog tretmana na teniserima iz eksperimentalne grupe dovela je do statistički značajnih razlika u odnosu na kontrolnu grupu tenisera u finalnom merenju. Zahvaljujući velikom broju skokova (u visinu, u dubinu, skokovi preko prepona različitih visina (od 50-80 cm), uskoka i iskoka iz kutija različitih visina) koji su realizovani na dodatnim pliometrijskim treninzima, ispitanici iz eksperimentalne grupe su znatno poboljšali svoje eksplozivne sposobnosti nogu.

Istraživanja su pokazala da je za razvoj eksplozivne snage najefikasniji stimulus naprezanje mišića u tzv. „udarnom“ režimu naprezanja koji je bio primenjen kroz pliometrijske treninge tenisera eksperimentalne grupe (Bačić, Ivanišević i Tudor, 2006.). Ovaj režim naprezanja muskulature karakteriše naglo istezanje mišića izazvano spoljašnjim faktorima koje smenjuje nagla amortizacija kretanja i prelazak iz pliometrijskog u miometrijski režim naprezanja muskulature. Osnovni smisao ovog metoda je da izazove naglo istezanje mišića pojačavanjem voljne mišićne kontrakcije i uključivanjem miotatičkog refleksa mišića na istezanje. Kroz skokove u dubinu, njihovi mišići nogu su se istezali i skraćivali što je rezultiralo uključivanjem većeg broja mišićnih jedinica u rad, pa se i snaga mišića povećala. Veća snaga nogu je rezultat intenziteta impulsacije kore velikog mozga, tj. slanja većeg broja impulsa ka mišićima koji izvršavaju rad, propustljivosti motoričkih sinapsi, koja je veća pod uticajem intenzivnih vežbi, brzini transmisije impulsa od centra do periferije efektor-mišića (brža transmisija je posledica boljeg stanja treniranosti mišića tenisera eksperimentalne grupe), broja aktivnih motoričkih jedinica (ako je snaga mišića veća, pretpostavlja se da je uključen i veći broj motorinih vlakana čime se obezbeđuje i veća snaga tog mišića), biohemijskog stanja mišića. Sigurno je da su se ove pojave i reakcije u mišićima poboljšale kroz primenu pliometrijskog treninga na teniserima eksperimentalne grupe.

Trenažni pliometrijski program u kombinaciji sa redovnim treningom za razvoj i unapređenje tehnike teniske igre je pokazao odlične rezultate u cilju poboljšanja eksplozivne snage nogu, agilnosti tenisera eksperimentalne grupe, te se pokazao kao pozitivnim u radu sa teniserima juniorskog uzrasta. Ovo istraživanje će omogućiti pravilnu i pravovremenu primenu pliometrijskih programa vežbanja za poboljšavanje motoričkih sposobnosti (eksplozivnoj snazi i agilnosti) tenisera, kao i pravovremenu selekciju talentovanih tenisera u klubovima iz Srbije. Rezultati ovog istraživanja su ukazali na potrebu primene pliometrijskog treninga kod tenisera, jer im se na taj način podiže i

nivo igre, snage, brzine, agilnosti, što može dovesti samo do poboljšanja njihovih rezultata u narednom periodu, ako nastave da se bave tenisom.

Promene motoričkih sposobnosti tenisera iz eksperimentalne grupe su posledica primenjenog eksperimentalnog tretmana. Rezultati istraživanja će ukazati mnogim trenerima koje odgovarajuće vežbe koristiti za razvoj eksplozivne snage nogu i agilnosti tenisera.

Iz svega navedenog se može zaključiti da brzina, agilnost i eksplozivna snaga imaju visok uticaj na uspeh u tenisu, te zbog toga pliometrijski trening treba da bude sastavni deo svakog kondicionog treninga koji mora biti kvalitetno programiran u odnosu na nivo sposobnosti tenisera, te na period sezone u kojem se nalazi.

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## EFFECTS OF PLYOMETRIC TRAINING ON THE MOTOR ABILITIES OF TENNIS PLAYERS

*Plyometrics is a training method that uses an overload (Chu, 1983), and the main purpose of plyometric training is the development of greater reactive force (Allerheiligen & Rogers, 1995). Sports physiologists agree that plyometrics includes specific exercises that cause significant stretching of muscles located below the eccentric contraction, and followed by a strong concentric contraction, which is used for the development of a strong movement in a short period of time (LaChance, 1995). The aim of the research is to identify and analyze the transformative effects of plyometric exercise program on the manifest dimension of strength, agility and speed running experimental group of players. Methods: The sample consisted of 50 players were aged 17.5 years ( $\pm 6$  months), who were divided into two groups. One group consisted of 25*



tennis players TK “Gemaks” from Belgrade who represented the experimental group, while the second group included 25 tennis players TK “AS” from Belgrade who represented the control group. For evaluation of motor abilities used 4 tests. In order to determine differences between groups of respondents in the initial and final measurement was applied univariate analysis of variance. In order to determine the effect of training programs between the two tests was applied univariate analysis of covariance. Results: This research has proven that plyometric training, which was applied to a group of tennis players from Belgrade for a period of three months, youth ages contributed to improving their motor abilities, primarily explosive strength and agility. Discussion: Practice plyometric program combined with regular training for the development and improvement of techniques tennis game showed excellent results in order to improve the explosive leg strength, agility tennis players eksperimentatne groups, and proved to be positive in working with junior tennis players ages. Research have shown that the development of explosive energy efficient stimulus muscle strain in the so-called. “Primetime” regime of strain that was applied through plyometric training tennis players experimental groups (Bacic et al, 2006).

„Vijesti“, 13. april 2015.

CRNOGORSKA SPORTSKA AKADEMIJA

## Časopis na posmatranju za prestižnu bazu

Naučni časopis Crnogorske sportske akademije „Montenegrin journal of sports science and medicine“, koji uređuje prof. dr **Duško Bjelica**, nalazi se u 48 indeksnih međunarodnih baza i na posmatranju je kod Kobsona za ulazak u Tomson rojters bazu. Savjetodavni i uređivački odbor časopisa čini 48 međunarodnih naučnika. **Sv. M.**



UDK 796.09.298-055.1

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## **RAZLIKE U STATISTIČKIM POKAZATELJIMA FINALISTA AUSTRALIJAN OPENA 2014. GODINE U MUŠKOJ POJEDINAČNOJ KONKURENCIJI**

### **UVOD**

Motorička znanja u tenisu (motoričke informacije) predstavljaju određen skup dobro uvežbanih osnovnih elemenata (držanja reketa, osnovni stav, predviđanje), osnovnih udaraca i kretanja, naprednih udaraca i kretanja, specijalnih udaraca i kretanja (kretanja napred-nazad, napred i u stranu sa udarcima) i taktičkih elemenata (Roetert, 2003). Motorička znanja u tenisu moraju se razvijati u skladu sa biomehaničkim zakonitostima, te konstitucionalnim tipom, motoričkim sposobnostima i psihološkim karakteristikama svakog pojedinog tenisera i teniserke, u skladu sa senzitivnim periodima razvoja određene motoričke sposobnosti i jasno je da se razlikuju po polnoj pripadnosti zbog vremena sazrevanja osobe. Razvoj i napredak vrhunskog tenisa uslovljen je naučnim pristupom toj problematici (Crespo & Miley, 1999). Interdisciplinarni pristup rešavanja problema vrhunskog bavljenja tenisom postaje sve više uslov za dalji razvoj i planiranje tog sporta, jer bez saradnje sa drugim naukama, upoređivanjem rezultata između grupa i pojedinaca, ne postoji mogućnost postizanja vrhunskih rezultata (Roetert, 2003). U vrhunskom tenisu većina trenera obraća pažnju i na statističke pokazatelje svojih igrača nakon meča, kao i za vreme njegovog toka.

Dijagnostika u sportu, pa tako i u tenisu podrazumeva prikupljanje upotrebljivih informacija o inicijalnom, tranzitivnom i finalnom stanju sportista u prostoru sposobnosti i osobina koje su bitne za uspešnost u takmičarskom sportu (Guyton, 2006). Važno je izmeriti i utvrditi one sposobnosti, osobine ali i parametre uspešnosti, koje znatno učestvuju u jednačini uspeha u određenoj sportskoj disciplini, a u našem slučaju vezano za tenis, koji postaje sve popularniji u našoj zemlji, kako bismo što ranije otkrili talentovu decu i omladinu za ovaj sport.

Poslednjih nekoliko godina se sa posebnom pažnjom sprovede statističke analize sportskih takmičenja, kako sa aspekta samih mečeva, tako i sa aspekta učesnika u njima, a sve u cilju poboljšanja finalnih rezultata i učinka igrača.

Cilj istraživanja je bilo utvrđivanje razlika u statističkim pokazateljima između tenisera Stanislasa Vavrinke i Rafaela Nadala, finalista Australijan opena 2014. godine u muškoj pojedinačnoj konkurenciji.

## METODE

Uzorak entiteta je predstavljalo 7 mečeva i njihovi statistički pokazatelji za dva teniska igrača, Stanislava Vavrinku i Rafaela Nadala sa Australijan opena 2014. godine u muškoj pojedinačnoj konkurenciji. Kao uzorak posmatranih parametara analizirano je 12 statističkih pokazatelja najbitnijih u toku teniskih mečeva koje su odigrala dvojica finalista na Australijan openu 2014. godine (broj aseva na meču, broj duplih servis grešaka, procenat ubačenog prvog servisa, procenat osvojenih poena nakon prvog servisa, procenat osvojenih poena nakon drugog servisa, procenat osvojenih poena na mreži, procenat iskorišćenosti brejk šansi, procenat osvojenih poena nakon prijema servisa, broj vinera na meču, broj neprisiljenih grešaka, ukupno osvojenih poena, vreme provedeno na terenu za vreme meča). Statistički značajna razlika između dva najbolja tenisera analizirana je t testom za dve nezavisne grupe, na nivou statističke značajnosti od  $p \leq 0,01$  za sve parametre.

## REZULTATI

U skladu sa metodologijom istraživanja, analizirani su deskriptivni statistički parametri za procenu uspešnosti tenisera (Tabela 1.), i uočena je homogenost rezultata Rafaela Nadala u parametrima *Procenat ubačenog prvog servisa*, *Procenat osvojenih poena nakon ubačenog prvog servisa*, koja ga je odlikovala na svim mečevima na Australijan openu te godine. Na osnovu konstatnosti ubačenog prvog servisa (70%), njegove veće brzine, porastao je i procenat osvojenih poena nakon ubačenog prvog servisa. Uočava se uzročno posledična veza ova dva parametra koja je doprinela i osvajanju velikog broja osvojenih poena na meču.

Vavrinka je posedovao ujednačenost *Procenta ubačenog prvog servisa*, *Procenta osvojenih poena nakon ubačenog prvog servisa*, *Procenta osvojenih poena nakon drugog servisa* i *Procentu osvojenih poena na mreži*, dok se u preostalim parametrima uočava povećan varijabilitet rezultata. Kao i kod Nadala, srednje vrednosti procenta ubačenog prvog servisa (58,33%), je prouzrokovalo i više osvojenih poena na prvi servis, što je posledica jako dobre igre Vavrinke nakon ubačenog prvog servisa. Uočen je i mnogo veći broj ostvarenih as servisa osmoplasiranog igrača sveta. Odlikovao ga je i izlazak na mrežu i broj osvojenih poena. Ovo je sve posledica dobrog prvog servisa, koji je bio snažan, precizan i omogućavao je dominaciju igre u nastavku poena.

**Tabela 1.** Deskriptivni statistici analiziranih parametara za procenu uspešnosti

| Parametar                           | Igrač    | MIN | MAKS | AS    | S    | Sk    | Kurt  |
|-------------------------------------|----------|-----|------|-------|------|-------|-------|
| Broj ostvarenih aseva (frek.)       | Nadal    | 1   | 12   | 4.71  | 3.77 | 1,38  | 1.75  |
|                                     | Vavrinka | 4   | 21   | 14.00 | 6.45 | -0,70 | -0.70 |
| Broj servis grešaka (frek.)         | Nadal    | 0   | 7    | 2.00  | 2.38 | 1,97  | 2.99  |
|                                     | Vavrinka | 1   | 7    | 3.67  | 2.16 | 0,46  | -0.30 |
| Procenat ubačenog prvog servisa (%) | Nadal    | 62  | 78   | 70.00 | 5.60 | -0,08 | -1.01 |
|                                     | Vavrinka | 55  | 63   | 58.33 | 3.39 | 0,68  | -1.79 |

|                                                   |          |    |     |        |       |       |       |
|---------------------------------------------------|----------|----|-----|--------|-------|-------|-------|
| Procenat osvojenih poena nakon prvog servisa (%)  | Nadal    | 60 | 88  | 74.86  | 8.76  | -0,24 | 1.03  |
|                                                   | Vavrinka | 72 | 91  | 84.17  | 6.97  | -1,13 | 1.27  |
| Procenat osvojenih poena nakon drugog servisa (%) | Nadal    | 39 | 77  | 59.43  | 14.57 | -0,41 | -1.27 |
|                                                   | Vavrinka | 46 | 57  | 51.83  | 4.07  | 0,02  | -0.68 |
| Procenat osvojenih poena na mreži (%)             | Nadal    | 30 | 100 | 70.29  | 24.84 | -0,35 | -0.38 |
|                                                   | Vavrinka | 67 | 93  | 78.83  | 11.44 | 0,42  | -2.00 |
| Procenat iskorišćenih brejk šansi (%)             | Nadal    | 29 | 100 | 51.57  | 24.42 | 1,50  | 2.42  |
|                                                   | Vavrinka | 0  | 50  | 33.67  | 17.60 | -1,78 | 2.73  |
| Prijem servisa (%)                                | Nadal    | 29 | 50  | 40.00  | 6.71  | -0,21 | 0.49  |
|                                                   | Vavrinka | 28 | 50  | 38.50  | 7.99  | 0,15  | -0.70 |
| Broj vinera na meču (frek.)                       | Nadal    | 13 | 42  | 28.29  | 11.07 | -0,08 | -1.77 |
|                                                   | Vavrinka | 23 | 61  | 50.83  | 14.18 | -1,05 | 3.01  |
| Broj neprisljenih grešaka na meču (frek.)         | Nadal    | 5  | 47  | 24.86  | 13.08 | 0,30  | 1.00  |
|                                                   | Vavrinka | 11 | 60  | 42.00  | 16.88 | -1,46 | 2.68  |
| Ukupno osvojenih poena na meču (poen)             | Nadal    | 30 | 147 | 99.57  | 37.08 | -0,94 | 1.83  |
|                                                   | Vavrinka | 54 | 153 | 120.83 | 35.86 | -1,60 | 2.82  |
| Ukupno vreme provedeno na meču (poen)             | Nadal    | 41 | 217 | 139.57 | 57.71 | -0,40 | 0.57  |
|                                                   | Vavrinka | 65 | 240 | 161.17 | 61.18 | -0,38 | 0.27  |

Legenda: AS – aritmetička sredina; S – standardna devijacija; MIN – minimalni zabeleženi rezultat merenja; MAKS – maksimalni zabeleženi rezultata merenja; Sk- skjunis; Kurt – kurtosis

Na osnovu dobijenih rezultata u ostalim analiziranim parametrima može se konstatovati izuzetan varijabilitet kod oba ispitanika, što je posledica malog uzorka odigranih mečeva (7 Nadala i 6 Vavrinke) ali i velikih oscilacija tenisera na mečevima koje su odigrali na turniru. Bilo je mečeva koji su igrani samo jedan set, što je moglo uzrokovati ovakvu pojavu.

Takođe, na osnovu vrednosti kurtosisa i skjunisa, može se konstatovati da se radi o mezokurtičnim krivama koje ne odstupaju značajno od normalne distribucije.

Na osnovu t-testa za dve nezavisne grupe, analiziranih parametara (Tabela 2.), može se konstatovati postojanje statistički značajne razlike u parametrima:

- 1) *Broj ostvarenih aseva* ( $p=0.01$ );
- 2) *Broj vinera na meču* ( $p=0.01$ ) u korist Stanislasa Vavrinke;
- 3) *Procenata ubačenog prvog servisa* ( $p=0.00$ ) u korist Rafaela Nadala.

U parametru *Broj ostvarenih aseva*, Stanislas Vavrinka je bio prosečno bolji za sa nešto više od 9 aseva po meču (9,27 aseva), u parametru *Broj vinera na meču*, bio je bolji za prosečno 22,5 vinera. Nadal je prosečno bio bolji u parametru *Procenat ubačenog prvog servisa* za 11.67% u odnosu na Vavrinku.

Iako u parametrima tipa *Broj neprisljenih grešaka* i *Procenat osvojenih poena nakon prvog servisa* nije uočena statistički značajna razlika, može se uočiti i vredno je

pomenuti jasno vidljivo manje neprisiljenih grešaka kod Nadala, što govori o kontinuitetu igre ovog igrača i vrhunske tehničke i fizičke pripremljenosti.

U drugom navedenom parametru uočljiva je razlika u korist Vavrinke, koji se odlikovao većim procentom osvojenih poena nakon prvog servisa što je i posledica većeg broja as servisa na turniru. U ostalim analiziranim parametrima statistički značajne razlike nisu uočene pošto su ispitanici ostvarili prosečno slične vrednosti rezultata na odigranim mečevima.

**Tabela 2.** Razlike u analiziranim parametrima za procenu uspešnosti na Australijan openu 2014.

| Parametar                                         | t            | df        | p            | Razlika AS    |
|---------------------------------------------------|--------------|-----------|--------------|---------------|
| <b>Broj ostvarenih aseva (frek.)</b>              | <b>-3.23</b> | <b>11</b> | <b>0.01*</b> | <b>-9.27</b>  |
| Broj servis grešaka (frek.)                       | -1.31        | 11        | 0.,22        | -1.67         |
| <b>Procenat ubačenog prvog servisa (%)</b>        | <b>4.44</b>  | <b>11</b> | <b>0.00*</b> | <b>11.67</b>  |
| Procenat osvojenih poena nakon prvog servisa (%)  | -2.09        | 11        | 0.06         | -9.31         |
| Procenat osvojenih poena nakon drugog servisa (%) | 1.32         | 11        | 0.23         | 7.60          |
| Procenat osvojenih poena na mreži (%)             | -0.77        | 11        | 0.46         | -8.55         |
| Procenat iskorišćenih brejk šansi (%)             | 1.49         | 11        | 0.16         | 17.91         |
| Prijem servisa (%)                                | 0.37         | 11        | 0.72         | 1.50          |
| <b>Broj vinera na meču (frek.)</b>                | <b>-3.22</b> | <b>11</b> | <b>0.01*</b> | <b>-22.55</b> |
| Broj neprisiljenih grešaka na meču (frek.)        | -2.06        | 11        | 0.06         | -17.14        |
| Ukupno osvojenih poena na meču (poen)             | -1.05        | 11        | 0.32         | -21.26        |
| Ukupno vreme provedeno na meču (poen)             | -0.65        | 11        | 0.53         | -21.60        |

t-vrednost t testa; p-nivo statističke značajnosti t-testa; df – stepeni slobode; \* signifikantna razlika ( $p \leq 0,01$ )

## DISKUSIJA

Rezultati istraživanja ukazali su na postojanje statistički značajnih razlika između dvojice tenisera na Australijan openu, prvog niosoca R. Nadala i osmog nosioca S. Vavrinke u 3 analizirana parametra, od kojih je u dva dominirao Vavrinka (*Broj ostvarenih aseva*, *Broj vinera na meču*) dok je u parametru *Procenat ubačenog prvog servisa* bolje vrednosti zabeležio Nadal.

Stanislas Vavrinka je poznat kao dobar servis igrač, koji poseduje izuzetno snažan i precizan servis, sa kojim počinje da preuzima dominaciju u toku poena. Jak servis je njegovo osnovno oružje. Ovakvim pristupom igri, ovaj igrač zna da napravi veliki broj servis grešaka po meču, što je bio slučaj i na ovom turniru. Iako nije uočena statistički značajna razlika u varijabli za procenu *Broj servis grešaka*, moglo se uočiti veći prosečna broj grešaka u odnosu na Nadala. Nadal je igrač koji pravi manji broj neiznuđenih grešaka po meču što je bio slučaj i na ovom turniru, jer su njegove tehničke i fizičke sposobnosti na jako visokom nivou i slabo greši prilikom odigravanja

udaraca, sem ako mu greške nisu nametnute nekim dobrim potezom protivnika. Na šljaci je u stanju da napravi i više od 40 izmena i naravno osvoji poena.

Vavrinka svojim agresivnim pristupom igri, kada je u formi, kao što je to bio slučaj na Australijan openu, on ostvaruje i veći broj vinera. Njegovi razorni jednoručni bekend udarci i siloviti forhendi doneli su prevagu u igri na ovom turniru. O varijabli za procenu broja ostvarenih direktnih poena, je imao statistički značajnije i bolje rezultate u odnosu na drugog finalistu. Tih dana, kroz ceo turnir, Vavrinku je krasila silovita igra i poeni koje je besprekorno pogađao. U prilog ovoj rečenici ide i činjenica da je u parametru *Ukupno osvojenih poena na meču*, Stanislas Vavrinka imao bolji prosečni skor u odnosu na prvog nosioca turnira. Možda je i ova činjenica mogla da donese prevagu u igri koja ga je dovela do osvajanja Australijan opena.

Nadal je imao veći procenat ubačenog prvog servisa što je posledica slabijeg servisa Nadala (po brzini su njegovi servisi slabiji u odnosu na Vavrinkine). Sigurnijim pristupom igre kroz servis, u prvom servisu, Nadal „otvara teren“ i lako uz pomoć izuzetnih i preciznih spinova izbacuje protivnika iz igre i lako osvaja teren koji mu otvara mogućnost lakog poentiranja.

Preuzimajući više rizika u svojim poenima, igrajući opuštenije, izlazeći više na mrežu i poentirajući, servirajući snažnije i pogađajući as servise, Vavrinka je napravio prevagu i osvojio ovaj prestižni turnir prvi put u svojoj karijeri. Šampion Australijan opena za 2014. godinu, imao je i prosečno više ukupno osvojenih poena za vreme turnira, proveo je više vremena na terenu, posedovao slabiji prijem servisa od Nadala. Vavrinka je imao manji procenat iskorišćenih brejk lopti na mečevima, odigrao jedan meč manje, imao manji procenat osvojenih poena nakon drugog servisa, imao veći procenat osvojenih poena nakon prvog servisa i veći broj servis grešaka.

Možda je prevagu prilikom osvajanja turnira mogla da preuzme i činjenica manjeg broja vinera i ukupnog broja poena Rafaela Nadala kroz ceo turnir, što upućuje na činjenicu da je poene i osvajao zahvaljujući greškama protivnika, što se može i nasluti posmatrajući vrednosti u parametru *Broj neprisiljenih grešaka na meču*, kojih je Nadal imao manje nego Vavrinka iako statistički značajna razlika nije uočena. Ne treba zaboraviti da se Nadal u toku finalnog meča žalio i na bolove u leđima, što je moglo da implicira pobedu Vavrinke tj. slabiju igru prvog nosioca. U daljim istraživanjima, potrebno je analizirati širi spektar parametara koji bi mogli da dovedu do osvajanja nekog turnira, broj povreda igrača, žalbe na tegobe, stanje igrača pre turnira i broj odigranih turnira i mečeva na njima koji bi mogli da dovedu do povreda igrača (preopterećenost).

Vavrinka je tako postao prvi teniser koji je posle nešto više od 20 godina uspeo da savlada i prvog i drugog tenisera planete (u ovom slučaju je drugi nosilac bio Novak Đoković koga je Vavrinka pobedio u četvrtfinalu).

Rezultati istraživanja se donekle podudaraju sa rezultatima istraživanja koje je sprovedeno gde je utvrđeno da se statističke razlike između pobednika i poraženih ogledaju u sledećim parametrima: broju osvojenih ukupnih poena, broju neiznuđenih grešaka, broju osvojenih vinera, kao i da ne postoji statistički značajna razlika u parametru broj duplih grešaka (Klassen i Magnus, 2001).

Takođe, rezultati ovog istraživanja se donekle podudaraju i sa istaživanjem gde je utvrđeno da postoji statistički značajna razlika između igrača koji su pobeđivali i onih koji su gubili mečeve i da su igrači koji su pobeđivali na mečevima osvojili i više setova, što je i logično, kao i da su imali veći broj osvojenih vinera, manje su imali neiznuđenih grešaka od igrača koji su gubili mečeve, veći broj ukupno osvojenih poena i više su postizali asova na mečevima. Jedina varijabla koja se pokazala da nije statistički značajna bila je broj duplih grešaka (Kovačević, 2013).

Može se izvući generalni zaključak da je Vavrinka zahvaljujući boljoj fizičkoj formi, većem broju as servisa, većim brojem vinera na mečevima, uspeo da osvoji ovaj prestižni turnir. Kroz ceo turnir ga je pratio veliki broj osvojenih poena koji je bio veći zahvaljujući odigranom većem broju setova Vavrinke. Ovaj rad treba je trebao da skrene pažnju na potrebu konciznije i detaljnije analize teniskih mečeva sa aspekta statističkih pokazatelja koji se prate na svakom grend slem turniru. Želi se istaći neophodnost praćenja statističkih pokazatelja u današnjem vrhunskom sportu. To ujedno treba da predstavlja i praktičnu vrednost rada.

Rezultati rada su ukazali na potrebu stalnog praćenja statističkih parametara kod vrhunskih igrača koja je danas neophodna kako bi se jedna turnir iz ove serije osvojio. Svaki teniski trener danas želi da ima pristup statističkim pokazateljima za svog tenisera ali i za protivnika sa kojim je njegov igrač igrao ili će tek igrati, kako bi postavio strategiju igre i unapredio kompletnu igru svog igrača i podigao je na veći nivo.

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## DIFFERENCES IN STATISTICAL INDICATORS OF AUSTRALIAN OPEN 2014 FINALISTS IN MEN'S SINGLES COMPETITION

*An interdisciplinary approach to solving the problem of dealing with top tennis is becoming increasingly a requirement for further development and planning of the sport, because without cooperation with other sciences, comparing results between groups and individuals, there is no possibility of achieving top results (Roetert, 2003). The aim of the research is to determine the difference between a tennis player Wawrinka and Nadal finalist of Australian Open in 2014 in the singel competition. Methods: A sample of entities represented the seven matches and their statistics for the*

two tennis players, Wawrinka and Nadal with the last Australian Open is played open in tennis men's single competition. As a sample of observed parameters given of the 12 statistical indicators. Statistically significant difference between the two tennis players was analyzed by t-test. Results: Based on the analyzed parameters can be concluded that statistically significant differences in parameter Number of achieved aces, where Wawrinka was on average better for a little over nine aces per game (9.27 aces), in parameter Number of winners in the match, he was better for an average of 22.5 winners, while Nadal average was better in the parameter Percent of the loaded first service for 11.67% compared to Wawrinka. Discussion: Results of the research are somewhat consistent with the results of research carried out where it was determined that the statistical differences between the winners and losers are reflected in the following variables: the number of total points won, the number of unforced errors, the number of won winners, and there was no statistically significant difference in the variable number of double faults (Klassen et al, 2001).

„Dan“, 2. april 2015.



Све је спремно за Међународну научну конференцију о трансформационим процесима у спорту „Спортека достигнућа“, 12. по реду, која ће се од данас до недеље одржати у Подгорици, уз учешће преко 160 водећих светских аутора и истраживача у области спортских, друштвених и хуманистичких наука и медицине из 22 земље. Конференцију организују Црногорска спортска академија и Факултет за спорт и физичко васпитање, под покровитељством Универзитета Црне Горе. Регистрација учесника обавиће се вечерас и сјутра, са радом се почиње у петак у 9 часова, док је свечано отварање заказано за 16 часова у плавој сали на приземљу ректората УЦГ. Након добродошлице председника Конференције проф. др Душка Ђељинице, присутне ће поздравити министар просвете мр Предраг Бошковић, након чега ће се додијелити награда проф. др Мирославу Антали, председнику ФИЕП Европе, уз поздравну ријеч проф. др Георгија Балинта, потпред-

седника ФИЕП Европе, који ће примити награду умјесто њега. У оквиру конференције ће се уприличити потписивање споразума Факултета за спорт и физичко васпитање из Никшића и Института за спортске науке Универзитета у Минстеру у Њемачкој, чији ће потписници бити декан проф. др Добрислав Вујовић и проф. др Маике Тистјенс. За учеснике конференције организатори су за недељу планирали екскурзију – посјету манастиру Острог.

Т.Б.



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## KRVNI DOPING I RIZICI

### Uvod

Institucionalizovanje sporta, tehnološki napredak i razvoj mas-medija su od sporta napravili društveni fenomen, ali su ga pretvorili i u posao sa velikim obrtom kapitala. Razvoj tehnologije je unapredio rekvizite vezane za sport koji pomažu postizanju boljih rezultata, kao što je doprineo razvoju medicine i sredstava koji pomažu pacijentima da se brže oporave ili prikriju tragove povrede. Međutim, telo čoveka je ograničeno svojim mogućnostima i ne može preko toga. Primenom određenih materija i metoda u sportu, dolazi do psihološke i fizičke prednosti sportista nad drugima. Zato je doping u sportu za mnoge sportiste ostao jedina nada da postignu vrhunske rezultate, jer su telesno slabiji od drugih, a novac vezan uz vrhunske sportove je sve veći, iako postoje i negativne posledice po zdravlje.

Doping je veštački način povećanja sposobnosti sportiste putem farmakoloških preparata ili metoda. Jedna od tih metoda je i povećanje krvne oksigenacije, koja se može postići na dva načina: veštačkim prenosnicima kiseonika i dopingom krvi. Svaki od ovih načina povećava sposobnosti sportista za transportom kiseonika iz krvi u tkiva u cilju povećanja izdržljivosti i podizanja radne sposobnosti na maksimalni nivo. Sličan efekat se postiže tokom priprema sportista na umerenim visinama, od 1500-2500 metara nadmorske visine. Telo sportista, usled razređenog kiseonika, proizvodi veću količinu crvenih krvnih zrnaca. Broj eritrocita u krvi ostaje na povišenom nivou otprilike tri meseca nakon ovakvih priprema.

**Cilj** ovog rada je analiza dopinga putem krvi, kao i prikaz negativnih posledica koje ovaj način povećanja sposobnosti nosi sa sobom.

### Zabranjene metode - krvni doping

Termin krvni doping prvenstveno podrazumeva doping, odnosno veštačko povećanje radne sposobnosti putem krvi. Ovaj postupak podrazumeva transfuziju crvenih krvnih zrnaca, koja su pogodna za ovaj postupak, jer se mogu koncentrisati, zamrznuti i kasnije po potrebi odmrznuti i dati putem transfuzije.

Doping krvi - podrazumeva transfuziju krvi, koja sadrži eritrocite. Ovi proizvodi se mogu dobiti od iste osobe (autotransfuzija) ili druge (heterotransfuzija). U principu, ne postoji medicinska opravdanost za njenu upotrebu, sem u slučaju teškog krvarenja ili akutne anemije.

Homologna transfuzija podrazumeva uzimanje crvenih krvnih zrnaca od odgovarajućeg donora, koja se koncentrisana ubacuje u cirkulatorni sistem sportiste pre

određene aktivnosti tipa izdržljivosti. Kod autolognog tipa transfuzije, vađenjem krvi sportisti se izoluju crvena krvna zrnca unapred i potom se ponovo vrate u organizam pre takmičenja. Treba naglasiti, da neko vreme nakon vađenja krvi iz organizma, sportisti mogu biti anemični.

Postoje postupci davanja veštačkih prenosilaca kiseonika ili plazma ekspandera koji se smatraju manipulatorima sistema, jer ne postoji porast crvenih krvnih zrnaca, koji bi normalno bio uočen određivanjem hemoglobina ili hematokrita. Zatim, farmakološka, hemijska i fizička manipulacija npr. upotreba supstanci i metoda koji utiču na sastav i validnost uzorka urina je zabranjena, kao što su: kateterizacija, zamena urina, inhibicija bubrežne ekskrecije.

Dr Bjorn Ekblom sa štokholmskog instituta za gimnastiku i sport sedamdesetih godina prošlog veka je počeo primenjivati krvni doping. Vadio je krv sportistima, te centrifugiranjem eritrocite odvojio od plazme, čuvao ih u frižideru, da bi ih mesec dana kasnije svakom sportisti vratio u krvotok. Prva ispitivanja aerobnog kapaciteta posle toga su pokazala da dotični sportisti mogu, u proseku, na tredmilu trčati 25% duže nego ranije. Kasnija praksa je pokazala da je krv najbolje vaditi jednom do četiri puta po 450 ml, 6-8 nedelja pre, a odvojene eritrocite vratiti jedan do sedam dana od glavnog takmičenja. Pravilno učinjena procedura može povećati nivo hemoglobina i eritrocita za čak 20%. Takvo povećanje može trajati i do tri meseca, kada se hemoglobin i eritrociti u krvi postupno počinju vraćati u normalu.

### **Kako otkriti doping putem krvi**

Često se u otkrivanju krvnog dopinga koristi nasumično i ponavljajuće ispitivanje kuća sportista i objekata gde treniraju, sa ciljem pronalaska dokaza o zabranjenim supstancama. Moderniji pristup u otkrivanju krvnog dopinga je testiranje krvi ili urina sportiste za dokaze o prisustvu zabranjenih supstanci, obično eritropoetina. Postoje dva načina otkrivanja krvnog dopinga: merenjem nivoa hematokrita i nivoa hemoglobina koji pokazuju da li sportista koristi krvni doping. Hematokrit predstavlja odnos zapremine eritrocita u odnosu na zapreminu pune krvi (normalne vrednosti 0.40-0.54 M; 0.37-0.47 Ž). Hemoglobin je protein koji sadrži gvožđe koje vezuje kiseonik u crvenim krvnim zrcima (normalne vrednosti 130-170 g/l M; 120-150 g/l Ž). Ukoliko su visoke vrednosti oba pokazatelja, pretpostavka je krvnog dopinga. Međunarodna biciklistička unija nameće 15-dnevnu suspenziju sa trka bilo kom sportisti kod koga se pronađe vrednost hematokrita preko 0.54 i vrednost hemoglobina preko 170g/l.

Naprednije metode analiza krvi upoređuju broj zrelih i nezrelih crvenih krvnih zrnaca u krvotoku sportiste. Ako veliki broj zrelih crvenih krvnih zrnaca nije praćen velikim brojem nezrelih, retikulocita, onda se zaključuje da su zrela crvena krvna zrnca veštački uneta u krvotok putem transfuzije.

Test za otkrivanje homologne krvne transfuzije je u upotrebi od dvehiljadite godine. Ispitivanjem tragova na površini krvnih ćelija određuje se da li je u krvotoku prisutna krv više od jedne osobe. Ni dan danas nema pouzdane metode za otkrivanje autologne transfuzije. Postoji test koji meri nivo bifosfoglicerata u crvenim krvnim zrcima, čiji nivo opada vremenom, te zato, skladištena krv korišćena u autolognoj

transfuziji ima manji nivo ovog jedinjenja nego sveža. Stoga, manji nivo koncentracije bifosfoglicerata od normalnog ukazuje na autolognu transfuziju, odnosno doping. Još jedan dokaz krvnog dopinga je i prisustvo aditiva za plastiku u krvi, koji se čuvaju u kesama u kojima se skladišti krv.

### **Otkrivanje upotrebe eritropoetina**

Izvestan uspeh je postignut u upotrebi testova koji otkrivaju korišćenje eritropoetina. Problem kod ovih testova predstavlja činjenica da je farmaceutski eritropoetin nemoguće otkriti u krvi nekoliko dana nakon unosa. Jedan ovakav test, otkriva farmaceutski eritropoetin tako što ga razlikuje od skoro identičnog prirodnog hormona, koji se normalno nalazi u urinu sportista.

Dodatno unošenje eritropoetina moguće je otkriti i običnom analizom krvi, jer ovaj hormon povećava samo količinu crvenih krvnih zrnaca. To znači da količina leukocita i trombocita ostaje ista pa je tako poređenjem odnosa nivoa krvnih ćelija moguće ustanoviti da je došlo do zloupotrebe nedozvoljenih sredstava.

Naučna dostignuća pomeraju granice postignuća sportista do nesagledivih razmera. Najbolji primer je genski doping, iako se još uvek ne koristi u praksi. Teorijski, moguće je da se u DNK lanac ubaci i dodatna kopija gena za eritropoetin čime bi se udvostručila produkcija eritrocita. Ipak, to za sada ostaje u domenu teorije, jer eksperimenti na laboratorijskim životinjama nisu dali željene rezultate.

### **Prednosti upotrebe krvnog dopinga**

Sportisti, posebno oni u disciplinama izdržljivosti, nesumnjivo će ga i dalje uveliko praktikovati, jer ga doping kontrola može teško ustanoviti, a daje izuzetna povećanja sposobnosti. Ispitivanja su pokazala da krvni doping, u proseku, kod vrhunskih sportista daje sledeće rezultate: u trci na 1500m poboljšanje iznosi 3-5 sekundi, 10000m trči se oko jedan minut brže, a maraton čak četiri minuta brže. Tradicionalne metode, poput priprema na velikim nadmorskim visinama, pa čak i boravak u barokomorama, gube bitku s agresivnim i vrlo efikasnim farmakološkim sredstvima.

Poznato je da u sportovima izdržljivosti, kao što je maraton, biciklizam, veslanje itd. krajnji rezultat u velikoj meri zavisi od aerobnog kapaciteta, odnosno sposobnosti tela da u što većoj meri koristi kiseonik za proizvodnju energije. Da bi kiseonik dospeo do mišića, gde je neophodan za proizvodnju energije, zadužen je transportni sistem krvi koji se uglavnom sastoji od eritrocita, odnosno crvenih krvnih zrnaca te hemoglobina koji se nalazi u njima.

Opisani slučaj (Ero Mantiranta) Finca koji se šezdesetih godina prošlog veka takmičio u skijaškom trčanju, je i obeležio tu dekadu, jer je osvojio sedam olimpijskih medalja. Svoje rezultate ovaj skijaš dobrim delom duguje genetskom poremećaju - urođenoj porodičnoj policitemiji. Mutacija koju je imao Mantiranta dovodi do povećane proizvodnje crvenih krvnih zrnaca u organizmu, a samim tim se povećava i količina hemoglobina zaduženog za prenošenje kiseonika do svake pojedinačne ćelije u telu, pa tako i do mišićnih ćelija. Pri pojačanoj fizičkoj aktivnosti dolazi do ubrzanja pulsa, jer srce brže pumpa krv kako bi veći broj krvnih zrnaca uspešno doturio kiseonik do

mišića. Ovaj kiseonik se kroz mitohondrije u ćelijama pretvara u energiju. Prednost koju je ovaj sportista imao je u tome što mu je povišeni nivo crvenih krvnih zrnaca omogućavao da bez povećanja pulsa i krvnog pritiska postiže iste performanse koje ljudi sa normalnim nivoom crvenih krvnih zrnaca postižu uz drastično povišeni puls i krvni pritisak.

### **Diskusija o rizicima krvnog doping**

Sam postupak veštačkog povećanja crvenih krvnih zrnaca u krvi dovodi do porasta viskoznosti krvi, što može dovesti do zgrušavanja ili koagulacije, što povećava šanse za infarkt, šlog, plućne embolije. Sve ove bolesti se mogu desiti u slučaju da se previše krvi vrati veštačkim putem u krvotok, što uzrokuje stanje poznato kao policitemija, koje može imati i smrtni ishod.

Zaražavanje krvi tokom pripreme za čuvanje takođe predstavlja veliki problem, dešava se jednom na 500.000 transfuzija, što može izazvati pojavu sepse ili neku infekciju koja može dovesti do posledica (Blajkman, 2002).

Takođe, pojedini lekovi koji se koriste kako bi se povećao broj crvenih krvnih zrnaca dovode do oštećenja jetre i hipofize, kao i povećanja holesterola u krvi (Urhauzen; Torsten; 2003).

Eritropoetin nosi sa sobom mnoštvo opasnosti: prekomerna upotreba ovog hormona može promeniti vrednosti krvne slike, što može prouzrokovati policitemiju. Kod pojedinih vrhunskih sportista, najčešće atletičara i biciklista, koji su umrli od infarkta, obično za vreme sna, pronađene su neprirodno visoke koncentracije crvenih krvnih zrnaca u krvi (Kejn; 2006).

Procedure krvnog dopinga donose sa sobom i vrlo ozbiljne rizike. Neretko se događa da prevelika količina eritrocita preterano zgusne krv, odnosno poveća hematokrit, što može dovesti do smanjene sposobnosti srca da je pumpa prema periferiji. Time se postiže upravo suprotan učinak od željenog. Jako se smanji dotok krvi u mišiće, što rezultuje velikim padom radne sposobnosti. Pošto je protok takve krvi kroz krvne sudove otežan postoji velika opasnost da dođe do zastoja u cirkulaciji. U zavisnosti od mesta gde se takvo začepljenje dogodi, posledice mogu biti srčani zastoj, moždani udar, pulmonalni edem te ostale komplikacije koje najčešće završavaju fatalno. Zbog ove opasnosti, kao i zbog testa na eritropoetin, sportisti istovremeno koriste i razne ekspandere plazme (Albumex, Gelufozine i Haemaccel), kako bi razredili krv odnosno smanjili hematokrit.

Osim opasnosti manipulacije krvlju, transfuzije uvek nose određeni rizik od infekcije hepatitisom, AIDS-om i ostalim uzročnicima teških zaraznih bolesti. Sportiste izgleda ne zabrinjava mnogo činjenica, da je od posledica krvnog dopinga i eritropoetina od 1987. godine do danas umrlo najmanje 30-40 vrhunskih sportista, uglavnom biciklista. Štaviše, veliki broj njih se okrenuo novijoj generaciji preparata, koji su, prema svedočenju stručnjaka, još opasniji. Radi se o dodacima krvi koji sadrže ultrapročišćeni i modifikovan hemoglobin uzet iz goveda. Hemopure, kako se zove najnoviji preparat, ima vrlo brz i drastičan učinak na krv sportista. Međutim, i najmanja greška pri njegovoj upotrebi može prouzrokovati fatalne posledice.

Eritropoetin se uveliko koristi za lečenje različitih anemija, uključujući i one povezane sa rakom. U nekoliko kliničkih studija došlo je do neočekivanih rezultata u kojima se eritropoetin koristio za lečenje pacijenata obolelih od karcinoma, pa se pokrenulo pitanje potencijalnog direktnog uticaja eritropoetina na rast ćelija raka (Sytkowski, 2007).

Naravno, ovaj vid podizanja sposobnosti nosi sa sobom rizike koji mogu dovesti čak i do smrtnog ishoda. Jedan od načina da se stane na put korišćenju nedozvoljenih sredstava jeste uvođenje bioloških pasosa pomoću kojih je moguće pratiti promene koje nastaju u organizmu sportista.

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### BLOOD DOPING AND RISKS

*Doping is the way in which athletes misuse of chemicals and other types of medical interventions (eg, blood replacement), try to get ahead in the results of other athletes or their performance at the expense of their own health. The aim of this work is the analysis of blood doping and the display of negative consequences that this way of increasing capabilities brings. Method: The methodological work is done descriptively. Results: Even in 1972 at the Stockholm Institute for gymnastics and sport, first Dr. Bjorn Eklblom started having blood doping. Taken from the blood, athletes through centrifuge separating red blood cells from blood plasma, which is after a month of storage in the fridge, every athlete back into the bloodstream. Tests aerobic capacity thereafter showed that the concerned athletes can run longer on average for 25% of the treadmill than before. Discussion: Blood doping carries with it serious risks, excessive amount of red cells "thickens the blood," increased hematocrit, which reduces the heart's ability to pump blood to the periphery. All this makes it difficult for blood to flow through blood vessels, and there is a great danger that comes to a halt in the circulation, which can cause cardiac arrest, stroke, pulmonary edema, and other complications that can be fatal.*

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## OSNOVNE KOMPONENTE USPJEŠNOG VOĐSTVA U SPORTU

### UVOD

U vezi sa pojmovima vođa i vođstvo susreću se brojne terminološke nedoumice, kao što su rukovodilac i rukovođenje, a sve češće i termine lider i liderstvo. Ne tako davno bila je ustaljena upotreba termina rukovodilac umjesto vođa, odnosno rukovođenje umjesto vođstvo. Poneko pomodno umjesto naših termina upotrebljava termine potekle s anglosaksonskog govornog područja: lider umjesto rukovodilac (vođa), odnosno liderstvo umjesto rukovođenje (vođstvo). U našoj psihološkoj literaturi (Rot, Čokorilo, Bojanović i dr.) razjašnjena je distinkcija između ovih pojedinih termina. Termin vođa označava pojedinca koji zauzima poseban položaj i ima posebnu ulogu u grupi što mu omogućava poseban uticaj na ostale članove. Termin rukovođenje označava i osnovne aktivnosti vođe i osnovne funkcije koje rukovođenje ima i koje su važne za održavanje i funkcionisanje grupe, kao i ostvarivanje njenih ciljeva.

Vođstvo se najčešće definiše kao *proces u kojem pojedinac utiče na grupu u svrhu postizanja zajedničkog cilja*. Ključne riječi definicije su: *proces, uticaj, grupa, cilj*. Kao *proces* vođstvo podrazumijeva da ono nije samo obilježje vođe, nego je to transakcijski događaj koji se uspostavlja između vođe i njegovih sljedbenika – vođa utiče na sljedbenike i pod njihovim je uticajem. Kao *uticaj* vođstvo je povezano s time kako vođa djeluje na sljedbenike. Bez uticaja vođstvo suštinski i ne može postojati. *Grupa* je mjesto gdje se odvija vođstvo. Ono dakle, uključuje uticaj na skup pojedinaca koji imaju zajednički cilj (mala radna grupa ili velika radna grupa koja obuhvata čitavu organizaciju. Kao usmjerenost na *ciljeve* vođstvo podrazumijeva usmjeravanje grupe pojedinaca prema postizanju nekog zadatka ili svrhe. Dakle, vođe usmjeravaju svoju energiju prema pojedincima koji zajednički nastoje nešto postići (up. Northaus, 2010: 341). Analiza definicije pokazuje da je fenomen vođstva prevashodno međuljudski interakcijski proces koji ima višedimenzionalan karakter i da ga većina savremenih teoretičara na taj način shvata i proučava. U tom smjeru je fokusirana pažnja u osvjetljavanju ove važne problematike u savremenom sportu.

### 2. METOD

U ovom radu primarno je korišćen deskriptivni, analitičko-sintetički metod, koji je omogućio kritičku analizu fenomena vođstva u sportskoj grupi i različite pristupe u njegovom proučavanju. U vezi s tim vođstvo je opservirano kao važan višedimenzionalno uslovljen interaktivni proces u sportskoj grupi. Pored proučavanja i analize reprezentativnih naučnih i stručnih radova iz ove oblasti, korišćeno je iskustvo (bilo vođe bilo sljedbenika) iz prakse vođstva u sportskim, političkim i školskim uslovima.

### 3. REZULTATI

#### 3. 1. Različiti pristupi u proučavanju vođstva

Vođstvo predstavlja značajan faktor strukture i dinamike svake sportske grupe. U najširem smislu može se posmatrati kao *proces djelovanja* koji utiče na pojedince i grupe u ostvarivanju postavljenih ciljeva. Ovako određeno vođstvo sadrži u sebi nekoliko bitnih elemenata kao što su *usmjeravanje grupe, davanje informacija, odlučivanje, izgrađivanje međusobnih odnosa, motivisanje* i sl. Vođa je onaj koji zna u kom pravcu ide grupa. On određuje strategije kako da se ostvari cilj. Nije to samo vizija cilja, nego i svakodnevna konceptualizacija i motivacija da se vizija pretvori u realnost, da se kroz pojedinačni uspjeh ostvari uspjeh kompletnog tima.

Psiholozi su pokušali otkriti ove vještine vođstva u nadi da predvide i izaberu one koji mogu postati kvalitetne vođe. Pored ostalog, istraživani su i pojedini faktori koji su zaslužni za uspješno vođstvo, kao što su, na primjer, faktori okruženja.

Postoje veoma različita shvatanja vođstva koja su nastala u brojnim naukama sa veoma različitim polazišnim stanovištima. Baveći se tom problematikom, Stogdil (1974) je sve teorije vođstva podijelio u šest velikih grupacija:

- 1) teorije o osobinama vođe,
- 2) situacione teorije,
- 3) teorije koje naglašavaju značaj ličnosti vođe i situacionih faktora,
- 4) teorije interakcije (očekivanja),
- 5) teorije razmjene i
- 6) teorije humanističke.

Vajnberg i Guld (1999) su obradili četiri modaliteta istraživanju vođstva u sportu:

- 1) karakterni,
- 2) bihejvioristički,
- 3) interakcijski i
- 4) višedimenzionalni.

Stogdil je u dva svoja rada (1948. i 1974) dao solidan pregled istraživačkih radova koji tretiraju ulogu osobina ličnosti u vođstvu.

U prvom radu je analizirao više od 124 istraživanja osobina vođe koja su provedena između 1904. i 1947. godine. Utvrdio je da se prosječan pojedinac u ulozi vođe razlikuje od prosječnog člana grupe u sljedećim osobinama: inteligencija, pripravnost, uvid, odgovornost, inicijativa, istrajnost, samopouzdanje i društvenost. Rezultati analize su pokazali da pojedinac ne postaje vođa samo zato što posjeduje određene osobine, nego zato što osobine koje posjeduje moraju biti važne u situaciji u kojoj vođa djeluje.

U drugom radu Stogdil je analizirao 163 istraživanja koja su provedena između 1948. i 1970. godine. Ovaj rad u odnosu na prvi je uravnoteženiji u opisu uloge osobina u vođstvu. Dok je u prvom radu podrazumijevao da je vođstvo najviše određeno situacijskim faktorima, a ne faktorima ličnosti, u drugom radu je zaključio da su i faktori ličnosti i situacije determinante vođstva.

Men (1959) je proveo slično istraživanje u kojem je proučio više od 1 400 podataka o ličnosti i vođstvu u malim grupama. Stavio je manji naglasak na to kako situacijski faktori utiču na vođstvo. Zaključio je da se osobine ličnosti mogu primjenjivati

kako bi se vođe razlikovale od osoba koje nisu vođe. Te osobine su: inteligencija, muževnost, prilagodljivost, dominantnost, ekstravertiranost, konzervativizam.

Ogilvi i Tutko (1966. i 1970) su utvrdili, istina bez dovoljno dokumentovanih dokaza, da u profilu tipičnog trenera postoje sljedeće osobine: odlučan, autoritativan, sposoban da podnese pritisak obožavalaca i medija, emocionalno zreo, nezavisan u svom mišljenju i realan kad se radi o budućnosti.

Lord, Di Vader i Eliger (1986) na bazi pregleda 27 studija ustanovili da je pojava vođstva čvrsto povezana sa tri centralne osobine: inteligencija, tipična muževnost (agresivnost, odlučanost) i dominantnost. Do ovih rezultata su došli koristeći savršenije metode (metaanalizu) i čvrsto su zastupali stanovište da se osobine ličnosti mogu sistematski upotrebljavati u razlikovanju vođa i osoba koje to nisu u različitim situacijama.

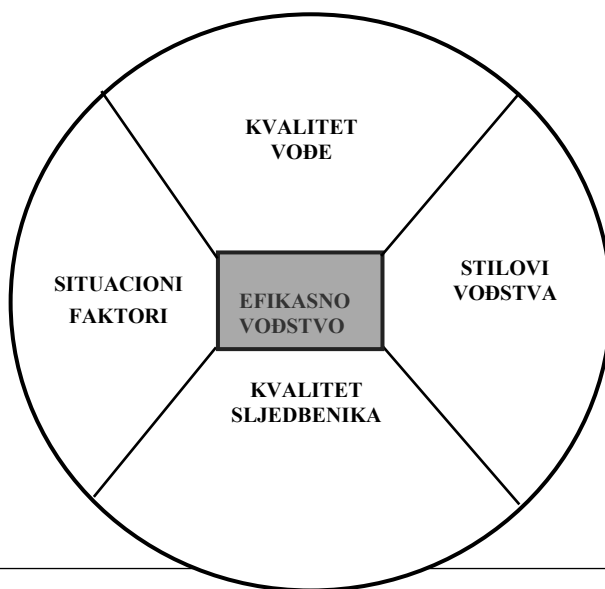
Kirkpatrik i Lok (1991) su na temelju kvalitativne sinteze ranijih istraživanja zaključili da vođe nisu poput ostalih ljudi i da se od njih razlikuju u šest osobina: inicijativa, motivacija, poštenje, pouzdanje, kognitivne sposobnosti i znanje o zadatku.

### 3.2. Osnovne komponente uspješnog vođstva u sportu

Na osnovu sumiranja različitih pristupa u proučavanju vođstva u sportu možemo zaključiti da nijedan od njih ponaosob nije najbolji, ali poznavanje dobrih strana i nedostataka pojedinih pristupa omogućava da bolje razumijemo i prihvatimo ono što vođstvo čini uspješnim.

Vajnberg i Guld (1999) izdvajaju i objašnjavaju četiri najvažnije komponente vođstva koje u skladu sa interakcijskim modelom pokazuju da se ponašanje najbolje razumije kao interakcija između ličnih i situacionih faktora. Te komponente su prikazane na slici 1.

*Slika 1. Komponente uspješnog vođstva u sportu (prema Martenesu, 1987)*





**1. Kvalitet vođe.** Bez obzira na to što ne postoji jedan određen niz suštinskih ličnih karakteristika koje omogućuju da osoba postane vođa, uspješne vođe imaju mnogo zajedničkih kvaliteta. Tako je Bil Parsels, uspješni fudbalski trener, u knjizi *Pronaći put do pobjede* (1995) izložio u nekoliko tačaka svoje stanovište o ključu uspješnog vođstva:

- *Integritet.* Filozofija vođe mora biti veoma upečatljiva i utemeljena na njegovim osnovnim vrijednostima.
- *Fleksibilnost.* Tradicije su stvorene da bi se rušile. Ukoliko nešto radite samo zbog toga što se to oduvijek tako radilo, možda propuštate šansu da nešto uradite mnogo bolje.
- *Lojalnost.* Prvi zadatak vođstva je da unaprijedi i insistira na kolektivnoj lojalnosti (timski rad).
- *Povjerenje.* Dajte trenerima i igračima odgovorne zadatke i zaduženja kao i mogućnost da sami donose odluke i podržite ih u njihovim naporima da ih izvrše.
- *Odgovornost.* Ne gradi se odgovorna organizacija bez vođe koji preuzima punu odgovornost.
- *Iskrenost.* Kad šaljete neku poruku, nije dovoljno samo da budete iskreni. Prijem poruke će zavisiti i od onoga ko je prima i šta je on voljan da prihvati tog trenutka.
- *Spremnost.* Dobro pripremljen plan vođe je onaj koji obuhvata i najsitniji detalj, pa čak i one koji ne izgleda da će se desiti ili koji im se ne sviđaju.
- *Iznalaženje snage.* Na osnovnom nivou nalaženje snage predstavlja neku vrstu fleksibilnosti, odbijanja da se odustane ili preda čak i kada sve izgleda neizvodljivo.
- *Samodisciplina.* Uvijek postoji način da se takmiči, čak i protiv superiornijih snaga, ali to zahtijeva strogo proračunati plan.
- *Strpljenje.* Strpljenje je najrjeđe i istovremeno najznačajnije, pogotovo onda kada organizacija funkcioniše loše.

Kao dopunu Parselsovih suštinskim karakteristikama uspješnog vođe, drugi istraživači su identifikovali još neke atribute, kao što su: inteligencija, optimizam, suštinska motivacija i razumijevanje. Treba naglasiti da su ovo neophodni i zahtijevani kvaliteti da bi neko postao vođa, ali ipak nisu dovoljni, odnosno, posjedovanje svih ovih kvaliteta ne garantuje da će neko biti dobar vođa. Spomenuti kvaliteti će biti neophodni u većoj ili manjoj mjeri u zavisnosti od onoga šta članovi grupe preferiraju i od specifičnosti situacije.

**2. Stilovi vođstva.** U literaturi se najčešće spominju demokratski i autokratski stil vođstva u sportu. Treneri koji njeguju demokratski stil treniranja su tipično sportski orijentisani, kooperativni su i dopuštaju sportistima da učestvuju u donošenju odluka koje se tiču ciljeva grupe, metoda, taktike i strategije igre. Nasuprot njima, autokratski treneri su orijentisani na pobjedu, strogo određeni i usmjereni ka osnovnom zadatku, obično donose nezavisne odluke, pri tom naglašavaju sopstveni autoritet i ne uvažavaju sugestije sportista. Uspješan trener ne bi smio biti strogo usmjeren ni ka jednom od ova

dva pravca. Bilo bi poželjno da u sebi integriše oba stila i pokaže fleksibilnost u trenajnom procesu. Izazov okolnosti određuje koji je stil najbolji. Kao što smo već kazali, odgovarajući stil treniranja uglavnom zavisi od situacionih faktora i karakteristika članova grupe.

**3) Situacioni faktori.** Dobar vođa mora da osjeti određenu situaciju kao i okruženje u kome se nalazi. Martins i sar. (1987) navode nekoliko situacionih faktora koji su relevantni za planiranje i uspješno vođstvo u sportu:

- sportisti timskog sporta obično više vole autokratske vođe nego što to vole sportisti u individualnom sportu;
- sportisti interaktivnog tima (košarka) više vole vođe koji daju zadatke nego što to vole sportisti koaktivnog tima (kuglanje);
- kako brojnost tima raste, postaje sve teže da se uspješno koristi demokratski stil vođstva;
- kada ima malo vremena na raspolaganju, bolji je vođa koji daje zadatke;
- grupa koja ima ustaljenu tradiciju stila vođenja imaće poteškoća u prilagođavanju drugom stilu vođstva.

**4. Kvalitet sljedbenika (članova grupe).** Karakteristike sljedbenika su veoma značajne u određivanju uspješnosti vođe tima. Potreba za miješanjem stila vođe i učesnika ukazuje na važnost interakcijskih procesa za uspješno vođstvo. Pokazano je već da stariji i iskusniji sportisti obično preferiraju autokratski trenerski stil, dok su žensportisti uglavnom naklonjenije demokratskom stilu vođenja ekipe. Ovamo spadaju još neke posebne karakteristike (iskustvo, zrelost, nacionalnost, sposobnosti, ličnost...), o čemu smo takođe govorili na prethodnim stranicama.

## DISKUSIJA

Preglednom analizom i sistematizacijom najvažnijih modela i empirijskih istraživanja u proučavanju vođstva u različitim organizacionim oblicima, moguće je zaključiti da u najvećem broju slučajeva kao najvažniji faktori se posmatraju brojne osobine i obilježja ličnosti, s jedne strane te različiti situacioni faktori, s druge strane. Pojedini od ovih faktora se često neopravdano preneglašavaju. Najdugovječniji i najtipičniji je slučaj sa karakternim modelom koji polazi od teze da se uspješne vođe rađaju, tj. imaju određene karakteristike ličnosti koje im omogućavaju da budu uspješne vođe bez obzira na situaciju koja vlada u određenoj sportskoj ili bilo kojoj drugoj grupi.

Za praksu vođstva u sportu afirmisane su četiri dominantne komponente (kvalitet vođe, situacioni faktori, kvalitet sljedbenika i stil vođstva) koje u međusobnoj interakciji obezbjeđuju njegovu efikasnost. Najvažniji kvaliteti vođe su integritet, fleksibilnost, odgovornost, samodisciplina i strpljenje. Njih treba posmatrati u kontekstu demokratskog i autokratskog stila vođstva. Situacioni faktori i faktori okruženja (vrsta sporta, brojnost tima, interaktivnost tima...) nameću važnost pojedinih osobina ličnosti i izbor stila vođstva. Na kraju, kvalitet članova grupe umnogome determiniše uspješnost vođe. Potreba za miješanjem stila vođe i učesnika potvrdila je važnost interakcijskih procesa za uspješno vođstvo.

Iskustvene su i naučne činjenice da treneri koji su dobri stručnjaci i pedagozi (demokratski se ponašaju, podržavaju i uvažavaju sportiste, umiju napraviti poželjnu interakciju i komunikaciju u svakoj situaciji), uspijevaju stvoriti timove koji posjeduju veću povezanost i jedinstvo, što im omogućava da postižu veće rezultate. Za buduća istraživanja uspješnog vođstva u sportu bilo bi neophodno naše analitičko-sintetičke i heurističke rezultate potkrijepiti empirijskim istraživanjima u praksi vođstva u različitim sportovima, na različitim uzrasnim nivoima i nivoima takmičenja.

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## THE BASIC COMPONENTS OF SUCCESSFUL LEADERSHIP IN SPORT

*Leadership is the most important factor of the structure and dynamics of each organised group. In the broadest sense of the word it is studied as an interactive process which affects individuals and groups in terms of implementation of the set goals. Leadership affects orientation of a group, exchange of information, decision making, establishment of interpersonal relations, people's motivation, etc. One should make a difference between a leader and a manager who is primarily in charge of planning, organisation, budget, assistant personnel etc. Even though leaders sometimes have the same obligations, leadership involves the skill of vision embodiment and not only a specific style of behaviour. Methods: It is primarily used descriptive, analytical and synthetic methods by which we critically analyzed phenomenon of leadership in the sports group and different approaches in his study. In connection with this leadership was observed as an important multidimensional conditioned interactive process in the sports group. Results: Summative results of different approaches to analysis of leadership in sport have shown that no single approach can be labelled as the best one however, understanding advantages of each one of them enables us to understand better the thing that makes leadership successful. The most important proven factors of efficient leadership are leader's quality, followers' quality, situational factor and leadership style. Discussion: The most important quality leaders have proven integrity, flexibility, responsibility, self-discipline and patience. They should be viewed in the context of democratic and autocratic style of leadership. Situational and environmental factors (type of sport, the number of teams, team interactivity ...) determine the importance of certain personality traits and leadership style choice. Finally, the quality of group members largely determines the success of a leader. The need for blending style leaders and participants highlighted the importance of the interaction processes for successful leadership.*

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## **ANTHROPOMETRICAL STATUS AND GENDER DIFFERENCES AT 12 YEARS OF AGE**

### **Introduction**

The assessment of growth in children is important for monitoring health status, identifying deviations from normality and determining the effectiveness of interventions<sup>1</sup>.

The significance of timely detection of poor growth in early life resides in its association with adverse functional consequences, including poor cognition and educational performance, low adult wages, lost productivity and, when accompanied by excessive weight gain later in childhood, increased risk of nutrition-related chronic diseases<sup>2</sup>.

Kosovo is one of the countries that do not show on the WHO map which shows the standards of rising and development of children of different ages. The aim of this study is to verify the current anthropometric status of pupils of age  $12 \pm 6$  months, of both genders, as well as their comparison with the WHO standards of raising and development.

### **Methods**

In this study was included 62 students (42 male, 20 female), primary school "Iliria" in Prishtina. All measurements were made in the gym during June 2014. From anthropometrical tests were made: body height (BH); body weight (BW); biacromial breadth (BB); elbow breadth (EB); adipose tissue at: suprailiac skinfold (SIS); subscapular skinfold (SSS); triceps skinfold (TRS); and abdominal circumference (AC). It was expected to gain significant correlations between circular parameters, subcutaneous adipose tissue and body mass. Also is achieved significant value between body height, body weight and width parameters. Used methods for the analysis of data are the standard methods from the SPSS, respectively basic statistical parameters, Correlation method and we have compared them.

## Results

**Table 1.** Basic statistical parameters for male and female – morphological status

|     | Descriptive Statistics |       |        |          |        |      |        |          |
|-----|------------------------|-------|--------|----------|--------|------|--------|----------|
|     | MALE                   |       |        |          | FEMALE |      |        |          |
|     | Min                    | Max   | Mean   | Std. Dev | Min    | Max  | Mean   | Std. Dev |
| BH  | 140                    | 175.5 | 151.81 | 8.32     | 141    | 168  | 151.98 | 6.85     |
| BW  | 28                     | 70    | 43.14  | 9.79     | 30.5   | 56   | 44.35  | 7.04     |
| SIS | 4                      | 62    | 16.63  | 13.71    | 5      | 35   | 16.64  | 8.69     |
| SSS | 3.5                    | 38    | 10.80  | 8.30     | 5      | 36   | 11.18  | 6.96     |
| TRS | 5                      | 38    | 13.86  | 8.41     | 5.5    | 25   | 13.53  | 5.57     |
| AC  | 50.5                   | 95.5  | 67.90  | 10.06    | 53.5   | 79   | 66.25  | 7.71     |
| EB  | 5.3                    | 7     | 5.98   | 0.44     | 5      | 6.4  | 5.84   | 0.41     |
| BB  | 29.3                   | 36.9  | 33.19  | 1.84     | 23.5   | 36.5 | 32.86  | 2.97     |
| N   | 42                     |       |        |          | 20     |      |        |          |

Based on the basic statistical parameters, it was noticed that there is a heterogeneous distribution of results, especially as regards body weight with the male pupils. The distribution of results between the maximal and minimal result are 28-70 kg, with the average of  $43.14 \pm 9,79$  standard deviation, while with the female pupils is noticed a homogenous group, where the standard deviation is significantly lower than with the male pupils at all variables. It is characteristic the fact that except with the body breadth, in all other female variables is more developed in the sense of rising.

**Table 2.** Correlations between variables - males

|     | Correlations MALE |        |        |        |        |        |        |    |
|-----|-------------------|--------|--------|--------|--------|--------|--------|----|
|     | BH                | BW     | SIS    | SSS    | TRS    | AC     | EB     | BB |
| BH  | 1                 |        |        |        |        |        |        |    |
| BW  | .713**            | 1      |        |        |        |        |        |    |
| SIS | .384*             | .858** | 1      |        |        |        |        |    |
| SSS | .386*             | .834** | .958** | 1      |        |        |        |    |
| TRS | .473**            | .849** | .900** | .903** | 1      |        |        |    |
| AC  | .485**            | .902** | .915** | .904** | .848** | 1      |        |    |
| EB  | .648**            | .815** | .573** | .509** | .559** | .649** | 1      |    |
| BB  | .700**            | .774** | .551** | .506** | .562** | .647** | .781** | 1  |

\*\* . Correlation is significant at the 0.01 level (2-tailed).

\* . Correlation is significant at the 0.05 level (2-tailed).

**Table 3.** *Correlations between variables - females*

|            | Correlations FEMALE |        |        |        |        |       |       |    |
|------------|---------------------|--------|--------|--------|--------|-------|-------|----|
|            | BH                  | BW     | SIS    | SSS    | TRS    | AC    | EB    | BB |
| <b>BH</b>  | 1                   |        |        |        |        |       |       |    |
| <b>BW</b>  | .585**              | 1      |        |        |        |       |       |    |
| <b>SIS</b> | -0.024              | .709** | 1      |        |        |       |       |    |
| <b>SSS</b> | -0.176              | .499*  | .791** | 1      |        |       |       |    |
| <b>TRS</b> | -0.088              | .666** | .894** | .843** | 1      |       |       |    |
| <b>AC</b>  | 0.388               | .906** | .783** | .706** | .786** | 1     |       |    |
| <b>EB</b>  | .577**              | .663** | 0.264  | 0.202  | 0.392  | .557* | 1     |    |
| <b>BB</b>  | 0.421               | 0.326  | 0.176  | -0.068 | -0.107 | 0.287 | 0.222 | 1  |

**\*\*.** Correlation is significant at the 0.01 level (2-tailed).

**\*.** Correlation is significant at the 0.05 level (2-tailed).

In the tables 2 and 3, are obtained high correlative values between the tests of circular parameters and tests of subcutaneous adipose tissue with the body weight, and between the body height, body weight and the breadth parameters on the level 0.01 of significance.

### Discussion

The findings show that female, almost in all measured variables, are more developed at this age. This can be noticed in the variables of body height, body weight and subcutaneous adipose tissue, while in the variables of abdominal circumference and body breadth is noticed that male are more developed (differentiated). This fact is also confirmed by the following authors: Group, W. M. (2006), WHO, (2007), and Baghianimoghadam, et al. (2012), which made a comparison with the WHO data where the indicators of rising and development of this age confirm the same in the variables of body weight and height. But, comparison of the such a symbolic sample of this research with the WHO data shows an approximate trend of rising and development of children included in the research, and the difference is as follows: Female pupils, age 12 in this research – body height=151.97 cm<OBSH=151.2 – 156.00 cm, the male pupils of the age 12, body height=151.80 cm<OBSH=149.1 – 155.40 cm, which means the difference is minimal.

Based on the data of correlations, it is noticed that male gender obtained higher correlative values in the level of significance 0.01 between the variables of height and the body weight and breadth, as well as between the body weight and all other variables which have realized high correlative values between themselves. Such a situation is not the same at the female gender. High correlative values have been achieved between the body height and body weight, and the elbow breadth. The body weight has achieved high correlative value with the variables of adipose tissue and the elbow breadth. But, the characteristics for the female pupils of this age are the variables of

biacromial breadth, which did not achieve high correlative value with none of other variables. Based on this, we can conclude that the sample of our research shows that we have children developed in accordance with the world standards, but within the surveyed group, differences, especially in the parameters of adipose tissue, are very much expressed. This may be a result of dynamic changes and economic categories of the population. In the next research, we will increase this sample numerically and in the aspect of geographical outreach within the Kosovo.

### Acknowledgement

This work is part of a research which is expected to be continued with much bigger sample, and intends to extract results of different ages, which will present standard values of rising and development of the population in Kosovo. In addition, the future orientations will be focused on identification of health of the Kosovo population based on the age and to find out possible health problems among the various ages aiming to avoid them as earlier as possible. This research is expected to have interdisciplinary character, which means it will include experts from various field.

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ANTHROPOMETRICAL STATUS AND GENDER DIFFERENCES  
AT 12 YEARS OF AGE

The aim of this study is to verify the current anthropometrical status of the pupils of the age 12, as well as to compare it with the standards of raising in accordance with the WHO standards. In the study were included 62 pupils (42 male and 20 female). The anthropometric tests were done in height, weight, biacromial and elbow breadth, subcutaneous adipose tissue at the : suprailiac skinfold; subscapular skinfold; triceps skinfold, as well as the abdominal circumference. The results showed that there was a heterogenic distribution of results, especially in the body weight, where the distribution between the minimal and maximal results is 28-70 kg, with the average  $43, 14 \pm 9, 78$  of standard deviation with the male pupils, while with the female pupils was noticed more homogenous group and the standard deviation was significantly lower than with the female pupils in all variables. The findings show that almost in all measured variables female pupils are more developed at this age, especially the body high, body weight and subcutaneous adipose tissue, while as regards the abdominal circumference and body breadth, the male pupils are more developed. Comparison of symbolic sample of this research with the WHO data shows an approximate trend of raising and development of children which were included in this research, and the difference is as follows: Female pupils age 12, body height=151.97 cm<OBSH=151.2 – 156.00 cm, male pupils of the same age included in the research, body height=151.80 cm<OBSH=149.1 – 155.40 cm, which means the difference is minimal.

**Key words:** anthropometric status, standards for raising, differences, gender

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## OUTCOME OF PHYSICAL THERAPY IN LOW BACK PAIN

### Introduction

Back pain in the lumbar part are very common problem in the whole world's population, approximately 70-80% of them experience at least once in their life such a pain. The back pain could be caused due to degenerative processes of spinal cord, different traumas, sitting positions (forced positions), congenital malformations. The risk factors for causing and worsening of BP are: age, life style, smoking and physical state. We should add that the weight gain is the risk factor of back pain (Kraeme 2009), and very often we encounter literature that a very often cause is weight gain or obesity (Wong 2007, Szpalski 2010). The most affected part of the spinal cord by the OB is the lumbar part. Physical exercise can be helpful for the patients with chronically low back pain to return to work and to normal activities. Application of regular exercise 3-4 times a week is the best solution for reducing the back pain. The aim of physical exercise is alleviating the pain and deformities, preserving and education for better posture, return of full function and prevention of disabilities (Clare 2004).

The main purpose of this paper is to define whether obesity affects the duration of treatment with physical therapy according to McKenzie techniques.

### Objectives

The aim of this study is to define: is the obesity one of the factors which prolongs the healing time in rehabilitation of physical therapy?

### Materials and methods

The research was carried out at the Physiatrist Service at Medicine of Occupation Institute (MOI) covering one year period during 2013. The total number of the patients that were subject to research was 101, who were adults within the Kosovo Energy Corporation (KEK). The research was long-term and retrospective. The material received was protocolled. The subjective data were taken: age, gender, working experience, clinical signs and occupation. The objective data, such as specific tests and diagnostics were performed by physiatrist, while the radiography was examined by the radiologist. From the records of the systematic visits the weight and height were taken. Out of these diseased, 32 of them had normal IBM, 38 of them were overweight and 31 obese. In order to easier review the statistical data, the groups were divided into not-obese and obese. In non-obese group were included the ones with IBM smaller than 29.9, in obese group were included diseased whose IBM was higher than 30. In our case 70 of them belonged to the first group (non-obese) and 31 to the second group

(obese). Their average age was 50 years old, while the average working experience was 24 years. The following were criteria for including them in the research: confirmed diagnostics of illness as final diagnose Sy. Lumbalae, and to be an employee of KEK. The diseased were treated with physical therapy in MOI. All 101 patients underwent physical therapy. Physical exercises were applied according to McKenzie protocol, with standardized protocol and adjusted for each case. In general, according to McKenzie standard protocol adjusted individually according to each case, physical exercises were focused on static and dynamic exercises for strengthening the muscles of the lumbar region, abdominal region, pelvis muscles, spinal cord muscles and in general of extremities, especially lower ones.

### Ethical clearance

The study was approved by the Regional Ethical Board of the Institute of Occupational Medicine, Kosovo. Written informed consent of each participant was obtained along with the approval of the Kosovo Energy Corporation administration.

### Statistical analysis

Data was presented using tables and graphs. Data processing was done with Office 2007 Excel. From statistical parameters the following were calculated: the index structure, the arithmetic mean, standard deviation, and minimum/maximum values. For parametric data the T-test was used. Verification of tests for confidence level of 99% is ( $p < 0.01$ ).

### Results

101 patients with low back pain were included in the research, wherein 70 or 69.3% according to the of body mass index were classified as non-obese (that according to body mass index were normal and overweight) and 31 or 30.7% obese. Non-obese patients were included in the group 1, while obese ones in the group 2 (Table 1).

**Table 1.** Statistical data of according to age parameters according to groups

| Data            | Group 1<br>(Non-obese) | Group 2<br>(Obese) | Total |
|-----------------|------------------------|--------------------|-------|
| N               | 70                     | 31                 | 101   |
| Average         | 49.11                  | 50.97              | 49.68 |
| Dev. Standard   | 8.67                   | 7.28               | 8.28  |
| Min             | 26                     | 34                 | 26    |
| Max             | 64                     | 60                 | 64    |
| T-test, P-value | t=1.042, P=0.299       |                    |       |

The average age of the respondents was 49.68 years (SD  $\pm 8.28$ ) from 26 - 64 years of age. The average age of non-obese patients was 49.11 years (SD  $\pm 8.67$ ), whereas for obese group it was 50.97 years (SD  $\pm 7.28$  years) from 34 – 60. By t-test

we didn't find high statistical significance between average age according to groups (Table 2., picture 1).

**Table 2.** *The number of physical therapy sessions by groups*

| Physical therapy<br>(sessions) | Group 1<br>(not obese) |              | Group 2<br>(Obese) |              | Total      |              |
|--------------------------------|------------------------|--------------|--------------------|--------------|------------|--------------|
|                                | N                      | %            | N                  | %            | N          | %            |
| <10                            | 25                     | 35.7         | 6                  | 19.4         | 31         | 30.7         |
| 10+                            | 45                     | 64.3         | 25                 | 80.6         | 70         | 69.3         |
| <b>Total</b>                   | <b>70</b>              | <b>100.0</b> | <b>31</b>          | <b>100.0</b> | <b>101</b> | <b>100.0</b> |

More than half of the patients involved in research (70 or 69.3%) had 10 or more sessions of physical therapy out of which with higher structure Obese Group compared with the Not Obese Group (80.6% vs. 64.3%), (Table 3).

**Tabela 3.** *The average number of physical therapies according to groups*

|                 | Group 1<br>(not obese) | Group 2<br>(Obese) | Total |
|-----------------|------------------------|--------------------|-------|
| N               | 70                     | 31                 | 101   |
| Average         | 9.39                   | 12.42              | 10.32 |
| Standard Dev.   | 4.81                   | 5.57               | 5.22  |
| Min             | 2                      | 5                  | 2     |
| Max             | 30                     | 25                 | 30    |
| T-test, P-value | t=2.78, P=0.0065       |                    |       |

The average number of physical therapy sessions of the research patients was 10.32 (SD  $\pm$  5.22). The smaller number of sessions applied to the patients involved in the research was two sessions and 30 sessions the largest.

The average number of physical therapy sessions of the research patients of Group 1 (not obese) was 9.39 sessions (SD  $\pm$  4.81). The smaller number of sessions applied to the patients of this group was two sessions and the largest 30 sessions.

## Discussion

The physical exercises might be of great help for the patients who suffer chronic back pain in order to return to normal daily activities and work (Mauris 2000). Regular exercises 3-4 times a week is the best solution for the reduction of the back pain level (Kwon 2006). The factors that contribute to the pain healing prolongation are: age, location of pain and socio-economic factors as well as psychological (Anderson 1999). (Mangvani 2010) presented in his work that obesity didn't present any important factor in elimination of pain by physical therapy. On contrary, we got high statistical significance during the application of physical therapy according to the groups (t-test=2.78, P=0.0065, P<0.01) wherein the obese group had higher average of physical therapies undergone in comparison with non-obese group. So, the obese group has

undergone more physical therapy sessions in comparison with non-obese group. Similarly to our research, (Miranda 2002) came to a conclusion that weight loss influences the treatment of the patients with osteo-muscular problems. There is a huge unclearness in the theory of obesity influence on low back pain. The loss of muscle mass of the trunk and low extremity muscles and central OB might be the risk factors for the low back pain (Al-Shamari 2003).

### Conclusions

The treatment with physical therapy according to McKenzie is longer at obese patients with low back pain for many reasons, especially because of the weight as the full amplitude of movement is not possible, in comparison with the ones whose IBM is normal or who are overweight.

The obesity and age do not have direct impact on the low back pain, but it affects the duration of healing process. The weight loss helps the treatment of the diseased with osteo-muscular problems.

### Conflict of interest

None declared

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## OUTCOME OF PHYSICAL THERAPY IN LOW BACK PAIN

**Introduction:** Physical exercise can be helpful for patients with chronic low back pain to return to work and to normal activities. Application of regular exercise 3-4 a week is the best solution for reducing the pain of back. Low back pain pose significant problem in clinics and public health. It presents one of the main problems with adults, since 70-80% of adults face this problem at least once in their life. The aims of this study is to define: is the obesity one of factors which prolong the healing time in rehabilitation of physical therapy? **Materials and Methodology:** This study has been conducted in Physiatry Service of the Institution of Occupational Medicine, in a one year timeframe. The total was 101 patients that were employed with Kosovo Energy Corporation (KEK). The study was long-term and retrospective, whereas the material was gathered particularly. The patients were treated with physical therapy in IMP, all of 101 patients have applied physical therapy. **Results** – from 101 sick individuals from the research, based on the body weight index, 69.3% are classified as non-obese and 30.7% obese.

There are no major statistical differences found in comparison between the groups in relation to age, job position, pain localization, sick leave and radiography. With the T-Test we have found a difference with high statistical significance between the average number of applied physical therapy sessions in relation to examined groups ( $T\text{-Test}=2.78$ ,  $P=0.0065$ , so,  $P<0.01$ ). **Conclusion:** Obesity and age have no direct influence in the back pain, but affects in prolonging healing. Loss of weight helps in treating with physical therapy the sick suffering from osteomuscular problems, also psycho-social factors affect in prolonging the healing process.

**Key words:** Low back, pain, physical therapy.

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## **LEASURE TIME ACTIVITY AMONG STUDENTS FROM UNIVERSITY OF PRISHTINA, FAKULTY OF PHYSICAL EDUCATION AND SPORT**

### **Introduction**

This paper presents the results of the research on the physical activity of students in the Faculty of Physical Education and Sport, University of Prishtina “Hasan Prishtina” Kosovo.

The purpose of this cognitive study was to diagnose certain aspects of the lifestyle of students, with a particular focus on physical activity and, the difference in sex. The practical aim, however, was an indication of such actions, arising from past experience that would aim at improving lifestyle, and could lead to the elimination or prevention of certain causes of not making movement activity by the students in our country.

For the sake of clarity, first the term of leisure requires clarification. For the first time it was defined during an international conference held in 1956. As defined by the French sociologist Dumazedier [4], leisure means all the activities taken by an individual for entertainment or self-development, voluntary participation in social life out of social and family responsibilities. This definition considers only adults. In turn, according to Dąbrowski [3] leisure time of young people is the time, which remains to their own disposal after fulfilling their body's needs and completing the duties at home and school, in which they can conduct their activities in agreement with their tastes, related to leisure, entertainment and satisfying their own interests.

### **Aims of the research**

The cognitive objective [2] of the present study was to diagnose certain aspects of the lifestyle of student aged 18-37 from Kosovo, with particular focus on physical activity. The practical aim was an indication of such actions, arising from past experience that would improve lifestyle, and could lead to the elimination or prevention of certain causes of not making movement activity by the students of our country.

### **Material and method**

The research was conducted by the use of diagnostic survey in the group of 120 people aged 18-37, among whom there were 56 girls and 64 boys. The research material

was collected in 2014. A research tool used to evaluate the lifestyle of students was a questionnaire "My lifestyle", developed by a team of European researchers [Telama R., Naul R., Nupponen H., Rychtecky A., Vuolle P., 2002]. *To determine the strength of the relationship between variable non-parametric test chi was used. Significance level  $p < 0,05$  was accepted.*

## Results

Table 1 show the way on spending leisure time of students. Frequently provided response was *listening to music*. This considers both students (girls 58.9 boys 39.1%). Other activities taken mostly in leisure time include *watching television, video* (girls 44.6%, boys 37.5%), *meeting friends* (girls 21.4%, boys 39.1%, and do sport in a club or sports section (boys 42.2%, boys 57.1%). No girls answered play instruments, sing in a choir, play cards or board games, play computer.

**Table 1.** Distribution of responses of students to the question: "How do you spend your free time?".[%]

| Leisure activities                         | KOSOVO |       |
|--------------------------------------------|--------|-------|
|                                            | GIRLS  | BOYS  |
| a) listen to music                         | 58.9%  | 39.1% |
| b) play instruments, sing in a choir, etc. | 0.0%   | 4.7%  |
| c) watch TV, video                         | 44.6%  | 37.5% |
| d) meet friends                            | 21.4%  | 39.1% |
| e) play cards or board games               | 0.0%   | 1.6%  |
| f) play computer                           | 0.0%   | 6.3%  |
| g) read books, magazines                   | 48.2%  | 29.7% |
| h) do sport in a club or sports section    | 57.1%  | 42.2% |
| i) do sport individually                   | 7.1%   | 18.8% |
| j) attend sports events                    | 3.6%   | 20.3% |
| k) go to discos                            | 1.8%   | 6.3%  |
| l) go to the cinema, theatre, concerts     | 5.4%   | 4.7%  |
| m) spend time on hobbies                   | 10.7%  | 12.5% |
| n) visit family                            | 21.4%  | 18.8% |
| o) do nothing                              | 7.1%   | 4.7%  |
| p) play with peers                         | 8.9%   | 9.4%  |
| q) others                                  | 1.8%   | 7.8%  |

Answers do not sum up to 100% because respondents could choose three answers.

In the next table are presented (Tab.2) forms of physical activity taken in leisure time most often were presented. To those that are taken *often* by the surveyed boys belong: soccer (71.9% of the boys. Among the forms of physical activity that are *never* taken by the boys we can find roller-skating (93.8%).



**Table 2.** Distribution of responses of boys to the question: "In which forms of leisure-time physical activity do you participate?"

| Frequency of taking various forms of physical activity by boys in % |       |       |        |       |       |       |
|---------------------------------------------------------------------|-------|-------|--------|-------|-------|-------|
| Forms of motor activity                                             | Never |       | Rarely |       | Often |       |
|                                                                     | Boys  | Girls | Boys   | Girls | Boys  | Girls |
| a) running                                                          | 1.6   | 0,0   | 40.6   | 28.6  | 57.8  | 71.4  |
| b) swimming                                                         | 18.8  | 25.0  | 78.1   | 71.4  | 3.1   | 3.6   |
| c) cycling                                                          | 29.7  | 23.2  | 46.9   | 58.9  | 23.4  | 17.9  |
| d) roller-skating                                                   | 93.8  | 94.6  | 4.7    | 5.4   | 1.6   | 0.0   |
| e) basketball                                                       | 12.5  | 19.6  | 71.9   | 64.3  | 15.6  | 16.1  |
| f) volleyball                                                       | 15.6  | 10.7  | 65.6   | 50.0  | 18.8  | 39.3  |
| g) football                                                         | 4.7   | 50.0  | 23.4   | 23.2  | 71.9  | 26.8  |
| h) handball                                                         | 50.0  | 33.9  | 42.2   | 35.7  | 7.8   | 30.4  |
| i) aerobics                                                         | 68.8  | 48.2  | 23.4   | 37.5  | 7.8   | 14.3  |
| j) gym                                                              | 35.9  | 82.1  | 43.8   | 14.3  | 20.3  | 3.6   |
| k) skateboarding                                                    | 84.4  | 89.3  | 12.5   | 7.1   | 3.1   | 3.6   |
| l) skating                                                          | 90.6  | 85.7  | 7.8    | 12.5  | 1.6   | 1.8   |
| m) skiing                                                           | 39,2  | 60.7  | 28.1   | 39.3  | 1.6   | 0,0   |
| n) snowboarding                                                     | 84.4  | 82.1  | 15.6   | 12.5  | 0,0   | 5.4   |
| p) gymnastics                                                       | 54.7  | 26.8  | 29.7   | 57.1  | 15.6  | 16.1  |
| q) martial arts                                                     | 40.6  | 48.2  | 51.6   | 33.9  | 7.8   | 17.9  |
| r) others                                                           | 3.1   | 7.1   | 70.3   | 73.2  | 26.6  | 19.6  |

Respondents could choose three answers. The results do not sum up to 100%.

Table 3 show the motivation to participate in physical activities. Respondents had the option of choosing three of the 13 responses given. The factor that motivates the surveyed most to participate in a variety of physical activities is that girls want to make a sports career (62.5% of the girls, 34.4% of boys), and for boys the most factor that motivates was I participate for health (76.6% of boys, 53.6% of girls). Quite often there were also answers: want to be in good shape (39.3% of girls, 70.3% of boys), for relaxation (23.2% of the girls and, 25.0% of the boys), for fun (21.4% of girls and 17.2% of boys) and can realize in sport (41.1% of girls and 31.3% of boys).

**Table 3.** Distribution of responses to the question: "I participate in a variety of motor activities because: . . . " [%]

| Motives to take motor activity in free time   | KOSOVO |       |
|-----------------------------------------------|--------|-------|
|                                               | Girls  | Boys  |
| a) want to make a sports career               | 62.5%  | 34.4% |
| b) this is an opportunity to make new friends | 14.3%  | 6.3%  |
| c) like competing                             | 39.3%  | 21.9% |
| d) want to be in good shape                   | 39.3%  | 70.3% |
| e) this is relaxation for me                  | 23.2%  | 25.0% |
| f) for health                                 | 53.6%  | 76.6% |
| g) to gain nice appearance                    | 0,0    | 3.1%  |

|                                                  |       |             |
|--------------------------------------------------|-------|-------------|
| <b>h) want to get a better mark in PE</b>        | 1.8%  | 4.7%        |
| <b>i) this is an opportunity to meet friends</b> | 1.8%  | 4.7%        |
| <b>j) for fun</b>                                | 21.4% | 17.2%       |
| <b>k) can realize in sport</b>                   | 41.1% | 31.3%       |
| <b>l) I'm encouraged by friends and siblings</b> | 0,0   | <b>1.6%</b> |
| <b>m) to gain material profits</b>               | 1.8%  | 3.1%        |

Respondents could choose three answers. The results do not sum up to 100%.

Table 4 present with whom respondents take physical activities in their leisure time and they had five options to choose from. The vast majority of the respondents take movement activities in their leisure time *with her friends*. There is no statistically significant differences between girls and boys ( $p < 0,05$ )

**Table 4.** *The distribution of responses to the question: I take motor activities in my free time mostly with..."*

| I take motor activities most often with... | Kosovo BOYS |                                     | Kosovo GIRLS |              |
|--------------------------------------------|-------------|-------------------------------------|--------------|--------------|
|                                            | N           | %                                   | N            | %            |
| <b>a) with parents</b>                     | 2           | 1.7%                                | 3            | 2.5%         |
| <b>b) with siblings</b>                    | 1           | .8%                                 | 5            | 4.2%         |
| <b>c) with friends</b>                     | <b>51</b>   | <b>42.5%</b>                        | <b>38</b>    | <b>31.7%</b> |
| <b>d) on my own</b>                        | 8           | 6.7%                                | 6            | 5.0%         |
| <b>e) with other persons</b>               | 3           | 1.7%                                | 2            | 2.5%         |
| Chi square                                 |             | Chi <sup>2</sup> = .332<br>p < 0,05 |              |              |

Table 5 concern the evaluation of one's health condition, where respondents had five options to choose from. Most of the of respondents have evaluated their health condition *as healthy*. There is no statistically significant differences between girls and boys ( $p < 0,05$ )

**Table 5.** *The distribution of responses to the question: "I believe I am... "*

| I believe I am                                | Kosovo BOYS |                                     | Kosovo GIRLS |              |
|-----------------------------------------------|-------------|-------------------------------------|--------------|--------------|
|                                               | N           | %                                   | N            | %            |
| <b>a) very healthy</b>                        | 26          | 21.7%                               | 16           | 13.3%        |
| <b>b) healthy</b>                             | <b>30</b>   | <b>25.0%</b>                        | <b>37</b>    | <b>30.8%</b> |
| <b>c) ill</b>                                 | 1           | .8%                                 | 0            | 0,0          |
| <b>d) cannot evaluate my health condition</b> | 7           | 5.8%                                | 3            | 2.5%         |
| Chi square                                    |             | Chi <sup>2</sup> = .158<br>p < 0,05 |              |              |

Respondents were asked also to assess their physical condition. Table 6 show the results. The boys (30,0%), girls (21,7%) appreciated their agility as *very good*. However, the boys (13,3%), and girls (17,5%), rated their physical condition as *good*. There is no statistically significant differences between girls and boys ( $p < 0,05$ )

**Table 6.** Distribution of responses to the question: " I evaluate my physical condition as... "

| My motor skills evaluate<br>as... | Kosovo<br>BOYS |                                    | Kosovo<br>GIRLS |       |
|-----------------------------------|----------------|------------------------------------|-----------------|-------|
|                                   | N              | %                                  | N               | %     |
| a) outstanding                    | 9              | 7.5%                               | 6               | 5.0%  |
| b) very good                      | 36             | 30.0%                              | 26              | 21.7% |
| c) good                           | 16             | 13.3%                              | 21              | 17.5% |
| d) sufficient                     | 3              | 2.5%                               | 3               | 2.5%  |
| e) poor                           | 0              | 0,0                                | 0               | 0,0   |
| Chi square                        |                | Chi <sup>2</sup> =.500<br>p < 0,05 |                 |       |

### Discussion and Conclusion

Leisure time is one of the major social problems. It is getting more difficult for the youth to resist the temptation of spending their leisure time in the form of passively. Therefore, a very important aspect is providing a good model to imitate by parents, teachers and education. The educational system steps into man's life in the period of its greatest spiritual and physical plasticity when one's lifestyle is created. According to Nowak-Starz [6]: "lifestyle is the image of the individual's and/or group's functioning. The process of forming a given lifestyle runs parallel to the educational activity.

The information collected in the course of our research has made it possible to specify the leisure time, the frequency of the physical activity of the enterprise, participation in various activities, and physical activity it's taken by parents of students.

Obtained results indicate that the factors encouraging the surveyed to physical activity are: relaxation, good physical condition, the benefits for health, the desire to have fun and sports career. What's interesting for girls at this age getting a nice appearance is not the most motivating factor. The girls as a motivator enumerated: their health, can realize in sport, want to be in good shape, like competing, the desire to be in good physical condition, an opportunity to meet new people, and the desire to have fun.

The results of the research correspond to the results of the studies carried out among young people by Skawiński et al. [9], who pointed out that young people spend most of their leisure time in front of the computer or TV. The results of the research indicate that most frequently taken forms of spending leisure time by young people in my country surveyed are listening to music, meeting friends, playing computer and watching TV, video.

Among the forms of activity never taken by students from Kosovo there are: roller-skating, snowboarding, skateboarding and skating, those forms can be classified

as “expensive” (due to the need to have an expensive sports equipment), and poorly available (fitness club, sports halls, skating rinks). Only 1,8% of the girls indicated going to discos, which may result from cultural and religious differences. On the other hand, while a higher percentage of respondents among girls 48.2%, pointed out to reading books. Interestingly, also the youth girls 21.4% and boys 18.8% in their free time are more likely to visit family,

The need for action for the health of young people is noted in the important international documents in which a lot of joint recommendations are repeated on the need to improve the health care system, promote healthy lifestyles, improving life skills as a modern strategy for prevention of many disorders. An important idea is to enable young people to prioritize and build programs for the benefit of their health [11]. It seems that in some countries as (Kosovo) this is difficult to achieve at the present stage of the development of local education systems and often very authoritarian educational traditions.

Bridging social inequality can contribute to the overall improvement of health of the public and selected age groups. The relationship between social factors and teenagers’ health is complex. Drawing conclusions on differences depends on accepted health and social factors. It is believed that during the period of adolescence health differences show not that strongly as in early childhood and during adulthood [8]. The reason for that is smaller influence of factors connected with family life, and bigger of those connected with a peer group.

The research shows that the level of wealth (not so much of the family, but of the whole country) may have an impact on the choice of activities taken by the youth. Undoubtedly, socio-cultural factors and the tradition of social life also have great impact and impose certain ways of spending leisure time (as can be seen particularly in the case of the Kosovo youth). The above observations, the future taking into account local conditions, should be a starting point for those planning changes in physical education systems in our country.

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#### *LEASURE TIME ACTIVITY AMONG STUDENTS FROM UNIVERSITY OF PRISHTINA, FAKULTY OF PHYSICAL EDUCATION AND SPORT*

*This paper presents the results of the research on the physical activity of students in the Faculty of Physical Education and Sport, University of Prishtina “Hasan Prishtina” Kosovo. The purpose of this cognitive study was to diagnose certain aspects of the lifestyle of students, with a particular focus on physical activity and, the difference in sex. The practical aim, however, was an indication of such actions, arising from past experience that would aim at improving lifestyle, and could lead to the elimination or prevention of certain causes of not making movement activity by the students in our country.*

*The research was conducted by the use of diagnostic survey in the group of 120 people aged 18-37, among whom there were 56 girls and 64 boys. The research material was collected in 2014.*

*The research that was carried out shows that the level of wealth (not so much of the family, but of the whole country) may have an impact on the choice of activities taken by the students. Undoubtedly socio-cultural factors and the tradition of social life, which impose certain ways of spending leisure time (particularly in the case of the students) as we can see in the results obtained, where we also found that there is no statistically significant differences between girls and boys.*

*The above observations, taking into account local conditions, should be the starting point for all those who plan introducing changes in physical education systems in my country in the future.*

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## **A NEW METHOD HIGHLIGHTING PSYCHOMOTOR SKILLS AND COGNITIVE ATTRIBUTES IN ATHLETE SELECTIONS**

### **Introduction**

Talent is explained as a set of traits in a field that is above normal but has not developed fully yet (Yasar, 2010). These traits in sports reflect a rather wide field and are explained as anthropometric, conditional, techno-motoric and cognitive traits and psychological factors. It is of significant importance that these traits to be determined specifically for each branch of sports in selecting and training athletes (Yasar, 2010; Muratlı, 2003). Thus, the initial tests implemented in selecting athletes are generally motoric sports tests based on the measurement of conditional traits of the athletic performance of the individual. Today, these performance criteria highly prevalent in elite level athletic training applications are inevitable for physical and conditional development (Dundar, 2013; Yasar, 2010; Muratlı, 2003).

However, it is no longer possible to attain athletic achievement only through known and determined few aspects. Because training applications have started to include new approaches that compel the limits of an athlete in many respects, different from the classical methods of the science of sports. Especially countries and athletes that compete for the Olympic and world records endeavor for superiority continuously by using different methods in a quick manner. Sports is a very dynamic concept in this respect and change rapidly in accordance with the requirements of every age (Dundar, 2013; Yasar, 2010).

Within this rapid change, the need to approach the selection, development and training applications of athletes with a different method and to evaluate the close relationship between and the effects of cognitive attributes on the performance become prevalent. In Turkey, selection processes ignoring the cognitive attributes and psychomotor skills are widely used. Although the significance and effectiveness of the processes of attention and perception to attain learning based on movement in initial and secondary selection of the athletes are known, appropriate methods are not used. During training and competition, focusing the attention to the appropriate stimulus and sustaining the attention is a significant factor for achievement in sports. Thus, it is important for the performance to evaluate the attention levels of the athletes in early stages and to provide necessary training (Caglar, 2006). Visual perception, skills and reaction time are the most significant criteria in the evaluation of athletic performance. It is known that visual skills are trainable and visual perception education programs are effective on the development of visual skills (Allen, 2006; Allen, 2004; Polat, 2000; Bellenkes, 1997).

Thus, a new method that could be used in selection and follow up of the development levels of the athletes and could be used in evaluation of their reaction times was scrutinized in this study with the participation of athlete and non-athlete female students.

### Methods

The study was conducted in Uludağ University, Faculty of Sports Sciences in June 2014. The study was conducted under the approval of the “Uludağ University Faculty of Medicine Clinical Research Ethical Committee.”

9 female table tennis athletes between the ages of 10 and 14 and trained for an average of  $4.8 \pm 1.6$  (2-7) years and for an average of  $16.1 \pm 3.8$  (10-21) hours weekly, and 9 female students who never did sports before. Participants were informed about the study. After the briefing, demographical information, and general health data were collected for the participants.

Consequently, the participants' height, weight and full body impedance measurements were taken using Tanita BC-418 equipment. Afterwards, in a silent room participants' auditory reaction times were recorded using BioPack MP36 system. The click sound was transmitted in random intervals during the first 10 trials and then in fixed intervals during the latter 10 trials using headphones. The participants were asked to press a button using their dominant hands. Averages of these 10 trials were recorded as **random** and **fixed interval reaction (ms) times**.

### Selective Action Array

Selective Action Array was implemented using a table tennis robot (Butterfly/Amic-3000). The robot was set up to send 26 balls in different colors (6 white, 10 yellow, 10 pink) randomly to different spots on the table at a fixed speed and at a rate of 100 balls a minute. The participants were asked to ignore the white balls, to only touch the yellow balls and to catch the pink balls after the balls bounced from the table once. Participants did not witness the test before they participated. After they were explained about the details of the test, two consecutive trials were conducted. The whole test was recorded in video. Every action or the lack of it by the participants was transformed into points on a scoring scale (Table-1) between the points of 0 and 6. For each colored ball, a **color point** and **total points** for the total number of balls were assigned. Furthermore, the percentage of the achievement of the six points worth of action was designated as “**6 points achievement %**.”

### Statistics

The findings were presented as mean  $\pm$  standard deviation and minimum and maximum values. The comparison of intragroup dependent values were done using Wilcoxon rank sum test and for the intergroup comparisons, Mann-Whitney U-test was used. Significance level was accepted as  $p < 0.05$ .

## Results

There was no difference between the age and body mass index (BMI) scores of the athlete and non-athlete students. Random and fixed interval auditory reaction times for the athletes were found as  $230.8 \pm 23.8$  (207-272) and  $188.1 \pm 19.7$  (161-216) ms respectively. They were found as  $307.6 \pm 121.0$  (213-578) and  $271.4 \pm 132.9$  (172-588) ms for the students that were not athletes respectively. Both random interval and fixed interval reaction times were found to be significantly quicker in athletes when compared with the students that did not do any sports (Table-2). Furthermore, fixed interval reaction times were significantly lower than the random interval reaction times.

*Table 1. Scoring scale*

| <b>White Ball &gt; No reaction</b> | <b>Points</b> |
|------------------------------------|---------------|
| No reaction                        | <b>6</b>      |
| Reacted but did not touch          | <b>4</b>      |
| Touched                            | <b>2</b>      |
| Caught                             | <b>0</b>      |
| <b>Yellow Ball &gt; Touch</b>      |               |
| No reaction                        | <b>0</b>      |
| Reacted but did not touch          | <b>4</b>      |
| Touched                            | <b>6</b>      |
| Caught                             | <b>2</b>      |
| <b>Pink Ball &gt; Catch</b>        |               |
| No reaction                        | <b>0</b>      |
| Reacted but could not touch        | <b>2</b>      |
| Reacted but could not catch        | <b>4</b>      |
| Caught                             | <b>6</b>      |

*Table 2. Age, body mass index (BMI), random and fixed interval auditory reaction times of the athlete and non-athlete students*

|                               | <b>Non-Athletes (n=9)</b>   | <b>Athletes (n=9)</b>      | <b>p</b> |
|-------------------------------|-----------------------------|----------------------------|----------|
| <b>Age (year)</b>             | 12,4 $\pm$ 1,2 (10-14)      | 13,1 $\pm$ 1,1 (11-14)     | p>0,05   |
| <b>BMI (kg/m<sup>2</sup>)</b> | 17,5 $\pm$ 2,7 (13,8-21,2)  | 18,6 $\pm$ 2,5 (13,5-21,4) | p>0,05   |
| <b>RI-RT (ms)</b>             | 307,6 $\pm$ 121,0 (213-578) | 230,8 $\pm$ 23,8 (207-272) | p<0,05   |
| <b>FI-RT (ms)</b>             | 271,4 $\pm$ 132,9 (172-588) | 188,1 $\pm$ 19,7 (161-216) | p<0,05   |

BMI: Body mass index; RI-RT: Random interval-reaction time; FI-RT: Fix interval-reaction time.

In the selective action array, first trial total points for the athletes were  $103.3 \pm 17.3$  (72-124) and for the non-athletes were  $102.4 \pm 18.8$  (68-134), and the second trial points were  $122.4 \pm 10.5$  (110-142) and  $104.7 \pm 20.3$  (64-132), respectively. The increase in total points in the second trial was significant for the athletes, while the increase for non-athletes was insignificant. Non-athletes received 30% more points than the athletes with the white balls that required to no reaction, there was a very insignificant increase between the first and second trials with the white balls.



With the yellow balls, athletes and non-athletes got similar points in the first trial, while the table tennis players increased their points about 20% in the second trial; non-athletes scored almost the same points.

**Table 3.** Color points, 6 points achievement% and total points for the non-athlete students in the Selective Action Array's first and second trials

| Non-Athletes<br>(n=9)              | White (n=6)               |                         |        | Yellow (n=10)         |                        |        | Pink (n=10)           |                       |        |
|------------------------------------|---------------------------|-------------------------|--------|-----------------------|------------------------|--------|-----------------------|-----------------------|--------|
|                                    | First Trial               | Second Trial            | p      | First Trial           | Second Trial           | p      | First Trial           | Second Trial          | p      |
| <b>Color Point (max: 36,60,60)</b> | 29,6 ± 6,0<br>(22-36)     | 32,9 ± 3,0<br>(28-36)   | p<0,05 | 44,7 ± 6,5<br>(32-52) | 44,4 ± 13,4<br>(14-58) | p>0,05 | 28,2 ± 9,3<br>(14-46) | 27,3 ± 6,5<br>(20-40) | p>0,05 |
| <b>6 points achievement %</b>      | 68,9 ± 26,5<br>(16,7-100) | 83,0 ± 14,5<br>(50-100) | p<0,05 | 31,9 ± 23,6<br>(0-60) | 47,5 ± 26,5<br>(0-90)  | p>0,05 | 8,9 ± 13,6<br>(0-40)  | 5,6 ± 7,3<br>(0-20)   | p>0,05 |

| Non-Athletes<br>(n=9)          | First Trial                 | Second Trial                | p      |
|--------------------------------|-----------------------------|-----------------------------|--------|
| <b>Total Points (max: 156)</b> | 102,4<br>± 18,8<br>(68-134) | 104,7<br>± 20,3<br>(64-132) | p>0,05 |

**Table 4.** Color points, 6 points achievement% and total points for the athlete students in the Selective Action Array's first and second trials

| Athletes<br>(n=9)                  | White (n=6)             |                          |        | Yellow (n=10)          |                           |        | Pink (n=10)             |                        |        |
|------------------------------------|-------------------------|--------------------------|--------|------------------------|---------------------------|--------|-------------------------|------------------------|--------|
|                                    | First Trial             | Second Trial             | p      | First Trial            | Second Trial              | p      | First Trial             | Second Trial           | p      |
| <b>Color Point (max: 36,60,60)</b> | 21,6 ± 8,7<br>(10-36)   | 23,8 ± 3,5<br>(16-28)    | p>0,05 | 45,6 ± 6,7<br>(32-52)  | 53,3 ± 5,6<br>(44-60)     | p<0,05 | 36,2 ± 11,2<br>(20-56)  | 45,3 ± 8,7<br>(32-58)  | p<0,05 |
| <b>6 points achievement %</b>      | 30,4 ± 30,5<br>(0-83,3) | 37,4 ± 13,3<br>(16,7-50) | p>0,05 | 63,0 ± 11,0<br>(40-80) | 84,7 ± 11,4<br>(66,7-100) | p<0,05 | 29,1 ± 22,4<br>(0-72,7) | 45,3 ± 28,9<br>(10-90) | p<0,05 |

| Athletes (n=9)                 | First Trial                 | Second Trial                 | p      |
|--------------------------------|-----------------------------|------------------------------|--------|
| <b>Total Points (max: 156)</b> | 103,3<br>± 17,3<br>(72-124) | 122,4<br>± 10,5<br>(110-142) | p<0,05 |

With the pink balls, non-athletes scored the lowest points, and their points decreased slightly in the second trial. Table tennis athletes demonstrated the highest increase in points in the second trial with the pink balls.

On 6 points achievement % values, non-athletes were significantly more successful with the white balls; athletes demonstrated the highest achievement with the

yellow balls. Athletes demonstrated an increase in their achievements with all balls in the second trial, but non-athletes demonstrated a decrease in achievement with the pink balls. In total, the highest achievement was observed with the yellow balls, the lowest achievement was observed with the pink balls (Table-3-4).

Since the table tennis players were accustomed to reacting every single ball due to their training, adaptation to the white balls were more difficult for them. While the non-athletes were most successful with the white balls, they failed in the catching task with the pink balls. They could not transfer their learning even in the second trial.

### **Discussion**

Reaction time is the time necessary to recognize the stimulus, select and program the appropriate response and activate the related muscle group. Reaction time has premotor and motor components. Reaction time is a simple measurement method that is thought to provide direct or indirect information on cognitive functions, perception and other neurological functions (Sisman, 2007). In humans, reaction time is directly related to the speed of neural transmission. Although this speed reaches 100-120 m/s, the voyage of the transmission from the sense organs to the brain and from there to the muscle groups takes a certain time (Welford, 1980; Laming, 1968). Laming (1968) designated the “average basic reaction time” as 220 ms (12). Lapszo (1999) tested the psychomotor efficiency of the table tennis players specific to the game under simulator conditions and determined that the expectation of hitting and directing the ball was high. In different sports played using a racket, when the athletes were given 3 stimuli in different speed levels (low, medium, high) and their actions towards the ball were scrutinized, it was found that badminton players reacted the best in medium speeds and table tennis players reacted the best in high speed levels (Menevse, 2011). In the study, both the random interval and fixed interval reaction times were significantly lower for the athletes when compared with the students with no athletic past. Furthermore, it was expected that the fixed interval reaction times to be found as significantly lower than the random interval reaction times.

The fact that athletes were 80 ms quicker than non-athletes in auditory reaction times could not only be explained by their increased speeds of neural transmission. The effects of sports on cognitive functions such as attention, focusing, sustaining the concentration, in other words its effects on premotor reaction time, are more dominant in this context. After an average of 5 years of playing sports, 30% increase in the speed of auditory reaction time, especially in premotor section, proves the necessity to evaluate the cognitive functions in sports from a different angle.

In selective action array, the significant increase in points in the second trial for the athletes and the failure of the non-athletes to accomplish this, demonstrated that athletes had higher learning and adaptation skills. Thus, the two trials in the selective action array and the increase in points in the second trial demonstrate that cognitive function was evaluated quantitatively.

The findings of the selective action array developed for this study showed that in the first phase, and especially in sports played using rackets, the development of the

sustenance of attention and visual perception could be attained rapidly. Furthermore, it was predicted that by developing the automatism of hitting the ball in a desired fashion during the game, the learning skill would be developed positively as well. Thus, by making changes in the number and color of the balls as well as differences in actions to be taken, this method could be a new approach to be used for other sports to include the cognitive attributes in selection process of the athletes.

### Aknoledgeent

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#### A NEW METHOD HIGHLIGHTING PSYCHOMOTOR SKILLS AND COGNITIVE ATTRIBUTES IN ATHLETE SELECTIONS

*Talents are extraordinary but not completely developed characteristics in a field. These attributes cover a relatively wide range in sports. Tests perused in selection of athletes are generally motoric sports tests and measure predominantly conditional attributes. It is known that in sports, performance is related to cognitive skills as well as physical features and motor skills. This study explored a new method that could be utilized in the selection and tracking the level of improvement of athletes, and evaluate their attention, perception and learning levels, on athlete and other female students.*

*9 female table tennis athletes that trained for 16 hours per week for the last 5 years and 9 female students that never played in any sports, aged between 10 and 14 years, were participated in our study. For the Selective Action Array, developed for this study, a table tennis robot was utilized. Robot was set up to send a total of 26 balls in 3 different colors (6 whites, 10 yellows, 10 pinks) to different areas of the table, in random colors and at the rate of 90 balls per minute. The participants were asked to ignore the white balls, to touch the yellow balls and to grab the pink balls using their dominant hands. Pursuant to explaining the task to the participants, two consecutive trials were executed and recorded using a camera. Every action performed/not performed by the participants was transformed into points in the scoring system.*

*First trial total points in the Selective Action Array were  $104 \pm 17$  for athletes and  $102 \pm 19$  for non-athletes, whereas on the second trial total points were  $122 \pm 11$  and  $105 \pm 20$ , respectively. The higher scores obtained in the second trial were significant for the athletes; the difference in the scores for non-athletes was minor. Non-athletes scored 33% better for the white balls as compared to the table tennis athletes. For the yellow balls, athletes and non-athletes scored similar points on the first trial, whereas the athletes improved their points around 20%, while no improvement was observed for the non-athletes. Non-athletes scored the worst points for the pink balls and during the second trial a minor decrease in their points was observed. Table tennis athletes demonstrated the highest improvement in points in the second trial for the pink balls.*

*The findings of the selective action array developed for this study showed that in the first phase, and especially in sports played using rackets, the development of the sustenance of attention and visual perception could be attained rapidly. Thus, by making changes in the number and color of the balls as well as differences in actions to be taken, this method could be a new approach to be used for other sports to include the cognitive attributes in selection process of the athletes.*

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## **THE STATUS OF STUDENTS OF THE FACULTY OF PHYSICAL EDUCATION AND SPORTS IN COMPARISON WITH STANDARD PARAMETERS OF THE ILLINOIS AGILITY TEST**

### **Introduction**

High-speed actions during soccer competition can be categorized into actions requiring acceleration, maximal speed, or agility. Acceleration is the rate of change in velocity that allows a player to reach maximum velocity in a minimum amount of time. Maximum speed is the maximal velocity at which a player can sprint. Agility does not have a global definition, but it is often recognized as the ability to change direction and start and stop quickly (Little & Williams, 2005). Preparation for agility and other performance training should involve both long- and short-term preparations. Longterm preparation may include a well-developed agility training program, while short-term preparation should include a warm-up (3,4,32). Often, stretching is performed as part of a warm-up prior to physical exertion. Stretching can be defined as the act of applying tensile force to lengthen muscles and connective tissues. Typically, stretching is used to enhance the range of motion (ROM) about a joint (flexibility). There are various techniques of stretching, such as static, ballistic, proprioceptive neuromuscular facilitation, and dynamic stretching (15,16) (Amiri-Khorasani, Sahebozamani, & Tabrizi, 2010). The aim of this study was to verify current gility status of the students of the Faculty of Education and Sports in the University of Prishtina. Alos, another aim of the study was to compare results of the students with the international notms of gility testing.

### **Methods**

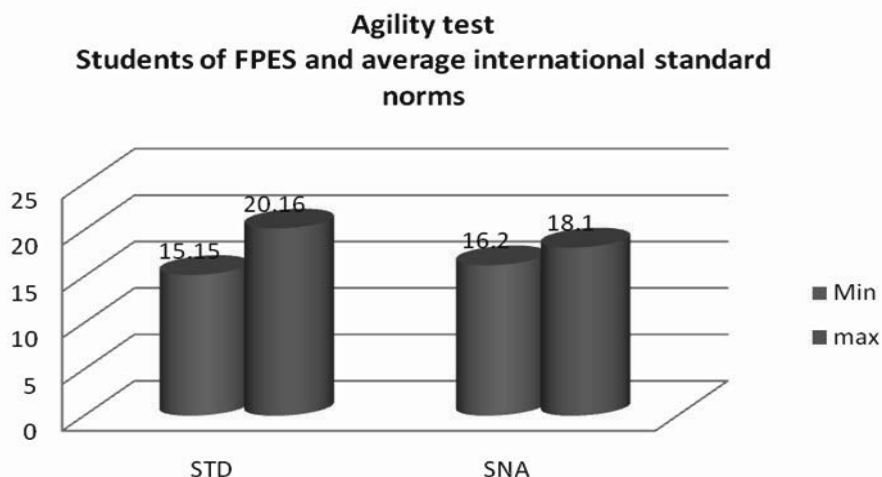
In this study were included 92 students of FPES. All tested studnets completed the second year of studies and pursued regular courses in the faculties and during the testing were informed about the nature of testing by obtaining prior consent. The tested students were in a good health situation. The measurements were taken during the regular courses in the sports hall of the physical education during the June 2014. The agility test was conducted through the "Illinois agility test", as a standard test. It was expected that the obtained results will present a homogenous group (based on their selection in the admission exam). Also, it was expected that the obtained results during the comparison with the standard results of the agility test to show a better result comparing to international norms. The methods applied for the analyses of the research data are also international methods, basic statistical parameters and comparative methods are from the SPSS package.

## Results

**Table 1.** Descriptive statistics of the students te FPES

| Descriptive Statistics |    |         |         |         |                |
|------------------------|----|---------|---------|---------|----------------|
|                        | N  | Minimum | Maximum | Mean    | Std. Deviation |
| Illinois               | 87 | 15.15   | 20.16   | 16.5420 | .92183         |
| Valid N (listwise)     | 87 |         |         |         |                |

Based on the basic statistical parameters, it was noticed that there was a homogenous distribution of results, where the distribution between the distribution between the minimal and maximal results is 15.15 – 20.16, with the average 16.54 and standard deviation 0.92. Based on the distribution parameters, it is noticeable that the values of skew parameters are 1.43, which proves about a very homogeneous group. Comparing to international norms of acility testing, it is noticable that the students of the Faculty of Physical Eduation and Sports have a satisfactory level and they belong to the category of average results. 16.54 sec, while the international standard norms are 16.2 - 18.1 sec.



**Figure 1.** Comparison chart between students and international standards

From the graphic presentation it is noticable that the defference between the minimal and maxial results of the group of students is bigger than the group of international group for Illionis Agility Test, because in this group of students are included all sample results. While, in the group of Illinoiis Agility Test are included only results of the norms on the level of average achievements. But, the average result of students belongs to the vategory of average achievemnts based on the international standards for this test.

### Discussion

The findings show that the obtained result is the expected one, taking into account that we have a sample to select students of the FPES. The measurements took place in adequate conditions and after the students understood their task very well. Based on the results, it is noticeable that compared to international norms for the athletes, we obtained average results 16.54 sec. The international norms show that this result belongs to the group of average results (<http://www.brianmac.co.uk/illinois.htm>). Comparing to the other groups of students, our sample showed the best results. For example, the group of tested students from Mehmet Kutlu, Hakan Yapıcı, Oğuzhan Yoncalık, Serkan Çelik, 2012, where the result of their students is  $16.54 \pm 0.41$ , but the testing took place in the synthetic carpet. This is a sample selected through the various physical tests and which is needed to them to qualify for admission to FPES.

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### THE STATUS OF STUDENTS OF THE FACULTY OF PHYSICAL EDUCATION AND SPORTS IN COMPARISON WITH STANDARD PARAMETERS OF THE ILLINOIS AGILITY TEST

The aim of this study is to verify the current agility status of students of the Faculty of Physical Education and Sports of the University of Prishtina. Also, another aim was to compare results of the students with the international norms of the standard agility test. In this study were included 92 students of FPES. The agility tests were done based on the Illinois Agility Test as a standard test. The methods used to analyse the data obtained from the research, are standard methods, basic statistical parameters and comparative methods. From the basic statistical parameters, it was noticed a homogenous distribution of results. The distribution between the minimal and maximal result is from 15.15 – 20.16, with the average 16.54 and standard deviation 0.92. Based on the parameters on distribution, it is noticeable that the value of skew parameters is 1.43. Comparing to international norms of standard agility test, it is noticeable that the students of the Faculty of Physical Education and Sports have a satisfactory

level and belong to the category of average results 16.54 sec, the international standard norms are 16.2 - 18.1 sec, which implies average achievement of results.

Comparing to other groups of students, our sample showed almost the same results with the group of students tested by Mehmet Kutlu, Hakan Yapıcı, Oğuzhan Yoncalık, Serkan Çelik, 2012, where the results of their students are  $16.54 \pm 0.41$ , but the testing took place in synthetic carpet.

**Key words:** agility, student, status. comparison, performance.

„Dan“, 11. mart 2015.

МЕЂУНАРОДНА КОНФЕРЕНЦИЈА „СПОРТСКА  
ДОСТИГНУЋА“ ОКУПИЋЕ У ПОДГОРИЦИ СПОРТСКЕ  
НАУЧНИКЕ ИЗ ЦИЈЕЛОГ СВИЈЕТА

## Ђукановић на челу Почасног одбора

Црногорска спортска академија и Факултет за спорт и физичко васпитање, под покровитељством Универзитета Црне Горе, организују 12. Међународну научну конференцију о трансформационим процесима у спорту „Спортска достигнућа“, која ће се одржати од 2. до 5. априла у Подгорици. Конференција ће окупити преко 250 учесника из 27 земаља, а своје радове представиће водећи свјетски аутори и истраживачи у области спортских, друштвених и хуманистичких наука и медицине. У сусрет конференцији, најзначајнијем регионалном научном скупу у области спорта, формиран је и Почасни одбор, којим ће предједавати предједник Владе Црне Горе **Мило Ђукановић**. Чланови Почасног одбора су угледни спортски радници и истакнуте јавне личности: ректорка Универзитета Црне Горе **проф. Радмила Војводић**, министарка науке **Сања Влаховић**, градоначелник Подгорице **Славољуб Стијеповић**, предједник Црногорског олимпијског комитета **мр Душан Симоновић**, директор Управе за младе и спорт **Игор Вушуровић**, предједник Рукометног савеза **Предраг Бошковић**, предједник Атлетског савеза **Милорад Вулетић**, предједница Скијашког савеза **Весна Меденица**, предједник Ватерполо и пливачког савеза **Срђан Ковачевић**, рукометни тренер **Драган Ашић**, ватерполо тренер **Петар Поробић**, фудбалски тренер **Миодраг Божовић**, рукометашица **Катарина Булатовић**, ватерполиста **Никола Јановић**, дудиста **Срђан Мрваљевић**, спортски директор ЖРК Будућност **Бојана Поповић**, професор новосадског Факултета спорта и физичког васпитања **Зоран Милошевић** и професор Факултета спорта и тјелесног одгоја из Сарајева и потпредједник Олимпијског комитета Босне и Херцеговине **Изет Рађо**.

Окупљањем истакнутих личности друштвене и спортске заједнице, Почасни одбор допринијеће још већем међународном угледу који је конференција стекла за више од деценије преданог рада на афирмацији спортске науке. Прва сједница Одбора требало би да буде ускоро заказана, о чему ћемо обавијестити јавност. Учеснике овогодишње конференције очекује богат програм, уз предавања и радионице специјалних гостију са партнерских универзитета. Т.Б.



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## **STRUCTURE OF ISOLATED PRECISION FACTORS OF THE MALE STUDENT ON VOLLEYBALL**

### **Introduction**

It is not difficult to understand volleyball, if we recognize the fact that in its basis, it is a sport which due to its characteristic peculiarities, dominates in the volleyball movements. However, the modern volleyball belongs to phenomenon in which the movement forms are extremely complex and diverse, taking into account first of all the structural positioning of the volleyball technique (Karelic, Vujmirlovic, & Savic, 2012).

The successful team should be composed of active volleyball players. They should possess, amongst others, adequate motoric abilities, which can be divided as a factor of success in the tactic and technique structures of the volleyball game. Because, in whatever situation during the game must be realized through the adequate movements assisted by general force of the loco motoric device through the speed, accuracy, coordination, agility etc. It is precisely this factor of motoric abilities which is key factor for the success of the volleyball game.

Volleyball coaches use skill tests to assess the skill level of their players and to predict the players' future success (Lidor R, 2007). The purpose of the study was to verify the current precision status of the students of the Faculty of Physical Education and Sport at the University of Prishtina. All they have completed the Volleyball course, and that the four variables tested were an integral part of the practical lessons from the subject of volleyball.

### **Methods**

In this study have participated 70 respondents, students of FEFS (35 in first year and 35 in fourth year). All tested have completed volleyball course from the study program in the FEFS. Measurements are held in the hall of the faculty. Standard tests were conducted to measure the precision in volleyball with separate emphasis on: overhand pass (SOP), underhand pass (SUP), as well as underhand serve (SUT) on the six positions and float (tactic) serve (SFS) on three positions. It is expected to achieve average results of the maximum scores in all tested variables. It is also expected that the group of students who have heard the fourth year was expected to give better results in the applied tests. Used methods for the analysis of data are the standard methods from the SPSS, respectively basic statistical parameters, Correlation method and T-test.

## Results

**Table 1.** Descriptive Statistics first year

| Descriptive Statistics first year |         |         |         |                   |          |          |
|-----------------------------------|---------|---------|---------|-------------------|----------|----------|
|                                   | Minimum | Maximum | Mean    | Std.<br>Deviation | Skewness | Kurtosis |
| SFIP                              | 9.00    | 36.00   | 21.6571 | 6.48955           | .255     | -.526    |
| SFAP                              | 2.00    | 22.00   | 10.6286 | 5.65210           | .337     | -.764    |
| SBUS                              | 8.00    | 23.00   | 14.7429 | 3.67298           | .206     | -.484    |
| STOS                              | 3.00    | 21.00   | 13.3143 | 3.95394           | -.420    | .539     |

**Table 2.** Descriptive Statistics fourth year

| Descriptive Statistics fourth year |         |         |         |                   |          |          |
|------------------------------------|---------|---------|---------|-------------------|----------|----------|
|                                    | Minimum | Maximum | Mean    | Std.<br>Deviation | Skewness | Kurtosis |
| SFIP                               | 13.00   | 33.00   | 23.6857 | 5.76894           | -.085    | -.996    |
| SFAP                               | 6.00    | 26.00   | 15.1714 | 5.18190           | -.003    | -.803    |
| SBUS                               | 7.00    | 22.00   | 15.1143 | 3.41081           | -.198    | .188     |
| STOS                               | 8.00    | 21.00   | 16.6000 | 3.13613           | -1.196   | 1.208    |

In the tables 1 and 2, the basic statistical parameters show that grouping of results of the students of forth academic year is more homogenous. Also, results for this group showed they are even better. While, as regards variables, better results are noticed with the students of fourth academic year. Also, the standard deviation is smaller at this group, which shows that the group is more homogenous, for example, SFIP the first academic year with  $21.65 \pm 6.48$  SD, the fourth academic year  $23.68 \pm 5.76$  SD. Based on the obtained results in the other variables as well, the group of fourth academic year appears to be a group with better achievements of quality comparing to the students of first academic year in the aspect of the achieved results and homogeneity as a group.

Table 3.

| Independent Samples Test |        |    |                 |                                                 |                       |                                           |          |
|--------------------------|--------|----|-----------------|-------------------------------------------------|-----------------------|-------------------------------------------|----------|
|                          | t      | df | Sig. (2-tailed) | t-test for Equality of Means<br>Mean Difference | Std. Error Difference | 95% Confidence Interval of the Difference |          |
|                          |        |    |                 |                                                 |                       | Lower                                     | Upper    |
| SFIP                     | -1.382 | 68 | .171            | -2.02857                                        | 1.46770               | -4.95732                                  | .90018   |
| SFAP                     | -3.505 | 68 | .001            | -4.54286                                        | 1.29613               | -7.12924                                  | -1.95647 |
| SBUS                     | -.438  | 68 | .662            | -.37143                                         | .84725                | -2.06210                                  | 1.31924  |
| STOS                     | -3.852 | 68 | .000            | -3.28571                                        | .85304                | -4.98794                                  | -1.58349 |

Also, results obtained through T-test show significant statistical changes in both applied tests. This is noticed especially in the variables SFAP where the average results of the students of the fourth academic year are  $15.77 \pm 5.18$  DS, and average  $10.63 \pm 5.65$  DS for the students of the first academic year with the significance 0.001. The same is with the variable STOS, the results vary from the average  $16.6 \pm 3.13$  DS for the students of the fourth academic year with the average result  $13.31 \pm 3.95$  DS for the students of the first year with the significance 0.000. So, the level of results which appear to be statistically important is clearly projected in both abovementioned variables and with better results achieved by students of fourth academic year. In both other variables, the level of statistical significance does not represent significant changes, although these changes are obvious in the favour of the students of the fourth academic year.

### Discussion

Based on the obtained results, the group of students of the fourth academic year have scored better results comparing to results obtained by the group of students from the first academic year in all variables. Also, it has been noticed that the group of students of the fourth academic year appeared to be more homogenous because in all four variables, where the obtained results are even better and the difference between the minimal and maximal result is smaller, as well as the value of standard deviation. Thus, better realization of variables extracted from the processing of results shows the advantages of the professional and continuous work in the faculties of the students of fourth year, which confirms the role of exercises (repetitions) in strengthening the movement information, because except pursuing the volleyball obligatory module during the year of studies, the students of the fourth passed another two modules of hand sports. They also had the opportunity to strengthen their volleyball skills through the possibility of selecting the volleyball as an optional subject. Through the analysis of T-test, it has been noticed statistically significant change in the variables SFAP and STOS, but the change which reaches the significant statistical change is also noticed in the variables SFIP and SBUS,

but it did not present significant statistical change. This presupposes that although when discussing about the sample as an entirety, the achieved results speak about a sample selected through testing in sample in the beginning of selection of the candidates for students, respectively through the addition exam.

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### STRUCTURE OF ISOLATED PRECISION FACTORS OF THE MALE STUDENT ON VOLLEYBALL

The aim of this study was to verify the current status of accuracy in volleyball for the students of the faculty of physical education and sports of the University of Prishtina, all the students have successfully passed the exam in volleyball. Also, another aim was to observe whether there are significant statistical differences between both generations of students. In the study were included 70 students (35 from the second academic year and 35 from the fourth academic year). To achieve the aim, were applied standard tests of accuracy in volleyball, such as the following: overhand pass, underhand pass, as well as underhand serve on the six positions and float (tactic) serve on three positions. From the basic statistical parameters, it has been noticed that grouping of results from the students of fourth academic year are more homogenous, and for this group showed that results are better than at students of first year of studying. While, in the variables is noticed that there are changes into the favour of students of fourth academic year, it has also been noticed that standard deviation is smaller with this group, which shows even more homogenous groups, for example, SFIP group 1 mean  $21.65 \pm 6.48$  SD, gr.2 mean  $23.68 \pm 5.76$  SD. The obtained results from T-test shows significant statistical changes in both applied tests. The best obtained results are a derivate of continuous professional work, as well as result of “selection”, due to continuous testing and completion of both exams of hand sports and volleyball as an elective subject.

**Key words:** accuracy, services, passing, differences, students.

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## **INCIDENCA POSTURALNIH POREMEĆAJA U ODNOSU NA STEPEN UHRANJENOSTI ADOLESCENATA**

### **UVOD**

Posturalni poremećaji predstavljaju loše držanje pojedinih segmenata tela u odnosu na pravilno, fiziološko. Zastupljenost funkcionalnih poremećaja posturalnog statusa, srećom, procentualno je veća u odnosu na strukturalne. Poremećaji pravilnog držanja tela nastaju dejstvom različitih urođenih ili stečenih faktora, koji se odražavaju na mišićnom i ligamentarnom aparatu i koštano-zglobnom sistemu. Važan faktor koji utiče na pojavu lošeg držanja tela je i način savremenog života sa smanjenom fizičkom aktivnošću. U svakodnevnom bivstvovanju mnoštvo je uzročnika koji ostavljaju i dobre i loše posledice na posturu. Posturalni poremećaji mogu se javiti u bilo kom uzrastu, mada postoje izvesni periodi u kojima im je organizam podložniji (Savić, 2007). To su periodi u kojima su deca posebno osetljiva i treba biti krajnje obazriv prilikom učešća u njihovom odgoju. Jedan od tih perioda je i adolescencija.

Broj adolescenata sa posturalnim poremećajima i gojaznošću raste, što može rezultirati ozbiljnim problemima ukoliko se na vreme ne otklone. Gojaznost dece i mladih predstavlja predmet interesovanja mnogih istraživača širom sveta, zbog toga što gojazna deca mogu biti u opasnosti po dva osnova – mogu imati kratkoročne posledice po zdravlje (Freedman et al., 2007) ali ih gojaznost može pratiti tokom odrastanja i odraslog doba (Singh et al., 2008).

Značajna je prevalenca gojaznosti kod adolescenata oba pola tokom 80-tih i 90-tih godina (Ogden et al., 2002), ali između 1999-2000. i 2007-2008. značajna razlika uočava se samo na najvišoj tački BMI koja odgovara 97. percentilu (Ogden et al., 2010). Bitan činilac u nastanku ovog savremenog problema, koji zahvata sve generacije (a nažalost sve više mlade naraštaje), jeste gojaznost u porodici (Zdravković, 2001). Nesumnjivo je da na razvoj gojaznosti utiču kako genetski tako i faktori okoline. Poremećaji uhranjenosti dece nose rizike raznih zdravstvenih tegoba (Stanišić i sar., 2005), jer gojaznost predstavlja stanje patološkog nakupljanja masnog tkiva u organizmu i bolest koju treba lečiti.

### **METOD**

U istraživanje se pošlo od pretpostavke da postoji statistički značajna razlika u posturalnom statusu ispitanika u odnosu na posturalni i nutritivni status i u odnosu na polnu pripadnost. Istraživanje predstavlja transversalnu studiju preseka. Uzorak ispitanika činilo je 305 adolescenata uzrasta od 11 do 14 godina, oba pola (158 devojčica i 147 dečaka), učenika osnovne škole iz Vojvodine. Posturalni status procenjen je

metodom Napoleona Volanskog, gde je posmatrano osam indikatora posturalnog statusa: držanje glave, ramena, lopatica, razvijenost grudnog koša, držanje trbuha, bočne krivine kičmenog stuba (kifotično, lordotično i skoliotično držanje), oblik nogu i status stopala. Ocene se kreću u rasponu od 0 do 2, gde nula predstavlja dobro, jedinica malo odstupanje i dvojka veliko odstupanje od dobrog držanja. Sve ocene se sabiraju i predstavljaju zbir na osnovu kojeg su ispitanici svrstani u određenu kategoriju držanja tela (izvrsno, vrlo dobro, dobro slabo i izrazito loše). Bodi mas indeks izračunat je standardnom procedurom, a ispitanici su podeljeni u pothranjene (ispod 80 percentila), normalno uhranjene (80 do 85 percentila), predgojazne (85 do 95 percentila) i gojazne (preko 95 percentila). Za statističku obradu podataka korišćene su neparametrijske metode. Razlike u posturalnom statusu u odnosu na uzrast i pol, izračunate su  $\chi^2$  testom ( $p=0.05$ ), Mann-Whitney testom izračunate su kumulativne ocene posturalnog statusa ( $p=0.01$  i  $p=0.05$ ), a Kruskal-Wallis test razlike u posturalnom statusu ispitanika formiranih u grupe u odnosu na nutritivni status.

## REZULTATI I DISKUSIJA

Rezultati istraživanja pokazuju da ne postoje statistički značajne razlike u posturalnom statusu ispitanika formiranih prema zbirnoj oceni u odnosu na polnu pripadnost. Nisu utvrđene statistički značajne razlike u držanju pojedinih segmenata tela kod ispitanika muškog i ženskog pola uzrasta od 11 do 12 godina.

Rezultati koji se odnose na razlike u držanju pojedinih segmenata tela u grupi ispitanika uzrasta od 13 do 14 godina, prikazani su u Tabeli 1.

Rezultati pokazuju da su utvrđene statistički značajne razlike u segmentima držanje glave, držanje ramena i držanje lopatica, na nivou procene od  $p \leq 0.01$ , a u segmentu skoliotična krivina kičmenog stuba na nivou značajnosti  $p \leq 0.05$ . Slične rezultate pri analizi posturalnih poremećaja po segmentima dobio je Kosinac, Z. (2003), koji je istraživao paramorfične i dismorfične promene grudnog koša i torakalnog dela kičmenog stuba splitskih učenika uzrasta 15-16 godina. Promene su bile učestalije kod ispitanica ženskog pola kao i u našem istraživanju.

Distribucija stepena uhranjenosti ispitanika uzrasta od 11 do 12 godina prikazana je u Tabeli 2. Kod uzrasne kategorije od 11 do 12 godina uočena je statistički značajna razlika u odnosu na polnu pripadnost. Razlika je izražena u kategoriji rizična težina u korist dečaka i u kategoriji normalna težina u korist devojčica. Prema Ugarkoviću (2001), na našem podneblju početak puberteta kod devojčica vezujemo za period od 10 do 11 godine, a kod dečaka od 12 do 14 godine života. U tom periodu dolazi do pojačanog lučenja hormona rasta i polnih hormona što ima za posledicu uvećanje svih koštanih, mišićnih pa i masnih ćelija. U kategoriji ispitanika uzrasta od 13 do 14 godina nije uočena statistički značajna razlika u stepenu uhranjenosti u odnosu na polnu pripadnost.

U Tabeli 3 prikazana je distirbucija ocene uhranjenosti i posturalnog statusa ispitanika uzrasta od 11 do 12 godina. Uočava se statistički značajna razlika u zbirnoj oceni držanja tela ispitanika u odnosu na stepen uhranjenosti. Izvrsno (14,4%), vrlo

dobro (23,2%) i dobro držanje (60%) tela imaju ispitanici u kategoriji normalne uhranjenosti.

**Tabela 1.** Mann-Whitney test za utvrđivanje razlika u držanju pojedinih segmenata tela ispitanika uzrasta od 13 do 14 godina

| Varijabla           | Pol | Srednji rang | Z      | p              |
|---------------------|-----|--------------|--------|----------------|
| Držanje glave       | M   | 65,19        | -3,091 | <b>0,002**</b> |
|                     | Ž   | 84,17        |        |                |
| Držanje ramena      | M   | 67,28        | -2,410 | <b>0,014**</b> |
|                     | Ž   | 82,22        |        |                |
| Oblik grudnog koša  | M   | 77,06        | -,820  | 0,412          |
|                     | Ž   | 73,08        |        |                |
| Držanje lopatica    | M   | 65,69        | -2,906 | <b>0,004**</b> |
|                     | Ž   | 83,70        |        |                |
| Kifotična krivina   | M   | 74,35        | -,220  | 0,826          |
|                     | Ž   | 75,61        |        |                |
| Skoliotična krivina | M   | 69,06        | -2,044 | 0,041*         |
|                     | Ž   | 80,56        |        |                |
| Lordotična krivina  | M   | 77,89        | -1,262 | 0,207          |
|                     | Ž   | 72,30        |        |                |
| Držanje trbuha      | M   | 70,76        | -1,402 | 0,161          |
|                     | Ž   | 78,96        |        |                |
| Oblik nogu          | M   | 70,93        | -1,285 | 0,199          |
|                     | Ž   | 78,81        |        |                |
| Držanje stopala     | M   | 69,20        | -1,829 | 0,067          |
|                     | Ž   | 80,42        |        |                |

Z – standardna vrednost; p – nivo značajnosti; \*\* -  $p \leq 0.01$ ; \* -  $p \leq 0.05$

U subuzorku ispitanika uzrasta od 13 do 14 godina uočene su takođe statistički značajne razlike u zbirnoj oceni posturalnog statusa u odnosu na stepen uhranjenosti (Tabela 4).

Analiza posturalnog statusa ispitivane populacije učenika uzrasta od 11 do 14 godina ukazuje na to da najveći procenat dece ima izraženu pognutost ramena, mlohavo držanje trbuha, izraženo odstojanje lopatica i pognuto držanje glave, što nam okvirno daje model osnovca sa hipotonusom (slabošću) mišića vrata, leđa i trbuha, te skraćenih mišića zadnje lože buta. Ne treba posebno napominjati da je ovo posledica smanjene fizičke aktivnosti dece, sedenternog načina života, kao i neadekvatnog školskog nameštaja. Slučajevi strukturalnih promena su u ovom uzrastu sporadični, ali

ono što zabrinjava je visok procenat funkcionalnih poremećaja koji mogu prerasti u strukturalne i stvoriti buduće pacijente banjisko-fizijatrijskih i ortopedskih ustanova.

**Tabela 2.** Distribucija ocene uhranjenosti ispitanika uzrasta od 11 do 12 godina u odnosu na pol

| Grupe              | Statistici | Pol    |        | Ukupno |
|--------------------|------------|--------|--------|--------|
|                    |            | muški  | ženski |        |
| Pothranjeni        | Broj       | 4      | 3      | 7      |
|                    | % za pol   | 5,3%   | 3,7%   | 4,5%   |
| Normalno uhranjeni | Broj       | 54     | 72     | 126    |
|                    | % za pol   | 72,0%  | 88,9%  | 80,8%  |
| Predgojazni        | Broj       | 13     | 3      | 16     |
|                    | % za pol   | 17,3%  | 3,7%   | 10,3%  |
| Gojazni            | Broj       | 4      | 3      | 7      |
|                    | % za pol   | 5,3%   | 3,7%   | 4,5%   |
| Ukupno             | Broj       | 75     | 81     | 156    |
|                    | % za pol   | 100,0% | 100,0% | 100,0% |

$\chi^2 = 8,890$        $p = 0,031$

**Tabela 3.** Distirbucija ocene uhranjenosti i posturalnog statusa ispitanika uzrasta od 11 do 12 godina

| Ocena posture      | Statistici     | Grupe po BM indeksu |                    |             |         | Ukupno |
|--------------------|----------------|---------------------|--------------------|-------------|---------|--------|
|                    |                | Pothranjeni         | Normalno uhranjeni | Predgojazni | Gojazni |        |
| Izvršno držanje    | Broj           | 0                   | 18                 | 1           | 0       | 19     |
|                    | % za BMI grupe | 0,0%                | 14,4%              | 6,3%        | 0,0%    | 12,3%  |
| Vrlo dobro držanje | Broj           | 2                   | 29                 | 6           | 1       | 38     |
|                    | % za BMI grupe | 28,6%               | 23,2%              | 37,5%       | 14,3%   | 24,5%  |
| Dobro držanje      | Broj           | 5                   | 75                 | 2           | 3       | 85     |
|                    | % za BMI grupe | 71,4%               | 60,0%              | 12,5%       | 42,9%   | 54,8%  |
| Slabo držanje      | Broj           | 0                   | 3                  | 4           | 0       | 7      |
|                    | % za BMI grupe | 0,0%                | 2,4%               | 25,0%       | 0,0%    | 4,5%   |
| Vrlo loše držanje  | Broj           | 0                   | 0                  | 3           | 3       | 6      |
|                    | % za BMI grupe | 0,0%                | 0,0%               | 18,8%       | 42,9%   | 3,9%   |
| Ukupno             | Broj           | 7                   | 125                | 16          | 7       | 155    |
|                    | % za BMI grupe | 100,0%              | 100,0%             | 100,0%      | 100,0%  | 100,0% |

$\chi^2 = 69,044$        $r = 0,000$   
 $\gamma = 0,308$        $r = 0,062$



**Tabela 4.** Distribucija ocene uhranjenosti i posturalnog statusa ispitanika uzrasta od 13 do 14 godina

| Ocena posture      | Statistiki     | Grupe po BM indeksu |                    |             |         | Ukupno |
|--------------------|----------------|---------------------|--------------------|-------------|---------|--------|
|                    |                | Pothranjeni         | Normalno uhranjeni | Predgojazni | Gojazni |        |
| Izvršno držanje    | Broj           | 1                   | 40                 | 0           | 0       | 41     |
|                    | % za BMI grupe | 14,3%               | 33,3%              | 0,0%        | 0,0%    | 27,5%  |
| Vrlo dobro držanje | Broj           | 4                   | 22                 | 0           | 0       | 26     |
|                    | % za BMI grupe | 57,1%               | 18,3%              | 0,0%        | 0,0%    | 17,4%  |
| Dobro držanje      | Broj           | 2                   | 58                 | 6           | 2       | 68     |
|                    | % za BMI grupe | 28,6%               | 48,3%              | 40,0%       | 28,6%   | 45,6%  |
| Slabo držanje      | Broj           | 0                   | 0                  | 5           | 4       | 9      |
|                    | % za BMI grupe | 0,0%                | 0,0%               | 33,3%       | 57,1%   | 6,0%   |
| Vrlo loše držanje  | Broj           | 0                   | 0                  | 4           | 1       | 5      |
|                    | % za BMI grupe | 0,0%                | 0,0%               | 26,7%       | 14,3%   | 3,4%   |
| Ukupno             | Broj           | 7                   | 120                | 15          | 7       | 149    |
|                    | % za BMI grupe | 100,0%              | 100,0%             | 100,0%      | 100,0%  | 100,0% |
|                    |                | $\chi^2 = 106,854$  |                    | $r = 0,000$ |         |        |
|                    |                | $\gamma = 0,787$    |                    | $r = 0,000$ |         |        |

Rezultati dobijeni ovim istraživanjem navode na neophodnost kvalitetnijeg organizovanja fizičkih aktivnosti u koje bi bio uključen veliki broj učenika. Takođe, nameće se i obaveza edukovanja dece i mladih, kao i njihovih roditelja o zdravom načinu ishrane, u cilju prevencije gojaznosti.

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#### THE INCIDENCE OF POSTURAL DISORDERS WITH REGARD TO DEGREE OF NUTRITIONAL STATUS IN ADOLESCENTS

*The aim of this research was to analyze postural disorders and nutritional status in adolescents, and to compare the incidence of postural disorders with regard to category of nutritional status and gender dimorphism. The sample of subjects comprised of 305 adolescents, 11 - 14 years old, of both gender (158 girls and 147 boys), students of the primary school from Vojvodina. Postural status was assessed by Napoleon Volanski method. Body mass index (BMI) was calculated by standard procedure. In establishment of differences in postural status, with regard to the age and the gender dimorphism, we used  $\chi^2$  test. In order to establish differences in cumulative evaluation of postural status, we applied rank-sum Mann-Whitney Z-test. In establishment of differences in groups of subjects formed on the basis of degree of nutritional status, we applied Kruskal-Wallis test. The survey results show a statistically significant difference according to the gender, in subjects aged 13 to 14 years, and a significant positive correlation of these two indicators.*

**Key words:** postural status; obesity; BMI; adolescents.

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## **TOURISM EXPENDITURE FOR SPORT ACTIVITIES: THE CASE OF MONTENEGRO**

### **Introduction**

The tourists' expenditure is considered as one of the most important variables and indicators in the economic analysis of a quality of tourist experience that creates destination's tourism industry. Apart from expenditures on core or basic services and products (e.g. accommodation, food, transport), great attention has to be paid to diversification of spending on other services and product that construct so called supporting product (e.g. Kotler, Bowen and Makens, 2006; Middleton and Clarke, 2001; Bowie and Buttle, 2006). Its role is to add value to the core product and to differentiate it from competition. In other words, tourist's interests are very different from the time when people spent their holidays in the sun and near the sea. They are more eager for something new and exciting in destinations which offer active holiday with recreational sports facilities and programs. Therefore, sport and recreational activities represent very important tool for differentiation of the integral product (MSDT, 2010). Accordingly, the research in sports tourism has been growing importantly (Shipway and Jones, 2007; Higham and Hinch, 2002). However, several research underline that even that sport and tourism agencies collaborate effectively many sport tourism policy initiatives have been unsuccessful (Weed, 2003).

As far as the economic impact of sport and recreation is concerned, a considerable discussion has been developed about a number of different aspects of sport/recreational tourism experiences (e.g. events, activities, tours, facilities) and its economic impacts. Recognition of the positive and negative effects associated with sporting mega events is widely acknowledged in the academic literature, especially following: positive image and identity, inward investment, tourism promotion through media coverage, as well as the short term character of mega events with long-term consequences for the local community (Higham, 1999, Solberg and Preuss, 2007). Much attention has been paid to the economic development potential of different sport activities and facilities. In this sense, Cropton (1995) point out the importance and purpose of an economic impact analysis as the way to measure the net economic change that accrue to a community as the result of direct expenditures to sport activities, facilities and events (Cropton, 1995). Also, Higham and Hitch claim that nowadays tourism has more clearly defined link with modern sport, especially in sense of the challenges and potential of overcoming seasonality and more stable economic effects during the year. Apart from this, the experiences from cycle network in the North East of England, UK, confirm

that incomes, group sizes and durations of activity are integrally linked determinants of cyclist's expenditure (Downward, Lumsdon, and Weston, 2009). Key findings indicate that planners of routes have to take into consideration mentioned implications in order to attract all types of cyclists from the most casual leisure trip to racing cyclists. Similarly, in the case of Swedish mountain region, key determinants of expenditures for downhill skiers, backpackers, snowmobilers and general visitors are: duration of stay, household income, choice of activity, occupation, participation in an organized trip and choice of accommodation (Fredman, 2008). Moreover, gender, mode of travel, distance and 'significance of activity' are additional determinants of expenditures outside the destination. In line with above mentioned, complex nature of tourist's expenditure for sport activities has important implications on strategic planning of an integral destination product, both in size and structure. This implies the need for more systematic and planned approach to economic impact analysis of sport and recreational activities in sense of recognition of its positive/negative impacts, as well as the identification of the size and structure of that impacts on the tourism industry, local community and destination.

Therefore, the purpose of this empirically based paper is to examine what type of sport activities generate more intensive tourists' spending in Montenegro in order to better formulate tourism policies related to sport activities. Therefore, this paper contributes to the strand of the literature on the determinants of individual tourist spending with particular focus on sport related activities. Noteworthy, tourist's spending is one of the most important variables of analysis for tourist destinations because it influences directly the tourism sector's profitability (Frechtling, 2006).

The study will help to our understanding of the economic impacts associated with the future development of Montenegrin tourism, including values beyond primary expenditures aimed to accommodation and transport.

The remainder of this paper is organized as follows. Section 2 presents the data and model specification. The results are provided in section 3. Section 4 concludes and suggests policy and managerial implications.

## Methods

The data is extracted from the Montenegrin survey called Guest Survey 2010. The creation of the database is financed and organized by the Ministry of Tourism of Montenegro, the National Tourism Organization of Montenegro and the German Organization for Technical Cooperation (GTZ). The main objective of the survey is to obtain representative view regarding tourist travel behavior and satisfaction during their stay in Montenegro. The questionnaire contains 34 questions and it was translated in seven languages (Montenegrin, English, French, Italian, Albanian, German and Russian). The survey was conducted in 21 municipalities in Montenegro, from July to the end of September 2010. After deleting observations that do not provide all necessary information for this research, we work with a sample of 197 tourists.

The dependent variable denoted *SPORT\_EXPENDITURE* is a continuous variable that measures the tourist spending on sport activities.

To examine our research question, we use three main independent variables. The first variable denoted *MOUNTAIN* is a binary variable equal to 1 if the tourist main sport activity during the stay in Montenegro is mountaineering, mountainbiking, rafting and kayaking. Furthermore, the second variable is named *WATER\_BEACH* since it equals 1 if the tourist main sport activity during the stay in Montenegro is associated to water beach sport activities such as swimming, sailing, waterskiing, etc. Finally, the variable *REST* indicates other sport activities such as riding, fishing, birth watching, etc.

In order to control for firm-level heterogeneity, our analysis includes a set of control variables representing tourist socio-demographic characteristics such as gender, age, wage, country of residence. Additionally, we include type of transportation as a control variable.

The relation between sport activities and tourist's spending is tested using an Ordinary Least Square (OLS) regression. The model can be written as:

$$y_i = X_i a + e_i$$

where  $y_i$ ,  $X_i$ ,  $a$ , and  $e_i$  represent tourist's spending, the vector of exogenous variables, estimated coefficients, and error term, respectively.

## Results

The results of the OLS regression are presented in Table 1. The fit is reasonable with an adjusted pseudo-R-squared of 7% ( $p < 0.001$ ).

**Table 1.** OLS estimates of the relation between of sport activities and tourist's spending

| Variables         | Coefficients and significance |
|-------------------|-------------------------------|
| Intercept         | 93.24***                      |
| MOUNTAIN          | 32.72**                       |
| WATER BEACH       | -11.03                        |
| MAN               | 4.99                          |
| AGE               | -0.47                         |
| EU                | -3.76                         |
| RUSSIA            | 16.89                         |
| REST OF THE WORLD | -3.35                         |
| LOWER INCOME      | -30.03***                     |
| HIGHER INCOME     | 19.92***                      |
| PLAIN             | -21.63*                       |
| BUS               | 1.37                          |
| CAR               | -7.54                         |
| Observations      | 197                           |
| R-squared         | 0.07                          |

(\*), (\*\*), (\*\*\*) indicate parameter significance at the 10, 5 and 1 per cent level, respectively.

References for country of residence is ex Yugoslavia, for income is middle income (from 900 to 3000 Euros) while reference for type of transportation is train.

The results show that findings indicate that mountain related sport activities such as mountaineering, mountainbiking, rafting and kayaking influence positively sport tourists' spending during their stay in Montenegro more than other sport activities ( $\alpha = 32.72$ ,  $p < 0.005$ ). Our findings are consistent with previous studies concluding that sport activities lead to tourist's spending. Moreover, the obtained findings show that variable representing water beach sport activities such as swimming, sailing, water-skiing, etc. is not significant. The results induce that different types of sport activities influence differently tourist's spending. The findings are in the same sense as those from Nogawa et al. (1996). The authors compare cross-country skiers with a walking group participating in a walking event. They find that the walking group respondents spend twice more as those in the cross-country skiing group.

Moreover, looking to control variables, Table 1 shows that the variables representing the type of transport and tourist's income are also significant. More precisely, we can observe that tourist that came by plane spent less on sport activities than tourist that came by train. Additionally, as expected, variable that representing tourist with low income is negatively associated with sport related spending while variable representing high income is positively related to tourist's spending.

### Discussion

The main findings of this study are that different sport activities influence differently on tourists' expenditures in Montenegro. More precisely, we found that continental related sport activities offer influence positively tourist's spending while water beach related sport activities are not significant.

Our results have several managerial implications for the tourism industry in Montenegro. Firstly, policy makers have to further improve existing offer concerning sport activities related to continental areas. Furthermore, the findings that beach water related sport activities are not significant for tourist's spending could represent a problem for Montenegrin tourism if it is not effectively managed. One reason that could explain these results could be due to the fact that cooperation between sport and tourism managers in this area is not well established. Therefore, both destination and sport managers have to work together in order to improve offer related to beach sport activities and due increase the value of experience and financial effect generated by sport activities. Additionally, managers from both tourism and sport organization have to devote sufficient time to developing effective sport tourism linkages (Weed, 2003).

Future research can extend the application to other countries. To go further, it will be very interesting to check how specific sport activities influence tourist's spending.

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#### TOURISM EXPENDITURE FOR SPORT ACTIVITIES: THE CASE OF MONTENEGRO

*The tourists' spending is considered as one of the most important variables in the economic analysis of a destination's tourism industry. Therefore, the purpose of this study is to examine what type of sport activities influence better sport tourists' spending in Montenegro. Working on the sample of 197 tourist, our findings indicate that mount-*

ain related sport activities such as mountaineering, mountainbiking, rafting and kayaking influence positively sport tourists' spending during their stay in Montenegro more than water beach sport activities. The contribution of this research is providing several possible directions for both destination and sport managers to improve existing sport activities related to mountain sport as well as to significantly improve beach sport activities in order increase the value and amounts generated by sport activities.

„Pobjeda“, 26. mart 2015

Konstitutivna sjednica Počasnog odbora konferencije „Sportska dostignuća“

## Đukanović: Sport je najprestižnija djelatnost u državi

**PODGORICA** - Veća međunarodna vidljivost i osnaživanje prepoznatljivog kvaliteta crnogorskog sporta je zajednički zadatak akademske, društvene i sportske zajednice, a udruženim i posvećenim radom ti ciljevi se mogu i ostvariti. To je poruka koju je na sinoćnjoj, konstitutivnoj sjednici uputio Počasni odbor međunarodne naučne konferencije „Sportska dostignuća“, kojim predsjedava predsjednik Vlade Crne Gore Milo Đukanović.

Najavljeno je da će 12. Međunarodna naučna konferencija o transformacionim procesima u sportu „Sportska dostignuća“ biti održana od 2. do 5. aprila u Podgorici, i okupiti više od 160 vodećih svjetskih autora i istraživača u oblasti sportskih, društvenih i humanističkih nauka i medicine iz 22 zemlje.

- Sport je najprestižnija djelatnost u našoj državi, kojom se Crna Gora danas najubjedljivije predstavlja na međunarodnom tržištu. Zato moramo obezbijediti šire razumijevanje i podršku ukupne društvene zajednice ljudima koji su velikim pregnućem doprinijeli tim rezultatima. Važno je razvijati i naučnu dimenziju u oblasti sporta, za šta su zaslužne institucije poput Fakulteta za sport i fizičko vaspitanje - rekao je Đukanović.

U razvoju sporta u Crnoj Gori veliku ulogu i odgovornost ima i akademska zajednica Univerziteta Crne Gore, kroz uspješni višegodišnji rad na promociji, izučavanju i istraživanju sportskih nauka. Posebnu pažnju, svakako, zavređuju događaji poput predstojeće

međunarodne konferencije, koja će okupiti vodeće svjetske autore iz oblasti sportskih nauka, saopšteno je iz PR službe Rektorata UCG.

Rektorka Univerziteta Crne Gore Radmila Vojvodić naglasila je da je sport značajan za vrijednosno ozdravljenje čitavog društva, a Univerzitet predano radi na razvoju njegovog naučno-istraživačkog aspekta.

Članovi odbora su i: ministarka nauke Sanja Vlahović, ministar prosvjete i predsjednik Rukometnog saveza Predrag Bošković, gradonačelnik Podgorice Slavoljub Stijepović, predsjednik Crnogorskog olimpijskog komiteta mr Dušan Simonović, direktor Uprave za mlade i sport Igor Vušurović, predsjednik Atletskog saveza Milorad Vuletić, predsjednica Skijaškog saveza Vesna Medenica, predsjednik Vaterpolo i plivačkog saveza Srdan Kovačević, rukometni trener Dragan Adžić, vaterpolo trener Petar Porobić, fudbalski trener Miodrag Božović, rukometašica Katarina Bulatović, vaterpolista Nikola Janović, džudista Srdan Mrvaljević, sportski direktor Ženskog rukometnog kluba „Budućnost“ Bojana Popović, profesor novosadskog Fakulteta sporta i fizičkog vaspitanja Zoran Milošević i profesor Fakulteta sporta i tjelesnog odgoja iz Sarajeva i potpredsjednik Olimpijskog komiteta Bosne i Hercegovine Izet Rado.

Konferenciju organizuju Crnogorska sportska akademija i Fakultet za sport i fizičko vaspitanje, pod pokroviteljstvom Univerziteta Crne Gore.

**RU-I**





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## **ASSESMENT OF A NEW METHOD HIGHLIGHTING COGNITIVE ATTRIBUTES WITH TABLE TENNIS ATHLETES**

### **Introduction**

Cognition could be defined as the totality of mental processes related to the intelligent functions such as perception, comprehension, learning, recognition and remembrance. In sports sciences it includes specifications such as attention, perception, the speed of comprehension, transformations in short term visual memory, motoric rationalism (game concept), creativity, interpretation, forming correct relationships, the skill to take initiative and tactical skills. These characteristics are extremely valuable in the exhibition of athletes' performances for their athletic achievements. Thus, it is inevitable for the top-tier coaches to implement applications to develop these characteristics in their athletes. In today's extreme athletic competition, it is anticipated that the achievements are only possible through advanced performance awareness in early stages and systematic training applications (Dundar, 2013; Yasar, 2010; Sharkey, 2006).

It is known that physical, anthropometric, conditional and techno-motoric specifications evaluated during the selection and training of athletes could be developed. However, in addition to these basic specifications, the rapid and permanent effects of cognitive features to the improvement of athletes are not paid sufficient attention. Thus, it would be valuable for time and application facilities to design new methods in addition to utilizing methods that determine the cognitive characteristics (Sharkey, 2006; Murat, 1997).

The applications to be implemented are significant for the development of continuous and selective attention levels, recognition and visual skills, reaction times and motoric rationalism (game concept) abilities especially in fast sports. One of the sports that these characteristics are significant is the table tennis. Table tennis is one of the sports that are difficult to learn and perform. An athlete has to perceive the ball that could land in different points on the table, in a short distance and in different rotations and has to perform the right action. The top table tennis athletes, in addition to performing technically difficult actions within a short period of time, have to study their opponents quickly during a competition and to implement a game style that would cover their own weaknesses and attack their opponents' weaknesses to win. Thus, evaluation and development of the attention, perception, learning and adaptation skills of the athletes is a significant factor in training top athletes. This study aims to evaluate the skills of rapid learning, visual perception and adaptation of national level table tennis athletes by developing the Selective Action Array.

### Methods

The study was conducted in table tennis national camp (Bursa, Mimar Sinan Sports Hall) in June 2014. The study was conducted under the approval of the “Uludağ University Faculty of Medicine Clinical Research Ethical Committee.”

11 male and 11 female athletes between the ages of 14 and 18 competing at 2014 Balkan Youth Table Tennis Championship participated in the study. Participants were informed about the study. After the briefing, demographical information, general health data, their years as a table tennis player, daily training times and athletic achievement data were collected.

Consequently, the participants' height, weight and full body impedance measurements were taken using Tanita BC-418 equipment. Afterwards, in a silent room participants' auditory reaction times were recorded. The click sound was transmitted in random intervals during the first 10 trials and then in fixed intervals during the second 10 trials using headphones. The participants were asked to press a button using their dominant hands. Averages of these 10 trials were recorded as **random** and **fixed interval reaction (ms) times**.

### Selective Action Array

Selective Action Array was implemented using a table tennis robot (Butterfly/Amic-3000). The robot was set up to send 26 balls in different colors (6 white, 10 yellow, 10 pink) randomly to different spots on the table in a fixed speed and at a rate of 100 balls a minute. The participants were asked to ignore the white balls, to only touch the yellow balls and to catch the pink balls after the balls bounced from the table once. Participants did not witness the test before they participated. After they were explained about the details of the test, two trials were conducted. The whole test was recorded in video. Every action or the lack of it by the participants was transformed into points on a scoring scale (Table-1) between the points of 0 and 6. For each colored ball, a **color point** and **total points** for the total number of balls were assigned. Furthermore, the percentage of the achievement of the six points worth of action was designated as “**6 points achievement %**.”

### Statistics

The findings were presented as mean  $\pm$  standard deviation and minimum and maximum values. The comparison of intragroup dependent values were done using Wilcoxon rank sum test and for the intergroup comparisons, Mann-Whitney U-test was used. Significance level was accepted as  $p < 0.05$ .

### Results

There was no difference between the age and BMI scores of female and male athletes. While female athletes trained for an average 15.5 hours weekly for an average of 6.4 years, male athletes trained for an average of 21.3 hours weekly for an average of 8.1 years.

**Table 1. Scoring scale**

| <b>White Ball &gt; No reaction</b> | <b>Points</b> |
|------------------------------------|---------------|
| No reaction                        | 6             |
| Reacted but did not touch          | 4             |
| Touched                            | 2             |
| Caught                             | 0             |
| <b>Yellow Ball &gt; Touch</b>      |               |
| No reaction                        | 0             |
| Reacted but did not touch          | 4             |
| Touched                            | 6             |
| Caught                             | 2             |
| <b>Pink Ball &gt; Catch</b>        |               |
| No reaction                        | 0             |
| Reacted but could not touch        | 2             |
| Reacted but could not catch        | 4             |
| Caught                             | 6             |

**Table 2. Age, body mass index (BMI), random and fixed interval auditory reaction times of the male and female table tennis athletes**

|                               | <b>Male Athletes (n=11)</b> | <b>Female Athletes (n=11)</b> | <b>p</b> |
|-------------------------------|-----------------------------|-------------------------------|----------|
| <b>Age (year)</b>             | 16,2 ± 1,3 (14-18)          | 15,6 ± 1,6 (14-18)            | p>0,05   |
| <b>BMI (kg/m<sup>2</sup>)</b> | 21,6 ± 3,0 (16,9-27,1)      | 20,3 ± 2,0 (17,4-24,5)        | p>0,05   |
| <b>RI-RT (ms)</b>             | 211,4 ± 24,3 (178-257)      | 216,7 ± 26,7 (178-264)        | p>0,05   |
| <b>FI-RT (ms)</b>             | 174,6 ± 18,2 (153-207)      | 185,0 ± 23,2 (138-209)        | p>0,05   |

BMI: Body mass index; RI-RT: Random interval-reaction time; FI-RT: Fix interval-reaction time.

Random and fixed interval auditory reaction times for female athletes were found as 216.7 ± 26.7 (178-264) and 185.0 ± 23.2 (138-209) ms respectively. They were found as 211.4 ± 24.3 (178-257) and 174.6 ± 18.2 (153-207) ms for male athletes respectively. There was no significant difference between the reaction times of female and male athletes.

Fixed interval reaction times were found to be significantly less than random interval reaction times.

First trial total points for males in selective action array were observed as 124.4 ± 14.9 (92-144) for males and 107.5 ± 13.0 (90-132) for females, while the second trial total points were 136.2 ± 14.7 (104-156), 118.9 ± 7.8 (108-130), respectively. The increase in the points for the second trial was significant, but the females could not reach the points of males on the first trial even on their second. The most increase in points based on the ball color was observed in the yellow ball.

**Table 3.** Color points, 6 points achievement% and total points for the male table tennis athletes in the Selective Action Array's first and second trials

| Male Athletes (n=11)               | White (n=6)            |                           |        | Yellow (n=10)             |                         |        | Pink (n=10)            |                         |        |
|------------------------------------|------------------------|---------------------------|--------|---------------------------|-------------------------|--------|------------------------|-------------------------|--------|
|                                    | First Trial            | Second Trial              | p      | First Trial               | Second Trial            | p      | First Trial            | Second Trial            | p      |
| <b>Color Point (max: 36,60,60)</b> | 28,0 ± 7,3<br>(16-36)  | 30,9 ± 6,2<br>(22-36)     | p>0,05 | 49,8 ± 5,8<br>(42-60)     | 55,3 ± 6,8<br>(38-60)   | p>0,05 | 46,5 ± 7,5<br>(34-56)  | 49,5 ± 4,4<br>(42-56)   | p>0,05 |
| <b>6 points achievement %</b>      | 61,3 ± 35,8<br>(0-100) | 73,3 ± 35,3<br>(16,7-100) | p>0,05 | 80,0 ± 14,4<br>(55,6-100) | 87,4 ± 17,7<br>(40-100) | p>0,05 | 45,6 ± 22,8<br>(10-80) | 63,8 ± 21,8<br>(30-100) | p>0,05 |

| Male Athletes (n=11)           | First Trial              | Second Trial              | p      |
|--------------------------------|--------------------------|---------------------------|--------|
| <b>Total Points (max: 156)</b> | 124,4 ± 14,9<br>(92-144) | 136,2 ± 14,7<br>(104-156) | p<0,05 |

**Table 4.** Color points, 6 points achievement% and total points for the female table tennis athletes in the Selective Action Array's first and second trials

| Female Athletes (n=11)             | White (n=6)             |                            |        | Yellow (n=10)          |                         |        | Pink (n=10)              |                        |        |
|------------------------------------|-------------------------|----------------------------|--------|------------------------|-------------------------|--------|--------------------------|------------------------|--------|
|                                    | First Trial             | Second Trial               | p      | First Trial            | Second Trial            | p      | First Trial              | Second Trial           | p      |
| <b>Color Point (max: 36,60,60)</b> | 22,4 ± 7,3<br>(14-32)   | 24,5 ± 5,3<br>(18-34)      | p>0,05 | 45,6 ± 7,6<br>(32-58)  | 52,0 ± 6,8<br>(42-60)   | p<0,05 | 39,5 ± 8,3<br>(28-54)    | 42,2 ± 6,2<br>(32-50)  | p>0,05 |
| <b>6 points achievement %</b>      | 37,3 ± 31,3<br>(0-83,3) | 42,5 ± 18,7<br>(16,7-83,3) | p>0,05 | 67,1 ± 13,2<br>(40-80) | 82,0 ± 12,3<br>(60-100) | p<0,05 | 29,5 ± 15,0<br>(10-54,5) | 38,4 ± 19,9<br>(10-70) | p>0,05 |

| Female Athletes (n=11)         | First Trial              | Second Trial             | p      |
|--------------------------------|--------------------------|--------------------------|--------|
| <b>Total Points (max: 156)</b> | 107,5 ± 13,0<br>(90-132) | 118,9 ± 7,8<br>(108-130) | p<0,05 |

On 6 points achievement % values, males accomplished better achievement levels both in the first and second trials than females. The biggest difference in achievement levels between the females and males was observed in white ball achievement rates. Males performed 23% better in the first trial and 31% in the second trial on white ball achievement rates. The highest increase in achievement between the first and second trials was observed in the yellow ball points among females, while it was observed in the pink ball among males.

The highest total achievement rate was observed in the yellow ball, while the lowest was observed with the pink ball.

### Discussion

In the study a new method to evaluate cognitive features and auditory reaction times in national team table tennis athletes were investigated. A gradual degree of difficulty was created in the selective action array, which was implemented by associating different actions to 3 balls in different colors sent to different areas of the table in the rate of 100 balls a minute by a table tennis robot. In the two trials conducted with the participants that did not see the trials before but only instructed what to do, cognitive concepts such as adaptation, learning and reinforcement were associated with motor skills. Visual perception and attention, continuity of the concentration and motor coordination were evaluated by different skill levels observed by catching and leaving some, by touching some and not reacting to some of the 26 table tennis balls with a diameter of 40 mm and 2.7 gr of weight that arrive successively and with speed.

The facts that table tennis athletes are subjected to visual and auditory stimuli during high-speed competition and the shortness of the length of the stimulus and the need to distinguish the direction of the stimulus demonstrate the significance of the cognitive features for the athletes who play table tennis (Bhabhor, 2013; Can, 2008). Lapszo (1999) tested psychomotor efficiency particular to table tennis in table tennis players in a simulator and determined that the expectation for the movement to hit the ball and leading expectations were high. Polat (2000) demonstrated that the reaction times of table tennis players to visual and auditory stimuli were sedentary and lower than the tennis players.

This study attempted to evaluate the achievement levels of the table tennis players in different actions based on visual perception, learning skills based on repetition and adaptation in the application using selective action array. Thus, since table tennis players react to every ball during a game, our request in the selective action array that players should not respond to the white balls was turned into a difficult task. The highest rate of success was observed with the yellow balls because it was identical to the sport of table tennis. The action that would be taken with the pink balls was significant in being the most difficult task in the array. Although the lowest achievement rates were observed with the pink balls, males displayed the highest adaptation to the pink balls in the second trial.

Fixed interval reaction time to be lower than random interval reaction time was an expected finding. Predictability of the timing of the sound results in a 35 ms quicker reaction, which is 16% quicker.

The Selective Action Array the study aimed to develop would help evaluate the adaptation skills of athletes to a new situation, their learning speed, the length of focusing their attention and their visual perceptions quantitatively. This method, which could be adjusted for other branches of sports, would introduce a different point of view to evaluate cognitive characteristics, which are as significant as condition and motor motions in sports.

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### ASSESSMENT OF A NEW METHOD HIGHLIGHTING COGNITIVE ATTRIBUTES WITH TABLE TENNIS ATHLETES

*Table tennis is among sports that are difficult to learn and perform. The athlete has to perceive the ball within a small space, in various rotations, aiming at different points on the table in various speeds, and perform the correct action. Thus, assessing and improving the athletes' attention, perception, fast learning and adaptation skills are of significant importance in training top-tier athletes. This study evaluates fast learning, visual perception and adaptation skills of national athletes using Selective Action Array, developed recently for the study.*

*Out of 2014 Balkan Youth Table Tennis Championship athletes, 11 male and 11 females participated in the study. For the Selective Action Array, a table tennis robot was utilized. Robot was set up to send a total of 26 balls in 3 different colors (6 whites, 10 yellows, 10 pinks) to different areas of the table, in random colors and at the rate of 90 balls per minute. The participants were asked to ignore the white balls, to touch the yellow balls and to grab the pink balls using their dominant hands. Pursuant to*

explaining the task to the participants, two consecutive trials were executed and recorded using a camera. Every action performed/not performed by the participants was transformed into points in the scoring system.

First trial total points in the Selective Action Array were  $124 \pm 15$  for males and  $108 \pm 13$  for females, whereas on the second trial the total points were  $136 \pm 15$  and  $119 \pm 8$ , respectively. The higher scores obtained in the second trial were significant. Success rates for males were higher than females in both trials based on the percentage of correct actions executed in relation with the ball. The most significant difference between males and females was observed in white ball success rates. The highest increase in the rate of success in the second trial as compared to the first was observed in yellow balls for females and in pink balls for males. Table tennis players usually react to every ball on their side of the table; hence our request to ignore the white balls in our selective action array became a handicap for all.

The study aimed to evaluate the adaptation skills, learning speed, focus span and visual perceptions of athletes quantitatively via our experimental Selective Action Array. This method, which could also be applied to other sports branches, could introduce a different point of view to evaluate cognitive attributes, which are as important as condition and motor skills in sports.

„Blic“, 27. mart 2015.



» Prva sjednica počasnog odbora „Sportska dostignuća“

## Vojvodić: Sport vodi ozdravljenju društva

» **PODGORICA** Veća međunarodna vidljivost i osnaživanje prepoznatljivog kvaliteta crnogorskog sporta zajednički je zadatak akademske, društvene i sportske zajednice, a udruženim i posvećenim radom ti ciljevi se mogu i ostvariti. To je poruka koju je na održanoj prvoj konstitutivnoj sjednici uputio Po-

časni odbor međunarodne naučne konferencije

„Sportska dostignuća“, kojim predsjedava predsjednik Vlade Crne Gore Milo Đukanović.

Sport predstavlja najprestižniju djelatnost u našoj državi, kojom se Crna Gora danas najubjedljivije predstavlja na međunarodnom

tržištu. Zato moramo obezbijediti šire razumijevanje i podršku ukupne društvene zajednice ljudima koji su velikim pregnućem doprinijeli tim rezultatima. Važno je razvijati i naučnu dimenziju u oblasti sporta, za šta su zaslužne institucije poput Fakulteta za sport i fizičko vaspitanje,

istakao je premijer Đukanović.

Rektorka Univerziteta Crne Gore Radmila Vojvodić naglasila je da je sport značajan za vrijednosno ozdravljenje čitavog društva, a Univerzitet predano radi na razvoju njegovog naučnoistraživačkog aspekta. ■

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## DIFFERENCES IN ANTHROPOMETRIC SPACE AND MOTOR DIMENSIONS BETWEEN THE TWO CLUBS IN THE CATEGORY OF CADETS

### Introduction

The game of handball as in other countries, is as well a very popular sport in our country, this game is one of the most advanced sports especially women's handball.

In recent time, contemporary handball possesses complex fast movements that require a high physical training, with high morphological and technical-tactical characteristics, strong kick on goal and very precise are elements that give high attractiveness to this sport.

Based on all the achievements in the sport of handball also in our country has raised great interest to females, especially in younger age groups.

Based on all these data we will try through this scientific research paper to verify and as base we took two clubs of Handball Super League in Kosovo in the category of cadets:

FHC Prishtina from Prishtina and FHC Vëllaznimi from Gjakova female players that are regular on their trainings organized by the club and take part in official competitions.

Group is consisted from 25 handball players of the club FHC "Prishtina" – and from 25 handball players of the club FHC "Vëllaznimi" females of age 15 ( $\pm 6$  months).

Through this paper we will try to make the difference of players from two clubs that in the ranking table in official competitions organized by the Kosovo Federation do not stay close to each other because handball team Prishtina is on the upper part of the table and competes for the Champion's title, whereas the team Vllaznimi is on the bottom part of the table.

### Methods

Methods of processing the results, *Sample of anthropometric variables:* APESHA – Body weight, ALARTE – Body height, AGJASHU- Length of palm, AGJRSHU- Width of palm.

**Group of measuring instruments of motor variables:** Jumping from place in length – MKVGJ, High jumping from place- MKVL, Running in resistance 3x1 – MVR3KU



, MVZ6-9m - Zigzag running between 6 and 9 meters.

Based on the purpose of the paper we will implement some of the processing methods from which will be provided information for realization of this study.

For the group of players of system variables will be calculated basic statistical parameters.

- T-test Analysis

## Results and Discussions

### T – Testi for anthropometric and motor variables between two handball clubs

In order to verify the differences between the two group, the first group FHC Prishtina team of cadets and second group, FHC Vëllaznimi from Gjakova both teams members of Handball Super League in Kosovo, where for each variable is T- Test. With this method we verify the differences how many times is higher the difference of two arithmetic averages and how is its standard error. In order to show statistical validation, through this test we must know the value of T-Testi, which is  $T > 1.96$ , in the level of statistical significance  $p < .05$ . We can say that the difference between these two groups in one variable is statistically valid.

In Table 1 are presented the data between two clubs which are processed by T-Test.

Based on the results presented in Table 1 we can see the changes for each variable separately and can conclude that there was a valid statistical difference in anthropometric space in three variables wherethe most successful players of Prishtina team were in variables of body weight, body height and length of palm.

In motor space also we have valid statistical differences, presented in table 2 also to the motor space relatively large differences and in favour of FHC Prishtina, are shown in tests of explosive force as in high jumping from place and jumping from place in length and in the test of zigzag running from 6 meters to 9 meters. With these results we can say that is verified the purpose of this paper because are presented large differences between these two clubs and FHC Prishtina is a club which is ranked in the top of the table whereas FHC Vëllaznimi in the bottom of the table.

**Table 1.** T- test of anthropometric variables of two clubs

|          | Club           | N  | Mean     | Std. Deviation | Sig. (2-tailed) |
|----------|----------------|----|----------|----------------|-----------------|
| Apesha   | Fhc Prishtina  | 25 | 56.8072  | 10.54462       | <b>.010</b>     |
|          | Fhc Vëllaznimi | 25 | 50.9760  | 2.45514        |                 |
| Alart    | Fhc Prishtina  | 25 | 163.8712 | 7.48064        | <b>.040</b>     |
|          | Fhc Vëllaznimi | 25 | 160.2800 | 4.05195        |                 |
| Agjashu  | Fhc Prishtina  | 25 | 18.5388  | 1.16678        | <b>.001</b>     |
|          | Fhc Vëllaznimi | 25 | 17.6200  | .61712         |                 |
| Agjershu | Fhc Prishtina  | 25 | 19.8536  | .85387         | .422            |
|          | Fhc Vëllaznimi | 25 | 19.6600  | .83815         |                 |

**Tabela 2.** *T- test of anthropometric variables of two clubs*

|         |                | N  | Mean     | Std. Deviation | Sig. (2-tailed) |
|---------|----------------|----|----------|----------------|-----------------|
| Mkgjv   | Fhc Prishtina  | 25 | 162.0800 | 13.68978       | .012            |
|         | Fhc Vëllaznimi | 25 | 149.8000 | 19.28946       |                 |
| Mklv    | Fhc Prishtina  | 25 | 31.0800  | 2.61279        | .000            |
|         | Fhc Vëllaznimi | 25 | 26.5600  | 2.06317        |                 |
| Mvr3ku  | Fhc Prishtina  | 25 | 17.3872  | 1.07069        | .312            |
|         | Fhc Vëllaznimi | 25 | 17.0316  | 1.37392        |                 |
| Mvz6m9m | Fhc Prishtina  | 25 | 11.2444  | 1.34585        | .000            |
|         | Fhc Vëllaznimi | 25 | 15.1216  | 1.05999        |                 |

### Conclusion

The main purpose of this research was to determine and evaluate the differences in anthropometric space and in motor dimensions between two handball clubs that compete in Kosovo Handball Super League in the Category of cadets for females of FHC "Prishtina" which is on the top of the table and FHC "Vëllaznimi" that is positioned in the bottom of the table.

Based on the importance of this paper to present the differences that exist between these two clubs in anthropometric and motor space, we have achieved that through the analysis of T-test we can verify these differences that are more significant in favour of FHC Prishtina which is in the top of the table and that during the week organizes 5 trainings and one official game during the weekend, larger differences were presented in anthropometric space in three variables where more successful were players of the Prishtina team in the variables of Body weight, Body height, and Length of palm, whereas in motor space in favor of FHC Prishtina are presented in the testings of explosive force as in high jump from place and length jump from place and in the test of zigzag running from 6 meters to 9 meters.

In the end of this paper we can say that the value of this work may be in the function of information for the development of anthropometric and motor characteristics of handball players in the category of cadets which with the work and a good program of training process by coaches of this category results will not be absent in Kosovo Handball Super League championship.

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#### *DIFFERENCES IN ANTHROPOMETRIC SPACE AND MOTOR DIMENSIONS BETWEEN THE TWO CLUBS IN THE CATEGORY OF CADETS*

*The purpose of the research has to do with the defining the differences in anthropometric variables and motor assessment skills of the female cadets between KHF. “Prishtina” from Prishtina and Khf Vëllaznimi, from Gjakova.*

*- The primary goal of this research is to certify differences of anthropometric and motor skills characteristics of female cadets of KHF. “Prishtina” and Khf Vëllaznimi also to be verified the parameters that affect in development and their training for handball game.*

*Besides this should be verified what impact had the implementation of program and teaching process of motor activities in the development of anthropometric characteristics, motor skills or the successful placement of Kosovo super league competitions in handball.*

*Based on the importance of this paper to present the differences that exist between these two clubs in anthropometric and motor space, we have achieved that through the analysis of T-test we can verify these differences that are more significant in favour of FHC Prishtina which is in the top of the table and that during the week organizes 5 trainings and one official game during the weekend, larger differences were presented in anthropometric space in three variables where more successful were players of the Prishtina team in the variables of Body weight, Body height, and Length of palm, whereas in motor space in favor of FHC Prishtina are presented in the testings of explosive force as in high jump from place and length jump from place and in the test of zigzag running from 6 meters to 9 meters.*

**Key Words:** *Anthropometric and motor dimensions, Handball players, two teams.*

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## **CANONIC RELATIONS OF ANTHROPOMETRIC AND MOTOR SPACE BETWEEN STUDENTS AS FOOTBALL PLAYERS AND NON FOOTBALL PLAYERS**

### **Introduction**

The game of football is one of the most popular sport on earth, including our country. Contemporary football possesses complex movements as in technical-tactical aspect, as well as in a high physical preparation, with eloquent and fast actions, then with effective and perfect kick. At certain situational moments these are giving to the game of football a dynamics and glow that awakens the curiosity of athletes and great interest among younger groups.

Even scientific researchers are looking for means and more professional methods and optimal consignments so that children from a young age to give accurate kinesiological information, that in sport, respectively in Kosovo football are scarce, and to them should be given special treatment and to work with young age groups in a professional way.

Considering the importance of anthropometric and motor parameters, then should be selected adequate operators in order to transform and do right orientation of achieving more qualitative results in football, respectively in sport in general.

Creativity and professional-scientific ability are the guidelines for the implementation and evaluation of the achieved results in sport. Based on this, the aim of this paper is supported on strengthening and in verification of scientific legalities in the achievements of qualitative results in football, as well as in changes in anthropometric and motor dimensions and situational movements. Also it will be reflected creativity and the work of physical education teachers in elementary school. With other words, I think that will be reflected statistical valid differences between students who attended football school with those that have not attended this school, but have attended physical education class during teaching process.

### **Methods**

The sample of the tested students in this research was chosen as a population of males, 13 – years old,  $\pm$  six months. The total number of the tested students is 142 entities, 71- are students from 13 - years  $\pm$  6 months, of the elementary school „Vëllëzerit Frashëri „ from Lypjan, that attended only regular hours of physical education at the school, whereas other group of the tested students 71- are regular students of the

football school at the football club „ULPIANA“ from Lypjan. But the majority of students gravitate from villages of the Lypjan Municipality, that besides the hours of physical education have attended 3-4 trainings during the week and one full year have attended football school.

In this research were applied 10- anthropometric variables which are considered to cover enough space for this research work, this method have used in their work (Hasangjekaj, B. & Kikaj Xh. 1999).: evaluation dimension of volume and body mass, definition of longitudinal dimension of skeleton and definition of transversal dimension of skeleton: APESHA–Body weight, ALARTE–Body height, AGJAKË–Length of the leg, AGJASHP–Length of the foot, APEGJO–Chest perimeter, APEBEL–Waist perimeter, APEKOF–Thigh perimeter, APEKËR–Shinbone perimeter, ADIGJU–Knee diameter, ADIZOG–Ankle diameter. BASIC MOVEMENT TESTS: MVR–Running on 20m, high start. MKGJV–Long jump from the place. MTAPKE–Taping legs.

SITUATIONAL MOVEMENT TESTS: MPUTOP–Work with the ball, MUSLLA–Ball leading in slalom 10m, MUGJRR–Ball leading in half a circle, MUTK 20m–Ball leading in the hall 20m.

### Results and discussion

For common explanation of both spaces, anthropometric and motor, must be used canonical analysis which is presented in tables 1,2,3,4,5,6.

In table 1, is presented matrix of canonical correlations of motor space by the right side (right set). Based on the values presented in correlation matrix in motor space is shown to have had connectivity in most of the tests, except test MUGJRR, which showed a weaker connectivity. Whereas in Table 2, was given matrix of canonical correlation in anthropometric space by the left side (left set). Based on the data in the table is shown that all variables indicated high correlative connectivity except variable ADIGJU. In Table 3, are given crosscorrelation values or interconnection between anthropometric and motor space for two groups (students as football players non football players). Based on the values shown on the table, it is seen not so stressed intercorrelation between the two spaces, only variables of circular dimension ADIGJU and ADIZOG which have connectivity with all motor tests except tests MTAPKË and MPUTOP.

**Table 1.** Canonical motor correlations

|          | Mkgjv | Mtapkë | Mvr 20m | Mputop | Muslla | Mugjrr | Mutk 20m |
|----------|-------|--------|---------|--------|--------|--------|----------|
| Mkgjv    | 1.000 |        |         |        |        |        |          |
| Mtapkë   | .252  | 1.000  |         |        |        |        |          |
| Mvr 20m  | -.575 | -.294  | 1, 000  |        |        |        |          |
| Mputop   | .057  | .447   | -.070   | 1,000  |        |        |          |
| Muslla   | -.336 | -.308  | ,312    | -.247  | 1,000  |        |          |
| Mugjrr   | -.118 | -.119  | ,174    | -.105  | ,247   | 1,000  |          |
| Mutk 20m | -.521 | -.252  | ,591    | -.030  | ,493   | ,161   | 1,000    |

**Table 2.** *Canonical anthropometric correlations*

|        | Apesha | Alarte | Agjake | Agjash | Apegjo | Apebel | Apekof | Apeker | Adigju | Adizog |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| Apesha | 1,0000 |        |        |        |        |        |        |        |        |        |
| Alarte | ,789   | 1,0000 |        |        |        |        |        |        |        |        |
| Agjake | ,709   | ,854   | 1,0000 |        |        |        |        |        |        |        |
| Agjash | ,711   | ,756   | ,721   | 1,0000 |        |        |        |        |        |        |
| Apegjo | ,824   | ,713   | ,612   | ,664   | 1,0000 |        |        |        |        |        |
| Apebel | ,776   | ,561   | ,556   | ,571   | ,781   | 1,0000 |        |        |        |        |
| Apekof | ,818   | ,574   | ,568   | ,585   | ,743   | ,797   | 1,0000 |        |        |        |
| Apeker | ,796   | ,658   | ,605   | ,614   | ,774   | ,714   | ,783   | 1,0000 |        |        |
| Adigju | ,191   | ,114   | ,249   | ,137   | ,175   | ,301   | ,223   | ,258   | 1,0000 |        |
| Adizog | ,257   | ,261   | ,389   | ,283   | ,251   | ,331   | ,250   | ,315   | ,855   | 1,0000 |

**Table 3.** *Canonical Crosscorrelations*

|        | Mkgjv | Mtapkē | Mvr 20m | Mputop | Muslla  | Mugjrr | Mutk 20m |
|--------|-------|--------|---------|--------|---------|--------|----------|
| Apesha | .138  | .111   | -.130   | .115   | .003    | .030   | .057     |
| Alarte | .153  | .126   | -.027   | .196   | .075    | .026   | .181     |
| Agjake | .165  | .097   | -.041   | .268   | -.034.. | .041   | .056     |
| Agjash | .154  | .170   | -.033   | .183   | .021    | .110   | .085     |
| Apegjo | .193  | .167   | -.068   | .118   | -.083   | .014   | .016     |
| Apebel | .156  | .097   | -.123   | .082   | -.137   | -.062  | -.102    |
| Apekof | .125  | .134   | -.167   | .094   | -.045   | -.007  | -.031    |
| Apeker | .248  | .104   | -.173   | .060   | -.033   | .078   | -.078    |
| Adigju | .410  | -.012  | -.357   | .089   | -.375   | -.209  | -.492    |
| Adizog | .390  | .004   | -.278   | .044   | -.290   | -.226  | -.387    |

For the explanation of anthropometric and motor space is shown in Table 4, which presents the correlation coefficient (R), level of validity (P), level of freedom (Df) and roots of canonical correlation and common variance ( $R^2$ ). In this case are shown two canonical factors: First canonical factor has the value of canonical coefficient  $R=.66$  which explains with 44% of the common variance  $R^2=.44$  in the level of validity  $p=0.00$ .

Second canonical factor has weaker connectivity of canonical correlation with  $R=.40$  which explains common variance with 16% and with validity  $P=0.04$ , table 4.

Whereas, in table 5, are given the values of first canonical factor in morphological space, whose vectors go in two directions with bipolar dimension which projects the variable vector ALARTE in positive pole .55, while in negative pole is projected variable vector ADIGJU -.74. In motor space in table 6, is distinguished the test for leading the ball corridor 20 meters MUTK .70, whereas weak correlative links are showing the tests MUSLLA and MUTK 20m. We can conclude that between anthropometric and motor space is a relation between knee diameter and body height, which have effect in performing the test of leading the ball in corridor 20 meters.

Second canonical factor presents a weaker relation of canonical correlation with  $R=.40$  which explains common variance with 16% and with validity  $P=0.04$ , table 4. As well as vectors of second canonical factor table 5 are distributed in two directions with bipolar

dimension where in morphological space, variables give the value of different poles as: ALARTE-.68, AGJASHE -.51, APEGJO -.41, APEKER -.74, ADIGJU -.43, vectors of which are projected in negative pole, while in positive pole are projected the variables: AGJAKE .54, and APEBEL .57. Whereas in motor space is distinguished only the test of explosive force, vectors of which are projected in negative pole MKGJV -1.03 which has a great relation with the variables of anthropometric space that enable realization of a long jump from place. As a conclusion we can say that from the structure of first canonical relation from the factors of anthropometric and motor space, it is shown that the test of leading the ball in corridor is dependent from a knee diameter and body height, while in second canonical factor test of explosive force is dependent on many anthropometric variables: ALARTE, AGJASHE, APEBEL, APEKER, ADIGJU, AGJAKE, APEGJO.

**Table 4. Canonical relation**

|   | R-Canonic | R <sup>2</sup> -Canonic | h <sup>2</sup> | Df | P – level  | λ   |
|---|-----------|-------------------------|----------------|----|------------|-----|
| 0 | ,66       | ,44                     | 150,85         | 70 | <b>,00</b> | ,31 |
| 1 | ,40       | ,16                     | 73,18          | 54 | <b>,04</b> | ,57 |
| 2 | ,39       | ,15                     | 50,00          | 40 | ,13        | ,68 |
| 3 | ,29       | ,08                     | 27,81          | 28 | ,47        | ,81 |
| 4 | ,24       | ,06                     | 15,83          | 18 | ,60        | ,88 |
| 5 | ,20       | ,04                     | 7,34           | 10 | ,69        | ,94 |
| 6 | ,10       | ,01                     | 1,58           | 4  | ,81        | ,98 |

**Table 5. Canonical anthropometric factors**

|        | Root 1      | Root 2      |
|--------|-------------|-------------|
| Apesha | .07         | .39         |
| Alarte | <b>.55</b>  | <b>-.68</b> |
| Agjake | -.07        | <b>.54</b>  |
| Agjash | .19         | <b>-.51</b> |
| Apegjo | -.10        | <b>-.41</b> |
| Apebel | -.26        | <b>.57</b>  |
| Apekof | .23         | .22         |
| Apeker | -.21        | <b>-.74</b> |
| Adigju | <b>-.74</b> | <b>-.42</b> |
| Adizog | -.11        | .07         |

**Table 6. Canonical motor factors**

|          | Root 1     | Root 2       |
|----------|------------|--------------|
| Mkgjv    | -.09       | <b>-1.03</b> |
| Mtapkë   | .32        | -.24         |
| Mvr 20m  | -.04       | .00          |
| Mputop   | .25        | .02          |
| Muslla   | .35        | -.06         |
| Mugjrr   | .18        | -.36         |
| Mutk 20m | <b>.70</b> | -.38         |

### Conclusion

Based on the main purpose of this paper is the verification of anthropometric and motor relations of situational movements between students as football players and non football players of 13 years of age.

In this paper was treated the sample of 142 entities students of elementary school “Vëllëzërit Frashëri” in Lipjan, 71 students non active in football and 71 followers of football school.

In order to verify relations between anthropometric and motor space, we conducted also canonical analysis, in this case we have obtained two matrixes of correlations, to the right and to the left side, where are shown the correlations in anthropometric and motor space.

Crosscorrelations or interconnections altogether (right with left set), it is noticed that circular dimension factor has connectivity with all the situational tests in football with what once again confirms the importance of this factor for football players and it remains that in the future this process to continue and to have further researches.

Whereas from the structure of first canonical relations from the factors of anthropometric and motor variables, it is shown that test of leading the ball in corridor is dependent by anthropometric variables, knee diameter and body height, whereas in second canonical factor, the test of explosive force is dependent by more than anthropometric variables as: ALARTE, AGJASHE, APEBEL, APEKER, ADIGJU, AGJAKE, APEGJO.

Such research studies are good predictors and should be considered especially during the selection of children and youth for football school, for orientation and their selection in sport.

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#### *CANONIC RELATIONS OF ANTHROPOMETRIC AND MOTOR SPACE BETWEEN STUDENTS AS FOOTBALL PLAYERS AND NON FOOTBALL PLAYERS*

*Taking into account morphological and motor status of students as football players and non-football players of the same age is meant by certain basic system of anthropometric latent dimensions which are developed with the influence of endogenous and exogenous factors and the impact of physical education classes and various trainings outside education process.*

*While in motor space the research is defined in basic motorics and in situational motorics of the game of football, where in some of them important factor is genetic factor or born.*

*Through this study we can conclude that the work that students do in schools during the physical education classes, will change with the students as football players who had the opportunity that except physical education classes to work as well during the week 3 to 4 trainings in the football school.*

*For this research was taken the sample of 142 entities of 13 – years of age from them 71 students and active football players and 71 students that aren't active football players.*

*The main purpose of this paper is verification of canonic relations, anthropometric space, and basic motor tests and situational motorics to the students as football players and non-football players and also we will verify the implementation and impact of anthropometric space parameters in basic motor tests and situational motorics.*

**Key Words:** *Anthropometric and motor dimensions, Football players and non football players.*

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## STAVOVI SRBIJANSKIH POTROŠAČA PREMA REKLAMIRANJU U SPORTU U ODNOSU NA UČESTALOST KUPOVINE SPORTSKIH PROIZVODA

### UVOD

Vodeći se činjenicom da ulaganje u reklamiranje u sportu kao dijelu korporativne marketinške strategije neprestano raste, nastala je i potreba za istraživanjima u ovoj oblasti kako bi se utvrdile najbolje perspektive (Muratović, Bjelica i Popović, 2014). Sistematsko proučavanje odnosa potrošača prema reklamiranju je ukorijenjeno u studiji Bauer-a i Greyser-a (1968), dok je kapitalan doprinos nauci kada je reklamiranje u sportu u pitanju dao Pyun (2006), koji je sproveo istraživanje u kojem je konstruisao novi model za ispitivanje odnosa potrošača prema reklamiranju u sportu i time omogućio raznim istraživačima da ispituju stavove i uvjerenja potrošača prema reklamiranju u sportu širom svijeta i upoređuju svoje zaključke. U studijama koje su istraživale stavove prema opštem reklamiranju četrdesetih i pedesetih godina prošlog vijeka, prema Bauer-u i Greyser-u (1968), došlo se do zaključaka da potrošači, generalno imaju pozitivne stavove. Međutim, upoređujući navedene rezultate sa novijim istraživanjima (citirano u Popović, Jakšić, Matić, Bjelica i Maksimović, 2014), utvrđeno je da potrošači imaju sve više negativne stavove prema reklamiranju. Nakon određenog vremena, Shavitt i saradnici (1998) su utvrdili da ispitanici u njihovoj studiji imaju mnogo pozitivnije stavove prema reklamiranju nego što je to bio slučaj u prethodnim studijama. Pretpostavljalo se da stavovi variraju kod ispitanika, budući da oni razlikuju određene vrste reklamnih poruka, a Mittal (1994) je utvrdio da imaju mnogo negativnije stavove prema reklamiranju na televiziji u poređenju sa opštim reklamiranjem. Sa povećanjem negativnih stavova prema reklamiranju na televiziji marketari su morali osmisliti način kako da povrate povjerenje gledalaca kada je reklamiranje na televiziji u pitanju, a to su učinili na taj način što su počeli primjenjivati nove tehnološke izume kao što su video rekorderi koji su nezainteresovanim gledaocima za reklamne poruke omogućavali da preskoče djelove koji im nijesu bili interesantni (citirano u Popović, Jakšić, Matić, Bjelica i Maksimović, 2014). Uporedo sa navedenim izumom, stavovi prema reklami-

ranju u sportu su, ponovo postali mnogo pozitivniji. Međutim, ovaj izum nije omogućavao gledaocima da preskoče dio programa sa reklamnim porukama kada je praćenje sportskih događaja u pitanju, budući da su se oni odvijali u trenutku vremena. Ipak, bliskost i privrženost sportu i sportskim subjektima su, vremenom nadvladale negativne stavove gledalaca koje su oni imali kada je opšte reklamiranje u pitanju i dovele do toga da reklamiranje u sportu zauzme mjesto koje mu danas pripada. Samim tim se postavilo i pitanje, kako kupovina sportskih proizvoda utiče na stavove potrošača prema reklamiranju u sportu kao jedan od niza problema sa kojim se nije susrećalo mnogo autora do sada (Bjelica i Popović, 2011; Popović, 2011; Popović, Molnar i Radovanović, 2011), a to upravo i jeste cilj ove studije.

## METOD

Populaciju u ovoj studiji su činili studenti Fakulteta za sport i fizičko vaspitanje na Univerzitetu u Novom Sadu koji su, u vrijeme anketiranja, imali boravište na teritoriji Srbije, dok je uzorak ispitanika organizovan putem kombinovanja ili raslojavanja, tako da su obrađena različita svojstva navedene populacije i različiti prostori na kojima je ona egzistirala.

Anketa je sprovedena tokom ljetnjeg semestra 2012/2013 školske godine a upitnici su distribuirani studentima osnovnih studija u štampanom i elektronskom obliku. Ukupno je prikupljeno 150 upitnika, ali je 29 upitnika bilo isključeno iz analize, budući da nisu bili adekvatno popunjeni, tako da je u istraživanju učestvovalo, ukupno 121 ispitanika. Instrument istraživanja je predstavljao standardizovani upitnik koji je bio prilagođen prema Pyun-u (2006) i sastojao se iz dva dijela, generalnih stavova prema reklamiranju u sportu i socio-demografskih karakteristika ispitanika kada je učestalost bavljenja sportskim aktivnostima tokom mjeseca u pitanju. Sistem promjenljivih u ovom upitniku je sadržao tri tvrdnje koje su ispitanici trebali da ocijene prema sedmo-stepenoj Likertovoj skali vrijednosti, kao i šest socio-demografskih karakteristika ispitanika (uopšte ne kupuju, manje od jednom mjesečno, 1–3 puta mjesečno, 4–6 puta mjesečno, 7–9 puta mjesečno, i više od 10 puta mjesečno). Popunjavanje upitnika nije trajalo, u prosjeku više od 10 minuta a ispitanici su u anketi učestvovali na dobrovoljnoj osnovi. Važno je naglasiti da je anketa bila anonimna i da su svi odgovori bili strogo poverljivi. Vrijedno je napomenuti i da su ispitanici, i pored svega navedenog imali mogućnost da u svakom momentu opozovu svoje učešće u anketi, ali se niko od njih nije na tako nešto odlučio.

Empirijski podaci su analizirani putem statističkog paketa za društvene nauke (SPSS 20.0), a kao prvi korak, bila je primijenjena deskriptivna statistika kojom su izračunati, prije svega frekvencija, zatim aritmetička sredina, standardna devijacija, kao i mjere asimetrije (Skewness) i spljoštenosti (Kurtosis) za svaku od tvrdnji. Budući da su se promjenljive u ovoj studiji nalazile na naparametrijskim skalama, radi detaljnijih analiza koje su slijedile, bilo je neophodno da se one, primjenom Blomovog postupka, transformišu u skale višeg reda. Zatim su primjenom multivarijatne analize varijanse (MANOVA), univarijatne analize varijanse (ANOVA) i LSD Post Hoc testa utvrđene

razlike u generalnim stavovima ispitanika prema reklamiranju u sportu u odnosu na učestalost bavljenja sportskim aktivnostima tokom mjeseca.

## REZULTATI

U prvoj tabeli su prikazani deskriptivni statistici za sve tri tvrdnje koje su se odnosile na generalne stavove ispitanika prema reklamiranju u sportu. Prije svega, prikazana je aritmetička sredina koja oslikava pozitivne vrijednosti stavova kada su sve tri tvrdnje u pitanju, dok vrijednosti standardne devijacije pokazuju da elementi skupa, u prosjeku ne odstupaju značajno od aritmetičke sredine. Kada se govori o mjerama asimetrije (Skewness) i spljoštenosti (Kurtosis), negativne vrijednosti asimetrije kod svih promjenljivih pokazuju da je većina rezultata desno od srednje vrijednosti, među većim vrijednostima, dok negativne vrijednosti spljoštenosti kod jedne promjenljive (GSS1) pokazuju da je raspodjela pljosnatija od normalne tj. da ima više slučajeva na repovima a pozitivna vrijednost kod jedne promjenljive (GSS2 i GSS3) da je raspodjela šiljatija od normalne tj. da ima više rezultata nagomilanih oko centra raspodjele.

**Tabela 1.** Generalni stavovi prema reklamiranju u sportu

|      | Mean | S.D.  | Skewness  |      | Kurtosis  |      |
|------|------|-------|-----------|------|-----------|------|
|      |      |       | Statistic | S.E. | Statistic | S.E. |
| GSS1 | 5,09 | 1,845 | -,834     | ,215 | -,380     | ,427 |
| GSS2 | 5,39 | 1,840 | -1,203    | ,217 | ,387      | ,430 |
| GSS3 | 4,93 | 1,430 | -,693     | ,219 | ,259      | ,435 |

Legenda:

Mean – Aritmetička sredina

S.D. – Standardna devijacija

Skewness – Mjera asimetrije

Kurtosis – Mjera spljoštenosti

Statistic – Statistička vrijednost

S.E. – Standardna greška

GSS1 – Moje opšte mišljenje je naklonjeno reklamiranju u sportu

GSS2 – Sveobuhvatno, smatram da je reklamiranje u sportu dobra stvar

GSS3 – Sveobuhvatno, da li ne volite ili volite reklamiranje u sportu

U nastavku ove studije su prikazani komparativni statistici generalnih stavova prema reklamiranju u sportu koji su dobijeni primjenom multivarijatne analize varijanse (MANOVA), univarijatne analize varijanse (ANOVA) i LSD Post Hoc testa, a u cilju utvrđivanja razlike u generalnim stavovima ispitanika prema reklamiranju u sportu u odnosu na učestalost kupovine sportskih proizvoda.

Inspekcijom druge tabele u kojoj su prikazani rezultati multivarijatne analize, jasno se uočava da u cijelom sistemu upoređivanih parametara postoji statistički značajna razlika u generalnim stavovima prema reklamiranju u sportu u odnosu na učestalost kupovine sportskih proizvoda ( $p=,046$ ).

**Tabela 2.** Multivarijantna značajnost razlika u sistemu generalnih stavova prema reklamiranju u sportu kod ispitanika sa različitim navikama kada je kupovina sportskih proizvoda u pitanju

|      |           | N   | Mean | S.D.  |
|------|-----------|-----|------|-------|
| GSS1 | Ne kupuju | 6   | 4,33 | 1,366 |
|      | > 1       | 55  | 5,44 | 1,537 |
|      | 1 – 3     | 45  | 5,22 | 1,782 |
|      | 4 – 6     | 8   | 4,63 | 2,446 |
|      | 7 – 9     | 4   | 2,25 | 2,500 |
|      | < 10      | 3   | 5,33 | ,577  |
|      | Ukupno    | 121 | 5,14 | 1,786 |
| GSS2 | Ne kupuju | 6   | 5,50 | 1,378 |
|      | > 1       | 55  | 5,56 | 1,596 |
|      | 1 – 3     | 45  | 5,58 | 1,751 |
|      | 4 – 6     | 8   | 5,00 | 2,563 |
|      | 7 – 9     | 4   | 2,50 | 2,380 |
|      | < 10      | 3   | 6,00 | 1,000 |
|      | Ukupno    | 121 | 5,44 | 1,793 |
| GSS3 | Ne kupuju | 6   | 4,17 | 1,169 |
|      | > 1       | 55  | 4,80 | 1,297 |
|      | 1 – 3     | 45  | 5,22 | 1,550 |
|      | 4 – 6     | 8   | 5,00 | 1,195 |
|      | 7 – 9     | 4   | 3,50 | 1,915 |
|      | < 10      | 3   | 5,67 | ,577  |
|      | Ukupno    | 121 | 4,92 | 1,424 |

F=1,723; p=,046

Inspekcijom treće tabele u kojoj su prikazani rezultati univarijantne analize, jasno se uočava da je došlo do statistički značajnih razlika u generalnih stavovima prema reklamiranju u sportu u odnosu na učestalost kupovine sportskih proizvoda kod dvije promjenljive (GSS1 i GSS2), dok kod treće varijable navedena diskriminacija nije utvrđena.

**Tabela 3.** Univarijantna značajnost razlika u sistemu generalnih stavova prema reklamiranju u sportu kod ispitanika sa različitim navikama kada je kupovina sportskih proizvoda u pitanju

|      | F     | p           |
|------|-------|-------------|
| GSS1 | 3,041 | <b>,013</b> |
| GSS2 | 2,570 | <b>,030</b> |
| GSS3 | 1,849 | ,109        |

Inspekcijom naredne tri tabele gdje su prikazani rezultati Post Hoc testa koji ukazuju na značajnost razlika između parova pojedinačnih entiteta sa različitim navikama kada je kupovina sportskih proizvoda u pitanju za svaku promjenljivu, budući na

rezultate koji su se pojavili na univarijatnom nivou, očekivano se uočava da postoje statistički značajne razlike pojedinih parametara.

Rezultati Post Hoc testa u četvrtoj tabeli ukazuju da postoje statistički značajne razlike između određenih parova pojedinačnih entiteta sa različitim navikama kada je kupovina sportskih proizvoda u pitanju za promjenljivu „moje opšte mišljenje je naklonjeno reklamiranju u sportu“, i to između ispitanika koji se kupuju sportske proizvode sedam do devet puta mjesečno i svih ostalih entiteta. S obzirom da entitet koji se izvo- jio sadrži odgovore samo četiri ispitanika, i da su stavovi istih značajno negativniji nego svih ostalih ispitanika, veoma je važno ovu činjenicu prihvatiti sa rezervom.

**Tabela 4.** Utvrđivanje značajnih razlika u sistemu generalnih stavova prema reklamiranju u sportu primjenom Post Hoc testa između pojedinačnih entiteta sa različitim navikama kada je kupovina sportskih proizvoda u pitanju kod tvrdnje „moje opšte mišljenje je naklonjeno reklamiranju u sportu“

| vs    | Ne kupuju   | > 1         | 1 – 3       | 4 – 6       | 7 – 9       |
|-------|-------------|-------------|-------------|-------------|-------------|
| > 1   | ,137        |             |             |             |             |
| 1 – 3 | ,235        | ,536        |             |             |             |
| 4 – 6 | ,753        | ,214        | ,366        |             |             |
| 7 – 9 | <b>,062</b> | <b>,000</b> | <b>,001</b> | <b>,026</b> |             |
| < 10  | ,411        | ,919        | ,914        | ,543        | <b>,020</b> |

Rezultati Post Hoc testa u petoj tabeli ukazuju da postoje statistički značajne razlike između određenih parova pojedinačnih entiteta sa različitim navikama kada je kupovina sportskih proizvoda u pitanju za promjenljivu „sveobuhvatno, smatram da je reklamiranje u sportu dobra stvar“, takođe između ispitanika koji se kupuju sportske proizvode sedam do devet puta mjesečno i svih ostalih entiteta. Kao i kod prethodne tvrdnje, navedeni intetit iskazuje drastično negativnije stavove, a zbog malog broja istitanika, takođe je značajno ne uzeti utvrđenu razliku kao činjenicu.

**Tabela 5.** Utvrđivanje značajnih razlika u sistemu generalnih stavova prema reklamiranju u sportu primjenom Post Hoc testa između pojedinačnih entiteta sa različitim navikama kada je kupovina sportskih proizvoda u pitanju kod tvrdnje „sveobuhvatno, smatram da je reklamiranje u sportu dobra stvar“

| vs    | Ne kupuju   | > 1         | 1 – 3       | 4 – 6       | 7 – 9       |
|-------|-------------|-------------|-------------|-------------|-------------|
| > 1   | ,932        |             |             |             |             |
| 1 – 3 | ,918        | ,968        |             |             |             |
| 4 – 6 | ,595        | ,393        | ,388        |             |             |
| 7 – 9 | <b>,009</b> | <b>,001</b> | <b>,001</b> | <b>,020</b> |             |
| < 10  | ,685        | ,673        | ,684        | ,397        | <b>,009</b> |

Rezultati Post Hoc testa u šestoj tabeli ukazuju da postoje statistički značajne razlike između određenih parova pojedinačnih entiteta sa različitim navikama kada je kupovina sportskih proizvoda u pitanju za promjenljivu „sveobuhvatno, da li ne volite ili volite reklamiranje u sportu“, samo kod ispitanika već pomenutog entiteta tj. ispita-

nika koji kupuju sportske proizvode od sedam do devet puta mjesečno. Kod ove promjenljive, stavovi ispitanika ovog entiteta su za nijansu pozitivniji te se ne razlikuju značajno od svih ostalih entiteta, već samo od ispitanika koji sportske proizvode kupuju manje od tri puta mjesečno, i onih koje sportske proizvode kupuju više od deset puta.

**Tabela 6.** Utvrđivanje značajnih razlika u sistemu generalnih stavova prema reklamiranju u sportu primjenom Post Hoc testa između pojedinačnih entiteta sa različitim navikama kada je kupovina sportskih proizvoda u pitanju kod tvrdnje „sveobuhvatno, da li ne volite ili volite reklamiranje u sportu“

| vs    | Ne kupuju | > 1         | 1 – 3       | 4 – 6 | 7 – 9       |
|-------|-----------|-------------|-------------|-------|-------------|
| > 1   | ,295      |             |             |       |             |
| 1 – 3 | ,085      | ,136        |             |       |             |
| 4 – 6 | ,272      | ,706        | ,680        |       |             |
| 7 – 9 | ,462      | <b>,075</b> | <b>,020</b> | ,083  |             |
| < 10  | ,132      | ,298        | ,595        | ,483  | <b>,045</b> |

## DISKUSIJA

S obzirom da su rezultati pokazali da ispitanici imaju veoma pozitivan stav prema reklamiranju u sportu što potvrđuje visoka vrijednost aritmetičke sredine za sve tri promjenljive, kao i da skoro dvije trećine ispitanika imaju pozitivan stav prema reklamiranju u sportu što se ogleda u izrazito negativnim vrijednostima mjera asimetrije, trebalo bi naglasiti da su ovi rezultati usaglašeni sa rezultatima prethodnih istraživanja (Pyun, 2006; Akpınar i sar. 2010), i da ne postoje značajne razlike koje bi trebalo pomenuti. Dobijeni rezultati, takođe jasno ukazuju da ispitanici koji žive na različitim lokacijama, kao što su Sjedinjene države, Turska i Crna Gora, imaju pozitivne stavove prema reklamiranju u sportu, dok je, ipak, poređenja radi, vrijedno navesti da, prema Mittal-u (1994), različite studije ukazuju na negativne stavove kada je reklamiranje proizvoda u tradicionalnim industrijama u pitanju. Prema tome, više je nego evidentno da je primjena sporta u savremenoj poslovnoj komunikaciji uticala da se generalni stavovi potrošača značajno promijene kada je reklamiranje u pitanju, a prepoznavanje privlačnosti sporta je omogućilo poslovnim organizacijama da se približe sportskim potrošačima i na mnogo bezbolniji način utiču na njihovo ponašanje.

Utvrđivanjem razlike u generalnim stavovima ispitanika prema reklamiranju u sportu u odnosu na učestalost kupovine sportskih proizvoda, u ovoj studiji su pronađene razlike u stavovima između ispitanika koji imaju različite navike kada je kupovina sportskih proizvoda u pitanju. Ove razlike su se pojavile kod dvije promjenljive na univarijantnom nivou. Kod promjenljive „moje opšte mišljenje je naklonjeno reklamiranju u sportu“, ispitanici sa jednog entiteta tj. onih koji kupuju sportske proizvode sedam do devet puta mjesečno imaju značajno negativnije stavove od svih ostalih ispitanika, što bi bilo veoma interesantno ukoliko bi uzorak ovo entiteta bio veći. Na žalost, samo četiri ispitanika u ovoj grupi nisu dovoljna da se izvede čvrst zaključak da ovaj entitet ima specifično drugačije stavove od ostalih ispitanika. Ipak, trebalo bi imati u vidu ovu činjenicu prilikom narednih istraživanja tj. posvetiti posebnu pažnju kada uzorak bude

bio veći. Kod sljedeće promjenljive „sveobuhvatno, smatram da je reklamiranje u sportu dobra stvar“, situacije identična kao i kod prve promjenljive, tj. ispitanici koji kupuju sportske proizvode sedam do devet puta mjesečno iskazuju značajno negativnije generalne stavove prema reklamiranju u sportu; međutim, zaključak je isti kao i kod prethodne promjenljive i mora se uvažiti sa velikim prezmom zbog malog broja ispitanika koji su se prepoznali u ovom entitetu. Kod posljednje promjenljive „sveobuhvatno, da li ne volite ili volite reklamiranje u sportu“, iako nije bilo značajnih razlika na univarijantnom nivou, primjećuju se značajno negativniji stavovi kod istog entiteta; međutim, kod ove promjenljive, situacija nije dramatična kao u prethodna dva slučaja, tako da se značajne razlike pojavljuju samo u diskriminaciji sa ispitanicima koji kupuju sportske proizvode manje od tri puta i više od deset puta mjesečno.

Na osnovu statističkih analiza, utvrđeno je da se značajne razlike javljaju na multivarijantnom nivou, kao između dvije varijable na univarijantnom nivou značajnosti od  $p=0.05$ . Takođe, interesantno je istaći da se većina ispitanika identifikovala sa dva ponuđena entiteta, njih 99 (od 121), prije svega sa potrošačima koji sportske proizvode kupuju manje od jedanput mjesečno i sa potrošačima koji sportske proizvode kupuju jedanput do tri puta mjesečno. Sa druge strane ostali entiteti nisu uspjeli da izdvoje dovoljan broj ispitanika, tako da ni trebalo potencijalne razlike uzimati kao činjenice, najviše iz razloga što nisu u skladu sa prethodnim istraživanjima (Popović, 2011). Sa druge strane buduća istraživanja bi trebalo usmjeriti na veći broj ispitanika, budući da se značajan broj ispitanika grupiše u kategoriji sa manjim brojem kupovine sportskih proizvoda tokom mjeseca, posebno iz razloga što se očekuje da bi neke razlike mogle biti interesantne i korisne kako za teoriju tako i za praksu.

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#### ATTITUDES OF SERBIAN CONSUMERS TOWARD ADVERTISING THROUGH SPORT WITH REGARD TO THE FREQUENCY OF PURCHASING SPORTING GOODS

*Unlike other advertising mediums such as TV commercials or online advertising, people involuntarily receive sport advertisements; they prefer to watch the game for the sporting action than the regular advertisements (Pyun, 2006; Pyun & James, 2009). However, it is important to analyse their general attitudes among various questions, mostly due to the reason it varies from demographic group to demographic group (Popović et al., 2011). Hence, this research was aimed at gaining relevant knowledge about the attitudes of Serbian consumers toward advertising through sport among the question how often consumers purchase sporting goods. Methods: The sample included 127 respondents, divided into six subsample groups: consumers who do not purchase sport goods at all, then consumers who purchase sport goods less than ones a month, next 1–3 a month, 4–6 a month, 7–9 a month, as well as consumers who purchase sport goods more than 10 times a months. The sample of variables contained the system of three general attitudes which were modeled by seven-point Likert scale. The results of the measuring were analyzed by multivariate analysis (MANOVA) and univariate analysis (ANOVA) and Post Hoc test. Results: Based on the results it was concluded that significant differences occur at multivariate level, as well as between two of three variables at univariate level ( $p=.05$ ). It is interesting to highlight that most of potential consumers ( $n=99$ ) have recognized themselves in two categories: as consumers who purchase sport goods less than ones a months and one to three times a months, while there were no differences among the groups. Discussion: If we take a look into previous investigations, it is interesting there are some opposite results (Popović, 2011), and the recommendations for further research are directed to spread out the sample a little bit more, mostly due to the reason the consumers who had a more negative attitudes were in the group that didn't purchase the sport goods at all (Popović, 2011).*

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## STAVOVI SRBIJANSKIH POTROŠAČA PREMA REKLAMIRANJU U SPORTU U ODNOSU NA UČESTALOST POSMATRANJA SPORTSKIH DOGAĐAJA

### UVOD

Reklamiranje je nekada predstavljalo isključivo dio marketinga. Međutim, ono danas ne predstavlja samo dio marketinga, već se sve više razvija u disciplinu koja predstavlja, takođe i kombinaciju komunikacije, sociologije i psihologije (citirano u Muratović, Bjelicu i Popović, 2014). Samim tim, reklamiranje postaje sve više multidisciplinarna i multimedijalna oblast koja svoje potrošače putem sredstava javnog informisanja usmjerava na različite aktivnosti i radnje koje ih čine veoma srećnim i zadovoljnim (Bjelica i Popović, 2011; Popović, 2011; Popović, Molnar i Radovanović, 2011). Dok su, sa druge strane, savremeni potrošači u stalnoj potrazi za novim komunikacijskim svojstvima proizvoda i usluga, te sa velikim nestrpljenjem i pažnjom iščekuju nove identitete i značenja koja im se nude u reklamnim porukama koje se mijenjaju paralelno sa promjenama modnih trendova (Bjelica i Popović, 2011; Popović, 2011; Popović, Molnar i Radovanović, 2011). Ove promjene neprestano podstiču potrošače da budu aktivni i podložni promjenama pa se, samim tim i javlja neophodnost za praćenjem i analiziranjem njihovih aktivnosti i potreba. Shodno ovoj potrebi javljaju se brojna istraživanja koja su pokazala da stavovi potrošača prema proizvodima ili njihovim robnim markama na osnovu reklamiranja u sportu utiču na odluke za kupovinu određenog proizvoda (Crimmins, & Horn, 1996; Turco, 1996; Shilbury, & Berriman, 1996; Pope, & Voges, 1997; Harshaw, & Turner, 1999; Nicholls, Roslow, & Dublsh, 1999; Pope, & Voges, 2000; Stotlar, & Bennett, 2000; Turley, & Shannon, 2000). Međutim, mali je broj onih koje ispituju razlike u stavovima potrošača u odnosu na određene socio-demografske karakteristike (Bjelica i Popović, 2011; Popović, 2011; Popović, Molnar i Radovanović, 2011). Stoga, cilj ove studije se odnosi na provjeravanje i analiziranje stavova potrošača prema reklamiranju u sportu u odnosu na učestalost posmatranja sportskih događaja.

## METOD

Populaciju u ovoj studiji su činili studenti Fakulteta za sport i fizičko vaspitanje na Univerzitetu u Novom Sadu koji su, u vrijeme anketiranja, imali boravište na teritoriji Srbije, dok je uzorak ispitanika organizovan putem kombinovanja ili raslojavanja, tako da su obrađena različita svojstva navedene populacije i različiti prostori na kojima je ona egzistirala.

Anketa je sprovedena tokom ljetnjeg semestra 2012/2013 školske godine a upitnici su distribuirani studentima osnovnih studija u štampanom i elektronskom obliku. Ukupno je prikupljeno 150 upitnika, ali je 29 upitnika bilo isključeno iz analize, budući da nisu bili adekvatno popunjeni, tako da je u istraživanju učestvovalo, ukupno 121 ispitanika. Instrument istraživanja je predstavljao standardizovani upitnik koji je bio prilagođen prema Pyun-u (2006) i sastojao se iz dva dijela, generalnih stavova prema reklamiranju u sportu i socio-demografskih karakteristika ispitanika kada je učestalost bavljenja sportskim aktivnostima tokom mjeseca u pitanju. Sistem promjenljivih u ovom upitniku je sadržao tri tvrdnje koje su ispitanici trebali da ocijene prema sedmostepenoj Likertovoj skali vrijednosti, kao i šest socio-demografskih karakteristika ispitanika (uopšte ne posmatraju, 1–30 minuta, 31–60 minuta, 61–90 minuta, 91–120 minuta, i više od 120 minuta). Popunjavanje upitnika nije trajalo, u prosjeku više od 10 minuta a ispitanici su u anketi učestvovali na dobrovoljnoj osnovi. Važno je naglasiti da je anketa bila anonimna i da su svi odgovori bili strogo poverljivi. Vrijedno je napomenuti i da su ispitanici, i pored svega navedenog imali mogućnost da u svakom momentu opozovu svoje učešće u anketi, ali se niko od njih nije na tako nešto odlučio.

Empirijski podaci su analizirani putem statističkog paketa za društvene nauke (SPSS 20.0), a kao prvi korak, bila je primijenjena deskriptivna statistika kojom su izračunati, prije svega frekvencija, zatim aritmetička sredina, standardna devijacija, kao i mjere asimetrije (Skewness) i spljoštenosti (Kurtosis) za svaku od tvrdnji. Budući da su se promjenljive u ovoj studiji nalazile na naparametrijskim skalama, radi detaljnijih analiza koje su slijedile, bilo je neophodno da se one, primjenom Blomovog postupka, transformišu u skale višeg reda. Zatim su primjenom multivarijatne analize varijanse (MANOVA), univarijatne analize varijanse (ANOVA) i LSD Post Hoc testa utvrđene razlike u generalnim stavovima ispitanika prema reklamiranju u sportu u odnosu na učestalost bavljenja sportskim aktivnostima tokom mjeseca.

## REZULTATI

U prvoj tabeli su prikazani deskriptivni statistici za sve tri tvrdnje koje su se odnosile na generalne stavove ispitanika prema reklamiranju u sportu. Prije svega, prikazana je aritmetička sredina koja oslikava pozitivne vrijednosti stavova kada su sve tri tvrdnje u pitanju, dok vrijednosti standardne devijacije pokazuju da elementi skupa, u prosjeku ne odstupaju značajno od aritmetičke sredine. Kada se govori o mjerama asimetrije (Skewness) i spljoštenosti (Kurtosis), negativne vrijednosti asimetrije kod svih promjenljivih pokazuju da je većina rezultata desno od srednje vrijednosti, među većim vrijednostima, dok negativne vrijednosti spljoštenosti kod jedne promjenljive (GSS1) pokazuju da je raspodjela pljosnatija od normalne tj. da ima više slučajeva na repovima

a pozitivna vrijednost kod jedne promjenljive (GSS2 i GSS3) da je raspodjela šiljatija od normalne tj. da ima više rezultata nagomilanih oko centra raspodjele.

**Tabela 1. Generalni stavovi prema reklamiranju u sportu**

|      | Mean | S.D.  | Skewness  |      | Kurtosis  |      |
|------|------|-------|-----------|------|-----------|------|
|      |      |       | Statistic | S.E. | Statistic | S.E. |
| GSS1 | 5,09 | 1,845 | -,834     | ,215 | -,380     | ,427 |
| GSS2 | 5,39 | 1,840 | -1,203    | ,217 | ,387      | ,430 |
| GSS3 | 4,93 | 1,430 | -,693     | ,219 | ,259      | ,435 |

Legenda:

Mean – Aritmetička sredina

S.D. – Standardna devijacija

Skewness – Mjera asimetrije

Kurtosis – Mjera spljoštenosti

Statistic – Statistička vrijednost

S.E. – Standardna greška

GSS1 – Moje opšte mišljenje je naklonjeno reklamiranju u sportu

GSS2 – Sveobuhvatno, smatram da je reklamiranje u sportu dobra stvar

GSS3 – Sveobuhvatno, da li ne volite ili volite reklamiranje u sportu

U nastavku ove studije su prikazani komparativni statistici generalnih stavova prema reklamiranju u sportu koji su dobijeni primjenom multivarijatne analize varijanse (MANOVA), univarijatne analize varijanse (ANOVA) i LSD Post Hoc testa, a u cilju utvrđivanja razlike u generalnim stavovima ispitanika prema reklamiranju u sportu u odnosu na učestalost bavljenja sportskim aktivnostima.

Inspekcijom druge tabele u kojoj su prikazani rezultati multivarijatne analize, jasno se uočava da u cijelom sistemu upoređivanih parametara ne postoji statistički značajna razlika u generalnim stavovima prema reklamiranju u sportu u odnosu na učestalost bavljenja sportskim aktivnostima ( $p=,199$ ).

**Tabela 2. Multivarijatna značajnost razlika u sistemu generalnih stavova prema reklamiranju u sportu kod ispitanika sa različitim navikama kada je posmatranja sportskih događaja u pitanju**

|      |               | N   | Mean | S.D.  |
|------|---------------|-----|------|-------|
| GSS1 | Ne posmatraju | 11  | 4,55 | 1,753 |
|      | 1 – 30        | 27  | 5,56 | 1,423 |
|      | 31 – 60       | 32  | 4,97 | 2,055 |
|      | 61 – 90       | 26  | 5,62 | 1,472 |
|      | 91 – 120      | 8   | 5,13 | 1,642 |
|      | < 120         | 17  | 4,47 | 2,125 |
|      | Ukupno        | 121 | 5,14 | 1,786 |
| GSS2 | Ne posmatraju | 11  | 5,64 | 1,286 |
|      | 1 – 30        | 27  | 5,81 | 1,442 |
|      | 31 – 60       | 32  | 5,22 | 2,028 |
|      | 61 – 90       | 26  | 5,65 | 1,696 |
|      | 91 – 120      | 8   | 5,50 | 1,414 |
|      | < 120         | 17  | 4,76 | 2,333 |
|      | Ukupno        | 121 | 5,44 | 1,793 |

|      |               |     |      |       |
|------|---------------|-----|------|-------|
| GSS3 | Ne posmatraju | 11  | 4,27 | 1,009 |
|      | 1 – 30        | 27  | 5,00 | 1,301 |
|      | 31 – 60       | 32  | 5,03 | 1,576 |
|      | 61 – 90       | 26  | 5,38 | 1,267 |
|      | 91 – 120      | 8   | 5,13 | 1,246 |
|      | < 120         | 17  | 4,18 | 1,590 |
|      | Ukupno        | 121 | 4,92 | 1,424 |

F=1,302; p=,199

Inspekcijom treće tabele u kojoj su prikazani rezultati univarijatne analize, jasno se uočava da, takođe nije došlo do statistički značajnih razlika u generalnih stavovima prema reklamiranju u sportu u odnosu na posmatranja sportskih događaja kod sve tri promjenljive.

**Tabela 3.** Univarijatna značajnost razlika u sistemu generalnih stavova prema reklamiranju u sportu kod ispitanika sa različitim navikama kada je posmatranje sportskih događaja u pitanju

|      | F     | p    |
|------|-------|------|
| GSS1 | 1,470 | ,205 |
| GSS2 | ,915  | ,474 |
| GSS3 | 2,120 | ,068 |

Inspekcijom naredne tri tabele gdje su prikazani rezultati Post Hoc testa koji ukazuju na nepostojanje značajnih razlika između parova pojedinačnih entiteta sa različitim navikama kada je posmatranje sportskih događaja u pitanju za skoro svaku promjenljivu (pojavljuje se nekoliko nesimetričnih razlika). Iako je, sudeći po rezultatima koji su se pojavili na univarijatnom nivou, očekivano da ne postoje statistički značajne razlike pojedinih parametara, ipak je analizirano gdje postoje razlike u jednom pojedinačnom odnosu svih analiziranih parametara.

**Tabela 4.** Utvrđivanje značajnih razlika u sistemu generalnih stavova prema reklamiranju u sportu primjenom Post Hoc testa između pojedinačnih entiteta sa različitim navikama kada je posmatranje sportskih događaja u pitanju kod tvrdnje „moje opšte mišljenje je naklonjeno reklamiranju u sportu“

| vs       | Ne posmatraju | 1 – 30      | 31 – 60 | 61 – 90     | 91 – 120 |
|----------|---------------|-------------|---------|-------------|----------|
| 1 – 30   | ,113          |             |         |             |          |
| 31 – 60  | ,495          | ,207        |         |             |          |
| 61 – 90  | ,095          | ,902        | ,169    |             |          |
| 91 – 120 | ,482          | ,546        | ,824    | ,494        |          |
| < 120    | ,913          | <b>,050</b> | ,350    | <b>,040</b> | ,390     |

Došlo se do zaključka da se kod prve tvrdnje „moje opšte mišljenje je naklonjeno reklamiranju u sportu“ javljaju razlike između ispitanika koji gledaju sportske događaje više od 120 minuta dnevno i ispitanika koji to čine između jedan i 30 minuta i

ispitanika koji to čine pet između 61 i 90 minuta, u korist ispitanika koji manje posmatraju sportske događaje.

**Tabela 5.** Utvrđivanje značajnih razlika u sistemu generalnih stavova prema reklamiranju u sportu primjenom Post Hoc testa između pojedinačnih entiteta sa različitim navikama kada je posmatranje sportskih događaja u pitanju kod tvrdnje „sveobuhvatno, smatram da je reklamiranje u sportu dobra stvar“

| vs       | Ne posmatraju | 1 – 30 | 31 – 60 | 61 – 90 | 91 – 120 |
|----------|---------------|--------|---------|---------|----------|
| 1 – 30   | ,782          |        |         |         |          |
| 31 – 60  | ,507          | ,207   |         |         |          |
| 61 – 90  | ,978          | ,745   | ,361    |         |          |
| 91 – 120 | ,871          | ,664   | ,693    | ,833    |          |
| < 120    | ,212          | ,062   | ,401    | ,115    | ,342     |

Takođe se došlo do činjenice da se kod treće tvrdnje „sveobuhvatno, da li ne volite ili volite reklamiranje u sportu“ pojavljuju razlike između ispitanika koji pomstraju sportske događaje između 61 i 90 minuta i ispitanika koji ne posmatraju sportske događaje u korist prvih, kao i između ispitanika koji posmatraju sportske događaje više od 120 minuta dnevno i ispitanika koji ih posmatraju od 1 do 31 minut i oni koji to čine između 61 i 90 minuta, sve u korist onih koji manje posmatraju sportske događaje.

**Tabela 6.** Utvrđivanje značajnih razlika u sistemu generalnih stavova prema reklamiranju u sportu primjenom Post Hoc testa između pojedinačnih entiteta sa različitim navikama kada je posmatranje sportskih događaja u pitanju kod tvrdnje „sveobuhvatno, da li ne volite ili volite reklamiranje u sportu“

| vs       | Ne posmatraju | 1 – 30 | 31 – 60     | 61 – 90     | 91 – 120 |
|----------|---------------|--------|-------------|-------------|----------|
| 1 – 30   | ,147          |        |             |             |          |
| 31 – 60  | ,122          | ,932   |             |             |          |
| 61 – 90  | <b>,028</b>   | ,317   | ,338        |             |          |
| 91 – 120 | ,190          | ,824   | ,865        | ,645        |          |
| < 120    | ,858          | ,058   | <b>,043</b> | <b>,006</b> | ,115     |

## DISKUSIJA

S obzirom da su rezultati pokazali da ispitanici imaju veoma pozitivan stav prema reklamiranju u sportu što potvrđuje visoka vrijednost aritmetičke sredine za sve tri promjenljive, kao i da skoro dvije trećine ispitanika imaju pozitivan stav prema reklamiranju u sportu što se ogleda u izrazito negativnim vrijednostima mjera asimetrije, trebalo bi naglasiti da su ovi rezultate usaglašeni sa rezultatima prethodnih istraživanja (Bjelica i Popović, 2011; Muratović, Bjelicu i Popović, 2014; Popović, 2011; Popović, Molnar i Radovanović, 2011; Popović, Jakšić, Matić, Bjelica i Maksimović, 2014; Pyun, 2006), i da ne postoje značajne razlike koje bi trebalo pomenuti. Dobijeni rezultati, takođe jasno ukazuju da ispitanici koji žive na različitim lokacijama, kao što su Sjedinjene države, Turska, Crna Gora i Srbiji, imaju pozitivne stavove prema reklamiranju u sportu, dok je, ipak, poređenja radi, vrijedno navesti da, prema Mittal-u (1994),

različite studije ukazuju na negativne stavove kada je reklamiranje proizvoda u tradicionalnim industrijama u pitanju. Prema tome, više je nego evidentno da je primjena sporta u savremenoj poslovnoj komunikaciji uticala da se generalni stavovi potrošača značajno promijene kada je reklamiranje u pitanju, a prepoznavanje privlačnosti sporta je omogućilo poslovnim organizacijama da se približe sportskim potrošačima i na mnogo bezbolniji način utiču na njihovo ponašanje.

Na osnovu statističkih analiza, utvrđeno je da se značajne razlike ne javljaju na multivarijatom nivou, kao ni između sve tri varijable na univerzijatom nivou značajnosti od  $p=.05$ . Takođe, interesantno je istaći da nisu pronađene značajne razlike između potrošača koji posmatraju sportske aktivnosti u različitim vremenskim intervalima, što nije bio slučaj u prethodnim studijama (Bjelica i Popović, 2011). Ovi rezultati su veoma važni za poslovne subjekte u Srbiji, uglavnom iz razloga što mogu spojiti sve potencijalne potrošače koji posmatraju sportske aktivnosti u različitim vremenskim intervalima u jednu homogenu grupu, ali i za naučnu javnost, najviše iz razloga što do sada to nije bio slučaj u prethodnim istraživanjima (Bjelica i Popović, 2011).

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#### *ATTITUDES OF SERBIAN CONSUMERS TOWARD ADVERTISING THROUGH SPORT WITH REGARD TO THE FREQUENCY OF WATCHING SPORTS EVENTS*

*It is proposed that potential consumers form attitudes based on advertising through sport can influence decisions to purchase a particular advertiser's product (Pyun, 2006). From this reason, it is important to analyse their general attitudes toward advertising through sport among various questions, and this investigation was aimed at gaining relevant knowledge about the attitudes of Serbian consumers toward advertising through sport among. Methods: The sample included 127 respondents, divided into six subsample groups: consumers, who do not watch sports events at all, then consumers who watch sports events 1-30 minutes, next 31-60 minutes, 61-90 minutes, 91-120 minutes, as well as consumers who watch sports events more than 120 minutes during the typical day. The sample of variables contained the system of three general attitudes which were modeled by seven-point Likert scale. The results of the measuring were analyzed by multivariate analysis (MANOVA) and univariate analysis (ANOVA) and Post Hoc test. Results: Based on the statistical analyses it was found that significant differences didn't occur at multivariate level, as well as between all three variables at a significance level of  $p=.05$ . Hence, it is interesting to highlight that it was found there are no significant differences showed up between the attitudes of consumers toward advertising through sport among the frequency of watching sports events. Discussion: These results are so important for the marketers, mostly due to the reason they can merge all the potential consumers regarding the frequency they watch the sports events. On the other hand, this wasn't the case in previous investigations (Bjelica and Popović, 2011) and this observation presents relevant information.*



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## **BODY HEIGHT AND ITS ESTIMATION UTILIZING ARM SPAN MEASUREMENTS IN FEMALE ADOLESCENTS FROM CENTRAL REGION IN MONTENEGRO**

### **INTRODUCTION**

The unusual tallness of Montenegrin highlanders was a fact recognized by European anthropologists more than 100 years ago (cited in Bjelica, Popović, Kezunović, Petković, Jurak & Grasgruber, 2012). Pineau et al. (2005) have contributed to an update of average body heights among European populations. Although this study doesn't contain the exact data of Montenegrin female population, it represents the most recent study related to the average body height of modern female Montenegrins. Pineau et al.'s investigation showed that, contrary to the general belief, the female population in the Dinaric Alps, with an average height of 171 centimeters come a close second to girls in the Netherlands (Pineau et al. 2005). Thus, this study has challenged many scientists to believe that female Montenegrins might be the tallest population in Europe and Bjelica and his collaborators (2012) confirm they are very tall but not the tallest with 168.37 centimeters. From the reason the sample in this study was created by university students, a more recent study was conducted as a national survey (Popović, Bjelica & Hadžić, 2014) and it confirmed the results of the previous study and found the average body height of Montenegrin female adolescents were 169.48 centimeters tall. From this reason the modern Montenegrins fall partly into the Dinaric racial classification (Bjelica et al., 2012), the authors did believe the female population that live in the central region might be taller than average Dinaric Alps subjects. Hence, the purpose of this study was twofold. The first purpose was to examine the body height in Montenegrin female adolescents from central region as the authors did believe this is the place where the population can reach the full potential of the Dinaric Alps, while the second purpose was to examine the relationship between body height and arm span as an alternative to estimating the body height, which would vary from region to region in Montenegro.

### **METHOD**

The nature and scope of this study qualifies 593 female adolescents from the central region in Montenegro to be subjects. The average age of the male subject was  $18.54 \pm 0.56$  years old (range 17-20 yrs). It is also important to emphasize that the authors could not accept adolescents with physical deformities that could affect body

height or arm span, and without informed consent were excluded from the study. The exclusion criterion was also being non-Montenegrin.

According to Marfell-Jones, Olds, Stew & Carter (2006), the anthropometric measurements, including body height and arm span were taken according to the protocol of the International Society for the Advancement of Kinanthropometry (ISAK). The trained anthropometrist (the same one for each measure) whose quality of performance was evaluated against prescribed “ISAK Manual” prior to the study performed these measurements. The age of the individuals was determined directly from their reported date of birth.

The body height presents the perpendicular distance between the top of the head (the vertex) and the bottom of the feet. It was measured using stadiometer to the nearest 0.1 centimeters in bare feet with the participants standing upright against a stadiometer. The respondents had to put their feet together and move back until their heels touched the bottom of the stadiometer upright. Their buttocks and upper part of their back have also been touching the stadiometer upright while their head didn't have to touch the stadiometer. The respondent's head had to be in the Frankfort horizontal plane. This was achieved when the lower edge of the eye socket (the orbitale) is horizontal with the tragon. The vertex was the highest point on their head, otherwise the respondents had to raise or lower their chin until it was in the Frankfort horizontal plane to align their head properly.

The arm span is the anthropometric measurement of the length from the tip of the middle fingers of the left and right hands when raised parallel to the ground at shoulder height at a one-hundred eighty degree angle. It was measured using a calibrated steel tape to the nearest 0.1 centimeters in bare feet on a level concrete floor with their upper backs, buttocks and heels against the wall which provide support. The participant's head was also in the Frankfort horizontal plane and the arms were outstretched at right angles to the body with palms facing forwards. The measurement were taken from one middle fingertip to the other middle fingertip, with the tape passing in front of the clavicles while two field workers supported the elbows. The measurements were taken twice, and an average of the two readings was calculated. When the two measurements agreed within 0.4 centimeters, their average was taken as the best estimate for the true value. When the two initial measures didn't satisfy the 0.4 centimeters criterion, two additional determinations were made and the mean of the closest records was used as the best score.

The analysis was carried out using Statistical Package for Social Sciences (SPSS) version 20.0. Means and standard deviations (SD) were obtained for both anthropometric variables. A comparison of means of body heights and arm spans within this gender group was carried out using a t-test. The relationships between body height and arm span were determined using simple correlation coefficients and their 95% confidence interval. Then linear regression analyses was performed to examine the extent to which arm span can reliably predict body height. Finally these relationships were plotted as scatter diagrams and regression lines. Statistical significance was set at  $p < 0.05$ .

## RESULTS

A summary of the anthropometric measurements is shown in Table 1. The mean of the arm span for female subjects was  $168.03 \pm 10.34$  centimeters, which was  $1.21 \pm 1.27$  centimeters more than the body height and statistically insignificant ( $t=1.899$ ,  $p<0.058$ ).

**Table 1.** *Anthropometric Measurements of the Population*

| Body Height Range<br>(Mean $\pm$ SD) | Arm span Range<br>(Mean $\pm$ SD)   |
|--------------------------------------|-------------------------------------|
| 154.8-197.7<br>(169.24 $\pm$ 11.61)  | 150.6-193.6<br>(168.03 $\pm$ 10.34) |

The simple correlation coefficient and their 95% confidence interval analysis between the anthropometric measurements are presented in Table 2. The relationships between body height and arm span was high and significant in the applied sample.

**Table 2.** *Correlation Between Body Height and Arm Span of the Study Subjects*

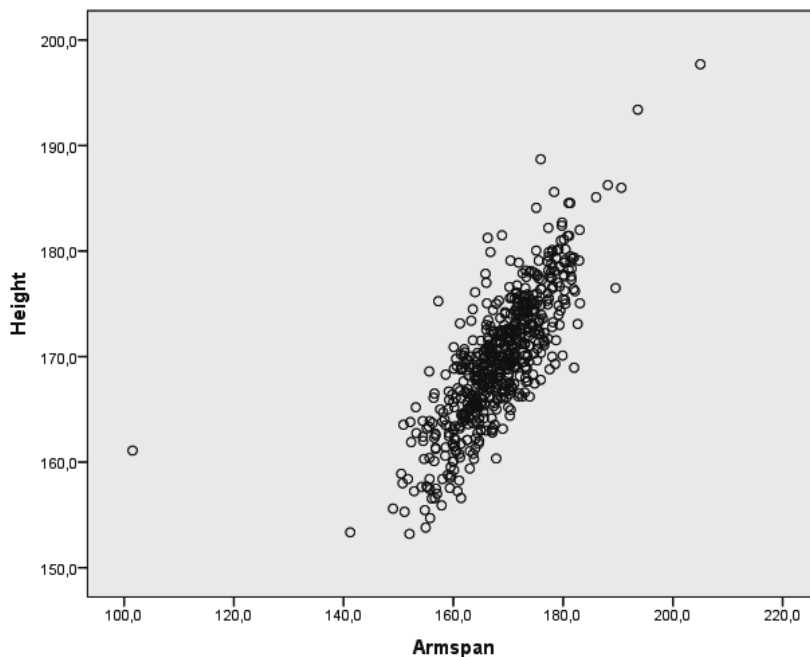
| Correlation<br>Coefficient | 95% confidence<br>interval | Significance<br>p-value |
|----------------------------|----------------------------|-------------------------|
| 0.790                      | 0.732-0.845                | <0.000                  |

The results of the linear regression analysis are shown in Table 3. The first of all models were derived by including age as a covariate. However, it was found that the contribution of age was insignificant and therefore the age was dropped and estimates were derived as univariate analysis. The high values of the regression coefficient signify that arm span significantly predicts body height in the applied sample.

**Table 3.** *Results of Linear Regression Analysis Where the Arm Span Predicts the Body Height*

| Regression<br>Coefficient | Standard Error<br>(SE) | R-square (%) | t-value | p-value |
|---------------------------|------------------------|--------------|---------|---------|
| 0.790                     | 3.314                  | 62.3         | 19.988  | 0.000   |

The relationships between arm span measurements and body height among the above subjects is plotted as a scatter diagram.



**Figure 1.** Scatter Diagram and Relationship Between Arm Span Measurements and Body Height Among the Applied Sample

## DISCUSSION

The results of this study proved that Central-Macedonian females are very tall with an average of 169.24 centimeters, taller than general male population in Montenegro with 168.37 centimeters (cited in Popović, Bjelica, Molnar, Jakšić, & Akpinar, 2013; Quanjier et al., 2014). The results proved that Montenegrin females are tall on average but not as tall as 171.1 centimeters of the female population in the Dinaric Alps (Pineau et al., 2005) and 170.7 centimeters of the Netherlands (TNO, 2010). However, there is a hypothesis that Montenegrin females did not reach their full genetic potential yet, since they have been influenced by various environmental factors (wars, poor economic situation, etc.) in the last few decades (Popović, Bjelica & Hadžić, 2014; Popović, Bjelica, Petković, Muratović, & Georgiev, 2014). Therefore, the authors believe that these circumstances had a negative bearing on the secular trend in Montenegro, while it is expected that the secular changes affecting height will go up in the following 20 years, comparing it to developed countries where this trend has already stopped.

On the other hand, expectably, the arm span reliably predicts body height in this gender. However, the estimation equations which have been obtained in Central-Montenegrins are, different alike in general population, since arm span was shorter than the body heights ( $1.21 \pm 1.27$  centimetres), much more than in general population (Bjelica et

al., 2012). This confirms the necessity for developing separate height models for each region in Montenegro.

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## BODY HEIGHT AND ITS ESTIMATION UTILIZING ARM SPAN MEASUREMENTS IN FEMALE ADOLESCENTS FROM CENTRAL REGION IN MONTENEGRO

*Anthropologists recognized the tallness of nations in the Dinaric Alps long time ago (Popovic et al, 2013). As the modern Montenegrins fall partly into the Dinaric racial classification (Bjelica et al., 2012), the purpose of this study was to examine the body height in Montenegrin female adolescents from central region as well as the relationship between arm span as an alternative to estimating the body height, which would vary from region to region in Montenegro. Method: Our investigation analyses 593 female adolescents from the central region in Montenegro. The anthropometric measurements were taken according to the protocol of the International Society for the*

*Advancement of Kinanthropometry (ISAK). Means and standard deviations regarding the anthropometric measurements were obtained. A comparison of means of body heights and arm spans within this gender group were carried out using a t-test. The relationships between body height and arm span were determined using simple correlation coefficients and their 95% confidence interval. Then a linear regression analysis was performed to examine the extent to which the arm span can reliably predict body height. Results: The results displayed that female Central-Montenegrins are  $169.24 \pm 11.61$  cm tall and have an arm span of  $168.03 \pm 10.34$  cm. Discussion: Compared to other studies, the results of this study have shown that this gender made Central-Macedonians the tall population, taller than general female population in Montenegro (Bjelica et al., 2012). On the other hand, expectably, the arm span reliably predicts body height in this gender. However, the estimation equations which have been obtained in Central-Montenegrins are, different alike in general population, since arm span was shorter than the body heights ( $1.21 \pm 1.27$  centimetres), much more than in general population (Bjelica et al., 2012). This confirms the necessity for developing separate height models for each region in Montenegro.*

„Dan“, 4. april 2015.

МЕЂУНАРОДНА НАУЧНА КОНФЕРЕНЦИЈА  
ЦРНОГОРСКЕ СПОРТСКЕ АКАДЕМИЈЕ ОКУПИЛА  
НАУЧНИКЕ ИЗ 22 ЗЕМЉЕ СВИЈЕТА

## Са четири континента 166 стручњака



UDC 572.087-055.15(497.16)

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## **BODY HEIGHT AND ITS ESTIMATION UTILIZING ARM SPAN MEASUREMENTS IN MALE ADOLESCENTS FROM CENTRAL REGION IN MONTENEGRO**

### **INTRODUCTION**

The unusual tallness of Montenegrin highlanders was a fact recognized by European anthropologists more than 100 years ago (cited in Bjelica, Popović, Kezunović, Petković, Jurak & Grasgruber, 2012). A sample of 800 Montenegrin men measured by Robert W. Ehrich (Coon, 1975) at the beginning of the 20th century gave the highest average in all of Europe (177 cm), with some districts approaching 178 centimeters. Furthermore, a contemporary research study conducted by Pineau, Delamarche, & Božinović (2005), showed that the male population of the Dinaric Alps is on average, the tallest in whole Europe. Thus, this study has challenged many scientists to believe that Montenegrins are still the tallest population in Europe and Bjelica and his collaborators (2012) confirm they are very tall but not the tallest with 183.2 centimeters. From the reason the sample in this study was created by university students, a more recent study was conducted as a national survey (Popović, Bjelica & Hadžić, 2014) and it confirmed the results of the previous study and found the average body height of Montenegrin male adolescents were 183.74 centimeters tall. From this reason the modern Montenegrins fall partly into the Dinaric racial classification (Bjelica et al., 2012), the authors did believe the male population that live in the central region might be much taller than average Dinaric Alps subjects. Hence, the purpose of this study was twofold. The first purpose was to examine the body height in Montenegrin male adolescents from central region as the authors did believe this is the place where the population can reach the full potential of the Dinaric Alps, while the second purpose was to examine the relationship between body height and arm span as an alternative to estimating the body height, which would vary from region to region in Montenegro.

### **METHOD**

The nature and scope of this study qualifies 546 male adolescents from the central region in Montenegro to be subjects. The average age of the male subject was  $18.67 \pm 0.49$  years old (range 18-20 yrs). It is also important to emphasize that the authors could not accept adolescents with physical deformities that could affect body height or arm span, and without informed consent were excluded from the study. The exclusion criterion was also being non-Montenegrin.

According to Marfell-Jones, Olds, Stew & Carter (2006), the anthropometric measurements, including body height and arm span were taken according to the protocol of the International Society for the Advancement of Kinanthropometry (ISAK). The trained anthropometrist (the same one for each measure) whose quality of performance was evaluated against prescribed “ISAK Manual” prior to the study performed these measurements. The age of the individuals was determined directly from their reported date of birth.

The body height presents the perpendicular distance between the top of the head (the vertex) and the bottom of the feet. It was measured using stadiometer to the nearest 0.1 centimeters in bare feet with the participants standing upright against a stadiometer. The respondents had to put their feet together and move back until their heels touched the bottom of the stadiometer upright. Their buttocks and upper part of their back have also been touching the stadiometer upright while their head didn't have to touch the stadiometer. The respondent's head had to be in the Frankfort horizontal plane. This was achieved when the lower edge of the eye socket (the orbitale) is horizontal with the tragon. The vertex was the highest point on their head, otherwise the respondents had to raise or lower their chin until it was in the Frankfort horizontal plane to align their head properly.

The arm span is the anthropometric measurement of the length from the tip of the middle fingers of the left and right hands when raised parallel to the ground at shoulder height at a one-hundred eighty degree angle. It was measured using a calibrated steel tape to the nearest 0.1 centimeters in bare feet on a level concrete floor with their upper backs, buttocks and heels against the wall which provide support. The participant's head was also in the Frankfort horizontal plane and the arms were outstretched at right angles to the body with palms facing forwards. The measurement were taken from one middle fingertip to the other middle fingertip, with the tape passing in front of the clavicles while two field workers supported the elbows. The measurements were taken twice, and an average of the two readings was calculated. When the two measurements agreed within 0.4 centimeters, their average was taken as the best estimate for the true value. When the two initial measures didn't satisfy the 0.4 centimeters criterion, two additional determinations were made and the mean of the closest records was used as the best score.

The analysis was carried out using Statistical Package for Social Sciences (SPSS) version 20.0. Means and standard deviations (SD) were obtained for both anthropometric variables. A comparison of means of body heights and arm spans within this gender group was carried out using a t-test. The relationships between body height and arm span were determined using simple correlation coefficients and their 95% confidence interval. Then linear regression analyses was performed to examine the extent to which arm span can reliably predict body height. Finally these relationships were plotted as scatter diagrams and regression lines. Statistical significance was set at  $p < 0.05$ .



## RESULTS

A summary of the anthropometric measurements is shown in Table 1. The mean of the arm span for male subjects was  $184.99 \pm 8.30$  centimeters, which was  $1.33 \pm 1.37$  centimeters more than the body height and statistically significant ( $t = -2.876$ ,  $p < 0.004$ ).

**Table 1.** *Anthropometric Measurements of the Population*

| Body Height Range<br>(Mean $\pm$ SD) | Arm span Range<br>(Mean $\pm$ SD)  |
|--------------------------------------|------------------------------------|
| 160.9-204.2<br>(183.66 $\pm$ 6.93)   | 156,7-208,3<br>(184.99 $\pm$ 8.30) |

The simple correlation coefficient and their 95% confidence interval analysis between the anthropometric measurements are presented in Table 2. The relationships between body height and arm span was high and significant in the sample.

**Table 2.** *Correlation Between Body Height and Arm Span of the Study Subjects*

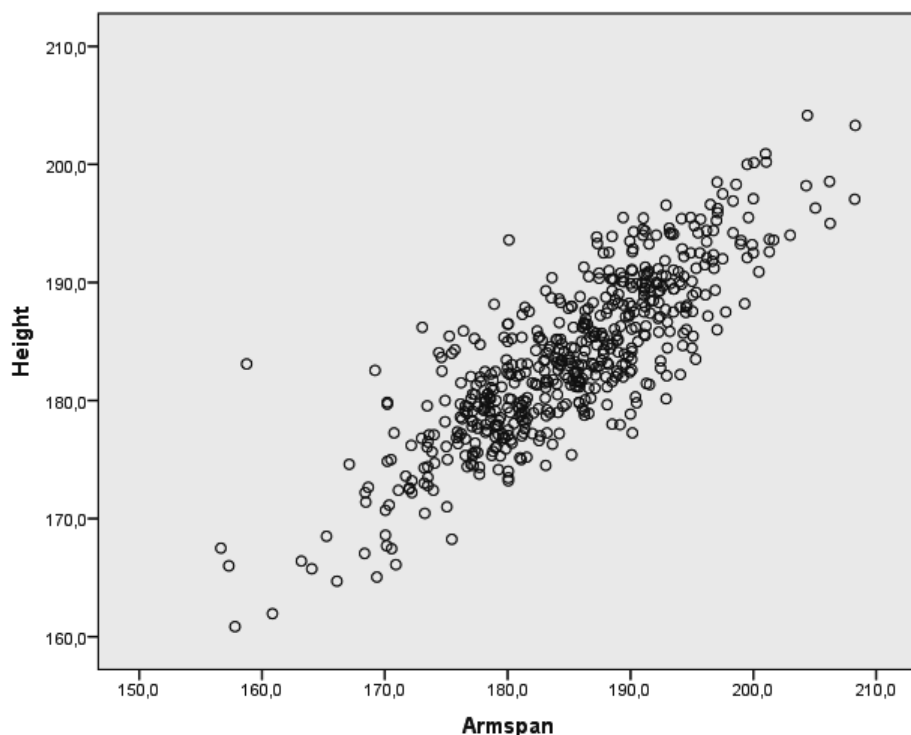
| Correlation<br>Coefficient | 95% confidence<br>interval | Significance<br>p-value |
|----------------------------|----------------------------|-------------------------|
| 0.837                      | 0.804–0.865                | <0.000                  |

The results of the linear regression analysis are shown in Table 3. The first of all models were derived by including age as a covariate. However, it was found that the contribution of age was insignificant and therefore the age was dropped and estimates were derived as univariate analysis. The high values of the regression coefficient signify that arm span significantly predicts body height in the applied sample.

**Table 3.** *Results of Linear Regression Analysis Where the Arm Span Predicts the Body Height*

| Regression<br>Coefficient | Standard Error<br>(SE) | R-square (%) | t-value | p-value |
|---------------------------|------------------------|--------------|---------|---------|
| 0.837                     | 3.793                  | 70.1         | 35.741  | 0.000   |

The relationships between arm span measurements and body height among the above subjects is plotted as a scatter diagram.



**Figure 1.** Scatter Diagram and Relationship Between Arm Span Measurements and Body Height Among the Applied Sample

## DISCUSSION

The results of this study proved that Central-Macedonian males are very tall with an average of 183.66 centimeters, taller than general male population in Montenegro with 183.21 centimeters (cited in Quanjier et al., 2014). On the other hand, they are not the tallest in Europe. It does not come close to 184.6 centimeters documented by Pineau et al. (cited in Popović, Bjelica, Petković, Muratović, & Georgiev, 2014) and 183.8 centimeters of the Dutch male population measured in the last nationwide survey in 2010 (TNO, 2010). On the other hand, the 183.66 centimeters average height of Central-Montenegrin men is taller than 181.3 centimeters of the Lithuanians (cited in Popović, Bjelica & Hadžić, 2014), 180.9 centimeters of the Serbs (J. Grozdanov, personal communication, 1 December 2011), 180.6 centimeters of the Icelanders (cited in Bjelica et al., 2012) et cetera. However, there is a hypothesis that Montenegrin males did not reach their full genetic potential yet, since they have been influenced by various environmental factors (wars, poor economic situation, etc.) in the last few decades (Popović, Bjelica, Molnar, Jakšić, & Akpinar, 2013). Therefore, the authors believe that these circumstances had a negative bearing on the secular trend in Montenegro,

while it is expected that the secular changes affecting height will go up in the following 20 years, comparing it to developed countries where this trend has already stopped.

On the other hand, expectably, the arm span reliably predicts body height in this gender. However, the estimation equations which have been obtained in Central-Montenegrins are, different alike in general population, since arm span was closer to body heights ( $1.33 \pm 1.37$  centimetres), more than in general population (cited in Bjelica et al., 2012). This confirms the necessity for developing separate height models for each region in Montenegro.

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**BODY HEIGHT AND ITS ESTIMATION UTILIZING ARM SPAN MEASUREMENTS  
IN MALE ADOLESCENTS FROM CENTRAL REGION IN MONTENEGRO**

*Anthropologists recognized the tallness of nations in the Dinaric Alps long time ago (Popovic et al, 2013). As the modern Montenegrins fall partly into the Dinaric racial classification (Bjelica et al., 2012), the purpose of this study was to examine the body height in Montenegrin male adolescents from central region as well as the relationship between arm span as an alternative to estimating the body height, which would vary from region to region in Montenegro. Method: Our investigation analyses 548 male adolescents from the central region in Montenegro. The anthropometric measurements were taken according to the protocol of the International Society for the Advancement of Kinanthropometry (ISAK). Means and standard deviations regarding the anthropometric measurements were obtained. A comparison of means of body heights and arm spans within this gender group were carried out using a t-test. The relationships between body height and arm span were determined using simple correlation coefficients and their 95% confidence interval. Then a linear regression analysis was performed to examine the extent to which the arm span can reliably predict body height. Results: The results displayed that male Central-Montenegrins are  $183.66 \pm 6.93$  cm tall and have an arm span of  $184.99 \pm 8.30$  cm. Discussion: Compared to other studies, the results of this study have shown that this gender made Central-Macedonians the tall population, taller than general male population in Montenegro (Bjelica et al., 2012). On the other hand, expectably, the arm span reliably predicts body height in this gender. However, the estimation equations which have been obtained in Central-Montenegrins are, different alike in general population, since arm span was closer to body heights ( $1.33 \pm 1.37$  centimetres), more than in general population (Bjelica et al., 2012). This confirms the necessity for developing separate height models for each region in Montenegro.*

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## **HIGH ALTITUDES EFFECTS ON HEMATOLOGIC BLOOD PARAMETERS**

### **1. Introduction**

Currently, training in high altitudes is a known effective method for improving essential human body systems efficacy by increasing the level of competitiveness of professional athletes for competitions.

Even though, many aspects of such trainings are little informed in the sport gyms.

Sport performance after training at high altitudes (2000 meters above sea level) effects the morphologic aspect of blood parameters.

This type of training stimulates the production of erythrocytes (responsible for transport of oxygen in blood), hemoglobin and blood circulation volume ( in order that bigger amount of oxygen transfers to cells).

Training in high altitudes has become a decisive element in preparing the athletes for big competitions. The effects of high altitude environments in the athletes varies from many factors of which the most desirable and encouraging are the decreased level of oxygen pressure in the air. This combined with other factors such as difference in ambient temperature and increased air humidity combines for a real challenge for athletes.

Specifically, the aim and the objective of this study consists on verifying the change in blood parameters after a stay in a high altitude environment (1800-2300m) of student population during a 10 day ski-trip.

### **1.1 Changes of blood parameters during high altitude training**

Physical exercise in high altitudes have started emerging in year 1950 when many experiments began by performing sport exercises in different atmospheric conditions and carrying analyses and also by studying the load effects of athletes in these conditions.

Such kind of training has reached its peak during the preparations for Mexico City Olympic games in 1968 that were held at an altitude of 2240 meters above sea level.

Exercises in conditions where the amount of oxygen in the air is smaller (hypoxia) result in many positive effects in physical preparation of athletes by which is made possible to achieve better sport results. With the increase of altitude, the atmospheric density decreases and the amount of oxygen the athlete intakes during exercise at high altitudes falls.

## 2. Aim of the study

This study was realized in order to verify the change of hematologic blood parameters that we will identify at the students before and after sport activities at high altitudes of 1800-2000 meters above sea level. As such for the purpose of verifying of this objective we have set two basic hypothesis:

H1: We expect to identify important changes of hematologic blood parameters after the exercise in high altitude.

H2: We expect to identify same changes of hematologic blood parameters according to gender.

### 2.1. Material and work methodology

In this work, 64 students of the faculty of sports sciences in University of Prishtina aged between 19 and 25 with an average of 21 were included. All students preliminary attended a medical test to check TA, Arterial Pulse and Respiratory frequency. A thorough medical test was carried to evaluate the subjects health condition. Additionally, blood tests of each subject were carried.

12 females and 52 male subjects were treated in this experiment all of the above-mentioned age group. Initially the blood samples of all the students were taken moments before leaving for the trip to Brezovica mountains where the exercises were to be carried and the blood test results for erythrocyte, hematocrit and hemoglobin were determined at an altitude of 560m.

For the tests, the blood was taken from the finger.

The same procedure was carried after the return from Brezovica Mountains. Blood samples were taken by the participants in the “Biotechnic Lab” in Prishtina and were processed in SWELAB ALFA machine.

Swelab Alfa Analyzer Standard is used in big laboratories with the main objective being the quality of the gained results by using high level hematologic systems, it is an automatic machine which helps us carry exact measures in short times.

### 2.2. Analysis and interpretation of experiment results

Based on the research material and processed data we have arrived to conclusion of this results which we expressed in table forms and graphics.

In table 1 the average level of hemoglobin in the female subjects of the experiment is expressed in table form and graphic before and after the 10 day stay at high altitude.

Before leaving for Brezovica Mountains the average level of hemoglobin in the female subjects was 123,9 g/L while the average level of Hemoglobin after returning from the trip was 127,8g/L

*Table 1. Average level of hemoglobin of female subjects:*

|                                  |          |
|----------------------------------|----------|
| Hemoglobin level before the trip | 123.9g/L |
| Hemoglobin level after the trip  | 127.8g/L |

Based on this result we can notice that the level of hemoglobin has increased by 3,9g/L

In table number 2 the average level of erythrocytes for female subjects is expressed before and after the trip. The average level of erythrocytes was calculated for the female gender, before the trip this value was  $4.09 \times 10^{12}/L$  while after the trip the value was  $4.42 \times 10^{12}/L$ .

**Table 2.** *Average levels of erythrocytes of female subjects:*

|                                              |                                           |
|----------------------------------------------|-------------------------------------------|
| <b>Value of erythrocytes before the trip</b> | <b><math>4.09 \times 10^{12}/L</math></b> |
| <b>Value of erythrocytes upon return</b>     | <b><math>4.42 \times 10^{12}/L</math></b> |

Based on this result we notice an increase of erythrocyte level by  $0.33 \times 10^{12}/L$

In similar manner as in the previous to tables, in table number 3 we have expressed the average values of hematocrit for the female subjects before and after the trip to Brezovica. Before the trip the hematocrit level was 33.98% while upon return the level was 36.61%

**Table 3.** *Average level of hematocrit for the female subjects*

|                                                    |               |
|----------------------------------------------------|---------------|
| <b>Average level of hematocrit before the trip</b> | <b>33.98%</b> |
| <b>Average level of hematocrit upon return</b>     | <b>36.61%</b> |

Based on this result we notice an increase of hematocrit level by 2.63%.

**In table number 4 we have expressed the increase in percentage of hemoglobin, erythrocytes and hematocrit in female subjects**

**Table 4.**

|                     |              |
|---------------------|--------------|
| <b>Hemoglobin</b>   | <b>3.15%</b> |
| <b>Erythrocytes</b> | <b>8.60%</b> |
| <b>Hematocrit</b>   | <b>7.74%</b> |

While comparing the three parameters, the gained differences are analyzed in statistical manner and it's been concluded that the erythrocytes have encountered the bigger change out of the three tested parameters.

In Table 5 the average level of hemoglobin in the male subjects before and after the 10 day trip at altitude above 2000m is expressed

Before the trip to Brezovica the average level of hemoglobin for the male subjects was 147,6g/L while upon return from Brezovica the hemoglobin level was 152,7g/L

**Table 5.** *Average level of hemoglobin for male subjects.*

|                                         |                 |
|-----------------------------------------|-----------------|
| <b>Hemoglobin level before the trip</b> | <b>147.6g/L</b> |
| <b>Hemoglobin level upon return</b>     | <b>152.7g/L</b> |

Based on this result, we notice that hemoglobin level increased by 5,1 g/L.

In Table 6, the average level of erythrocytes in the male subjects of the experiment before and after the 10 day trip to Brezovica is expressed. The average level of erythrocytes for the male gender before and after the trip is calculated. Before the trip the value was  $4,84 \times 10^{12}/L$  while upon return the value was  $5,22 \times 10^{12}/L$ .

**Table 6.** *Average level of erythrocytes for male subjects*

|                                          |                                           |
|------------------------------------------|-------------------------------------------|
| <b>Erythrocyte level before the trip</b> | <b><math>4.84 \times 10^{12}/L</math></b> |
| <b>Erythrocyte level upon return</b>     | <b><math>5.22 \times 10^{12}/L</math></b> |

In similar way as in the previous two tables, in table 7 we have expressed the average value of hematocrit for male subjects. Before the trip to Brezovica the average level of hematocrit for the male subjects was 40,77% while upon return the level was 43,14%.

**Table 7.** *Average level of hematocrit of male subjects.*

|                                         |               |
|-----------------------------------------|---------------|
| <b>Hematocrit level before the trip</b> | <b>40.77%</b> |
| <b>Hematocrit level upon return</b>     | <b>43.14%</b> |

Based on this result we notice that the average hematocrit level has increased by 2,37%..

In table 8 we have expressed the increase in percentage of hemoglobin, erythrocytes and hematocrit for the male gender

|                     |              |
|---------------------|--------------|
| <b>Hemoglobin</b>   | <b>3.44%</b> |
| <b>Erythrocytes</b> | <b>7.85%</b> |
| <b>Hematocrit</b>   | <b>5.81%</b> |

While comparing the three parameters, the gained differences are analyzed in statistical manner and it's been concluded that the erythrocytes have encountered the bigger change out of the three tested parameters.

In table 9, the average levels of hemoglobin, erythrocytes and hematocrit for all the students are expressed (M & F).

**Table 9.**

| <b>Parametres</b>   | <b>Before</b> | <b>After</b>  |
|---------------------|---------------|---------------|
| <b>Hemoglobin</b>   | <b>143.19</b> | <b>148.04</b> |
| <b>Erythrocytes</b> | <b>4.7</b>    | <b>5.0</b>    |
| <b>Hematocrit</b>   | <b>39.5</b>   | <b>41.9</b>   |



### 2.3. Verification of hypothesis

By the experiment carried in the sample of students of the faculty of Sports Sciences we fully confirm the verification of the main objectives while the verification of hypothesis set in the beginning of this study is possible in this manner:

H1: Hypothesis 1 is fully confirmed as important changes between the first and second measurements are confirmed

H2: Hypothesis 2 is not confirmed the way it was set out in the beginning because the changes noticed were not the same for both genders

### 3. Conclusion

With the research carried in this study we have noticed an increase in the average level of erythrocytes, hemoglobin and hematocrit even in shorter stays.

During the exercise in high altitude a stimulation of bone marrow can be noticed and increasing number of reticulocytes circulation which happens during the adaption of body organism in the new environment at this high altitude. This phenomenon was also confirmed with blood parameters tests upon return from the high altitude trip.

Based on many researchers it takes a minimum of 3 weeks stay at an altitude above 2000 meters in order to notice increase in erythrocytes, hemoglobin and hematocrit levels. Based on other authors in their studies we have noticed that in stays over 3 weeks in altitudes above 2000 meters erythrocytes have increased by 12,4% while with our research we have noticed that after a 10 day stay at an altitude above 2000 meters the level of erythrocytes has increased by 8,23% which is a significantly larger increase if compared to other studies because the students during this stay have engaged in sport activities.

1. Based on this research and discussions with other authors we have concluded that: during a 10 days stay of the students at an altitude above 2000 meters we have the following increase in blood parameters:

- Number of erythrocytes (red blood cells)
- Hemoglobin concentration(Hb)
- Overall blood volume(HCT)

With our research we concluded that even with a shorter stay than three weeks the increase of these parameters (erythrocytes, hemoglobin and hematocrit) is possible with a considerable increase.

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#### HIGH ALTITUDES EFFECTS ON HEMATOLOGIC BLOOD PARAMETERS

*The approach and the objective of this experiment are consistent with the determination of changes of blood parameters after the stay of the students at an altitude of 1800-2300 meters, for a ten-day long ski course. In this paper are included a total of 64 students of the Faculty of Sport Sciences in Prishtina, of the age group of 19-25 (the average age is 21). All students previously have undergone a medical check for TA, arterial pulse and respiratory rate.*

*In particular, the health situation of subjects was examined, then, all students, at the same time, gave blood for analysis. In this experiment, three main hematologic parameters were taken in consideration: such as hemoglobin, hematocrit and red blood cells.*

*The same analyses were carried out after the 10-day stay at a high altitude. The results of the experiment have shown significant changes after the ten-day stay at high altitude, despite the previous results that show changes only after the twenty-day stay in such elevations.*

**Key words:** *elevation, blood parameters, sports science students, high altitudes.*

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## **DIAGNOSING OF BASIC AND SPECIFIC MOTORIC CAPABILITIES AT THE YOUTH OF THE BASKETBALL SCHOOL**

### **Introduction**

We are aware that well known worldwide authors have written and also given a lot in terms of the experimental research in diagnosing of problematics in movement information of the basic and specific character of the basketball game. It should be emphasized that first analysis of technical-tactical elements in the basketball game we face in the previous terms, respectively since 1891 in Massachusetts, known as Springfield College. Wissel (1994) in its project, he distinct seven main steps in achieving success in the basketball game which contains and comprehends: movement without the ball, movement with ball, shoot a basketball, pass a basketball etc. In our region we pick out the projects produced by the authors from the Kinesiology Institute in Zabreb, such as; Jukic, Milanovic, Matkovic, T. Dokman, Tirninic etc.

Purpose of the paperwork: Due to the existing fact, as I have been part of this kind of sport activity and that I'm a staff member of the Faculty of Sport Sciences, is what urged me to at least research the circumstances and conditions of the upcoming sportspersons. In this paperwork I have tried to summarize in few brief points, where you will be able to see in the following text:

- Verification of the quantitative characteristics above the level of the basic motoric capabilities of the respective age.
- Documenting the changes in the motoric capability between the respective basketball schools.
- Volume of the various effect in the curricula of both basketball schools in developing the motoric capabilities and the specific once.
- Documenting of the important relations between the basic motoric tests with the typical movements in the basketball game.

### **Methods**

In this aspect there will be treated the youth population from 13-14 years old, where total number of them will be defined, out of 87 youngsters, belonging to two basketball schools.

- First group /G1/, approximately half of the population of this research was defined by the youngsters, attendees of the basketball school "**Prishtina probasket**", that have been regular attendees of the basketball school.
- Second Group /G2/ is consisted of the same number of youth and respective ages, that are a regular attendees of the basketball school "**Alba basket**"

First group of variables is consisted of motoric capability movement tests of the basic character, while the second group is consisted of typical specific variables of the basketball game.

Testing manner/Method – description of the measurement technique of the situational variable group related to the basketball game.

### **I. Motoric basic variables**

1. Jump from place in length - JPL
2. High jump in length - HJL
3. 20 m run from starting point - R 20m
4. Throw of medicine ball - THMB
5. Agility Test. - AGIL Test

### **II Motoric situational Variables (specific)**

1. Run with ball in 20 m distance. R 15B
2. Shot in basket for 30"sek. Going - coming Sh2STEP
3. Sustainable anaerobe with ball (kamikaze) - KAM
4. Shot in the basket from five positions - Sh5POS
5. Manipulation with ball around the body - BHANDEL

Processing method:

### **G 2 – Basketball School "ALBA BASKET"**

**Table 3 Statistical basic parameters of the basic motoric variables**

|        | N  | Min  | Maks  | Ma     | SD    | Skew   | Kurt   | Kv%   |
|--------|----|------|-------|--------|-------|--------|--------|-------|
| 1.JPL  | 42 | 140  | 210   | 171.74 | 19.76 | 0.159  | -0.991 | 11.5  |
| 2.HJL  | 42 | 15   | 48    | 31.19  | 7.44  | 0.175  | -0.188 | 23.25 |
| 3.V20M | 42 | 3.39 | 4.72  | 4.13   | 0.29  | -0.068 | 0.030  | 7.02  |
| 4.HMED | 42 | 4.9  | 8.9   | 6.93   | 1.07  | -0.058 | -0.831 | 15.44 |
| 5.AGIL | 42 | 9.39 | 12.55 | 10.68  | 0.58  | 0.689  | 1.455  | 5.43  |

**Table 4 Statistical basic parameters of the situational variables**

|          | N  | Min   | Maks  | Ma    | SD   | Skew   | Kurt   | Kv%   |
|----------|----|-------|-------|-------|------|--------|--------|-------|
| V15mT1   | 42 | 14.7  | 18.36 | 16.59 | 0.87 | -0.182 | -0.166 | 5.18  |
| KAM1     | 42 | 31.86 | 41.26 | 36.10 | 2.49 | 0.341  | -0.797 | 6.89  |
| FGJ5PO1  | 42 | 2     | 8     | 5.00  | 1.59 | -0.190 | -0.828 | 31.8  |
| HAP21    | 42 | 2     | 7     | 4.00  | 1.10 | 0.342  | 0.223  | 27.5  |
| B.HANDE1 | 42 | 4     | 13    | 8.19  | 1.85 | 0.143  | 0.376  | 22.58 |

- In both groups of variable system will be calculated **basic statistic parameters** and distribution for each variable, as well as asymmetry measure and normal distribution.
- Cross **correlation** will be calculated through the variable system.
- To document the changes between the two groups there will be applied discriminative analysis of T –Test.

**Results:**G 1 – Basketball school "**PRISHTINA PROBASKET**"**Table 1 Statistical basic parameters of the basic motoric variables**

|        | N  | Min  | Max   | Ma     | SD    | Skew  | Kurt   | Kv%   |
|--------|----|------|-------|--------|-------|-------|--------|-------|
| 1.JPL  | 45 | 142  | 250   | 179.84 | 28.42 | 0.857 | 0.211  | 15.80 |
| 2.HJL  | 45 | 20   | 52    | 33.98  | 7.68  | 0.206 | -0.731 | 22.60 |
| 3.R20M | 45 | 3.28 | 4.8   | 4.04   | 0.34  | 0.234 | 0.086  | 8.41  |
| 4.HMED | 45 | 4.45 | 10.2  | 7.26   | 1.47  | 0.138 | -0.760 | 16.11 |
| 5.AGIL | 45 | 8.9  | 12.91 | 10.48  | 0.88  | 0.442 | -0.030 | 8.39  |

**Table 2 Statistical basic parameters of the situational variables**

|             | N  | Min   | Max   | Ma    | SD   | Skew   | Kurt   | Kv%   |
|-------------|----|-------|-------|-------|------|--------|--------|-------|
| 1.Ball 15m  | 45 | 13.67 | 19.10 | 15.20 | 1.11 | 1.038  | 1.917  | 7.3   |
| 2.Kamik     | 45 | 30.51 | 47.00 | 36.89 | 4.12 | 0.300  | -0.758 | 11.14 |
| 3.Thr.5 pos | 45 | 1     | 7     | 3.40  | 1.74 | 0.489  | -0.588 | 51.17 |
| 4.2 step    | 45 | 1     | 7     | 4.33  | 1.60 | -0.226 | -0.525 | 36.95 |
| 5.B.handl   | 45 | 4     | 12    | 8.00  | 2.11 | -0.076 | -0.580 | 26.37 |

Cross Correlations between variables

**Table 5**

Gr.I

| 1 |      | V15mT  | KAM    | GJ5PO  | GJ2HAP | BHANDEL |
|---|------|--------|--------|--------|--------|---------|
|   | KVGJ | -0.479 | -0.674 | -0.073 | 0.338  | 0.246   |
| 2 |      | 0.001  | 0.000  | 0.632  | 0.023  | 0.104   |
|   | KVL  | -0.557 | -0.790 | -0.032 | 0.470  | 0.464   |
| 3 |      | 0.000  | 0.000  | 0.836  | 0.001  | 0.001   |
|   | V20M | 0.524  | 0.726  | 0.006  | -0.313 | -0.355  |
| 4 |      | 0.000  | 0.000  | 0.970  | 0.036  | 0.017   |
|   | HMED | -0.410 | -0.652 | -0.030 | 0.396  | 0.194   |
| 5 |      | 0.005  | 0.000  | 0.846  | 0.007  | 0.201   |
|   | AGIL | 0.585  | 0.739  | 0.080  | -0.430 | -0.506  |
|   |      | 0.000  | 0.000  | 0.601  | 0.003  | 0.000   |

\*\* Validity in the level 0.01

\* Validity in the level 0.05

**Table 6**

Cross relations between variables

Gr. II

|   |      | V15mT  | KAM    | GJ5PO  | GJ2HAP | BHANDEL |
|---|------|--------|--------|--------|--------|---------|
| 1 | KVGJ | -0.452 | -0.244 | 0.342  | -0.097 | 0.135   |
|   |      | 0.003  | 0.119  | 0.027  | 0.540  | 0.395   |
| 2 | KVL  | -0.354 | -0.251 | 0.342  | -0.050 | 0.027   |
|   |      | 0.021  | 0.109  | 0.027  | 0.751  | 0.863   |
| 3 | V20M | 0.395  | 0.253  | -0.215 | -0.038 | -0.033  |
|   |      | 0.010  | 0.106  | 0.171  | 0.813  | 0.836   |
| 4 | HMED | -0.529 | -0.237 | 0.340  | 0.141  | 0.058   |
|   |      | 0.000  | 0.130  | 0.028  | 0.373  | 0.716   |
| 5 | AGIL | 0.737  | 0.364  | -0.257 | -0.111 | -0.273  |
|   |      | 0.000  | 0.018  | 0.101  | 0.484  | 0.080   |

\*\* Validity in level 0.01

\* Validity in level 0.05

Table 7:

Differences between average values – between two groups,

Groups:

G1 – Basketball school " PRISHTINA PROBASKET"

G2 - Basketball school "ALBA BASKET"

|    | Tests | Ma     | N  | D.st  |
|----|-------|--------|----|-------|
| G1 | KVGJ  | 177.57 | 42 | 26.94 |
| G2 | KVGJ  | 171.74 | 42 | 19.76 |
| G1 | KVL   | 33.43  | 42 | 7.53  |
| G2 | KVL   | 31.19  | 42 | 7.44  |
| G1 | V20M  | 4.04   | 42 | 0.35  |
| G2 | V20M  | 4.13   | 42 | 0.29  |
| G1 | HMED  | 7.14   | 42 | 1.42  |
| G2 | HMED  | 6.93   | 42 | 1.07  |
| G1 | AGIL  | 10.53  | 42 | 0.89  |
| G2 | AGIL  | 10.68  | 42 | 0.58  |

Values of the T - test

|       |             | Difference |       |    |         |
|-------|-------------|------------|-------|----|---------|
|       | Pairs       | Mean       | t     | df | Propab. |
| G1-G2 | KVGJ - KVGJ | 5.83       | 1.11  | 41 | 0.272   |
| G1-G2 | KVL - KVL   | 2.24       | 1.50  | 41 | 0.140   |
| G1-G2 | V20M - V20M | -0.09      | -1.25 | 41 | 0.217   |
| G1-G2 | HMED - HMED | 0.21       | 0.76  | 41 | 0.452   |
| G1-G2 | AGIL - AGIL | -0.15      | -0.80 | 41 | 0.426   |

### Discussion:

Paper work organized in the group of age 13-+6 months that in a systematic manner attending the basketball school in two different centers, in an experimental manner and sustainable arguments, is proving the importance of the systematic assessment, in order to solve the following problems such as; planning, programming, development as well as effective control of the training work.

In particular, important is the continual control of the motoric movement development level at the youth.

Based on the gained results, we can conclude that in basic motoric parameters there is no existence of highlighted changes between two groups G1 and G2, more due to the affect of the systematic training rather than the changes as regards to the school belonging.

In a clear manner were ascertained the changes and difference between the groups in the situational typical movement capabilities of the basketball game.

Also while comparing the gained values of the students in this paper work, with the same age of the normal population; we can notice much more development of the movement capabilities.

Therefore we can conclude that the accomplishment level of the specific motoric duties are not yet at the required level, in the other side even these motoric duties for this age and small experience of some subjects are presenting difficult and complicated duties for successful realization.

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#### *DIAGNOSING OF BASIC AND SPECIFIC MOTORIC CAPABILITIES AT THE YOUTH OF THE BASKETBALL SCHOOL*

*In general the problematic of diagnosing basic and specific motor movement information of the basketball game is found in the works of a considerable number of world authors. In this work a youth population of an age group between 13-14 years old will be treated. The total number of participants is defined to an amount of 100 of young basketball players, members of two different basketball academies.*

*In this experiment, the subjects will conduct tests in 5 basic motor skills variables such as: 1. Steady*

*Jump in length, 2 steady jump in height 3.20 meters run from a steady start, 4. Medicine Ball Throw and 5. Agility Test. While from specific motor movement skills are included: 1.20 Meters run with ball, 2. Basketball shooting for 30 seconds round-trip, 3. Anaerobic durability with and without ball (kamikaze) 4. Intensive Basketball shots-Change of direction and 5. Basketball shots from five positions. Subject to the number of subjects that will be treated in this study, their age and the amount of variables tested, the main objectives of this study, will be limited to achieving the goals.*

*On the basis of test results, it can be concluded that there are no significant differences in the basic motor parameters between the two groups G1 and G2, mostly due to systematic practice effects rather than academy belonging itself. In the other hand, differences between the two groups in the specific basketball motor skills can be clearly identified, in typical basketball movement situations. In addition, comparing the test results of the individuals tested for the purpose of this diagnosis, a much more advanced level of motor skills compared to the normal population of the same group age that don't engage in basketball activities can be identified.*

*This experimental study carried out in the youth in the age group of 13(+ 6 months) years old that continuously in a systematic manner attend basketball academies in two different basketball schools, with sustainable arguments can show the importance of systematic valuation in order to solve many arising problems such as: planning, programming, developing and controlling the effects of training work. Specifically, it is of utmost importance to continuously monitor the development of motor skills in the youth.*



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**ZNAČAJ MEĐUNARODNOG PRIZNANJA CRNE GORE NA BERLINSKOM KONGRESU ZA RAZVOJ SPORTA U CRNOJ GORI****Uvod**

Održavanje Berlinskog kongresa predstavlja jedan od najznačajnijih događaja u evropskoj istoriji XIX vijeka, a njegovim odlukama utemeljen je jedan od najvažnijih datuma u istoriji državnosti Crne Gore, kojima je definitivno riješeno pitanje njenog međunarodnog statusa. Odlukama donesenim na Berlinskom kongresu, Crna Gora je 13. jula 1878. godine postala 27. međunarodno priznata država na svijetu.

Upravo u ovom periodu, zbog ostvarenja međunarodnog priznanja crnogorske državnosti, dolazi do intenziviranja diplomatske aktivnosti sa evropskim državama, što je izuzetno doprinijelo modernizaciji naše države u svakom pogledu i razvoju sporta u njoj. Orijentacija Crne Gore i prema istoku i prema zapadu doprinijela je upoznavanju, prihvatanju i razvijanju sportskog pokreta na ovim prostorima. Sportske aktivnosti koje su se ranije uglavnom odvijale stihijno, dobile su u ovom razdoblju pravu formu i preduslove za intenzivniji razvoj koji je potom uslijedio.

**Metode**

U ovom radu je korišten istorijski metod, kao skup istraživačkih postupaka o događajima koji su se zbili u prošlosti. Njegova primjena je ostvarena u nekoliko faze:

U prvoj fazi primjenjena je metoda heuristike, u smislu prikupljanja relevantnih podataka i obuhvatanja istorijske građe kroz primarne, sekundarne i tercijalne pisane izvore. U drugoj fazi - *kritici izvora*, pristupilo se pažljivoj analizi pomenutih istorijskih izvora, jer svaki pisani izvor, osim primarnog pisanog je u većoj ili manjoj mjeri podložan subjektivnom tumačenju istraživača, dok je u trećoj, odnosno završnoj fazi - *ekspoziciji*, izvršen prikaz rezultata do kojih se analizom pomenutih istorijskih izvora došlo.

**Istorijski izvori**

Mada se veliki broj ljudi u ovom periodu bavio fizičkim aktivnostima koje danas obuhvata pojam fizička kultura, na raspolaganju je veoma malo arhivske građe i naučne publicistike na osnovu koje bi mogli dati obuhvatan prikaz razvoja fizičke kulture u ovom periodu. Najznačajniji izvori tih informacija bili su listovi: Cetinjski vjesnik, zatim Ustavnost, Onogošt kao i neke druge povremene publikacije. Značajne podatke o nastavnim planovima i programima za crnogorske škole, moguće je naći u Državnom muzeju na Cetinju. Oni sadrže program nastave za predmet Gimnastika i vojničko vježbanje, koji su bili poseban predmet u crnogorskim osnovnim i srednjim školama.

Zapisi o sportu i fizičkoj kulturi, za koje je najčešće korišten termin „gimnastika“, mogli su se naći u novinama, koje su tada povremeno izlazile.

Iako štamparska djelatnost u Crnoj Gori ima viševjekovnu tradiciju, koja počinje sa *Oktoihiom Crnojevića*, izdavaštvo se obnavlja u doba Njegoša, a procvat doživljava sredinom druge polovine devetnaestog vijeka. Štampaju se knjige, novine i časopisi. Izlaze prve crnogorske novine *Crnogorac* (1871.godine), da bi dvije godine kasnije nastavio pod novim imenom *Glas Crnogorca*. Izlaze i drugi listovi u ovom periodu i to ne samo na Cetinju. Ovdje je najpoznatiji *Cetinjski vjesnik*, *Ustavnost*, a uoči Prvog svjetskog rata u prijestonici izlazi više dnevnih listova i biltena u kojima se, povremeno objavljuju članci iz oblasti sporta i uopšte fizičke kulture, za koju je tada bio uobičajeni naziv gimnastika. Pored pomenutih, u Cetinju izlaze i časopisi (kalendar *Orlić* - obnovljeno izdanje iz 1885.godine), zatim časopisi: *Luča*, *Književni list*, *Zeta*, *Nova Zeta*, *Crnogorka*. Izlaze stručni časopisi: *Zdravlje* i *Prosvjeta* – u kojoj se povremeno pojavljuju zabilješke o gimnastici kao školskom predmetu. Osim u Cetinju, izdavači novina se javljaju u Podgorici, Nikšiću. I u tim novinama se periodično pojavljuju zapis o sportu i fizičkoj kulturi.

Veliki broj značajnih informacija o razvoju sporta u Crnoj Gori ostavili su strani putopisci, diplomate, vojni zvaničnici i književnici koji su sa puno nadahnuća pisali o običajima, načinu života i sportskim nadmetanjima na ovim prostorima.

Putopisac Viala de Somier, inače Francuski visoki činovnik, ostavio je zanimljive zapise o sportu u Crnoj Gori. O tome, sa puno nadahnuća govori u knjizi „Istorijsko i političko putovanje u Crnu Goru“, u kojoj njegovi zapisi obiluju opisima viteških nadmetanja Crnogoraca, u kojima su prisutni elementi takmičenja, sporta i fizičke kulture u širem smislu.

Značajne podatke o crnogorskom sportu s kraja XIX vijeka ostavio je norveški kapetan Henrich Augusto Angell, koji je proputovao je kroz Crnu Goru 1893. godine i iza sebe ostavio putopis „Kroz Crnu Goru na skijama“ Naslov originala glasi: *Gjennem Montenegro pas ski*. Knjiga je izdata u Kristianii (Oslo) 1895.godine i bila je "omiljeno štivo norveške omladine", a kod nas je prvi put izdata 1991. godine. Knjiga je obima 132 strane, od čega je 125 strana originalnog Angellovog teksta, a 7 strana je predgovor čiji je autor dr Novak Jovanović. Knjiga je ilustrovana crtežima crnogorskih, uglavnom zimskih pejzaža, i ličnosti iz crnogorske istorije, a na prednjoj korici je slika skijaša u pokretu, u crnogorskoj narodnoj nošnji. Angel je bio očaran crnogorskim pejzažima i ljudima, bio je impresioniran izgledom tada vjerovatno najmanje metropole u Evropi, a konačno dočekom i odlikovanjem na crnogorskom dvoru, gdje ga je knez odlikovao Danilovim krstom četvrtog reda.

U Kraljevini Crnoj Gori je često boravio književnik Simo Matavulj ( 18 ), koji je u svom djelu „Bilješke jednog pisca“ dao slikovit opis atletskog takmičenja, upriličenog u čast dolaska jednog inostranog vladara u Cetinje. U navedenom djelu govori o stepenu razvijenosti sporta na ovim prostorima i zavidnom nivou fizičke pripremljenosti crnogoraca.

Književnik Ljubomir Nenadović, često je boravio u Crnoj Gori i sa puno zanosu pisao o njoj. Između ostalog, pisao je i o sportskim nadmetanjima koja su bila sastavni dio života crnogorskog naroda i koja se mogu pratiti vjekovima kroz istoriju.

Veoma respektivan izvor informacija o razvoju fizičke kulture u ovom periodu predstavljaju knjige istoričara dr Novaka Jovanovića, (1994): Sport u Crnoj Gori do 1914. godine, i Đura Lakovića, (2000): Momačke viteške igre, koje su nam, takođe poslužile kao vrijedan izvor informacija.

### Rezultati i diskusija

Neposredno po održavanju Berlinskog kongresa, čijim je odlukama utemeljen jedan od najvažnijih datuma u istoriji državnosti Crne Gore, crnogorska prijestonica Cetinje, u ovom periodu ima intenzivan kulturni razvoj zahvaljujući uspješnoj diplomatskoj saradnji sa evropskim državama, što je pratilo osnivanje raznih sportskih društava i klubova. Posebno treba istaći da je 1906. godine na Cetinju formiran prvi golf klub na prostorima današnjeg zapadnog Balkana. Na tu činjenicu, s ponosom ukazuje i istoričar sporta Rastko Radunović, u svom načnom radu: "Pojave i razvoj modernog sporta u Crnoj Gori do 1914. godine" (Fizička kultura, časopis za stručna i društvena pitanja, Titograd, br.1/1980, str 109-116) Ovaj značajni događaj uveliko je upotpunio bogatu sportsku istoriju Crne Gore, na čijem prostoru su počeli da se razvijaju moderni sportovi, poput klizanja, tenisa, fudbala, mačevanja, skejtinga, itd. Osnivanjem ovih klubova, naša zemlja se i na sportskom planu povezala sa evropskim državama i time formirala temelje bogate crnogorske istorije sporta.

Na Cetinju je 1906. godine osnovan Golf klub, a pokrovitelj ove organizacije je bila knjeginja Milica, supruga prestolonasljednika Danila, najstarijeg sina knjaza Nikole.

Igralište se nalazilo u tzv. Novoj Varoši - novom dijelu grada, u neposrednoj blizini zgrade Italijanskog poslanstva, čija je izgradnja započeta te godine. Na svečanom otvaranju prvu partiju je odigrala knjeginja, a onda su se do sumraka redali članovi kluba. Ova igra je ipak bila i ostala privilegija uzanog kruga ljudi, uglavnom diplomatskog kora i članova vladajuće kuće.

Za tenis se može reći da je išao ukorak sa zapadnim djelovima Balkana ili je zanemarljivo malo, kasnio. Podaci za ovakvu tvrdnju nađeni su u jednoj reviji izdatoj u Engleskoj, zemlji koja je ovom sportu izuzetno naklonjena. Tamo je objavljena fotografija koju prati tekst "*Crnogorski knjaz igra tenis na Cetinju*". A ispod toga naslova, u tekstu piše da je crnogorski knjaz za vrijeme ljetne posjete britanske mediteranske flote kotorskom zalivu primio gospodina Majkla Kalta Zejtiga i njegove oficire na Cetinju. Poslije prijema uslijedila je partija tenisa koju je priredila supruga britanskog poslanika. Dalje se navodi podatak da je knjaz bio obučen u kompletnu crnogorsku nošnju i da je dobro igrao tenis. Tenisom su se u prvo vrijeme bavili diplomati i dvorska svita. On nije bio dostupan širim narodnim slojevima. Tek od 1906. godine, od osnivanja teniskog kluba, otvaraju se kapije teniskih igrališta za građanstvo, ali je i dalje bio mali broj onih koji su se bavili ovim sportom. U krugu poslanstva na Cetinju, ostao je do danas sačuvan jedan broj teniskih igrališta.

Na Cetinju je 1891. godine formirano Društvo za gimnastiku, točiljanje i bore-nje. A to borenje je, ustvari, mačevanje. Dakle, rano se počelo sa organizovanjem mačevanja, preko odgovarajuće asocijacije. Tu sekciju su činili uglavnom oficiri (Crna Gora još nije imala stajaću vojsku), i studenti koji su se školovali na strani, po raznim evropskim zemljama. Tri godine kasnije osnovano društvo Gorski vijenac ima sekciju za mačevanje. Časopis *Luča*, pokrenut na Cetinju sljedeće 1895. godine, u prvom broju donosi informaciju o formiranju društva Gorski vijenac, a u njoj se pominju Književni odbor i Učiteljski zbor. Dalje se kaže da Učiteljski zbor ima učitelje za mačevanje i gimnastiku.

Kad je u pitanju pojava klizanja na ledu, sporta koji se u to doba naziva točiljanje citiraćemo vjeran opis klizališta koji nam je ostavio Norvežanin Angell u svoj knjizi *Kroz Crnu Goru na skijama*. Evo toga opisa:

„Ko bi vjerovao da će se u Crnoj Gori naći klizalište, i to zaista izvrsno? Kad se prođe pokraj Dvora, i probije nekako između zbilja zastrašujućih smetova, i krene još malo stazom duž Kneževskog parka, prvo ugledate Kneževsku kapelu sa starim Manastirirom, a odmah kraj nje, na poljani, predivno malo klizalište. A što da ovdje nema klizališta? Dovoljno je hladno i sportista ima dovoljno, a vodovod je prvoklasan. Uz klizalište je odaja za presvlačenje, kao i prostorija za bife kad su prijemi, tu je i novi vatrogasni šmrk da se poliva led svake večeri, pa orijentalne svjetiljke i luče - a ponekad svira i kneževski orkestar. Teško da se može napraviti bolje klizalište“.

Prve skije u Crnu Goru donio je norveški putnik, pustolov, već pomenuti kapetan norveške kraljevske vojske, Henrich Augusto Angell, koji je januara 1893. godine stigao u Kotor, a zatim se, za strance uobičajenom maršrutom, uputio preko Njeguša za Cetinje. Ovom svjetskom putniku je ukazana izuzetna čast time što je bio primljen i na dvoru knjaza Nikole gdje je zadržan na večeri. Sa Cetinja je, i dalje na skijama, nastavio put do sjeverozapadne granice Crne Gore, da posjeti Dugu Nikšićku i u njoj mjesta sa kojih je Crna Gora branila i odbranila svoju nezavisnost. Usput je prošao Rijeku Crnojevića, Podgoricu, Danilovgrad i Nikšić i svuda demonstrirao vještinu skijanja, a to je imalo izvanredan značaj za propagandu ovog sporta. Ljudi su se počeli naglo interesovati gdje se mogu nabaviti liže (skije); iz Crne Gore su u Beč pošle prve porudžbine neobičnih dasaka. Krajem januara 1893. godine Angell je završio svoju misiju po Crnoj Gori, vratio se na Cetinje, i ovdje je ostavio lijepu uspomenu - svoje skije.

Pregledom materijala iz rukopisnog fonda Centralne biblioteke Republike Crne Gore "Đurđe Crnojević" na Cetinju, mogu se zapaziti zabilješke o fudbalu, koje je ostavio dr Niko Simov Martinović, naučni radnik, dugo godina direktor ove institucije. Kod njega nalazimo podatak da je prva fudbalska lopta na Cetinje donesena 1904. godine, i da je donio Mihailo-Mišo Vuković, iz Pariza, sin vojvode Gavra Vukovića, poznatog crnogorskog diplomate. Inače, za Vukovićevo ime vezani su počeci fudbala u Beranama. Prema navodima Huseina Tuzovića, prvu loptu je u Podgoricu donio 1909. godine Aleksin Miler, predstavnik jedne trgovačke firme iz Engleske. Miler se bavio fudbalom i hokejom. Igralište je bilo na mjestu gdje se danas nalazi zgrada Republičke zajednice zdravstvenog osiguranja Crne Gore, u neposrednoj blizini stadiona Budućnosti. Tada su počele da se igraju prve utakmice. Znatno ranije, Angell, (1991) je zabilje-

žio da se Crnogorci ljeti bave svim vrstama gimnastike i sportskih igara: loptanjem, skokovima, trčanjem, penjanjem, rvanjem . . . što ukazuje da se fudbalska igra pojavila u Crnoj Gori vjerovatno ranije.

Kad je Olimpizam u pitanju, možemo konstatovati da je najava prvih modernih Olimpijskih igara u Atini 1896. godine, pobudila je živo interesovanje u Crnoj Gori. Časopis *Luča*, koji je izlazio na Cetinju, iz 1895.godine, zalaže se za učešće Crne Gore na ovoj Olimpijadi, pa u tekstu *Olimpijske igre* autor nastoji da pored informisanja čitalaca i propagandno djeluje, da kod svojih čitalaca pobudi interes za ono što će se, godinu dana kasnije, dešavati u grčkoj metropoli. I za Olimpijske igre u Londonu u Crnoj Gori je postojalo određeno interesovanje, pa i spremnost za učešće, jer su bili uspostavljeni vrlo prisni odnosi Crne Gore sa Engleskim kraljevstvom, posredstvom kneza Batenberga za kojim je bila udata princeza Ana, kćerka Knjaza Nikole. Naročito su bili dobri odnosi knjaza sa Edvardom VII, koji je pokušao da nagovori crnogorskog suverena da Crna Gora učestvuje na četvrtim Olimpijskim igrama modernog doba u Londonu 1908. godine. Knjaz Nikola to nije mogao prihvatiti zbog velikih materijalnih troškova, a uz to sport u Crnoj Gori još nije dostigao olimpijske visine da bi se mogli nadati nekim uspjesima.

Zato i danas, s pravom, možemo reći da je tadašnja Knjaževina, odnosno kasnije Kraljevina Crna Gora pripadala redu zemalja koje su prihvatile, pratile i njegovale nove evropske sportske trendove i među prvima na ovim prostorima, brzo prihvatala sve evropske tekovine s kraja devetnaestog i početka dvadesetog vijeka. Spletom istorijskih okolnosti, ovako značajni istorijski događaji, kojih je u istoriji Crne Gore bilo dosta, sve do skoro su bili zaboravljeni ili zapostavljeni, uprkos tome što bi se tako važnim datumima i događajima podičile i mnogo veće i razvijenije nacije i države.

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### IMPORTANCE OF INTERNATIONAL RECOGNITION OF MONTENEGRO AT BERLIN CONGRESS FOR SPORTS DEVELOPMENT IN MONTENEGRO

*The Berlin Congress has been one of the greatest events in the European history since the nineteenth century and its decisions founded one of the most important dates in the history of Montenegro, which definitely solve the question of its international status. By decisions made at the Berlin Congress, Montenegro became the internationally recognized country in the world on the 13<sup>th</sup> July 1878. In that time period, for achieving international recognition of Montenegrin statehood, it comes to intensifying of diplomatic activity with European countries, which has extremely contributed to modernization of our country and sports development in it. Focus of Montenegro towards east and west was suitable for breakthrough of sports movement in here. Many activities got a real form and preconditions for more intensive development. We used historical method in this paper, as an aggregate of research procedures on events happened in the past. The most important sources of the data were the newspapers: Cetinjski vjesnik, then Ustavnost, Onogošt as well as some occasional publications. In the National Museum in Cetinje it is possible to find some data on teaching plans and programs for Montenegrin schools which at that time had a teaching program for subject called Gymnastics and military practicing, which was an independent subject in elementary and secondary schools. The capital of Montenegro, Cetinje, had an accelerated development in that time period thanks to intensive diplomatic activities with European countries. Many sports clubs and associations were founded. We must emphasize that the first golf club in Montenegro was founded in Cetinje in 1906. This important event has completed a rich sports history of Montenegro, whose space had been already taken by a wave of modern sports, such as ice skating, tennis, football, swordplay, skating, etc.*

**Key words:** international recognition, sports development, diplomatic activities.

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## **ADAPTIRANI PLIVAČKI TRENING USMJEREN NA POBOLJŠANJE PSIHOFIZIČKOG STATUSA STARIJIH OSOBA**

### **Uvod**

Danas smo suočeni s činjenicom da se, na globalnom nivou produžava čovjekov životni vijek a samim tim povećava broj starih ljudi. U razvijenim zemljama hronološko doba od 65 godina se najčešće pominje kao granica kada počinje starije životno doba, koje karakterišu značajne promjene u organskim funkcijama ali i u socijalnim relacijama. Zahvaljujući kvalitetnijoj medicinskoj njezi i savremenim načinima liječenja, prosječni životni vijek stanovništva je produžen. Tako, prema nekim podacima u razvijenim zemljama zapadne Evrope svaki četvrti stanovnik je stariji od 65 godina.

U naučnoj disciplini koja se bavi problemima starosti (gerontologiji), sve više se odbacuje teorija o istrošenosti organizma a njeno mjesto zauzima teza da su starenje i život isti procesi, čime se unosi u cijeli problem više optimizma, (Corbin B. C. et all, 2002) Takav više filozofski stav preokreće situaciju koja je vladala do sada, da se starost tretirala kao nesposobnost uz pratnju drugih neželjenih pokazatelja. Međutim, većina autora danas smatra da opšte slabljenje i naglo starenje nije direktan odraz tih objektivno smanjenih fizičkih sposobnosti koliko činjenica da se čovjek u tim godinama opusti i time postane manje spreman za aktivnosti kojima bi mogao usporiti procese starenja. Iz ovakvog pristupa proizilazi da bi centralni problem poboljšanja kvaliteta života starih ljudi mogao biti problem psihičke adaptacije na promjene koje prate pomenuo životno razdoblje.

### **Psihička adaptacija na starost**

Psihičke poteškoće adaptacije ogledaju se u tome što u starosti dolazi do osporavanja niza socijalnih motiva. Nemogućnost afirmacije ličnosti predstavlja gubitak autonomije, njeno izdvajanje iz društvenog konteksta, ili, u najblažem obliku, „pomjeranje na sporedni kolosjek“, (Zvonarević, M.,1981) sa kojeg se, često ne može postići više ništa osim fizičkog opstanka. A čovjek, kao ličnost, kako mnogi navodi Anderson, E.J. (1996), i u starosti može postići ispunjeniji i zdraviji način života, koji će pratiti odgovarajući kvalitet. Prema tome, starost koju srećemo u našem svakidašnjem životu ispod je nivoa koji čovjek može doseći kada bi uspio prebroditi krizu adaptacije, što predstavlja širi društveni problem. Osnovno je pravilo da se u starosti čovjek ne „predaje“, već da i nadalje bude aktivan, da radi i da stalno uči nešto novo.

Ono što je zajedničko većini gledišta na ovu životno doba (Bromley,D. 1998), je da u starosti nastupa prirodno opadanje energije i psihofizičkih sposobnosti uslijed biološke degeneracije. Upravo zbog toga je i zdravlje starih osoba često narušeno, što direktno utiče na smanjenje kvaliteta života. Kod velikog broja starijih radnika fizička radna sposobnost opada do kritičnog nivoa ukoliko se ne bave redovnim fizičkim

vježbanjem. U tom životnom dobu smanjena je i sposobnost uživanja u mnogim aktivnostima slobodnog vremena, ali i sposobnost za obavljanje elementarnih svakodnevnih fizičkih poslova. Jedna od najvažnijih mjera kod savjetovanja starih osoba je težnja da se oni sačuvaju od pada u potpunu pasivnost.

Značajan zadatak svakog naprednog društva je podrška starim ljudima da svoju starost pretvore u što aktivniji i produktivniji životni period, što će svakako pratiti i subjektivni osjećaj potpunijeg i kvalitetnijeg života.

### **Psihofizičke sposobnosti starih osoba**

Običnim posmatranjem zapaža se tendencija da se sa starenjem usporava cjelokupno ponašanje. Sistematska ispitivanja registruju to usporavanje mjerenjem vremena reakcije, kao i proučavanjem složenih motornih reakcija i raznih psihomotornih vještina kod ispitanika raznih uzrasta. U traženju uzroka usporavanja reakcija sa starenjem išlo se u tri pravca. Postavljena su pitanja koliki udio u tom usporavanju ima opadanje oštine čula kojima se opaža signal, zatim da li je usporavanje sa starenjem djelimično rezultat smanjenja brzine provođenja impulsa perifernih nerava i, na kraju, da li osnovu za ovo usporavanje čine promjene u centralnom nervnom sistemu. Poznato je da sa starenjem dolazi do smanjenja čulne osjetljivosti i prirodno je pretpostaviti da to može usporiti reakciju. Ispitivanjima je provjeravana i druga pretpostavka da smanjenje provodljivosti perifernih nerava starih ljudi dovodi do povećanja vremena reakcije. Rezultati brojnih ispitivanja su pokazali da se provodljivost nerava sa starenjem zaista smanjuje. Ali, to smanjenje je tako malo da se njime ne može objasniti veće opadanje vremena reakcije kod starih ljudi. Nije ostalo ništa drugo nego da se zaključi da usporavanje reakcija u toku starenja odražava starenje centralnog nervnog sistema. Jedan element je propadanje nervnih ćelija do koga izgleda dolazi u cijelom nervnom sistemu u toku starenja. Drugi element su fizičko-hemijske promjene u sinapsama koje ograničavaju brzinu provođenja impulsa kroz nerve i smanjuju pobuđivanje nervnog sistema.

S obzirom da stariji ljudi postaju sporiji oni moraju da mijenjaju svoje ponašanje reagujući na okolinu. Međutim, to smanjenje brzine ne djeluje direktno na efikasnost starih osoba. Stare osobe uglavnom su u stanju da kompenzuju ovo smanjenje brzine, prije svega povećanjem pažnje prilikom rada. U nekim slučajevima pri ovakvom smanjenju brzine stare osobe mogu čak postići i poboljšanje vještine, što je čest slučaj kad se radi o osobama koje su se redovno bavile fizičkim vježbanjem. Mišićna vlakna kod starijih gube na elastičnosti, kao i brzina reagovanja na draž, a takva promjena svojstva može da izazove štetan uticaj na mišićni sistem u smislu anatomske oštećenja različitog stepena. Ni dugotrajni naponi ne bi trebalo da se upražnjavaju u staračkom periodu zbog smanjenja aerobne moći. Zbog toga stare osobe nisu sposobne za dugotrajne napore umjerenog i većeg intenziteta. Promjene na kostima i zglobovima nisu vezane za ranu starost, ali počeci ovih promjena nastaju već kod mnogih oko 50 godine života. One su naročito naglašene na dugim kostima i kičmenim pršljenovima, te je lomljenje ovih kostiju češće u tom dobu. Regresivne promjene se ispoljavaju smanjenjem sadržaja Kalijuma i Fosfora koji daju čvrstinu kostima, pa dolazi do osteomalacije. Na zglobovima često dolazi do degenerativnih promjena na djelu hrskavice. One mogu



dovesti do ogolićenja zglobnih površina u dubljoj starosti, što može rezultirati znatnim smetnjama u pogledu pokretljivosti zahvaćenih zglobova. One su naročito izražene na zglobovima gornjih i donjih ekstremiteta i kičmenog stuba. Poseban etiološki činilac ovih promjena je fizičko opterećenje zglobova nogu i kičmenog stuba pri obavljanju određenih profesija.

### **Unapređenje psihofizičkog statusa starih osoba kroz fizičku aktivnost**

Jedan od prioriteta ciljeva savremenog čovječanstva je produženje aktivnog ljudskog života. Produženjem života stanovništva i produženjem njihovog aktivnog perioda pomjeraju se granice starosti. Činjenica je, da se psihomotorne sposobnosti duže održavaju i sporije opadaju kod osoba koje se redovno bave fizičkim aktivnostima. Starijim ljudima je potrebno obezbijediti takvu vrstu i karakter vježbi, kao i obim opterećenja, koje će odgovarati njihovom hronološkom dobu i polu, ali i opštem zdravstvenom stanju kao i nivou motoričkih i funkcionalnih mogućnosti. Fizičko vježbanje starijih osoba može se organizovati individualno ili, što je češći slučaj grupno. Ono što se u svakom slučaju podrazumijeva je nadzor i vođstvo stručne osobe. Pri određivanju vrste i dužine trajanja fizičkog vježbanja starijih osoba prvenstveno mora da se vodi računa o:

- starosti i polu svakog pojedinca;
- navikama starijih ljudi u pogledu fizičkog vježbanja;
- zdravstvenom stanju;
- motoričko-funkcionalnim mogućnostima i
- stepenu izraženosti eventualnih degenerativnih promjena.

Ne smije se zanemariti činjenica da doba starosti, koje najčešće prate atrofični i degenerativni procesi, a naročito arteroskleroza, čine da izvjesni tjelesni naponi mogu da budu štetni po organizam starijih, ukoliko nijesu pravilno odabrani i dozirani. Često se među starijim osobama nalaze oni kod kojih se ne podudaraju kalendarska i biološka starost. U takvim slučajevima je potrebno posebnu pažnju obratiti na procjenu odstupanja biološkog od hronološkog doba i sa tim u skladu vršiti izbor vježbi i homogenizaciju grupa.

Prilikom formiranja grupa prvo je potrebno izvršiti zdravstvenu kontrolu kojom se mogu evidentirati pojedinci kod kojih je fizička aktivnost kontra-indikovana, što ih isključuje iz procesa fizičkog vježbanja. Homogenizacija grupa starih osoba za uključivanje u organizovanu fizičku aktivnost vrši se prema sljedećim karakteristikama:

- grupa starijih osoba bez ozbiljnijih funkcionalnih i zdravstvenih smetnji, kod kojih se podudaraju biološko i hronološko doba;
- grupa koju čine starije osobe sa prisutnim funkcionalnim i zdravstvenim poremećajima blažeg oblika;
- grupa sa izraženim funkcionalnim smetnjama koje su u biološkom smislu “starije” od svog hronološkog doba.

### **Prednosti fizičkog vježbanja u vodenoj sredini**

Prednosti koje fizičko vježbanje u vodenoj sredini pruža starijim osobama u odnosu na vježbanje na suvom su brojne. Prema Arhimedovom zakonu svako tijelo uronjeno u tečnost prividno gubi onoliko od svoje težine koliko teži tim tijelom istisnuta tečnost. Voda djeluje na tijelo tako da ga potiskuje ka površini kada je ono djelimično ili potpuno uronjeno u nju. Sila potiska podupire vježbača u toku trenažnog programa i omogućava izvođenje raznih vrsta vježbi kao i precizno doziranje obima i intenziteta opterećenja. Zbog rasterećenja pritiska na koštano zglobovi sistem, a posebno na kičmeni stub smanjena je mogućnost povređivanja u odnosu na adekvatne vježbe koje se mogu raditi na tlu. U isto vrijeme, kada podupire tijelo, voda daje i otpor pokretu. Svaki pokret bilo koje brzine ili smjera je usporen u vodi usljed ovog otpora. Pošto je voda gušća od vazduha mišići ulažu veći napor krećući se kroz vodu nego kroz vazduh. Upravo gustina vode omogućava bolju kontrolu pokreta i omogućava starijim osobama sigurnije i raznovrsnije vježbanje. Osim toga, samim ulaskom u vodu nastupa potpuna promjena raspoloženja. Prethodno akumuliran stres iščezava, pa se ove prednosti vježbanja u vodi mogu koristiti u okviru raznih anti-stres programa.

Trening u vodi se može primjeniti na osobe različitog nivoa fizičkih sposobnosti i nivoa treniranosti (Counsleman, J.E.1978). Pritom, ako se fizičko vježbanje organizuje u plitkoj vodi, znanje plivanja nije neophodno, ali je, svakako poželjno. Većina situacija koje zahtijevaju terepeutski tretman rezultat su uticaja koji djeluje na tijelo usljed lošeg držanja. Upravo adekvatan izbor vježbi u vodi jača posturalne mišiće i time pospešuje pravilno držanje tijela.

Svaka vježba u vodi može da omogući podjednako opterećenje mišića agonista i antagonista. Poslije svakog pokreta usmjerenog protiv otpora vode treba da slijedi pokret vraćanja nazad u početnu poziciju. Pritom je potrebno voditi računa da se svaki pokret izvodi približno istom silom.

Vježbanje u plitkoj vodi je naročito pogodno je za starije osobe, različitih tjelesnih sposobnosti i nivoa treniranosti. Vježbe u plitkoj vodi omogućavaju kvalitetan i uravnotežen aerobni i anaerobni trening kao i uticaj na svaku regiju, a posebno na: mišiće ruku i ramena, grudi, leđa, trbuha, i nogu. Adaptirani trening u vodi treba da bude u skladu sa trenutnim sposobnostima, tako da su da su za različite nivoe sposobnosti predviđeni i različiti programi vježbanja.

### **Zaključak**

Adaptirane fizičke aktivnosti u vodi prilagođene mogućnostima i zdravstvenom stanju starijih osoba pružaju niz prednosti u odnosu na vježbe koje mogu da se izvode izvan vodene sredine. One proističu prije svega, iz fizičkih karakteristika vode u kojoj se prema Arhimedovom zakonu prividno umanjuje težina tijela za vrijednost zapremine vode koji tijelo istisne. Iz tog razloga se u tim uslovima smanjuje dejstvo gravitacije pa zglobovo-koštani sistem trpi znatno manje opterećenje nego prilikom vježbanja na suvom. S obzirom da je posturalni sistem kod starijih osoba često oslabljen, veliku prednost prilikom vježbanja u vodi omogućava horizontalni položaj koji osoba zauzima. To

blagotvorno djeluje prije svega na kičmeni stub, koji je u horizontalnom položaju rasterećen uticaja gravitacije što blagotvorno djeluje na kičmena oboljenja.

Rezultari brojnih istraživanja ukazuju da se vježbanjem u vodi harmonično opterećuje organizam tako da je moguće aktivirati skoro cjelokupnu muskulaturu gornjih i donjih ekstremiteta, trupa i karličnog pojasa. Pritom se ubrzava cirkulacija krvi, srce dobija veću količinu krvi što direktno utiče na povećanje minutnog volumena. Vremenom sve to dovodi do povećanja snage srčanog mišića i do trajnijeg povećanja njegovih funkcionalnih sposobnosti. Uz to, aktivirana muskulatura omogućava brzu relaksaciju organizma, poboljšanje i održavanje kardiorespiratornih funkcija. Mogućnost tačnog doziranja opterećenja prilikom vježbanja u vodi omogućava izvanredan vid terapije kod starijih ljudi kao i onih sa oslabljenim funkcionalnim sposobnostima.

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### ADAPTED SWIMMING TRAINING FOCUSED ON IMPROVEMENT OF PSYCHOPHYSICAL STATUS OD ELDERLY PEOPLE

*In developed countries chronological age from 65 is often mentioned as age limit when elderly age begins, and which is featured by significant changes in organic functions as well as social relations. Thanks to qualitative medical care and contemporary ways of treatment, an average lifetime has been prolonged. Therefore, according to some data in developed countries of Western Europe, each fourth inhabitant is older than 65. However, there psychophysical abilities are very weak, and health is bad, which directly influences reduction in life quality. Adapted physical activities in water adjusted to abilities and health state of elderly people give a series of advantages in comparison to exercises that can be done outside water. For that reason, in those conditions effect of gravity is reduced, so PTO-skeletal system suffers a lot less burden than at practicing on dry. Taking into account that postural system with elderly people is often damaged, great advantage at exercising in water enables horizontal*

position that a person takes. That beneficially influences first of all the spinal cord, which is in horizontal position disburdened from gravity influence which beneficially affects spinal disease. Results of numerous researches indicate that swimming harmonically burdens organism so that almost all musculature of upper and lower extremities, belly and pelvis area. Swimming accelerates blood circulation, heart gets more blood which directly affects increase in minute volume. Eventually it all brings to fortification of heart muscle and more permanent increase of its functional abilities. Possibility of precise dosing of burden in swimming enables excellent way of therapy with elderly people and those with damaged heart. Besides, activated musculature enables fast relaxation of organism and restore of all vital cardio respiratory functions to the level of standard work ability of a man.

**Key words:** adapted training, psychophysical status, elderly people

„Dan“, 13. januar 2015.



Проф. др Душко Бјелица

МЕЂУНАРОДНА НАУЧНА КОНФЕРENCIЈА ЦРНОГОРСКЕ СПОРТСКЕ АКАДЕМИЈЕ ОДРЖАЊЕ СЕ ОД 2. ДО 5. АПРИЛА У ПОДГОРЦИ

## Највећи регионални научни скуп у спорту

Међународна научна конференција, коју традиционално организује Црногорска спортска академија у сарадњи са УЦГ и Факултетом за спорт и физичко васпитање из Никшића, под називом „Трансформациони процеси у спорту“, која ће се одржати од 2. до 5. априла 2015. године у Подгорици постаје све препознатљивија у научној јавности на свим континентима.

– Прошле године су стандарди за пријаву радова унапријеђени према европским стандардима, а структура учесника се драстично промијенила, будући да су захтеви за прихватање радова на много вишем нивоу, те је на протеклој конференцији учешће узео велики број научника из већине европских земаља, али и аутора из азијских земаља, као што су Јапан, Тај-

ланд и Тајван, рекао је доц. др Стево Поповић, један од организатора предстојећег научног скупа. – Са друге стране, број аутора из региона се смањно, прије свега из разлога што је постављен највиши научни праг за прихватање пријављених радова. Ове године, је тренд сличан, те очекујемо да ће предстојећа конференција бити још квалитетнија и посјећенија, судећи по пресеку

пријављених радова на данашњи дан, рекао је Поповић. Председник приређивачког одбора проф. др Душко Бјелица је истакао да се ради о највећем научном скупу у региону.

– Са задовољством могу да кажем да је пријављено до сада 116 радова, од тог броја за сад је прихваћено након рецензије 70 радова. Остали су одбијени или су враћени на дораду. За сад су пријављени

аутори из 19 земаља свијета, а за сад их је највише из Турске. Пријављивање за учешће се завршава 15. јануара и сви прихваћени и презентовани радови биће објављени у стручном часопису Спорт Монт који се налази у пет индексних база, а три најбоља рада објавиће се у међународном научном часопису „Montenegrin Journal Science Sports and Medicine“, који се налази у чак 30 међународних индексних база, рекао је проф. др Душко Бјелица.

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## PSIHOLOŠKO-PEDAGOŠKI ASPEKTI OBUKE NEPLIVAČA

### Uvod

Potpuna priprema plivača od početne obuke do postizanja vrhunskih rezultata predstavlja dugotrajan proces, koji radi svoje složenosti zahtijeva od učitelja plivanja ili trenera poznavanje niza naučnih disciplina kao što su psihologija i pedagogija, anatomija i fiziologijai sl. Iako priprema plivača ima za svrhu postizanje dobrih rezultata, ne smijenio zaboraviti da tokom ove pripreme, isto kao i za vrijeme početne obuke, trebamo i vaspitno uticati na mlade plivače. Bez vaspitnog uticaja trenera, ni početna obuka plivanja ni priprema za ozbiljnije bavljenje plivanjem ne može biti kompletna. Vaspitanje i psihofizičku pripremu treba ukomponovati u jedinstvenu cjelinu i samo treneri koji s ovog stanovišta pristupe organizaciji obuke plivanja i kasnije trenažnog procesa u plivanju mogu imati uspjeha u radu. Vaspitna strana procesa učenja plivanja često se zanemaruje, ali isto tako i treningu se često prilazi jednostrano, jer se smatra da je dovoljno da plivač obavi intenzivan rad na suvom i u vodi da bi savladao određenu tehniku a potom dugotrajnim treningom konačno postigao zapažene rezultate.

U toku obuke neplivača ne smiju se zanemariti ni dva bitna područja psihologije, a to su kognitivno (spoznajno) i afektivno (osjećajno) područje. Kognitivno područje se odnosi na usvajanje svih znanja koja omogućavaju plivačima da svjesno, stvaralački prilaze čitavoj problematici sopstvenog usavršavanja. Plivači, konačno, trebaju zauzimati pozitivne etičke i estetske stavove, a pritom trebaju usmjeriti u jedinstven sopstveni sistem sve vrijednosti stečene tokom obuke i usavršavanja tehnike plivanja.

Kao i svaku drugu motornu aktivnost tako i plivanje treba shvatiti kao jedinstven proces učenja, usavršavanja i automatizacije specifične motorike, (Volčanšek, B.1996). Posebno treba shvatiti da je to proces koji ima svoje programsko-vremenske etape ili faze koje se smjenjuju određenim redoslijedom. Prema Kerlović, A.1999., toku procesa učenja neplivači prolaze kroz nekoliko logičnih etapa. Prvu etapu čini samo učenje plivanja, odnosno, ovladavanje sposobnošću održavanja tijela na površini vode. Drugu etapu čini učenje različitih tehnika plivanja. Treću etapu čini sposobnost da se naučene tehnike plivaju maksimalnom brzinom bez narušavanja tehnike. Ove tri navedene faze učenja plivanja su tipični metodičko-empirijski konstrukti, ali zasnovani na teorijskim osnovama psiho-fiziološke teorije učenja.

### Proces formiranja motornih navika

Dijete po rođenju počinje da vrši određene motorne pokrete i radnje a na osnovu urođenih refleksnih pokreta uči mnoge nove pokrete. Učenje novih kretanja iz oblasti motorike odvija se spontano, neorganizovano, a kasnije, kroz nastavu fizičkog vaspitanja, to se postiže realizacijom organizovanog procesa, budući da je nastava najorganizovaniji sistem obrazovanja. Proces učenja motornih radnji u vodi je znatno teži nego

na tlu, jer voda za čovjeka nije prirodna sredina u kojoj živi. Boravak u vodi treba razumjeti kao nešto potpuno novo i nepoznato, tako da sva kretanja koja čovjek nauči u vodi pripadaju kategoriji stečenih znanja i navika. Kretanje kroz tečni fluid otežavaju sile i otpori koji se ne susreću na tlu. Znanje plivanja se ogleda i u sposobnosti da se te sile iskoriste. Uz to, kod većine ljudi koji ne plivaju dovoljno dobro prisutan je strah od utapanja (hidrofobija), što se negativno manifestuje na proces učenja kretanja u vodi. S obzirom na specifičnosti koje se susreću u procesu stvaranja motornih navika u vodi, proces učenja mora da se dobro osmisli i organizuje. Kako za formiranje motornih navika treba da prođe određeno vrijeme i pošto je to proces koji treba da se sprovodi po određenim pedagoškim principima, to svaki nastavnik i trener treba da zna osnovne karakteristike i zakonitosti koje važe u tom procesu.

### **Psihološki aspekti obuke neplivača**

Generalno definisano, učenje predstavlja složen proces koga, prema Lazareviću (1987) i Dunđeroviću (1999), čine odgovarajuća misaona i motorna aktivnost koja prolazi kroz četiri faze.

*Prva faza* predstavlja sam početak obuke, kada se onaj koga treba naučiti vještini plivanja upoznaje sa određenim kretanjem. Tada učitelj plivanja nastoji da adekvatnim postupkom stvori kod učenika jasnu predstavu o kretanju. Koristeći sva raspoloživa nastavna sredstva, pedagoško znanje i iskustvo, nastavnik nastoji da ta predstava bude što jasnija. Zatim učenici izvode dato kretanje. U ovoj fazi ti prvi pokreti su obično netačni, grčeviti, sa dosta povećanim mišićnim tonusom, sa velikim brojem nepotrebnih i suvišnih pokreta. Osnovni uzrok nastanka ove pojave je činjenica da u fazi prvih pokušaja u motorne centre kore velikog mozga dolazi veliki broj nadražaja. Iz tih centara se šalju impulsi u određene lokomotorne strukture, što uslovljava da se javi veći broj pokreta, potrebnih ali i onih koji su nepotrebni i koji negativno utiču na tehniku kretanja. Zato ovu fazu nazivamo *fazom široke iradijacije*. Kretanja su u ovoj fazi po formi i karakteru neprecizna, sa kraćim ili prevelikim amplitudama, sa povećanom ili umanjenom brzinom. Praktično izvođenje vježbi u ovom razdoblju zato se i odlikuje odsustvom usklađenosti pokreta, suvišnim mišićnim naprezanjem, prisustvom mnogih suvišnih pokreta i grčevitošću čitavog organizma, što sve zajedno daje utisak neumiješnosti i uslovljava mnoge teškoće.

Treba istaći da je u ovoj fazi učenja motornih navika učešće svijesti vrlo izraženo i značajno. Pravilnom i dobrom organizacijom nastavnog procesa, mnogobrojnim ponavljanjima, postepeno dolazi do selektiranja pokreta i eliminisanja onih suvišnih, a povećani mišićni tonus postepeno opada. Poslije izvjesnog broja održanih časova kojima je uvijek bavano određeno kretanje dolazi do kvalitetnih promjena.

*Druga faza* formiranja motornih navika odlikuje se diferencijacijom pravih (potrebnih) od suvišnih pokreta, pa se ta faza naziva *fazom diferencijacije*. Eliminise se veći broj suvišnih pokreta koji su ometali pravilno kretanje i izazivali poremećaje u položaju tijela. Koordinacija kretanja postaje bolja i kvalitetnija, dolazi do pravilnijeg smjenjivanja mišićne napetosti (povećanog tonusa) sa mišićnom opuštenošću (relaksacijom), što uslovljava manju potrošnju energije i ekonomičniji mišićni rad.

U ovoj etapi učenja dolazi do saznanja o napredovanju, već se uočavaju rezultati rada, što pozitivno utiče na dalju angažovanost i mobilnost učenika. Učešće svijesti vježbača se u odnosu na prethodnu fazu smanjuje, ali je i dalje neophodno. Osnovno sredstvo obuke u ovoj fazi je višekratno plivanje u cjelini, dok je vježbanje pojedinih delova tehnike od manjeg značaja.

*Treća faza*, koju u psihologiji učenja nazivamo *fazom koncentracije*, karakteriše se kvalitetnijim i intenzivnijim nadražajem i istovremeno ekonomičnijim pokretima što uslovljava još uvek vrlo svjesno izvođenje osnovnih pokreta uz maksimalnu aktivnost piramidalne motorne zone. Izvođenje pojedinih djelova ciklusa je još uvijek svjesno, ali znatno manje nego u prethodnoj fazi. Usljed ponavljanja, redosljed pojedinih pokreta postaje sve više nesvjestan pa zato sada i govorimo o automatizaciji pokreta. Nastavnik na ovom stepenu učenja koriguje tehniku, i ispravlja greške. U radu sada više preovlađuje vježbanje elemenata u cjelini. Poželjno je otežavati uslove, ali, pritom biti obazriv. To, upravo, znači povećavati brzinu i tempo i organizovati neki oblik takmičenja. U ovoj fazi, kada se potpuno i definitivno kod svakog pojedinca formira određeno kretanje, dolazi do ugrađivanja elemenata individualnog u tehniku plivanja. Veći broj pedagoga i sportskih trenera smatra da treba zadržati u tehnici pojedinca onaj segment individualnog koji se pokazuje efikasnim.

*Četvrti*, zadnja faza kroz koju mora da prođe motorna navika je *faza stabilizacije*. Na ovom stepenu učenja se praktično već automatizovana navika učvršćuje i definitivno oblikuje što se najbolje ostvaruje permanentnom promjenom uslova vježbanja (isplivavanje različitih dužina u različitom tempu, raznim brzinama itd.). Mentalni obrasci pokreta prenose se u ekstrapiramidalne motorne centre i dijete-plivač, na neki način, "bez razmišljanja" izvodi kompletnu plivačku tehniku.

Na ovoj fazi obuke svi elementi određene tehnike plivanja se povezuju u jednu cjelinu, upravo onako kako se traži na takmičenju (start, plivanje, okret, dolazak na cilj). Više nego na prethodnim etapama učenja plivanja, ovdje se koristi metod takmičenja. Međutim, i nadalje treba biti obazriv, jer ako nije došlo do potpune stabilizacije, moguća je pojava starih grešaka pri realizaciji tehnike. Ovo se obično javlja ukoliko su zahtjevi veći od mogućnosti plivača pa dolazi do zamora.

Poznato je da se pri realizaciji pravilne tehnike plivanja zamor javlja kao činilac koji najviše utiče nepovoljno na pravilnu koordinaciju. Kad se pojavi zamor treba smanjiti intenzitet vježbanja, napraviti pauzu u radu, promijeniti aktivnost i sl. Kretanja koja dostignu maksimalni stepen automatizacije ne mogu se izjednačiti sa kretanjima koja čovjek izvodi refleksno (urođeni pokreti). Kod naučenih motornih navika, iako minimalno, ipak se osijeća prisustvo svijesti, a potrošnja energije je veća nego kod refleksnih radnji. Dok prve tri etape treba shvatiti kao etape kada se "uči" kretanje, kao period edukacije, četvrtu treba razumjeti kao fazu kada se naučeno uvježbava i učvršćuje, pa se iz tih razloga rad u ovom periodu podudara sa radom i zahtjevima koji se susreću na treninzima sportista.

Primjenjujući navedenu shemu formiranja motornih navika u plivanju, valja imati na umu da početnik ima niz poteškoća koje nastaju u ovoj novoj sredini, (Matković, I.1992). Ovo nije slučaj u drugim sportskim disciplinama. Početnik na suvom

Iako može oponašati plivačke pokrete, ali to ne može učiniti u vodi jer mozak dobiva niz nepoznatih nadražaja koji otežavaju kontrolu pokreta.

Iako teško možemo precizirati sve faze učenja plivanja, praksa nas upućuje na sljedeću podjelu:

1. Faza početnog obučavanja;
  - a) Potfaza upoznavanje vodene sredine,
  - b) Potfaza obučavanja jednostavnih tehnika plivanja.
2. Faza obučavanja sportskog plivanja. U ovoj fazi se formira dinamički stereotip;
3. Faza usavršavanja tehnika plivanja.

*Kognitivni zadaci:*

- saznavanje specifičnih podataka:
  - a) sposobnost njihovog prepoznavanja,
  - b) sposobnost njihove obnove.
- razumijevanje dobijenih podataka sa sposobnošću njihovog interpretiranja i razvrstavanja;
- osposobljavanje učenika da primjeni svoja znanja u novim područjima;
- sposobnost analiziranja već poznate situacije kao i sinteze na osnovu poznatih elemenata;
- sticanje sposobnost za procjenu vrijednosti sredstava i metoda za postizanje određenog cilja.

*Afektivni zadaci:*

- jedinstvena recepcija s pasivnom pažnjom;
- svojevolutna reakcija na stimulanse;
- valorizacija sopstvene aktivnost sa stanovišta svojevolutnog reagovanja;
- konceptualizacija svake vrijednost na koju je već reagovao;
- organizacija vrijednosti u koherentne sisteme koji postaju dio njegovog sopstvenog sistema vrijednosti.

*Psihomotorički zadaci:*

Učenika treba osposobiti da:

- oponaša pokazane aktivnosti;
- racionalno koristi naučena motorna znanja u situacijama analognim onima tokom učenja;
- svoje reakcije prilagodi potrebama novih situacija i sagledava priliku da tu vrstu aktivnosti praktikuje;
- preinači naučene motorne sheme radi njihovog prilagođavanja sopstvenim sposobnostima i tako stvori svoj sopstveni stil;
- konceptualizuje svaku vrijednost na koju je već prepoznao i reagovao;
- Učenik je sposoban da shvati nova motorna znanja i da ih izvodi prikladno u adekvatnim situacijama.



### **Pedagoški aspekti obuke neplivača**

Nastavnici i treneri pronalaze najpogodnije načine usvajanja tehnike plivanja. Prema Ahmetović, Z. i Matković, I. (1995), pri učenju plivanja koristi se sintetička, analitička i kompleksna metoda. Osim ovih, koje se odnose na aktivan rad plivača postoje i pomoćne metode koje pomažu da se plivanje bolje savlada. Pomoćne metode u plivanju su: metoda usmenog izlaganja, metoda demonstracije i metoda igre.

#### *Sintetička metoda učenja plivanja*

Ova metoda se najviše primjenjuje u sportskom vježbanju, a sastoji se u tome da se kretanje odvija u cjelini. Opšta forma i suština tehnike zadržavaju se, a pojedini elementi se usavršavaju i tako se stvara individualna tehnika koja odgovara anatomskim i fiziološkim mogućnostima plivača. Sintetička metoda ne može se primijeniti na početku plivačke aktivnosti i zbog toga što u toj fazi generalizacije, kao što je već rečeno, u kori velikog mozga prevladavaju procesi nadraživanja motornih centara pa početnik pokrete izvodi nespretno. Prema tome, rad s početnicima u početnoj fazi učenja plivanja (upoznavanje vodene sredine i učenje najjednostavnijih tehnika plivanja) mora biti drukčiji. Početnicima dajemo veliku slobodu u radu, a sintetička metoda svodi se na slobodna kretanja u plićaku. Na taj način početnik savladava pripremne vježbe i najjednostavnije tehnike plivanja.

#### *Analitička metoda učenja plivanja*

Raščlanjivanje pokreta na sastavne dijelove (elemente) zovemo analitičkom metodom. Ovu metodu plivači naročito koriste u drugoj potfazi početnog podučavanja kada se pojedini elementi tehnike uče odvojeno, a zatim se postepeno povezuju u koordiniranu cjelinu. Analitičku metodu ne možemo koristiti kao cjelovitu i samostalnu metodu, već nam ona služi samo za podučavanje pojedinih elemenata tehnike što rezultira konačnim dobrim savladavanjem čitave tehnike. Valja odmah istaći da analitička metoda može i negativno uticati na formiranje tehnike jer usporava savladavanje potpunog kretanja i stvara velike pauze između elemenata.

Analitičkom metodom izvode se slijedeće vježbe:

1. Imitacione vježbe koje daju plivaču osnovnu predstavu o pokretu.
2. Specijalne vježbe koje se izvode istom brzinom kao i prilikom sinteze pokreta. Ove vježbe ne stvaraju samo predstavu o pokretu nego se stiče i potpunija slika karakteristikama tog pokreta.
3. Vježbe za korigovanje pokreta mogu biti kontrastne ili pomoćne. Kontrastne vježbe uklanjaju greške svojim suprotnim djelovanjem.
4. Uvodne vježbe su kombinacija imitacionih i specijalnih vježbi koje se postupno usaglašavaju povećanjem broja elemenata i brzine pokreta te povećanjem veličine mišićnog naprezanja. Pokrete nije dovoljno analizirati samo u vodi. Poželjno je da se imitacioni plivački pokreti izvode, radi sticanja predstave, najprije na suvom, u plićaku i na kraju u dubokoj vodi

Raščlanjene plivačke pokrete ne valja učiti do potpune automatizacije jer će ona nastati u koordinisanom plivanju. Izvođenje odvojenih pokreta ne odvija se, ipak, u uslovima potpune tehnike (drukčiji položaj tijela, drukčija brzina pokreta i dr.) te bi

želja za postizanjem automatizacije elemenata pokreta stvorila poteškoće pri formiranju kompletnog kretanja u vodi.

#### *Kompleksna metoda učenja plivanja*

Već je istaknuto da se analitička metoda zapravo ne može koristiti kao samostalna metoda. Početnici najbolje savladavaju tehniku plivanja tako da kroz pripremne vježbe u cijelosti vrše plivačke pokrete. Pripremne vježbe moraju sadržavati i neke prirodne elemente plivačkih tehnika (npr.-udarci nogama kraulom u vježbama klizanja). Plivači, nadalje, izučavaju elemente plivačkih tehnika (analitička metoda) i koordinišu ih, da bi konačno mogli slobodno plivati jednostavnom tehnikom (sintetička metoda). Ovakav način rada kada se od sintetičke metode prelazi na analitičku, a zatim se opet vraća na sintetičku, nazivamo kompleksnom ili analitičko-sintetičkom metodom. Poslije usvojenih jednostavnih tehnika plivanja plivač lako usavršava sportske tehnike plivanja.

#### *Faze u obučavanju plivanja*

Za kvalitetnu obuku učitelji plivanja i treneri moraju odabrati pravilan redoslijed učenja plivačkih tehnika. Početna obuka u modernim školama plivanja bazira se na učenju kraula i leđnog plivanja jer su naizmjenični pokreti ovih tehnika prirodni i slični hodanju i trčanju. Ipak se može preporučiti da prije obučavanja kraula i leđnog plivanja početnici usvoje različite jednostavne pokrete u vodi (jednostavne tehnike). Ove pokrete treba izvoditi bez velikih tehničkih zahtjeva, ali ipak u sebi moraju sadržavati niz elemenata kraula i leđnog plivanja. U početnom obučavanju modernog plivanja nije primarna određena analitička razrada tehnike koja odmah teži savršenstvu. Budućem plivaču valja dati mogućnost da usvajanjem raznolikih, u početku jednostavnih pokreta, stekne okretnost u vodi i razvije početnu plivačku motoriku.

#### **Zaključak:**

Proces učenja vještine plivanja je veoma složena i dugotrajna aktivnost. Pored pukog ponavljanja zadate motorne radnje neophodno je uvažiti i niz psihološko-pedagoških zakonitosti, počev od onih opštih, pa do krajnje individualnih koje su specifične za svakog vaspitanika ponaosob. Kao i svaku drugu motornu aktivnost tako i plivanje treba shvatiti kao jedinstven proces učenja, usavršavanja i automatizacije specifične motorike. Posebno treba imati na umu da je to proces koji ima svoje programsko-vremenske etape ili faze koje se smjenjuju određenim redoslijedom. U toku procesa učenja neplivači prolaze kroz nekoliko logičnih etapa. Prvu etapu čini samo učenje plivanja, odnosno, ovladavanje sposobnošću održavanja tela na površini vode. Drugu etapu čini učenje različitih tehnika plivanja. Treću etapu sposobnost da se naučene tehnike plivaju maksimalnom brzinom bez narušavanja tehnike. Ove tri navedene faze učenja plivanja su tipični metodičko-empirijski konstrukti, ali zasnovani na teorijskim osnovama psiho-fiziološke teorije učenja.

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### PSYCHOLOGICAL-PEDAGOGICAL TRAINING ASPECTS FOR NON-SWIMMERS

*In developed countries chronological age from 65 is often mentioned as age limit when elderly age begins, and which is featured by significant changes in organic functions as well as social relations. Thanks to qualitative medical care and contemporary ways of treatment, an average lifetime has been prolonged. Therefore, according to some data in developed countries of Western Europe, each fourth inhabitant is older than 65. However, there psychophysical abilities are very weak, and health is bad, which directly influences reduction in life quality. Adapted physical activities in water adjusted to abilities and health state of elderly people give a series of advantages in comparison to exercises that can be done outside water. For that reason, in those conditions effect of gravity is reduced, so PTO-skeletal system suffers a lot less burden than at practicing on dry. Taking into account that postural system with elderly people is often damaged, great advantage at exercising in water enables horizontal position that a person takes. That beneficially influences first of all the spinal cord, which is in horizontal position disburdened from gravity influence which beneficially affects spinal disease. Results of numerous researches indicate that swimming harmonically burdens organism so that almost all musculature of upper and lower extremities, belly and pelvis area. Swimming accelerates blood circulation, heart gets more blood which directly affects increase in minute volume. Eventually it all brings to fortification of heart muscle and more permanent increase of its functional abilities. Possibility of precise dosing of burden in swimming enables excellent way of therapy with elderly people and those with damaged heart. Besides, activated musculature enables fast relaxation of organism and restore of all vital cardio respiratory functions to the level of standard work ability of a man.*

**Key words:** adapted training, psychophysical status, elderly people.

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## KVANTITATIVNE I KVALITATIVNE RAZLIKE MORFOLOŠKIH KARAKTERISTIKA I EKSPLOZIVNE SNAGE NOGU ODBOJKAŠICA I ODBOJKAŠA KADETSKOG UZRASTA

### UVOD

Odbojka je polistrukturalna, aciklična kompleksna aktivnost, kolektivna sportska igra koja pred igrače postavlja brojne zahteve definisane pravilima i strukturom igre. Kroz fazu organizacije odbrane i napada dolazi do razmene poena koje karakterišu visoko-intenzivne aktivnosti uz periode kratkotrajnog odmora. Smatra se anaerobnim sportom sa metaboličkim zahtevima za energijom zadovoljenim dominantno fofsfokreatinskim sistemom (Kunstlinger et al., 1987). Da bi postigli visok nivo izvođenja, od igrača se zahteva ne samo usavršavanje tehničko-taktičkog umeća već i ispoljavanje visokog nivoa motoričkih sposobnosti eksplozivne snage, brzine i agilnosti. Dinamička snaga je dominantna u odbojci jer je ovaj sport saturiran skokovima i brzim promenama pravca kretanja. Da bi ostvarili prednost u odnosu na protivnika, svi elementi kojim je to moguće, teži se izvođenju u skoku (Ziv & Lidor, 2010). Veći broj istraživanja je utvrdio jasnu povezanost nivoa dinamičke snage i sportskog postignuća. (Spence et al., 1980; 15% Fleck et al., 1985; 7-11% Smith et al., 1992; 10% Forthomme et al., 2005; Sheppard et al., 2008). Na osnovu trajanja poena od oko prosečnih 1,2 s (Sheppard et al., 2007), igrači u zoni napada izvedu do 3 skoka u smeč i 4 skoka u bloku. Dizači izvedu 11-21 skokova, srednji blokeri 2-15 skokova u smeču i 3-19 skokova u bloku, primači i korektori 1-15 skokova u smeču i 1-13 skokova u bloku (Sheppard et al., 2007). Eksplozivna svojstva zauzimaju najviše mesto u hijerarhiji motoričkih sposobnosti (Bosco & Pittera, 1982), a odmah iza nje se nalazi repetitivna snaga mišića nogu, trupa i ruku i ramenog pojasa (Kostić, 1995). Telesna visina je u pozitivnim korelacijama sa eksplozivnom snagom nogu, dok je težina, pa i potkožno masno tkivo, pokazala u mnogim istraživanjima negativne korelacije sa eksplozivnom snagom nogu (Blanksby et al., 1986; Baxter-Jones, 1995; Kostić i Klajdžić, 1995; Bomp, 2000, Ujsasi, 2014).

Problem istraživanja je usmeren na detekciju razlika između odbojkaša i odbojkašica u morfološkim karakteristikama i eksplozivnoj snazi nogu adolescentnog uzrasta, kao i utvrđivanje veličine uticaja morfoloških karakteristika na eksplozivnu snagu donjih ekstremiteta. Predmet istraživanja su morfološke karakteristike i eksplozivna snaga nogu. Osnovni cilj istraživanja je bolje planiranje i modelovanje odbojkaškog treninga kod oba subuzorka odbojkaša. Pošlo se od pretpostavke da će nivo eksplozivne snage biti viši kod odbojkaša i da će uticaji morfoloških karakteristika na manifestaciju eksplozivne snage nogu biti pozitivni i statistički značajni.

## METOD

Uzorak ispitanika izveden je iz populacije odbojkaša OK „RFU“ iz Futoga, N=27 (uzrasta  $14,67 \pm 0,67$ SD) i odbojkašica ŽOK „Futog“ iz Futoga, N=38 (uzrasta  $14,53 \pm 0,71$ SD). Ispitanici se bave odbojkom najmanje 3 godine. Pre početka realizacije istraživanja, s obzirom da se radi o maloletnim licima, roditeljima dece su bili podeljeni upitnici, uz poštovanje etičkih principa, a roditelji su svojim potpisom odobriti učešće njihove dece u istraživanju (Helsinška deklaracija).

Morfološke karakteristike su bile merene prema IBP standardima (predstavljale prediktorske varijable u radu), a izabrane su one za koje smo smatrali da mogu da imaju uticaj na ispoljavanje eksplozivne snage: Telesna visina (mm), Telesna masa (kg), Dužina noge (mm), Obim potkolenice (mm), Obim natkolenice (mm). Merene su prema standardima IBP-a.

Hipotetski motorički faktor eksplozivne snage bio je procenjen pomoću sledećih standardizovanih testova (Bašćevan, Antekolović, 2008): Skok za smeč iz zaleta (cm), Troskok iz mesta (cm) i Skok uvis iz mesta (kao rezultat uzeta je razlika između dohvatne visine i ostvarenog skoka iz mesta izraženo u cm).

Metodologija obrade podataka je podrazumevala izračunavanje deskriptivnih karakteristika varijabli (AS, S, CV). Primenom MANOVA i ANOVA utvrđene su razlike između dva subuzorka ispitanika formirana na osnovu polnog dimorfizma u analiziranim morfološkim i motoričkim varijablama. U cilju utvrđivanja uticaja sistema prediktorskih varijabli na kriterijumske varijable, bila je primenjena Linearna regresiona analiza (biće komentarisana prilagođen koeficijent determinacije zbog malog uzorka ispitanika).

## REZULTATI

Na osnovu analiziranih vrednosti rezultata deskriptivnih statistika (tabela 1) može se zaključiti da su odbojkaši i odbojkašice bili izrazito homogeni u svim ispitivanim morfološkim i motoričkim varijablama. Uočen je ujednačen rast i razvoj organizma po pitanju longitudinalnosti skeleta, volumena i mase tela, što je posledica dobre selekcije koja je izvršena pre svega na osnovu longitudinalnosti skeleta i činjenice da su ispitanici sličnog uzrasta.

Na osnovu Wilks lambda F odnosa (tabela 1) može se zaključiti da postoji statistički značajna razlika između ispitanika različitog pola pa u pogledu morfoloških karakteristika pri vrednosti  $F=33,03$ . Pojedinačnom analizom svake morfološke varijable, zaključuje se da statistički značajne razlike postoje u varijablama za procenu longitudinalnosti skeleta, *Telesna visina*, *Dužina noge* i varijabli za procenu volumena tela, *Obim potkolenice* u korist odbojkaša. Uzimajući u obzir vrednosti Wilks lambda F koeficijenta (tabela 1) za motoričke varijable može se zaključiti da postoji statistički značajna razlika ( $P=0,00$ ) dva subuzorka u pogledu eksplozivne snage nogu ( $F=69,48$ ). Ako analiziramo svaku motoričku varijablu pojedinalno, zaključuje se da te razlike postoje u sve tri ispitane varijable varijabli: *Skok za smeč iz zaleta*, *Troskok iz mesta* i *Skok uvis iz mesta* u korist odbojkaša.

**Tabela 1.** Deskriptivni statistici analiziranih varijabli i razlike

| Varijabla                   | Grupa | AS      | S     | CV    | f      | p    | F     | P    |
|-----------------------------|-------|---------|-------|-------|--------|------|-------|------|
| Telesna visina (mm)         | M     | 1747,11 | 51,12 | 2,93  | 43,41  | 0,00 | 33,03 | 0,00 |
|                             | F     | 1657,58 | 56,74 | 3,42  |        |      |       |      |
| Telesna masa (kg)           | M     | 57,64   | 4,53  | 7,86  | 0,69   | 0,41 |       |      |
|                             | F     | 56,34   | 7,23  | 12,83 |        |      |       |      |
| Dužina noge (mm)            | M     | 1064,07 | 41,81 | 3,93  | 130,82 | 0,00 |       |      |
|                             | F     | 906,15  | 47,58 | 5,25  |        |      |       |      |
| Obim potkolnice (mm)        | M     | 372,26  | 23,84 | 6,40  | 6,48   | 0,01 |       |      |
|                             | F     | 353,73  | 32,35 | 9,15  |        |      |       |      |
| Obim natkolenice (mm)       | M     | 530,04  | 40,69 | 7,68  | 2,14   | 0,15 |       |      |
|                             | F     | 515,47  | 39,44 | 7,65  |        |      |       |      |
| Skok za smeč iz zaleta (cm) | M     | 273,81  | 10,25 | 3,75  | 117,57 | 0,00 |       |      |
|                             | F     | 249,98  | 7,70  | 3,08  |        |      |       |      |
| Troskok iz mesta (cm)       | M     | 532,78  | 44,92 | 8,43  | 75,17  | 0,00 | 69,48 | 0,00 |
|                             | F     | 453,38  | 30,14 | 6,65  |        |      |       |      |
| Skok uvis iz mesta (cm)     | M     | 43,59   | 4,47  | 10,25 | 86,14  | 0,00 |       |      |
|                             | F     | 34,30   | 3,91  | 11,40 |        |      |       |      |

Legenda: AS – aritmetička sredina; S – standardna devijacija; CV – koeficijent varijacije; f – univarijantni f test; p – nivo statističke značajnosti f testa; F – multivarijantni Wilksov F test; P – statistička značajnost multivarijantnog F testa

Utvrđeno je da postoji statistički značajan uticaj ( $P=0,00$ ) sistema prediktorskih varijabli na kriterijum *Skok za smeč iz zaleta* (tabela 2) kod oba subuzorka, pri vrednosti koeficijenta multiple korelacije  $R=0,75$  kod odbojkaša i  $R=0,81$  kod odbojkašica, što objašnjava od 46% zajedničkog varijabiliteta kod odbojkaša do 65% kod odbojkašica, dok se ostali procenat može pripisati nekim drugim karakteristikama i sposobnostima koje nisu bile obuhvaćene primenjenim sistemom prediktora (npr. dužina koraka prilikom zaleta, dužini potkolenice, dužina natkolenice, trenutnom stanju mišića, prethodnoj istegnutosti istih (načinu istezanja - dinamička gipkost), potkožno masnom tkivu potkolenice i natkolenice, itd.). Posmatrajući varijable pojedinačno, uočava se da prediktorska varijabla *Telesna visina* pokazuje pozitivan statistički značajan uticaj na posmatrani kriterijum ( $p_{\text{beta}}=0,02$  i  $p_{\text{beta}}=0,01$ ) kod oba subuzorka. Viši ispitanici su ostvarivali veće vrednosti skoka za smeč na datom uzorku ispitanika. Ona se izdiferencirala kao varijabla koja najviše determiniše visinu skoka za smeč što se može videti ina osnovu njene parcijalizacije.

Iz podataka o linearnim korelacijama prediktorskih varijabli sa *Skok za smeč* kod odbojkaša uočava se da je ta povezanost najizraženija i statistički značajna ( $p=0,00$ ) kod varijable *Telesna visina* ( $r=0,66$ ). Nešto niži koeficijent korelacije koji je takođe bio statistički značajan ( $p=0,00$ ) je uočen u varijabli *Telesna masa* kod odbojkaša ( $r=0,56$ ). Viši i teži odbojkaši su ostvarivali bolje rezultate u varijabli *Skok za smeč* u odnosu na niže i lakše odbojkaše, a ovo je posledica uzročno-posledične (interkorelacije) veze između ove dve varijable. Vrednosti Pirsonovog koeficijenta korelacije kod odbojkašica ukazuju da najveću statističku značajnu i pozitivnu povezanost sa kriterijumom ( $p=0,00$ ) pokazuje varijabla *Telesna visina*, nešto nižu, prediktorska varijabla *Dužina noge*, potom *Telesna težina*). Od dve

mere za procenu voluminoznosti skelata, prediktorska varijabla *Obim potkolenice* imala je veću povezanost u odnosu na obim natkolenice sa istim prediktorom. Obe varijable su takođe statistički značajne u odnosu na isti ispitivani kriterijum.

**Tabela 2. Rezultati regresione analize Skok za smeč iz zaleta**

| Varijabla                 | Odbojkaši   |             |             |             | Odbojkašice |             |             |             |
|---------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                           | r           | p           | Beta        | pbeta       | r           | p           | Beta        | pbeta       |
| Telesna visina            | <b>0,66</b> | <b>0,00</b> | <b>0,55</b> | <b>0,02</b> | <b>0,77</b> | <b>0,00</b> | <b>0,56</b> | <b>0,01</b> |
| Telesna masa              | <b>0,56</b> | <b>0,00</b> | 0,14        | 0,53        | <b>0,56</b> | <b>0,00</b> | -0,18       | 0,67        |
| Dužina noge               | -0,16       | 0,22        | -0,20       | 0,20        | <b>0,72</b> | <b>0,00</b> | 0,25        | 0,23        |
| Obim potkolenice          | -0,23       | 0,13        | -0,24       | 0,13        | <b>0,47</b> | <b>0,00</b> | 0,29        | 0,17        |
| Obim natkolenice          | -0,21       | 0,15        | -0,19       | 0,23        | <b>0,34</b> | <b>0,01</b> | 0,04        | 0,85        |
| R                         |             |             | <b>0,75</b> |             |             |             | <b>0,81</b> |             |
| R <sup>2</sup>            |             |             | <b>0,46</b> |             |             |             | <b>0,65</b> |             |
| Prilagođen R <sup>2</sup> |             |             | <b>0,46</b> |             |             |             | <b>0,59</b> |             |
| P                         |             |             | <b>0,00</b> |             |             |             | <b>0,00</b> |             |

Legenda: r - Pirsonov koeficijent korelacije; p - nivo statističke značajnosti za r; Beta – regresioni koeficijent; pbeta - nivo značajnosti regresionog koeficijenta; R - koeficijent multiple korelacije; R<sup>2</sup> - koeficijent determinacije; Prilagođen R<sup>2</sup> – koeficijent determinacije koji je prilagođen za male uzorke; P - značajnost koeficijenta multiple korelacije

Rezultati regresione analize *Troskoka iz mesta* (tabela 3), ukazuju da ne postoji statistički značajan uticaj sistema morfoloških (prediktorskih) varijabli na kriterijum *Troskok iz mesta* kod odbojkaša (P=0,78) i odbojkašica (P=0,34), pri vrednosti koeficijenta multiple korelacije R=0,32 i R=0,38, što je objašnjavalo svega od 1-3% zajedničkog varijabiliteta, dok se ostali procenat može pripisati nekim drugim karakteristikam i sposobnostima koje nisu bile obuhvaćeni ovim prediktorskim sistemom (dužina svakog koraka, intramuskularna koordinacija, motivacija, konativne karakteristike, druge morfološke karakteristike, tehnika izvođenja troskoka, kordinacija pokreta, prethodno iskustvo i druge).

**Tabela 3. Rezultati regresione analize Troskoka iz mesta**

| Varijabla                 | Odbojkaši |      |       |       | Odbojkašice |      |       |       |
|---------------------------|-----------|------|-------|-------|-------------|------|-------|-------|
|                           | r         | p    | Beta  | pbeta | r           | p    | Beta  | pbeta |
| Telesna visina            | -0,06     | 0,38 | 0,18  | 0,55  | 0,08        | 0,32 | -0,23 | 0,50  |
| Telesna masa              | -0,18     | 0,19 | -0,32 | 0,30  | 0,18        | 0,14 | 0,23  | 0,48  |
| Dužina noge               | 0,15      | 0,23 | 0,20  | 0,35  | 0,13        | 0,21 | 0,11  | 0,74  |
| Obim potkolenice          | -0,08     | 0,34 | -0,12 | 0,58  | 0,24        | 0,07 | 0,60  | 0,07  |
| Obim natkolenice          | 0,13      | 0,27 | 0,13  | 0,55  | 0,05        | 0,37 | -0,57 | 0,09  |
| R                         |           |      | 0,32  |       |             |      | 0,38  |       |
| R <sup>2</sup>            |           |      | 0,10  |       |             |      | 0,15  |       |
| Prilagođen R <sup>2</sup> |           |      | 0,01  |       |             |      | 0,03  |       |
| P                         |           |      | 0,78  |       |             |      | 0,34  |       |

Legenda: r - Pirsonov koeficijent korelacije; p - nivo statističke značajnosti za r; Beta – regresioni koeficijent; pbeta - nivo značajnosti regresionog koeficijenta; R - koeficijent multiple korelacije; R<sup>2</sup> - koeficijent determinacije; Prilagođen R<sup>2</sup> – koeficijent determinacije koji je prilagođen za male uzorke; P - značajnost koeficijenta multiple korelacije

Takođe i rezultati regresione analize *Skoka uvis iz mesta* (tabela 4), ukazali su da ne postoji statistički značajan uticaj sistema morfoloških (prediktorskih) varijabli na kriterijum *Skok uvis iz mesta* kod odbojkaša ( $P=0,67$ ) i odbojkašica ( $P=0,32$ ), pri vrednosti koeficijenta multiple korelacije  $R=0,36$  i  $R=0,39$ , što je objašnjavalo od 2-3% zajedničkog varijabiliteta. Preostali procenat se može pripisati nekim drugim karakteristikama i sposobnostima koje nisu bile obuhvaćeni ovim prediktorskim sistemom (motivacija, intramuskularna i intermuskularna koordinacija, prethodno zagrevanje, dubina čučnja prilikom skoka, konativne karakteristike, druge morfološke karakteristike, pre svega dužina ruke, potkožno masno tkivo, itd.).

**Tabela 4. Rezultati regresione analize Skok uvis iz mesta**

| Varijabla                 | Odbojkaši |      |       |       | Odbojkašice |      |       |       |
|---------------------------|-----------|------|-------|-------|-------------|------|-------|-------|
|                           | r         | p    | Beta  | pbeta | r           | p    | Beta  | pbeta |
| Telesna visina            | -0,24     | 0,11 | -0,06 | 0,84  | 0,29        | 0,04 | 0,03  | 0,94  |
| Telesna masa              | -0,30     | 0,07 | -0,23 | 0,44  | 0,37        | 0,01 | 0,39  | 0,23  |
| Dužina noge               | -0,21     | 0,14 | -0,21 | 0,33  | 0,27        | 0,05 | 0,01  | 0,97  |
| Obim potkolenice          | 0,05      | 0,41 | 0,06  | 0,77  | 0,26        | 0,05 | 0,13  | 0,69  |
| Obim natkolenice          | 0,04      | 0,41 | -0,01 | 0,97  | 0,20        | 0,11 | -0,19 | 0,57  |
| R                         |           |      | 0,36  |       |             |      | 0,39  |       |
| R <sup>2</sup>            |           |      | 0,13  |       |             |      | 0,15  |       |
| Prilagođen R <sup>2</sup> |           |      | 0,02  |       |             |      | 0,03  |       |
| P                         |           |      | 0,67  |       |             |      | 0,32  |       |

Legenda: r - Pirsonov koeficijent korelacije; p - nivo statističke značajnosti za r; Beta – regresioni koeficijent; pbeta - nivo značajnosti regresionog koeficijenta; R - koeficijent multiple korelacije; R<sup>2</sup> - koeficijent determinacije; Prilagođen R<sup>2</sup> – koeficijent determinacije koji je prilagođen za male uzorke; P - značajnost koeficijenta multiple korelacije

## DISKUSIJA

U kadetskom uzrastu odbojkaša i odbojkašica iz Futoga utvrđena je diferencijacija morfoloških karakteristika (longitudinalnosti skeleta i voluminoznosti tela) i eksplozivne snage nogu. Drugačiji zahtevi muške odbojke, strožija selekcija, veća visina mreže više hvatanje lopte u smeču, zahtevniji viši blok nego što je to kod odbojkašica prouzrokovali su drugačiju selekciju sportista. Možda i činjenica viših zahteva u vidu telesne visine zbog visine mreže kod dečaka (238 cm prema 224 cm kod odbojkašica) je prouzrokovala i pojavu statistički značajnih razlika u korist odbojkaša u varijablama za procenu longitudinalnosti skeleta, *Telesnoj visini* i *Dužini noge*, jer je selekcija u muškoj odbojci strožija nego u ženskoj odbojci. Veći poprečni presek mišića m. tricepsa surea kod odbojkaša je posledica treninga, prirasta mišićne mase i androgenih hormona (tetsosterona i hormona rasta - somatotropina, kao i faktora rasta koji je sličan insulinu). ne treba zaboraviti ni dejstvo kortizola koji pomaže u oporavku na razvoj mišića. U ovom periodu razvoja, dečaci se odlikuju višim nivoom seruma navedenih hormona u krvi u odnosu na devojčice što se kasnije povezuje sa većom snagom (relativnom) kod odbojkaša. Na osnovu dobijenih vrednosti rezultata može se zaključiti da su se u ovom uzrastu na ovim grupama sportista izdiferencirale razlike u pogledu



eksplozivne snage donjih ekstremiteta, te da su odbojkaši snažnijih mišića nogu, sa većim obimom potkolenice (većim poprečnim presekom verovatno masom mišića) koji obezbeđuju veću silu prilikom ostvarivanja rada, kao i pod većim uticajem dejstva androgenih hormona. Veći obim potkolenice i veća angažovanost motoričkih jedinica (intramuskularna koordinacija - regrutovanje motornih jedinica, frekvencija pražnjenja, odnosno okidanja motornih jedinica i sinhronizacija, odnosno aktiviranje motornih jedinica sa većom sinhronizacijom) u mišićima kod odbojkaša, kao i veća telesna visina odbojkaša je doprinela boljim rezultatima testovima tipa skočnosti u odnosu na obojkašice sličnog uzrsta. Muška i ženska odbojka se razlikuju u visini mreže i snažnijim smečevim kod odbojkaša koja je prouzrokovana i „hvatanjem“ lopte za smeč u višoj tački u odnosu na odbojkašice. Ove razlike su evidentne već u kadetskom uzrastu.

Rezultati istraživanja, takođe, ukazuju da prediktorski sistem koji je bio sačinjen od morfoloških varijabli, statistički značno je utiče na kriterijum *Skok za smeč iz zaleta* kod mladih odbojkašica i odbojkaša iz Futoga. One su opisivale sa kriterijumom od 46 do 59% zajedničkog varijabiliteta, dok se preostali procenat može pripisati nekim drugim karakteristikama i sposobnostima kao što je motivacija, trenutno emotivno stanje, stanje mišića, uzrast, potkožno masno tkivo potkolenice i natkolenice. Telesna visina se izdiferencirala kao jedan od najbitniji pokazatelja dohvatne visine u smeču. Statistički značajno pozitivno je uticala na manifestaciju snage skoka za smeč kod obe grupe sportista. Odbojkaši i odbojkašice koje su bili višlji ostvarivali su bolje rezultate u testu u odnosu na niže ispitanice. Ovo je logičan zaključak ako se u obzir uzme i sama dohvatna visina viših odbojkašica koja je sigurno veća nego kod nižih ispitanica (i pogotovo ako se pretpostavi da niže odbojkaši i odbojkašice možda i poseduju kraće gornje ekstremitete u odnosu na više odbojkašice, koji su direktno mogli da utiču na dohvatnu visinu). Ovi rezultati se poklapaju sa jednim delom istraživanja stranih autora kao što su Baxter-Jones (1995), Blanksby et al. (1986). Kod ovih autora je telesnamasa negativno uticala na rezultat eksplozivne snage nogu, a ovde ona nije imala statistički značajan uticaj.

Sve antropometrijske prediktorske varijable kod odbojkašica su bile u pozitivnim statistički značajnim korelacijama sa istim kriterijumom dok kod odbojkaša su to samo bile *Telesna visina* i *Telesna masa*. Ovo ukazuje da je za doskočnu visinu skoka za smeč neophodna i određena težina (koja bi trebalo da bude na račun mišićne mase), određena dužina noge i svakako obima nakolenice i potkolenice. Svakako da odbojkašice sa većim obimom m. tricepsa surea i m. quadricepsa postižu bolje rezultate u postavljenom zadatku, jer je snaga mišića direktno proporcionalna njegovom poprečnom preseku (Fukunaga et al, 1992). Prilikom odraza, m. quadriceps i m. triceps surae odlaze u koncentričnu kontrakciju, opružaju maksimalno potkolenicu i na taj način se ostvaruje skok (uključuje se veći broj motornih jedinica čime se povećava snaga odraza, koja je sigurno veća kod utreniranih osoba – osoba podvrgnutih trenažnim procesima, jer se radi o neuromišićnim adaptacijama (Santos et al., 2003). Što je ta sila mišića veća, lakše će se savladati spoljašnji otpor (sila gravitacije) i odbojkašice i odbojkaši će ostvariti bolje rezultate. Ovo je i dokazano ovim istraživanjem. Ako pri tome odbojkašice poseduju i dugačke donje ekstremitete, koji su podržani određenom snagom mišića

nogu (i poseduju minimalne vrednosti potkožnog masnog tkiva na mišićima natkolenice i potkolenice) ostvarivaće bolje rezultate u skoku kao što je to bio slučaj u ovom odbojkaškom klubu. Rezultati parcijalne korelacije su potvrdili ove gore navedene tvrdnje, jer je vrednost koeficijenta parcijalne korelacije opao u odnosu na vrednost Pirsonovog koeficijenta korelacije. Dužina noge i obimi potkolenice i natkolenice, kao i telesna težina obezbeđivali su višim odbojkašicama postizanje većih rezultata u testu za procenu visine skoka za smeč u odnosu na niže odbojkašice u ovom odbojkaškom klubu. Postavljena hipoteza se može donekle prihvatiti, jer je utvrđena povezanost samo kroz jednu analiziranu varijablu.

Ovaj rad treba da pomogne u daljem selekcionisanju odbojkaša i odbojkašica u Futogu, ali se za te potrebe preporučuje i merenje drugih bitnih morfoloških karakteristika, poput sedeće visine, dužine gornjih ekstremiteta, potkožnih masnih tkiva, načina pripreme za skok (načina istezanja). Takođe je potrebno i proceniti druge motoričke sposobnosti poput repetitivne snage trupa, ruku i ramenog pojasa. S obzirom na mali broj ispitanika ovi rezultati se ne mogu generalizovati, pa su stoga potrebna istraživanja na većim uzorcima ispitanika, kao i longitudinalna praćenja istih grupa u budućem periodu kako bi se utvrdile i promene u ispoljavanju snage nakon određenog trenažnog procesa.

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**QUANTITATIVE AND QUALITATIVE DIFFERENCES OF MORPHOLOGICAL CHARACTERISTICS AND EXPLOSIVE STRENGTH OF LEGS IN UNDER-16 FEMALE AND MALE VOLLEYBALL PLAYERS**

*In order to achieve an advantage over the opponent, all the game elements, for which it is possible, are tended to be performed by jumping (Ziv & Lidor, 2010). This is supported by the fact that research of vertical jumping ability of volleyball players of different levels, found that players at a higher level of competition also achieve better results on tests of the assessment of vertical jumping ability (Forthomme, Croisier, Ciccarone, Crielaard, & Cloes, 2005). The research subject was aimed at detecting the difference between male and female volleyball players in morphological characteristics and explosive strength of legs, as well as determining the size of the impact of morphological characteristics of the explosive power of the lower limbs between*

groups formed on the basis of gender dimorphism. Method: The sample in this study was derived from a population of volleyball players of OK "RFU" from Futog, N=27 and female volleyball players of ŽOK "Futog" from Futog, N=38. For the purposes of this research morphological characteristics were measured. By using MANOVA and ANOVA differences were determined between the two sub-samples of respondents formed on the basis of gender dimorphism regarding the analyzed morphological and motor variables. In order to determine effects of the system of predictor variables on the criteria variables we used Linear Regression Analysis. Results: Statistically significant differences were observed for the variables Body height, Legs length and Lower-leg circumference, as well as for all three motor variables: Spike jump, Standing triple jump and Standing vertical jump in favor of volleyball players. Linear Regression Analysis revealed the impact of the predictor system of morphological characteristics on the criterion Spike jump in both subsamples. It was found that Body height gives the highest contribution to achieving better results regarding the height reached during spike jump in both groups. Common variability for both groups ranged from 46% in male volleyball players to 59% in female volleyball players. Discussion: Based on the values of obtained results it can be concluded that at this age and in this groups of athletes there were differences in terms of explosive power of the lower limbs, and that male volleyball players have stronger leg muscles and higher lower leg circumference (larger cross-section and probably also larger mass of muscles) that provide greater force during the play, and are also under the higher influence of androgen hormones.

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## TELESNO DRŽANJE ODBOJKAŠICA RAZLIČITIH UZRASNIH GRUPA

### UVOD

Sport pored svoji dobrih strana ima i svojih negativnosti, a većinom se te negativnosti upravo reflektuju na zdravlje sportistkinja. Odbojkašku igru karakteriše dinamičnost i pokretljivost, uz stalne i brze promene položaja tela i elemenata tehnike uz korišćenje jedne ruke (smeč ili servis) i oba gornja ekstremiteta - odbijanje lopte prstima i podlakticama, ili skok u blok koji je najčešće izvedene sa dve ruke. Ta posebnost odbojkaškog sporta na profesionalnom nivou omogućuje da estetska komponenta igre bude naglašena i uočljiva što se reflektuje i na specifične morfološke karakteristike odbojkašica (Grigoris et al., 2006). Odbojka je sport kompleksnih polistrukturalnih kretanja u kojima postoji čitav niz različitih kretnji (dokorak, ukršten korak), skokova (u smeču, bloku, servisu), bacanja i padova (upijač, rolanje, povaljke), sprintova, statičkih izdržaja u stavovima, udaraca (Janković i Marelić, 2003). Zbog određenih karakteristika odbojke kao sporta (delovi tehnike koji su asimetrični zbog igranja jednom rukom) postoji osnovana sumnja, da su odbojkašice izložene faktorima koje mogu uzrokovati razvoj posturalnih deformiteta (najviše skolioza). Asimetričan oblik opterećenja u kombinaciji s velikim brojem skokova i doskoka na tvrdim podlogama vrlo nepovoljno utiče na lokomotorni aparat, pre svega mišiće, ligamente i kosti (Aagarda & Jorgensen, 1996; Kosinac & Katić, 1999).

Kod odbojkašica, nepravilna telesna držanja mogu izazvati brojne hronične zdravstvene probleme i povrede koje utiču na njihovu uspešnost u odbojkaškom sportu. Iako je, generalno gledajući, jasno da telesna aktivnost višestruko pozitivno deluje na pojedine organske sisteme i ljudske organizme u celini, lekari opšte prakse i lekari sportske medicine ukazuju na povećanu učestalost nepravilnih držanja i deformiteta, i kod dece i omladine, aktivno uključenih u sportske aktivnosti (Kosinac 2002). Janda (1994) navodi da se nepravilna telesna držanja (skoliotično, kifotično ili lordotično držanje) smatraju anomalijom držanja tela koja su fleksibilna (labava) i koja se promenom položaja tela ili voljnom kontrakcijom muskulature mogu korigovati. Problematikom narušenih držanja tela kod odbojkaša i odbojkašica bavili su se Leko i sar., (2009); Dudek, et al, 2011; Krneta i sar., (2012) i utvrdili postojanje određenih odstupanja od normalnih držanja. Vodeći se dosadašnjim istraživanjima domaćih i stranih autora, želi se ispitati postojanje razlika u pojedinim segmentima posturalnog statusa odbojkašica različitog uzrasta iz Sombora i utvrditi da li ovaj sport utiče na pojavu narušenog telesnog držanja. U istraživanje se pošlo sa pretpostavkom da neće biti uočene

statistički značajne razlike između odbojkašica različitog uzrasta u segmentima posturalnog statusa.

## METOD

Uzorak ispitanika činilo je 60 odbojkašica ŽOK „CEV“ iz Sombora. Prva grupa (N=20) ispitanica su bile odbojkašice uzrasta 8-10 godina ( $\pm 6$  meseci), koje su u stvari početnice u ovom sportu i koje ne treniraju odbojku više od godinu dana. Drugu grupu (N=20) su činile odbojkašice uzrasta 11-13 godina ( $\pm 6$  meseci) koje treniraju najmanje 2 godine. A u trećoj grupi (N=20) su bile takođe odbojkašice uzrasta 14-16 godina ( $\pm 6$  meseci) koje se bave odbojkom najmanje 3 godine.

Kao uzorak mernih instrumenata za procenu posturalnog statusa odbojkašica bila je korištena modifikovana metoda Napoleona Wolanskog po Radisavljeviću (2001) za procenu posturalnog statusa tela, gde su u obzir bili uzeti sledeći segmenti:

1. Držanje glave u sagitalnoj ravni,
2. Držanje ramena u sagitalnoj ravni,
3. Držanje ramena u frontalnoj ravni,
4. Držanje lopatica,
5. Odstupanje kičmenog stuba u frontalnoj ravni,
6. Držanje trbuha.

Po ovoj metodi postoje tri ocene (kriterijuma) posturalnog statusa - 0, 1 i 2:

- 1) Ocena 0 je kada se svi parametri nalaze u normalnim odnosima – normalan status.
- 2) Ocena 1, predstavlja odstupanje od normalnog držanja tela, funkcionalni stadijum deformiteta.
- 3) Ocenu 2, karakterišu značajna odstupanja od normalnog statusa – strukturalne promene.

Ocenjivanje segmenata posturalnog statusa će se vršiti na sledeći način:

Procenu posturalnog statusa odbojkašica je sproveo lekar fizijatar na treninzima odbojkašica u Somboru u 2014. godine. Stanje pojedinih segmenata tela je izraženo procentualno i brojačno, a rezultati su predstavljeni i grafički. Za dalju statističku obradu podataka bila je korištena neparametrijska metoda Hi-kvadrat test, radi donešenja konkretnih zaključaka. Radi utvrđivanja veličine efekata manifestovanih razlika korišćen je Kramerov pokazatelj veličine efekata razlika<sup>1</sup> (V). Klasifikacija efekata razlika koeficijenta prema Cohen (1988) izgleda ovako: 0.10 -0.30 mali efekat razlika; od 0.30 - 0.50 srednji efekat razlika i >0.50 veliki efekat razlika.

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<sup>1</sup> Meri interkorelacije varijabli, ali je pristrasan budući da se povećava sa brojem ćelija u tabeli kontigencije.

## REZULTATI

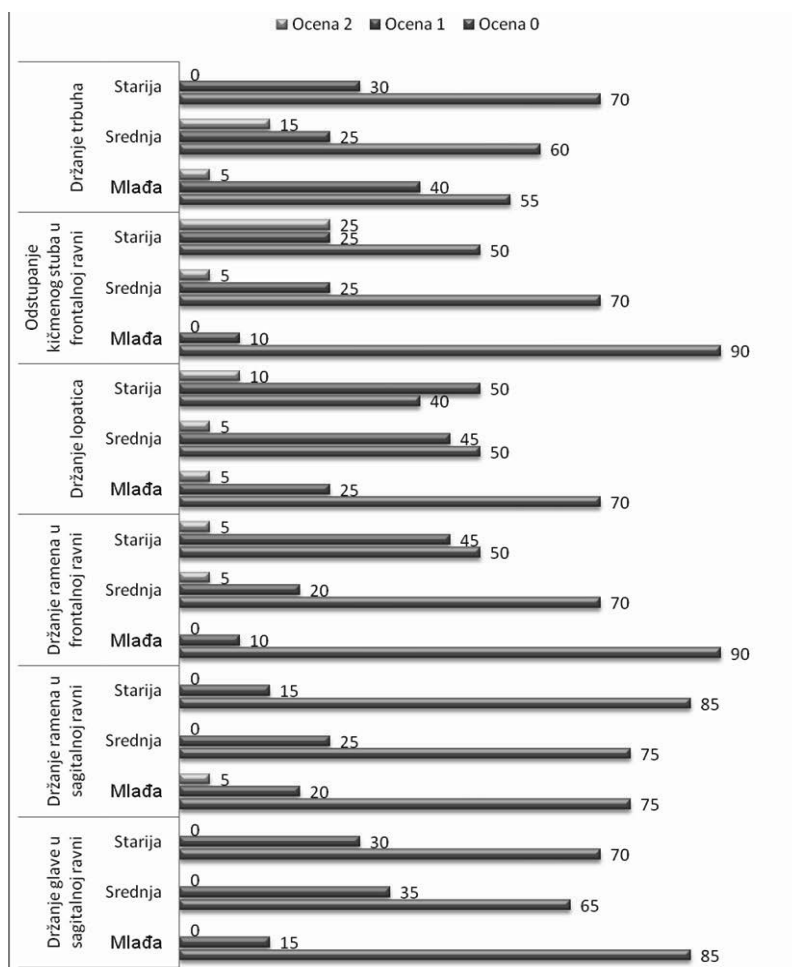
Frekvencije rezultata segmenta *Držanja glave u sagitalnoj ravni* (grafikon 1) koja može da postojanje kifotičnog lošeg držanja, ukazuje da 85% ispitnika uzrasta 8 do 10 godina, 65% uzrasta 11-13 godina, i 70% odbojkašica uzrasta 14-16 godina, poseduju dobro držanje glave. Ovaj segment se nalazi o normalnom položaju. Za razliku od ovih odbojkašica postoje odbojkašice kod kojih je ovaj segment posturalnog statusa pomerena ka napred (dobile ocenu 1); 15% najmlađe grupe odbojkašica, 35% srednje grupe odbojkašica i 30% najstarije grupe. Na osnovu ovih rezultata može se konstatovati da postoji određen procenat dece kod kojih ovaj stadijum deformiteta može lako da preraste u strukuralni (ocena 2).

U segmentu posturalnog statusa *Držanje ramena u sagitalnoj ravni* (grafikon 1), veliki broj odbojkašica sve tri ispitivane grupe poseduje dobro držanje ramena u sagitalnoj ravni (75%; 75% i 85%), četvrtina ispitanica (25%) srednje uzrasne grupe, petina ispitanica najmlađe uzrasne grupe (20%) i 15% ispitanica najstarije grupe su dobile ocenu 1, što ukazuje na pomeranost ovog segmenta unapred i mogućnost kifotičnog lošeg držanja tela. Samo 5% ispitanica najmlađe grupe je dobila ocenu 2, što može da ukaže na postojanje kifotičnog lošeg držanja tela.

Rezultati frekvencije ocena segmenta posturalnog statusa *Držanja ramena u frontalnoj ravni*, koji je trebao da ukaže na dislokaciju ramena (grafikon 1), tj projekcije gornje ivice acromiona odbojkašica (gore ili dole) i mogućnost postojanja skoliotičnog lošeg držanja tela, pokazao je da 90% odbojkašica najmlađe grupe ima odlično držanje ramena, 70% srednje grupe, dok se taj procenat kod najstarije grupe odbojkašica smanjio na 50%. Kod ove grupe odbojkašica 45% ispitanica ima podignuto jedno ili drugo rame. Ova pojava se povezuje sa uticajem smečiranja i serviranja u odbojci, gde se smeč ili servis (koji može biti smeč servis) izvodi sa jednom dominantnom rukom (levom ili desnom) (Leko i sar., 2009). Ova pojava se i najviše manifestovala kod najstarije grupe koja ujedno i najduže trenira, pa se može pretpostaviti da duže bavljenje odbojkom može izazvati asimetrična držanja pojedinih segmenta posturalnog statusa (npr. držanje ramena ili lopatica, jer pomereno rame može da sa sobom povuče i lopaticu gore ili dole).

Rezultati frekvencije *Držanja loptice* ukazali su na rapidno opadanje dobrog držanja lopatica kod odbojkašica starijih uzrasnih grupa (70% prema 50%, da bi se taj procenat smanjio na 40% kod najstarije grupe) (grafikon 1). Samim tim se i povećavao procenat lošeg držanja u vidu funkcionalnih promena (ocena 1) i strukturalnih promena (ocena 2). Po jedna odbojkašica najmlađeg i srednjeg uzrasta je dobila ocenu 1 (5%), i dve odbojkašice najstarije grupe (10%). Ovi podaci ukazuju da, što su odbojkašice starije grupe više, dolazi do izražaja asimetrije držanja lopatica, a verovatno i ramena u frontalnoj ravni (što se moglo videti u Tabeli 3), što može da ukaže na pojavu skoliotičnih promena na koštanoj i mišićnoj aparatu odbojkašica. Ne treba zaboraviti da je odbojka sport koji u većini akcija podrazumeva odigravanje lopte sa dve ruke (odigravanje lopte prstima, „čekičem“-odbijanje lopte suručno, skok u blok), ali i smečiranje lopte jednom rukom. Ova pojava je mogla i doprineti asimetriji lopatica odbojkašica koje su starije i koje se duže vremenom bave odbojkom.

U varijabli *Odstupanje kičmenog stuba u frontalnoj ravni*, potvrđuje se da i u ovom segmentu koji upućuje na postojanje skoliozičnog držanja kičmenog stuba (grafikon 1), najlošiji rezultati se uočavaju kod odbojkašica najstarijeg uzrasta, 25% je dobilo ocene 1 i 2, što je alarmantno za ovako mali uzorak ispitanika. Samo 50% je posedovalo dobro držanje kičmenog stuba u frontalnoj ravni. Ovaj procena je mnogo veći kod srednje uzrasne grupe, 70% i još veći kod najmlađe, 90%. Ocenu 1 kod najmlađe grupe, odnosno funkcionalni stadijum deformiteta (skolize) je dobilo 10 %, kod srednje 25% i kod najstarije takođe 25%. Verovatno je ovo posledica jednostranog korišćenja dominantne ruke prilikom smeča, servisa koja je sa sobom povukla i jedno rame i lopaticu (dominantne ruke prema gore, pa se projekcija acromiona pomerila u odnosu na suptorno rame), što je prouzokovalo nastajanje ovog deformiteta.



**Grafikon 1.** Posturalni status odbojkašica različitog uzrasta



Frekvencija rezultata varijable *Držanja trbuha* ukazala je na postojanje dobrog držanja trbuha odbojkašica svih uzrasta (jer 55% mlađeg uzrasta i 60 % srednjeg uzrasta poseduje dobro držanje, i 70% najstarijeg uzrasta (grafikon 1). Ocenu 1, što upućuje na slabiju muskulaturu je dobilo 40% najmlađeg uzrasta, 25% srednjeg uzrasta i 30% najstarijeg uzrasta. Odbojkašice najstarije grupe su bile nešto lošije, odnosno trbušna muskulatura je slabija.

Rezultati tabele 1 ukazuju na postojanje statistički značajnih razlika u varijabli *Odstupanja kičmenog stuba u frontalnoj ravni* ( $p=0,03$ ) između odbojkašica različitih uzrasnih grupa. Što su odbojkašice bile starije, njihovo odstupanje kičmenog stuba u frontalnoj ravni bilo je izraženije (grafikon 1), tako da se sa 8,3% ocenom 2 (strukturnalnog stepena deformiteta) ističu odbojkašice treće uzrasne grupe. U njihovoj grupi 8,3% odbojkašica poseduju funkcionalni stadijum deformiteta kičmenog stuba u frontalnoj ravni (nekog oblika skolioze, cervikalna, torakalna, slabinska ili multiple skolize). Ove odbojkašice su takođe ugrožene, jer njihov stadijum može veoma brzo da preraste u strukturnalni koji zahteva lečenje u specijalizovanim ustanovama. Najbolje rezultate u navedenoj varijabli su posedovale odbojkašice prve grupe, a nešto slabije odbojkašice druge grupe. Ova pojava se objašnjava vremenom treniranja odbojke, jer očito da što su se odbojkašice više vremenski bavile ovim dvoranskim sportom, njihov kičmeni stub se više krivio. Vrednosti  $\chi^2$  testa za preostale analizirane varijable posturalnog statusa odbojkašica upućuju na slično stanje držanja analiziranih segmenata ( $p>0,05$ ). Efekat razlika prikazan i izračuna preko Krametrovog V pokazatelja upućuje na mali ili najviše srednji efekat veličine razlika uočeni u varijabli *Odstupanje kičmenog stuba u frontalnoj ravni* izračunat preko Cohena (1988).

**Tabela 1.** Analiza držanja segmenata posturalnog statusa u odnosu na uzrasta odbojkašica

| Varijabla                                    | $\chi^2$ | p    | df | V    |
|----------------------------------------------|----------|------|----|------|
| Držanje glave u sagitalnoj ravni             | 2,22     | 0,33 | 2  | 0,19 |
| Držanja ramena u sagitalnoj ravni            | 2,67     | 0,61 | 4  | 0,15 |
| Držanja ramena u frontalnoj ravni            | 7,91     | 0,10 | 4  | 0,26 |
| Držanja lopatica                             | 4,00     | 0,41 | 4  | 0,18 |
| Odstupanje kičmenog stuba u frontalnoj ravni | 10,79    | 0,03 | 4  | 0,30 |
| Držanje trbuha                               | 4,62     | 0,33 | 4  | 0,20 |

Legenda:  $\chi^2$  – vrednost hi-kvadrat testa; p – nivo statističke značajnosti hi kvadrat testa; df - stepeni slobode; V Krametrov pokazatelj veličine efekata razlika

## DISKUSIJA

Rezultati istraživanja na uzorku od 60 odbojkašica ukazali su na postojanje pojedinih tipova nepravilnih telesnih držanja kod svih uzrasnih grupa. Može se pretpostaviti da je problem narušenih telesnih držanja kod odbojkašica povezan sa redovnim bavljenjem odbojkom. Odbojka može imati i korektivan uticaj na nepravilna telesna držanja, međutim najintenzivniji elementi u odbojci (smeč, servis, smeč servis) mogu uticati na razvoj tipova nepravilnog telesnog držanja pogotovo kod dece koja se nalaze u adolescentnom razdoblju (11-13 godina i 14-16 godina).

Odbojkašice u prvoj posmatranoj grupi su izložene treningu uvežbavanja osnovnih tehničkih elemenata (odigravanje lopte prstima i „čekićem“), te se može pretpostaviti da se u njihovoj dobi nije primarno uticalo na faktore pravilnog telesnog držanja određenim

kineziološkim operaterima (vežbama). Devojčice u tom uzrastu susreću se sa osnovnim elementima odbojke za koje je dokazano da pomažu u razvitku pravilnog telesnog držanja (Grigoris et al, 2006), ali pri tom pojedine elemente devojčice ne izvode pravilno te dolazi do učestalog ponavljanja pogrešaka što može dovesti do nepravilnog telesnog držanja (verovatno samo pojedinih odbojkašica kao što je to bio slučaj u ovom radu). Osim toga, pri izvođenju određenih tehničkih elemenata (podlaktično odbijanje - „čekić“), gde je naglašeno kifotično držanje tela, može dovesti do negativnog uticaja na pravilnu posturu takve dece. U ovom istraživanju je bilo dece različitih uzrasta koje su posedovale promene na rameno-lopaticnoj regiji, ali je taj procenat bio izuzetno mali i zanemarljiv.

U drugoj grupi (od 11 do 13 godina) primetno je da devojčice imaju priličan procenat nepravilnog telesnog držanja u svakom pojedinom tipu (od 25% do 50%). U toj uzrasnoj dobi događaju se velike promene na sveukupan organizam devojčica. Fazu ubrzanog rasta i razvoja pored pojačane hormonalne aktivnosti obeležava i brzi rast – izduživanje kostiju te na nepripremljen organizam, devojčica može imati negativan uticaj, a posledica može biti nastajanje različitih tipova nepravilnog telesnog držanja (kifoza i skolioza). U trećoj grupi (od 14-16 godina) kod koji je i uočene da postoji statistički značajna razlika u odnosu na srednju i namladu grupu u varijabli Odstupanje kičmenog stuba u frontalnoj ravni može se konstatovati da te razlike postoje i nastale su usled dugotrajnog bavljenja odbojkom i jednostranog korišćenja jedne ruke prilikom smeča i servisa. Obično se ova pojava dovodi u vezu sa hipotoničnom trbušnom muskulaturom i prekomernom uhranjenošću (gojaznošću), ali to ovde nije bio slučaj. Može se pretpostaviti da su devojčice imale ozbiljnije treninge sa treninzima kondicione pripreme te su dodatno jačale trbušne mišiće te na taj način korigovale telesno držanje, ali su se ipak pojave manifestovale na kičmeni stub u vidu skolioze. Posledica velikog procenta skolioze najstarije grupe (8,3% koji su dobili ocenu 1 i 8,8% koji su dobili ocenu 2, što ukupno čini 30% dece najstarije grupe), može nastati zbog najintenzivnijih pokreta u odbojci koji su asimetrični (servis, smeč smeč ili servis sa tla). Kod izvođenja pojedinih tehničkih elemenata koji su asimetrični prevladava dominacija jedne strane tela, a kao posledica toga može doći do narušavanja statodinamičkih odnosa, te nastajanja nepravilnih telesnih držanja (jedna stran mišića se izdužuje i postaje slabija, a druga strana se skraćuje i postaje jača i pod većim je tonusom). Iz dobijenih rezultata grafikona 1 može se pretpostaviti da odbojka osim pozitivnih učinaka na nepravilno telesno držanje može imati i negativan uticaj.

U varijabli držanje trbuha, uočen je veliki procenat ispitanica sa lošim držanjem, slabom muskulaturom trbuha (ocenom 1) od 40% kod mlađe grupe do 30% kod starije grupe. Ovakvi rezultati ukazuju da bi se morala posvetiti dosta veća pažnja razvoju trbušne muskulature odbojkašica svih uzrasta. Od snage trbušne muskulature direktno zavisi snaga izvođenja smeča (Stojanović i sar., 2006) pa se u ovim uzrasnim grupama preporučuje posvećivanje veće pažnje jačanju muskulature trbušnog zida različitim vežbama. Samo 15% odbojkašice srednjeg uzrasta i 5% odbojkašica mlađeg uzrasta posedovale su jako slabu muskulaturu trbuha, koja se može povezati sa gojaznošću te dece i pomeranjem gornjih ivica karlice prema dole-inklinaciju karlice. Ovakvoj deci treba preporučiti vežbe jačanja mišića trbuha, tj. svim odbojkašicama mlađeg uzrasta, da se ova pojava ne bi manifestovala u starijem uzrasnom dobu.

Potrebno je da treneri već od najranijeg uzrasta posvete pažnju opštoj pripremi lokomotornog aparata, prvenstveno dajući prednost jačanju posturalnih (antigravitacionih) mišića (leđa, trbuha i nogu), jačanju celog trupa, te posebnu pažnju posvete vežbama istezanja (pogotovo m. iliopsoasa). Trening treba prilagoditi uzrastu odbojkašica, sa naglaskom na opštu fizičku pripremu, a ne samo specifičnostima vezanima za taj sport. Pravilan i sistematičan razvoj mladih odbojkašica može u velikoj meri sprečiti negativna delovanja na posturu i u konačnici smanjiti rizik od nastajanja telesnih deformiteta.

Rezultati ovog istraživanja potvrđuju istraživanja Leka i sar. (2009) koji su takođe utvrdili da postoje narušena telesna držanja kod odbojkašica strajnih uzrasnih grupa. Pogotovo se to odnosi na odbojkašice adolescentnog perioda. Preporuka za dalja istraživanja je praćenja odbojkašica kroz duži vremenski period, uzeti u obzir faktore rasta i razvoja i obim i intenzitet treninga na nedeljnom ili mesečnom nivou.

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#### BODY POSTURE IN FEMALE VOLLEYBALL PLAYERS OF DIFFERENT AGE GROUPS

*Improper postures due to the weakened structure of postural muscles, detected during the period of growth and development, especially those discovered at an early age, can be corrected with additional programs of physical exercise. Early, i.e. timely diagnosis is the most important element of successful treatment (Paušić, 2007) of physical deformities. Method: In a sample of 60 female volleyball players of ŽOK "CEV" from Sombor, which were divided into three subsamples based on age (8-10 years, 11-13 years and 14-16 years) the evaluation of postural status was performed using a modified method of Napoleon Wolanski according to Radosavljević (2001). Condition of individual body segments was expressed in percentage and numerically. For further statistical analysis of the data a nonparametric chi-square test was used, namely Cross-tabulation. Results: The results of the chi-square test indicated a statistically significant differences between the groups tested regarding the variable Deviation of the spine in the frontal plane ( $p=0.03$ ) in favor of young female volleyball players. Regarding other studied variables of postural status no statistically significant differences were observed. Discussion: Female volleyball players from the first tested group are exposed to the training of basic technical elements (playing the ball with fingers and digging), and can be assumed that at their age, factors of proper physical posture were not primarily influenced by certain kinesiological operators (exercises). Girls at this age are faced with the basic elements of volleyball that are proven to help in the development of proper body posture (Grigoris, Malousaris, Nikolaos, Bergeles, Barzouka & George, 2006), but during training girls do not perform properly some elements which results in frequent repetition of errors which can lead to irregular body posture (perhaps only in a certain number of female volleyball players as was the case in this paper). In addition, during the performance of certain technical elements (forearm play, digging) where kyphotic body posture is emphasized, it can lead to a negative impact on the correct posture in such children. This study included children of different age with changes in shoulder-spatula region, but the percentage of those children was very small and negligible.*

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## RELACIJE MORFOLOŠKIH KARAKTERISTIKA I PRECIZNOSTI UDARCA PO LOPTI NOGOM MLADIH FUDBALERA

### UVOD

Udarci po lopti nogom ili glavom osnovni su elementi fudbalske igre, kojima se može podići efikasnost igre i u pogledu brzine akcije i u pogledu postizanja golova. Ujedno to je i razlog zašto udarac treba ostvariti pravovremeno, korisno ga uputiti, ali i najčešće brzo izvesti. Najčešće korišćeni udarac u fudbalu je udarac sredinom hrpta stopala koji je i najprecizniji i najkorišćeniji među fudbalerima (Cabri, De Prof, Dufour, & Clarys, 1998). Uspešnost u izvođenju preciznih radnji u velikoj meri zavisi od preciznog lociranja cilja u prostoru, pa je prema tome uloga naših receptora od presudne važnosti za uspešnu preciznost (Sekereš, Santrač, Radosav, Toplak, Miljanić, 1994). Kod igrača, preciznost se ispoljava prilikom predaje lopte, dodavanja, centaršuteva i udaraca prema голу. A krajnji cilj preciznosti u fudbalu je postizanje pogotka. Preciznost zavisi od dobrog prijema i analize vidnih signala, od određenog nivoa snage, od brzine protoka informacije ka mozgu i nazad u mišiće, emotivnog stanja, koncentracije pažnje, nivoa treniranosti, uzrasta, udaljenosti ispitanika od mete, pola. Problematikom relacija morfoloških karakteristika i motoričke sposobnosti – preciznosti kod fudbalera mladih kategorija bavili su se domaći (Mekić, 1985, Joksimović, 2008; Smajić i Molnar, 2008a; Smajić i Molnar, 2008b; Smajić i Molnar, 2008c; Smajić, Molnar, Radoman, Tomić & Kapidžić, 2010) i strani autori (Panajotis, Panajotis, Dulgeridis & Debelnogić, 2005), te došli do zajedničkog zaključka, da postoji statistički značajna povezanost morfoloških dimenzija, te da one mogu da utiču na preciznost udarca po lopti nogom. Morfološke karakteristike imaju značajan uticaj na rezultate preciznosti pogađanja horizontalnog i vertikalnog cilja kod fudbalera mlađeg uzrasta. Problem rada predstavljalo je utvrđivanje prediktivnih uticaja morfoloških karakteristika na elevacijsku preciznost udarca po lopti nogom kod mladih fudbalera.

Povezanost između dinamičke mišićne snage ekstenzora kolena i šutiranja lopte možda je zavisna i od nivoa stečenih veština u trenutku testiranja. Istraživanja koja su sproveli Trolle i sar. (1993), a koji su merili izokinetičku snagu mišića ekstenzora kolena kod vrhunskih fudbalera pri ugaonom ubrzanju od 0 do 4.18 rad/s nisu pokazala zavisnost između ovih parametara i ubrzanja lopte zabeležene tokom standardizovanog šutiranja. Ubrzanje lopte je bilo nepromenjeno nakon 12 nedelja treninga snage. Moguće je da motorna kontrola prevladava nad mišićnom snagom kod vrhunskih fudbalera mada i izbor kriterijuma za merenje parametara šuta može biti značajan faktor. Rame i

mišići trupa uključeni su u aktivnosti ubacivanja lopte iz auta, a dugačka ubacivanja u protivnički kazneni prostor mogu biti izvor izglednih šansi za pogodak.

Distanca izbacivanja kod fudbalera, pokazalo se, zavisi od snage povlačenja i fleksione snage trupa. Metod treniranja, korišćenjem medicinskih lopti, povećava parametre snage ali bez odgovarajućeg porasta distance izbacivanja. Ovo pokazuje određeni stepen specifičnosti u veštini izbacivanja i sugerise da pojedinci trebaju biti predselektovani za ove aktivnosti. Pošto fizički zahtevi variraju sa ulogama i mestima u timu, vrednosti mišićne snage mogu zavisiti od pozicije koju igrač zauzima. Golmani i odbrambeni igrači imaju veću maksimalnu ekstenzije kolena na 0.52 rad/s nego napadači ili vezni igrači. Rezultati su pod znakom pitanja zbog toga što nisu korigovani za telesnu površinu kako bi se uklonio pozicioni efekat (Lees & Nolan, 1998). Slična zapažanja izneta su od Togari i sar. (1988) u testu na japanskim fudbalerima. Autori su pokazali da su golmani značajno „jači“ od napadača u pokretima spore brzine (1.05 rad/s), a vezni igrači se nalaze između ove dve grupe. Razlike postaju neznajne pri ugaonom ubrzanju od 3.14 rad/s. Danas je čest slučaj praćenja mišićne snage putem izokinetičke aparature (Cybex, Kin-Com, Bio-Dex, Lido). Ovi aparati nude mogućnost za determinisanje moment-ubrzanje krivulja u izokinetičkim pokretima i formiranje kriva zglobova u serijama izometrijskih kontrakcija. Kompleksniji sistemi dopuštaju merenje mišićne aktivnosti u ekscentričnim kao i koncentričnim i izometrijskim modelima. U ekscentričnoj aktivnosti, muskulatura udova se suprostavlja otporu mašine; mišići se produžuju i ostvaruju ekscentričnu kontrakciju. Izokinetički testovi su gotovo isključivo izvođeni na donjim ekstremitetima i pri koncentričnim kontrakcijama. Dok je snaga ekstenzije kolena u koncentričnoj kontrakciji u korelaciji sa izvođenjem šuta, još veća korelacija je pronađena za snagu fleksije kolena u ekscentričnoj kontrakciji (Cabri et al, 1998). Posedovanje snažnih mišića zadnje lože buta, naročito u ekscentričnom obrascu, je veoma važan zahtev za vrhunski fudbal. Balans između snage mišića zadnje lože buta i kvadricepsa može uticati na frekvencu povreda kod fudbalera. Na niskim brzinama i pod izometrijskim režimom preporučuje se odnos fleksije i ekstenzije kolena od 60-65%. Ovaj odnos je povećan pri porastu ugaonog ubrzanja, mada se preciznost merenja gubi sa povećanjem brzine. Izokinetička testiranja snage dopuštaju poređenje leve i desne noge kako bi se identifikovao mišićni disbalans i slabija strana kao podložnija povredi. Test profili su takođe značajni tokom praćenja oporavka korišćenjem nepovredene strane kao referentne. Ova poređenja za identifikovanje asimetrije ili slabosti jednog ekstremiteta jednog sportiste su verovatno značajnija od komparacije među timovima i članovima tima. Ova napomena je ključna pri poređenjima različitih podataka iz drugih laboratorija koje su koristile alternativne test protokole.

Predmet istraživanja su bile morfološke karakteristike i preciznost fudbalera.

Cilj rada treba je bilo utvrđivanje relacija morfoloških karakteristika i preciznosti fudbalera, uzrasta 13-14 godina iz Novog Sada. U istraživanje se pošlo sa pretpostavkom da su relacije telesne visine, obima natkolenice, obima potkolenice, dužine noge, kožnog nabora trbuha, kožnog nabora natkolenice, kožnog nabora leđa, kožnog nabora sprailjačnog i preciznosti fudbalera pozitivne i statistički značajne.

## METOD

Testiranju je bilo podvrgnuto 47 fudbalera uzrasta 13-14 godina, FK „Kabel“ iz Novog Sada. Na njima je sprovedeno je merenje morfoloških karakteristika: telesne visine (mm), telesne mase (kg), obima natkolenice (cm), obima potkolenice (cm), dužine noge (mm), kožnih nabora trbuha (mm), kožnog nabora leđa (mm), kožnog nabora natkolenice (mm) i suprailijačnog kožnog nabora (mm).

Pošto se radi o maloletnim sportistima dobijena je sglasnost roditelja za njihovo testiranje, pridržavajući se Helšinske deklaracije o pravima deteta i učešća u testiranjima.

Testirana je specifična (elevacijska) preciznost udarca po lopti nogom u vertikalni (poen) i horizontalni cilj (poen).

U statističkoj obradi podataka prvo su utvrđeni osnovni deskriptivni statistici varijabli (AS-aritmetička sredina, S-standardna devijacija, MIN-minimalne vrednosti rezultata merenja, MAX-maksimalne vrednosti rezultata merenja). Za određivanje veličine uticaja antropometrijske karakteristike na motoričku sposobnost - preciznost, bila je korišćena linearna regresiona analiza (komentarisan je prilagođen koeficijent determinacije zbog malog uzorka ispitanika).

## REZULTATI

Rezultati deskriptivnih statistika morfoloških varijabli (tabela 1) zaključuje se da su ispitanici bili slične longitudinalnosti skeleta (*Telesne visine* i *Dužine noge*), voluminoznosti tela (*Telesne mase*, *Obima natkolenice* i *Obima potkolenice*). Sličan nivo morfoloških karakteristika ukazuje na homogenost grupe, tj. na sličan rast i razvoj navedenih karakteristika što je posledica ujednačenog rasta i razvoja organizma, a veće oscilacije se mogu očekivati u narednom periodu kada će nastupiti intenzivan telesni rast. Veći varijabilitet rezultata uočen je u svim analiziranim varijablama za procenu potkožnog masnog tkiva (*Kožni nabor trbuha*, *Kožni nabor leđa*, *Kožni nabor natkolenice* i *Kožni nabor suprailijačni*). Ovakav varijabilitet rezultat je posledica uticaja socio-ekonomskih činilaca, vrste ishrane fudbalera i nivoa fizičke aktivnosti. Najveći uticaj ima porodica i faktori sredine (društvo i sredina u kojem se mladi fudbaler kreće). U motoričkim varijablama uočava se veći varijabilitet rezultata, koji je posledica neujednačenog nivoa ispoljavanja preciznosti koja se može dovesti u vezu sa razvojem snage nogu, propriorepcije i stanjem mišića. Može se pretpostaviti da se radi i o različitom ispoljavanju nivoa snage mišića nogu koji se manifestovao i na brzinu šuta i njenu preciznost. Treba navesti da na preciznost može uticati i sportski staž mladih fudbalera koji se može dovesti u korelaciju sa tehnikom izvođenja šuta. Tehnika šuta se trenira i usavršava godinama. Na njoj se uvek može raditi kako bi šut bio što savršeniji, pas precizniji, čime se povećava „tačnost u igri“ jedne ekipe.

**Tabela 1.** Deskriptivni statistici analiziranih varijabli

| Varijabla                                                   | AS      | S     | MIN   | MAX   |
|-------------------------------------------------------------|---------|-------|-------|-------|
| Telesna visina (mm)                                         | 1516,20 | 63,93 | 1438  | 1730  |
| Telesna masa (kg)                                           | 39,19   | 4,04  | 25,40 | 47,60 |
| Dužina noge (mm)                                            | 990,30  | 51,99 | 879   | 1129  |
| Obim natkolenice (mm)                                       | 431,28  | 38,01 | 376   | 511   |
| Obim potkolenice (mm)                                       | 296,78  | 23,74 | 248   | 360   |
| Kožni nabor trbuha (mm)                                     | 90,36   | 19,00 | 58    | 185   |
| Kožni nabor leđa (mm)                                       | 83,48   | 25,68 | 41    | 170   |
| Kožni nabor natkolenice (mm)                                | 64,70   | 18,17 | 35    | 122   |
| Kožni nabor suprailijačni (mm)                              | 89,11   | 44,91 | 32    | 282   |
| Preciznost udarca po lopti nogom u horizontalni cilj (poen) | 11,74   | 2,62  | 5     | 17    |
| Preciznost udarca po lopti nogom u vertikalni cilj (poen)   | 11,60   | 2,81  | 0     | 16    |

Legenda: AS – aritmetička sredina; S – standardna devijacija; MIN–minimalne vrednosti rezultata merenja; MAX–maksimalne vrednosti rezultata merenja

Analizirajući rezultate regresione analize *Preciznosti udarca po lopti nogom u horizontalni cilj* (tabela 2) uočava se da ne postoji statistički značajna povezanost sistema prediktorskih varijabli na dati kriterijum ( $P=0,64$ ). Koeficijent multiple korelacije je iznosio  $R=0,36$  što je objašnjavalo 3% zajedničkog varijabiliteta između sistema prediktora i kriterijske varijable. Može se doneti zaključak da neke druge karakteristike i sposobnosti imaju veći uticaj na specifičnu (elevacijsku) preciznost udarca po lopti nogom (stepen treniranosti mišića, tehnika šuta, svakodnevna aktivnost i trening, stanje proprioreceptora – mišićnih vretena, goldžijevog tetivnog organa). Može se pretpostaviti da najviše uticaja ima trenažni proces i tehnika šuta, pa bi u narednim istraživanjima bilo neophodno ocenjivanje tehnike izvođenja šuta, uzimanje u obzir pozicije u timu na kojima igrači igraju i vodeći računa o sportskom stažu fudbalera.

**Tabela 2.** Regresiona Preciznosti udarca po lopti nogom u horizontalni cilj

| Varijabla                      | r     | p    | $r_{part}$ | $p_{part}$ | Beta  | $p_{bete}$ |
|--------------------------------|-------|------|------------|------------|-------|------------|
| Telesna visina (mm)            | -0,02 | 0,43 | -0,01      | 0,92       | -0,01 | 0,92       |
| Dužina noge (mm)               | 0,18  | 0,10 | 0,11       | 0,43       | 0,11  | 0,43       |
| Obim natkolenice (mm)          | 0,03  | 0,41 | -0,01      | 0,95       | -0,01 | 0,95       |
| Obim potkolenice (mm)          | 0,02  | 0,47 | -0,08      | 0,49       | -0,08 | 0,49       |
| Kožni nabor trbuha (mm)        | 0,02  | 0,50 | 0,13       | 0,41       | 0,15  | 0,41       |
| Kožni nabor leđa (mm)          | -0,16 | 0,12 | -0,18      | 0,23       | -0,22 | 0,23       |
| Kožni nabor natkolenice (mm)   | -0,22 | 0,06 | -0,23      | 0,12       | -0,29 | 0,12       |
| Kožni nabor suprailijačni (mm) | 0,07  | 0,30 | 0,23       | 0,12       | 0,26  | 0,12       |

$R=0,36$

Prilagođen  $R^2=0,03$

$P=0,64$

Legenda: r - Pearsonov koeficijent korelacije; p - nivo statističke značajnosti za r;  $r_{part}$  - vrednost koeficijenta parcijalne korelacije;  $p_{part}$  - nivo statističke značajnosti za  $r_{part}$  Beta - regresijski koeficijent;  $p_{bete}$  - nivo značajnosti regresijskog koeficijenta; R - koeficijent multiple korelacije; Prilagođen  $R^2$  – koeficijent determinacije koji je prilagođen za male uzorke; P - značajnost koeficijenta multiple korelacije



Takođe, koristeći linearnu regresionu analizu druge kriterijske varijable, *Preciznosti udarca po lopti nogom u vertikalni cilj* (tabela 3), utvrđeno je da ne postoji statistički značajna povezanost sistema prediktorskih varijabli na drugi ispitivani kriterijum. Značajnost koeficijenta multiple korelacije  $P=0,83$ . Vrednost koeficijenta multiple korelacije iznosila je  $R=0,30$  što je objašnjavalo svega 4% zajedničkog varijabiliteta prediktorskog sistema i kriterijuma. Može konstatovati da i u ovom slučaju neke druge karakteristike i sposobnosti više determinišu rezultate u testu *Preciznosti udarca po lopti nogom u vertikalni cilj* (motivacija, koncentracija stepen treniranosti). Posmatrajući varijable pojedinačno, uočava se da nijedna prediktorska varijabla ne pokazuje statistički značajan uticaj na ispitivani kriterijum. I u ovom slučaju potrebno je analizirati tehniku šuta mladih fudbalera iz Novog Sada. Potrebno je analizirati i brzinu šuta kao i stanje snage mišića natkolenice, potkolenice i trupa.

**Tabela 3.** Regresiona Preciznosti udarca po lopti nogom u vertikalni cilj

| Varijabla                               | r     | p    | $r_{part}$ | $p_{part}$ | Beta  | $p_{bete}$ |
|-----------------------------------------|-------|------|------------|------------|-------|------------|
| Telesna visina (mm)                     | 0,01  | 0,45 | 0,03       | 0,82       | 0,04  | 0,82       |
| Dužina noge (mm)                        | 0,19  | 0,10 | 0,23       | 0,11       | 0,26  | 0,11       |
| Obim natkolenice (mm)                   | 0,02  | 0,43 | -0,02      | 0,89       | -0,02 | 0,89       |
| Obim potkolenice (mm)                   | -0,02 | 0,43 | -0,08      | 0,60       | -0,08 | 0,60       |
| Kožni nabor trbuha (mm)                 | 0,06  | 0,36 | 0,09       | 0,53       | 0,11  | 0,53       |
| Kožni nabor leđa (mm)                   | 0,01  | 0,50 | -0,07      | 0,65       | -0,09 | 0,65       |
| Kožni nabor natkolenice (mm)            | 0,02  | 0,43 | 0,03       | 0,91       | 0,03  | 0,91       |
| Kožni nabor suprailijačni (mm)          | 0,18  | 0,11 | 0,17       | 0,20       | 0,19  | 0,20       |
| $R=0,30$ Prilagođen $R^2=0,04$ $P=0,83$ |       |      |            |            |       |            |

## DISKUSIJA

Rezultati istraživanja ukazali su da ne postoji statistički značajan uticaj prediktorskog sistema sačinjenog od antropometrijskih karakteristika za procenu longitudinalnosti skeleta i voluminoznosti tela na ispitivane kriterijume. Rezultati o nepostojanju statistički značajnih uticaja antropometrijskih karakteristika na izvođenje specifične preciznosti mladih fudbalera ukazuju da druge sposobnosti više determinišu ovu specifičnu motoričku sposobnost, a tu se pre svega misli na iskustvo u sličnim situacijama, stajne mišića, biološka starost ispitanika, uticaji stanja motorinog sistema na preciznost udaraca po lopti, stanja proprioreceptora, i drugih mehanizama odgovornih za ispoljavanje snage – koordinacija tela.

Nervni sistem osigurava prilagođavanje tela na spoljašnju sredinu. Pod uslovima koji se stalno menjaju u toku našeg života nervni sistem reguliše aktivnost svih tkiva i organa, povezuje međusobno organe i delove tela, omogućava povratne informacije, o potrebama tela povezanim sa spoljašnjom sredinom. Na preciznost fudbalera, uticala je proprioreceptivna i receptorna-senzorna sposobnost dečaka koja je nedovoljno razvijena u ovom uzrastu (Cabri et al., 1988). Nervni sistem u ovom periodu se nalazi u periodu sazrevanja, pojačava se protok impulsa, pa su oscilacije od individue do individue veoma različite, a treba uzeti u obzir i ispoljavanje snage i brzine šuta mladih fudbalera.

ra, kao dva bitna faktora preciznosti udarca polopti nogom (Cabri et al., 1988). Autori su utvrdili značajnu vezu između snage nogu, merene kao vrh krive izometrijskog pokreta, i šutiranja lopte pokazane kroz dostignutu distancu. Veza je bila značajna i za koncentrične i za ekscentrične kontrakcije zgloba kuka i kolena u fleksiji i ekstenziji. Povezanost između snage noge i šutiranja lopte pokazuje da trening snage može biti efektivan u poboljšanju parametara fudbalskog šuta (Reilly, 1990). Uz upotrebu tehničkih elemenata, izgleda da trening snage pridodat normalnom režimu treninga, poboljšava i mišićnu snagu i šutiranje lopte. Fudbaleri koji duže treniraju imaju veću brzinu pokreta u poređenju sa kontrolnom grupom koja je kraće u trenažnom procesu, što takođe može biti važna determinanta tehnike u šutiranju lopte.

Istraživanja ovoga tipa mogu dobiti na važnosti jer se danas svetu traže egzaktni podaci. Istraživanja i analize preciznosti, u svetu su već dugo važan segment kako u praksi tako i u teoriji fudbalske igre, pa se čak i prate srčane frekvencije tokom izvođenja preciznih šuteva, metabolička stanja organizma nakon situacionih akcija u kojima se prati preciznost pasa. Dobijeni rezultata se mogu koristiti kao uvid u trenutno stanje i nivo ispoljavanja fudbalskih elemenata pojedinih igrača (skauting), iskoristiti za korekciju i poboljšanje trenažnog procesa ovih fudbalera i poslužiti kao smernice koje pokazuju tendencije razvoja savremene fudbalske igre. Istraživanjem je odbačena postavljena istraživačka hipoteza  $H_1$ , a u daljim radovima potrebno je dati osvrt na trenažno iskustvo, broj treninga nedeljno i snagu fudbalera.

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#### RELATIONS BETWEEN MORPHOLOGICAL CHARACTERISTICS AND ACCURACY OF KICKING A BALL BY YOUNG FOOTBALL PLAYERS.

*Kicking a ball with foot or head are the basic elements of football, which can raise the efficiency of the game in terms of speed of action in terms of scoring goals. This is also the reason why the shot needs to be performed in a timely manner, usefully, and also quickly. The most commonly used kick in football is a inner-side instep kick, which is also the most accurate and the most used among the players (Cabri, De Prof, Dufour, & Clarys, 1998). Success in performing precise actions largely depends on the precision of locating targets in space, and therefore the role of our receptors is crucial for a successful precision (Szekeres, Santrač, Radosav, Toplak, Miljanić, 1994). The research problem was establishing the predictive impacts of morphological characteristics on the elevation accuracy of hitting the ball in young players. The subjects of the study were morphological characteristics and accuracy of the players. Aim of the paper was to determine the relation between morphological characteristics and precision of players aged 13-14 years from the Municipality of Loznica. Method: 50 players of the FK "Kabel" from Novi Sad, aged 13-14 years, were subjected to testing. Measurement of morphological characteristics was conducted. Specific (elevation) accuracy of hitting the ball towards the vertical and horizontal objective was also tested. By using the statistical analysis firstly the basic descriptive statistics of variables were determined. In order to determine the size of the impact of anthropometric characteristics on*

*motor skills – precision, linear regression analysis was used. Results: Regression analysis showed that there was no statistically significant effect of the predictor system of morphological variables on the criterion variables tested: Precision of kicking the ball towards a horizontal target and Precision of kicking the ball towards a vertical target. The results pointed to the fact that some other characteristics and capabilities have greater influence on the manifestation of elevation precision of kicking the ball towards the horizontal and vertical target, in young players. Discussion: Research results indicated that there was no statistically significant effect of the predictor system composed of anthropometric characteristics on assessment of longitudinallity of skeleton and body volume regarding the test criteria.*

„Blic“, 2. april 2015.

### **Konferencija „Sportska dostignuća“**

**PODGORICA** Svečano otvaranje XII MEĐUNARODNE

konferencije „Sportska dostignuća“ održaće se sutra u 16.00

sati u Rektoratu Univerziteta Crne Gore. Na otvaranju će

govoriti predsjednik konferencije prof. dr Duško Bjelica i

ministar prosvjete **PREDRAG BOŠKOVIĆ**. Konferencija

predstavlja najveći regionalni skup u oblasti sporta i ove

godine okuplja preko 160 vodećih svjetskih autora i istraživača

u oblasti sportskih, društvenih i humanističkih nauka i

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## LEISURE TIME OF TEACHERS' TRAINING FACULTY STUDENTS IN SUBOTICA

### Introduction

Health and the life style are connected with physical activity which should be part of everyone's daily routine. Physical activity is considered today to be more important than ever before, especially having in mind increased obesity and following diseases. Insufficient physical activity is related to the progress of different diseases (Kohl, 2001; Blair et al., 2004; Hernelahti et al, 2004) causing 1.9 million premature deaths per year globally (WHO, 2002). Physical activity is closely related to mental health and well-being (Sjogren et al, 2006). It is well known that the type and the amount of physical exercise that one is engaged in plays major role in preserving one's health and body weight. Those people that are constantly engaged in physical activities are not doing that just to preserve their own healthy body weight, but to lower the risk of creating some chronic diseases (Blair & Hardman, 1995). Intensive practice of physical activity is clearly connected with maintaining the body weight. Body engaged in physical activity responds to changes by almost all physiological systems, primarily by bone-muscular, cardiovascular, respiratory, endocrine and immune system (Mikalački, 2005; Mišigoj- Duraković, 2006). The environment and socio-economic conditions have significant role in the occurrence of overweight and obesity related to insufficient physical activity (Dunton, 2005). Exceeding amount of fat is harmful for the body and it presents more load on the joint and on the tissue around the joint which increases the risk of the diseases (tumor of the cologne, kidney, diabetes, heart dieses). Modern society is making fast and constant progress in the area of human activity. Due to automatization and robotization man of today doesn't have major needs for movement during life and work, which at the end brings to lower level of his psycho-physical capabilities. Lack of movement reflects time during man of today lives and works, and it is known that moving represents one of the important health factors (Pantelić et al., 2008).

The purpose of this study was to collect data from students of Teachers' Training Faculty in Hungarian (TTFH) in Subotica about behaving at their free time, focusing on physical activities and sedentary behavior.

### Methods

The TTFH carried out a survey to assess the leisure time habits of their students in academic year 2013/2014. This study included a total of 116 voluntary undergraduate students (male: 23, female: 93). The questionnaire included 31 items, distributed in 5 blocks: descriptive data (3 items), healthy habits (5 items), feeding habits (5 items), sedentary behavior (14 items) and unhealthy behaviors (4 items). The anthropometric

measures body mass and body height were taken using standard procedures and instruments, and accordingly, BMI ( $\text{height/weight}^2$ ) values were calculated. Descriptive statistics and frequency distributions were taken into consideration by gender. Parametric data with t-test, non-parametric data with Mann-Whitney U test were calculated.

## Results

Students reported that they had spent almost one day in a typical week engaged in sedentary behaviors, but they also indicated they had participated in some exercise about 2 hours a week (Table 1).

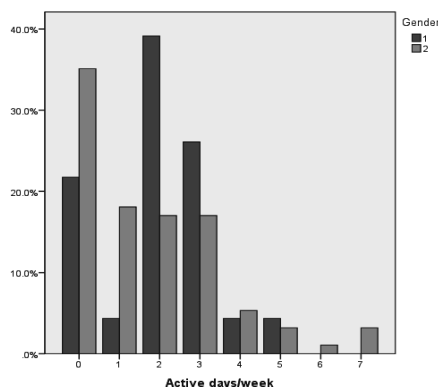
**Table 1.** Gender differences

|                         | Gender | M      | SD     | t     |
|-------------------------|--------|--------|--------|-------|
| BMI ( $\text{kg/m}^2$ ) | M      | 26.54  | 7.14   | 4.97* |
|                         | F      | 21.81  | 2.90   |       |
| TV/DVD (hour/week)      | M      | 8.83   | 6.00   | -1.28 |
|                         | F      | 9.69   | 6.28   |       |
| PC (hour/week)          | M      | 14.83  | 10.84  | 0.43  |
|                         | F      | 13.81  | 10.00  |       |
| Total TV/DVD+PC         | M      | 23.65  | 12.88  | -0.29 |
|                         | F      | 23.50  | 12.45  |       |
| Sport time (min/week)   | M      | 120.00 | 80.91  | 0.89  |
|                         | F      | 99.57  | 101.31 |       |

Legend: M-mean; SD-standard deviation; t-independent-samples t-test;

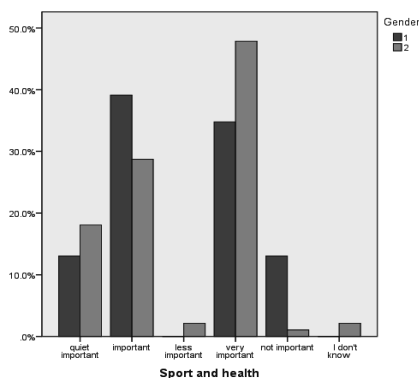
\* $p \leq 0.05$ .

The male students reported greater participation in healthy habits than did female students, and they also spent more time with sedentary behavior, but the difference were not significant. Significant differenced appeared only in BMI variables, where the results pointed out that males at the faculty are inclined towards obesity according to WHO references (2000).



**Figure 1.** How many days are you active a week?

Results showed that 35.1% of female students are not involved in any physical activities during free time, while 65.2% male students are engaged in some physical activities at least 2-3 times a week.



**Figure 2.** Sport is important to be healthy

According to students' opinion, 76.1% of them consider sport as important or very important factor in maintaining health.

### Discussion

The excess body mass and low physical fitness were associated with several metabolic risk factors that increase students' risk of chronic disease (Sasceck et al., 2010). Research has consistently revealed that lifestyle behaviours of the young adult population are putting individuals at an increased risk for the development of numerous chronic diseases later in life (Sparling, 2003). Obesity and higher body weight are strongly associated with a sedentary lifestyle and lack of physical activity in the adult population of the European Union (Martinez-González et al., 1999). During student years, students become more sedentary and as their physical activity levels decrease. Watching television, using the computer or playing video games and finishing tasks for the university, these all belong to the sedentary behavior, and students forget to be physically active. They must find the balance between those areas. The majority of the negative health outcomes that arise from physical inactivity are largely preventable with lifestyle changes. Physical activity patterns during college are important influences on habitual physical activity during the full span of the adult life and, consequently, have significant implications for short- and long- term health outcomes (Sparling, 2003). Living an active lifestyle by integrating physical activity into an individual's daily routine can be an effective way to increase personal fitness (Edwards & Tsouros, 2006).

Students spent little time on physical activities in their free time and a lot of time with sedentary behaviour. It has to be changed, because by time they will become teachers and if they are bad examples to children we can't expect from the society to be

healthier. The role and importance of physical activities should be taught on the universities. With the change of the university curriculum in all year there should be a course related to physical activity for which students could get credits and they would attend it weekly. It would help to make it possible for students to lead not only a sedentary but on active way of life organized by the university.

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#### LEISURE TIME OF TEACHERS' TRAINING FACULTY STUDENTS IN SUBOTICA

*Physical activity contributes to physical, mental and social health and improves the quality of life of people of all ages. Physical inactivity and sedentary lifestyle are a public health problems in Serbia. The purpose of study was to collect a data from students of Teachers' Training Faculty on Hungarian (TTFH) in Subotica about behaving at their free time, focusing on physical activities and sedentary behavior. The TTFH carried out a survey to assess the leisure time habits of their students in academic year 2013/2014. This study included a total of 116 voluntary undergraduated students (male:23, female:93). The questionnaire included 31 items, distributed in 5 blocks: descriptive data (3 items), healthy habits (5 items), feeding habits (5 items), sedentary behavior (14 items) and unhealthy behaviors (4 items). The anthropometric measures body mass and body height were taken using standard procedures and instruments, and accordingly, BMI (height/weight<sup>2</sup>) values were calculated. Surveyed male students reported greater participation in healthy habits than did female students, and they also spent more time with sedentary behavior, but the difference weren't significant. Students spent little time on physical activities in their free time and a lot of time with sedentary behaviour. It has to be changed, because by time they will become teachers and if they are bad examples to children we can't expect from the society to be healthier*

**Key words:** leisure time, physical activity, students.

UDC 796.011

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## THE TIPOLOGY OF PHYSICAL STRUCTURES OF PROGRAM FACILITIES FOR GAMES AND SPORT

### Introduction

This paper presents the analytical research of spatial facilities for playing/games and sport in Montenegro with comparative indicators of solutions for these in the area of Europe.

The categories of the following activities have been included through the typology – classification: children's games, games and sport in schools, free activities of extracurricular youths and adults, top sports and activities and exercises for medical therapeutic purposes.

### Methods

In this example (the covered area of the coastal region of Montenegro) the typology of physical structures will be analysed through program starting points, by observing the total population in Montenegro, which according to the census from 2011 was 620,029 inhabitants, of which 293,509 in the Central Region, in the Southern Region 148,683 and in the Northern region 177,837 persons.<sup>1,)</sup>

The method used is based on the methods of observation, research and analysis that are based on Remplajn's division into life stages (children from three to fourteen years, youths from fifteen to eighteen, the age form nineteen to forty five and over forty five years).

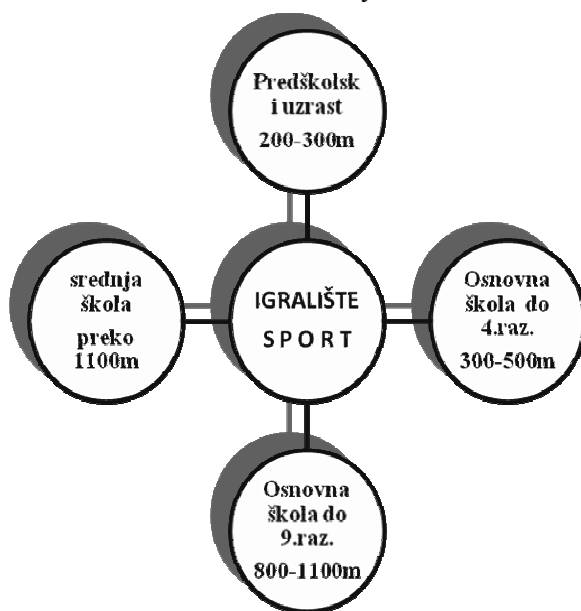
In this division the classification of age categories by period of human life can also be displayed, namely: childhood (0 -1.3 years is the first and second stage of early childhood; from 1.3 - 3.6 years is the third stage of early childhood and from 3.6 - 6 0 years is the first stage of middle to late childhood), education time (6 -10 years is the second stage of middle to late childhood; from 10 -12.5 years is the third phase of the middle to late childhood; from 12.5 - 14.5 years is the age of maturing; from 14.5-17 years is puberty and from 17-20 years is adolescence), the period of maturity-active life and the period of decline of overall human activities – the period of ageing.<sup>2,)</sup>

Typologies of physical structures are in close connection with forms of activities of certain age categories, so these activities are related to *children's play* in the period up to the age of fourteen; *games and sport in schools* from elementary school to high schools up to the age of twenty four; *free and programmed activities of extracurricular youths and adults* from the age of fifteen to forty five years and over forty five years; *sport* (top sport for school and employed youths as well as for the rest of the population in sport) and *activities and exercises for medical therapeutic purposes* (preventive, therapeutic and rehabilitation). Such a typology presented as a program indicator determines the development of planned spatial distribution of facilities which includes the

building plan and plan of equipping facilities on the basis of which spatial organization-grouping is carried out, their function and gravitation zone within certain settlement and region zones.

According to European standards, for example Germany<sup>3.)</sup> (Buddatsch, 2001) the maximum distance from the area for physical activity for children is 300 meters, and for adults taking a straight way is 400 meters or 15 minutes or a maximum of 800 meters or 30 minutes, if we are talking about larger open spaces or recreational areas for the entire city. The norm is for children up to the age of six a minimum of 150m<sup>2</sup> and for children and youth a minimum of 2000m<sup>2</sup>. According to the law of Germany for building playgrounds the minimum is 4 m<sup>2</sup>, compared to the 30's of the 20<sup>th</sup> century when this surface was 1.6m<sup>2</sup>.

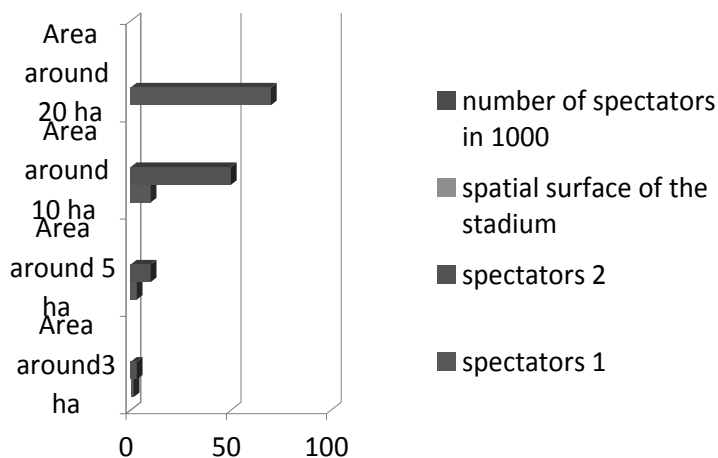
In our country the given base is as shown schematically in Picture 1 or described with the destination time data as follows: as the distance a child covers on foot for 1 minute is about 70 meters, it can practically be taken that the time that a child walks from home to school amounts the most to as many minutes as is the age of the child.



**Picture 1.** Schematic display of the distance from home to the space for outdoor recreation  
Preschool 200-300m; Elementary school to the 4th grade 300-500m; Elementary  
school to the 9th grade 800-1100m; High school over 1100m

Normative equipping of recreation areas for *Facilities intended for outdoor sports* is 3m<sup>2</sup>/inhabitant of which usable area is 1.3m<sup>2</sup>/inhabitant, while supporting area is 1.7m<sup>2</sup>/inhabitant. For *Facilities intended for indoor sports* the standards of 0.50m<sup>2</sup>/inhabitant are applicable.

In addition to outdoor areas, the standard for sports facilities in physical structures has also been given as shown in Picture 2 where the proportional ratio of the number of spectators is set in relation to the surface area of the sports complex in which green areas make about 30% of greenery<sup>4.)</sup>.



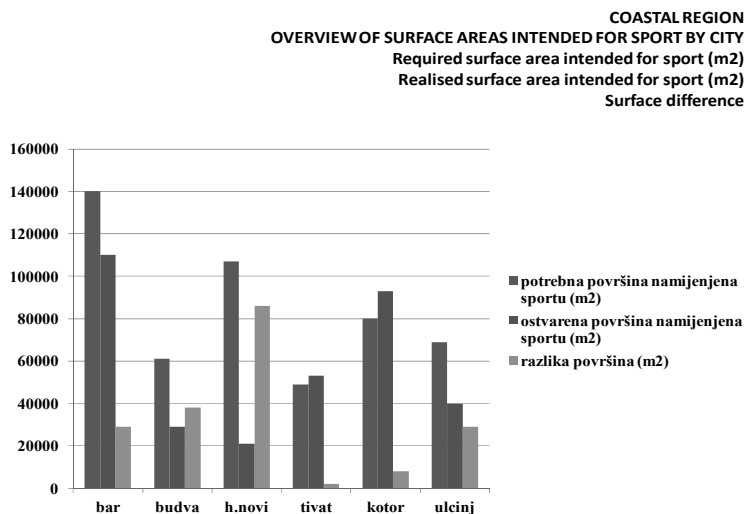
**Picture 2.** Schematic display of the stadium surface in relation to the number of spectators

## Results

Based on the conducted research and analysis on the proposed sample of the coastal region of Montenegro, which are manifold and complex because they include the fields of public, social, medical, urban architectural and other research results are obtained that are based on the classification and the influence of age categories by main periods of human life, the activities, the typology of physical structures and division of facilities for games and sports. By the comparative method in relation to European standards the possibility of grading the analyzed sample of quality of typology of the facility is obtained.

Norms that are approaching European standards for free recreational areas are being applied for educational institutions. Around schools and children's institutions the size of the schoolyard outside the city centre should be 25-35 m<sup>2</sup> per student, taking into consideration only one shift. In the dense tissue of block construction the optimal area per student may be 10-15 m<sup>2</sup> and in no way less than 4 m<sup>2</sup>. In that case, teaching of physical education is held at the nearest sports centre.

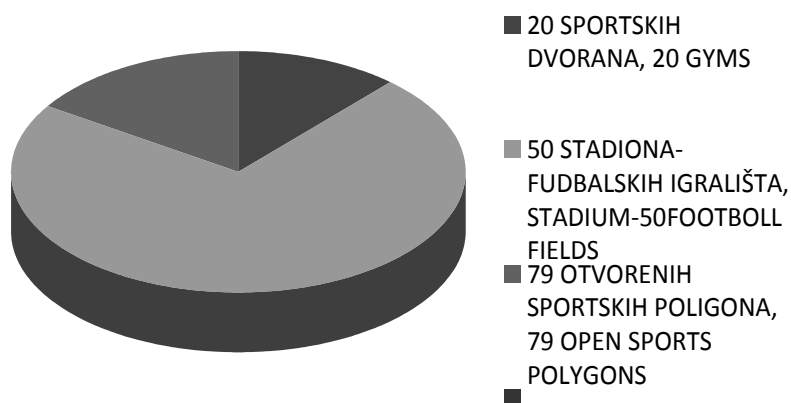
Large fields (sports fields, running tracks, courts for basketball, soccer, volleyball, etc.) for school children, from the age of 7 to 14 years are made within the standard of 6m<sup>2</sup> per child. School playgrounds are also available for other users, out of school activities.



**Picture 3.** Overview of surface areas intended for sport in the coastal region

In **Pictures 3 and 4** sports facilities in the Coastal Region are presented, as well as facilities registered in Montenegro, where in the example of one coastal city – Tivat, one can see that it is best equipped with facilities of sports infrastructure.

### SPORTSKI OBJEKTI U CRNOJ GORI SPORTS FACILITIES IN MONTENEGRO



**Picture 4.** Sports facilities in Montenegro

In the area of the Municipality of Tivat there is a total of 41 sports facilities with developed space of nearly 30,000 m<sup>2</sup> as follows: 1 large hall with 3,200 m<sup>2</sup> and five smaller ones with 1,426 m<sup>2</sup>; 3 sports stadiums with 25,574 m<sup>2</sup>; 9 open fields for small sports with an area of 17.804m<sup>2</sup>; 7 tennis courts with an area of 3,644 m<sup>2</sup>; 5 basketball courts with a surface area of 300m<sup>2</sup>; 6 other indoor terrains with an area of 720m<sup>2</sup>; and 15 bowling areas with a surface of 1.300m<sup>2</sup>. Compared to other sports facilities in Montenegro, Tivat sports facilities represent a significant sports construction fund.

In Montenegro there are: 20 sports halls, as independent facilities or within sports centres with a total surface of about 74,878 m<sup>2</sup> and 27.560 m<sup>2</sup> of spectator seats, 50 stadiums - football fields with the total surface of 464,539 m<sup>2</sup> with the capacity of 81,600 seats; 79 open sports polygons with the total area of 100,720 m<sup>2</sup>; 14 swimming pools, of which 5 indoor and 9 outdoor pools; 66 tennis courts and 7 running tracks with supporting athletic arenas. So, Montenegro has a fund of sports facilities, the density of which in space represents a significant potential and basis for further development of sport.

### Discussion

Here we have an example of research of the coastal region that actually meets the need considering its economic and social status.

However, the question arises with regard to European standards in relation to facilities intended for physical activities whether the other regions in Montenegro can meet them.

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### *THE TIPOLOGY OF PHYSICAL STRUCTURES OF PROGRAM FACILITIES FOR GAMES AND SPORT*

*This paper presents the analytical research of spatial facilities for games and sport in Montenegro with comparative indicators of solutions for these in the area of Europe. The categories of the following activities have been included through the typology – classification: children's games, games and sport in schools, free activities of extracurricular youths and adults, top sports and activities and exercises for medical therapeutic purposes.*

**Keywords:** sport, research, grading, typology, structure

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## **EFEKTI PROGRAMIRANOG RADA TOKOM PRIPREMNOG PERIODA NA TRANSFORMACIJU AGILNOSTI KOD FUDBALERA KADETSKOG UZRASTA**

### **Uvod**

Fudbal je sport koga odlikuju mnogobrojne i raznovrsne složene dinamičke kinetiološke aktivnosti koje se odlikuju velikim brojem cikličnih i acikličnih kretanja. Evidentno je da sva četiri momenta igre, i posjed lopte, i posjed lopte protivnika, i transformacija po osvojenoj lopti i transformacija po izgubljenoj lopti zavisi od mogućnosti igrača da izvede određeno kretanje različitog intenziteta, u različitim pravcima i različitim dionicama igrališta. Oni moraju imati razvijene i bazične i specifične motoričke sposobnosti. Jedna od bazičnih motoričkih sposobnosti koja treba biti na visokom nivou je agilnost.

Agilnost, ili brza promjena pravca kretanja, pravi razliku između prosječnih i najboljih fudbalera. Agilnost se tretira kao kombinacija temeljnih sposobnosti brzine i koordinacije. Pearson (2001) agilnost stavlja u kontekst brzine i eksplozivnosti (speed, agility and quickness – SAQ). Agilnost se definiše kao sposobnost usporavanja, ubrzavanja i promjene smjera kretanja uz održavanje dobre kontrole tijela (ravnoteže) bez smanjivanja brzine. Svaki vrhunski fudbaler koji zagospodari ovom sposobnošću može se nadati najboljim rezultatima, dominaciji na samom terenu. Fudbaleri koji razvijaju brzo kretanje u više smjerova, moći će da izvode sve tehničko-taktičke elemente sa najvećom mogućom eksplozivnošću, brzinom i kontrolom, i sa najmanjim mogućim utroškom energije bez suvišnih i nepotrebnih pokreta.

Problem ovog istraživanja moguće je posmatrati sa dva aspekta:

U primarnom smislu, problemska orijentacija istraživanja odnosila se na utvrđivanje efekata ponuđenog modela trenažnih stimulusa u okviru pripremnog perioda u trajanju od četrdeset dva dana i njihovog uticaja na kvantitativne promjene agilnosti fudbalera kadetskog uzrasta.

U sekundarnom smislu, problem istraživanja je predstavljao i sagledavanje nivoa agilnosti fudbalera kadetskog uzrasta prije i poslije pripremnog perioda.

Predmet ovog istraživanja je bila agilnost fudbalera kadeta, kao i njena varijabilnost izazvana ponuđenim modelom treninga u pripremnom periodu.

Osnovni cilj ovog istraživanja je bio da se utvrdi nivo kvantitativnih promjena agilnosti kod fudbalera kadetskog uzrasta, pod uticajem programiranog fudbalskog treninga koji je obuhvatio jedan pripremini period u trajanju od četrdeset dva dana.

## Metode

Ovo je bilo longitudinalno istraživanje sa ciljem da se u vremenski različite dvije tačke utvrde kvantitativne promjene agilnosti kod fudbalera kadetskog uzrasta (15–godishnjaka  $\pm$  6 mjeseci) pod uticajem programiranog trenažnog rada, koji je obuhvatio ljetnji pripremni period za takmičarsku sezonu u jedinstvenoj kadetskoj ligi Crne Gore i kadetskoj ligi srednje regije Crne Gore. Trenažni program je trajao 42 dana i sproveden je na pomoćnom terenu FK Sutjeska u Nikšiću. Trenažni program je obuhvatio 44 trenažne jedinice, u sklopu kojih je odigrano i 8 prijateljskih utakmica.

Za obradu podataka su uzeti samo rezultati onih ispitanika koji su prošli kompletni program rada i koji su pristupili inicijalnom i finalnom mjerenju.

U ovom istraživanju je obuhvaćen uzorak od 120 mladih fudbalera kadetskog uzrasta, članova 4 kluba, svih iz Nikšića. Svi ispitanici su prije programiranog rada uredno prošli sistematske preglede da bi sa sigurnošću mogli pristupiti trenažnom procesu.

Prilikom izbora mjernih instrumenata (testova) vodilo se računa da oni zadovoljavaju osnovne metrijske karakteristike, da su prikladni uzrastu i objektivnim materijalnim i prostornim uslovima. Za procjenu agilnosti upotrijebljeni su slijedeći testovi:

1. Trčanje sa promjenom pravca pod pravim uglom (MAGTPP)
2. Vijugavo trčanje (MAGVTR)
3. Čunasto trčanje 10x5m (MAGČUT)

Autori ovog rada su odlučili da najveći dio trenažnog programa čini situacioni trening.

Po Michelsu (2001), situacioni trening zasnovan je na modernoj viziji fudbalskog treninga mlađih uzrasta koju propagira UEFA, odnosno usvajanja elemenata fudbalske igre kroz igru.

Vremenska struktura treninga je iznosila od 60 do 120 minuta, zavisno od cilja i zadataka trenažne jedinice i podijeljena je u 3 faze:

- Uvodno-pripremni dio (od 25–30% trajanja treninga)
- Osnovni dio (od 60–65% trajanja treninga)
- Završni dio (do 10% trajanja treninga)

U uvodno-pripremnom dijelu treninga akcenat je bio na podizanju radne temperature kod djece. Kao sredstvo korišćene su razne elementarne igrice sa loptom koje su omogućile rad na elementarnoj tehnici i osnovama taktike, zatim korišćeni su i razni poligoni na vježbama koordinacije. Razne igre i vježbe za povećanje pokretljivosti zglobova i jačanja muskulature takođe su primjenjivane u ovoj fazi.

U prvoj fazi osnovnog dijela treninga intenzitet se nešto povećavao u odnosu na fazu zagrijavanja i trenažni program je realizovan kroz razne igre sa loptom (3:0 u ograničenom prostoru; 3:1 u ograničenom prostoru; 4:2 u ograničenom prostoru; 5:2 u ograničenom prostoru; 6:2 u ograničenom prostoru; igre sa tri boje; 1:1 sa završnicom na gol; 2:1 sa završnicom na gol, 2:2 sa završnicom na gol, razne elementarne igrice sa pomoćnim igračima i džokerima u ograničenim prostorima ili sa završnicom na gol; i



druge). Metodom igre ispitanici su učili, uvježbavali i usavršavali fudbalske vještine kroz veliki broj ponavljanja.

U drugoj fazi osnovnog dijela treninga igrači su imali uglavnom slobodnu igru na dva gola koja im je omogućila kreativno djelovanje i isticanje pojedinca, maštu, slobodno mišljenje i zalaganje, primjenjivanje elemenata koje uče metodom igre u prvoj fazi osnovnog dijela, a samim tim jačanje voljnih kvaliteta. U ovoj fazi treninga intenzitet je bio najveći.

U završnom dijelu treninga zadatak je bio spuštanje fiziološke krive na optimalan nivo, a korišćeni su sadržaji niskog intenziteta: vježbe istezanja i relaksacije, takmičarske igre izvođenja penala, slobodnih udaraca.

Podaci dobijeni istraživanjem obrađeni su postupcima deskriptivne i komparativne statistike.

U segmentu deskriptivne statistike, za svaku varijablu i u inicijalnom i u finalnom stanju obrađeni su centralni i disperzioni parametri kao i mjere asimetrije i spljoštenosti. Hipoteza o normalnoj distribuciji rezultata testirana je na osnovu Kolmogorov i Smirnov testa.

U segmentu komparativne statistike, za utvrđivanje razlika primijenjenih varijabli za procjenu agilnosti na početku (inicijalno stanje) i kraju (finalno stanje) trenažnog programa u pripremnom periodu, korišćena je diskriminativna parametrijska procedura Studentov t-test za velike zavisne uzorke.

## Rezultati

U tabelama 1. i 2. prikazani su osnovni deskriptivni statistički parametri varijabli za procjenu agilnosti u inicijalnom i finalnom mjeranju, gdje su izračunate vrijednosti mjera centralne i disperzione tendencije i to: aritmetička sredina (Mean), standardna devijacija (Std. Dev.), standardna greška aritmetičke sredine (Std. Error), koeficijent varijacije (CV%), minimalne (Minimum) i maksimalne (Maximum) vrijednosti, raspon rezultata (Range), koeficijenti zakrivljenosti (Skewness) i izduženosti (Kurtosis), kao i vrijednosti Kolmogorov i Smirnov testa (K-S test).

Prvo će biti analizirani centralni i disperzioni parametri varijabli za procjenu agilnosti u inicijalnom stanju (tabela 1.)

**Tabela 1.** Centralni i disperzioni parametri varijabli za procjenu agilnosti u inicijalnom stanju

| R.br. | Varijable | Mean  | Std. Dev. | Std. Error | CV%  | Minimum | Maximum | Range | Skewness | Kurtosis | K-S test |
|-------|-----------|-------|-----------|------------|------|---------|---------|-------|----------|----------|----------|
| 1.    | MAGTPPI   | 7,69  | 0,29      | 0,03       | 3,83 | 7       | 8,38    | 1,38  | 0,01     | -0,35    | 0,28     |
| 2.    | MAGVTRI   | 6,71  | 0,27      | 0,02       | 3,96 | 6,22    | 7,40    | 1,18  | 0,46     | 0,27     | 0,47     |
| 3.    | MAGČUTI   | 18,69 | 0,41      | 0,04       | 2,17 | 17,88   | 19,65   | 1,77  | -0,29    | -0,34    | 0,27     |

Analizirajući centralne i disperzione parametre varijabli za procjenu *agilnosti* u inicijalnom stanju – vidi se po vrijednosti standardne devijacije i koeficijenata varijacije da su rezultati u sve tri varijable izrazito homogeni, da im je distribucija normalna.

Vrijednost skewnessa kod varijable trčanje sa promjenom pravca pod pravim uglom (MAGTPPI) govori nam o skoro idealnoj simetričnosti krive, sa vrijednošću kurtosisa koja oslikava blago zaobljenu krivu. Pozitivan skewness kod varijabli kod kojih se mjeri vrijeme, kao kod varijable vijugavo trčanje (MAGVTRI), nam govori o nagnutosti krive ka boljim rezultatima, i obratno, negativan skewness kod varijable čunasto trčanje (MAGČUTI) govori o nagnutosti ka lošijim rezultatima, ali ne i statistički značajnoj nagnutosti u onosu na normalan raspored.

Centralni i disperzioni parametri varijabli za procjenu agilnosti u finalnom mjeranju pokazali su slijedeće vrijednosti (tabela 2.)

**Tabela 2.** Centralni i disperzioni parametri varijabli za procjenu agilnosti u finalnom stanju

| R.Br. | Varijable | Mean  | Std. Dev. | Std. Error | CV%  | Mini-mum | Maxi-mum | Range | Skewness | Kurtosis | K-S test |
|-------|-----------|-------|-----------|------------|------|----------|----------|-------|----------|----------|----------|
| 1.    | MAGTPPF   | 7,44  | 0,25      | 0,02       | 3,35 | 6,93     | 7,99     | 1,06  | 0,05     | -0,47    | 0,10     |
| 2.    | MAGVTRF   | 6,49  | 0,30      | 0,03       | 4,62 | 5,93     | 7,20     | 1,27  | 0,37     | -0,16    | 0,78     |
| 3.    | MAGČUTF   | 18,23 | 0,40      | 0,04       | 2,21 | 17,32    | 19,11    | 1,79  | -0,22    | -0,59    | 0,30     |

Analizirajući centralne i disperzione parametre varijabli za procjenu *agilnosti* u finalnom stanju – vidljivo je da i ovdje aritmetičke sredine kod sve tri varijable za procjenu agilnosti imaju bolje vrijednosti u odnosu na inicijalno stanje. Koeficijenti varijacije kod sve tri varijable govore da rezultati pripadaju izrazito homogenim skupovima, odstupanja od aritmetičke sredine su mala, raspon rezultata je mali, a vrijednost skewnessa i kurtosisa su u granicama male asimetričnosti i zakrivljenosti krive, statistički neznačajnoj. Vrijednosti Kolmogorov i Smirnov testa govore o normalnoj distribuciji rezultata kod sve tri varijable za procjenu agilnosti pokreta.

Za utvrđivanje statističke značajnosti (signifikantnosti) razlika aritmetičkih sredina (parcijalne kvantitativne promjene) varijabli za procjenu agilnosti, primijenjen je **t-test** za velike zavisne uzorke.

Vrijednosti t-testa su bile na nivou značajnosti ili signifikantnosti (Sig.) od 0.01 ( $p \leq .01$ ) u svim varijablama za procjenu agilnosti.

Razlike aritmetičkih sredina inicijalnog i finalnog mjerenja varijabli za procjenu agilnosti, prikazane su u tabeli 3.

**Tabela 3.** Vrijednosti t-testa između aritmetičkih sredina inicijalnog i finalnog mjerenja varijabli za procjenu agilnosti

|       | Varijable | Mean  | Std. Deviation | Std. Error | Correlation | T-test | Sig. |
|-------|-----------|-------|----------------|------------|-------------|--------|------|
| Par 1 | MAGTPPI   | 7,69  | 0,29           | 0,03       | 0,77        | 14,15  | 0,00 |
|       | MAGTPPF   | 7,44  | 0,25           | 0,02       |             |        |      |
| Par 2 | MAGVTRI   | 6,71  | 0,27           | 0,02       | 0,89        | 17,75  | 0,00 |
|       | MAGVTRF   | 6,49  | 0,30           | 0,03       |             |        |      |
| Par 3 | MAGČUTI   | 18,69 | 0,41           | 0,04       | 0,90        | 27,94  | 0,00 |
|       | MAGČUTF   | 18,23 | 0,40           | 0,04       |             |        |      |

Na osnovu dobijenih rezultata može se primijetiti da postoje statistički značajne razlike kod svih varijabli za procjenu agilnosti, te se stoga može reći da je došlo do pozitivnih statistički značajnih parcijalnih efekata trenažnog programa u pripremnom periodu, a vrijednosti t-testa bile su značajne na nivou pouzdanosti  $p < .01$  kod svih varijabli za procjenu agilnosti.

### Diskusija

Na osnovu uvida u dobijene parametre može se zaključiti da su statistički značajni parcijalni kvantitativni efekti (promjene) kod svih varijabli za procjenu agilnosti dobijeni kao rezultat primijenjenog trenažnog programa u pripremnom periodu.

Statistički značajna poboljšanja koja su se dogodila kod ovih varijabli posljedica su situacionog metoda rada koji je primijenjen u ovom trenažnom programu koji obiluje vježbama u kojima dominiraju kretanja sa promjenama pravca.

Na osnovu rezultata t-testa za velike zavisne uzorke, kod varijabli za procjenu agilnosti utvrđene su statistički značajne razlike u svim parovima varijabli između inicijalnog i finalnog stanja, na nivou statističke značajnosti (signifikantnosti)  $p < .01$ . Može se zaključiti da je trenažni program rada u pripremnom periodu doveo do pozitivnih transformacija kod svih varijabli koje su procjenjivale, po strukturi hipotetski postavljenog modela, agilnost.

U ovom istraživačkom radu autori su se rukovodili činjenicom da je ovakav trenažni program rada u pripremnom periodu, gdje uglavnom dominira situacioni model treninga, veoma efikasan način rada u pogledu podizanja nivoa agilnosti fudbalera kadeta. Autori zaključuju da je ljetnji period od 42 dana kod fudbalera kadeta, sa ovakvim trenažnim programom rada, optimalan za podizanje agilnosti na nivo potreban za takmičenje. Dobijeni rezultati se mogu usmjeriti u pravcu inoviranja planova i programa rada u pripremnom periodu, te prilagođavanju istih potrebama dotične populacije.

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Zaciorski, V. M. (1975). *Fizička svojstva sportiste*. Beograd: NIP Partizan.

#### THE EFFECTS OF THE TRAINING IN THE PREPARATION PERIOD ON THE AGILITY TRANSFORMATION WITH CADET LEVEL FOOTBALL PLAYERS

The main aim of the research was to identify a level of quantitative changes of the agility with fifteen years old football players under the influence of the programmed football training of a six weeks preparation period. Methods: According to the time orientation this was a longitudinal study with the aim to define in a two timely different points a quantitative changes of the agility under the influence of the programmed football training with fifteen years old football players under the influence of the programmed training process, which involved a summer preparation period of forty-two days. The training programme covered forty-four training units. The research was made on a sample of 120 cadet level football players. To estimate the agility three tests have been used: running with change of direction under the angle of ninety degrees, winding running and pin running 10x5m. In the area of comparative statistics, for determining differences in the variables used to estimate the agility at the beginning (initial state) and at the end (final state) training program in the preparation period we used discriminant parametric procedure t-test for big paired samples. Results: Based on the numerical values of the t-test it can be concluded that there are no statistically significant differences in all three variables to estimate the agility. This confirmed the hypothesis that the expected significant positive quantitative changes of situational-motor abilities influenced by the proposed model of training in preparation period with fifteen years old football players. Discussion: In this research the authors were guided by the fact that this kind of training program in preparation period, where dominates the situational model training is very effective in terms of raising the agility level with fifteen years old, because the model that is used in this training period abounds in exercises in which dominates change of direction with and without the ball. The obtained results can be directed towards innovation plans and programs in the preparation period, and the adaptation of the same needs of the respective population. References: Bajramovic I, Talovic M, Alic H, Jeleskovic E (2008). Nivo kvantitativnih promjena specifčno-motoričkih sposobnosti nogometaša pod uticajem situacionog treninga, *Sport Mont*, (15, 16, 17./VI), 104–109. Bjelica D (2003). Uticaj fudbalskog treninga na biomotorni status kadeta Crne Gore, *Doktorska disertacija*, Fakultet sporta i fizickog vaspitanja Univerziteta u Beogradu. Bjelica D (2004). Zavisnost tjelesnih sposobnosti od sportskog treninga kod populacije fudbalskih kadeta Crne Gore, *Sport Mont*, (4/II), 58-71. Bjelica D (2005). Sportski trening i njegov uticaj na antropomotoricke sposobnosti fudbalera cetnaestogodisnjaka mediteranske regije u Crnoj Gori, *Sport Mont*, (8-9), 26-41. Michels R (2001). *Teamcoaching: Der Weg zum erfolg durch Teambuilding*, BpF Versand-onli Verlag.

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## **ROLE OF SPORTS MANAGEMENT IN PREPARATION FOR FOOTBALL REFEREES IN THE CZECH REPUBLIC**

### **Introduction**

Football is the most watched sport in the world. It is obvious that the primary factor in the success of the teams is a good workout. It should be noted that football referees have been in this field neglected topic for a long time.

Successfully refereed football game is a guarantee of satisfaction for all stakeholders, players, officials, spectators, shareholders and especially football referees. Today, when the football incredibly accelerated and is under constant surveillance cameras and media, it is necessary to minimize errors in football referees. In football games managed at the certain level, it is necessary to work with referees constantly, i.e. prepare them both physically and mentally, to educate referees in perfect knowledge of the rules and game management.

Term game management can be seen as a technique and tactics to control the match and solve critical situations. In recent years, efforts to establish correct communication with all stakeholders before during and after the match becomes increasingly apparent. As I mentioned in an interview a couple of years ago two situations are never the same ([www.brnenskydenik.cz](http://www.brnenskydenik.cz)). There are many factors that enter the decision making process of a referee, all of which must be correctly analyzed and make a correct decision. These are ones of many reasons to train referees at increasing level to make constant improvement. (That is why several training programs for referees were created). For reasons stated above we implement findings from areas of sport management and human resources management.

According to Armstrong (2007) is HR management seen as a strategicly and logically made approach to controlling the most precious possession within the organization - workers. It is people who work in organization and contribute to achieving the goals.

Another definition by Storey (1989) sees HR management as completion of mutually connected policies deriving from certain ideology and philosophy. Armstrong (2007) considers the main aim of management to control and determine the development, facilitate changes and reaching goals while exploiting certain sources through effective creative and responsible means. The foremost and crucial aim of HR management is the ability of a company to achieve its goals by employees through managing the company in effective ways (Armstrong 2007).

It is obvious that the aims described above are clearly applicable also for particular referee commissions whose aims are correctly carried out matches and education of

referees on the top level. Education programs, which will be presented, were created for this purpose.

### **Methods**

In this article we will use the description of basic education programmes, which are used for the training of football referees in the Czech Republic. It will also describe various groups according to the level of referees in competitions in which they manage football games and we will also describe various requirements that are imposed on these referees, their methods of training and testing their skills and abilities.

### **Results**

Individual football associations and the Union of European Football Associations UEFA are now significantly involved in the preparation resp. work with football referees. The UEFA conducts an educational programme UEFA CORE (Centre of Refereeing Excellence) and also contributes to the financial means to pursue other educational programmes within individual football associations.

### **Referee education**

#### **UEFA education**

UEFA education is carried out on the basis of UEFA Referee Convention which aims to improve the level of referees in all of 54 member countries of UEFA. This Convention consists of several parts: to establish a referee organization within each national association, this association has to be independent from government, league management, clubs, it must guarantee unified level of referees, referees-specialists, by presenting complex educational programme of referees from the basic level to professional footballers; they must constantly improve the state and quality of referees, referees-specialists; defines legal and professional position of referees in various member clubs.

(UEFA Referee Convention: Convention on Referee Education and Organization. *UEFA.com*).

#### **UEFA CORE programme**

UEFA CORE (Centre of Refereeing Excellence) is an educational programme for young international referees in European countries. Each country is a part of UEFA (54 countries) and has an opportunity to send three referees into this programme (one main referee and two assistants). The referees are usually younger than 32 years of age, they are in charge of the highest or the second highest league in their country, they have potential to become FIFA referees within a couple of years and they speak good English. At the end of the training each of the participants receives the UEFA diploma or the UEFA certificate of participation.

Many experts and advisors are committed to the programme. The programmes are managed by internationally recognised experts. Referee-specialist is a person who has been active for last five years and has various experience from areas, referee, instructor, observer, mentor. The programme uses assistance of fitness trainers who supervise the physical training. The experts offer counselling on injury prevention and treatment and healthy lifestyle. These programmes aim to bring up, prepare and improve referees (delegates, mentors) on international level.

The programme seeks to unify the interpretation of rules, so as the decisions are recognised in the manner of football rules and the spirit of the rules. The referees are like a team, The members support each other and their decision should be irreversible and unanimous (Elleray, D., 2011).

### **Programme Talent and Mentor in Czech competitions**

The state programme Talent and Mentor was introduced in Fall 2009. This programme is focused on development of young talented referees who are given the opportunity to develop and take advice from experienced top referees. The programme is assigned for referees at all levels from OFS, KFS to top competitions (Legiersky, B., 2009).

Talent and Mentor should have enough qualities to achieve better referee performance. Both, talent and mentor, influence each other and consult not only the field situations but also possible problems concerning fitness and psychical readiness. The programme should lead to complex development in referees' personality (Legiersky, B., 2010).

The programme consists of: theoretical training, physical training, movement and position, techniques of match management, tactics of match management, game situations. Collina (2003) states that he prefers an active form of regeneration to relax.

An ideal mentor should be interested in young referee growth, pass on his experience, be an expert, have an individual approach and natural authority. At the same time, he should be objective, strict, fair and friendly, and take the referees opinion into account.

### **Educational programme for referees**

This educational programme is for young referees, ideally between 21-30 yrs. For further condition of performance growth in season 2013-2014 the lowest age limit was set on 21 years for the young referees in order to have had some experience from regional competitions. The candidates, who show interest in personal development and have clear vision of their future, are chosen by committee of referees

### **Programme plan**

The basic programme structure of the seminars for referees incorporated into regional competitions and those at different types such as Czech Steering Committee and Moravian Steering Committee is similar. It takes into consideration different level each referee is dealing with.

The programme is made to test football rules, theory of football rules, orders and football norms, the most frequent injuries in football, social network - threat for the future, football violence, racist behaviour, specialized English movement and game positions referees, nutrition and regeneration.

Bahr (2008) describes what nutrition composition, resp. time-frame of nutrition is necessary to use before, during and after the match. The theoretical lectures are completed with video recordings UEFA, SYNOT league, practice sessions in the field for theoretical feedback. UEFA DVD teaching materials are used called FUTURO III, which is an interactive tool for referee skills development. In several sections issues such as fouls judging, handling the ball, advantage, movement of referees, tactical misconducts, are presented.

First they deal with theoretical definitions which are later confronted with rules and consequently, when watching clips from matches, they confront their opinions with reality and correct solutions as offered by FIFA and UEFA. Practical exercises in the field with feedback on the routines of referees and their assistants, cooperation of referees in the field, physical readiness, training programmes.

The participants who are divided into three groups go through eight whole-day seminars. This programme is aiming to improve the qualities of referees. Newly, this year the referees drafted from Czech Steering Committee and Moravian Steering Committee manage chosen matches in the junior league. After that, a match video analysis follows.

The capacity of these programmes is limited, therefore referees should not prolong their stay extensively, in order for the others to have the opportunity to take part. Taking part twice should be the top limit, provided there are other referees willing to enrol.

### **University education at the Masaryk University Brno**

In 2011 a new study programme was established at Masaryk University in Brno. It is an undergraduate programme Physical Education and Sport, specialisation Referee in Collective Sports; Football Referee and Ice-Hockey Referee. The graduates should find their place in sports organizations dedicated to free time activities. Students should have basic abilities and skills necessary for optional graduate programmes as well as managing sport matches.

(<http://www.fsps.muni.cz>).

### **Referees in the Czech Republic**

There are over 4000 football referees registered in the Czech Republic (Nietsche, 2014). These referees are, according to their level, subdivided under individual managing organizations whose competitions they attend to. The football referees can be divided into four performance levels:

1. Referees in professional competitions - first and second league (FAČR)



2. Referees under the authority of Czech commission and managing commission of Moravia - Czech Football League and Moravian-Silesian Football League and Divisions (ŘKČ, ŘKM)

3. Referees under authority of regional commissions - regional championships class 1.A, 1.B (KFS)

4. Referees under authority of local commissions - local championships, classes 3 and 4 (OFS)

Management and education of referees depends on many factors. One of the main factors is implementing referees according to their licence and their participations in educational programmes. Referees at the local level are mostly experienced individuals - former active players who, given the age, do not expect any promotion to higher league (great deal of these referees have previous playing experience from these leagues in their former careers)

Their match referring does not typically involve too much mobility but their experience and forethought helps them to solve any situation accordingly.

The authority of these players has been gradually built up through the years and they know which personality in individual teams to cooperate with to ensure smooth game.

On the other hand, the young beginning referees are those who have only started building their authorities as they slowly gain experience. Field movement in these referees corresponds with their age, they take advantage of their condition as they try to be always in a good position. Most players try to put them under pressure in order to take advantage of them. It is always up to the referees to resist this pressure and keep their head.

If there are more experienced colleagues present at the match they tend to give advice to these referees and point out troublesome players, personalities in teams and useful tactics.

After putting referees into higher level they fall under KFS management, more requirements are made on referees. they should focus on diminishing the mistakes and deficiencies during the match.

At the adult matches, often a delegate is present who can often point out the mistakes the referee fails to notice and often gives good advice and solution to a particular problem, so he can focus on minimizing mistakes or reduce them completely.

On regional level, there is a rule of compulsory video-recordings of the matches which gives referees even better feedback on their performance. Referees of regional championship should be proficient enough to set the strategy and management of the match along with movement and positioning, leading the team of referees and avoiding possible conflicts.

Referees falling under Czech Steering Committee and Moravian Steering Committee are to be physically prepared as the teams are expected to be. Comparing to KFS referees, where there are two seminars and one physical test yearly, these referees

have four physical tests and two seminars each season. Also the preparation for matches tends to be more thorough. TOP-KP referees undergo the same procedure as KFS and additionally they take part in seminars and physical tests on the level of Czech Steering Committee and Moravian Steering Committee. They also have a chance to be present at matches as referee - assistants on the level of adults and youths.

The referees of Czech Steering Committee and Moravian Steering Committee do not take part in Football National League but they are present at junior and youth matches. Referees of OFS, KFS and Czech Steering Committee and Moravian Steering Committee only take part in seminars and physical testing. During seminars they get familiar with findings of sport-technical commission, disciplinary commission, changes in rules, problems and how to solve them, i.e. hosting services, unsuitable behaviour of officials, spectators leaving the game.

During seminars there are tests from rules that differ in difficulty according to ones level (OFS, KFS, Czech Steering Committee and Moravian Steering Committee). These referees do not have much chance for further education unless put into some of the educational programmes (VPR, Talent a mentor). Each of them is responsible for own individual training as they must meet requirements at the testing.

They have seldom any experience with psychological preparation. It would be suitable to put lectures on psychology on the programme, such as preparation for matches in long, middle and short term perspective, self-regulation of psychical state along with relaxation techniques. In physical tests a pre-match warm up should be scheduled or have a short demonstration with an expert.

The SYNOT League and Football National League referees are divided into two categories - the main referee and the assistant referee. The referees meet twelve-times a year at seminars and four times at physical tests. These referees are treated by specialists in particular areas i.e. UEFA-bound lecturers, fitness trainers, sport science specialists, tactical and mental preparation specialists. The seminars focal points are: match video analysis, taking lessons from mistakes, fitness decision tactics (theory, practise), mental preparation, media communication, nutritional habits and lifestyle. Referees - assistants practise offside situations via camera recordings. Referees on this level sign an ethical codex. They are under pressure of psychological demands as many of the matches are broadcasted. They feel they are under constant surveillance where any bigger mistake is analysed in media. That is the reason why suitable psychological preparation is scheduled at seminars and the top experts are present.

### **Discussion**

Referees who are involved in education programmes, unlike the referees attending merely seminars, have the advantage of professional growth. It is also up to an individual referee if he exploits these offers. If he is already taking apart in these programmes he should be initiative and work on his carrier to use this chance because there are other referees who are deprived of this. These have only a limited chance for further education. If they want educate themselves they must wait to work up the opportu-

nity to slip into these programmes i.e. pass the exams and to be accepted to Masaryk University and find the necessary information on the internet.

The education of referees abroad (those who are part of UEFA) and derive their approach from UEFA conventions about football referees support, have four main tasks in education when using the best findings of referee education. Recruitment and keeping referees, system of work with referee observers, upbringing of talented referees and educating referees from beginners to top level. Education of referees in the Czech Republic and abroad follows direction and rules of UEFA convention. There are differences in number of referees and allocated finances which are proportional to the size of individual football competitions.

The aim of the education is also arranging meetings of experts, UEFA members, who can mutually learn from each other, share their experience from various areas which can be generally beneficial for upbringing the level of their expertise. The Czech Republic has taken the education of referees far ahead as it has been made part of university education and it is only Masaryk University in Brno that can take pride in presenting the unique study programme - The Referee of Collective Sports at Faculty of Sport Studies

([www.rozhodci-ks.cz](http://www.rozhodci-ks.cz), [www.rozhodci-fotbal.cz](http://www.rozhodci-fotbal.cz)).

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### **ROLE OF SPORTS MANAGEMENT IN PREPARATION FOR FOOTBALL REFEREES IN THE CZECH REPUBLIC**

*In our article we present the preparation of football referees, who are duly registered under the banner of the FACR - Football Association of the Czech Republic. We deal with referees who participate in various educational programmes and seminars led by teachers and other practitioners.*

*The main communication represents the structure of competition in the Czech Republic, a way of organizing football referee and course preparation at individual levels of performance, which is unique within Europe. Introducing individual education programmes that the FACR used for education and training of referees. We also describe the content of curricula for football referees. An integral part of the article is also showing the claims that are arbitration subject to performance levels.*

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## **ANALIZA USVOJENOSTI TEHNIKE OSNOVNOG ZAOKRETA U ALPSKOM SKIJANJU U ODNOSU NA MOTORIČKE SPOSOBNOSTI ISPITANIKA**

### **1. UVOD**

Razvoj i usavršavanje skijaške opreme, prije svega skija, uslovio je pojavu i proizvodnju prvih karving skija, koje su omogućile i razvoj nove, karving tehnike skijanja (Hirano i Tada, 2005; Horterer, 2005). Karving tehnika skijanja omogućava cijelo vrijeme vožnju po rubnicima u zaokretu i izvođenje zaokreta bez otklizavanja (Horterer, 2005). Takođe, pojava karving skijanja, uslovlila je određene inovacije u samom procesu učenja skijaške karving tehnike.

U procesu učenja skijaške tehnike, kao prelazna faza između elemenata pluzne i paralelne skijaške tehnike danas se često upotrebljava tzv. klinasti položaj skija, kod kojega su zadnji djelovi skija manje rašireni u odnosu na pluzni položaj (Lešnik i sar. 2002). Osnovni zaokret je element skijaške tehnike, zastupljen je u programu osnovnog oblika skijanja, za koji se može reći da predstavlja kombinaciju klinastog zaokreta i paralelne tehnike skijanja (Hadžić, 2008). Koristeći klinasti položaj skija u procesu učenja, bilo kao metodičku vježbu ili kao element skijaške tehnike, postepeno se vrši priprema za izvođenje zaokreta u cjelosti pomoću paralelne skijaške tehnike (Cigovski i sar. 2010). Pored pomenutog, postoji i pristup koji u procesu učenja u potpunosti izostavlja elemente pluzne skijaške tehnike (Murovec, 2006). Može se reći da uspjeh u alpskom skijanju primarno zavisi od nivoa usvojenih specifičnih motoričkih znanja. (Kuna i sar., 2008). Zbog toga je važno, u cilju uspješnog usvajanja skijaške tehnike, prepoznati faktore koji imaju značajnu ulogu u usvajanju tehnike skijanja, kod skijaša početnika (Joksimović i sar., 2009).

Upravo zbog novog pristupa u procesu učenja alpskog skijanja, cilj u ovom radu je da se utvrdi razlika u stepena usvojenosti tehnike osnovnog zaokreta u odnosu na nivo motoričkih sposobnosti ispitanika (skijaških početnika).

### **2. METOD RADA**

Kao što je već u uvodnom izlaganju predočeno, ključni problem, kao i celokupna problemska orijentacija ovog istraživanja, vezuje se za analizu izvođenja osnovnog zaokreta u alpskom skijanju u odnosu na nivo motoričkih sposobnosti ispitanika.

Procjena motoričkih sposobnosti ispitanika, izvršena je na Fakultetu za sport i fizičko vaspitanje, neposredno prije odlaska studenata na obaveznu praktičnu nastavu skijanja. Procjena tehnike skijanja izvršena na padinama Kopaonika nakon realizova-

nog sedmodnevnog programa. Nastavu skijanja, u trajanju od sedam dana, ukupno 42 časa, izvodila su tri nastavnika po identičnom programu. U toku dana nastava je realizovana sa fondom od 6 časova gdje je na prva tri časa bilo primijenjeno učenje elementarnih skijaške tehnike a na tri časa poslije podne uvježbavanje istih.

Ocenjivači su bili nastavnici koji posjeduju ogromno iskustvo u ovom poslu. Svaki ispitanik, ocijenjen je od strane sva tri ocenjivača iz različitih pozicija. Tri nezavisna ocenjivača dala su ocjene svakom ispitaniku za demonstraciju odabranog elementa skijaške tehnike – osnovni zaokret a kao konačna ocjena uzimala se srednja vrijednost tri ocjene. Procjena usvojenosti nivoa skijaškog znanja kod ispitanika valorizovana je ocenom od 1 do 5 koji se odnosi na vođenje skija lukom zaokreta, kontrola brzine na osnovu završavanja zaokreta, odgovarajući stav koji osigurava težište u sredini stopala, lakoća i mekoća demonstracije. U postupku obrade podataka, na osnovu konačnih (formiranih) ocjena u izvođenju osnovnog zaokreta, uzorak ispitanika je podijeljen na tri subuzorka koji su nazvani: **loši**, **umjereni** i **dobri**. Prvi subuzorak (**loši**) su činili ispitanici koji su dobili ocjenu 1 za prikazanu tehniku osnovnog zaokreta, drugi subuzorak (**umjereni**) su činili ispitanici koji su dobili ocjene 2 i 3 za prikazanu tehniku osnovnog zaokreta i treći subuzorak (**dobri**) su činili ispitanici koji su dobili ocjene 4 i 5 za prikazanu tehniku osnovnog zaokreta

U uzorak ispitanika ušlo je 30 studenata Fakulteta za sport i fizičko vaspitanje, skijaških početnika muškog pola, prosječne starosti 22 godine, podijeljenih na 3 subuzorka u odnosu na tehniku izvođenja osnovnog zaokreta i to: prvi subuzorak broji 14 ispitanika svrstanih u kategoriju (loši), drugi subuzorak broji 9 ispitanika svrstanih u kategoriju (umerena) i treći subuzorak čini 7 ispitanika svrstanih u kategoriju (dobra).

U uzorak mjenih instrumenata iz prostora motoričkih sposobnosti uzeti su: skok u dalj iz mjesta, bacanje medicinke sa grudi iz sjedanja na stolici, trčanje na 20 m iz visokog starta, okretnost u vazduhu, koraci u stranu, 20 iskoraka sa provlačenjem palice).

### **Kriterijska varijabla u ovom istraživanju je bila osnovni zaokret.**

U skladu sa ciljem istraživanja izabrana je odgovarajuća metoda statističke obrade podataka. U prvom dijelu prikazani su centralni i disperzioni parametri, mere asimetrije i spljoštenosti u odnosu na praćene parametre. U drugom dijelu analizirana je razlika između ispitanika u tehnici izvođenja osnovnog zaokreta u odnosu na motoričke sposobnosti ispitanika.

Da bi se izbeglo gubljenje informacija, pronalaženjem najfinijih veza i saznanja, na neparametrijskim veličinama, izvršeno je skaliranje podataka na tabelama kontingenције. Na osnovu izloženog vidi se da je na skaliranim podacima moguća primjena multivarijantne analize varijanse (MANOVA), i diskriminativne analize (DISKRA).

## **3. REZULTATI I DISKUSIJA**

### **3.1 Osnovni parametri motoričkih sposobnosti ispitanika u odnosu na osnovni zaokret**

U ovom istraživanju, analiza je sprovedena na (6) šest varijabli za procjenu motoričkih sposobnosti i to: skok u dalj iz mjesta, bacanje medicinke sa grudi iz sjedanja

na stolici, trčanje na 20 m iz visokog starta, okretnost u vazduhu, koraci u stranu, 20 iskoraka sa provlačenjem palice.

Centralni i disperzioni parametri, mere asimetrije i spljoštenosti praćenih obeležja motoričkih sposobnosti reprezentuju osnovni zaokret i usmjeravaju na mogućnost primjene parametrijskih postupaka.

**Tabela 1.** Centralni i disperzioni parametri i mjere asimetrije i spljoštenosti motoričkih sposobnosti subuzorka - „loši“ (14)

|         | M      | SD    | Min   | Max   | CV    | Range  | Sk     | Ku    | p     |      |
|---------|--------|-------|-------|-------|-------|--------|--------|-------|-------|------|
| MSDM    | 223.07 | 33.95 | 135.0 | 256.0 | 15.22 | 203.46 | 242.68 | -1.35 | 1.20  | .763 |
| MBACMED | 401.50 | 63.04 | 280.0 | 485.0 | 15.70 | 365.09 | 437.91 | -.63  | -.74  | .960 |
| MTRČ20M | 3.49   | .40   | 2.8   | 4.4   | 11.33 | 3.26   | 3.72   | .81   | .61   | .854 |
| MOKRVAZ | 4.26   | .61   | 3.7   | 5.6   | 14.20 | 3.91   | 4.61   | 1.12  | -.07  | .412 |
| MKORSTR | 22.90  | 3.24  | 15.2  | 29.0  | 14.16 | 21.03  | 24.77  | -.52  | .90   | .966 |
| MISKP   | 53.78  | 29.05 | 19.7  | 100.0 | 54.02 | 37.00  | 70.55  | .46   | -1.42 | .425 |

Legenda: M (aritmetička sredina), SD (standardna devijacija), Min (minimalna vrijednost), Max (maksimalna vrijednost), CV (koeficijent varijacije), Range (raspon), Sk (standardizovani koeficijent asimetrije, zakrivljenost), Ku (standardizovani koeficijent izduženosti ili spoljaštenosti).

**Minimalne** (min) i **maksimalne** (max) vrijednosti motoričkih sposobnosti subuzorka „loši“ ukazuju da se vrijednosti nalaze u očekivanom rasponu. Veće vrijednosti **koeficijenta varijacije** (CV) koje ukazuju na heterogenost subuzorka „loši“ mogu se uočiti kod testa 20 iskoraka sa provlačenjem palice (MISKP) (54.02). Vrijednosti **koeficijenta varijacije** (CV) koje ukazuju na homogenost subuzorka uočavaju se kod skoka u dalj iz mjesta (MSDM) (15.22), bacanja medicine sa grudi iz sjedanja na stolici (MBACMED) (15.70), trčanja na 20 m iz visokog starta (MTRČ20M) (11.33), okretnost u vazduhu (MOKRVAZ) (14.20), koraci u stranu (MKORSTR) (14.16).

Distribucija vrijednosti uglavnom se kreće u okviru normalne raspodjele (p) kod primijenjenih varijabli. Pregledom dobijenih mjera distribucije standardizovanog koeficijenta asimetrije (skjunisa), uočava se da povećane vrijednosti **Skjunisa** (sk) ukazuju da je raspodjela *negativno asimetrična*, to znači da kriva raspodjele rezultata naginje ka većim vrijednostima, odnosno da ima više većih vrijednosti u odnosu na normalnu raspodjelu kod sledećih varijabli: trčanje na 20 m iz visokog starta (MTRČ20M) (.81), okretnost u vazduhu (MOKRVAZ) (1.12), 20 iskoraka sa provlačenjem palice (MISKP) (.46).

Smanjene vrijednosti **Skjunisa** (sk) ukazuju da je raspodjela *pozitivno asimetrična*, to znači da kriva raspodjele rezultata naginje ka manjim vrijednostima, odnosno da ima više manjih vrijednosti u odnosu na normalnu raspodjelu kod sledećih varijabli: skok u dalj iz mjesta (MSDM) (-1.35), bacanje medicine sa grudi iz sjedanja na stolici (MBACMED) (-.63), koraci u stranu (MKORSTR) (-.52).

Veće vrijednosti **Kurtozisa** (ku) ukazuju da je kriva izdužena, kod sledećih varijabli: skok u dalj iz mjesta (MSDM) (1.20), trčanje na 20 m iz visokog starta (MTRČ20M) (.61), koraci u stranu (MKORSTR) (.90). Negativne vrijednosti **Kurtozisa** (ku) ukazuju da je kriva spljoštena, kod sledećih varijabli: bacanje medicine sa grudi iz sjedanja na stolici (MBACMED) (-.74).

danja na stolici (MBACMED) (-.74), okretnost u vazduhu (MOKRVAZ) (-.07), 20 iskoraka sa provlačenjem palice (MISKP) (-1.42).

**Tabela 2.** Centralni i disperzioni parametri i mjere asimetrije i spljoštenosti Motoričkih sposobnosti subuzorka - umerena (9)

|                | M      | SD    | Min   | Max   | CV    | Range         | Sk    | Ku    | p    |
|----------------|--------|-------|-------|-------|-------|---------------|-------|-------|------|
| <b>MSDM</b>    | 237.89 | 14.07 | 208.0 | 255.0 | 5.92  | 227.07 248.71 | -1.09 | .34   | .857 |
| <b>MBACMED</b> | 440.00 | 46.84 | 355.0 | 490.0 | 10.65 | 403.99 476.01 | -.83  | -.65  | .993 |
| <b>MTRČ20M</b> | 3.37   | .32   | 2.8   | 3.9   | 9.47  | 3.12 3.61     | -.01  | -.89  | .952 |
| <b>MOKRVAZ</b> | 4.19   | .20   | 3.8   | 4.5   | 4.82  | 4.04 4.35     | -.40  | -.23  | .981 |
| <b>MKORSTR</b> | 22.21  | 1.28  | 20.5  | 24.2  | 5.77  | 21.22 23.19   | .39   | -1.08 | .931 |
| <b>MISKP</b>   | 68.99  | 32.75 | 24.3  | 98.5  | 47.47 | 43.81 94.17   | -.67  | -1.49 | .675 |

Legenda: M (aritmetička sredina), SD (standardna devijacija), Min (minimalna vrijednost), Max (maksimalna vrijednost), CV (koeficijent varijacije), Range (raspon), Sk (standardizovani koeficijent asimetrije, zakrivljenost), Ku (standardizovani koeficijent izduženosti ili spljoštenosti).

**Minimalne** (min) i **maksimalne** (max) vrijednosti motoričkih sposobnosti subuzorka “umerena” ukazuju da se vrijednosti nalaze u očekivanom rasponu. Veće vrijednosti **koeficijenta varijacije** (CV) ukazuju na heterogenost subuzorka “umerena” kod testa 20 iskoraka sa provlačenjem palice (MISKP) (47.47). Vrijednosti **koeficijenta varijacije** (CV) koje ukazuju na homogenost subuzorka, uočavaju se kod skok u dalj iz mjesta (MSDM) (5.92), bacanje medicine sa grudi iz sjedanja na stolici (MBACMED) (10.65), trčanje na 20 m iz visokog starta (MTRČ20M) (9.47), okretnost u vazduhu (MOKRVAZ) (4.82), koraci u stranu (MKORSTR) (5.77). Distribucija vrijednosti uglavnom se kreće u okviru normalne raspodjele (p) kod primijenjenih varijabli. Povećane vrijednosti **Skjunisa** (sk) ukazuju da je raspodjela *negativno asimetrična*, to znači da kriva raspodjele rezultata naginje ka većim vrijednostima, odnosno da ima više većih vrijednosti u odnosu na normalnu raspodjelu, kod varijable koraci u stranu (MKORSTR) (.39). Smanjene vrijednosti **Skjunisa** (sk) ukazuju da je raspodjela *pozitivno asimetrična*, to znači da kriva raspodjele rezultata naginje ka manjim vrijednostima, odnosno da ima više manjih vrijednosti u odnosu na normalnu raspodjelu, kod sledećih varijabli: skok u dalj iz mjesta (MSDM) (-1.09), bacanje medicine sa grudi iz sjedanja na stolici (MBACMED) (-.83), okretnost u vazduhu (MOKRVAZ) (-.40), 20 iskoraka sa provlačenjem palice (MISKP) (-.67). Vrijednosti **Skjunisa** (sk) ukazuju da raspodjela nije asimetrična kod varijable trčanje na 20 m iz visokog starta (MTRČ20M) (-.01). Veće vrijednosti **Kurtozisa** (ku) ukazuju da je kriva izdužena, kod varijable skok u dalj iz mjesta (MSDM) (.34). Negativne vrijednosti **Kurtozisa** (ku) ukazuju da je kriva spljoštena, kod sledećih varijabli: bacanje medicine sa grudi iz sjedanja na stolici (MBACMED) (-.65), trčanje na 20 m iz visokog starta (MTRČ20M) (-.89), okretnost u vazduhu (MOKRVAZ) (-.23), koraci u stranu (MKORSTR) (-1.08), 20 iskoraka sa provlačenjem palice (MISKP) (-1.49).



**Tabela 3.** Centralni i disperzioni parametri i mjere asimetrije i spljoštenosti  
Motoričkih sposobnosti subuzorka - dobra (7)

|                | M      | SD    | Min   | Max   | CV    | Range  | Sk     | Ku    | p     |       |
|----------------|--------|-------|-------|-------|-------|--------|--------|-------|-------|-------|
| <b>MSDM</b>    | 231.14 | 19.64 | 194.0 | 247.0 | 8.50  | 212.97 | 249.31 | -1.10 | -.25  | .896  |
| <b>MBACMED</b> | 408.29 | 64.04 | 290.0 | 490.0 | 15.68 | 349.05 | 467.52 | -.72  | -.17  | 1.000 |
| <b>MTRČ20M</b> | 3.47   | .26   | 3.1   | 3.8   | 7.42  | 3.23   | 3.71   | .01   | -1.14 | .974  |
| <b>MOKRVAZ</b> | 4.33   | .36   | 3.9   | 4.9   | 8.42  | 3.99   | 4.66   | .38   | -.91  | .993  |
| <b>MKORSTR</b> | 22.15  | 3.01  | 16.5  | 26.0  | 13.58 | 19.36  | 24.93  | -.74  | .06   | 1.000 |
| <b>MISKP</b>   | 69.37  | 31.95 | 21.9  | 94.0  | 46.06 | 39.81  | 98.92  | -.89  | -1.12 | .870  |

Legenda: M (aritmetička sredina), SD (standardna devijacija), Min (minimalna vrijednost); Max (maksimalna vrijednost), CV (koeficijent varijacije), Range (raspon), Sk (standardizovani koeficijent asimetrije, zakrivljenost), Ku (standardizovani koeficijent izduženosti ili spljoštenosti).

**Minimalne** (min) i **maksimalne** (max) vrijednosti motoričkih sposobnosti subuzorka “dobara” ukazuju da se vrijednosti nalaze u očekivanom rasponu. Veće vrijednosti **koeficijenta varijacije** (CV) ukazuju na heterogenost subuzorka “dobara” kod testa 20 iskoraka sa provlačenjem palice (MISKP) (46.06). Vrijednosti **koeficijenta varijacije** (CV) koje ukazuju na homogenost subuzorka, uočavaju se kod skok u dalj iz mjesta (MSDM) (8.50), bacanje medicine sa grudi iz sjedanja na stolici (MBACMED) (15.68), trčanje na 20 m iz visokog starta (MTRČ20M) (7.42), okretnost u vazduhu (MOKRVAZ) (8.42), koraci u stranu (MKORSTR) (13.58).

Distribucija vrijednosti uglavnom se kreće u okviru normalne raspodjele (p) kod primijenjenih varijabli. Povećane vrijednosti **Skjunisa** (sk) ukazuju da je raspodjela *negativno asimetrična*, to znači da kriva raspodjele rezultata naginje ka većim vrijednostima, odnosno da ima više većih vrijednosti u odnosu na normalnu raspodjelu, kod varijable okretnost u vazduhu (MOKRVAZ) (.38). Smanjene vrijednosti **Skjunisa** (sk) ukazuju da je raspodjela *pozitivno asimetrična*, to znači da kriva raspodjele rezultata naginje ka manjim vrijednostima, odnosno da ima više manjih vrijednosti u odnosu na normalnu raspodjelu, kod sledećih varijabli: skok u dalj iz mjesta (MSDM) (-1.10), bacanje medicine sa grudi iz sjedanja na stolici (MBACMED) (-.72), koraci u stranu (MKORSTR) (-.74), 20 iskoraka sa provlačenjem palice (MISKP) (-.89). Vrijednosti **Skjunisa** (sk) ukazuju da raspodjela nije asimetrična kod varijable trčanje na 20 m iz visokog starta (MTRČ20M) (.01).

Veće vrijednosti **Kurtozisa** (ku) ukazuju da je kriva izdužena, kod varijable koraci u stranu (MKORSTR) (.06). Negativne vrijednosti **Kurtozisa** (ku) ukazuju da je kriva spljoštena, kod sledećih varijabli: skok u dalj iz mjesta (MSDM) (-.25), bacanje medicine sa grudi iz sjedanja na stolici (MBACMED) (-.17), trčanje na 20 m iz visokog starta (MTRČ20M) (-1.14), okretnost u vazduhu (MOKRVAZ) (-.91), 20 iskoraka sa provlačenjem palice (MISKP) (-1.12).

### 3.2 Analiza razlika između ispitanika na osnovu izvođenja osnovnog zaokreta u odnosu na motoričke sposobnosti

U ovom podpoglavlju će se dokazati ili odbaciti tvrdnja da postoji značajna razlika između ispitanika u tehnici izvođenja osnovnog zaokreta, u odnosu na motoričke sposobnosti.

**Tabela 4.** Značajnost razlike između ispitanika u tehnici izvođenja osnovnog zaokreta u odnosu na motoričke sposobnosti

| Analiza | n | F    | p    |
|---------|---|------|------|
| MANOVA  | 9 | .711 | .779 |
| DISKRA  | 9 | .667 | .818 |

U tabeli 4 prikazani su rezultati Multivarijatne analize varijanse (MANOVA) i diskriminativne analize (DISKRA) da bi se utvrdilo da li postoji statistički značajna razlika i jasno definisana granica između subuzoraka u tehnici izvođenja osnovnog zaokreta u odnosu na motoričke sposobnosti.

Kada se posmatraju motoričke sposobnosti pojedinačno uz primjenu multivarijatne analize varijanse (MANOVA), može vidjeti na osnovu vrijednosti  $p=.779$  da ne postoji statistički značajna razlika između ispitanika u tehnici izvođenja osnovnog zaokreta.

Takođe, kada se posmatraju motoričke sposobnosti kao sistem uz primjenu diskriminativne analize (DISKRA), na osnovu vrijednosti  $p=.818$ , jasno se pokazuje da nije uočena razlika i jasno definisana granica između ispitanika u tehnici izvođenja osnovnog zaokreta.

**Tabela 5.** Značajnost razlike između ispitanika u tehnici izvođenja osnovnog zaokreta u odnosu na motoričke sposobnosti

|         | F     | p    |
|---------|-------|------|
| MSDM    | .880  | .427 |
| MBACMED | 1.221 | .311 |
| MTRČ20M | .372  | .693 |
| MOKRVAZ | .170  | .844 |
| MKORSTR | .256  | .776 |
| MISKP   | .928  | .407 |

Pregledom tabele 5, uz primjenu multivarijatne analize varijanse (MANOVA), dobijena vrijednost  $p > .1$ , to znači da nije uočena značajna razlika između ispitanika u tehnici izvođenja osnovnog zaokreta u odnosu na motoričke sposobnosti kod skok u dalj iz mjesta (.427), bacanje medicine sa grudi iz sedanja na stolicu (.311), trčanje na 20 m iz visokog starta (.693), okretnost u vazduhu (.844), koraci u stranu (.776), 20 iskoraka sa provlačenjem palice (.407) tj. kod svih 6 primijenjenih varijabli.

Na osnovu dosadašnjih razmatranja i analize uzorka od 30 ispitanika, u skladu sa primenjenom metodologijom, logički slijed istraživanja je određivanje karakteristika

i homogenosti svakog subuzorka na osnovu izvođenja tehnike osnovnog zaokreta i distance između njih.

Činjenica da je  $p = .818$ , diskriminativne analize (DISKARA), znači da ne postoji jasno definisana granica između subuzoraka na osnovu izvođenja tehnike osnovnog zaokreta, odnosno nije moguće odrediti karakteriske svakog subuzorka na osnovu izvođenja osnovnog zaokreta u odnosu na motoričke sposobnosti.

S obzirom na broj ispitanika pojedinih subuzoraka (loša 14, umerena 9, dobra 7), bili smo obazrivi prilikom tumačenja rezultata i pored toga što je kod svih varijabli normalna raspodjela svakog subuzorka. Mali broj uzorka je doprinio nešto većoj disperziji i više izraženim latentnim razlikama.

#### 4. ZAKLJUČAK

Analiziran je uzorak od 30 ispitanika koji je podeljen u 3 subuzorka, na osnovu ocjena dobijenih za prikazanu tehniku izvođenja osnovnog zaokreta

U skladu sa ranije utvrđenim ciljevima istraživanja, metodološkim pristupom, u ovom istraživanju analizirala se razlika između ispitanika u tehnici izvođenja osnovnog zaokreta u odnosu na motoričke sposobnosti.

**Na osnovu dobijenih rezultata, njihove interpretacije mogu se izvesti sledeći zaključci:**

Kod motoričkih sposobnosti **ispitanika uz primjenu multivarijatne analize varijanse** (MANOVA .779) **i diskriminativne analize** (DISKARA .818), dobijeni rezultati ukazuju da ne postoji razlika između 3 subuzorka u tehnici izvođenja osnovnog zaokreta u odnosu na primijenjene motoričke sposobnosti.

*Kako nijesu ustanovljene razlike i jasno definisane granice između subuzoraka, shodno tome, nijesu određene karakteristike i homogenost svakog subuzorka.*

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*ACHIEVEMENT DIFFERENCES OF BASIC MEANDERING OVER MOTOR ABILITIES OF SKIERS*

*This research analyzes the techniques involved in basic winding in relation to motor abilities of subjects. The aim of this study was to determine the difference in the technique of primary windings in relation to motor abilities of subjects. Methods: In a sample of 30 students, average age 22 years, male, measuring by 6 measures of motor abilities and one situational-motor test was carried out. The data were analyzed by SPSS 20.0 version. Results and discussion: Alpine skiing technique was assessed through primary windings, the technical element of skiing which is present in the basic form of skiing. Based on these results, we can conclude that, the differences are established and boundaries are clearly defined in the level of adoption of the basic winding techniques between subsamples in relation to motor abilities.*

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**REPORT OF THE 12TH INTERNATIONAL SCIENTIFIC CONFERENCE  
“TRANSFORMATIONAL PROCESSES IN SPORT – SPORTS  
ACHIEVEMENTS”**

The Montenegrin Sports Academy, in cooperation with the Faculty of Sport and Physical Education and the University of Montenegro, traditionally organises, in the first week of April, an annual scientific conference on multidisciplinary areas of sport science. Aside from the above-mentioned institutions, the organisation of this year's 12th International Scientific Conference on Transformational Processes in Sport titled “Sports Achievements”, held from 2 – 5 April 2015, was joined by similar international organisations, primarily by the Faculty of Sport and Physical Education of the University of Novi Sad (Serbia), and also by the Faculty of Sports Studies of the Masaryk University in Brno (Czech Republic), the Institute of Sport Science of the University of Münster (Germany), Pansport Medical from Bucharest (Romania), the Institute for Combat Sports and Science from Boston (England), and the European College of Sport Science from Köln (Germany). Traditional support for organising the conference came from the Ministry of Education and the Ministry of Science.

The Montenegrin Sports Academy, in cooperation with the Faculty of Sport and Physical Education, and under the auspices of the University of Montenegro, has for a few years been trying to raise the standards as far as the scientific conference is concerned, as well as when serial publications are concerned, and to attain the highest world standards. This year's scientific conference published on its official website a call for submissions in August last year, and the deadline for the submission of papers passed on 15 January 2015. Simultaneously with receiving the papers, the Organising and Scientific Committees worked hard, headed by the president of the conference prof. dr Duško Bjelica, and the presidents of the Scientific Committee doc. dr Stevo Popović and doc. dr Selçuk Akpınar from the Nevşehir University in Cappadocia (Turkey), and, as was determined, concluded the list of accepted papers for presentation on the first day of February. On that day, 110 scientific papers were accepted for presentation, for which 196 names, i.e. authors were registered, from 32 different countries (Albania, Australia, Bosnia and Herzegovina, Bulgaria, Ghana, Croatia, Czech Republic, Egypt, Germany, Hungary, India, Indonesia, Iran, Japan, South Korea, Kosovo, Macedonia, Malaysia, Montenegro, Nigeria, Romania, Saudi Arabia, Serbia, Slovakia, Slovenia, Spain, Taiwan, Tunisia, Turkey, Ukraine and the United Kingdom) and 4 continents. The areas allowed by the call for submissions were: biomechanics, motor learning, adapted physical education, biochemistry, health and fitness, molecular biology, neuromuscular physiology, nutrition, physiology, physiotherapy, rehabilitation, sports medicine, training and testing, traumatology, computer sciences, economics,

history, philosophy and ethics, physical education and pedagogy, psychology, sociology, sports management, analysis and statistics in sport. However, for objective reasons, a number of authors did not complete their registration, so the final number of papers that entered the official programme of the conference was reduced to 94 papers, for which 166 names, i.e. authors were registered, from 22 different countries (Albania, Australia, Bosnia and Herzegovina, Bulgaria, Ghana, Croatia, Czech Republic, Germany, India, Indonesia, Japan, South Korea, Kosovo, Montenegro, Saudi Arabia, Serbia, Slovakia, Slovenia, Taiwan, Turkey and the United Kingdom) and 4 continents.

It is very important to mention the authors that constituted, or are still constituting the Scientific Committee, bearing in mind that they represent eminent names in multidisciplinary areas of sport science: Ana Maria Botelho Teixeira, Antonino Bianco, Asterios Deligiannis, Beat Knechtle, Bojan Jošt, Carlos Goncaves, Daniela Dasheva, Dejan Madić, Elena Taina Avramescu, Ferman Konukman, Feza Korkusuz, George Giatsis, Georgi Georgiev, Gregor Jurak, Guilherme Giannini Artioli, Haris Pojskić, Hugo Sarmiento, Jan Konarski, Jonatan Ruiz Ruiz, Jorge Knijnik, Joško Sindik, Juan Manuel Cortell Tormo, Julio Calleja-Gonzalez, Lateef O. Amusa, Mahboub Sheikhali-zadeh Heris, Maike Tietjens, Manuel J. Coelho-e-Silva, Michael Chia, Michal Vit, Milan Čoh, Morteza Jourkesh, Mustafa Levent Ince, Nejc Šarabon, Praveen Kumar, Raúl Arellano, Rusu Ligia, Sandra Mandić, Sergej Ostojić and Yang Zhang. On the other hand, organising the conference at such a high level would not have been possible without the strong contribution of the members of the Organising Committee: Jovica Petković, Dobrislav Vujović, Rašid Hadžić, Dragan Krivokapić, Kemal Idrizović, Miroslav Kezunović, Aldijana Muratović, Ivan Vasiljević, Danilo Bojanić, Marija Bujanja and Jovan Gardašević, as well as the Honorary Committee presided by the Montenegrin Prime Minister Mr Milo Djukanovic, and constituted by: Bojana Petrović-Popović, Dragan Adžić, Dušan Simonović, Igor Vušurović, Izet Rado, Katarina Bulatović, Milorad Vuletić, Miodrag Božović, Nikola Janović, Petar Porobić, Predrag Bošković, Radmila Vojvodić, Sanja Vlahović, Slavoljub Stijepović, Srdjan Kovačević, Srdjan Mrvaljević, Vesna Medenica and Zoran Milošević.

The opening ceremony of the 12th International Scientific Conference on Transformational Processes in Sport titled “Sports Achievements” was held on Friday, 3 April at 4 pm. The foreword was given by the president of the Conference who, in a moving speech, initiated a two-day activity that was conducted in two oral sessions (consisting of two panels each, with eleven and thirteen presentations, respectively) and two poster sessions (consisting of two panels each, with ten and twelve presentations, respectively). The special guest of the Conference was supposed to be prof. dr Branislav Antala who is a full professor at the Comenius University in Slovakia, who also holds the office of president of FIEP Europe – an organisation founded in 1923 in Brussels, which aims to promote a wide range of activities in the area of education, sport and physical education and school sport, sport for everyone, fitness and health and recreation for people of all ages in all countries of the world, and to cherish and contribute to international cooperation in these areas. His federation is recognised by the International Olympic Committee, and has strong links with the International

Council of Sport Science and Physical Education, the Arabian Confederation of Sport – L'Union Arabe de l'Éducation Physique et du sport, and other international organisations. However, due to his busy schedule, professor Antala cancelled his attendance and sent the vice president of his organisation prof. dr Gheorghe Balint, who on his behalf received the golden plaguette which the Montenegrin Sports Academy awarded to prof. dr Branislav Antala for his personal contribution to the development of the science of sport at the international level. The plaquette was also awarded during the opening ceremony.

As is the case with every scientific conference, our conference also involved guest lecturers. This year's guests at the 12th International Scientific Conference on Transformational Processes in Sport titled "Sports Achievements" held from 2 – 5 April 2015, represent eminent names coming from the partner institutions, primarily prof. dr Henk Erik Meier from the University of Münster in Germany (who talked about the relationship between sport and national identity in unified Germany), prof. dr Zdenko Reguli from the Masaryk University in the Czech Republic (who talked about the falls in sport), prof. dr Višnja Djordjić from the University of Novi Sad (who talked about the promotion of physical education in sport), and prof. dr Martin Zvonar from the Masaryk University in the Czech Republic (who talked about the specific biomechanical shoes for pregnant women).

It is also worth emphasising that two workshops were also organised, which allowed those present at the Conference to meet one of the most prominent scientists in the area of sport science in former Yugoslavia, prof dr Nejc Šarabon from the University of Primorska from Slovenia, who presented a modern approach to narrowing the gap between research in the area of sport science and the needs of industry, through practical examples. The workshops also allowed those present at the conference to meet the representative of the world-renowned company Desys, which produces EMG systems (systems for electromyography) dr Stephen Lindley, who represented the newest achievements in this area.

It is worth emphasising that the Conference participants have the opportunity to publish their papers in the reputable journal *Sport Mont* which represents an international journal registered at four index bases, among which is the EBSCO base (section: Academic Journal) which the Faculty of Sport and Physical Education at the University of Belgrade recognises as an international journal in the M24 category, which puts it at the same level of journals from the SCI, SSCI and SCIE, as far as the selection of their academic staff is concerned. This journal has an ascending trend of recognisability in the world, so we do expect its further improvement and faster inclusion in a larger number of important index bases. On the other hand, the authors of the best papers received a special invitation to prepare the text for the other journal published by the Montenegrin Sports Academy titled "Montenegrin Journal of Sports Science and Medicine", which is already highly ranked and represents one of the best journals in Montenegro. It is important to emphasise that over 46 index bases recognised the quality of this journal and introduced it into their index pages. The greatest success represents the impact factor of 6.17 which the Index Copernicus assigned to this journal. By compari-

son, certain journals in the same area, that already exist in the Thomson Reuters bases, never reached above the impact factor of 6.00, which gives the hope to the editorial board of this journal that it will soon enter the Scopus and ProQuest, as well as the Thomson Reuters bases.

During the closing ceremony of the Conference, prof. dr Dobrislav Vujović, the dean of the Faculty of Sport and Physical Education of the University of Montenegro, and prof. dr Maïke Tietjens, the representative of the Institute of Sport Science of the University of Münster, signed an agreement of international cooperation between these two institutions.

**REPORT OF THE 12TH INTERNATIONAL SCIENTIFIC CONFERENCE  
“TRANSFORMATIONAL PROCESSES IN SPORT – SPORTS  
ACHIEVEMENTS”**

*The article discusses the highlights of the 12th International Scientific Conference on Transformation Processes in Sport "Sports Performance", which was held in Podgorica, Montenegro. The event comprised oral and poster sessions. Professor, Duško Bjelica, President of Montenegrin Sports Academy, gave one of the opening speeches.*